UG-2257
ADAU186x Hardware Reference Manual

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Evaluating the ADAU186x Three ADCs, One DAC, Low Power Codec with Audio DSPs

SCOPE

This user guide provides a detailed description of the ADAU186x functionality and features, includes ADAU1860 and ADAU1861.

Note: the HPVDD pin in ADAU1860 is same as the AVDD pin in the ADAU1861. Package differences information referred to it their relative data sheets.

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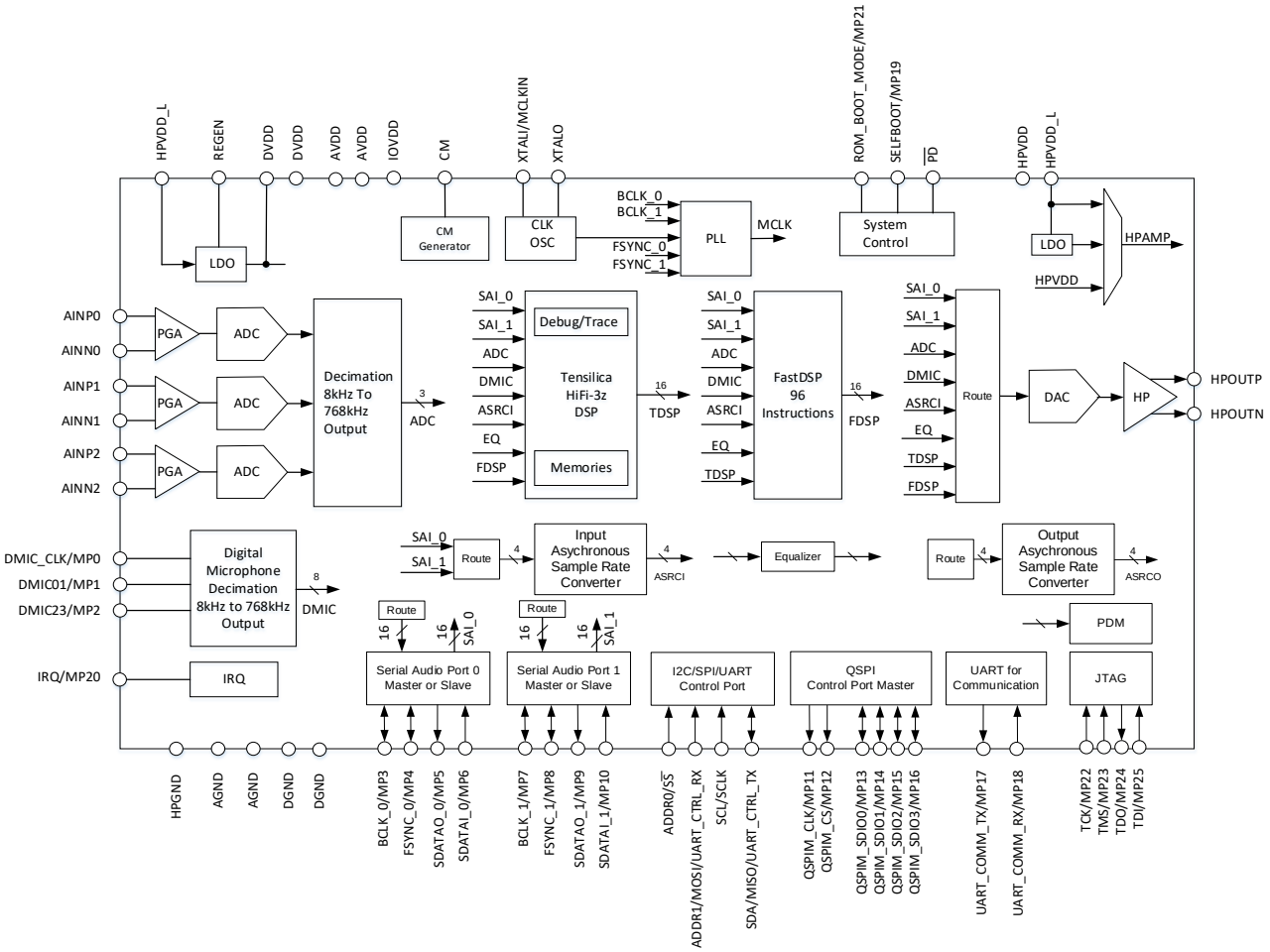


Figure 1. ADAU186x Block Diagram

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REVISION HISTORY

7/2024—Revision 0: Initial Version

USING THE ADAU186x HARDWARE USER GUIDE

NUMBER NOTATIONS

Table 1. Number Notations

Notation	Description
Bit N	Bits are numbered in little endian format, that is, the least significant bit of a number is referred to as Bit 0.
0xNN	Hexadecimal (Base 16) numbers are preceded by the prefix 0x.
0bNN	Binary (Base 2) numbers are preceded by the prefix 0b.

REGISTER ACCESS CONVENTIONS

Table 2. Register Access Conventions

Access	Description
RW	Memory location has read and write access.
R	Memory location is read access only. A read always returns 0, unless otherwise specified.
W	Memory location is write access only.

ACRONYMS AND ABBREVIATIONS

Table 3. Acronyms and Abbreviations

Acronym/Abbreviation	Description
Σ-Δ	Sigma delta
ADC	Analog-to-digital converter
ADP	Audio Data Path
AFE	Analog front end
DMA	Direct memory access
JTAG	Joint test action group
LSB	Least significant byte/bit
MMR	Memory mapped register
MSB	Most significant byte/bit
RMS	Root mean square
Rx	Receive
SPI	Serial peripheral interface
Tx	Transmit
UART	Universal asynchronous transmitter

INTRODCUTION TO THE ADAU186X

The ADAU186x is a low power audio codec with an optimized audio processing core, making it ideal for noise cancelling applications that require high quality audio, low power, small size, and low latency. The three ADC and one DAC channels have the SNR of at least 105 dB and the THD + N of at least −90 dB. The two serial audio ports are compatible with I²S, left justified, right justified, and TDM modes, with tristating for interfacing to digital audio data. The analog operating voltage is 1.8 V, and an optional internal regulator can be used to generate the digital supply voltage. If desired, the regulator can be powered down and the digital supply voltage can be supplied externally which is determined by the REG_EN pin.

The input signal path includes flexible configurations that can accept differential or single-ended analog microphone inputs as well as up to four digital microphone inputs. Each input signal has its own programmable gain amplifier (PGA) for volume adjustment.

The ADCs and DAC are high quality, 24-bit Σ-Δ converters that operate at a selectable 12kHz to 768kHz sampling rate and the ADCs also support 8kHz or 16kHz sampling rate in the voice wake up mode. The ADCs and DAC have an optional high-pass filter with a cutoff frequency of 1 Hz and fine-step digital soft volume controls.

The DAC output is capable of differentially driving a headphone earpiece speaker with 16 Ω impedance or higher. There is also the option to change to line output mode when the output is lightly loaded.

The Tensilica HiFi-3z core (TDSP) is optimized for low-power audio processing. It allows the ADAU186x to provide more flexible solutions to meet more complicated applications.

The FastDSP core has a reduced instruction set that optimizes this codec for noise cancellation. The program and parameter RAMs can be loaded with custom audio processing signal flow built using the **Lark Studio.** The values stored in the parameter RAM control individual signal processing blocks.

The ADAU186x also has a self-boot function that can load the program and parameter RAMs of both cores along with the register settings on power-up using an external flash memory over the quad SPI interface. The external flash memory is fully memory mapped to the HiFi-3z bus fabric.

The ADAu186x architecture is shown in Figure 2.

Use the **Lark Studio** to program and control the cores through the control port. Along with designing and tuning a signal flow, the tool can configure all ADAU186x registers. The graphical user interface (GUI) of the tool allows anyone with digital or analog audio processing knowledge to design the DSP signal flow and export the flow to a target application. The interface also provides enough flexibility and programmability for an experienced DSP programmer to have control of the design. In ADAU186x development tool, the user can connect graphical blocks (such as biquad filters, volume controls, and arithmetic operations), compile the design, and load the program and parameter files into the ADAU186x memory through the control port. The tool also allows the user to download the design to an external flash memory for self-boot operation.

The ADAU186x can generate the internal clocks from a wide range of input clocks by using the on-board bypassable fractional PLL. The PLL accepts inputs from 30 kHz to 36 MHz. For standalone operation, the clock can be generated using the on-board crystal oscillator.

The ADAU186x is provided in a small, 56-ball, 2.980 mm × 2.679 mm WLCSP package.

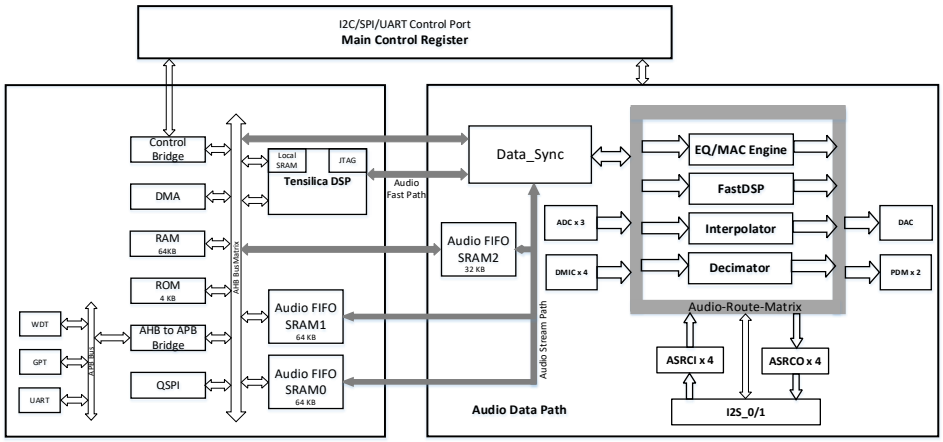


Figure 2 ADAU186x Architecture

SYSTEM CLOCKING AND POWER-UP

POWER-UP/POWER-DOWN OPERATION AND OPTIONS

When pulled low, the $\overline{PD}$ pin puts the chip in the lowest power state, hardware full chip power-down. If the regulator is being used, it also powers down during this state. The chip will fully reset in this state and retain no state memory. It's impossible to communicate with the ADAU186x when the $\overline{PD}$ pin is low.

By default, out of reset, the chip is in its lowest power state (hibernate 2 mode) that can be entered via a control interface, software full chip power-down. To enter or exit this power state, use the PWR_MODE bits in the CHIP_PWR register. When PWR_MODE is 2'b00, the I2C/SPI/UART control ports are operational, and everything else is powered down except the DVDD regulator, the crystal and the input interrupt pins if used. The digital portion of the chips has several power domains. By default, only the domain that powers the control ports and their associated registers are powered on, the rest of the digital design has its power supplies gated and its state is lost.

There are several options to retain various additional state memory during software full chip power down. The SOC_MEM_RETAIN and ADP_MEM_RETAIN bits can be used to retain the state of the memory in the Tensilica-HiFi-3z core, FDSP and the EQ engine. The control register map always retains its state when PWR_MODE is 2'b00.

When PWR_MODE is 2'b00, the CM pin or the common-mode output can either maintain its state or not by using the CM_KEEP_ALIVE bit. If CM_KEEP_ALIVE = 0, the CM voltage will be lost when PWR_MODE = "00", thus producing the lowest possible software power-down current. However, if CM_KEEP_ALIVE = 0, the ADAU186x will have a longer turn on time because the PLL and other analog blocks rely on the CM voltage. A wait time of 35 ms is needed for CM to charge before any analog blocks, such as the PLL, can be enabled.

Conversely, with CM_KEEP_ALIVE=1, the power-down current is higher, but the start-up time is faster because the 35ms wait time can be omitted.

If CM_KEEP_ALIVE=0, use the CM_STARTUP_OVER register to fast charge the CM voltage to have the lowest turn on time. This can be done by setting CM_STARUP_OVER=0 before PWR_MODE changes from hibernate 2 mode. Then after the 35ms wait time, set CM_STARUP_OVER=1 to keep power consumption low. The reset state of CM_STARTUP_OVER is 0. Therefore, if the $\overline{PD}$ pin is used to power down the part, the step of setting to 0 can be omitted.

When PWR_MODE is not 2'b00, the power supplies on the rest of the digital portion of the chip are enabled. Therefore, this register must be set first during the power-up sequence.

The PLL, crystal or frequency multiplier must be configured and enabled after CM_STARTUP_OVER sequencing is completed.

After all the internal digital power supplies are powered up, and other needed sequencing is completed, the POWER_UP_ COMPLETE bit or an interrupt request (IRQ) indicates this. The IRQx_POWER_ UP_COMPLETE bit indicates the power up complete event. If the IRQs are used to request an interrupt after POWER_UP_ COMPLETE, the IRQs must be unmasked. The IRQx_POWER_ UP_COMPLETE_MASK bit must be cleared. By default, the IRQs for POWER_UP_COMPLETE are masked.

After POWER_UP_COMPLETE=1, the memories can be programmed.

The ADAU186x has highly flexible block level power controls. Each individual channel of each block can be powered on or off separately. There is a control bit, MASTER_BLOCK_EN, that by default is 0 and that overrides all block level gating except PLL, XTAL, TDSP and FDSP. The PLL, XTAL, Tensilica core, and FastDSP can be enabled, even when MASTER_ BLOCK_EN = 0. All other blocks are always in power-down in this state, allowing the PLL to be enabled and locked and the DSP memories to be initialized before all other signal path blocks are enabled.

When configuring the device, it is recommended to fully set up all control registers and block level power controls to their desired state, to allow the PLL to lock, to initialize the DSP memories to be used, and then to enable the blocks by setting MASTER_BLOCK_EN = 1.

Block level power controls and other settings can be changed on-the-fly while the chip is active. However, care must be taken when enabling or disabling blocks other than the DAC and/or headphone mode blocks that are actively routed out to the DAC and/or headphone mode as audible artifacts may occur.

To power down the chip, set MASTER_BLOCK_EN=0 and PWR_MODE= 2'b00. The device then powers down all blocks and performs any required power-down sequencing.

When change the power mode from Active Mode or Hibernate 0 mode to Hibernate 1 mode or Hibernate 2 mode, which means to power off SOC domain, the SOC domain will be powered off directly if the TDSP_WAIT_MODE_BYP bit is set to 1, and the SOC domain will be powered off after WAITI instruction is executed if the TDSP_WAIT_MODE_BYP bit is cleared.

An overview of the power-up sequencing follows:

1. Applied AVDD, HPVDD, IOVDD, DVDD and HPVDD_L power supplies to ADAU186x following the power sequence in Power Supply Sequencing section.
2. Pull the $\overline{PD}$ pin high to enable the whole chip and the DVDD LDO if used.
3. There need at least 35ms delay for the CM voltage setting before any further operation of the ADAU186x.
    a) If the SELF_BOOT pin is low (ADAU186x works in Non-self-boot mode), the external MCU need wait at least 35ms after pull high the $\overline{PD}$ pin and then set the STARTUP_DLYCNT_BYP bit to 1.
    b) If the SELF_BOOT pin is high (ADAU186x works in self-boot mode), the ADAU186x will count 38282 clock cycles based on the clock on the MCLKIN pin to make sure the CM voltage is ready and then starts to download program/data from the external Flash through the QSPI interface. The counter can be bypassed by setting the STARTUP_DLYCNT_BYP bit to 1. Note that the program in the external Flash also need to follow the steps below.
4. Set CM_STARUP_OVER=1 for power saving if need.
5. Configure the master clock and other frequencies in the ADAU186x through the CLK_CTRLx registers and set the PLL_EN bit if use.
6. Configure all other setup bits while the PLL is locking (or at any other time after $\overline{PD}$ pulled high).
7. Ensure that all digital power domains are finished powering up, and the clocks are stable by reading the PLL_LOCK bit if PLL is used. Verify POWER_UP_COMPLETE bit and if this bit is set to 1, proceed further or wait until this bit is set to 1.

DVDD LDO REGULATOR

There is a DLDO (DVDD LDO regulator) that can optionally generate the DVDD power with using the HPVDD_L supply as the DLDO input. If the REG_EN pin is tied to ground, this regulator is disabled, and an appropriate DVDD voltage must be supplied externally on the DVDD pin. If the REG_EN pin is tied to AVDD, the DLDO regulator is enabled and generates the required DVDD voltage. If the DLDO is used, necessary decoupling capacitors need to be added on the DVDD pins. The DLDO_CTRL bits determine the voltage of the LDO output as shown in Table 4. By default, the output is 0.9V.

The DLDO requires the CM voltage to operate. Therefore, even if CM_KEEP_ALIVE = 0, the CM output remains present as long as the REG_EN=1. Therefore, to achieve the lowest possible power-down power when REG_EN = 1, set $\overline{PD}$ low.

Table 4. DVDD Output Setting

DLDO_CTRL[2:0]	Output Voltage of DVDD [V]
000	0.85
001	0.9
010	0.95
011	1.0
100	1.05
101	1.1

HPLDO REGULATOR

The on-chip HPLDO regulator is provided as an option for the headphone amplifier power supply. A better power supply noise will be achieved with using this HPLDO.
The HPLDO outputs voltage can be controlled with the HPLDO_VOUT_SET bits. Customer can disable and bypass the HPLDO by using the HPLDO_BYPASS bit and HPLDO_EN bit in the HPLDO_CTRL register.
On power up, the HPLDO will be disabled by default, and the headphone amplifier will work with the HPVDD power supply.

CLOCK INITIALIZATION

The master clock of the ADAU186x can be generated from the PLL, the frequency multiplier or directly from the input source frequency. The input frequency source can be selected by the PLL_SOURCE bits.

When a crystal oscillator is used, the crystal oscillator function must be enabled in the XTAL_EN and XTAL_MODE bits. Even if the crystal oscillator isn't used, the XTAL_EN bit shall be enabled for the system clock consideration.

If the PLL is used, it must always be set to output clock at 24.576MHz, 49.152MHz, 73.728MHz or 98.304MHz.

If frequency multiplier is used, a fixed 24.576MHz clock should be provided on the input source frequency pin.

The PLL and the frequency multiplier can be bypassed by setting the PLL_FM_BYPASS bit as 1 and using the input source frequency to provide the master clock: providing a fixed clock like 24.576MHz, 49.152MHz, 73.728MHz or 98.304MHz via the PLL_SOURCE bits and appropriate MCLKIN/BCLK_x/FSYNC_x pins.

All the other clocks in the ADAU186x are derived from the master clock and they can be configured though the registers.

- TDSP_CLK_RATE bits determine the clock of the Tensilica-HiFi 3z DSP core
- BUS_CLK_RATE bits determine the bus clock in the Tensilica-HiFi 3z DSP, including the AHB and APB bus
- FFSRAM_RATE bits determine the clock of the FIFO and SRAM

- UART_CTRL_CLK_RATE bits determine the clock used in the UART block. Note that the clock to the UART block shall not exceed 49.152MHz.
- AON_CLK_RATE bits determine the clock for the always on blocks in the ADAU186x. It recommends setting the AON_CLK_RATE as 0.

There are some rules need to follow during the clock configuration:

- The clock of the Tensilica-HiFi 3z DSP core shall higher than or equal to the bus clock. And the bus clock shall higher than or equal to the clock of the FIFO and SRAM.
- The ratio between the clock of Tensilica-HiFi 3z and bus clock shall keep at 1:1, 2:1, 3:1 or 4:1.
- Bus clock shall be configured at 24.576MHz, 49.152MHz, 73.728MHz or 98.304MHz.

The configuration of the MCLK_FREQ_INDEX bits shall match with the setting of the master clock, otherwise the ADAU186x will not work correctly.

All internal sampling rates specified within the datasheet is 48kHz which is based on the assumption that a master clock frequency is set at one of the below options: 24.576MHz, 49.152 MHz, 73.728MHz or 98.304MHz and the MCLK_FREQ_INDEX setting matches with the master clock. If the master clock is not in the four options, all internal sampling rates adjust accordingly. For example, if the master clock is 32.768 MHz while the MCLK_FREQ_INDEX setting is "01" (49.152MHz), all internal sampling rates must be adjusted as 32.768MHz÷ 49.152MHz × 48 kHz, which is 32kHz.

The ADAU186x provides clock gating to save power. Each function block's clock can be gated through the appropriate bit in the ADP_CLOCK_EN bits.

If the master clock is higher than 49.152MHz (not include 49.152MHz), 1.1V voltage should be provided on the DVDD.

Clocking Architecture Block Diagram

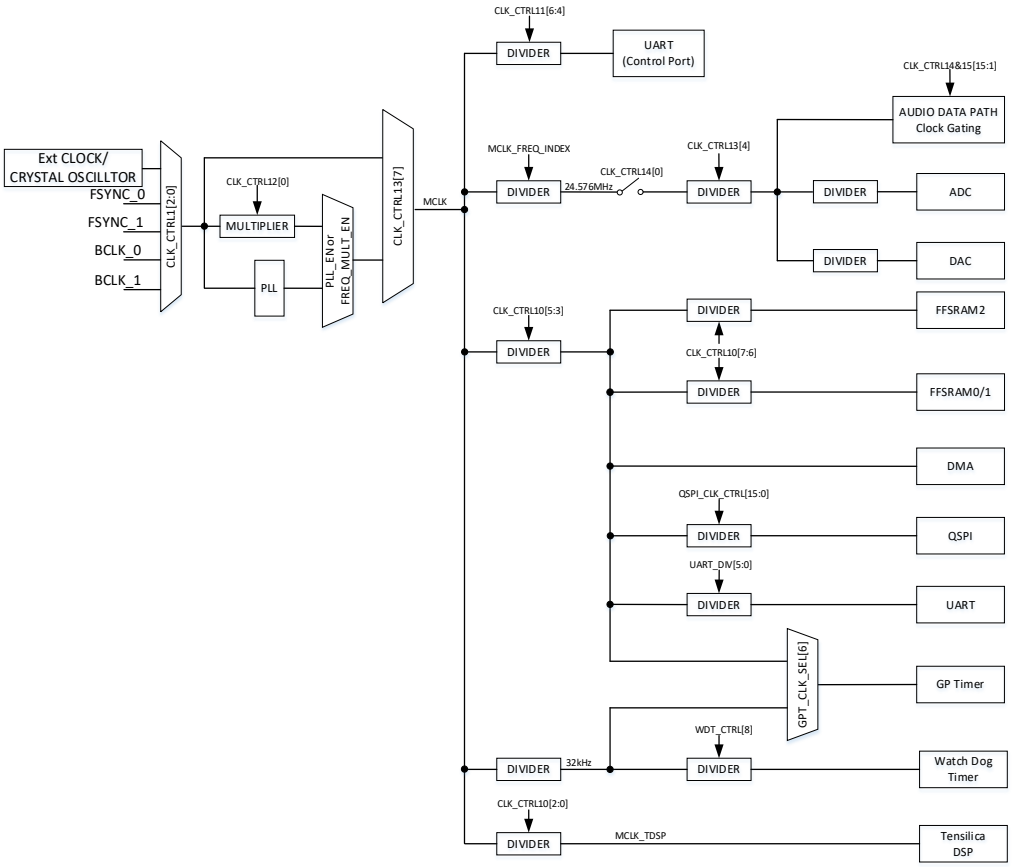


Figure 3. ADAU186x Clocking Architecture Block Diagram

PLL Enabled Setup

To program the PLL during initialization or reconfiguration of the ADAU186x, take the following steps:

1. Ensure that PLL_EN =0 and PLL bypassed.
2. Set the PLL control registers correctly.
3. Write 1 to PLL_UPDATE bit to propagate the PLL settings.
4. Make sure to clear the PLL_FM_BYPASS bit in the CLK_CTRL13 register.
5. Enable the PLL using the PLL_EN bit.

Other blocks can be powered up while the PLL is not enabled or locked. However, if PLL is enabled and not locked, all other circuitry will wait until the PLL is locked to begin the power-up sequences.

Control Port Access During Initialization

Any control registers can be accessed at any time during initialization, before PLL is enabled, or during PLL lock.

To access Tensilica core memories, PWR_MODE[1] must be set to 1, and the PLL must be locked, if in use.

To access FastDSP memories, FDSP_EN must be set to 1, and the PLL must be locked, if in use.

PLL

The PLL can use any of the BCLK_x, FSYNC_x, or MCLKIN signals as a reference to generate the master clock, and the source is selected via the PLL_SOURCE bits. Depending on the input clock frequency, the PLL must be set for either integer or fractional mode. The PLL can accept input frequencies in the range of 30 kHz to 36 MHz. PLL_EN bit should be set as 1 if PLL is used.

The PLL output frequency can be set between 24 MHz and 100 MHz, but the ADAU186x recommends the output frequency of the PLL be set at an integer multiple of 24.576MHz like 24.576MHz, 49.152MHz, 73.728MHz or 98.304MHz.

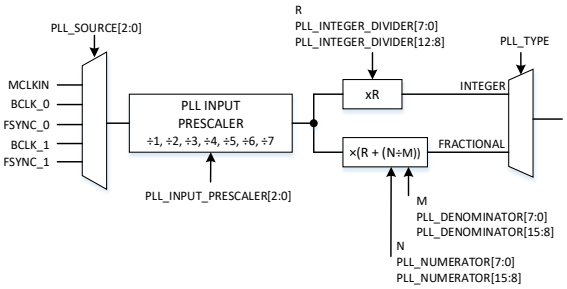


Figure 4. PLL Block Diagram

Input Clock Divider

Before reaching the PLL, the input clock signal goes through an integer clock divider to ensure that the clock frequency is within a suitable range for the PLL. The PLL_INPUT_PRESCALER bits set the PLL input clock divide ratio.

The input frequency limits of the PLL are specified after this input pre-scale divider. Therefore, the frequency after division must fall within specified range, which is 30kHz to 36MHz.

Integer Mode

Integer mode is used when the PLL output is an integer multiple of the PLL input clock. Configuring the PLL_TYPE as 0 to let the PLL works in the integer mode.

For example, if the PLL input clock = 12.288 MHz and the PLL_INPUT_PRESCALER = 1, the PLL required output = 49.152 MHz. Therefore, R = 49.152 MHz/12.288 MHz = 4, where R is PLL_INTEGER_DIVIDER.

Another example is as follows, if PLL input clock = 48 kHz, the PLL required output = 49.152 MHz, then R = 49.152 MHz/ 48 kHz = 1024.

In integer mode, the values set for N and M are ignored. Table 5 lists common integer PLL parameter settings for 48 kHz sampling rates.

Fractional Mode

Fractional mode is used when the clock input is a fractional multiple of the PLL output. Setting the PLL_TYPE as 1 to let the PLL works in the fractional mode.

For example, if MCLKIN = 13 MHz and f_S = 48 kHz, the PLL required output = 49.152 MHz and

$(R + (N/M)) = 49.152\ \text{MHz}/13\ \text{MHz} = (3 + (1269/1625))$

where:

$R = 3$.

$N = 1269$.

$M = 1625$.

Table 6 lists common fractional PLL parameter settings for 48 kHz sampling rates. When the PLL is used in fractional mode, it is important that the N/M fraction be kept within the $0.1 \leq N/M \leq 0.9$ range to ensure correct operation of the PLL.

When used in fractional mode, the input to the PLL after the input divider must be ≥1 MHz.

FREQUENCY MULTIPLIER

The frequency multiplier also can be used to provide the master clock. Setting the FREQ_MULT_EN as 1 to enable the frequency multiplier which will double the input source frequency. And the input source frequency is selected via the PLL_SOURCE bits. A fixed 24.576MHz should be provided on the input source frequency when frequency multiplier is used, and which gives a 49.152MHz master clock.

ADAU186x recommends using the MCLKIN as the source frequency when the frequency multiplier is used.

TRANSITION BETWEEN PLL, FREQUENCY MULTIPLIER, AND BYPASS

The ADAU186x provides three ways to generate the master clock: PLL, frequency multiplier and bypass mode (directly

from the input source frequency). Some procedures need to follow when transition between one way to another.

From PLL to Frequency Multiplier

When transition from the PLL mode to the frequency multiplier mode, below procedures should be followed:

1. Configure the PLL_FM_BYPASS=1 to bypass the PLL and frequency multiplier.
2. Configure the PLL_EN=0 to disable the PLL.
3. Configure the FREQ_MULT_EN=1 to enable the frequency multiplier.
4. Configure the PLL_FM_BYPASS=0 and using the frequency multiplier to provide the master clock once it's locked.

From Frequency Multiplier to PLL

When transition from the frequency multiplier to the PLL mode, below procedures should be followed:

1. Configure the PLL_FM_BYPASS=1 to bypass the PLL and frequency multiplier
2. Configure the FREQ_MULT_EN=0 to disable the frequency multiplier
3. Configure the PLL_EN=1 to enable the PLL
4. Configure the PLL_FM_BYPASS=0 and using the PLL to provide the master clock once it's locked.

From PLL/Frequency Multiplier to Bypass Mode

When transition from the PLL/frequency multiplier to the bypass mode, below procedures should be followed:

1. Configure the PLL_FM_BYPASS=1 to bypass the PLL and frequency multiplier.
2. Configure the PLL_EN=0, or FREQ_MULT_EN=0 to disable the PLL or frequency multiplier.

From Bypass Mode to PLL/Frequency Multiplier

When transition from bypass mode to the PLL/frequency multiplier, below procedures should be followed:

1. Configure the PLL_EN=1 or FREQ_MULT_EN=1 to enable the PLL or the frequency multiplier.
2. Waiting the PLL lock or the frequency multiplier lock
3. Configure the PLL_FM_BYPASS=0, using the PLL or the frequency multiplier to provide master clock.

MULTICHIP PHASE SYNCRONIZATION

Multiple ADAU186x devices can be ensured to remain in phase synchronization across the respective audio channels of the devices by setting the SYNC_SOURCE bits to use the same signal that both chips share. SYNC_SOURCE can be set to derive the phase synchronization signal from FSYNC_x. If only the shared serial ports between the two ICs are asynchronous to the core clock, then the SYNC_SOURCE must use the input asynchronous sample rate converter (ASRC). Alternatively, if no serial port is used, an internal synchronization source can be used.

CLOCK OUTPUT

A clock output of varying divisions of the PLL output can be generated on any of the MPx pins. The rate of the clock output is determined by the MCLKO_RATE bits.

POWER SUPPLY SEQUENCING

AVDD, HPVDD, and IOVDD are nominally 1.8 V, and DVDD is set at 0.9 V or 1.1V when using the on-board regulator which is powered by the HPVDD_L.

During power-up, AVDD and HPVDD shall be powered up no later than other power supplies.

During power-down, AVDD and HPVDD shall be powered off no earlier than other power supplies.

Enabling the $\overline{\text{PD}}$ pin (setting it low) powers down all analog and digital circuits and resets the devices to its default state. Before enabling the $\overline{\text{PD}}$ pin, mute the outputs to avoid any pops when the ADAU186x is powered down.

The $\overline{\text{PD}}$ pin can be tied directly to IOVDD for normal operation.

Power-Down Considerations

When powering down the ADAU186x, mute or power down the outputs before the power supplies are removed. Otherwise, pops or clicks may be heard.

Table 5. Integer PLL Parameter Settings for PLL Output = 49.152 MHz

PLL Input	Input Prescaler (X)	Integer (R)	Denominator (M)	Numerator (N)
32.768 kHz	0	1500	Don't care	Don't care
48 kHz	0	1024	Don't care	Don't care
12.288 MHz	0	4	Don't care	Don't care

Table 6. Fractional PLL Parameter Settings for PLL Output = 49.152 MHz

PLL Input	Input Prescaler (X)	Integer (R)	Denominator (M)	Numerator (N)
13 MHz	0	3	1625	1269
19.2 MHz	0	2	25	14

SIGNAL ROUTING

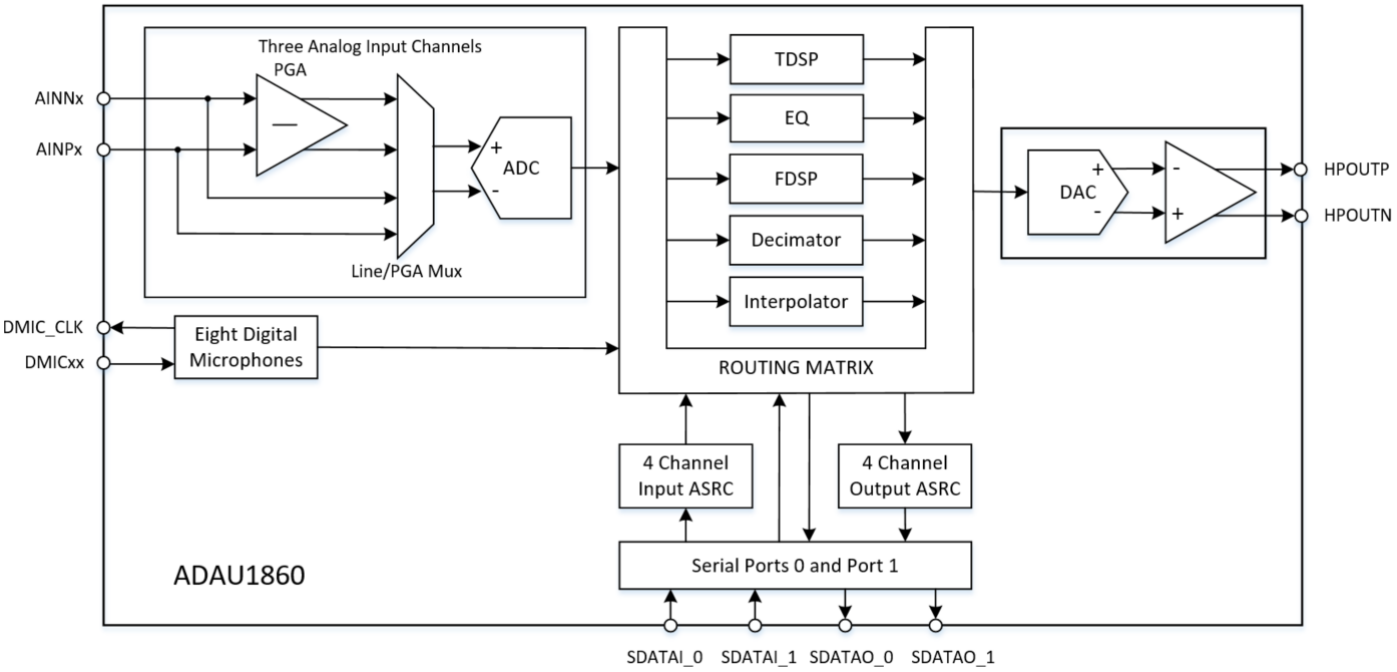


Figure 5. Input and Output Signal Routing

INPUT SIGNAL PATHS

ANALOG INPUTS

The ADAU186x can accept both line level and microphone inputs. Each of the three analog input channels can be configured in single-ended or differential mode with/without PGA. There are also inputs for up to four digital microphones. The analog inputs are biased at the CM voltage. Connect unused input pins to the CM pin or ac-couple the pins to ground.

Phase Difference Various Signal Path in ADAU186x

Figure 7 shows the phase variation between various blocks within the ADAU186x. The gray waveform shows the signal path from analog in to digital output or analog output, and the black waveform shows the signal path from digital in to analog output.

There is phase inversion from the analog input and the ADC, and similarly, from the DAC and headphone outputs (see Table 7). However, there is no phase inversion in the digital blocks.

Input Resistance

The input resistance of the analog inputs varies with the gain of the PGA. This resistance ranges from 1.2 kΩ to 41.2 kΩ with different gain setting, input mode configuration and input resistance configuration. The resistors inside the ADAU186x are precisely matched to each other, resulting in very little gain error. However, the exact value of the resistors depends on various conditions in the silicon manufacturing process and can vary by as much as ±20%.

With no PGA or line input mode, the input impedance is fixed at 9 kΩ in single ended input or 36 kΩ in differential input.

Analog Differential Inputs

For microphone signals, the ADAU186x analog inputs can be configured with differential input as shown in Figure 6 by configuring the AINx_INPUT_MODE bits as "0".

The PGA settings are controlled in Register PGAx_CTRLx. The PGA is enabled by setting the PGAx_EN bits.

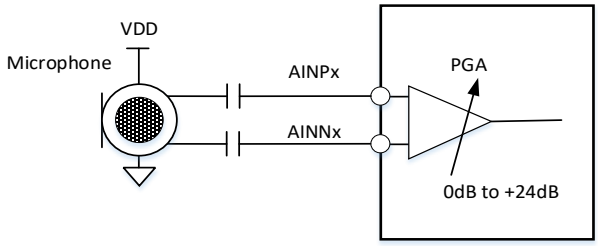


Figure 6. Differential Input with Analog Microphone Input

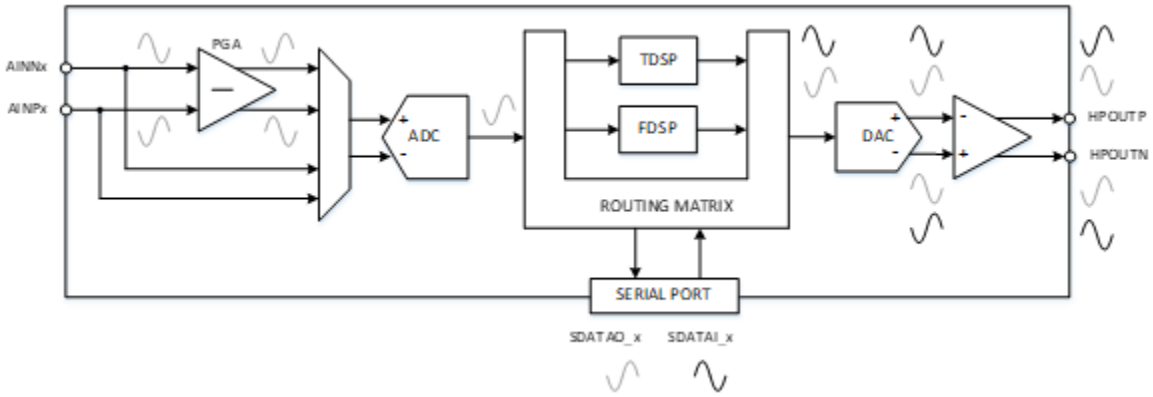


Figure 7. Phase Difference Between Input and Output

Table 7. Phase Difference Between the Input and Output Various Paths

Signal Path	Phase in Degrees
Analog In to ADC to Digital Output (Serial Port)	180
Analog In to PGA to ADC to Digital Output (Serial Port)	180
Analog In to ADC to DAC to HPOUT	0
Analog In to PGA to ADC to DAC to HPOUT	0
Digital In (Serial Port) to DAC to HPOUT	180

Analog Single-Ended Inputs

ADAU186x supports single-ended input when AINx_INPUT_MODE bits are set to "1". Figure 8 shows a single-ended input using the AINNx pins. The AINPx pin is connected to the CM pin or floating.

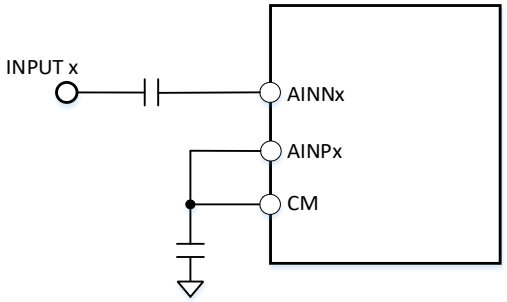


Figure 8. Single-Ended Line Inputs

Precharging Input Capacitors

Precharge amplifiers are enabled by default to quickly charge large series capacitors on the analog inputs. Precharging these capacitors prevents pops in the audio signal. The precharge circuits are powered up by default when an ADC channel is enabled and remain on for an amount of time determined by the ADC_AIN_CHRG_TIME bits register control. The internal impedance for the AINPx, AINNx pins is 750 Ω in this mode. However, at startup, the internal impedance is governed by the time constant of the reference voltage at the CM pin because the input precharge amplifiers use the CM voltage as a reference.

Programmable Gain Amplifiers

The PGAs have a programmable gain from 0 dB to 24 dB with 0.75dB step. The gain is controlled via the PGAx_GAIN registers.

The gain of the PGAs can be configured separately or can be linked to the gain setting of the PGA0 when the PGA_GAIN_LINK bit set to 1.

The slew between gain steps is performed automatically when the PGAx_SLEW_DIS register is 0. The automatic slew rate can be configured by the register bits PGA_SLEW_RATE.

When the PGAx_SLEW_DIS register is set to 1, the slew can be performed manually with the 5 LSBs of the PGAx_GAIN register. These bits are intended only for controlling smoother transitions between the 0.75 dB steps of the 6 MSBs (PGAx_GAIN[10:5]) and must only be set to a 0 when not transitioning the gain.

There are two input resistance mode for the PGAx which can be selected through the PGAx_RIN register. When the PGAx_RIN is set, the PGAx works in the low input resistance mode while the PGAx_RIN is configured as "0", it works in the high input resistance mode.

The PGAx in the ADAU186x also support different power mode to tradeoff the performance and power, which can be programmed through the PGAx_POWER_MODE. When the PGAx_POWER_MODE is configured as "0", the PGAx work in normal performance mode. When the PGAx_POWER_MODE is set to 1, the PGAx works in the enhanced performance mode, it achieves high performance with high power consumption.

The PGAx can work in three mode with combination of the PGAx_POWER_MODE and PGAx_RIN: Normal operation, Enhanced performance I and Enhanced performance II.

Table 8.

PGA Working Mode	PGAx_POWER_MODE Bit	PGAx_RIN Bit	Performance
Normal Operation	0	0	High input resistance, normal power
Enhanced Performance I	1	1	Low input resistance, highest power
Enhanced Performance II	1	0	High input resistance, high power

DIGITAL MICROPHONE INPUTS

When using digital microphones connected to the DMICx pins, the corresponding DMICx_EN registers must be set to enable the digital microphone signal paths. The digital microphone channels DMIC45 and DMIC67 can be swapped (left/right swap) by writing to the DMICxx_EDGE bits. But DMIC0 and DMIC2 always sample on rising edge, DMIC1 and DMIC3 always sample on falling edge.

DMIC01 or DMIC23 have the option of selecting DMIC0/1 or DMIC2/3 share same data or not through setting DMICxx_DAT_WIRE bits. But DMIC45 or DMIC67 must share same data line. DMIC 45 and DMIC 67 can be set trough multi-purpose pins MP0~MP25.

The digital microphone inputs have two clock sources. DMIC_CLKO is outputted from the DMIC_CLK/MP0 pin, MP8, MP9, MP10 and MP22 can output inverted DMIC CLK0 through setting MP_CTRLx. DMIC_CLK1 can be outputted from MP7, MP11, MP17, MP22 through setting MP_CTRLx The digital microphone data stream must be clocked by the pins can output DMIC CLK0 and DMIC CLK1 and not by a clock from another source, such as another audio IC. The frequency of the DMIC_ CLKx output can be programmed via the DMIC_CLKx_RATE bits and should be enabled by setting the DMIC_CLKx_EN as "1".

Each digital microphone input pair has separate sample rate controls that determine the sampling ratio. These controls are set via the DMICxx_FS bits. The output sample rate can be set between 12 kHz and 768 kHz. The initial decimation filter order can be selected between fourth-order or fifth-order via the DMICxx_DEC_ORDER bits. The fourth-order selection yields the lowest propagation delay, and the fifth-order selection may be needed to maintain full performance with some high dynamic range microphones. The DMICxx_FCOMP bits control whether or not the high frequency roll-off of the decimation filter is compensated for. No compensation gives the lowest propagation delay but slight attenuation in the pass-band. There are separate digital volume controls and 1 Hz high-pass filters for each digital microphone channel.
The input pulse density modulation (PDM) is mapped directly to the relative pulse code modulation (PCM) full-scale. For example, a 50% PDM density input generates a −6 dBFS output with a volume control setting of 0 dB.

The digital microphone signals and the ADCs are completely independent and do not share decimation filters.

Digital DMIC Volume Control

The volume setting of each digital microphone channel can be digitally attenuated in the DMICx_VOL registers. The volume can be set between +24 dB and −71.25 dB in 0.375 dB steps.

The digital microphone volume can also be digitally muted in the DMICx_MUTE bits. By default the volume control performs a soft ramp when changed, which can be bypassed for instantaneous change of volume via the DMIC_HARD_VOL bit. The volume control for every channel can be set to use the Channel 0 volume via the DMIC_VOL_LINK bit. When a digital microphone channel is enabled, it starts immediately at the volume level set by its DMICx_VOL register. When a digital microphone channel is disabled, it disables immediately and does not wait to ramp down the volume.

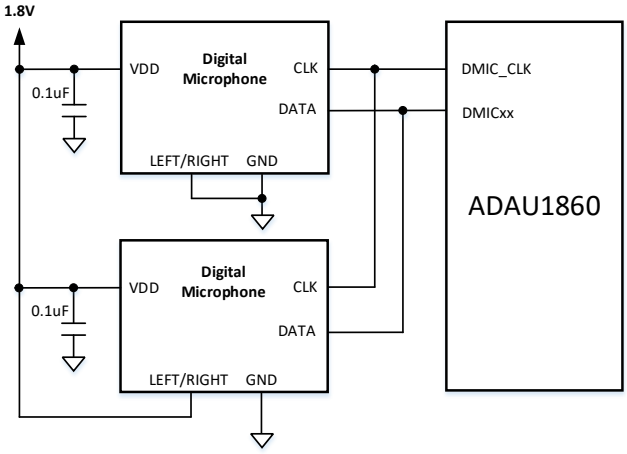


Figure 9. Digital Microphone Interface Block Diagram

ANALOG-TO-DIGITAL CONVERTERS

The ADAU186x includes three 24-bit Σ-Δ analog-to-digital converters (ADCs) with a selectable sample rate of 8 kHz to 768 kHz.

ADC Full-Scale Level

The full-scale input to the ADCs (0 dBFS) is nominally 0.49Vrms in single ended input and 0.98Vrms in differential input. Signal levels above the full-scale value cause the ADCs to clip.

In voice wake up mode, The full-scale input to the ADCs (0 dBFS) is nominally 0.49Vrms in both single ended input mode and differential input mode.

Digital ADC Volume

The volume setting of each ADC can be digitally attenuated in the ADCx_VOL registers. The volume can be set between +24 dB and −71.25 dB in 0.375 dB steps. The ADC volume can also be digitally muted in the ADCx_MUTE bits. By default, the volume control performs a soft ramp when changed, which can be bypassed for instantaneous change of volume via the ADC_HARD_VOL bit. The volume control for every channel can be set to use the Channel 0 volume via the ADC_VOL_LINK bit. When an ADC channel is enabled, it starts immediately at the volume level set by its ADCx_VOL register. When an ADC channel is disabled, it disables immediately and does not wait to ramp down the volume.

Filtering

The ADCx_DEC_ORDER bit controls the order of the decimation filter. Low order decimation filter gets low delay and high order decimation filter gets long delay.

The ADCx_FCOMP bits control whether the high frequency roll-off of the decimation filter is compensated for or not. No compensation gives the lowest propagation delay but with slight attenuation in the passband.

A high-pass filter is available on the ADC path to remove dc offsets. This filter can be programmed by using the ADCx_HPF_EN bits to setting the corner frequency with the operation: 1Hz, 4Hz, 8Hz or disable this high pass filter.

ADC Working Mode

The ADCx provide a mechanism to trade off the performance and power consumption. It can be programmed through the ADCx_IBIAS bits as shown in Table 11.

The ADCx can work in voice wake up mode to further reduce power consumption. Set the ADC_VOC_WKUP bit to enable the voice wake up mode. During the voice wake up mode, the ADCx sample rate can be 8kHz or 16kHz with setting the ADC_FREQ bit to 0, and the ADCx data width is 16 bits (16 MSBs of the ADC output).

ADC Sample Rate

The sample rate of the ADCx depends on the ADC_FREQ and ADCx_FS bits setting. When the ADC_FREQ sets to 1, the ADCx can support sample rate from 12kHz to 768kHz which is 12kHz based. When the ADC_FREQ configures as 0, the ADCx support 8kHz and 16kHz: 8kHz with ADCx_FS is “01” and 16kHz with ADCx_FS is “10”.

OUTPUT SIGNAL PATHS

Data can be routed to the output DAC path from the serial ports, the Tensilica core, the Fast DSP core, the ADCs, the digital microphones, the input ASRCs, the EQ, or the interpolator.

The analog output pins are capable of driving headphone or earpiece speakers. The line output can drive a load of at least 10 kΩ or can be put into headphone mode to drive headphones or earpiece speakers. The analog output pins are biased at the CM voltage.

ANALOG OUTPUTS

Headphone Output

The headphone output is differential, available at HPOUTP and HPOUTN. The output pins can be set as a headphone driver by configuring the HP_MODE bit to 0 in the HP_CTRL register. The headphone outputs can drive a minimum load of at least 10 Ω. To mute or unmute the headphone output, use the DAC_MUTE bit.

The Headphone can be operated in different performance level with different power consumption, which can be configured through the PB_POWER_CTRL bits.

Line Outputs

Set the output to line output mode by setting the HP_MODE bits to 1. The analog output pins (HPOUTP, HPOUTN) can drive differential loads of ≥10 kΩ. To mute or unmute the line output, use the DAC_MUTE bits

Headphone Output Power Supply

The headphone output stage of the ADAU186x can be powered by either HPVDD or HPVDD_L.

When HPVDD provides the power, the ADAU186x works in the headphone normal voltage mode and the HPVDD is fixed at 1.8V.

When HPVDD_L provides the power, the ADAU1850 works in the headphone low voltage mode and the HPVDD_L support voltage between 1.2V to the voltage no more than the HPVDD.

Follow the below procedure when the transition is happened from the headphone normal voltage mode to headphone low voltage mode.

1. Setting the HP_LVMODE_EN=1 to enable the related circuit that works in low voltage mode.
2. Configure the HP_LVMODE_CM_DLY bits to make sure the common voltage on the headphone output is stable before switch the power stage.
3. Setting the HP_LVMODE_CM_EN=1 to enable the HPVDD_L/2 generation circuit, which will be used as the headphone output common voltage in the low voltage mode.
4. Configure the HP_LVMODE_AUTO_DLY bits to set the delay time between HP_LVMODE_SWITCH change and unmute the DAC output.
5. Configure the HP_LVMODE_AUTOSW_MODE.
   a) If HP_LVMODE_AUTOSW_MODE=1: once the HP_LVMODE_SWITCH=1, the ADAU186x automatically transition to the low voltage mode.
   b) If HP_LVMODE_AUTOSW_MODE=0: User need to mute the DAC output before setting the HP_LVMODE_SWITCH=1. After setting the HP_LVMODE_SWITCH=1 and waiting for a delay time (need longer than HP_LVMODE_CM_DLY+ HP_LVMODE_AUTO_DLY), user need to unmute the DAC output.
6. Setting the HP_LVMODE_SWITCH=1, switch the headphone power from the HPVDD to HPVDD_L.

If the transition is from the headphone low voltage mode to headphone normal voltage mode, follow the below steps:

1. Configure the HP_LVMODE_CM_DLY bits to make sure the common voltage on the headphone output is stable before unmuting the DAC output.
2. Configure the HP_LVMODE_AUTO_DLY bits to set the delay time between HP_LVMODE_SWITCH change and switch the common voltage.
3. Configure the HP_LVMODE_AUTOSW_MODE.
   a) If HP_LVMODE_AUTOSW_MODE=1: once the HP_LVMODE_SWITCH=0, the ADAU186x automatically transition to the normal voltage mode.
   b) If HP_LVMODE_AUTOSW_MODE=0: User need to mute the DAC output before setting the HP_LVMODE_SWITCH=0. After setting the HP_LVMODE_SWITCH=0 and waiting for a delay time (need longer than HP_LVMODE_CM_DLY+ HP_LVMODE_AUTO_DLY), user need to unmute the DAC output.
4. Setting the HP_LVMODE_SWITCH=0 to switch the headphone power from the HPVDD_L to HPVDD.
5. Setting the HP_LVMODE_EN=0 to disable the related circuit that works in low voltage mode.
6. Setting the HP_LVMODE_CM_EN=0 to disable the HPVDD_L/2 generation circuit.

There is an internal LDO on the HPVDD_L, which gives an option to use this internal LDO to provide the clean power for the headphone output stage.

In the headphone normal voltage mode, set the HPLDO_EN=0 and HPLDO_BYPASS=0 to disable this LDO, the HPVDD powers the headphone output stage.

In the headphone low voltage mode, if configure the HPLDO_EN=1 and HPLDO_BYPASS=0 to enable the internal LDO, the internal LDO powers the headphone output stage and the output voltage of the internal LDO is determined by the HPLDO_VOUT.

If configure the HPLDO_EN = 0 to power down the HPLDO and configure the HPLDO_BYPASS=1 to bypass the LDO, the HPVDD_L will power the headphone output stage directly.

Pop and Click Suppression

To avoid clicks and pops, mute all analog outputs that are in use while changing any register settings that may affect the signal path. These outputs can then be unmuted after the changes have been made.

DIGITAL-TO-ANALOG CONVERTERS

The ADAU186x includes one 24-bit, Σ-Δ DAC. This converter can operate with input sampling frequencies of 12 kHz, 24 kHz, 48 kHz, 96 kHz, 192 kHz, 384 kHz, or 768 kHz. The sample rate is selectable via the DAC_FS bit.

DAC Full-Scale Level

The full-scale output from the DAC (0 dBFS) is nominally 1 V rms for a differential output.

Digital DAC Volume Control and Filtering

The volume of the DAC channel can be digitally attenuated using the DAC_VOL register. The volume can be set to be between +24 dB and −71.25 dB in 0.375 dB steps. The DAC volume can also be digitally muted in the DAC_MUTE bit. By default, the volume control performs a soft ramp when changed, which can be bypassed for instantaneous change of volume via the DAC_HARD_VOL bit. When the DAC channel is enabled, it starts at the lowest volume setting and ramps, if DAC_HARD_VOL = 0, to the volume level set by the DAC_VOL register. When the DAC channel is disabled, it ramps the volume from its current setting, if DAC_HARD_VOL = 0, to mute and then turns off.

A high-pass filter is available on the DAC path to remove dc offsets. This filter can be enabled or disabled using the DAC_HPF_EN bits. The corner frequency of this high-pass filter is set to 1 Hz.

The DAC_FCOMP bit control whether the high frequency roll-off of the decimation filter is compensated for or not. No compensation gives the lowest propagation delay but with slight attenuation in the passband.

The DAC linear interpolation filter can be selected via the DAC_MORE_FILT bit. Setting DAC_ MORE_FILT = 0 results in lower propagation delay at the expense of lower attenuation of out of band components.

DAC working mode

When DAC is working in the normal mode that PB_POWER_CTRL bits is 2'b00, there are another two special mode called low power mode (LPM) and low power mode Ⅱ (LPM_Ⅱ) can be used to trade off the DAC performance and power. When DAC is working in the enhanced mode that PB_POWER_CTRL bits is 2'b10, the low power mode and low power mode Ⅱ will have no effect on DAC.

PDM OUTPUTS

The ADAU186x includes two channels of high performance, 1-bit PDM outputs suitable for driving an external amplifier or other peripheral with low latency.
Data can be routed to the PDM output path from the serial ports, the Tensilica core, the Fast DSP core, the ADCs, the digital microphones, the input ASRCs, the EQ, or the interpolator.

These PDM outputs can operate with input sampling frequencies of 12 kHz, 24 kHz, 48 kHz, 96 kHz, 192 kHz, 384 kHz, or 768 kHz. The sample rate is selectable via the PDM_FS bit. Ensure that all channels routed to the PDM outputs are at the same sample rate. The PDM output modulators can run either at 3.072 MHz or 6.144 MHz, which is selected via the PDM_RATE bit. This bit also determines the rate of the PDM output clock. The PDM output is sent over a 2-wire (PDM clock and PDM data) dual data rate interface. These two signals can be routed to any multipurpose (MPx) pin output via the respective MPx_MODE bits for each pin.

PDM Outputs Full-Scale Level

The full-scale PDM input results in the full-scale PDM outputs. The PDM modulator performance reduces at an output amplitude greater than −7.5 dBFS.

PDM Outputs Volume Control and Filtering

The volume of each PDM channel can be digitally attenuated using the PDMx_VOL registers. The volume can be set to be between +24 dB and −71.25 dB in 0.375 dB steps. The PDM volume can also be digitally muted in the PDMx_MUTE bits. By default, the volume control performs a soft ramp when changed, which can be bypassed for instantaneous change of volume via the PDM_HARD_VOL bit. The volume control for both channels can be set to use the Channel 0 volume via the PDM_VOL_LINK bit. When a PDM channel is enabled, it starts at the lowest volume setting and ramps, if PDM_HARD_VOL = 0, to the volume level set by its PDM_VOLx register. When a PDM channel is disabled, it ramps the volume from its current setting, if PDM_HARD_VOL = 0, to mute and then turn off. A high-pass filter is available on the PDM path to remove dc offsets. This filter can be enabled or disabled by using the PDMx_HPF_EN bits. The corner frequency of this high-pass filter is set to 1 Hz.

The order of the final interpolation filter can be selected via the PDM_MORE_FILT bit. Selecting the lower order filter results in lower propagation delay at the expense of lower attenuation of out of band components.

ASYNCHRONOUS SAMPLE RATE CONVERTERS

The ADAU186x includes ASRCs to enable asynchronous full-duplex operation of the serial ports. Four channels of ASRC are available for the digital outputs, and four channels of ASRC are available for the digital input signals.

The ASRCs can convert serial output data from the internal rate of up to 192 kHz back down to 12 kHz. All intermediate frequencies and ratios are also supported.

ASRCI can receive its inputs from one of the serial audio ports via the ASRCI_SOURCE bit. Each channel of the input ASRC can select its source from any of the 16 channels on the selected serial audio port via the ASRCIx_ROUTE bits. The output (internal) sample rate of the input ASRC is set via the ASRCI_OUT_FS bit.

The output ASRC channels can receive their inputs from many internal sources via the ASRCOx_ROUTE bits: the Tensilica core, the Fast DSP core, the ADCs, the digital microphones, the EQ, or the decimator. Ensure that the sample rate of all sources to all of the channels of the output ASRC are at the same sample rate. The source of Channel 0 determines the internal sample rate of the output ASRC. The source of the channels to the output ASRC are set via the ASRCOx_ROUTE bits.

The output sample rate of the output ASRC must be linked to one of the serial ports and can be selected via the ASRCO_SAI_SEL bit.

The input and output ASRCs are fully independent and can operate on different serial ports.

The ASRCs automatically mute their outputs to zero data when the outputs are not locked. The state of each ASRC lock can be monitored via the ASRCI_LOCK and ASRCO_LOCK read only status bits. In addition, unlocked to locked or locked to unlocked transitions of each ASRC can be used as an interrupt source to the four interrupt controllers.

By default, the ASRCs use the high-performance mode of operation. A lower power, lower performance mode of operation can be enabled via each ASRCs ASRCx_LPM bit control.

Additional filtering options are available to further customize the ASRCs to any application. Each ASRC has an ASRCx_VFILT bit that can enable a voice band filter that provides additional rejection at the Nyquist frequency, which can be useful when using traditional voice band sampling frequencies. There is also an ASRCx_MORE_FILT control bit for each ASRC that provides additionally filtering of out of band energy and may improve performance in some conditions.

INTERPOLATION AND DECIMATION BLOCKS

The ADAU186x includes blocks designed to convert audio from the fast sampling rate used for noise cancelling and the slow audio rate of the audio source. There are eight channels of fast to slow decimation and eight channels of slow to fast interpolation.

Every two channel pairs of each block can independently operate at different input and output rates than the other two channel pairs. Ensure that the sampling rate of each two channel pair inputs matches when selecting the inputs via the routing register controls. The input sampling rates are determined by the FDECxx_IN_FS and FINTxx_IN_FS bits and the output sampling rates are determined by the FDECxx_OUT_FS and FINTxx_OUT_FS bits. For the interpolation block, the output rate must be set higher than the input rate. For the decimation block, the output rate must be set lower than the input rate.

SIGNAL LEVELS

Full-scale digital or 0 dBFS maps to the analog full-scale of the various converters. The FastDSP cores can maintain up to 24 dBFS internally but clip symmetrically to 0 dBFS at their outputs. By default, there is no gain adjustment between any block.

EQUALIZER (EQ)

The ADAU1850 integrates a MAC (Multiply Accumulate) Engine which supports following features:

- Single cycle double precision MAC operation
- Modulo address in Data RAM space
- 3-stage pipeline

The EQ input source includes: any data register, any accumulator register, any serial port input channel, any ADC input, any ASRCI channel, any digital microphone input, FDSP output and Tensilica DSP output.

The EQ band number is affected by FDSP_MODE through setting FDSP_MODE[0]. With default value 0, EQ works in Normal Mode and supports 13-band for sampling rate equal or below 192kHz. When value is set as 1, EQ works in Turbo Mode and supports 24-band for sampling rate equal or below 96kHz.

POWER AND RUN CONTROL

All program and parameter memory for the EQ can be read or written from any control port when POWER_EN = 1, and the PLL is locked, if used. The EQ starts processing when EQ_RUN = 1.

EQ CLEAR

When EQ isn't running, the data ram can be cleared by writing 1 to the EQ_CLEAR bit. The EQ_CLEAR_DONE bit in the EQ_STATUS register can be used to check if the EQ clear operation is done.

READ/WRITE DATA FORMATS

The read/write formats of the control port are byte oriented to allow ease of programming of common microcontrollers. To fit the data into a byte oriented format, 0s are added to the data fields before the MSB to extend the data-word to a full 8 bits. For example, 28-bit words written to the parameter RAM are preceded by four leading 0s to create a 32-bit (4-byte) word.

MEMORY ADDRESS ALLOCATION

When FDSP_MODE in Turbo Mode, the program memory and parameter memory address are shown in the Table 8. When FDSP_MODE in Normal Mode, program memory only 0x4000 A000 ~0x4000 A1FF can be used.

Table 9. Memory Address Allocation

Name	Start & Stop Address	Size	Width
Program Memory	0x4000 A000 ~0x4000 A1FF , 0x4000 A600 ~0x4000 A7FF	256 Word	18bit
Parameter memory Bank0	0x4000 A200 ~ 0x4000 A3FF	128 Word	28bit
Parameter memory Bank1	0x4000 A400 ~0x4000 A5FF	128 Word	28bit

PARAMETER BANK SWITCHING

Two banks of parameter RAM are available: A and B. At any given time, the EQ only uses one of the two banks. The two banks allow parameters for filters and variables for other instructions to easily be switched between different processing scenarios. The bank used is selected with the EQ_BANK_SEL bit.

PARAMETER MEMORY ACCESS

Reads from any parameter bank from the I^2C, SPI, UART are unrestricted if the EQ isn't running. When EQ is running, only the unused parameter bank can be read from the I^2C, SPI, or UART port.

Direct reads of in use banks from the I^2C/SPI/UART are not allowed and return 0s. A read of the current bank returns all 0s. Writes to any parameter bank are possible when the EQ isn't running. When EQ is running, only the unused parameter bank can be written from the I^2C, SPI, or UART port.

FASTDSP CORE

The ADAU186x FastDSP core is optimized for ANC processing. The processing capabilities of the core include biquad filters, limiters, expanders, multipliers, bit wise operations, clippers, volume controls, and weighted mixing. The core has inputs from all sources and sixteen outputs. The core is controlled with a 27-bit program word, with a maximum of 96 instructions per frame.

The FDSP instruction number is affected by FDSP_MODE through setting FDSP_MODE[0]. With default value 0, FDSP works in Normal Mode and supports maximum 64 instructions per frame for sampling rates equal or below 384kHz. When value is set as 1, FDSP works in turbo mode and supports maximum 96 instructions per frame per frame for sampling rates equal or below 192kHz.

INSTRUCTIONS

The instructions in the ADAU186x include the following:

- Single precision (27-bit fractional precision) biquad/second-order filters
- Double precision (54-bit fractional precision) biquad/second-order filters
- Lower precision (19-bit fractional precision) biquad/second-order filters
- Limiter with/without external detector loop or side chain input
- Expander with/without external detector loop or side chain input
- Volume slider
- Two to four input scale and mix
- Two input multiply
- Absolute value
- Symmetrical clipper
- Min and Max
- Shift
- OR, AND, XOR, and INV
- Memory read or write

FILTER PRECISION

Different levels of fractional precision are available for filters in the FastDSP core. Using lower fractional precision results in lower power consumption than using higher precision. However, care must be taken to ensure that filters have enough precision to maintain stability.

FLAGS AND CONDITIONAL EXECUTION

Several flags can be set or not set based on the output of each instruction. These flags include the following:

- Output equals zero
- Output is not equal to zero
- Output is greater than zero
- Output is less than zero
- Output is greater than or equal to zero
- Output is less than or equal to zero
- Accumulator overflow

Each instruction can always execute or conditionally execute based on an individual flag or other states. The other states are.

- The logic state of MPx pins if used as GPIO input.
- The FDSP_REG_COND0 to FDSP_REG_COND7 bits.
- Modulo N Counter equals zero

The GPIOs can be used on the MPx pins which are MP0-2, MP7-10, MP17, MP20, MP22-25, . The state of the MPx pins used as GPIOs determines whether or not an instruction executes.

The FDSP_REG_CONDx bits are read/write bits that can be accessed via any of the control interfaces or via the Tensilica DSP. The state of these registers determines whether or not an instruction executes.

The Modulo N counter is a counter that increments every frame of the FastDSP. The counter is reset to 0 after the number of frames is set in the FDSP_MOD_N bit. Instructions can execute every N frames set by the FDSP_MOD_N bit, which provides a mechanism to easily run some instructions at a lower rate than the frame rate.

When an instruction does not execute based on a condition, the instruction can be set to either do nothing or pass its input to its output.

INPUT SOURCES

Any instruction can use any of the following as an input source: any data register, any accumulator register, any serial port input channel, any digital microphone input, any ADC input, any Tensilica DSP output, any ASRCI channel, any output from the interpolation block, or the EQ.

The frame rate of the FastDSP must be set and determines when the program counter starts counting again at 0, which must be set to the sample rate of the fastest source. The source that the frame rate is determined by is set via the FDSP_RATE_SOURCE bits. If desired, the frame rate can be set independent of any source, and the rate can be set via the FDSP_RATE_DIV bits.

POWER AND RUN CONTROL

All program, parameter, and data memories for the FastDSP can be read or written from any control interface or the Tensilica DSP when PWR_MODE[0]=1, FDSP_EN = 1, and the PLL is locked, if used.

A single register FDSP_EN powers up the FastDSP core to allow access to the memories. The FastDSP core starts processing when both FDSP_EN = 1 and FDSP_RUN = 1.

DATA MEMORY

The ADAU186x FastDSP datapath is 28 bits (5.23 format) and up to 24 dBFS is allowed. All inputs and outputs to FastDSP are 24 bits (1.23 format). The outputs are truncated to 24 bits so >0 dBFS on an output results in clipping. The data memory is 96 words. The double length memory enables the core to perform double precision arithmetic with double length data and single length coefficients.

Each instruction has four associated data/state memory locations. These locations can read at any time via the control port interface or from the Tensilica DSP.

PARAMETERS

Parameters, such as filter coefficients, limiter settings, and volume control settings, are saved in parameter memories. Each parameter is a 32-bit number. The format of this number depends on the associated instruction. The number formats of the different parameters are shown in Table 9 for the biquad instructions.

Table 10. Parameter Number Formats

Parameter Type	Format
Filter Coefficient (B0, B1, B2,A1,A2)	5.27

There are three parameter banks available. Each bank can hold a full set of 480 parameters (96 filters × 5 coefficients). Users can switch between Bank A, Bank B, and Bank C, allowing three sets of parameters to be saved in memory and switched on-the-fly while the core is running. Bank switching can be achieved by writing to the FDSP_BANK_SEL bits. Parameters in the active bank must only be updated via the FastDSP safeload registers while the core is running. If parameters are not updated in this way, a bad output may happen.

Parameters are assigned to instructions in the order in which the instructions are instantiated in the code.

PARAMETER BANK SWITCHING

Three banks of parameters are available: A, B, and C. At any given time, the FastDSP uses only one of these banks. The three banks allow coefficients for filters and variables for other instructions to easily be switched between different processing scenarios. The bank used is selected with the FDSP_BANK_SEL bits.

When the current bank is changed, the parameter values used for processing can either be changed on the next frame start or ramped via linear interpolation between the previously selected bank and the new bank indicated via the FDSP_BANK_SEL bits. To select this change or ramp, use the FDSP_RAMP_MODE bit. When the linear parameter ramp mode is selected, only the parameters associated with the three biquad instructions ramp. All parameters associated with other instructions change at the beginning of the next frame.

Parameters in banks that are actively ramping do not change during a bank switch.

It is possible to stop the linear ramp of parameters between the two values in the previous and current bank. The FDSP_LAMBDA bits are a 6-bit value representing the point along the linear interpolation curve between the two banks at which the bank ramp switch stops. The lambda value can be updated on-the-fly via the control interfaces but only increased after a ramped bank switch is initiated. To complete a bank switch, set a value of 63 (default setting). The actual current ramp point (0 to 63) can be read via the FDSP_CURRENT_LAMBDA bits. When this value reaches 63, the bank switch is complete, and the current parameters used match the current bank. Parameters in the two banks being ramped between cannot be modified while a ramped bank switch is occurring.

An interrupt can be triggered to the interrupt controllers via the IRQx_PRAMP interrupt source bits. This interrupt triggers on the first frame when a ramped bank switch is active and FDSP_CURRENT_LAMBDA equals FDSP_LAMBDA.

The rate at which the ramp between the two banks occurs is selectable via the FDSP_RAMP_RATE bits.

PARAMETER BANK COPYING

The parameters of any bank can be copied to any other bank with a single control write. There are six registers, FDSP_COPY_xy, for the six possible bank copy operations. Writing a 1 to one of these bits initiates a bank copy. After a bank copy initiates, the FastDSP waits until the start of the next frame, and then during the next frame copies the content of the banks while the associated instruction is executing. The bank copy completes at the start of the subsequent frame and takes at most two frames to complete from the initiation. Copying to the active bank is not permitted but results in no action being taken.

PARAMETER MEMORY ACCESS

Reads from any parameter memory bank from the I²C, SPI, UART or Tensilica DSP are unrestricted if the FastDSP core is enabled but not running. Reads of unused parameter banks from the I²C, SPI, UART or Tensilica DSP are unrestricted if the FastDSP core is enabled and running. While the core is running, if the control port (I²C/SPI/UART), or Tensilica DSP try to access the same location on the same cycle, the control port has priority, and the read from TDSP returns all 0s.

Direct reads of in use banks from the I²C/SPI/UART, mREAD instruction, or Tensilica DSP are not allowed and return 0s. A read of the current bank returns all 0s. Writes to all parameter banks are possible when the FastDSP core is enabled but not running. Writes to unused banks are possible at any time. While the core is running, if the control port (I²C/SPI/UART), or Tensilica DSP try to write to the same location on the same

cycle, the control port has priority, and the write from the Tensilica DSP does not occur.

The memory map from the perspective of the FDSP is different than the mapping of the memories to the control port or TDSP because the FDSP has its own address mapping, see the Table 10. When FDSP_MODE set as Turbo Mode, base address of instruction 0~95 can be used. When set as Normal Mode, only base address of instructions 0~63 can be used.

FASTDSP PARAMETER SAFELOAD

The parameter memory for a single instruction can be updated in real time on the active bank via the safeload mechanism over the control interface. Set the instruction number in the FDSP_SL_ADDR register, set the parameter values in the FDSP_SL_Py_x registers, and write a 1 to the FDSP_SL_UPDATE register. After these settings and write occur, all parameters for that instruction are updated at the same time with the values in the FDSP_SL_Py_x registers at the beginning of the next frame. Parameter safeload must be treated as an extra FastDSP instruction, and to enable using it, at least one instruction must be unused.

There is a second FastDSP safeload interface that is mapped to the data memory space of the Tensilica DSP, which allows the Tensilica DSP to have word addressable access.

Table 11. Memory Addressing for FastDSP core

Memory	Memory Size	Word Size	Base Address Instruction 0~63 (Hexadecimal, Control port)	Base Address Instruction 64~95 (Hexadecimal, Control port)
Program	96	32	4000 8000 ~ 4000 80FF	4000 A800 ~ 4000 A87F
Bank A Parameter 0	96	32	4000 8100 ~ 4000 81FF	4000 A900 ~ 4000 A97F
Bank A Parameter 1	96	32	4000 8200 ~ 4000 82FF	4000 AA00 ~ 4000 AA7F
Bank A Parameter 2	96	32	4000 8300 ~ 4000 83FF	4000 AB00 ~ 4000 AB7F
Bank A Parameter 3	96	32	4000 8400 ~ 4000 84FF	4000 AC00 ~ 4000 AC7F
Bank A Parameter 4	96	32	4000 8500 ~ 4000 85FF	4000 AD00 ~ 4000 AD7F
Bank B Parameter 0	96	32	4000 8600 ~ 4000 86FF	4000 AE00 ~ 4000 AE7F
Bank B Parameter 1	96	32	4000 8700 ~ 4000 87FF	4000 AF00 ~ 4000 AF7F
Bank B Parameter 2	96	32	4000 8800 ~ 4000 88FF	4000 B000 ~ 4000 B07F
Bank B Parameter 3	96	32	4000 8900 ~ 4000 89FF	4000 B100 ~ 4000 B17F
Bank B Parameter 4	96	32	4000 8A00 ~ 4000 8AFF	4000 B200 ~ 4000 B27F
Bank C Parameter 0	96	32	4000 8B00 ~ 4000 8BFF	4000 B300 ~ 4000 B37F
Bank C Parameter 1	96	32	4000 8C00 ~ 4000 8CFF	4000 B400 ~ 4000 B47F
Bank C Parameter 2	96	32	4000 8D00 ~ 4000 8DFF	4000 B500 ~ 4000 B57F
Bank C Parameter 3	96	32	4000 8E00 ~ 4000 8EFF	4000 B600 ~ 4000 B67F
Bank C Parameter 4	96	32	4000 8F00 ~ 4000 8FFF	4000 B700 ~ 4000 B77F
State 0 (A1 High)	96	32	4000 9000 ~ 4000 90FF	4000 B800 ~ 4000 B87F
State 1 (A2 High)	96	32	4000 9100 ~ 4000 91FF	4000 B900 ~ 4000 B97F
State 2 (A1 Low)	96	32	4000 9200 ~ 4000 92FF	4000 BA00 ~ 4000 BA7F
State 3 (A2 Low)	96	32	4000 9300 ~ 4000 93FF	4000 BB00 ~ 4000 BB7F

DIGITAL TURBO MODE

In digital block, to provide more powerful digital function, Turbo Mode can be enabled through setting FDSP_MODE[0] to 1. Default mode is Normal Mode. All the register description in Register Summary and Register Details are about Normal Mode. Detailed differences between Normal Mode and Turbo Mode are shown as following.

NORMAL MODE

- Serial Audio Ports supporting I2S, Left justified, Right justified, or up to TDM16.
- FDSP supports maximum 64 instructions per frame for sampling rates equal or below 384kHz.
- Equalizer supports 13 - band for sampling rate equal or below 192kHz.

TURBO MODE

- Serial Audio Ports supporting I2S, Left justified, Right justified, or up to TDM12. In TDM mode, SPTx_OUT_ROUTE0 ~ SPTx_OUT_ROUTE11 can be used normally. SPTx_OUT_ROUTE15 will output a mixing of SAI0_0 and SAI0_1 (SAI0_0_high 16 bits << 16 & SAI0_1_high 8 bits).
- FDSP supports maximum 96 instructions per frame for sampling rates equal or below 192kHz. Detailed memory information please refer to FastDSP Core section.
- Equalizer supports 24 - band for sampling rate equal or below 96kHz. Detailed memory information please refer to Equalizer section.

TENSILICA DSP

The Tensilica HiFi-3z DSP (TDSP) features a 32-bit, audio DSP engine optimized for audio and voice processing and other demanding DSP functions. The Tensilica HiFi-3z core combines quad 24-bit x 24-bit fixed point MACs or two 32-bit x 32-bit fixed point MACs with 64-bit accumulators. The processor provides on-chip debug support with
control of the software state of the processor through the JTAG port.

DMA

A DMA was provided to transfer data between memory to memory, memory to peripheral, peripheral to memory. The DMA has the lowest priority to access the memory compared to the control port and TDSP core.

FIFO

FIFO block is provided to access the SRAM.

DATA SYNC

Data Sync block is used to transmit and receive data between TDSP and audio peripherals.

UART

The ADAU186x has a standard UART serial interface that provides a simple mechanism to communicate with other serial devices.

QSPI

One dedicated QSPI interface intended for external flash memory connection, which features memory mapped registers with support for direct read of SPI flash memory from the TDSP core. Using the QSPI interface for program and data loading if self-boot mode is enabled. The QSPI interface can be configured through the QSPI registers during the boot up, details shown in the register map. And it supports the maximum clock up to 50MHz.

WATCHDOG

A watchdog monitors the Tensilica DSP execution. Once the watchdog timeout, it will generate interrupt or reset the Tensilica DSP. Watchdog timer can be programmed through the registers.

GENERAL-PURPOSE TIMER

The ADAU186x integrates a low frequency timer which running at 32.768kHz. This timer can be used to generate interrupt and provide periodic events.

INTERRUPT

There are fifteen level trigger interrupt sources coming from FFSRAM0, FFSRM1, FFSRAM2, QSPI, Data Sync, DMA, UART, SPI, four AON_IRQx and three MPI_IRQx.

DEBUG INTERFACE

The ADAU186x provides a JTAG interface for the debugging. The Figure 10 shows the connection between the ADAU186x and the JTAG. And make sure that the voltage level from the JTAG is 1.8V since the ADAU186x uses 1.8V as the IO voltage.

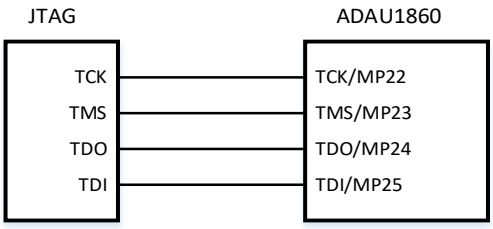


Figure 10. Connection between ADAU186x and JTAG

MEMORY MAPPING

The memory of the ADAU186x is shown in Figure 11.

An internal 4kB read only memory (ROM) contains boot code and the boot ROM executes at system reset.

The L1 memory system is the highest performance memory available to the TDSP core and includes 32kB L1 instruction RAM, 16kB L1 data RAM0 and 16kB L1 data RAM1. The TDSP core also contains 24kB instruction cache and 24kB data cache.

224kB L2 memory is available which includes: 64kB System RAM, 64kB FFSRAM0, 64kB FFSRAM1 and 32kB FFSRAM2.

The ADAU186x memory address space supports up to 16MB of external serial quad flash memory connected to the QSPI port. The TDSP core reads from flash memory are directly cached via internal cache, and direct memory mapped reads are permitted. The QSPI flash port is used for self-boot mode.

All the control and configuration registers are mapped into memory mapped registers through a 32 bit address which includes DMA registers, QSPI registers, SoC registers, SPI registers, UART registers, Timer registers, Watchdog registers, always on (AON) registers, FFSRAM0 registers, FFSRAM1 registers, FFSRAM2 registers, and DATA_SYNC registers.

5kB memory is allocated to FDSP, 1.5kB memory is allocated to EQ and 768B memory is allocated to Data sync function block.

Memories can be accessed by the TDSP core internally or externally from the control port, I2C/SPI/UART.

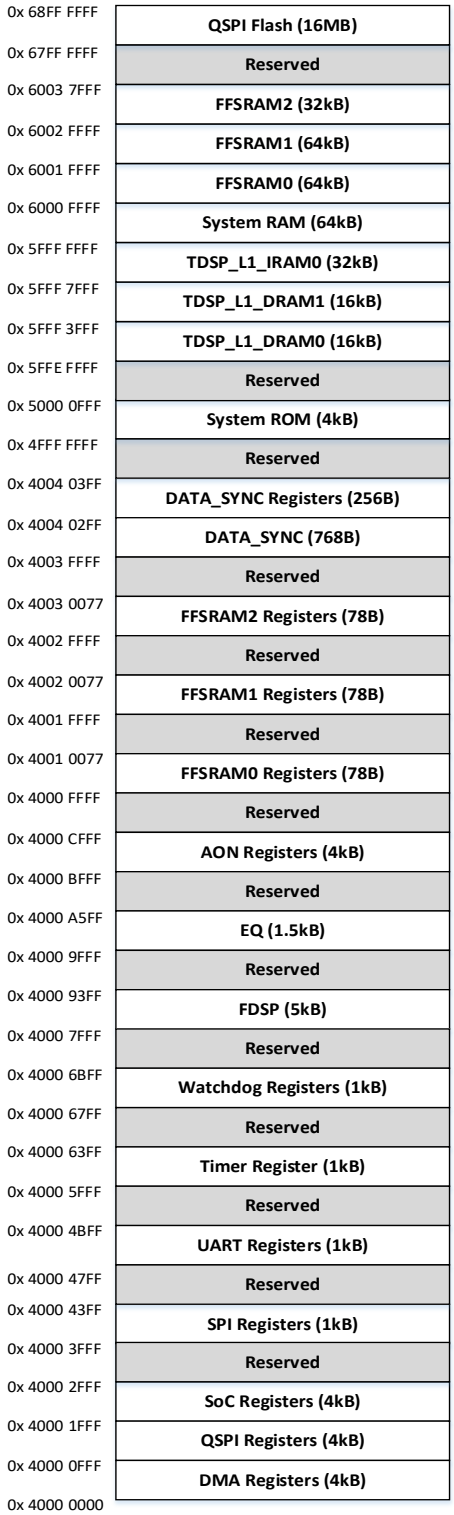


Figure 11. Memory Map of ADAU186x

POWER SAVING OPTIONS

The ADAU186x offers multiple options to balance the performance and power in some of the blocks.

ADC BIAS CURRENT CONTROL

The ADCs provide a mechanism to modify the each channel's bias current level independently via the ADCx_IBIAS and ADCx_IBIAS control bits. The ADC performance is very similar in different bias current level, but the power consumption is a little different. Please refer to the Table 11.

Playback Path Working Mode Control

The Playback path (DAC and headphone driver) provides a mechanism to tradeoff between performance and power consumption options for the user. Two possible settings can be set via the PB_POWER_CTRL control bit. Please refer to the Table 12.

PLL Bypass

Bypassing the PLL saves power. Please refer to the Table 13.

Table 12. ADC Power Consumption with Different ADCx_IBIAS Setting

ADCx_IBIAS[2:0]	ADC Work Mode Description	Change in Power Consumption Per ADC (mW) (Diff Mode, AVDD = 1.8V)
000	Normal Operation	0
010	Enhanced Performance	+0.23
011	Power Saving	-0.11

Table 13. Playback Path Performance, Power Consumption With Different PB_POWER_CTRL Configuration

PB_POWER_CTRL[1:0]	Work Mode	Change in SNR, A-Weighted (dB)	Change in THD+N at 1 kHz, 1 mW Output with 32 Ω Load (dB)	Change in Quiescent Power Consumption (mW)
10	Enhanced Performance	4	11	0.9
00	Normal Operation	0	0	0

Table 14. PLL Power Comparison

Configuration	PLL Operation	Relative Power Consumption (mW), AVDD = 1.8V
PLL_FM_BYPASS=0, PLL_EN=1, FREQ_MULT_EN=0	Used (output 24.576MHz)	0
PLL_FM_BYPASS=1, PLL_EN=0, FREQ_MULT_EN=0	Bypassed	-0.45

Tensilica DSP Clock Speed Control

The core and bus clock of the Tensilica DSP can be programmed through the TDSP_CLK_RATE and BUS_CLK_RATE register to balance the performance and the power consumption. Table 15 shows the power comparison with different speed where the core and bus clock running at the same clock.

Table 15 TDSP Power Comparison Running at Different Clock

TDSP Core and Bus Frequency	Relative Power Consumption (mW), DVDD = 1.1V, FTDP runs FFT2048
24.576 MHz	0
49.152 MHz	4.67
73.728 MHz	9.3
98.304 MHz	13.78

Asynchronous Sample Rate Converters Low Power Modes

The ASRCs offer two separate, selectable low power operating modes. These modes allow power vs. performance trade-offs when using the ASRCs. Generally, if the data being sourced or sinked to the ASRCs is from or to the ADC or DAC using the ASRCx_LPM_II setting provides the lowest power consumption and does not degrade the performance of the converters.

Table 16 Input ASRC Power and Performance Options for 44.1kHz to 48kHz Conversion

Mode	THD+N at 1kHz [dB]	THD+N at 20kHz [dBFS]	DNR AW [dB]	Relative power per channel [mW)]
Default	123	123	130	0
ASRCI_LPM=1	120	118	130	-0.041
ASRCI_LPM_II=1	112	108	130	-0.058

Table 17 Output ASRC Power and Performance options for 48kHz to 44.1kHz Conversion

Mode	THD+N at 1kHz [dB]	THD+N at 20kHz [dBFS]	DNR AW [dB]	Relative power per channel [mW]
Default	123	123	130	0
ASRCO_LPM=1	120	118	130	-0.045
ASRCO_LPM_II=1	112	108	130	-0.070

CONTROL PORT

The ADAU186x has a 4-wire SPI control port, a 2-wire I²C control port and an UART control port. These three types control port share four pins. Each port can set the memories and registers.

The IC defaults to I²C mode but can be put into SPI control mode or UART control mode. By pulling the $\overline{SS}$ pin low three times, in the meanwhile, toggling the SCLK pin three times, the IC can enter SPI mode. By toggling the UART RX pin 3 times, in the meanwhile, keeping the SCLK pin in low or high level, the IC can enter UART mode. The detailed methods to switch the control mode from I2C to SPI or UART are listed in Table 17.

The control port is capable of full read/write operation for all addressable memories and registers. Most signal processing parameters are controlled by writing new values to the parameter memories using the control port. Other functions, such as mute and input/output mode control, are programmed through the registers.

All addresses can be accessed in either single address mode or burst mode. The first byte (Byte 0) of a control port write contains the 7-bit IC address plus R/$\overline{W}$ bit. The next four bytes (Byte 1 to Byte 4) are the 32-bit subaddress of the memory or register location within the ADAU186x. All subsequent bytes (starting with Byte 5) contain the data, such as the register, program, or parameter data. The exact formats for specific types of writing are shown in Figure 14 and Figure 15.

If large blocks of data must be downloaded to the ADAU186x FastDSP core or Tensilica DSP, the output of the core can be disabled, new data can be loaded, and the core can then be restarted. This restart is typically done during the booting sequence at start-up or when loading a new program into memory. The control port pins are multifunctional, depending on the mode in which the device is operating. Table 18 describes these multiple functions.

BURST MODE COMMUNICATION

Burst mode addressing, in which the sub-addresses are automatically incremented at word boundaries, can be used for writing large amounts of data to contiguous memory locations. This increment happens automatically after a single-word write unless the control port communication is stopped (that is, a stop condition is issued for I²C, or $\overline{SS}$ is brought high for SPI, or a tail byte is issued for UART). The registers and RAMs in the ADAU186x are 4 bytes width, except the AON registers which is 1 byte width, so the auto-increment feature knows the mapping between subaddresses and the word length of the destination register (or memory location).

Table 18. Control Port Switch

Control Mode Switch	ADDR0/$\overline{SS}$	ADDR1/MOSI/UART_CTRL_RX	SCL/SCLK	SDA/MISO/UART_CTRL_TX
I²C to SPI	Toggle the pin three times from low to high	Keep high or low during ADDR0 toggle	Toggle the pin three times	MISO
I²C to UART	Tie low	Toggle the pin three times from low to high	Tie low	UART_CTRL_TX

Table 19. Control Port Pin Functions

Pin	I²C Mode	SPI Mode	UART Mode
SCL/SCLK	SCL—input	SCLK—input	Not applicable, connect to low
SDA/MISO/UART_CTRL_TX	SDA—open-drain output	MISO—output	UART_CTRL_TX—output
ADDR1/MOSI/UART_CTRL_RX	I²C Address Bit 1—input	MOSI—input	UART_CTRL_RX—input
ADDR0/$\overline{SS}$	I²C Address Bit 0—input	$\overline{SS}$—input	Not applicable, connect to low

Table 20 I2C/SPI/UART Control Pins Function Setup List

Mode	IOVDD (V)	I2C Address	ADDR1/MOSI/UART_CTRL_RX Pin	ADDR0/$\overline{SS}$ Pin	SCL/SCLK Pin	SDA/MISO/ UART_CTRL_TX Pin
I²C	1.2 to 1.8	0x64	0	0	SCL	SDA
I²C	1.2 to 1.8	0x65	0	1	SCL	SDA
I²C	1.2 to 1.8	0x66	1	0	SCL	SDA
I²C	1.2 to 1.8	0x67	1	1	SCL	SDA
SPI	1.2-1.8V	Not applicable	MOSI	$\overline{SS}$	SCLK	MISO
UART	1.2-1.8V	Not applicable	UART_CTRL_RX	Not applicable	Not applicable	UART_CTRL_TX

READING AND WRITING TO MEMORIES

All Tensilica DSP and FastDSP memory locations are larger than a single byte. While each byte occupies a single address when communicating over a control interface (I²C or SPI or UART), when writing to these memories, an entire memory word must be written starting with the lowest address and continuing sequentially to the highest address for a write to actually occur. Similarly, a read must begin at the lowest memory address. However, for reads, all locations must be read. The mapping of bytes over the control interface is the least significant byte, or a memory location is written or read first, and the most significant byte is written or read last. The memories can be read or written in burst mode or single byte mode so that the proceeding requirements are met.

Table 21. Example write to Tensilica DSP Program RAM Word 0

Address	Data
0x 5FFF 8000	data[7:0]
0x 5FFF 8001	data[15:8]
0x 5FFF 8002	data[23:16]
0x 5FFF 8003	data[31:24] (the memory actually gets written after this write)

I²C CONTROL PORT

The ADAU186x supports a 2-wire serial (I²C-compatible) microprocessor bus driving multiple peripherals. I²C uses two pins, serial data (SDA) and serial clock (SCL), to carry data between the ADAU186x and the system I²C master controller. In I²C mode, the ADAU186x is always a slave on the bus. The device supports fast mode plus I²C operation, but for most bus capacitances, the SDA_MISO_DRIVE bit must be set to 1 to support these operating speeds.

Each slave device is recognized by a unique 7-bit device address. The ADAU186x I²C address format is shown in Table 21. The LSB of this first byte sent from the I²C master sets either a read or write operation. Logic Level 1 corresponds to a read operation, and Logic Level 0 corresponds to a write operation.

Pin ADDR0 and Pin ADDR1 set the LSBs of the I²C address (see Table 21). Therefore, each ADAU186x can be set to one of four unique addresses, allowing multiple ICs to exist on the same I²C bus without address contention. The 7-bit I²C addresses are shown in Table 22.

An I²C data transfer is always terminated by a stop condition.

Both SDA and SCL must have 2.0 kΩ pull-up resistors on the lines connected to these pins. The voltage on these signal lines cannot be higher than IOVDD.

Table 22. I²C Address Format

Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
1	1	0	0	1	ADDR1	ADDR0

Table 23. I²C Addresses

ADDR1	ADDR0	Slave Address
0	0	0x64
0	1	0x65
1	0	0x66
1	1	0x67

Addressing

Initially, each device on the IP2PC bus is in an idle state and monitoring the SDA and SCL lines for a start condition and the proper address. The IP2PC master initiates a data transfer by establishing a start condition, defined by a high to low transition on SDA while SCL remains high, indicating that an address/data stream follows. All devices on the bus respond to the start condition and shift the next eight bits (the 7-bit address plus the R/$\overline{W}$ bit) MSB first. The device that recognizes the transmitted address responds by pulling the data line low during the ninth clock pulse. This ninth bit is known as an acknowledge bit. All other devices withdraw from the bus at this point and return to the idle condition. The R/$\overline{W}$ bit determines the direction of the data. A Logic 0 on the LSB of the first byte indicates that the master writes information to the peripheral, whereas a Logic 1 indicates that the master reads information from the peripheral after writing the subaddress and repeating the start address. A data transfer takes place until a stop condition is encountered. A stop condition occurs when SDA transitions from low to high while SCL is held high. Figure 12 shows the timing of an I²C write and Figure 13 shows an I²C read.

Stop and start conditions can be detected at any stage during the data transfer. If these conditions are asserted out of sequence with normal read and write operations, the ADAU186x immediately jumps to the idle condition. During a given SCL high period, the user can only issue one start condition, one stop condition, or a single stop condition followed by a single start condition. A no-acknowledge condition is where the SDA line is not pulled low on the ninth clock pulse on SCL. If an invalid subaddress is issued by the user, the ADAU186x issues an acknowledge, but no data write occurs, and a read returns zero. If the highest subaddress location is reached during in write mode, the data for the invalid byte is not loaded to any subaddress register.

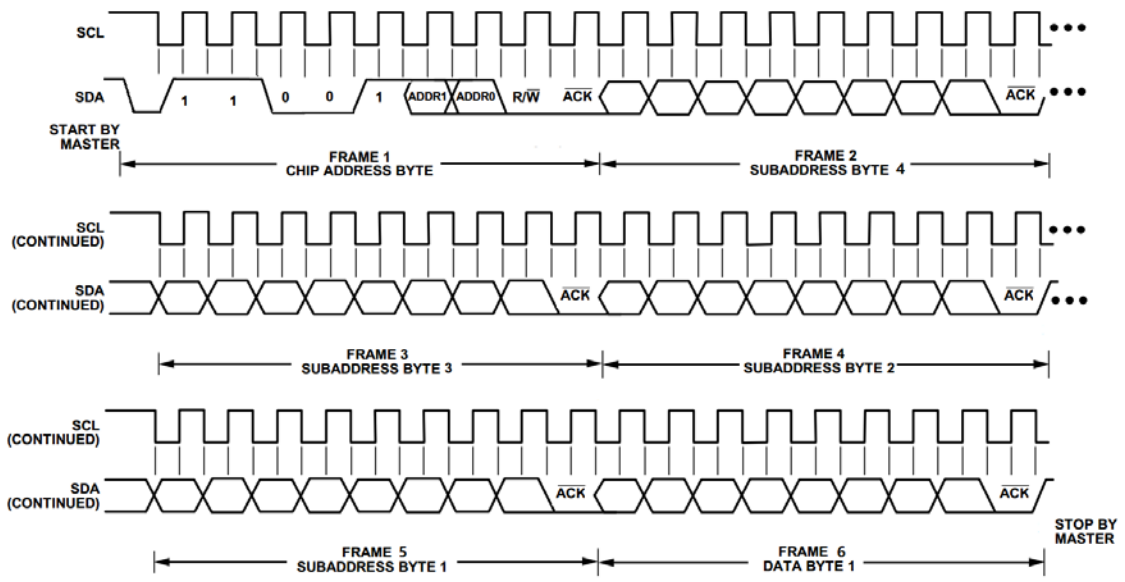


Figure 12. I²C Write to ADAU186x Clocking

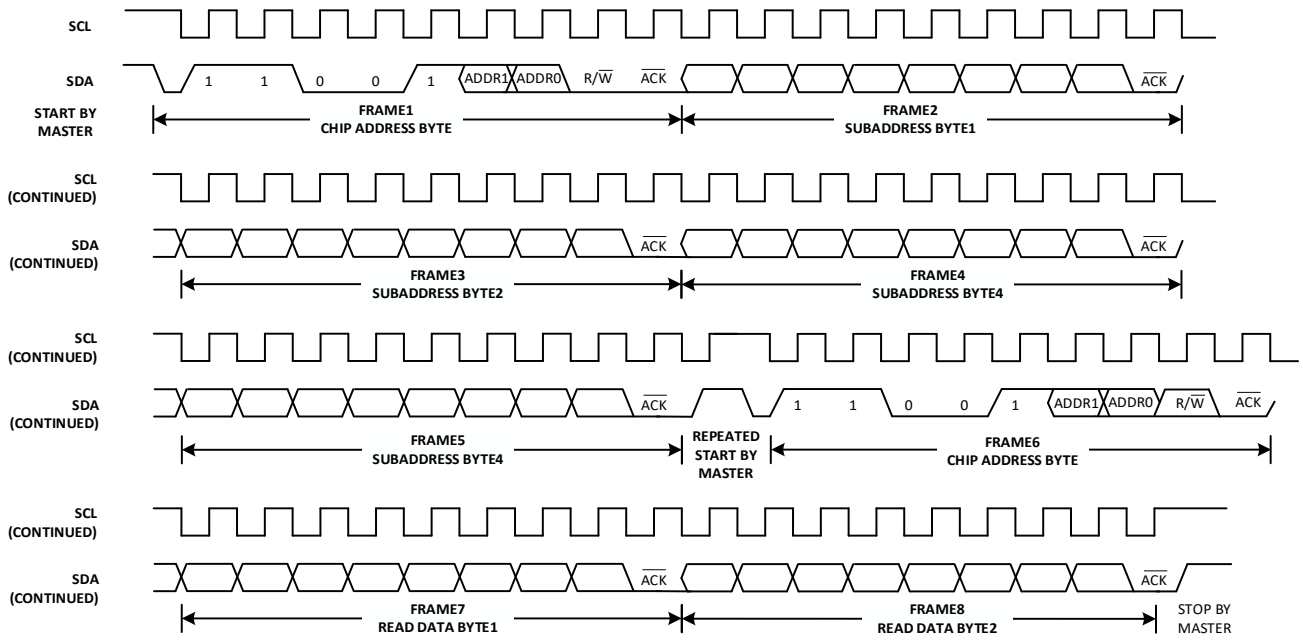


Figure 13. I²C Read from ADAU186x Clocking

I²C Read and Write Operations

Figure 14 shows the timing of a single word write operation. Every ninth clock pulse, the ADAU186x issues an acknowledge by pulling SDA low.

Figure 15 shows the timing of a burst mode write sequence where the target destination words are four bytes. The ADAU186x knows to increment its subaddress register every four bytes because the requested subaddress corresponds to a register or memory area with a 4-byte word length.

The timing of a single word read operation is shown in Figure 16. Note that the first R/$\overline{W}$ bit is 0, indicating a write operation because the subaddress still must be written to set up the internal address. After the ADAU186x acknowledges the receipt of the subaddress, the master must issue a repeated start command, followed by the chip address byte with R/$\overline{W}$ set to 1 (read), causing the ADAU186x SDA to reverse and begin driving data back to the master. The master then responds every ninth pulse with an acknowledge pulse to the ADAU186x.

Figure 17 shows the timing of a burst mode read sequence where the target read words are four bytes. The ADAU186x increments its subaddress every four bytes because the requested subaddress corresponds to a register or memory area with word lengths of four bytes. The ADAU186x always

decodes the subaddress and sets the auto-increment circuit so that the address increments after the appropriate number of bytes.

Figure 14 to Figure 17 use the following abbreviations:

- S is the start bit.
- P is the stop bit.
- AM is acknowledge by master.
- AS is acknowledge by slave.

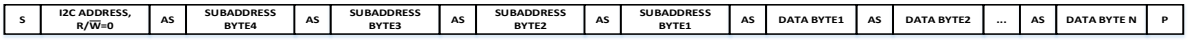


Figure 14. Single-Word I²C Write Format



Figure 15. Burst Mode I²C Write Format



Figure 16. Single-Word I²C Read Format

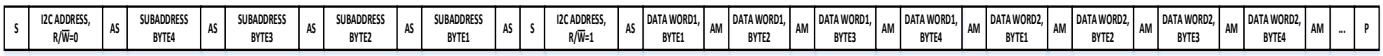


Figure 17. Burst Mode I²C Read Format

SPI CONTROL PORT

By default, the ADAU186x is in I2C mode, but the device can be put in SPI control mode by pulling $\overline{SS}$ low three times by issuing three SPI writes, which are in turn ignored by the ADAU186x. Recommend reading the VENDOR_ID register (address 0x 4000 C000) three times with SPI command from the external processor to enter to SPI mode. The next (fourth) SPI write is then latched in the SPI port.

The SPI port uses a 4-wire interface, consisting of $\overline{SS}$, SCLK, MOSI, and MISO signals, and is always a slave port. The $\overline{SS}$ signal must go low at the beginning of a transaction and high at the end of a transaction. The SCLK signal latches MOSI on a low to high transition. MISO data is shifted out of the ADAU186x on the falling edge of SCLK and must be clocked to a receiving device, such as a microcontroller, on the SCLK rising edge. The MOSI signal carries the serial input data, and the MISO signal is the serial output data. The MISO signal remains tristated until a read operation is requested, allowing other SPI-compatible peripherals to share the same readback line. The SCLK is low when clock is inactive.

All SPI transactions have the same basic format shown in Table 23. The timing diagrams for SPI write and SPI read are shown in Figure 18 and Figure 19, respectively. All data must be written MSB first. The ADAU186x can only be taken out of SPI mode by pulling the $\overline{PD}$ pin low or by powering down the IC.

Read/$\overline{Write}$

The first byte of an SPI transaction indicates whether the communication is a read or a write with the R/$\overline{W}$ bit. The LSB of this first byte determines whether the SPI transaction is a read (Logic Level 1) or a write (Logic Level 0).

Subaddress

The 32-bit subaddress word is decoded into a location in one of the memories or registers. This subaddress is the location of the appropriate memory location or register.

It is necessary to add an unused byte of zeros after the sub-address.

Data Bytes

The number of data bytes varies according to the register or memory being accessed. During a burst mode write, an initial subaddress is written followed by a continuous sequence of data for consecutive memory/register locations.

A sample timing diagram for a single-write SPI operation to the parameter RAM is shown in Figure 18. A sample timing diagram of a single-read SPI operation is shown in Figure 19. The MISO pin goes from tristate to being driven at the beginning of Byte 5. In this example, Byte 0 to Byte 4 contain the addresses and the R/$\overline{W}$ bit and subsequent bytes carry the data.

Table 24. Generic SPI Word Format

Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7 [1]
0000000, R/$\overline{W}$	Register/memory address Bits[31:24]	Register/memory address Bits[23:16]	Register/memory address Bits[15:8]	Register/memory address Bits[7:0]	Zeros Bits[7:0] (Dummy)	data	data

[1] Continues to end of data.

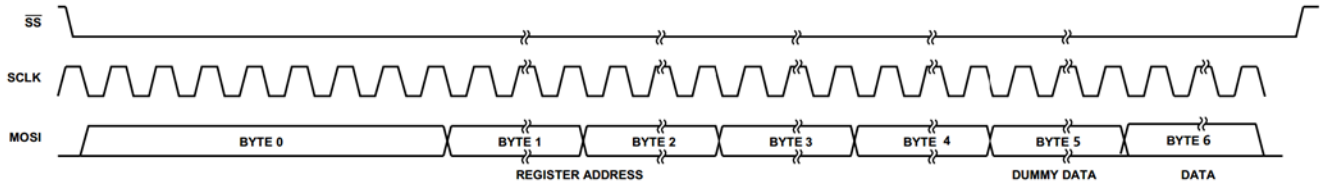


Figure 18. SPI Write to ADAU186x Clocking (Single-Write Mode)

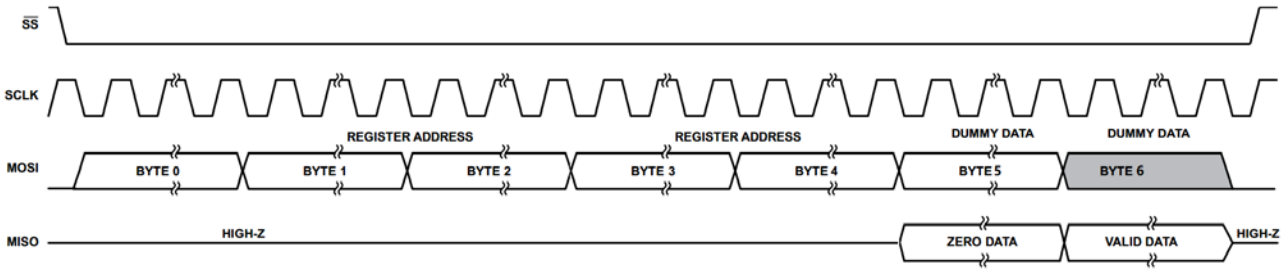


Figure 19. SPI Read from ADAU186x Clocking (Single-Read Mode)

UART CONTROL PORT

By default, the ADAU186x is in I²C mode, but the device can be put in UART mode by receiving a specific data which has three times of rising edge on the UART_CTRL_RX pin of the ADAU186x. Recommend reading the VENDOR_ID register (address 0x 4000 C000) with UART read command (ID must be 0x 00) from the external processor to enter to UART mode.

The UART port uses two pins, UART_CTRL_TX and UART_CTRL_RX, to transmit data and receive data. ADAU186x UART works in half duplex mode and supports auto baud rate from 1200 to 1M. Each data-word contains a start bit, eight data bits, a bit for even parity and one stop bit.

When using UART as the control port, it is necessary that an input clock on the MCLKIN pin is ready or the external crystal is provided before the first UART communication. It is recommended to use a lower baud rate at initial stage due to the limitation of the clock on the MCLKIN which is perhaps lower as 30kHz. After the PLL configuration by the UART port, higher baud rate can be used depending on the clock setting.

Write Frame

When a microprocessor sends a write frame to ADAU186x by UART control port, the frame must comply with the format which is shown in Figure 20. It includes one byte of Head, one byte of ID, one byte of write command, four bytes of the start address of writing data, two bytes of the data length which will be wrote to ADAU186x, N bytes of data, one byte of CRC, and one byte of Tail. After ADAU186x receives the byte of Tail, ADAU186x will send back a frame on the UART_CTRL_TX line following the byte of Tail on the UART_CTRL_RX line, starting with one byte of head and ending with one byte of Tail, includes one byte of ID, one byte of Status, and one byte of CRC.

Read Frame

When a microprocessor sends a read frame to ADAU186x by UART control port, the frame must comply with the format which is shown in Figure 21. It includes one byte of Head, one byte of ID, one byte of read command, four bytes of the start address of reading data, two bytes of the data length which will be read from ADAU186x, one byte of CRC, and one byte of Tail. After ADAU186x receives the byte of Tail, ADAU186x will send back a frame on the UART_CTRL_TX line following the byte of Tail on the UART_CTRL_RX line, starting with one byte of head and ending with one byte of Tail, includes one byte of ID, one byte of Status, N bytes of data read from ADAU186x, and one byte of CRC.

Head and Tail

When a microprocessor sends a frame to ADAU186x, each transaction must start with a byte of head (0x7F) and end with a byte of tail (0xF0) no matter a write or a read frame.

When ADAU186x sends back a frame, it will also comply with the format that start with a byte of head and end with a byte of tail.

ID

The number of the ID can be an arbitrary number and diverse in different frame. But the ADAU186x will send back the ID which is same with the ID that ADAU186x received from the microprocessor in one frame.

Read/$\overline{\text{Write}}$

The third byte of an UART transaction frame indicates whether the communication is a read or a write with the R/$\overline{\text{W}}$ bit. The MSB of this third byte determines whether the UART transaction is a read (Logic Level 1) or a write (Logic Level 0).

CRC

An 8-bit CRC validates the content of the UART frame. This CRC is strong enough to detect single error bursts of up to eight bits in size. The polynomial for the CRC is $x^8+x^7+x^3+x^2+1$ and the CRC seed is 0x FF.

Each write frame or read frame contains CRC on both UART_CTRL_RX line generated by the microprocessor and UART_CTRL_TX line generated by the ADAU186x. For a write frame, the one byte CRC generated by the microprocessor covers the content from the ID to the last number of Data. The one byte CRC generated by the ADAU186x covers the content of ID and Status (See Figure 20). For a read frame, the one byte CRC generated by the microprocessor covers the content from the ID to the byte of LENL. The one byte CRC generated by the ADAU186x covers the content from the ID to the last number of Data sent to microprocessor (See Figure 21).

If customer doesn't need the CRC feature, customer can write any data in the byte of CRC sent from microprocessor and ignore the byte of CRC and the CRC flag in the byte of Status sent from ADAU186x.

Status Information

When ADAU186x sends information to the microprocessor in a write frame or a read frame, a byte of status will be contained. The higher 4 bits show the previous transaction status which includes CRC, integrity and parity. Similar to the lower 4 bits which show the current transaction status (See Figure 22 for details).

If there is CRC error during a write frame or a read frame, the CRC bit in the byte of Status will be set to 1. Otherwise it will be 0.

If there is integrity error during a write frame or a read frame, the integrity bit in the byte of Status will be set to 1. Otherwise it will be 0.

If there is parity error during a write frame or a read frame, the parity bit in the byte of Status will be set to 1. Otherwise it will be 0.

Recover from an Abnormal Frame

If any abnormity happens when the microprocess writes a write or read frame to ADAU186x, the microprocess can send one byte of Head and then follow another byte of Head to start a new transaction.

Figure 23 shows an abnormal case, once the ADAU186x recognizes the second Head, the previous transaction will be discarded immediately no matter response or not yet. The event will be recorded, the status information will be sent in the higher 4 bits of the status data.

ESCAPE

During the write or read frame, escape is supported and performed during the UART frame between Head and Tail if the data is same with the Head or Tail. The escape formats are shown in below.

Escape Head: 0x7F = 0xFF + 0x5F

Escape Tail: 0xF0 = 0xFF + 0xD0

Escape None: 0xFF = 0xFF + 0xDF

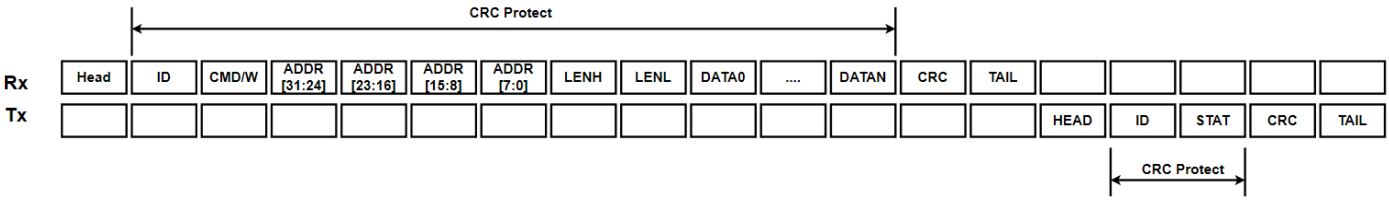


Figure 20. UART Write Frame

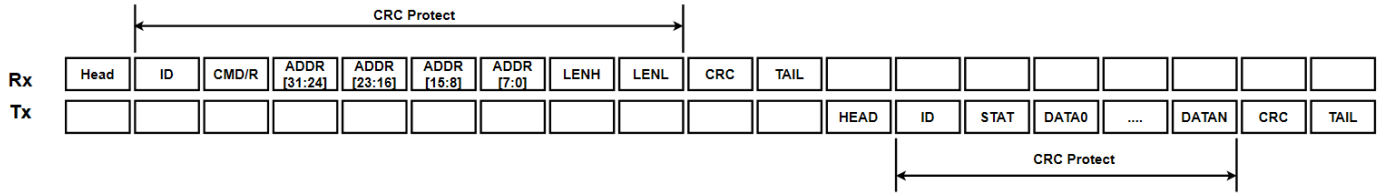


Figure 21. UART Read Frame

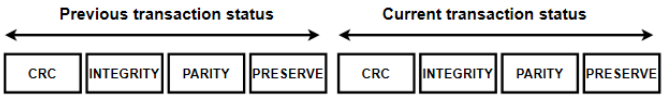


Figure 22. Content in the Byte of Status

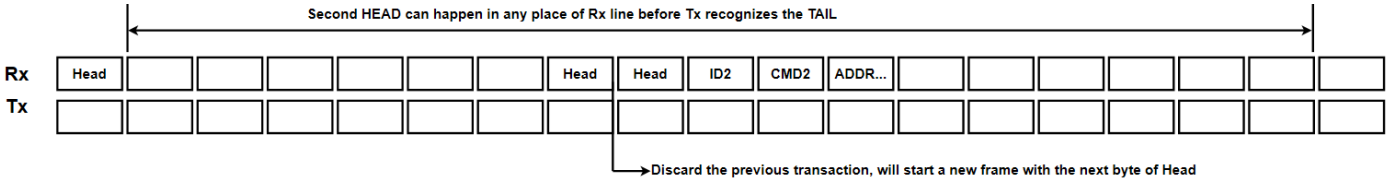


Figure 23. Recover from an Abnormal Frame

SELF BOOT

There are two boot mode in the ADAU186x: self-boot mode, or non-self-boot mode.

The ADAU186x works in self-boot mode when the SELFBOOT pin is set high at power up and $\overline{PD}$, otherwise it works in non-self-boot mode. The state of the SELFBOOT pin is checked internally only when the ADAU186x comes out of a reset via the $\overline{PD}$ pin going high or by applying power with $\overline{PD}$ already set high. When the device comes out of reset and there is a MCLK source present (on the MCLKIN pin), the state of the SELFBOOT pin is registered to determine whether to self-boot.

At reset, the state is set not to self-boot. Therefore, if there is no clock at the MCLKIN pin, the self-boot does not occur.

During booting, ensure that there is a stable DVDD in the system. The $\overline{PD}$ pin must remain high during self-boot operation. If the SELFBOOT pin is not used for a multipurpose pin function, tie the pin to either IOVDD or DGND.

In the self-boot mode, the TDSP actively load program and data through the QSPI. In the self-boot mode, ensure that the clock on the MCLK pin is higher than 11MHz. The ROM_BOOT_MODE pin determines the ROM boot up mode. when ROM_BOOT_MODE pin is high, the internal QSPI controller will using the un-packed mode to load program and data from external flash memory. When ROM_BOOT_MODE pin is low, the internal QSPI controller will using the packed mode to load program and data from external flash memory.

In the non-self-boot mode, the processors in the ADAU186x receives program and data from external host devices through control port, I^2C/SPI/UART.

MULTIPURPOSE PINS

The ADAU186x has twenty-six multipurpose (MPx) pins that can be used for general purpose input, general purpose output, clock outputs, interrupts out, interrupts input and PDM outputs. Each pin can be individually set to either its default or MPx setting. The function of each of these pins is set in using the MPx_MODE bits. By default, each pin is configured as its normal function. Care must be taken when using SELFBOOT/MP19 or ROM_BOOT_MODE/MP21 as multipurpose pins. The state of these two pins at power-up (later of either $\overline{PD}$ pin going high or power being applied with the $\overline{PD}$ pin already high) determine whether the device self boots and ROM boot up mode, which must still be followed even if the pin is used for another multipurpose function later.

When a MPx pin is set as a general-purpose input, the MPx pin can be read via all control interfaces via the GPIOx_IN bits, the pin can also be read and acted upon by the Tensilica DSP core, and the pin can be used to conditionally execute instructions or trigger the compressor in the FastDSP.

When an MPx pin is set as general-purpose output, the state of the pin can be set via all control interfaces using the GPIOx_OUT bits or by the Tensilica DSP core.

Any MPx pin can be used as a master clock output. The rate of the master clock output is determined by the MCLKO_RATE bits. Multiple pins can be used as this function if desired.

Any MPx pin can be used to output the PDM clock or data signal for the PDM output interface. Any MPx pin can be used to output the interrupt status from the four interrupt sources.

Any MPx pin can be used as the interrupt input and the event will be record in the MP_INPUT_IRQx_STATUSx registers.

Table 25. Multipurpose Pin Functions

MPx Pin Function	Direction
Normal Operation	Default
General Purpose Input (GPI)	In
General Purpose Output from GPIOx_OUT register (GPO_REG)	Out
General Purpose Output from Tensilica DSP (GPO_TDSP)	Out
MCLK Output (MCLKO)	Out
IRQ1 Output (IRQ1)	Out
IRQ2 Output (IRQ2)	Out
IRQ3 Output (IRQ3)	Out
IRQ4 Output (IRQ4)	Out
IRQ Input	In
PDM Clock Output	Out
PDM Data Output	Out
DMIC 45	In
DMIC67	In
DMIC_CLK0_INV	Out
DMIC_CLK1	Out

Interrupts

Each multipurpose pin can be used to output one of four interrupts that have various sources when selected for this function. The sources of the interrupts are for the AVDD undervoltage warning, PLL locked or unlocked, DAC and ADC channels clipping, input and output ASRCs locked or unlocked, EQ memory clear done, FDSP output not zero, parameter ramp up complete, DLDO DVS done, power up complete, serial port locked or unlocked, phase sync locked or unlocked and the generic Tensilica DSP interrupts. Each interrupt source can be individually masked with their respective IRQx_MASKx registers. Each interrupt output can be set to active low or active high output on the pin selected for the interrupt output via the IRQx_PINOUT_POL bits. The status of each interrupt source can be read via the IRQ status registers (IRQx_STATUSx). When an interrupt source is masked, if that interrupt becomes true, the interrupt is shown in the interrupt status registers but does not cause the MPx pin (if set as interrupt output) to show an interrupt. All sources of each interrupt are cleared via a write of 1 to the IRQx_CLEAR bits. The interrupt status bits are sticky, such that if an interrupt source becomes true, the status reads 1 until a clear occurs, even if that interrupt source is no longer true. The Tensilica DSP interrupts are initiated by the Tensilica DSP writing the SW_INTx bits.

The multipurpose pin can be used as interrupt input and the event will be record in the MP_INPUT_IRQx_STATUSx register which will trig an interrupt to the Tensilica.

Each interrupt input source can be individually masked with their respective MP_INPUT_IRQx_MASKx registers.

Each interrupt input can be set to active low, active high, rising edge or falling edge on the pin selected for the interrupt input via the MP_INPUT_IRQx_MODE bits.

The MP_INPUT_IRQx_STATUS value will be cleaned via a write of 1 to the MP_INPUT_IRQx_CLEAR bit.

Pin Controls

Each pin that can be used as a multipurpose pin has several control selections to set various setting. When the pin is used as an output, the drive strength can be selected at 2 mA, 4 mA, 8 mA, or 12 mA. In addition, a weak pull-up or pull-down can be selected. These settings are in their respective pin control register. These pin control settings affect the pins operation in both normal functional mode and when used in all multipurpose pin modes.

SERIAL DATA PORTS

The serial data input and output ports of the ADAU186x can be set to accept or transmit data in a 2-channel format such as I2S or up to 12 channels in a time division multiplexing (TDM) stream to interface to external ADCs, DACs, DSPs, and system on chips (SOCs). Data is processed in twos complement, MSB first format. The left channel data field always precedes the right channel data field in 2-channel streams.

The serial data clocks do not need to be synchronous with the ADAU186x master clock input, but the frame clock and bit clock must be synchronous to each other. The FSYNC_x and BCLK_x pins are used to clock both the serial input and output ports. The FSYNC_x and BCLK_x pins can also be used as a source to the PLL to provide the main chip clock. Each serial port can be set to be either the master or the slave in a system. Because there is only one set of serial data clocks, the input and output of a single port must always both be either master or slave.

The SPTx_SAI_MODE bits set whether the serial port is operating in stereo mode or TDM mode. In stereo modes, both edges of frame clock determine where data is placed, and the left channel maps to the output for Channel 0, while the right channel maps to the output for Channel 1. In TDM mode only, the rising edge of frame clock determine where data is placed. In TDM mode, each channel of data receives a slot that can be either 16, 24, or 32 BCLKs wide. The width of each slot is determined by the SPTx_SLOT_WIDTH bits. In TDM mode, the slot number is affected by register FDSP_MODE[0] value. Detailed information please refer to the Digital Turbo Mode section.

The serial data control registers allow control of the clock polarity and the data input modes. The valid data formats are I2S (delay by 1), left justified (delay by 0), or right justified (delay by 8, 12, or 16 BCLKs). The delay indicates the number of bit clocks BCLKs from the rising/falling edge of frame clock FSYNC_x where the MSB of the data is placed in stereo modes, and the number of bit clocks BCLKs from the rising of frame clock in TDM mode. In all modes except for the right justified mode, the serial port inputs an arbitrary number of bits up to a limit of 24. Extra bits do not cause an error, but the bits are ignored. The serial port can operate with an arbitrary number of bit clock BCLK_x transitions in each frame clock frame.

When using a high bit clock rate (12.288 MHz or higher), it is recommended to increase the drive strength settings for the output signal pins. The high drive strength effectively speeds up the transition times of the waveforms, thereby improving the signal integrity of the clock and data lines. The drive strength can be set in the pad drive strength registers.

Table 25 describes the proper serial port settings for standard audio data formats. More information about the settings in Table 25 can be found in the SPTx_CTRLx register descriptions.

The polarity of both frame clock and bit clock can be inverted via the SPTx_LRCLK_POL and SPTx_BCLK_POL bits. These bits do not need to be used to support the typical formats shown in Table 25. Setting either SPTx_LRCLK_POL or SPTx_BCLK_POL to 1 places an inverter at the input to the serial port on its respective signal. For example, while serial data and frame clock are normally sampled on the rising edge of bit clock, setting SPTx_BCLK_POL = 1 samples on the falling edge of bit clock.

Each serial port can be set to be a master, in which case BCLK_x and FSYNC_x are driven as outputs. The output rate and direction of these two signals are set via the SPTx_LRCLK_SRC and SPTx_BCLK_SRC bits. A bit clock rate higher than 24.576 MHz cannot be generated. Therefore, the settings of these registers that request this rate result in no bit clock.

Unused bit slots can be tristated so that multiple ICs can drive a single serial data bus, which is controlled via the SPTx_TRI_STATE bit. For example, in a 32-bit TDM frame with 24-bit data, the eight unused bits are tristated. Inactive channels are also tristated for one full frame each. Serial output channels are disabled when the SPTx_OUT_ROUTEy bits are set to 0x3F. Note that the timing for serial data output changes based on the minimum IOVDD voltage. While the serial ports can work for inputting a signal on SDATAI_x for any IOVDD and bit clock rate within the specification, the delay on SDATAO_x at 1.1 V excludes operating at higher bit clock rates.

Table 26. Serial Port Data Format Settings

Format	Frame Clock Mode (SPTx_SAI_MODE)	Slot Width (SPTx_SLOT_WIDTH)	MSB Position from Start of Frame Clock (SPTx_DATA_FORMAT)
I²S (see Figure 24)	0 (50% duty cycle)	XX	000 (One bit clock delay)
Left Justified (see Figure 24)	0	XX	001 (No delay)
Right Justified (see Figure 24)	0	XX	010 (Delay by 8 bit clocks)
	0	XX	011 (Delay by 12 bit clocks)
	0	XX	100 (Delay by 16 bit clocks)
TDM (see Figure 25)	1 (single bit clock wide pulse)	XX	000 (One bit clock delay)

[1] X = don't care.

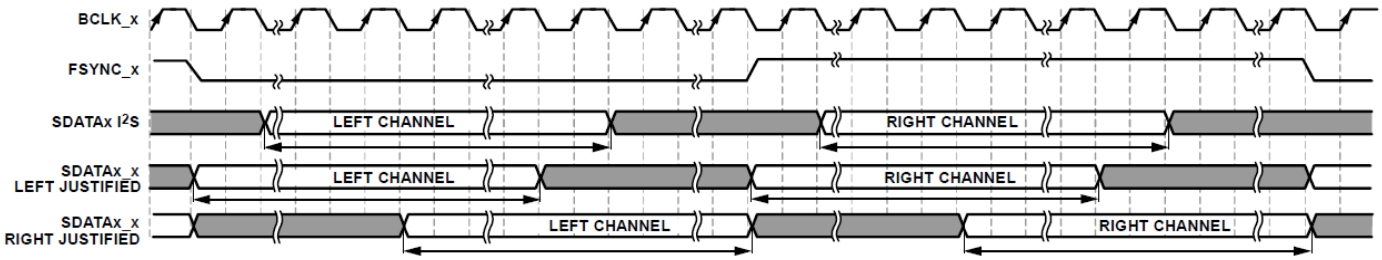


Figure 24. Stereo Modes : I²S, Left Justified, Right Justified Modes, 16 Bits to 24 Bits per Channel, any number of BCLKs are allowed

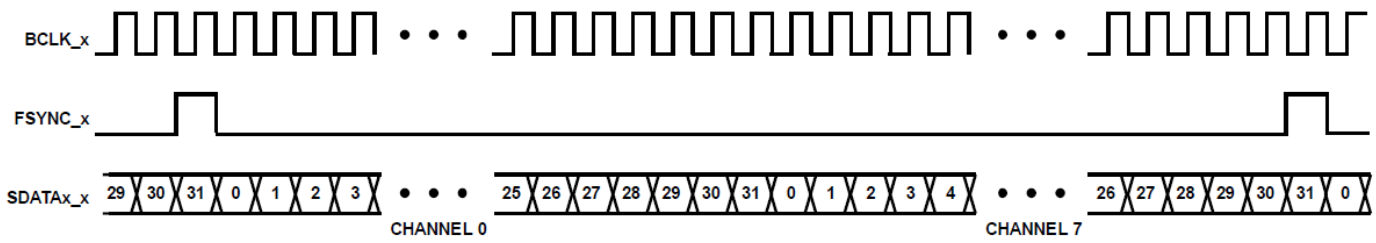


Figure 25. 8-Channel TDM Mode, default settings, except SPTx_SAI_MODE=1

APPLICATIONS INFORMATION

POWER SUPPLY BYPASS CAPACITORS

Each analog and digital power supply pin should be bypassed to its nearest appropriate ground pin with a single 0.1 µF capacitor. The connections to each side of the capacitor should be as short as possible, and the trace should be routed on a single layer with no vias. For maximum effectiveness, locate the capacitor equidistant from the power and ground pins or slightly closer to the power pin if equidistant placement is not possible. Thermal connections to the ground planes should be made on the far side of the capacitor.

Each supply signal on the board should also be bypassed with a single bulk capacitor (10 µF to 47 µF).

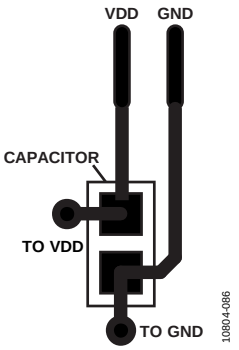


Figure 26. Recommended Power Supply Bypass Capacitor Layout

LAYOUT

The HPVDD, HPVDD_L supply is for the headphone amplifiers. If the headphone amplifiers are enabled, the PCB trace to this pin should be wider than traces to other pins to increase the current carrying capacity. A wider trace should also be used for the headphone output lines.

GROUNDING

A single ground plane should be used in the application layout. Components in an analog signal path should be placed away from digital signals.

INSTANCE TABLE: ADAU186X

Table 27. Instance Summary

Name	Module	Address
DMA	DMA	0x40000000
QSPI	QSPI	0x40001000
SOC	TENSILICA DSP	0x40002000
SPI	SPI	0x40004000
UART	UART	0x40004800
GPT	TMR	0x40006000
WDT	WDT	0x40006800
CONTROL_REGs	ADP	0x4000C000
FFSRAM0_CFG	FIFO_SRAM	0x40010000
FFSRAM1_CFG	FIFO_SRAM	0x40020000
FFSRAM2_CFG	FIFO_SRAM	0x40030000
DTSYNC	DATA_SYNC	0x40040300

REGISTER SUMMARY: DMA

Table 28. DMA(DMA) Register Summary

Address	Name	Description	Reset	Access
0x40000000	DMA_STATUS	DMA Status.	0x00130000	R
0x40000004	DMA_CTRL	DMA Configuration.	0x00000000	W
0x40000008	DMA_PRI_BASE_PTR	DMA Channel Primary Control Data Base Pointer.	0x00000000	R/W
0x4000000C	DMA_ALT_BASE_PTR	DMA Channel Alternate Control Data Base Pointer.	0x00000200	R
0x40000014	DMA_SW_REQ	DMA Channel Software Request.	0x00000000	W
0x40000020	DMA_REQ_MASK_SET	DMA Channel Request Mask Set.	0x00000000	R/W
0x40000024	DMA_REQ_MASK_CLR	DMA Channel Request Mask Clear.	0x00000000	W
0x40000028	DMA_CHNL_EN_SET	DMA Channel Enable Set.	0x00000000	R/W
0x4000002C	DMA_CHNL_EN_CLR	DMA Channel Enable Clear.	0x00000000	W
0x40000030	DMA_PRI_ALT_SET	DMA Channel Primary-alternate Set.	0x00000000	R/W
0x40000034	DMA_PRI_ALT_CLR	DMA Channel Primary-alternate Clear.	0x00000000	W
0x40000038	DMA_PRIORITY_SET	DMA Channel Priority Set.	0x00000000	W
0x4000003C	DMA_PRIORITY_CLR	DMA Channel Priority Clear.	0x00000000	W
0x40000040	DMA_INT_EN_SET	DMA Interrupt Enable Set.	0x00000000	R/W
0x40000044	DMA_INT_EN_CLR	DMA Interrupt Enable Clear.	0x00000000	R/W
0x4000004C	DMA_INT_STATUS_CLR	DMA Interrupt Status Clear.	0x00000000	R/W
0x40000054	DMA_BUSERR_IDX_CLR	DMA per Channel Error Clear.	0x00000000	R/W
0x4000005C	DMA_INVDESC_IDX_CLR	DMA per Channel Invalid Descriptor Clear.	0x00000000	R/W
0x40000800	DMA_BYTE_SWAP_SET	DMA Channel Bytes Swap Enable Set.	0x00000000	R/W
0x40000804	DMA_BYTE_SWAP_CLR	DMA Channel Bytes Swap Enable Clear.	0x00000000	W
0x40000810	DMA_SRCADR_DEC_SET	DMA Channel Source Address Decrement Enable Set.	0x00000000	R/W
0x40000814	DMA_SRCADR_DEC_CLR	DMA Channel Source Address Decrement Enable Clear.	0x00000000	W
0x40000818	DMA_DSTADR_DEC_SET	DMA Channel Destination Address Decrement Enable Set.	0x00000000	R/W
0x4000081C	DMA_DSTADR_DEC_CLR	DMA Channel Destination Address Decrement Enable Clear.	0x00000000	W
0x40000FE0	DMA_REV_ID	DMA Controller Revision ID.	0x00000002	R

REGISTER DETAILS: DMA

DMA STATUS REGISTER

Address: 0x40000000, Reset: 0x00130000, Name: DMA_STATUS

Table 29. Bit Descriptions for DMA_STATUS

Bits	Bit Name	Settings	Description	Reset	Access
[31:21]	RESERVED		Reserved.	0x0	R
[20:16]	DMA_STATUS_CHNL_NUM		Number of Available DMA Channels Minus 1. With 20 channels available, the register will read back 0x13.	0x13	R
[15:1]	RESERVED		Reserved.	0x0	R
0	DMA_STATUS_EN		Enable Status of the Controller.	0x0	R

DMA CONFIGURATION REGISTER

Address: 0x40000004, Reset: 0x00000000, Name: DMA_CTRL

Table 30. Bit Descriptions for DMA_CTRL

Bits	Bit Name	Settings	Description	Reset	Access
[31:1]	RESERVED		Reserved.	0x0	R
0	DMA_EN		Controller Enable.	0x0	W

DMA CHANNEL PRIMARY CONTROL DATA BASE POINTER REGISTER

Address: 0x40000008, Reset: 0x00000000, Name: DMA_PRI_BASE_PTR

The DMA_DMA_PRI_BASE_PTR register must be programmed to point to the primary channel control base pointer in the system memory. The amount of system memory that must be assigned to the DMA controller depends on the number of DMA channels used and whether the alternate channel control data structure is used. This register cannot be read when the DMA controller is in the reset state.

Table 31. Bit Descriptions for DMA_PRI_BASE_PTR

Bits	Bit Name	Settings	Description	Reset	Access
[31:0]	DMA_PRI_BASE_PTR		Pointer to the Base Address of the Primary Data Structure. 5 + log(2) M LSBs are reserved and must be written 0. M is number of channels.	0x0	R/W

DMA CHANNEL ALTERNATE CONTROL DATA BASE POINTER REGISTER

Address: 0x4000000C, Reset: 0x00000200, Name: DMA_ALT_BASE_PTR

The DMA_DMA_ALT_BASE_PTR read-only register returns the base address of the alternate channel control data structure. This register removes the necessity for application software to calculate the base address of the alternate data structure. This register cannot be read when the DMA controller is in the reset state.

Table 32. Bit Descriptions for DMA_ALT_BASE_PTR

Bits	Bit Name	Settings	Description	Reset	Access
[31:0]	DMA_ALT_BASE_PTR		Base Address of the Alternate Data Structure.	0x200	R

DMA CHANNEL SOFTWARE REQUEST REGISTER

Address: 0x40000014, Reset: 0x00000000, Name: DMA_SW_REQ

The DMA_DMA_SW_REQ register enables the generation of software DMA request. Each bit of the register represents the corresponding channel number in the DMA controller. M is the number of DMA channels

Table 33. Bit Descriptions for DMA_SW_REQ

Bits	Bit Name	Settings	Description	Reset	Access
[31:20]	RESERVED		Reserved.	0x0	R
[19:0]	DMA_SW_REQ_IDX		Generate Software Request. Set the appropriate bit to generate a software DMA request on the corresponding DMA channel. Bit 0 corresponds to DMA channel 0 … Bit M-1 corresponds to DMA channel M-1 When Written: DMA_DMA_SW_REQ.DMA_SW_REQ_IDX[C] = 0, Does not create a DMA request for channel C. DMA_DMA_SW_REQ.DMA_SW_REQ_IDX[C] = 1, Generates a DMA request for channel C. These bits are automatically cleared by the hardware after the corresponding software request completes.	0x0	W

DMA CHANNEL REQUEST MASK SET REGISTER

Address: 0x40000020, Reset: 0x00000000, Name: DMA_REQ_MASK_SET

Table 34. Bit Descriptions for DMA_REQ_MASK_SET

Bits	Bit Name	Settings	Description	Reset	Access
[31:20]	RESERVED		Reserved.	0x0	R
[19:0]	DMA_REQ_MASK_SET		Mask Requests from DMA Channels. This register disables DMA requests from peripherals. Each bit of the register represents the corresponding channel number in the DMA controller. Set the appropriate bit to mask the request from the corresponding DMA channel. Bit 0 corresponds to DMA channel 0 … Bit M-1 corresponds to DMA channel M-1 When Read: DMA_DMA_REQ_MASK_SET.DMA_REQ_MASK_SET[C] = 0, Requests are enabled for channel C. DMA_DMA_REQ_MASK_SET.DMA_REQ_MASK_SET[C] = 1, Requests are disabled for channel C. When Written: DMA_DMA_REQ_MASK_SET.DMA_REQ_MASK_SET[C] = 0, No effect. Use the DMA_DMA_REQ_MASK_SET register to enable DMA requests. DMA_DMA_REQ_MASK_SET.DMA_REQ_MASK_SET[C] = 1, Disables peripheral associated with channel C from generating DMA requests.	0x0	R/W

DMA CHANNEL REQUEST MASK CLEAR REGISTER

Address: 0x40000024, Reset: 0x00000000, Name: DMA_REQ_MASK_CLR

Table 35. Bit Descriptions for DMA_REQ_MASK_CLR

Bits	Bit Name	Settings	Description	Reset	Access
[31:20]	RESERVED		Reserved.	0x0	R
[19:0]	DMA_REQ_MASK_CLR		Clear REQ_MASK_SET Bits in DMA_REQ_MASK_SET. This register enables DMA requests from peripherals by clearing the mask set in DMA_DMA_REQ_MASK_SET register. Each bit of the register represents the corresponding channel number in the DMA controller. Set the appropriate bit to clear the corresponding DMA_DMA_REQ_MASK_SET.DMA_REQ_MASK_SET bit. Bit 0 corresponds to DMA channel 0 … Bit M-1 corresponds to DMA channel M-1 When Written: DMA_DMA_REQ_MASK_CLR.DMA_REQ_MASK_CLR[C] = 0, No effect. Use the DMA_DMA_REQ_MASK_SET register to disable DMA requests. DMA_DMA_REQ_MASK_CLR.DMA_REQ_MASK_CLR[C] = 1, Enables peripheral associated with channel C to generate DMA requests.	0x0	W

DMA CHANNEL ENABLE SET REGISTER

Address: 0x40000028, Reset: 0x00000000, Name: DMA_CHNL_EN_SET

Table 36. Bit Descriptions for DMA_CHNL_EN_SET

Bits	Bit Name	Settings	Description	Reset	Access
[31:20]	RESERVED		Reserved.	0x0	R
[19:0]	DMA_CHNL_EN_SET		Enable DMA Channels. This register allows for the enabling of DMA channels. Reading the register returns the enable status of the channels. Each bit of the register represents the corresponding channel number in the DMA controller. Set the appropriate bit to enable the corresponding channel. Bit 0 corresponds to DMA channel 0 … Bit M-1 corresponds to DMA channel M -1 When Read: DMA_DMA_CHNL_EN_SET.DMA_CHNL_EN_SET[C] = 0, Channel C is disabled. DMA_DMA_CHNL_EN_SET.DMA_CHNL_EN_SET[C] = 1, Channel C is enabled. When Written: DMA_DMA_CHNL_EN_SET.DMA_CHNL_EN_SET[C] = 0, No effect. Use the DMA_DMA_CHNL_EN_CLR register to disable the channel. DMA_DMA_CHNL_EN_SET.DMA_CHNL_EN_SET[C] = 1, Enables channel C.	0x0	R/W

DMA CHANNEL ENABLE CLEAR REGISTER

Address: 0x4000002C, Reset: 0x00000000, Name: DMA_CHNL_EN_CLR

Table 37. Bit Descriptions for DMA_CHNL_EN_CLR

Bits	Bit Name	Settings	Description	Reset	Access
[31:20]	RESERVED		Reserved.	0x0	R
[19:0]	DMA_CHNL_EN_CLR		Disable DMA Channels. This register allows for the disabling of DMA channels. This register is write only. Each bit of the register represents the corresponding channel number in the DMA controller. Note: The controller disables a channel automatically, by setting the appropriate bit, when it completes the DMA cycle. Set the appropriate bit to disable the corresponding channel. Bit 0 corresponds to DMA channel 0 … Bit M-1 corresponds to DMA channel M-1 When Written: DMA_DMA_CHNL_EN_CLR.DMA_CHNL_EN_CLR[C] = 0, No effect. Use the DMA_DMA_CHNL_EN_SET register to enable the channel. DMA_DMA_CHNL_EN_CLR.DMA_CHNL_EN_CLR[C] = 1, Disables channel C.	0x0	W

DMA CHANNEL PRIMARY-ALTERNATE SET REGISTER

Address: 0x40000030, Reset: 0x00000000, Name: DMA_PRI_ALT_SET

The DMA_DMA_PRI_ALT_SET register enables the user to configure the appropriate DMA channel to use the alternate control data structure. Reading the register returns the status of which data structure is in use for the corresponding DMA channel. Each bit of the register represents the corresponding channel number in the DMA controller.

Note: The DMA controller sets/clears these bits automatically as necessary for ping-pong, memory scatter-gather and peripheral scatter-gather transfers.

Table 38. Bit Descriptions for DMA_PRI_ALT_SET

Bits	Bit Name	Settings	Description	Reset	Access
[31:20]	RESERVED		Reserved.	0x0	R
[19:0]	DMA_PRI_ALT_SET		Control Structure Status / Select Alt Structure. Returns the channel control data structure status, or selects the alternate data structure for the corresponding DMA channel. Bit 0 corresponds to DMA channel 0 … Bit M-1 corresponds to DMA channel M-1 When Read: DMA_DMA_PRI_ALT_SET.DMA_PRI_ALT_SET[C] = 0, DMA channel C is using the primary data structure. DMA_DMA_PRI_ALT_SET.DMA_PRI_ALT_SET[C] = 1, DMA channel C is using the alternate data structure. When Written: DMA_DMA_PRI_ALT_SET.DMA_PRI_ALT_SET[C] = 0, No effect. Use the DMA_DMA_PRI_ALT_CLR register to set bit [C] to 0. DMA_DMA_PRI_ALT_SET.DMA_PRI_ALT_SET[C] = 1, Selects the alternate data structure for channel C.	0x0	R/W

DMA CHANNEL PRIMARY-ALTERNATE CLEAR REGISTER

Address: 0x40000034, Reset: 0x00000000, Name: DMA_PRI_ALT_CLR

The DMA_DMA_PRI_ALT_CLR write-only register enables the user to configure the appropriate DMA channel to use the primary control data structure. Each bit of the register represents the corresponding channel number in the DMA controller.

Note: The DMA controller sets/clears these bits automatically as necessary for ping-pong, memory scatter-gather and peripheral scatter-gather transfers.

Table 39. Bit Descriptions for DMA_PRI_ALT_CLR

Bits	Bit Name	Settings	Description	Reset	Access
[31:20]	RESERVED		Reserved.	0x0	R
[19:0]	DMA_PRI_ALT_CLR		Select Primary Data Structure. Set the appropriate bit to select the primary data structure for the corresponding DMA channel. Bit 0 corresponds to DMA channel 0 … Bit M-1 corresponds to DMA channel M-1 When Written: DMA_DMA_PRI_ALT_CLR.DMA_PRI_ALT_CLR[C] = 0, No effect. Use the DMA_DMA_PRI_ALT_SET register to select the alternate data structure. DMA_DMA_PRI_ALT_CLR.DMA_PRI_ALT_CLR[C] = 1, Selects the primary data structure for channel C.	0x0	W

DMA CHANNEL PRIORITY SET REGISTER

Address: 0x40000038, Reset: 0x00000000, Name: DMA_PRIORITY_SET

Table 40. Bit Descriptions for DMA_PRIORITY_SET

Bits	Bit Name	Settings	Description	Reset	Access
[31:20]	RESERVED		Reserved.	0x0	R
[19:0]	DMA_PRIORITY_SET		Configure Channel for High Priority. This register enables the user to you to configure a DMA channel to use the high priority level. Reading the register returns the status of the channel priority mask. Each bit of the register represents the corresponding channel number in the DMA controller. Returns the channel priority mask status, or sets the channel priority to high. Bit 0 corresponds to DMA channel 0, bit M-1 corresponds to DMA channel M-1 When Read: DMA_DMA_PRIORITY_SET.DMA_PRIORITY_SET[C] = 0, DMA channel C is using the default priority level. DMA_DMA_PRIORITY_SET.DMA_PRIORITY_SET[C] = 1, DMA channel C is using a high priority level. When Written: DMA_DMA_PRIORITY_SET.DMA_PRIORITY_SET[C] = 0, No effect. Use the DMA_DMA_PRIORITY_CLR register to set channel C to the default priority level. DMA_DMA_PRIORITY_SET.DMA_PRIORITY_SET[C] = 1, Channel C uses the high priority level.	0x0	W

DMA CHANNEL PRIORITY CLEAR REGISTER

Address: 0x4000003C, Reset: 0x00000000, Name: DMA_PRIORITY_CLR

Table 41. Bit Descriptions for DMA_PRIORITY_CLR

Bits	Bit Name	Settings	Description	Reset	Access
[31:20]	RESERVED		Reserved.	0x0	R
[19:0]	DMA_PRIORITY_CLR		Configure Channel for Default Priority Level. The DMA_DMA_PRIORITY_CLR write-only register enables the user to configure a DMA channel to use the default priority level. Each bit of the register represents the corresponding channel number in the DMA controller. Set the appropriate bit to select the default priority level for the specified DMA channel. Bit 0 corresponds to DMA channel and bit M-1 corresponds to DMA channel M-1 When Written: DMA_DMA_PRIORITY_CLR.DMA_PRIORITY_CLR[C] = 0, No effect. Use the DMA_DMA_PRIORITY_SET register to set channel C to the high priority level. DMA_DMA_PRIORITY_CLR.DMA_PRIORITY_CLR[C] = 1, Channel C uses the default priority level.	0x0	W

DMA INTERRUPT ENABLE SET REGISTER

Address: 0x40000040, Reset: 0x00000000, Name: DMA_INT_EN_SET

The DMA_DMA_INT_EN_SET register enables the user to configure the appropriate DMA channel to use the alternate control data structure. Reading the register returns the status of which data structure is in use for the corresponding DMA channel. Each bit of the register represents the corresponding channel number in the DMA controller.

Note: The DMA controller sets/clears these bits automatically as necessary for ping-pong, memory scatter-gather and peripheral scatter-gather transfers.

Table 42. Bit Descriptions for DMA_INT_EN_SET

Bits	Bit Name	Settings	Description	Reset	Access
31	INTE_SET_BUS_ERR		DMA Bus Error Interrupt Mask Set. This register is used to read and clear the DMA bus error status. The error status is set if the controller encountered a bus error while performing a transfer or when it reads an invalid descriptor (whose cycle control is 3'b000). If a bus error occurs or invalid cycle control is read on a channel, that channel is automatically disabled by the controller. The other channels are unaffected. Write one to clear bits. When Read: 0 – No bus_error/invalid cycle control occurred. 1 – A bus_error/invalid cycle control is pending. When Written: 0 – No effect. 1 – Bit is cleared.	0x0	R/W1C
30	DMA_INVDESC_INT_EN_SET		DMA Invalid Descriptor Interrupt Mask Set. This register is used to set DMA invalid-descriptor interrupt mask bit.	0x0	R/W1C
[29:20]	RESERVED		Reserved.	0x0	R
[19:0]	DMA_DONE_INT_EN_SET		DMA Transfer Done Interrupt Mask Set. This register is used to set the DMA transfer-done interrupt mask bit.	0x0	R/W

DMA INTERRUPT ENABLE CLEAR REGISTER

Address: 0x40000044, Reset: 0x00000000, Name: DMA_INT_EN_CLR

The DMA_DMA_INT_EN_CLR register enables the user to configure the appropriate DMA channel to use the alternate control data structure. Reading the register returns the status of which data structure is in use for the corresponding DMA channel. Each bit of the register represents the corresponding channel number in the DMA controller.

Note: The DMA controller sets/clears these bits automatically as necessary for ping-pong, memory scatter-gather and peripheral scatter-gather transfers.

Table 43. Bit Descriptions for DMA_INT_EN_CLR

Bits	Bit Name	Settings	Description	Reset	Access
31	DMA_BUSERR_INT_EN_CLR		DMA Bus Error Interrupt Mask Clear. This register is used to clear DMA bus-error interrupt mask bit.	0x0	R/W1C
30	DMA_INVDESC_INT_EN_CLR		DMA Invalid Descriptor Interrupt Mask Clear. This register is used to clear DMA invalid-descriptor interrupt mask bit.	0x0	R/W1C
[29:20]	RESERVED		Reserved.	0x0	R
[19:0]	DMA_DONE_INT_EN_CLR		DMA Transfer Done Interrupt Mask Clear. This register is used to clear the DMA transfer-done interrupt mask bit.	0x0	R/W

DMA INTERRUPT STATUS CLEAR REGISTER

Address: 0x4000004C, Reset: 0x00000000, Name: DMA_INT_STATUS_CLR

The DMA_INT_STATUS_CLR register enables the user to read and get the interrupt status. Each bit of the register represents the corresponding interrupt status. Write '1' to clear corresponding interrupt status.

Table 44. Bit Descriptions for DMA_INT_STATUS_CLR

Bits	Bit Name	Settings	Description	Reset	Access
31	DMA_BUSERR_INT_CLR		DMA Bus Error Interrupt Status Clear. This register is used to read and clear the DMA bus error status. The error status is set if the controller encountered a bus error while performing a transfer. If a bus error occurs, that channel is automatically disabled by the controller. The other channels are unaffected. Write one to clear bits. When Read: When Read: 0 – No bus_error occurred. 1 – A bus_error occurred. When Written: 0 – No effect. 1 – Bit is cleared.	0x0	R/W1C
30	DMA_INVDESC_INT_CLR		DMA Invalid Descriptor Interrupt Status Clear. This register is used to read and clear the DMA invalid descriptor interrupt status. This status is set if DMA transfer reads an invalid descriptor (whose cycle control is 3'b000). Write one to clear bits. When Read: 0 – No DMA invalid-descriptor interrupt occurred. 1 – The DMA invalid-descriptor interrupt occurred. When Written: 0 – No effect. 1 – Bit is cleared.	0x0	R/W1C
[29:20]	RESERVED		Reserved.	0x0	R

Bits	Bit Name	Settings	Description	Reset	Access
[19:0]	DMA_DONE_INT_CLR		DMA Transfer Done Interrupt Status Clear. This register is used to read and clear the DMA transfer done status. This status is set if the corresponding channel's DMA transfer is done.Write one to clear bits. When Read: 0 – No DMA transfer-done occurred. 1 – The DMA transfer is done. When Written: 0 – No effect. 1 – Bit is cleared.	0x0	R/W1C

DMA PER CHANNEL ERROR CLEAR REGISTER

Address: 0x40000054, Reset: 0x00000000, Name: DMA_BUSERR_IDX_CLR

Table 45. Bit Descriptions for DMA_BUSERR_IDX_CLR

Bits	Bit Name	Settings	Description	Reset	Access
[31:20]	RESERVED		Reserved.	0x0	R
[19:0]	DMA_BUSERR_IDX_CLR		DMA Channel Bus Error Index. DMA Channel Bus Error Index This register is used to read per channel DMA bus error status. The error status is set if the controller encountered a bus error while performing a transfer. If a bus error occurs on a channel, that channel is automatically disabled by the controller. The other channels are unaffected. The index value will be reset when the user clear the interrupt status by writting DMA_INT_STATUS_CLR[31]. When Read: 0 – No bus error occurred. 1 – A bus error control is pending.	0x0	R/W1C

DMA PER CHANNEL INVALID DESCRIPTOR CLEAR REGISTER

Address: 0x4000005C, Reset: 0x00000000, Name: DMA_INVDESC_IDX_CLR

Table 46. Bit Descriptions for DMA_INVDESC_IDX_CLR

Bits	Bit Name	Settings	Description	Reset	Access
[31:20]	RESERVED		Reserved.	0x0	R
[19:0]	DMA_INVDESC_IDX_CLR		DMA Invalid Descriptor Index. DMA Invalid Descriptor Index This register is used to read the per channel DMA invalid descriptor status. The per channel invalid descriptor status is set if the controller reads an invalid descriptor (whose cycle control is 3'b000). If it reads invalid cycle control for a channel, that channel is automatically disabled by the controller. The other channels are unaffected. The index value will be reset when the user clear the interrupt status by writting DMA_INT_STATUS_CLR[30]. When Read: 0 – No invalid cycle control occurred. 1 – An invalid cycle control is pending.	0x0	R/W1C

DMA CHANNEL BYTES SWAP ENABLE SET REGISTER

Address: 0x40000800, Reset: 0x00000000, Name: DMA_BYTE_SWAP_SET

Table 47. Bit Descriptions for DMA_BYTE_SWAP_SET

Bits	Bit Name	Settings	Description	Reset	Access
[31:20]	RESERVED		Reserved.	0x0	R
[19:0]	DMA_BYTE_SWAP_SET		Byte Swap Status. This register is used to configure a DMA channel to use byte swap. Each bit of the register represents the corresponding channel number in the DMA controller. Bit 0 corresponds to DMA channel 0 and bit M-1 corresponds to DMA channel M-1 When Read: DMA_DMA_BYTE_SWAP_SET.DMA_BYTE_SWAP_SET[C] = 0, Channel C Byte Swap is disabled. DMA_DMA_BYTE_SWAP_SET.DMA_BYTE_SWAP_SET[C] = 1, Channel C Byte Swap is enabled. When Written: DMA_DMA_BYTE_SWAP_SET.DMA_BYTE_SWAP_SET[C] = 0, No effect. Use the DMA_DMA_BYTE_SWAP_CLR register to disable byte swap on channel C DMA_DMA_BYTE_SWAP_SET.DMA_BYTE_SWAP_SET[C] = 1, Enables Byte Swap on channel C.	0x0	R/W

DMA CHANNEL BYTES SWAP ENABLE CLEAR REGISTER

Address: 0x40000804, Reset: 0x00000000, Name: DMA_BYTE_SWAP_CLR

Table 48. Bit Descriptions for DMA_BYTE_SWAP_CLR

Bits	Bit Name	Settings	Description	Reset	Access
[31:20]	RESERVED		Reserved.	0x0	R
[19:0]	DMA_BYTE_SWAP_CLR		Disable Byte Swap. The DMA_DMA_BYTE_SWAP_CLR write-only register enables the user to configure a DMA channel to not use byte swapping and use the default operation. Each bit of the register represents the corresponding channel number in the DMA controller. Bit 0 corresponds to DMA channel and bit M-1 corresponds to DMA channel M-1 When Written: DMA_DMA_BYTE_SWAP_CLR.DMA_BYTE_SWAP_CLR[C] = 0, No effect. Use the DMA_DMA_BYTE_SWAP_SET register to enable byte swap on channel C. DMA_DMA_BYTE_SWAP_CLR.DMA_BYTE_SWAP_CLR[C] = 1, Disables Byte Swap on channel C.	0x0	W

DMA CHANNEL SOURCE ADDRESS DECREMENT ENABLE SET REGISTER

Address: 0x40000810, Reset: 0x00000000, Name: DMA_SRCADR_DEC_SET

Table 49. Bit Descriptions for DMA_SRCADR_DEC_SET

Bits	Bit Name	Settings	Description	Reset	Access
[31:20]	RESERVED		Reserved.	0x0	R
[19:0]	DMA_SRCADR_DEC_SET		Source Address Decrement Status / Configure. The DMA_DMA_SRCADR_DEC_SET register is used to configure the source address of a DMA channel to decrement the address instead of incrementing the address after each access. Each bit of the register represents the corresponding channel number in the DMA controller. Bit 0 corresponds to DMA channel and bit M-1 corresponds to DMA channel M-1 When Read: DMA_DMA_SRCADR_DEC_SET.DMA_SRCADR_DEC_SET[C] = 0, Channel C Source Address decrement is disabled. DMA_DMA_SRCADR_DEC_SET.DMA_SRCADR_DEC_SET[C] = 1, Channel C Source Address decrement is enabled. When Written: DMA_DMA_SRCADR_DEC_SET.DMA_SRCADR_DEC_SET[C] = 0, No effect. Use the DMA_DMA_SRCADR_DEC_CLR register to disable source address decrement on channel C DMA_DMA_SRCADR_DEC_SET.DMA_SRCADR_DEC_SET[C] = 1, Enables source address decrement on channel C.	0x0	R/W

DMA CHANNEL SOURCE ADDRESS DECREMENT ENABLE CLEAR REGISTER

Address: 0x40000814, Reset: 0x00000000, Name: DMA_SRCADR_DEC_CLR

Table 50. Bit Descriptions for DMA_SRCADR_DEC_CLR

Bits	Bit Name	Settings	Description	Reset	Access
[31:20]	RESERVED		Reserved.	0x0	R
[19:0]	DMA_SRCADR_DEC_CLR		Disable Source Address Decrement. The DMA_DMA_SRCADR_DEC_CLR write-only register enables the user to configure a DMA channel to use the default source address in increment mode. Each bit of the register represents the corresponding channel number in the DMA controller. Bit 0 corresponds to DMA channel 0 … Bit M-1 corresponds to DMA channel M-1 When Written: DMA_DMA_SRCADR_DEC_CLR.DMA_SRCADR_DEC_CLR[C] = 0, No effect. Use the DMA_DMA_SRCADR_DEC_SET register to enable source address decrement on channel C DMA_DMA_SRCADR_DEC_CLR.DMA_SRCADR_DEC_CLR[C] = 1, Disables Address source decrement on channel C.	0x0	W

DMA CHANNEL DESTINATION ADDRESS DECREMENT ENABLE SET REGISTER

Address: 0x40000818, Reset: 0x00000000, Name: DMA_DSTADR_DEC_SET

Table 51. Bit Descriptions for DMA_DSTADR_DEC_SET

Bits	Bit Name	Settings	Description	Reset	Access
[31:20]	RESERVED		Reserved.	0x0	R
[19:0]	DMA_DSTADR_DEC_SET		Destination Address Decrement Status. The DMA_DMA_DSTADR_DEC_SET register is used to configure the destination address of a DMA channel to decrement the address instead of incrementing the address after each access. Each bit of the register represents the corresponding channel number in the DMA controller. Bit 0 corresponds to DMA channel and bit M-1 corresponds to DMA channel M-1 When Read: DMA_DMA_DSTADR_DEC_SET.DMA_DSTADR_DEC_SET[C] = 0, Channel C destination address decrement is disabled. DMA_DMA_DSTADR_DEC_SET.DMA_DSTADR_DEC_SET[C] = 1, Channel C destination address decrement is enabled. When Written: DMA_DMA_DSTADR_DEC_SET.DMA_DSTADR_DEC_SET[C] = 0, No effect. Use the DMA_DMA_DSTADR_DEC_CLR register to disable destination address decrement on channel C DMA_DMA_DSTADR_DEC_SET.DMA_DSTADR_DEC_SET[C] = 1, Enables destination address decrement on channel C.	0x0	R/W

DMA CHANNEL DESTINATION ADDRESS DECREMENT ENABLE CLEAR REGISTER

Address: 0x4000081C, Reset: 0x00000000, Name: DMA_DSTADR_DEC_CLR

Table 52. Bit Descriptions for DMA_DSTADR_DEC_CLR

Bits	Bit Name	Settings	Description	Reset	Access
[31:20]	RESERVED		Reserved.	0x0	R
[19:0]	DMA_DSTADR_DEC_CLR		Disable Destination Address Decrement. The DMA_DMA_DSTADR_DEC_CLR write-only register enables the user to configure a DMA channel to use the default destination address in increment mode. Each bit of the register represents the corresponding channel number in the DMA controller. Bit 0 corresponds to DMA channel 0 … Bit M-1 corresponds to DMA channel M-1 When Written: DMA_DMA_DSTADR_DEC_CLR.DMA_DSTADR_DEC_CLR[C] = 0, No effect. Use the DMA_DMA_DSTADR_DEC_SET register to enable destination address decrement on channel C DMA_DMA_DSTADR_DEC_CLR.DMA_DSTADR_DEC_CLR[C] = 1, Disables destination address decrement on channel C.	0x0	W

DMA CONTROLLER REVISION ID REGISTER

Address: 0x40000FE0, Reset: 0x00000002, Name: DMA_REV_ID

Table 53. Bit Descriptions for DMA_REV_ID

Bits	Bit Name	Settings	Description	Reset	Access
[31:8]	RESERVED		Reserved.	0x0	R
[7:0]	DMA_REV_ID		DMA Controller Revision ID.	0x2	R

REGISTER SUMMARY: SERIAL PERIPHERAL INTERFACE QSPI

Table 54. QSPI(QSPI) Register Summary

Address	Name	Description	Reset	Access
0x40001004	QSPI_CTRL	Control Register.	0x00000040	R/W
0x40001008	QSPI_RX_CTRL	Receive Control Register.	0x00000000	R/W
0x4000100C	QSPI_TX_CTRL	Transmit Control Register.	0x00000000	R/W
0x40001010	QSPI_CLK_CTRL	Clock Rate Register.	0x00000000	R/W
0x40001014	QSPI_DLY	Delay Register.	0x00000301	R/W
0x40001018	QSPI_SLV_SEL	Slave Select Register.	0x0000FE00	R/W
0x4000101C	QSPI_RX_CNT	Received Word Count Register.	0x00000000	R/W
0x40001020	QSPI_RX_RLD_CNT	Received Word Count Reload Register.	0x00000000	R/W
0x40001024	QSPI_TX_CNT	Transmitted Word Count Register.	0x00000000	R/W
0x40001028	QSPI_TX_RLD_CNT	Transmitted Word Count Reload Register.	0x00000000	R/W
0x40001030	QSPI_INT_MASK	Interrupt Mask Register.	0x00000000	R
0x40001034	QSPI_INT_MASK_CLR	Interrupt Mask Clear Register.	0x00000000	R/W
0x40001038	QSPI_INT_MASK_SET	Interrupt Mask Set Register.	0x00000000	R/W
0x40001040	QSPI_STATUS	Status Register.	0x00440001	R/W
0x40001044	QSPI_INT_LATCH	Masked Interrupt Condition Register?.	0x00000000	R
0x40001048	QSPI_INT_LATCH_CLR	Masked Interrupt Clear Register.	0x00000000	R/W
0x40001050	QSPI_RX_FIFO	Receive FIFO Data Register.	0x00000000	R
0x40001058	QSPI_TX_FIFO	Transmit FIFO Data Register.	0x00000000	R/W
0x40001060	QSPI_MMRD_HDR	Memory Mapped Read Header.	0x00000000	R/W
0x40001064	QSPI_MEMTOP_ADR	SPI Memory Top Address.	0x00000000	R/W

REGISTER DETAILS: SERIAL PERIPHERAL INTERFACE

CONTROL REGISTER

Address: 0x40001004, Reset: 0x00000040, Name: QSPI_CTRL

The QSPI_QSPI_CTRL register enables the QSPI and configures settings for operating modes, communication protocols, and buffer operations.

Table 55. Bit Descriptions for QSPI_CTRL

Bits	Bit Name	Settings	Description	Reset	Access
31	QSPI_MEMMAP_EN		Memory-Mapped SPI Enable. When the QSPI_QSPI_CTRL.QSPI_MEMMAP_EN bit is asserted, communication to a SPI memory device is automated such that the memory it contains is accessible directly through the read of processor address space assigned to it (As far as the SPI peripheral is concerned, this includes all read accesses received by the SPI peripheral's system crossbar slave port). Note that when memory-mapped access of SPI memory is enabled, attempts to communicate with the SPI device using legacy methods are blocked and receive fabric error responses are generated. Legacy methods include any direct access made to the TX and Rx FIFOs, whether initiated by DMA or processor MMR access. Blocked attempts also set the ?.:QSPI_STATUS:MMAE? bit notification bit.	0x0	R/W
		0	Hardware automated access of memory-mapped SPI memory disabled.		
		1	Hardware-automated access of memory-mapped SPI memory enabled.		
30	QSPI_MEMMAP_WR_ERR_MASK		Memory Mapped Write Error Mask. The QSPI_QSPI_CTRL.QSPI_MEMMAP_WR_ERR_MASK bit specifies whether an error response is returned to the fabric upon write attempts to address space reserved for memory-mapped reads of SPI memory. Whether or not a write error response is masked by this bit, the associated sticky notification bit ?.:QSPI_STATUS:MMWE? bit is still set.	0x0	R/W
		0	Write error response returned upon write attempts to memory-mapped SPI memory.		
		1	Write error response masked (not returned) upon write attempts to memory-mapped SPI memory.		
29	QSPI_MEMMAP_RD_ERR_MASK		Memory Mapped Write Error Mask. The QSPI_QSPI_CTRL.QSPI_MEMMAP_RD_ERR_MASK bit specifies whether an error response is returned to the fabric upon write attempts to address space reserved for memory-mapped reads of SPI memory. Whether or not a write error response is masked by this bit, the associated sticky notification bit ?.:QSPI_STATUS:MMWE? bit is still set.	0x0	R/W
		0	Write error response returned upon write attempts to memory-mapped SPI memory.		
		1	Write error response masked (not returned) upon write attempts to memory-mapped SPI memory.		
[28:23]	RESERVED		Reserved.	0x0	R

Bits	Bit Name	Settings	Description	Reset	Access
22	QSPI_MULTI_IO_ORDER		Start on MOSI. The QSPI_QSPI_CTRL.QSPI_MULTI_IO_ORDER bit is valid only when ?.::MIOM? is enabled for either DIOM or QIOM, and this bit selects the starting pin and the bit placement on pins for these modes. $[/p$]$[p$]In DIOM, by default (QSPI_QSPI_CTRL.QSPI_MULTI_IO_ORDER =0) SPI sends first bit on the DQ2_QMST pin and second bit on the DQ1_QMST pin. In QIOM, by default, the QSPI sends first bit on the D3 pin, second bit on the D2 pin, third bit on the DQ2_QMST pin and fourth bit on the DQ1_QMST pin. This order can be reversed by setting the QSPI_QSPI_CTRL.QSPI_MULTI_IO_ORDER bit. When this bit is set, the QSPI sends first bit on the DQ1_QMST pin. The first bit referred to here depends on the ?.::LSBF? bit setting (MSB bit or LSB bit).	0x0	R/W
		0	Start on MISO (DIOM) or start on SPIQ3 (QSPI).		
		1	Start on MOSI.		
[21:20]	QSPI_MULTI_IO		Multiple I/O Mode. The QSPI_QSPI_CTRL.QSPI_MULTI_IO bits enable QSPI operation in dual I/O mode (DIOM) or quad I/O mode (QIOM). $[/p$]$[p$]These bits may only be changed when the QSPI is disabled (?.::EN? =0).	0x0	R/W
		00	No MIOM (disabled).		
		01	DIOM operation.		
		10	QIOM operation.		
		11	Reserved.		
19	RESERVED		Reserved.	0x0	R
18	QSPI_FAST_MODE		Fast-Mode Enable. The QSPI_QSPI_CTRL.QSPI_FAST_MODE bit enables fast mode operation for QSPI receive transfers. QSPI transmit operations in fast mode are the same as normal mode.	0x0	R/W
		0	Disable.		
		1	Enable.		
[17:16]	QSPI_FLOW_CTRL_WATERMARK		Flow Control Watermark. The QSPI_QSPI_CTRL.QSPI_FLOW_CTRL_WATERMARK bits select the watermark level of the transmit channel (QSPI_QSPI_TX_FIFO buffer) or receive channel (QSPI_QSPI_RX_FIFO buffer) that triggers flow control operation. These bits are applicable only when the QSPI is a slave (?.::MSTR? = 0) and flow control is enabled (?.::FCEN? =1). When the watermark condition is met, the QSPI slave de-asserts the RDY pin.	0x0	R/W
		00	TFIFO empty or RFIFO full.		
		01	TFIFO 75% or more empty, or RFIFO 75% or more full.		
		10	TFIFO 50% or more empty, or RFIFO 50% or more full.		
		11	Reserved.		
15	QSPI_FLOW_CTRL_POL		Flow Control Polarity. The QSPI_QSPI_CTRL.QSPI_FLOW_CTRL_POL bit selects flow control polarity for the RDY pin when flow control is enabled. When the RDY pin is active, the QSPI is indicating it's ready for data transfer.	0x0	R/W
		0	Active-low RDY.		
		1	Active-high RDY.		

Bits	Bit Name	Settings	Description	Reset	Access
14	QSPI_FLOW_CTRL_CHSEL		Flow Control Channel Selection. The QSPI_QSPI_CTRL.QSPI_FLOW_CTRL_CHSEL bit selects whether the QSPI applies flow control to the transmit channel (QSPI_QSPI_TX_FIFO buffer) or receive channel (QSPI_QSPI_RX_FIFO buffer). This bit is applicable only when the QSPI is a slave and flow control is enabled.	0x0	R/W
		0	Flow control on RX buffer.		
		1	Flow control on TX buffer.		
13	QSPI_FLOW_CTRL_EN		Flow Control Enable. The QSPI_QSPI_CTRL.QSPI_FLOW_CTRL_EN bit enables QSPI flow control operation, which permits slow slave devices to interface with fast master devices. This bit controls operation of the RDY pin. Note that options for flow control operation are available using the ?.::FCCH?, ?.::FCPL?, and ?.::FCWM? bits.	0x0	R/W
		0	Disable.		
		1	Enable.		
12	QSPI_LSB_FIRST		Least Significant Bit First. The QSPI_QSPI_CTRL.QSPI_LSB_FIRST bit selects whether the QSPI transmits/receives data as LSB first (little endian) or MSB first (big endian). This bit may only be changed when the QSPI is disabled.	0x0	R/W
		0	MSB sent/received first (big endian).		
		1	LSB sent/received first (little endian).		
11	RESERVED		Reserved.	0x0	R
[10:9]	QSPI_SIZE		Word Transfer Size. The QSPI_QSPI_CTRL.QSPI_SIZE bits select the QSPI transfer word size as 8, 16 or 32 bits. To ensure correct operation, both the master and slave must be configured with the same word size. This bit may only be changed when the QSPI is disabled (?.::EN? =0).	0x0	R/W
		00	8-bit word.		
		01	16-bit word.		
		10	32-bit word.		
		11	Reserved.		
8	QSPI_MISO_EN		Enable MISO. The QSPI_QSPI_CTRL.QSPI_MISO_EN bit enables master-in-slave-out (MISO, DQ1_QMST) mode. This QSPI mode is applicable only when the QSPI is a slave.	0x0	R/W
		0	Disable.		
		1	Enable.		
7	QSPI_CS_POL_IN_TRAN		Slave Select Polarity Between Transfers. The QSPI_QSPI_CTRL.QSPI_CS_POL_IN_TRAN bit selects the state (polarity) for the SELn pin in-between QSPI transfers when the QSPI is a master and hardware slave select assertion is enabled (?.::ASSEL? =1). In slave mode, this bit affects the detection of both transmit collision (?.:QSPI_STATUS:TC? and under-run (?.:QSPI_STATUS:TUR?) errors.	0x0	R/W
		0	De-assert slave select (high).		
		1	Assert slave select (low).		

Bits	Bit Name	Settings	Description	Reset	Access
6	QSPI_CS_CTRL		Slave Select Pin Control. The QSPI_QSPI_CTRL.QSPI_CS_CTRL bit selects whether the QSPI hardware sets the SS_QMSTn pin output value (ignoring the slave select ?.:QSPI_SLV_SEL:SSEL1? - ?.:QSPI_SLV_SEL:SSEL7? bits) or whether software control of the slave select bits set the SS_QMSTn pin output value. This feature is applicable only when the QSPI is a master. When hardware control is enabled, the SS_QMSTn pin output is asserted during the transfers, and the pin polarity between transfers is selected by the ?.:.:SELST? bit. When software control is enabled, the SS_QMSTn pin output value is set through software control of the slave select bits, and as such, the pin may either remain asserted (low) or be deasserted between transfers.	0x1	R/W
		0	Software Slave Select Control.		
		1	Hardware Slave Select Control.		
5	QSPI_CPOL		Clock Polarity. The QSPI_QSPI_CTRL.QSPI_CPOL bit selects whether the QSPI uses an active-low or active-high signal for the QSPI clock (SCK_QMST). This bit works with the ?.:.:CPHA? bit to select combinations of clock phase and polarity for the SCK_QMST pin. This bit may only be changed when the QSPI is disabled.	0x0	R/W
		0	Active-high SPI CLK.		
		1	Active-low SPI CLK.		
4	QSPI_CPHA		Clock Phase. The QSPI_QSPI_CTRL.QSPI_CPHA bit selects whether the QSPI starts toggling the signal for the QSPI clock (SCK_QMST) from the start of the first data bit or from the middle of the first data bit. The QSPI_QSPI_CTRL.QSPI_CPHA bit works with the ?.:.:CPOL? bit to select combinations of clock phase and polarity for the SCK_QMST pin. This bit may only be changed when the QSPI is disabled.	0x0	R/W
		0	SPI CLK toggles from middle.		
		1	SPI CLK toggles from start.		
3	QSPI_OPEN_DRAIN		Open Drain Mode. The QSPI_QSPI_CTRL.QSPI_OPEN_DRAIN bit configures the data output pins (DQ2_QMST and DQ1_QMST to behave as open drain outputs, which prevents contention and possible damage to pin drivers in multi-master or multi-slave QSPI systems. $[/p$]$[p$]When QSPI_QSPI_CTRL.QSPI_OPEN_DRAIN is enabled and the QSPI is a master, the QSPI three-states the DQ2_QMST pin when the data driven out on DQ2_QMST is a logic-high. The QSPI does not three-state the DQ2_QMST pin when the driven data is a logic-low. $[/p$]$[p$]When QSPI_QSPI_CTRL.QSPI_OPEN_DRAIN is enabled and the QSPI is a slave, the QSPI three-states the DQ1_QMST pin when the data driven out on DQ1_QMST is a logic-high. $[/p$]$[p$]Note that an external pull-up resistor is required on both the DQ2_QMST and DQ1_QMST pins when QSPI_QSPI_CTRL.QSPI_OPEN_DRAIN is enabled.	0x0	R/W
		0	Disable.		
		1	Enable.		

Bits	Bit Name	Settings	Description	Reset	Access
2	QSPI_CS_PROT_EN		Protected Slave Select Enable. The QSPI_QSPI_CTRL.QSPI_CS_PROT_EN bit enables the SS_QMSTn pin to provide error detection input in a multi-master environment when the QSPI is in master mode. If some other device in the system asserts the SS_QMSTn pin while QSPI is enabled as master (and QSPI_QSPI_CTRL.QSPI_CS_PROT_EN is enabled), this condition causes a mode fault error.	0x0	R/W
		0	Disable.		
		1	Enable.		
1	QSPI_MASTER		Master/Slave. The QSPI_QSPI_CTRL.QSPI_MASTER bit toggles the QSPI between master mode and slave mode. This bit may only be changed when the QSPI is disabled.	0x0	R/W
		0	Slave.		
		1	Master.		
0	QSPI_EN		Enable. The QSPI_QSPI_CTRL.QSPI_EN bit enables QSPI operation.	0x0	R/W
		0	Disable SPI module.		
		1	Enable.		

RECEIVE CONTROL REGISTER

Address: 0x40001008, Reset: 0x00000000, Name: QSPI_RX_CTRL

The QSPI_QSPI_RX_CTRL register enables the QSPI receive channel, initiates receive transfers, and configures QSPI_QSPI_RX_FIFO buffer watermark settings.

Table 56. Bit Descriptions for QSPI_RX_CTRL

Bits	Bit Name	Settings	Description	Reset	Access
[31:19]	RESERVED		Reserved.	0x0	R
[18:16]	QSPI_RXFF_URG_WM		Receive FIFO Urgent Watermark. The QSPI_QSPI_RX_CTRL.QSPI_RXFF_URG_WM bits select the receive FIFO (QSPI_QSPI_RX_FIFO) watermark level for urgent data bus requests. The QSPI also uses this watermark level for generation of the ?.:QSPI_INT_LATCH:RUWM? interrupt. When an urgent QSPI_QSPI_RX_FIFO watermark is enabled with QSPI_QSPI_RX_CTRL.QSPI_RXFF_URG_WM, the ?.:.:RRWM? selection is used as the de-assertion condition for any ?.:QSPI_INT_LATCH:RUWM? interrupts that are latched.	0x0	R/W
		000	Disabled.		
		001	25% full RFIFO.		
		010	50% full RFIFO.		
		011	75% full RFIFO.		
		100	Full RFIFO.		
		101	Reserved.		
		110	Reserved.		
		111	Reserved.		
[15:14]	RESERVED		Reserved.	0x0	R

Bits	Bit Name	Settings	Description	Reset	Access
[13:12]	QSPI_RXFF_REG_WM		Receive FIFO Regular Watermark. The QSPI_QSPI_RX_CTRL.QSPI_RXFF_REG_WM bits select the receive FIFO (QSPI_QSPI_RX_FIFO) watermark level for regular data bus requests. When an urgent QSPI_QSPI_RX_FIFO watermark is enabled with ?.:.:RUWM?, the QSPI_QSPI_RX_CTRL.QSPI_RXFF_REG_WM selection is used as the de-assertion condition for any ?.:QSPI_INT_LATCH:RUWM? interrupts that are latched.	0x0	R/W
		00	Empty RFIFO.		
		01	RFIFO less than 25% full.		
		10	RFIFO less than 50% full.		
		11	RFIFO less than 75% full.		
[11:9]	RESERVED		Reserved.	0x0	R
8	QSPI_RXD_OVRUN		Receive Data Overrun. The QSPI_QSPI_RX_CTRL.QSPI_RXD_OVRUN bit selects handling for receive data requests when the receive buffer (QSPI_QSPI_RX_FIFO) is full. If enabled and QSPI_QSPI_RX_FIFO is full, the QSPI overwrites old data in the buffer with incoming data. If disabled and QSPI_QSPI_RX_FIFO is full, the QSPI keeps old data in the buffer and discards incoming data.	0x0	R/W
		0	Discard incoming data if SPI_RFIFO is full.		
		1	Overwrite old data if SPI_RFIFO is full.		
7	RESERVED		Reserved.	0x0	R
[6:4]	QSPI_RXD_REQ		Receive Data Request. The QSPI_QSPI_RX_CTRL.QSPI_RXD_REQ bits select receive FIFO (QSPI_QSPI_RX_FIFO) watermark conditions that direct the QSPI to generate a receive data request.	0x0	R/W
		000	Disabled.		
		001	Not empty RFIFO.		
		010	25% full RFIFO.		
		011	50% full RFIFO.		
		100	75% full RFIFO.		
		101	Full RFIFO.		
		110	Reserved.		
		111	Reserved.		
3	QSPI_RX_WCNT_EN		Receive Word Counter Enable. The QSPI_QSPI_RX_CTRL.QSPI_RX_WCNT_EN bit enables the decrement of the QSPI_QSPI_RX_CNT register when the count is not zero and ?.:.:RTI? is enabled. Enabling QSPI_QSPI_RX_CTRL.QSPI_RX_WCNT_EN prevents receive overrun errors from occurring. The QSPI_QSPI_RX_CTRL.QSPI_RX_WCNT_EN bit is valid only when the QSPI is a master.	0x0	R/W
		0	Disable.		
		1	Enable.		
2	QSPI_RX_INIT		Receive Transfer Initiate. The QSPI_QSPI_RX_CTRL.QSPI_RX_INIT bit enables initiation of receive transfers if the receive FIFO (QSPI_QSPI_RX_FIFO) is not full. The bit also enables this initiation if QSPI_QSPI_RX_CNT is not zero when ?.:.:RWCEN? is enabled. Enabling QSPI_QSPI_RX_CTRL.QSPI_RX_INIT prevents receive overrun errors from occurring. The QSPI_QSPI_RX_CTRL.QSPI_RX_INIT bit is valid only when the QSPI is a master.	0x0	R/W
		0	Disable.		
		1	Enable.		

Bits	Bit Name	Settings	Description	Reset	Access
1	RESERVED		Reserved.	0x0	R
0	QSPI_RX_EN		Receive Enable. The QSPI_QSPI_RX_CTRL.QSPI_RX_EN bit enables QSPI receive channel operation.	0x0	R/W
		0	Disable.		
		1	Enable.		

TRANSMIT CONTROL REGISTER

Address: 0x4000100C, Reset: 0x00000000, Name: QSPI_TX_CTRL

The QSPI_QSPI_TX_CTRL register enables the QSPI transmit channel, initiates transmit transfers, and configures QSPI_QSPI_TX_FIFO buffer watermark settings.

Table 57. Bit Descriptions for QSPI_TX_CTRL

Bits	Bit Name	Settings	Description	Reset	Access
[31:19]	RESERVED		Reserved.	0x0	R
[18:16]	QSPI_TXFF_URG_WM		FIFO Urgent Watermark. The QSPI_QSPI_TX_CTRL.QSPI_TXFF_URG_WM bits select the transmit FIFO (QSPI_QSPI_TX_FIFO) watermark level for urgent data bus requests. The QSPI also uses this watermark level for generation of the ?.:QSPI_INT_LATCH:TUWM? interrupt. When an urgent QSPI_QSPI_TX_FIFO watermark is enabled with QSPI_QSPI_TX_CTRL.QSPI_TXFF_URG_WM, the ?.:.:TRWM? selection is used as the de-assertion condition for any ?.:QSPI_INT_LATCH:TUWM? interrupts that are latched.	0x0	R/W
		000	Disabled.		
		001	25% empty TFIFO.		
		010	50% empty TFIFO.		
		011	75% empty TFIFO.		
		100	Empty TFIFO.		
[15:14]	RESERVED		Reserved.	0x0	R
[13:12]	QSPI_TXFF_REG_WM		FIFO Regular Watermark. The QSPI_QSPI_TX_CTRL.QSPI_TXFF_REG_WM bits select the transmit FIFO (QSPI_QSPI_TX_FIFO) watermark level for regular data bus requests. When an urgent QSPI_QSPI_TX_FIFO watermark is enabled with ?.:.:TUWM?, the QSPI_QSPI_TX_CTRL.QSPI_TXFF_REG_WM selection is used as the de-assertion condition for any ?.:QSPI_INT_LATCH:TUWM? interrupts that are latched.	0x0	R/W
		00	Full TFIFO.		
		01	TFIFO less than 25% empty.		
		10	TFIFO less than 50% empty.		
		11	TFIFO less than 75% empty.		
[11:9]	RESERVED		Reserved.	0x0	R
8	QSPI_TXD_UDRUN		Transmit Data Under-run. The QSPI_QSPI_TX_CTRL.QSPI_TXD_UDRUN bit selects handling for transmit data requests when the transmit buffer (QSPI_QSPI_TX_FIFO) is empty. If enabled and QSPI_QSPI_TX_FIFO is empty, the QSPI transmits zero as data. If disabled and QSPI_QSPI_TX_FIFO is empty, the QSPI transmits the last word in the buffer as data.	0x0	R/W
		0	Send last word when SPI_TFIFO is empty.		
		1	Send zeros when SPI_TFIFO is empty.		
7	RESERVED		Reserved.	0x0	R

Bits	Bit Name	Settings	Description	Reset	Access
[6:4]	QSPI_TXD_REQ		Transmit Data Request. The QSPI_QSPI_TX_CTRL.QSPI_TXD_REQ bits select transmit FIFO (QSPI_QSPI_TX_FIFO) watermark conditions that direct the QSPI to generate a transmit status interrupt.	0x0	R/W
		000	Disabled.		
		001	Not full TFIFO.		
		010	25% empty TFIFO.		
		011	50% empty TFIFO.		
		100	75% empty TFIFO.		
		101	Empty TFIFO.		
3	QSPI_TX_WCNT_EN		Transmit Word Counter Enable. The QSPI_QSPI_TX_CTRL.QSPI_TX_WCNT_EN bit enables the decrement of the transmit word count (QSPI_QSPI_TX_CNT) register when the count is not zero and ?.::.:TTI? is enabled. Enabling QSPI_QSPI_TX_CTRL.QSPI_TX_WCNT_EN prevents transmit under-run errors from occurring. The QSPI_QSPI_TX_CTRL.QSPI_TX_WCNT_EN bit is valid only when the QSPI is a master.	0x0	R/W
		0	Disable.		
		1	Enable.		
2	QSPI_TX_INIT		Transmit Transfer Initiate. The QSPI_QSPI_TX_CTRL.QSPI_TX_INIT bit enables initiation of transmit transfers if the transmit FIFO (QSPI_QSPI_TX_FIFO) is not empty. The bit also enables this initiation if QSPI_QSPI_TX_CNT is not zero when ?.::.:TWCEN? is enabled. Enabling QSPI_QSPI_TX_CTRL.QSPI_TX_INIT prevents transmit underrun errors from occurring. The QSPI_QSPI_TX_CTRL.QSPI_TX_INIT bit is valid only when the QSPI is a master.	0x0	R/W
		0	Disable.		
		1	Enable.		
1	RESERVED		Reserved.	0x0	R
0	QSPI_TX_EN		Transmit Enable. The QSPI_QSPI_TX_CTRL.QSPI_TX_EN bit enables QSPI transmit channel operation.	0x0	R/W
		0	Disable.		
		1	Enable.		

CLOCK RATE REGISTER

Address: 0x40001010, Reset: 0x00000000, Name: QSPI_CLK_CTRL

The QSPI_QSPI_CLK_CTRL register selects the baud rate for QSPI data transfers, relating this rate to the QSPI serial clock (SCK) and the system clock (SCLK).

Table 58. Bit Descriptions for QSPI_CLK_CTRL

Bits	Bit Name	Settings	Description	Reset	Access
[31:16]	RESERVED		Reserved.	0x0	R
[15:0]	QSPI_CLK_RATIO		Baud Rate. The QSPI_QSPI_CLK_CTRL.QSPI_CLK_RATIO bits set the QSPI baud rate according to the formula: $[/p$]$[p$]BAUD = (SCLK / SPI Clock) - 1	0x0	R/W

DELAY REGISTER

Address: 0x40001014, Reset: 0x00000301, Name: QSPI_DLY

The QSPI_QSPI_DLY register selects a transfer delay and the lead/lag timing between slave select signals and SPI clock edge assertion/de-assertion.

Table 59. Bit Descriptions for QSPI_DLY

Bits	Bit Name	Settings	Description	Reset	Access
[31:10]	RESERVED		Reserved.	0x0	R
9	QSPI_EXT_LAG_EN		Extended SPI Clock Lag Control. The QSPI_QSPI_DLY.QSPI_EXT_LAG_EN bit enables insertion of a 1-SPI_CLK cycle lag (extend lag) in the timing between the slave select (SS_QMSTn) assertion and first SPI Clock edge.	0x1	R/W
		0	Disable.		
		1	Enable.		
8	QSPI_EXT_LEAD_EN		Extended SPI Clock Lead Control. The QSPI_QSPI_DLY.QSPI_EXT_LEAD_EN bit enables insertion of a 1-SPI_CLK cycle lead (extend lead) in the timing between the slave select (SS_QMSTn) de-assertion and last SPI Clock edge.	0x1	R/W
		0	Disable.		
		1	Enable.		
[7:0]	QSPI_DLY_VAL		Transfer delay time in multiples of SPI clock period. The QSPI_QSPI_DLY.QSPI_DLY_VAL bits select a delay (number of stop bits in multiples of SPI Clock duration) at the end of each QSPI transfer. The default delay is the minimum value required to comply with the QSPI protocol (1-bit duration). The QSPI_QSPI_DLY.QSPI_DLY_VAL bits may be programmed with smaller delay values, resulting in continuous operation (for example, stop bits =0).	0x1	R/W

SLAVE SELECT REGISTER

Address: 0x40001018, Reset: 0x0000FE00, Name: QSPI_SLV_SEL

The QSPI_QSPI_SLV_SEL register enables the SS_QMSTn pins for input and indicates the state (high or low) of these pins when enabled.

Table 60. Bit Descriptions for QSPI_SLV_SEL

Bits	Bit Name	Settings	Description	Reset	Access
[31:16]	RESERVED		Reserved.	0x0	R
15	QSPI_SSEL7		Slave Select 7 Input. The QSPI_QSPI_SLV_SEL.QSPI_SSEL7 bit state indicates the value on the related SS_QMSTn pin.	0x1	R/W
		0	Low.		
		1	High.		
14	QSPI_SSEL6		Slave Select 6 Input. The QSPI_QSPI_SLV_SEL.QSPI_SSEL6 bit state indicates the value on the related SS_QMSTn pin.	0x1	R/W
		0	Low.		
		1	High.		
13	QSPI_SSEL5		Slave Select 5 Input. The QSPI_QSPI_SLV_SEL.QSPI_SSEL5 bit state indicates the value on the related SS_QMSTn pin.	0x1	R/W
		0	Low.		
		1	High.		
12	QSPI_SSEL4		Slave Select 4 Input. The QSPI_QSPI_SLV_SEL.QSPI_SSEL4 bit state indicates the value on the related SS_QMSTn pin.	0x1	R/W
		0	Low.		
		1	High.		
11	QSPI_SSEL3		Slave Select 3 Input. The QSPI_QSPI_SLV_SEL.QSPI_SSEL3 bit state indicates the value on the related SS_QMSTn pin.	0x1	R/W
		0	Low.		
		1	High.		

Bits	Bit Name	Settings	Description	Reset	Access
10	QSPI_SSEL2		Slave Select 2 Input. The QSPI_QSPI_SLV_SEL.QSPI_SSEL2 bit state indicates the value on the related SS_QMSTn pin.	0x1	R/W
		0	Low.		
		1	High.		
9	QSPI_SSEL1		Slave Select 1 Input. The QSPI_QSPI_SLV_SEL.QSPI_SSEL1 bit state indicates the value on the related SS_QMSTn pin.	0x1	R/W
		0	Low.		
		1	High.		
8	RESERVED		Reserved.	0x0	R
7	QSPI_SSEL7_EN		Slave Select 7 Enable. The QSPI_QSPI_SLV_SEL.QSPI_SSEL7_EN bit enables the related SELn pin for input. If disabled, the QSPI three-states the related SS_QMSTn pin. When the QSPI is a slave, the master (not the QSPI) asserts the input during the transfer. The input may be de-asserted or remain asserted between transfers. While the input is de-asserted, the QSPI ignores SPI Clock, ignores inputs, and three-states outputs.	0x0	R/W
		0	Disable.		
		1	Enable.		
6	QSPI_SSEL6_EN		Slave Select 6 Enable. The QSPI_QSPI_SLV_SEL.QSPI_SSEL6_EN bit enables the related SELn pin for input. See the ?.::.SSE7? bit description for more information.	0x0	R/W
		0	Disable.		
		1	Enable.		
5	QSPI_SSEL5_EN		Slave Select 5 Enable. The QSPI_QSPI_SLV_SEL.QSPI_SSEL5_EN bit enables the related SELn pin for input. See the ?.::.SSE7? bit description for more information.	0x0	R/W
		0	Disable.		
		1	Enable.		
4	QSPI_SSEL4_EN		Slave Select 4 Enable. The QSPI_QSPI_SLV_SEL.QSPI_SSEL4_EN bit enables the related SELn pin for input. See the ?.::.SSE7? bit description for more information.	0x0	R/W
		0	Disable.		
		1	Enable.		
3	QSPI_SSEL3_EN		Slave Select 3 Enable. The QSPI_QSPI_SLV_SEL.QSPI_SSEL3_EN bit enables the related SELn pin for input. See the ?.::.SSE7? bit description for more information.	0x0	R/W
		0	Disable.		
		1	Enable.		
2	QSPI_SSEL2_EN		Slave Select 2 Enable. The QSPI_QSPI_SLV_SEL.QSPI_SSEL2_EN bit enables the related SELn pin for input. See the ?.::.SSE7? bit description for more information.	0x0	R/W
		0	Disable.		
		1	Enable.		
1	QSPI_SSEL1_EN		Slave Select 1 Enable. The QSPI_QSPI_SLV_SEL.QSPI_SSEL1_EN bit enables the related SELn pin for input. See the ?.::.SSE7? bit description for more information.	0x0	R/W
		0	Disable.		
		1	Enable.		
0	RESERVED		Reserved.	0x0	R

RECEIVED WORD COUNT REGISTER

Address: 0x4000101C, Reset: 0x00000000, Name: QSPI_RX_CNT

The QSPI_QSPI_RX_CNT register holds a count of the number of words remaining to be received by the QSPI. To start the decrement of the word count in QSPI_QSPI_RX_CNT, enable the receive word counter (?.:QSPI_RX_CTRL:RWCEN? =1). The QSPI uses the word count to control the duration of transfers and to signal the completion of a burst of transfers with the receive finish interrupt (?.:QSPI_INT_LATCH:RF?). In DMA mode, the QSPI uses the QSPI_QSPI_RX_CNT to ensure that the number of frames received during a DMA transfer is equal to the number of words programmed in the DMA channel controller. The values programmed into the QSPI_QSPI_RX_CNT registers should match the word count in the DMA configuration. The QSPI_QSPI_RX_CNT maintains the number of frames to be received in a transfer. The QSPI_QSPI_RX_CNT should only be changed when the counter is disabled.

Table 61. Bit Descriptions for QSPI_RX_CNT

Bits	Bit Name	Settings	Description	Reset	Access
[31:16]	RESERVED		Reserved.	0x0	R
[15:0]	QSPI_RX_CNT		Received Word Count. The QSPI_QSPI_RX_CNT.QSPI_RX_CNT bits hold the receive transfer word count.	0x0	R/W

RECEIVED WORD COUNT RELOAD REGISTER

Address: 0x40001020, Reset: 0x00000000, Name: QSPI_RX_RLD_CNT

The QSPI_QSPI_RX_RLD_CNT register holds the receive word count value that the QSPI loads into the QSPI_QSPI_RX_CNT register when the transfer count decrements to zero. To prevent the QSPI from reloading the counter, use zero for the reload count value. The QSPI_QSPI_RX_RLD_CNT should only be changed when the counter is disabled.

Table 62. Bit Descriptions for QSPI_RX_RLD_CNT

Bits	Bit Name	Settings	Description	Reset	Access
[31:16]	RESERVED		Reserved.	0x0	R
[15:0]	QSPI_RX_RLD_CNT		Received Word Count Reload. The QSPI_QSPI_RX_RLD_CNT.QSPI_RX_RLD_CNT bits hold the receive transfer word count reload value.	0x0	R/W

TRANSMITTED WORD COUNT REGISTER

Address: 0x40001024, Reset: 0x00000000, Name: QSPI_TX_CNT

The QSPI_QSPI_TX_CNT register holds a count of the number of words remaining to be transmitted by the QSPI. To start the decrement of the word count in QSPI_QSPI_TX_CNT, enable the transmit word counter (?.:QSPI_TX_CTRL:TWCEN? =1). The QSPI uses the word count to control the duration of transfers and to signal the completion of a burst of transfers with the transmit finish interrupt. In DMA mode, the QSPI uses the QSPI_QSPI_TX_CNT to ensure that the number of frames transmitted during a DMA transfer is equal to the number of words programmed in the DMA channel controller. The values programmed into the QSPI_QSPI_TX_CNT registers should match the word count in the DMA configuration. The QSPI_QSPI_TX_CNT maintains the number of frames to be transmitted in a transfer. The QSPI_QSPI_TX_CNT should only be changed when the counter is disabled.

Table 63. Bit Descriptions for QSPI_TX_CNT

Bits	Bit Name	Settings	Description	Reset	Access
[31:16]	RESERVED		Reserved.	0x0	R
[15:0]	QSPI_TX_CNT		Transmitted Word Count. The QSPI_QSPI_TX_CNT.QSPI_TX_CNT bits hold the transmit transfer word count.	0x0	R/W

TRANSMITTED WORD COUNT RELOAD REGISTER

Address: 0x40001028, Reset: 0x00000000, Name: QSPI_TX_RLD_CNT

The QSPI_QSPI_TX_RLD_CNT register holds the transmit word count value that the QSPI loads into the QSPI_QSPI_TX_CNT register when the transfer count decrements to zero. To prevent the QSPI from reloading the counter, use zero for the reload count value. The QSPI_QSPI_TX_RLD_CNT should only be changed when the counter is disabled.

Table 64. Bit Descriptions for QSPI_TX_RLD_CNT

Bits	Bit Name	Settings	Description	Reset	Access
[31:16]	RESERVED		Reserved.	0x0	R
[15:0]	QSPI_TX_RLD_CNT		Transmitted Word Count Reload. The QSPI_QSPI_TX_RLD_CNT.QSPI_TX_RLD_CNT bits hold the transmit transfer word count reload value.	0x0	R/W

INTERRUPT MASK REGISTER

Address: 0x40001030, Reset: 0x00000000, Name: QSPI_INT_MASK

The QSPI_QSPI_INT_MASK register unmasks (enables) or masks (disables) QSPI interrupts. When a condition is indicated by a bit in the QSPI_QSPI_STATUS register and the corresponding interrupt is unmasked in QSPI_QSPI_INT_MASK, the QSPI latches the interrupt's bit in the QSPI_QSPI_INT_LATCH register, queuing the interrupt for service.

Table 65. Bit Descriptions for QSPI_INT_MASK

Bits	Bit Name	Settings	Description	Reset	Access
[31:12]	RESERVED		Reserved.	0x0	R
11	QSPI_TX_DONE_INTMASK		Transmit Finish Interrupt Mask.	0x0	R
		0	Disable (mask) interrupt.		
		1	Enable (unmask) interrupt.		
10	QSPI_RX_DONE_INTMASK		Receive Finish Interrupt Mask.	0x0	R
		0	Disable (mask) interrupt.		
		1	Enable (unmask) interrupt.		
9	QSPI_TX_START_INTMASK		Transmit Start Interrupt Mask.	0x0	R
		0	Disable (mask) interrupt.		
		1	Enable (unmask) interrupt.		
8	QSPI_RX_START_INTMASK		Receive Start Interrupt Mask.	0x0	R
		0	Disable (mask) interrupt.		
		1	Enable (unmask) interrupt.		
7	QSPI_MODE_FAULT_INTMASK		Mode Fault Interrupt Mask.	0x0	R
		0	Disable (mask) interrupt.		
		1	Enable (unmask) interrupt.		
6	QSPI_TX_COL_INTMASK		Transmit Collision Interrupt Mask.	0x0	R
		0	Disable (mask) interrupt.		
		1	Enable (unmask) interrupt.		
5	QSPI_TX_UDRUN_INTMASK		Transmit Underrun Interrupt Mask.	0x0	R
		0	Disable (mask) interrupt.		
		1	Enable (unmask) interrupt.		
4	QSPI_RX_OVRUN_INTMASK		Receive Overrun Interrupt Mask.	0x0	R
		0	Disable (mask) interrupt.		
		1	Enable (unmask) interrupt.		
3	RESERVED		Reserved.	0x0	R
2	QSPI_TX_URGWM_INTMASK		Transmit Urgent Watermark Interrupt Mask.	0x0	R
		0	Disable (mask) interrupt.		
		1	Enable (unmask) interrupt.		

Bits	Bit Name	Settings	Description	Reset	Access
1	QSPI_RX_URGWM_INTMASK		Receive Urgent Watermark Interrupt Mask.	0x0	R
		0	Disable (mask) interrupt.		
		1	Enable (unmask) interrupt.		
0	RESERVED		Reserved.	0x0	R

INTERRUPT MASK CLEAR REGISTER

Address: 0x40001034, Reset: 0x00000000, Name: QSPI_INT_MASK_CLR

The QSPI_QSPI_INT_MASK_CLR register permits clearing individual mask bits in the QSPI_QSPI_INT_MASK register without affecting other bits in the register. Use write-1-to-clear on a bit in QSPI_QSPI_INT_MASK_CLR to clear the corresponding bit in the QSPI_QSPI_INT_MASK register.

Table 66. Bit Descriptions for QSPI_INT_MASK_CLR

Bits	Bit Name	Settings	Description	Reset	Access
[31:12]	RESERVED		Reserved.	0x0	R
11	QSPI_TX_DONE_INTMASK_CLR		Clear Transmit Finish Interrupt Mask.	0x0	R/W1C
10	QSPI_RX_DONE_INTMASK_CLR		Clear Receive Finish Interrupt Mask.	0x0	R/W1C
9	QSPI_TX_START_INTMASK_CLR		Clear Transmit Start Interrupt Mask.	0x0	R/W1C
8	QSPI_RX_START_INTMASK_CLR		Clear Receive Start Interrupt Mask.	0x0	R/W1C
7	QSPI_MODE_FAULT_INTMASK_CLR		Clear Mode Fault Interrupt Mask.	0x0	R/W1C
6	QSPI_TX_COL_INTMASK_CLR		Clear Transmit Collision Interrupt Mask.	0x0	R/W1C
5	QSPI_TX_UDRUN_INTMASK_CLR		Clear Transmit Under-run Interrupt Mask.	0x0	R/W1C
4	QSPI_RX_OVRUN_INTMASK_CLR		Clear Receive Overrun Interrupt Mask.	0x0	R/W1C
3	RESERVED		Reserved.	0x0	R
2	QSPI_TX_URGWM_INTMASK_CLR		Clear Transmit Urgent Watermark Interrupt Mask.	0x0	R/W1C
1	QSPI_RX_URGWM_INTMASK_CLR		Clear Receive Urgent Watermark Interrupt Mask.	0x0	R/W1C
0	RESERVED		Reserved.	0x0	R

INTERRUPT MASK SET REGISTER

Address: 0x40001038, Reset: 0x00000000, Name: QSPI_INT_MASK_SET

The QSPI_QSPI_INT_MASK_SET register permits setting individual mask bits in the QSPI_QSPI_INT_MASK register without affecting other bits in the register. Use write-1-to-set on a bit in QSPI_QSPI_INT_MASK_SET to set the corresponding bit in the QSPI_QSPI_INT_MASK register.

Table 67. Bit Descriptions for QSPI_INT_MASK_SET

Bits	Bit Name	Settings	Description	Reset	Access
[31:12]	RESERVED		Reserved.	0x0	R
11	QSPI_TX_DONE_INTMASK_SET		Set Transmit Finish Interrupt Mask.	0x0	R/W1S
10	QSPI_RX_DONE_INTMASK_SET		Set Receive Finish Interrupt Mask.	0x0	R/W1S
9	QSPI_TX_START_INTMASK_SET		Set Transmit Start Interrupt Mask.	0x0	R/W1S
8	QSPI_RX_START_INTMASK_SET		Set Receive Start Interrupt Mask.	0x0	R/W1S
7	QSPI_MODE_FAULT_INTMASK_SET		Set Mode Fault Interrupt Mask.	0x0	R/W1S
6	QSPI_TX_COL_INTMASK_SET		Set Transmit Collision Interrupt Mask.	0x0	R/W1S
5	QSPI_TX_UDRUN_INTMASK_SET		Set Transmit Under-run Interrupt Mask.	0x0	R/W1S
4	QSPI_RX_OVRUN_INTMASK_SET		Set Receive Overrun Interrupt Mask.	0x0	R/W1S
3	RESERVED		Reserved.	0x0	R
2	QSPI_TX_URGWM_INTMASK_SET		Set Transmit Urgent Watermark Interrupt Mask.	0x0	R/W1S
1	QSPI_RX_URGWM_INTMASK_SET		Set Receive Urgent Watermark Interrupt Mask.	0x0	R/W1S
0	RESERVED		Reserved.	0x0	R

STATUS REGISTER

Address: 0x40001040, Reset: 0x00440001, Name: QSPI_STATUS

The QSPI_QSPI_STATUS register indicates QSPI status including FIFO status, error conditions, and interrupt conditions. When an interrupt condition from this register is unmasked (enabled) by the corresponding bit in the QSPI_QSPI_INT_MASK register, the interrupt is latched into the corresponding bit in the QSPI_QSPI_INT_LATCH register.

Table 68. Bit Descriptions for QSPI_STATUS

Bits	Bit Name	Settings	Description	Reset	Access
31	QSPI_MEMMAP_FF_ERR		Memory Mapped Access Error. The QSPI_QSPI_STATUS.QSPI_MEMMAP_FF_ERR bit =1 if an attempt is made to access either the Tx or Rx FIFO while memory-mapped access of SPI memory is enabled (see the SPI_CTL.MMSE) bit. The QSPI_QSPI_STATUS.QSPI_MEMMAP_FF_ERR bit =0 when a 1 is written to it. The QSPI_QSPI_STATUS.QSPI_MEMMAP_FF_ERR bit is provided for software notification only. Its state has no further effect.	0x0	R/W1C
30	RESERVED		Reserved.	0x0	R
29	QSPI_MEMMAP_RD_ERR		Memory Mapped Read Error. The QSPI_QSPI_STATUS.QSPI_MEMMAP_RD_ERR bit =1 if an attempt is made to read address space reserved for memory-mapped SPI memory while memory mapping is disabled (see the ?.:QSPI_CTRL:MMSE? bit. The QSPI_QSPI_STATUS.QSPI_MEMMAP_RD_ERR bit =0 when a 1 is written to it. This bit is provided for software notification only. Its state has no further effect.	0x0	R/W1C
28	QSPI_MEMMAP_WR_ERR		Memory Mapped Write Error. The QSPI_QSPI_STATUS.QSPI_MEMMAP_WR_ERR bit =1 if an attempt is made to write address space reserved for memory-mapped SPI memory. The QSPI_QSPI_STATUS.QSPI_MEMMAP_WR_ERR bit =0 when a 1 is written to it. This bit is provided for software notification only. Its state has no further effect.	0x0	R/W1C
[27:24]	RESERVED		Reserved.	0x0	R
23	QSPI_TX_FF_FULL		SPI_TFIFO Full. The QSPI_QSPI_STATUS.QSPI_TX_FF_FULL bit indicates whether the QSPI_QSPI_TX_FIFO is full or not full.	0x0	R
		0	Not full Tx FIFO.		
		1	Full Tx FIFO.		
22	QSPI_RX_FF_EMPTY		SPI_RFIFO Empty. The QSPI_QSPI_STATUS.QSPI_RX_FF_EMPTY bit indicates whether the QSPI_QSPI_RX_FIFO is empty or not empty.	0x1	R
		0	RX FIFO not empty.		
		1	RX FIFO empty.		
21	RESERVED		Reserved.	0x0	R
20	QSPI_FLOW_CTRL_STATUS		Flow Control Stall Indication. The QSPI_QSPI_STATUS.QSPI_FLOW_CTRL_STATUS bit indicates whether a slave has de-asserted the RDY pin to stall the QSPI master while the slave is unable to service the QSPI master's request. This bit is valid only when the QSPI is a master (?.:QSPI_CTRL:MSTR? =1 and flow control is enabled (?.:QSPI_CTRL:FCEN? =1).	0x0	R
		0	No Stall (RDY pin asserted).		
		1	Stall (RDY pin de-asserted).		
19	RESERVED		Reserved.	0x0	R

Bits	Bit Name	Settings	Description	Reset	Access
[18:16]	QSPI_TX_FF_STATUS		SPI_TFIFO Status. The QSPI_QSPI_STATUS.QSPI_TX_FF_STATUS bits indicate the status of the QSPI_QSPI_TX_FIFO. The QSPI uses this status when evaluating transmit watermark conditions.	0x4	R
		000	Full TFIFO.		
		001	25% empty TFIFO.		
		010	50% empty TFIFO.		
		011	75% empty TFIFO.		
		100	Empty TFIFO.		
15	RESERVED		Reserved.	0x0	R
[14:12]	QSPI_RX_FF_STATUS		SPI_RFIFO Status. The QSPI_QSPI_STATUS.QSPI_RX_FF_STATUS bits indicate the status of the QSPI_QSPI_RX_FIFO. The QSPI uses this status when evaluating receive watermark conditions.	0x0	R
		000	Empty RFIFO.		
		001	25% full RFIFO.		
		010	50% full RFIFO.		
		011	75% full RFIFO.		
		100	Full RFIFO.		
11	QSPI_TX_DONE		Transmit Finish Indication. The QSPI_QSPI_STATUS.QSPI_TX_DONE bit indicates that the QSPI has detected the finish of a transmit burst transfer (the QSPI_QSPI_TX_CNT count decrements to zero). This condition can only occur when ?.:QSPI_TX_CTRL:TTI? and ?.:QSPI_TX_CTRL:TWCEN? are enabled.	0x0	R/W1C
		0	No status.		
		1	Transmit finish detected.		
10	QSPI_RX_DONE		Receive Finish Indication. The QSPI_QSPI_STATUS.QSPI_RX_DONE bit indicates that the QSPI has detected the finish of a receive burst transfer (the QSPI_QSPI_RX_CNT count decrements to zero). This condition can only occur when ?.:QSPI_RX_CTRL:RTI? and ?.:QSPI_RX_CTRL:RWCEN? are enabled.	0x0	R/W1C
		0	No status.		
		1	Receive finish detected.		
9	QSPI_TX_START		Transmit Start. The QSPI_QSPI_STATUS.QSPI_TX_START bit indicates that the QSPI has detected the start of a transmit burst transfer. A transmit bursts starts with the load of QSPI_QSPI_TX_CNT from the QSPI_QSPI_TX_RLD_CNT. This condition can only occur when ?.:QSPI_TX_CTRL:TTI? and ?.:QSPI_TX_CTRL:TWCEN? are enabled.	0x0	R/W1C
		0	No status.		
		1	Transmit start detected.		
8	QSPI_RX_START		Receive Start. The QSPI_QSPI_STATUS.QSPI_RX_START bit indicates that the QSPI has detected the start of a receive burst transfer. A receive bursts starts with the load of QSPI_QSPI_RX_CNT from the QSPI_QSPI_RX_RLD_CNT. This condition can only occur when ?.:QSPI_RX_CTRL:RTI? and ?.:QSPI_RX_CTRL:RWCEN? are enabled.	0x0	R/W1C
		0	No status.		
		1	Receive start detected.		
7	QSPI_MODE_FAULT		Mode Fault Indication. The QSPI_QSPI_STATUS.QSPI_MODE_FAULT bit, when QSPI is a master and ?.:QSPI_CTRL:PSSE? is enabled, indicates that multiple masters have asserted slave select inputs.	0x0	R/W1C
		0	No status.		
		1	Mode fault occurred.		

Bits	Bit Name	Settings	Description	Reset	Access
6	QSPI_TX_COL		Transmit Collision Indication. The QSPI_QSPI_STATUS.QSPI_TX_COL bit, when QSPI is a slave, indicates that the load of data into the shift register has occurred too close to the first transmitting edge of the SPI Clock.	0x0	R/W1C
		0	No status.		
		1	Transmit collision occurred.		
5	QSPI_TX_UDRUN		Transmit Underrun Indication. The QSPI_QSPI_STATUS.QSPI_TX_UDRUN bit, when the transmit FIFO (QSPI_QSPI_TX_FIFO is empty, indicates that the last word in the transmit FIFO has been re-sent as transmit data. Alternately, it indicates that zero has been sent as transmit data.	0x0	R/W1C
		0	No status.		
		1	Transmit underrun occurred.		
4	QSPI_RX_OVRUN		Receive Overrun Indication. The QSPI_QSPI_STATUS.QSPI_RX_OVRUN bit, when the receive FIFO (QSPI_QSPI_RX_FIFO is full, indicates that a word in the receive FIFO has been overwritten with incoming receive data. Alternately, it indicates that incoming receive data has been discarded.	0x0	R/W1C
		0	No status.		
		1	Receive overrun occurred.		
3	RESERVED		Reserved.	0x0	R
2	QSPI_TX_URGWM		Transmit Urgent Watermark Breached. The QSPI_QSPI_STATUS.QSPI_TX_URGWM bit indicates that the transmit urgent watermark (?.:QSPI_TX_CTRL:TUWM?) has been reached. This condition is cleared when the transmit FIFO fills enough to reach the transmit regular watermark (?.:QSPI_TX_CTRL:TRWM?).	0x0	R
		0	TX Regular Watermark reached.		
		1	TX Urgent Watermark breached.		
1	QSPI_RX_URGWM		Receive Urgent Watermark Breached. The QSPI_QSPI_STATUS.QSPI_RX_URGWM bit indicates that the receive urgent watermark (?.:QSPI_RX_CTRL:RUWM?) has been reached. This condition is cleared when the receive FIFO empties enough to reach the receive regular watermark (?.:QSPI_RX_CTRL:RRWM?).	0x0	R
		0	RX Regular Watermark reached.		
		1	RX Urgent Watermark breached.		
0	QSPI_TRAN_DONE		SPI Finished. The QSPI_QSPI_STATUS.QSPI_TRAN_DONE bit indicates that a single word transfer is complete.	0x1	R
		0	No status.		
		1	Completed single-word transfer.		

MASKED INTERRUPT CONDITION REGISTER?

Address: 0x40001044, Reset: 0x00000000, Name: QSPI_INT_LATCH

The QSPI_QSPI_INT_LATCH register latches interrupts, queuing the interrupts for service. When a condition is indicated by a bit in the QSPI_QSPI_STATUS register and the corresponding interrupt is unmasked in QSPI_QSPI_INT_MASK, the QSPI latches the interrupt's bit in QSPI_QSPI_INT_LATCH.

Table 69. Bit Descriptions for QSPI_INT_LATCH

Bits	Bit Name	Settings	Description	Reset	Access
[31:12]	RESERVED		Reserved.	0x0	R
11	QSPI_TX_DONE_LATCH		Transmit Finish Interrupt Latch.	0x0	R
		0	No interrupt.		
		1	Latched interrupt.		
10	QSPI_RX_DONE_LATCH		Receive Finish Interrupt Latch.	0x0	R
		0	No interrupt.		
		1	Latched interrupt.		
9	QSPI_TX_START_LATCH		Transmit Start Interrupt Latch.	0x0	R
		0	No interrupt.		
		1	Latched interrupt.		
8	QSPI_RX_START_LATCH		Receive Start Interrupt Latch.	0x0	R
		0	No interrupt.		
		1	Latched interrupt.		
7	QSPI_MODE_FAULT_LATCH		Mode Fault Interrupt Latch.	0x0	R
		0	No interrupt.		
		1	Latched interrupt.		
6	QSPI_TX_COL_LATCH		Transmit Collision Interrupt Latch.	0x0	R
		0	No interrupt.		
		1	Latched interrupt.		
5	QSPI_TX_UDRUN_LATCH		Transmit Under-run Interrupt Latch.	0x0	R
		0	No interrupt.		
		1	Latched interrupt.		
4	QSPI_RX_OVRUN_LATCH		Receive Overrun Interrupt Latch.	0x0	R
		0	No interrupt.		
		1	Latched interrupt.		
3	RESERVED		Reserved.	0x0	R
2	QSPI_TX_URGWM_LATCH		Transmit Urgent Watermark Interrupt Latch.	0x0	R
		0	No interrupt.		
		1	Latched interrupt.		
1	QSPI_RX_URGWM_LATCH		Receive Urgent Watermark Interrupt Latch.	0x0	R
		0	No interrupt.		
		1	Latched interrupt.		
0	RESERVED		Reserved.	0x0	R

MASKED INTERRUPT CLEAR REGISTER

Address: 0x40001048, Reset: 0x00000000, Name: QSPI_INT_LATCH_CLR

The QSPI_QSPI_INT_LATCH_CLR register permits clearing individual mask bits in the QSPI_QSPI_INT_LATCH register without affecting other bits in the register. Use write-1-to-clear on a bit in QSPI_QSPI_INT_LATCH_CLR to clear the corresponding bit in the QSPI_QSPI_INT_LATCH register.

Table 70. Bit Descriptions for QSPI_INT_LATCH_CLR

Bits	Bit Name	Settings	Description	Reset	Access
[31:15]	RESERVED		Reserved.	0x0	R
[14:12]	QSPI_RX_FF_STATUS		SPI_RFIFO Status. The QSPI_QSPI_INT_LATCH_CLR.QSPI_RX_FF_STATUS bits indicate the status of the QSPI_QSPI_RX_FIFO. The QSPI uses this status when evaluating receive watermark conditions.	0x0	R
		000	Empty RFIFO.		
		001	25% full RFIFO.		
		010	50% full RFIFO.		
		011	75% full RFIFO.		
		100	Full RFIFO.		
11	QSPI_TX_DONE_LATCH_CLR		Clear Transmit Finish Interrupt Latch.	0x0	R/W1C
10	QSPI_RX_DONE_LATCH_CLR		Clear Receive Finish Interrupt Latch.	0x0	R/W1C
9	QSPI_TX_START_LATCH_CLR		Clear Transmit Start Interrupt Latch.	0x0	R/W1C
8	QSPI_RX_START_LATCH_CLR		Clear Receive Start Interrupt Latch.	0x0	R/W1C
7	QSPI_MODE_FAULT_LATCH_CLR		Clear Mode Fault Interrupt Latch.	0x0	R/W1C
6	QSPI_TX_COL_LATCH_CLR		Clear Transmit Collision Interrupt Latch.	0x0	R/W1C
5	QSPI_TX_UDRUN_LATCH_CLR		Clear Transmit Under-run Interrupt Latch.	0x0	R/W1C
4	QSPI_RX_OVRUN_LATCH_CLR		Clear Receive Overrun Interrupt Latch.	0x0	R/W1C
3	RESERVED		Reserved.	0x0	R
2	QSPI_TX_URGWM_LATCH_CLR		Clear Transmit Urgent Watermark Interrupt Latch.	0x0	R
1	QSPI_RX_URGWM_LATCH_CLR		Clear Receive Urgent Watermark Interrupt Latch.	0x0	R
0	RESERVED		Reserved.	0x0	R

RECEIVE FIFO DATA REGISTER

Address: 0x40001050, Reset: 0x00000000, Name: QSPI_RX_FIFO

The QSPI_QSPI_RX_FIFO register has an interface to the receive shift register in the QSPI and has an interface to the processor's data buses. The top level of the buffer is visible to programs as the 32-bit QSPI_QSPI_RX_FIFO register, but the size (number of word locations) of the receive FIFO is actually flexible with transfer word size. The size of the receive FIFO is 8 if word size is 8-bit, or the size is 4 if word size is 16-bit, or the size is 2 if word size is 32-bit.

Both masters and slaves may stop or stall receive transfers based on FIFO status. When the receive FIFO is full, the QSPI master stops initiating new transfers on the QSPI if ?.:QSPI_RX_CTRL:RTI? is enabled. A slave may stall the QSPI interface when the content of the FIFO crosses the selected watermark. If data reception continues after QSPI_QSPI_RX_FIFO is full, the data in the receive FIFO is invalid, and the QSPI indicates this condition with receive overrun (?.:QSPI_STATUS:ROR?). This condition is possible when ?.:QSPI_RX_CTRL:RTI? =0 and ?.:QSPI_RX_CTRL:REN? =1 for a master, or for a slave that does not exercise flow control.

Note that the receive FIFO is reset (cleared) when the QSPI is disabled after being enabled.

Table 71. Bit Descriptions for QSPI_RX_FIFO

Bits	Bit Name	Settings	Description	Reset	Access
[31:0]	QSPI_RX_DATA		Receive FIFO Data.	0x0	R

TRANSMIT FIFO DATA REGISTER

Address: 0x40001058, Reset: 0x00000000, Name: QSPI_TX_FIFO

The QSPI_QSPI_TX_FIFO register has an interface to the transmit shift register in the QSPI and has an interface to the processor's data buses. The top level of the buffer is visible to programs as the 32-bit QSPI_QSPI_TX_FIFO register, but the size (number of word locations) of the transmit FIFO is actually flexible with transfer word size. The size of the transmit FIFO is 8 if word size is 8-bit, or the size is 4 if word size is 16-bit, or the size is 2 if word size is 32-bit.

Both masters and slaves may stop or stall transmit transfers based on FIFO status. When the transmit FIFO is empty, the QSPI master stops initiating new transfers on the QSPI if ?.:QSPI_TX_CTRL:TTI? is enabled. A slave may stall the QSPI interface when the content of the FIFO crosses the selected watermark. If data transmit requests continue after QSPI_QSPI_TX_FIFO is empty, the data sent from the transmit FIFO is invalid, and the QSPI indicates this condition with transmit underrun (?.:QSPI_STATUS:TUR?). This condition is possible when ?.:QSPI_TX_CTRL:TTI? =0 and ?.:QSPI_TX_CTRL:TEN? =1 for a master, or for a slave that does not exercise flow control.

Note that the transmit FIFO is reset (cleared) when the QSPI is disabled after being enabled.

Table 72. Bit Descriptions for QSPI_TX_FIFO

Bits	Bit Name	Settings	Description	Reset	Access
[31:0]	QSPI_TX_DATA		Transmit FIFO Data.	0x0	R/W

MEMORY MAPPED READ HEADER REGISTER

Address: 0x40001060, Reset: 0x00000000, Name: QSPI_MMRD_HDR

The QSPI_QSPI_MMRD_HDR register enables the use of memory-mapped mode. This mode allows direct memory-mapped read accesses of a SPI memory device and is primarily used to directly execute instructions from a SPI FLASH memory without using a low-level software driver. All overhead tasks such as transmission of the read header, pin turnaround timing and receive data sizing are handled in hardware.

The memory-mapped access mode is enabled by setting the CTL.MMSE bit. The features within the QSPI_QSPI_MMRD_HDR register include a command skip mode, variable length byte addressing, and independent multi-pin support for command transmission, address transmission and data reception. In addition the command op-code and mode bytes are fully programmable.

Table 73. Bit Descriptions for QSPI_MMRD_HDR

Bits	Bit Name	Settings	Description	Reset	Access
[31:30]	RESERVED		Reserved.	0x0	R
29	QSPI_MMRD_CMD_PINS		Pins Used for Command. The QSPI_QSPI_MMRD_HDR.QSPI_MMRD_CMD_PINS bit specifies the number of pins to be used for command transmission. This bit must be set consistent with expectations established by read opcode. Hardware does not interpret the OPCODE, but rather relies on this bit to specify behavior. When cleared, overrides ?.:QSPI_CTRL:MIOM? bits). When set, uses bits specified by ?.:QSPI_CTRL:MIOM? bit setting.	0x0	R/W
		0	Use only one pin: MOSI (overrides ?.:QSPI_CTRL:MIOM? bits in register SPI_CTL).		
		1	Use pins specified by ?.:QSPI_CTRL:MIOM? bits in register SPI_CTL.		
28	QSPI_MMRD_CMD_SKIP_EN		Command Skip Enable. The QSPI_QSPI_MMRD_HDR.QSPI_MMRD_CMD_SKIP_EN bit enables command skip mode where the address is sent first and the OPCODE field is not sent (QSPI_QSPI_MMRD_HDR.QSPI_MMRD_CMD_SKIP_EN bit =1). This mode is useful for supporting XIP (Execute-In-Place) operation where only the address is sent and the same read command is assumed. The SPI flash device must be primed with an initial read command, before the QSPI_QSPI_MMRD_HDR.QSPI_MMRD_CMD_SKIP_EN bit is set.	0x0	R/W
		0	OPCODE field is sent first followed by address.		
		1	OPCODE field is not sent; address is sent first.		

Bits	Bit Name	Settings	Description	Reset	Access
27	QSPI_MMRD_WRAP		SPI Memory Wrap Indicator. The QSPI_QSPI_MMRD_HDR.QSPI_MMRD_WRAP bit must be set by software if software places a connected SPI memory device into a 8-byte, 16-byte or 32-byte wrap mode based on the ILINE and DLINE field setting of the cache configuration register address wrap mode. Software achieves this by transmitting a vendor specified command to the SPI memory device while the ?.:QSPI_CTRL:MMSE? bit =0. $[/p$]$[p$]If the QSPI_QSPI_MMRD_HDR.QSPI_MMRD_WRAP bit =1, the SPI does not need to deassert the SPI slave select signal and resend the read header in order to wrap to the cache line base when servicing misaligned cache fill requests. Although this improves cache fill efficiency, it requires that the SPI deassert the SPI slave select pin and resend the read header whenever a DMA burst requests crosses 32 byte alignments. Setting this bit improves cache throughput but decreases DMA throughput.	0x0	R/W
		0	SPI Memory auto increments address purely sequentially.		
		1	SPI Memory auto increments address but wraps within 32 Byte lines.		
26	QSPI_MMRD_MERGE_EN		Merge Enable. When the QSPI_QSPI_MMRD_HDR.QSPI_MMRD_MERGE_EN bit is set, SPI hardware combines the two successive transfers. This increases the throughput rate when accessing a large number of sequential memory locations. For more information refer to the Merged Read Accesses section.	0x0	R/W
[25:24]	QSPI_MMRD_TRISTATE_DMY		Tristate Dummy Timing. These bits specify whether and when output pins are tristated during the interval of time specified by DMYSIZE. Output pins potentially tristated include all pin which were used to transmit the address	0x0	R/W
		00	Tristate outputs immediately.		
		01	Tristate outputs after 4 bits of dummy/mode are transmitted.		
		10	Tristate outputs after 8 bits of dummy/mode are transmitted.		
		11	Never Tristate outputs never (previously specified output state is held).		
[23:16]	QSPI_MMRD_MODE		Mode Field. These bits specify up to a leading byte to be transmitted during the interval of time specified by DMYSIZE. This first byte, or a portion of it, is interpreted as mode bits when certain opcodes are used in conjunction with certain SPI memory device. Mode bits are sent using the same number of pins which were used to transmit the address. Once sent, output pins will be held in their final resultant state until the conclusion of all dummy byte periods, unless tristating of outputs is specified first by bits TRIDMY.	0x0	R/W
15	RESERVED		Reserved.	0x0	R
[14:12]	QSPI_MMRD_DMY_SIZE		Bytes of Dummy/Mode. These bits specify the number of bytes separating address transmission and read data return. Dummy bytes elapse assuming dummy bits are transmitted using the same number of pins which were used to transmit address.	0x0	R/W
		000	0 Bytes.		
		001	1 Bytes.		
		010	2 Bytes.		
		011	3 Bytes.		
		100	4 Bytes.		

Bits	Bit Name	Settings	Description	Reset	Access
		101	5 Bytes.		
		110	6 Bytes.		
		111	7 Bytes.		
11	QSPI_MMRD_ADR_PINS		Pins Used for Address. This bit specifies the number of pins to be used for address transmission. This bit must be set consistent with expectations established by read opcode. Hardware does not interpret the OPCODE, but rather relies on this bit to specify behavior .	0x0	R/W
		0	Use only one pin: MOSI (overrides ?.:QSPI_CTRL:MIOM? bits).		
		1	Use pins specified by ?.:QSPI_CTRL:MIOM? bits.		
[10:8]	QSPI_MMRD_ADR_SIZE		Bytes of Read Address. The QSPI_QSPI_MMRD_HDR.QSPI_MMRD_ADR_SIZE bit field defines the number of bytes used to specify the read address. The read address is sent immediately following the transmission of opcode. Unlike opcode bits, address bits may be sent using either one or multiple pins. The number of pins is selected using the ?.:.:ADRPINS? bit. The address sent to a connected SPI memory device is an echo of the read address received by the SPI peripheral slave port. Least significant bytes of address are sent when entire address is not sent.	0x0	R/W
		000	1 Byte.		
		001	1 Byte.		
		010	2 Bytes.		
		011	3 Bytes.		
		100	4 Bytes.		
[7:0]	QSPI_MMRD_OP_CODE		Read Opcode. The QSPI_QSPI_MMRD_HDR.QSPI_MMRD_OP_CODE bit field specifies initial bits transmitted in response to a read request of SPI memory. Although any opcode may be sent, values 0x03, 0x0B, 0x3B, 0x6B, 0xBB, and 0xEB are likely to be the most commonly used. Opcode is sent by the SPI without interpretation; the states of these bits have no affect beyond specifying what is initially shirted across the SPI interface.	0x0	R/W

SPI MEMORY TOP ADDRESS REGISTER

Address: 0x40001064, Reset: 0x00000000, Name: QSPI_MEMTOP_ADR

The QSPI_QSPI_MEMTOP_ADR register specifies the top populated address of a connected SPI memory device.

Table 74. Bit Descriptions for QSPI_MEMTOP_ADR

Bits	Bit Name	Settings	Description	Reset	Access
[31:0]	QSPI_MMRD_TOP_ADR		SPI Memory Top Address. The QSPI_QSPI_MEMTOP_ADR.QSPI_MMRD_TOP_ADR bit field specifies the top populated address of a connected SPI memory device. Attempts to access SPI memory are not blocked if this address is exceeded and an error is generated as part of the read response.	0x0	R/W

REGISTER SUMMARY: TENSILICA DSP (SOC)

Table 75. SOC Register Summary

Address	Name	Description	Reset	Access
0x40002000	SOC_CLKEN	SOC Clock Enable.	0x001F002F	R/W
0x40002004	SOC_DUMMY1	SOC Dummy Register 1.	0x00000000	R/W
0x40002008	SOC_DUMMY2	SOC Dummy Register 2.	0x00000000	R/W
0x4000200C	SOC_DUMMY3	SOC Dummy Register 3.	0x00000000	R/W
0x40002010	SOC_LOPRI_LVLINT_CON	SOC Low Priority Level Trigger Interrupt Connection.	0x0000FFFF	R/W
0x40002014	SOC_HIPRI_LVLINT_CON	SOC High Priority Level Trigger Interrupt Connection.	0xFFFFFFFF	R/W
0x40002018	SOC_LOPRI_EDGINT_CON	SOC Low Priority Edge Trigger Interrupt Connection.	0x03000000	R/W
0x4000201C	SOC_HIPRI_EDGINT_CON	SOC High Priority Edge Trigger Interrupt Connection.	0x000000FF	R/W
0x40002024	SOC_ERR_STATUS	SOC Error Status.	0x00000000	R/W
0x40002028	SOC_PFAULT_INFO	SOC Pfault Information.	0x00000000	R
0x40002040	SOC_MEM_LIGHT_SLEEP	SOC Memory Light-Sleep.	0x00000000	R/W

REGISTER DETAILS: TENSILICA DSP (SOC)

SOC CLOCK ENABLE REGISTER

Address: 0x40002000, Reset: 0x001F002F, Name: SOC_CLKEN

This register is used to enable or disable SOC sub-block clocks.

Table 76. Bit Descriptions for SOC_CLKEN

Bits	Bit Name	Settings	Description	Reset	Access
[31:21]	RESERVED		Reserved.	0x0	R
20	SOC_CLKEN_SOC2AON		SOC2AON Block Clock Enable. 1: Enable 0: Disable	0x1	R/W
19	SOC_CLKEN_WDT		WDT Block Clock Enable. 1: Enable 0: Disable	0x1	R/W
18	SOC_CLKEN_GPT		GPT Block Clock Enable. 1: Enable 0: Disable	0x1	R/W
17	RESERVED		Reserved	0x1	R
16	SOC_CLKEN_UART		UART Block Clock Enable. 1: Enable 0: Disable	0x1	R/W
[15:6]	RESERVED		Reserved.	0x0	R
5	SOC_CLKEN_DMA		DMA Block Clock Enable. 1: Enable 0: Disable	0x1	R/W
4	RESERVED		Reserved.	0x0	R
3	SOC_CLKEN_QSPI		QSPI Block Clock Enable. 1: Enable 0: Disable	0x1	R/W
2	SOC_CLKEN_FFSRAM1		FIFO_SRAM1 Block Clock Enable. 1: Enable 0: Disable	0x1	R/W
1	SOC_CLKEN_FFSRAM0		FIFO_SRAM0 Block Clock Enable. 1: Enable 0: Disable	0x1	R/W
0	SOC_CLKEN_SYSRAM		SYS_RAM Block Clock Enable. 1: Enable 0: Disable	0x1	R/W

SOC DUMMY REGISTER 1

Address: 0x40002004, Reset: 0x00000000, Name: SOC_DUMMY1

This is a Writable and Readable register inside SOC subsystem.

Table 77. Bit Descriptions for SOC_DUMMY1

Bits	Bit Name	Settings	Description	Reset	Access
[31:0]	SOC_DUMMY1		SOC Dummy Register 1. Dummy free read/write register	0x0	R/W

SOC DUMMY REGISTER 2

Address: 0x40002008, Reset: 0x00000000, Name: SOC_DUMMY2

This is a Writable and Readable register inside SOC subsystem.

Table 78. Bit Descriptions for SOC_DUMMY2

Bits	Bit Name	Settings	Description	Reset	Access
[31:0]	SOC_DUMMY2		SOC Dummy Register 2. Dummy free read/write register	0x0	R/W

SOC DUMMY REGISTER 3

Address: 0x4000200C, Reset: 0x00000000, Name: SOC_DUMMY3

This is a Writable and Readable register inside SOC subsystem.

Table 79. Bit Descriptions for SOC_DUMMY3

Bits	Bit Name	Settings	Description	Reset	Access
[31:0]	SOC_DUMMY3		SOC Dummy Register 3. Dummy free read/write register	0x0	R/W

SOC LOW PRIORITY LEVEL TRIGGER INTERRUPT CONNECTION REGISTER

Address: 0x40002010, Reset: 0x0000FFFF, Name: SOC_LOPRI_LVLINT_CON

In default, 'Level triggered' peripherals' interrupt sources are connected with TDSP input interrupt pin 0~14. All these interrupt are low priority interrupt from TDSP perspective. User is able to disconnect the connection by clearing these registers bits

Table 80. Bit Descriptions for SOC_LOPRI_LVLINT_CON

Bits	Bit Name	Settings	Description	Reset	Access
[31:16]	RESERVED		Reserved.	0x0	R
15	TDSP_INT15_RSVD		TDSP Interrupt Pin-15 Reserved for Future Use. Reserved	0x1	R/W
14	TDSP_INT14_CON		TDSP Interrupt Pin-14 Interrupt Connected. 1: Connected 0: Disconnected	0x1	R/W
13	TDSP_INT13_UART_CON		TDSP Interrupt Pin-13 and UART Interrupt Connected. 1: Connected 0: Disconnected	0x1	R/W
12	TDSP_INT12_MPIIRQ3_CON		TDSP Interrupt Pin-12 and MPI_IRQ3 Interrupt Connected. 1: Connected 0: Disconnected	0x1	R/W
11	TDSP_INT11_MPIIRQ2_CON		TDSP Interrupt Pin-11 and MPI_IRQ2 Interrupt Connected. 1: Connected 0: Disconnected	0x1	R/W
10	TDSP_INT10_MPIIRQ1_CON		TDSP Interrupt Pin-10 and MPI_IRQ1 Interrupt Connected. 1: Connected 0: Disconnected	0x1	R/W
9	TDSP_INT09_AONIRQ4_CON		TDSP Interrupt Pin-9 and AON_IRQ4 Interrupt Connected. 1: Connected 0: Disconnected	0x1	R/W
8	TDSP_INT08_AONIRQ3_CON		TDSP Interrupt Pin-8 and AON_IRQ3 Interrupt Connected. 1: Connected 0: Disconnected	0x1	R/W
7	TDSP_INT07_AONIRQ2_CON		TDSP Interrupt Pin-7 and AON_IRQ2 Interrupt Connected. 1: Connected 0: Disconnected	0x1	R/W
6	TDSP_INT06_AONIRQ1_CON		TDSP Interrupt Pin-6 and AON_IRQ1 Interrupt Connected. 1: Connected 0: Disconnected	0x1	R/W
5	TDSP_INT05_DMA_CON		TDSP Interrupt Pin-5 and DMA Interrupt Connected. 1: Connected 0: Disconnected	0x1	R/W
4	TDSP_INT04_DTSYNC_CON		TDSP Interrupt Pin-4 and Data_Sync Interrupt Connected. 1: Connected 0: Disconnected	0x1	R/W
3	TDSP_INT03_QSPI_CON		TDSP Interrupt Pin-3 and QSPI Interrupt Connected. 1: Connected 0: Disconnected	0x1	R/W
2	TDSP_INT02_FFSRAM2_CON		TDSP Interrupt Pin-2 and FIFO_SRAM2 Interrupt Connected. 1: Connected 0: Disconnected	0x1	R/W
1	TDSP_INT01_FFSRAM1_CON		TDSP Interrupt Pin-1 and FIFO_SRAM1 Interrupt Connected. 1: Connected 0: Disconnected	0x1	R/W
0	TDSP_INT00_FFSRAM0_CON		TDSP Interrupt Pin-0 and FIFO_SRAM0 Interrupt Connected. 1: Connected 0: Disconnected	0x1	R/W

SOC HIGH PRIORITY LEVEL TRIGGER INTERRUPT CONNECTION REGISTER

Address: 0x40002014, Reset: 0xFFFFFFFF, Name: SOC_HIPRI_LVLINT_CON

This register is used for user to re-configure or update 'Level Sensitive' peripherals' interrupt priority levels. In default, 'Level Sensitive' peripherals' interrupts are connected at TDSP interrupt input pin 0~14, which are 'low priority' interrupt from TDSP perspective. TDSP input interrupt pin 16~23 are 'high priority' interrupt and are available to be connected with specific block interrupt sources.

Table 81. Bit Descriptions for SOC_HIPRI_LVLINT_CON

Bits	Bit Name	Settings	Description	Reset	Access
[31:28]	TDSP_INT23_SEL		TDSP Interrupt 23 Input Select 4'h0: FFSRAM0 4'h1: FFSRAM1 4'h2: FFSRAM2 4'h3: QSPI 4'h4: Data_Sync 4'h5: DMA 4'h6: AON_IRQ1 4'h7: AON_IRQ2 4'h8: AON_IRQ3 4'h9: AON_IRQ4 4'ha: MPI_IRQ1 4'hb: MPI_IRQ2 4'hc: MPI_IRQ3 4'hd: UART 4'he: Reserved 4'hf: Tie Low	0xF	R/W

Bits	Bit Name	Settings	Description	Reset	Access
[27:24]	TDSP_INT22_SEL		TDSP Interrupt 22 Input Select 4'h0: FFSRAM0 4'h1: FFSRAM1 4'h2: FFSRAM2 4'h3: QSPI 4'h4: Data_Sync 4'h5: DMA 4'h6: AON_IRQ1 4'h7: AON_IRQ2 4'h8: AON_IRQ3 4'h9: AON_IRQ4 4'ha: MPI_IRQ1 4'hb: MPI_IRQ2 4'hc: MPI_IRQ3 4'hd: Tie Low 4'he: Tie Low 4'hf: Tie Low4'hd: UART 4'he: Reserved 4'hf: Tie Low	0xF	R/W
[23:20]	TDSP_INT21_SEL		TDSP Interrupt 21 Input Select 4'h0: FFSRAM0 4'h1: FFSRAM1 4'h2: FFSRAM2 4'h3: QSPI 4'h4: Data_Sync 4'h5: DMA 4'h6: AON_IRQ1 4'h7: AON_IRQ2 4'h8: AON_IRQ3 4'h9: AON_IRQ4 4'ha: MPI_IRQ1 4'hb: MPI_IRQ2 4'hc: MPI_IRQ3 4'hd: UART 4'he: Reserved 4'hf: Tie Low	0xF	R/W
[19:16]	TDSP_INT20_SEL		TDSP Interrupt 20 Input Select 4'h0: FFSRAM0 4'h1: FFSRAM1 4'h2: FFSRAM2 4'h3: QSPI 4'h4: Data_Sync 4'h5: DMA 4'h6: AON_IRQ1 4'h7: AON_IRQ2 4'h8: AON_IRQ3 4'h9: AON_IRQ4 4'ha: MPI_IRQ1 4'hb: MPI_IRQ2 4'hc: MPI_IRQ3 4'hd: UART 4'he: Reserved 4'hf: Tie Low	0xF	R/W
[15:12]	TDSP_INT19_SEL		TDSP Interrupt 19 Input Select 4'h0: FFSRAM0 4'h1: FFSRAM1 4'h2: FFSRAM2 4'h3: QSPI 4'h4: Data_Sync 4'h5: DMA 4'h6: AON_IRQ1 4'h7: AON_IRQ2 4'h8: AON_IRQ3 4'h9: AON_IRQ4 4'ha: MPI_IRQ1 4'hb: MPI_IRQ2 4'hc: MPI_IRQ3 4'hd: UART 4'he: Reserved 4'hf: Tie Low	0xF	R/W
[11:8]	TDSP_INT18_SEL		TDSP Interrupt 18 Input Select 4'h0: FFSRAM0 4'h1: FFSRAM1 4'h2: FFSRAM2 4'h3: QSPI 4'h4: Data_Sync 4'h5: DMA 4'h6: AON_IRQ1 4'h7: AON_IRQ2 4'h8: AON_IRQ3 4'h9: AON_IRQ4 4'ha: MPI_IRQ1 4'hb: MPI_IRQ2 4'hc: MPI_IRQ3 4'hd: UART 4'he: Reserved 4'hf: Tie Low	0xF	R/W
[7:4]	TDSP_INT17_SEL		TDSP Interrupt 17 Input Select 4'h0: FFSRAM0 4'h1: FFSRAM1 4'h2: FFSRAM2 4'h3: QSPI 4'h4: Data_Sync 4'h5: DMA 4'h6: AON_IRQ1 4'h7: AON_IRQ2 4'h8: AON_IRQ3 4'h9: AON_IRQ4 4'ha: MPI_IRQ1 4'hb: MPI_IRQ2 4'hc: MPI_IRQ3 4'hd: UART 4'he: Reserved 4'hf: Tie Low	0xF	R/W
[3:0]	TDSP_INT16_SEL		TDSP Interrupt 16 Input Select 4'h0: FFSRAM0 4'h1: FFSRAM1 4'h2: FFSRAM2 4'h3: QSPI 4'h4: Data_Sync 4'h5: DMA 4'h6: AON_IRQ1 4'h7: AON_IRQ2 4'h8: AON_IRQ3 4'h9: AON_IRQ4 4'ha: MPI_IRQ1 4'hb: MPI_IRQ2 4'hc: MPI_IRQ3 4'hd: UART 4'he: Reserved 4'hf: Tie Low	0xF	R/W

SOC LOW PRIORITY EDGE TRIGGER INTERRUPT CONNECTION REGISTER

Address: 0x40002018, Reset: 0x03000000, Name: SOC_LOPRI_EDGINT_CON

In default, 'Edge triggered' peripherals' interrupt sources are connected with TDSP input interrupt pin 24~25. All these interrupt are low priority interrupt from TDSP perspective. User is able to disconnect the connection by clearing these registers bits

Table 82. Bit Descriptions for SOC_LOPRI_EDGINT_CON

Bits	Bit Name	Settings	Description	Reset	Access
[31:26]	RESERVED		Reserved.	0x0	R
25	TDSP_INT25_WDT_CON		TDSP Interrupt Pin-25 and WDT Interrupt Connected. 1: Connected 0: Disconnected	0x1	R/W
24	TDSP_INT24_GPT_CON		TDSP Interrupt Pin-24 and GPT Interrupt Connected. 1: Connected 0: Disconnected	0x1	R/W
[23:0]	RESERVED		Reserved.	0x0	R

SOC HIGH PRIORITY EDGE TRIGGER INTERRUPT CONNECTION REGISTER

Address: 0x4000201C, Reset: 0x000000FF, Name: SOC_HIPRI_EDGINT_CON

This register is used for user to re-configure or update 'Edge Sensitive' peripherals' interrupt priority levels. In default, 'Edge Sensitive' peripherals' interrupts are connected at TDSP interrupt input pin 24~25, which are 'low priority' interrupt from TDSP perspective. TDSP input interrupt pin 26~27 are 'high priority' interrupt and are available to be connected with specific block interrupt sources.

Table 83. Bit Descriptions for SOC_HIPRI_EDGINT_CON

Bits	Bit Name	Settings	Description	Reset	Access
[31:8]	RESERVED		Reserved.	0x0	R
[7:4]	TDSP_INT27_SEL		TDSP Interrupt 27 Input Select. TDSP Interrupt 27 Input Select 4'h0: GPT 4'h1: WDT 4'h2 ~ 4'hf: Tie Low	0xF	R/W
[3:0]	TDSP_INT26_SEL		TDSP Interrupt 26 Input Select. TDSP Interrupt 26 Input Select 4'h0: GPT 4'h1: WDT 4'h2 ~ 4'hf: Tie Low	0xF	R/W

SOC ERROR STATUS REGISTER

Address: 0x40002024, Reset: 0x00000000, Name: SOC_ERR_STATUS

This register is used for indicating SOC internal and TDSP Error status.

Table 84. Bit Descriptions for SOC_ERR_STATUS

Bits	Bit Name	Settings	Description	Reset	Access
[31:10]	RESERVED		Reserved.	0x0	R
9	TDSP_WR_BUSY_ERR		TDSP Write Always-ON Busy Error. 1: Indicates an error status that caused by a 'busy' event. The current write-data cannot be written to Always-ON registers or FDSP/EQ memories. 0: No error. Read to get value. Write 0 to clear.	0x0	R/W
8	TDSP_WR_FDSPEQ_ERR		TDSP Write FDSP-or-EQ Error. 1: Indicates an error status that caused by TDSP was trying to write FDSP or EQ-engine memories with non-32 bit-width access. Only 32-bit word is allowed to write. 0: No error. Read to get the status value, write 0 to clear.	0x0	R/W
[7:2]	RESERVED		Reserved.	0x0	R
1	TDSP_DOUB_EXCP		Tensilica DSP Double Exception. 1: Tensilica DSP has ever been in Double-Exception state. 0: Tensilica DSP has never been in Double-Exception state. Read to get the value Write '0' or reset to clear.	0x0	R/W
0	TDSP_FATAL_ERROR		Tensilica DSP PFatal Error Indicator. 1: PFatal Error Status 0: No PFatal Error Read-only bit. Can only be reset when the SoC or TDSP is reset.	0x0	R

SOC PFAULT INFORMATION REGISTER

Address: 0x40002028, Reset: 0x00000000, Name: SOC_PFAULT_INFO

User can read and get TDSP output pin 'Pfault' value by reading this register.

Table 85. Bit Descriptions for SOC_PFAULT_INFO

Bits	Bit Name	Settings	Description	Reset	Access
[31:0]	TDSP_PFATAL_INFO		Tensilica DSP PFatal Information. Read this 32-bit field will get the detailed information of the PFatal state. Read-only bit. Can only be reset when the SoC or TDSP is reset.	0x0	R

SOC MEMORY LIGHT-SLEEP REGISTER

Address: 0x40002040, Reset: 0x00000000, Name: SOC_MEM_LIGHT_SLEEP

User can control memories 'LS' input pin values by writing this register.

Table 86. Bit Descriptions for SOC_MEM_LIGHT_SLEEP

Bits	Bit Name	Settings	Description	Reset	Access
[31:9]	RESERVED		Reserved.	0x0	R
[8:0]	MEM_LIGHT_SLEEP		Memory Light Sleep. User can control memories 'LS' input pin values by writing this register.	0x0	R/W

REGISTER SUMMARY: UNIVERSAL ASYNCHRONOUS RECEIVER/TRANSMITTER

Table 87. UART(UART) Register Summary

Address	Name	Description	Reset	Access
0x40004800	UART_RXTX	Receive/transfer Buffer Register.	0x0000	R
0x40004804	UART_IEN	Interrupt Enable.	0x0000	R/W
0x40004808	UART_IIR	Interrupt ID.	0x0001	R
0x4000480C	UART_LCR	Line Control.	0x0000	R/W
0x40004810	UART_MCR	Modem Control.	0x0000	R/W
0x40004814	UART_LSR	Line Status.	0x0060	R
0x40004818	UART_MSR	Modem Status.	0x00XX	R
0x4000481C	UART_SCR	Scratch Buffer.	0x0000	R/W
0x40004820	UART_FCR	FIFO Control.	0x0000	R/W
0x40004824	UART_FBR	Fractional Baud Rate.	0x0000	R/W
0x40004828	UART_DIV	Baudrate Divider.	0x0000	R/W
0x4000482C	UART_LCR2	Second Line Control.	0x0002	R/W
0x40004830	UART_CTL	UART Control Register.	0x0100	R/W
0x40004834	UART_RFC	RX FIFO Byte Count.	0x0000	R
0x40004838	UART_TFC	TX FIFO Byte Count.	0x0000	R
0x4000483C	UART_RSC	RS485 Half-duplex Control.	0x0000	R/W
0x40004840	UART_ACR	Auto Baud Control.	0x0000	R/W
0x40004844	UART_ASRL	Auto Baud Status (Low).	0x0000	R
0x40004848	UART_ASRH	Auto Baud Status (High).	0x0000	R

REGISTER DETAILS: UNIVERSAL ASYNCHRONOUS RECEIVER/TRANSMITTER

RECEIVE/TRANSFER BUFFER REGISTER

Address: 0x40004800, Reset: 0x0000, Name: UART_RXTX

Table 88. Bit Descriptions for UART_RXTX

Bits	Bit Name	Settings	Description	Reset	Access
[15:8]	RESERVED		Reserved.	0x0	R
[7:0]	UART_RBR		Receive Buffer Register.	0x0	R

INTERRUPT ENABLE REGISTER

Address: 0x40004804, Reset: 0x0000, Name: UART_IEN

Table 89. Bit Descriptions for UART_IEN

Bits	Bit Name	Settings	Description	Reset	Access
[15:6]	RESERVED		Reserved.	0x0	R
5	UART_EDMAR		DMA Requests in Receive Mode.	0x0	R/W
		0	DMA Requests Disabled.		
		1	DMA Requests Enabled.		
4	UART_EDMAT		DMA Requests in Transmit Mode.	0x0	R/W
		0	DMA Requests are Disabled.		
		1	DMA Requests are Enabled.		
3	UART_EDSSI		Modem Status Interrupt. Interrupt is generated when any of UART_UART_MSR.UART_DDCD, UART_UART_MSR.UART_TERI, UART_UART_MSR.UART_DDSR, or UART_UART_MSR.UART_DCTS are set.	0x0	R/W
		0	Interrupt Disabled.		
		1	Interrupt Enabled.		
2	UART_ELSI		Rx Status Interrupt.	0x0	R/W
		0	Interrupt Disabled.		
		1	Interrupt Enabled.		
1	UART_ETBEI		Transmit Buffer Empty Interrupt.	0x0	R/W
		0	Interrupt Disabled.		
		1	Interrupt Enabled.		
0	UART_ERBFI		Receive Buffer Full Interrupt.	0x0	R/W
		0	Interrupt Disabled.		
		1	Interrupt Enabled.		

INTERRUPT ID REGISTER

Address: 0x40004808, Reset: 0x0001, Name: UART_IIR

Table 90. Bit Descriptions for UART_IIR

Bits	Bit Name	Settings	Description	Reset	Access
[15:8]	RESERVED		Reserved.	0x0	R
[7:6]	UART_FEND		FIFO Enabled.	0x0	R
		00	FIFO Not Enabled, 16450 Mode.		
		11	FIFO Enabled, 16550 Mode.		
[5:4]	RESERVED		Reserved.	0x0	R

Bits	Bit Name	Settings	Description	Reset	Access
[3:1]	STAT		Interrupt Status. When UART_UART_IIR.UART_NIRQ is low (active-low), this indicates an interrupt and the UART_UART_IIR.STAT bit decoding below is used.	0x0	RC
		000	Modem Status Interrupt (Read MSR Register to Clear).		
		001	Transmit Buf Empty IRQ (Write COMTX or Read COMIIR to Clear). Transmit Buffer Empty Interrupt (Write to Tx Register or Read IIR Register to Clear)		
		010	Receive Buffer Full Interrupt (Read Rx Register to Clear).		
		110	Receive FIFO Time-out Interrupt (Read Rx Register to Clear).		
		011	Receive Line Status Interrupt (Read LSR Register to Clear).		
0	UART_NIRQ		Interrupt Flag.	0x1	RS

LINE CONTROL REGISTER

Address: 0x4000480C, Reset: 0x0000, Name: UART_LCR

Table 91. Bit Descriptions for UART_LCR

Bits	Bit Name	Settings	Description	Reset	Access
[15:7]	RESERVED		Reserved.	0x0	R
6	UART_BRK		Set Break.	0x0	R/W
5	UART_SP		Stick Parity. Used to force parity to defined values. When set, the parity will be based on the following bit settings : UART_UART_LCR.UART_EPS = 1 and UART_UART_LCR.UART_PEN = 1, parity will be forced to 0. UART_UART_LCR.UART_EPS = 0 and UART_UART_LCR.UART_PEN = 1, parity will be forced to 1. UART_UART_LCR.UART_EPS = X and UART_UART_LCR.UART_PEN = 0, no parity will be transmitted	0x0	R/W
		0	Parity Will Not Be Forced Based on EPS and PEN. Parity Will Not Be Forced Based on Parity Select and Parity Enable Bits.		
		1	Parity Forced Based on Parity Select and Parity Enable Bits.		
4	UART_EPS		Parity Select. This bit only has meaning if parity is enabled (UART_UART_LCR.UART_PEN set).	0x0	R/W
		0	Odd Parity Will Be Transmitted and Checked.		
		1	Even Parity Will Be Transmitted and Checked.		
3	UART_PEN		Parity Enable. Used to control of the parity bit transmitted and checked. The value transmitted and the value checked will be based on the settings of UART_UART_LCR.UART_EPS and UART_UART_LCR.UART_SP.	0x0	R/W
		0	Parity Will Not Be Transmitted or Checked.		
		1	Parity Will Be Transmitted and Checked.		
2	UART_STOPPED		Stop Bit. Used to control the number of stop bits transmitted. In all cases only the first stop bit will be evaluated on data received.	0x0	R/W
		0	Send 1 Stop Bit Regardless of the Word Length Select Bit.		
		1	Word Length of Stop Bits. Send a Number of Stop Bits Based on the Word Length as Follows: WLS = 00, 1.5 Stop Bits Transmitted (5-bit Word Length) WLS = 01 or 10 or 11, 2 Stop Bits Transmitted (6 or 7 or 8-bit Word Length).		
[1:0]	UART_WLS		Word Length Select. Selects the number of bits per transmission.	0x0	R/W
		00	5 Bits.		
		01	6 Bits.		
		10	7 Bits.		
		11	8 Bits.		

MODEM CONTROL REGISTER

Address: 0x40004810, Reset: 0x0000, Name: UART_MCR

Table 92. Bit Descriptions for UART_MCR

Bits	Bit Name	Settings	Description	Reset	Access
[15:5]	RESERVED		Reserved.	0x0	R
4	UART_LOOPBACK		Loop Back Mode. In loop back mode, the SOUT is forced high. The modem signals are also directly connected to the status inputs (UART_UART_MCR.UART_RTS to UART_UART_MSR.UART_CTS, UART_UART_MCR.UART_DTR to UART_UART_MSR.UART_DSR, UART_UART_MCR.UART_OUT1 to UART_UART_MSR.UART_RI, and UART_UART_MCR.UART_OUT2 to UART_UART_MSR.UART_DCD).	0x0	R/W
		0	Normal Operation - Loopback Disabled.		
		1	Loopback Enabled.		
3	UART_OUT2		Output 2.	0x0	R/W
		0	Force NOUT2 to a Logic 1.		
		1	Force NOUT2 to a Logic 0.		
2	UART_OUT1		Output 1.	0x0	R/W
		0	Force NOUT1 to a Logic 1.		
		1	Force NOUT1 to a Logic 0.		
1	UART_RTS		Request to Send.	0x0	R/W
		0	Force NRTS to a Logic 1.		
		1	Force NRTS to a Logic 0.		
0	UART_DTR		Data Terminal Ready.	0x0	R/W
		0	Force NDTR to a Logic 1.		
		1	Force NDTR to a Logic 0.		

LINE STATUS REGISTER

Address: 0x40004814, Reset: 0x0060, Name: UART_LSR

Table 93. Bit Descriptions for UART_LSR

Bits	Bit Name	Settings	Description	Reset	Access
[15:8]	RESERVED		Reserved.	0x0	R
7	UART_FIFOERR		FIFO Error. data byte(s) in RX FIFO have either parity error, frame error or break indication. only used in 16550 mode; Read-clear if no more error in RX FIFO	0x0	RC
6	UART_TEMT		Transmit and Shift Register Empty Status.	0x1	R
		0	Not Empty. COMTX Has Been Written to and Contains Data to Be Transmitted. Care Should Be Taken Not to Overwrite Its Value.		
		1	Empty. COMTX and the Transmit Shift Register are Empty and It is Safe to Write New Data to COMTX. Data Has Been Transmitted.		
5	UART_THRE		Transmit Register Empty. THRE is cleared when UART_UART_RXTX is read.	0x1	R
		0	Not Empty. Tx Register Has Been Written to and Contains Data to Be Transmitted. Care Should Be Taken Not to Overwrite Its Value.		
		1	Empty. Tx Register is Empty and It is Safe to Write New Data to Tx Register the Previous Data May Not Have Been Transmitted Yet and Can Still Be Present in the Shift Register.		
4	UART_BI		Break Indicator. If set, this bit will self clear after UART_UART_LSR is read.	0x0	RC
3	UART_FE		Framing Error. If set, this bit will self clear after UART_UART_LSR is read.	0x0	RC
		0	No Invalid Stop Bit Was Detected.		
		1	An Invalid Stop Bit Was Detected on a Received Word.		

Bits	Bit Name	Settings	Description	Reset	Access
2	UART_PE		Parity Error. If set, this bit will self clear after UART_UART_LSR is read.	0x0	RC
		0	No Parity Error Was Detected.		
		1	A Parity Error Occurred on a Received Word.		
1	UART_OE		Overrun Error. If set, this bit will self clear after UART_UART_LSR is read.	0x0	RC
		0	Receive Data Has Not Been Overwritten.		
		1	Receive Data Was Overwritten by New Data Before Read COMRX. Receive Data Was Overwritten by New Data Before Rx Register Was Read.		
0	UART_DR		Data Ready. This bit is cleared only by reading UART_UART_RXTX. This bit will not self clear.	0x0	RC
		0	Rx Register Does Not Contain New Receive Data.		
		1	Rx Register Contains Receive Data That Should Be Read.		

MODEM STATUS REGISTER

Address: 0x40004818, Reset: 0x00XX, Name: UART_MSR

Table 94. Bit Descriptions for UART_MSR

Bits	Bit Name	Settings	Description	Reset	Access
[15:8]	RESERVED		Reserved.	0x0	R
7	UART_DCD		Data Carrier Detect. This bit reflects the direct status complement of the nDCD pin.	0xX	R
		0	NDCD is Currently Logic High.		
		1	NDCD is Currently Logic Low.		
6	UART_RI		Ring Indicator. This bit reflects the direct status complement of the nRI pin.	0xX	R
		0	NRI is Currently Logic High.		
		1	NRI is Currently Logic Low.		
5	UART_DSR		Data Set Ready. This bit reflects the direct status complement of the nDSR pin	0xX	R
		0	NDSR is Currently Logic High.		
		1	NDSR is Currently Logic Low.		
4	UART_CTS		Clear to Send. This bit reflects the direct status complement of the nCTS pin	0xX	R
		0	NCTS is Currently Logic High.		
		1	NCTS is Currently Logic Low.		
3	UART_DDCD		Delta DCD. If set, this bit will self clear after UART_UART_MSR is read.	0xX	R
		0	NDCD Has Not Changed State Since UART_MSR Was Last Read. Data Carrier Detect Bit Has Not Changed State Since MSR Register Was Last Read		
		1	NDCD Changed State Since UART_MSR Last Read. Data Carrier Detect Bit Changed State Since MSR Register Last Read		
2	UART_TERI		Trailing Edge RI. If set, this bit will self clear after UART_UART_MSR is read.	0xX	R
		0	NRI Has Not Changed from 1 to 0 Since UART_MSR Last Read. Ring Indicator Bit Has Not Changed from 0 to 1 Since MSR Register Last Read.		
		1	NRI Changed from 1 to 0 Since UART_MSR Last Read. Ring Indicator Bit Changed from 0 to 1 Since MSR Register Last Read.		
1	UART_DDSR		Delta DSR. If set, this bit will self clear after UART_UART_MSR is read.	0xX	R
		0	NDSR Has Not Changed State Since UART_MSR Was Last Read. Data Set Ready Bit Has Not Changed State Since MSR Register Was Last Read		
		1	NDSR Changed State Since UART_MSR Last Read. Data Set Ready Bit Changed State Since MSR Register Last Read		
0	UART_DCTS		Delta CTS. If set, this bit will self clear after UART_UART_MSR is read.	0xX	R
		0	NCTS Has Not Changed State Since UART_MSR Was Last Read. Clear to Send Bit Has Not Changed State Since MSR Register Was Last Read		
		1	Clear to Send Bit Changed State Since MSR Register Last Read.		

SCRATCH BUFFER REGISTER

Address: 0x4000481C, Reset: 0x0000, Name: UART_SCR

Table 95. Bit Descriptions for UART_SCR

Bits	Bit Name	Settings	Description	Reset	Access
[15:8]	RESERVED		Reserved.	0x0	R
[7:0]	UART_SCR		Scratch. The scratch register is an 8-bit register used to store intermediate results. The value contained in Scratch Register does not affect UART functionality or performance. Only 8 bits of this register are implemented. Bits 15 to 8 are read only and always return 0x00 when read. Writable with any value from 0 to 255. A read will return the last value written.	0x0	R/W

FIFO CONTROL REGISTER

Address: 0x40004820, Reset: 0x0000, Name: UART_FCR

Table 96. Bit Descriptions for UART_FCR

Bits	Bit Name	Settings	Description	Reset	Access
[15:8]	RESERVED		Reserved.	0x0	R
[7:6]	UART_RFTRIG		RX FIFO Trig Level.	0x0	R/W
		00	1 Byte to Trig RX Interrupt.		
		01	4 Byte to Trig RX Interrupt.		
		10	8 Byte to Trig RX Interrupt.		
		11	14 Byte to Trig RX Interrupt.		
[5:4]	RESERVED		Reserved.	0x0	R
3	UART_FDMAMD		FIFO DMA Mode.	0x0	R/W
		0	In DMA Mode 0, Assert DMA Request. In DMA Mode 0, RX DMA Request Will Be Asserted Whenever There's Data in RBR or RX FIFO and De-assert Whenever RBR or RX FIFO is Empty; TX DMA Request Will Be Asserted Whenever THR or TX FIFO is Empty and De-assert Whenever Data Written to.		
		1	In DMA Mode 1, Assert DMA Request. In DMA Mode 1, RX DMA Request Will Be Asserted Whenever RX FIFO Trig Level or Time Out Reached and De-assert Thereafter When RX FIFO is Empty; TX DMA Request Will Be Asserted Whenever TX FIFO is Empty and De-assert Thereafter When TX FIFO is Completely Filled up Full.		
2	UART_TFCLR		Clear TX FIFO.	0x0	W
1	UART_RFCLR		Clear RX FIFO.	0x0	W
0	UART_FIFOEN		FIFO Enable as to Work in 16550 Mode.	0x0	R/W

FRACTIONAL BAUD RATE REGISTER

Address: 0x40004824, Reset: 0x0000, Name: UART_FBR

Table 97. Bit Descriptions for UART_FBR

Bits	Bit Name	Settings	Description	Reset	Access
15	UART_FBEN		Fractional Baud Rate Generator Enable. The generating of fractional baud rate can be described by the following formula and the final baud rate of UART operation is calculated as Baudrate = (UCLK / (2 * (UART_UART_FBR.UART_DIVM + UART_UART_FBR.UART_DIVN/2048) * 16 * UART_UART_DIV))	0x0	R/W
[14:13]	RESERVED		Reserved.	0x0	R
[12:11]	UART_DIVM		Fractional Baud Rate M Divide Bits 1 to 3. This bit should not be 0 when Fractional Baud Rate enabled.	0x0	R/W
[10:0]	UART_DIVN		Fractional Baud Rate N Divide Bits 0 to 2047.	0x0	R/W

BAUDRATE DIVIDER REGISTER

Address: 0x40004828, Reset: 0x0000, Name: UART_DIV

Internal UART baud generation counters are restarted whenever COMDIV register accessed by writing, regardless same or different value.

Table 98. Bit Descriptions for UART_DIV

Bits	Bit Name	Settings	Description	Reset	Access
[15:0]	UART_DIV		Baud Rate Divider. The UART_UART_DIV register should not be 0, which is not specified. The range of allowed UART_UART_DIV values is from 1 to 65535.	0x0	R/W

SECOND LINE CONTROL REGISTER

Address: 0x4000482C, Reset: 0x0002, Name: UART_LCR2

Table 99. Bit Descriptions for UART_LCR2

Bits	Bit Name	Settings	Description	Reset	Access
[15:2]	RESERVED		Reserved.	0x0	R
[1:0]	UART_OSR		Over Sample Rate.	0x2	R/W
		00	Over Sample by 4.		
		01	Over Sample by 8.		
		10	Over Sample by 16.		
		11	Over Sample by 32.		

UART CONTROL REGISTER

Address: 0x40004830, Reset: 0x0100, Name: UART_CTL

Table 100. Bit Descriptions for UART_CTL

Bits	Bit Name	Settings	Description	Reset	Access
[15:8]	UART_REV		UART Revision ID.	0x1	R
[7:5]	RESERVED		Reserved.	0x0	R
4	UART_RXINV		Invert Receiver Line.	0x0	R/W
		0	Don't Invert Receiver Line (Idling High).		
		1	Invert Receiver Line (Idling Low).		
[3:2]	RESERVED		Reserved.	0x0	R

Bits	Bit Name	Settings	Description	Reset	Access
1	FORCECLK		Force UCLK on.	0x0	R/W
		0	UCLK Automatically Gated.		
		1	UCLK Always Working.		
0	RESERVED		Reserved.	0x0	R

RX FIFO BYTE COUNT REGISTER

Address: 0x40004834, Reset: 0x0000, Name: UART_RFC

Table 101. Bit Descriptions for UART_RFC

Bits	Bit Name	Settings	Description	Reset	Access
[15:5]	RESERVED		Reserved.	0x0	R
[4:0]	UART_RFC		Current RX FIFO Data Bytes.	0x0	R

TX FIFO BYTE COUNT REGISTER

Address: 0x40004838, Reset: 0x0000, Name: UART_TFC

Table 102. Bit Descriptions for UART_TFC

Bits	Bit Name	Settings	Description	Reset	Access
[15:5]	RESERVED		Reserved.	0x0	R
[4:0]	UART_TFC		Current TX FIFO Data Bytes.	0x0	R

RS485 HALF-DUPLEX CONTROL REGISTER

Address: 0x4000483C, Reset: 0x0000, Name: UART_RSC

Table 103. Bit Descriptions for UART_RSC

Bits	Bit Name	Settings	Description	Reset	Access
[15:4]	RESERVED		Reserved.	0x0	R
3	UART_DISTX		Hold off TX When Receiving.	0x0	R/W
2	UART_DISRX		Disable RX When Transmitting.	0x0	R/W
1	UART_OENSP		SOUT_EN De-assert Before Full Stop Bit(s).	0x0	R/W
		0	SOUT_EN De-assert Same Time as Full Stop Bit(s).		
		1	SOUT_EN De-assert Half-bit Earlier Than Full Stop Bit(s).		
0	UART_OENP		SOUT_EN Polarity.	0x0	R/W
		0	High Active.		
		1	Low Active.		

AUTO BAUD CONTROL REGISTER

Address: 0x40004840, Reset: 0x0000, Name: UART_ACR

Table 104. Bit Descriptions for UART_ACR

Bits	Bit Name	Settings	Description	Reset	Access
[15:12]	RESERVED		Reserved.	0x0	R
[11:8]	UART_EEC		Ending Edge Count.	0x0	R/W
		0000	First Edge.		
		0001	Second Edge.		
		0010	Third Edge.		
		0011	Fourth Edge.		

Bits	Bit Name	Settings	Description	Reset	Access
		0100	Fifth Edge.		
		0101	Sixth Edge.		
		0110	Seventh Edge.		
		0111	Eighth Edge.		
		1000	Ninth Edge.		
7	RESERVED		Reserved.	0x0	R
[6:4]	UART_SEC		Starting Edge Count.	0x0	R/W
		000	First Edge (Always the Falling Edge of START Bit).		
		001	Second Edge.		
		010	Third Edge.		
		011	Fourth Edge.		
		100	Fifth Edge.		
		101	Sixth Edge.		
		110	Seventh Edge.		
		111	Eighth Edge.		
3	RESERVED		Reserved.	0x0	R
2	UART_TOIEN		Enable Time-out Interrupt.	0x0	R/W
1	UART_DNIEN		Enable Done Interrupt.	0x0	R/W
0	UART_ABE		Auto Baud Enable.	0x0	R/W

AUTO BAUD STATUS (LOW) REGISTER

Address: 0x40004844, Reset: 0x0000, Name: UART_ASRL

Table 105. Bit Descriptions for UART_ASRL

Bits	Bit Name	Settings	Description	Reset	Access
[15:4]	CNT		Auto Baud Counter Value.	0x0	R
3	UART_NEETO		Timed Out Due to No Valid Ending Edge Found.	0x0	RC
2	UART_NSETO		Timed Out Due to No Valid Start Edge Found.	0x0	RC
1	UART_BRKTO		Timed Out Due to Long Time Break Condition.	0x0	RC
0	UART_DONE		Auto Baud Done Successfully.	0x0	RC

AUTO BAUD STATUS (HIGH) REGISTER

Address: 0x40004848, Reset: 0x0000, Name: UART_ASRH

Table 106. Bit Descriptions for UART_ASRH

Bits	Bit Name	Settings	Description	Reset	Access
[15:8]	RESERVED		Reserved.	0x0	R
[7:0]	CNT		Auto Baud Counter Value.	0x0	R

REGISTER SUMMARY: GPT

Table 107. GPT(TMR) Register Summary

Address	Name	Description	Reset	Access
0x40006000	GPT_LD_VAL	16-bit Load Value Register.	0x0000	R/W
0x40006004	GPT_CURR_VAL	16-bit Current Timer Value Register.	0x0000	R
0x40006008	GPT_CTRL	Control Register.	0x000A	R/W
0x4000600C	GPT_INT_CLR	Clear Interrupt Register.	0x0000	R/W
0x4000601C	GPT_STATUS	Status Register.	0x0000	R

REGISTER DETAILS: GPT

16-BIT LOAD VALUE REGISTER

Address: 0x40006000, Reset: 0x0000, Name: GPT_LD_VAL

Table 108. Bit Descriptions for GPT_LD_VAL

Bits	Bit Name	Settings	Description	Reset	Access
[15:0]	GPT_LD_VAL		Load Value. The Up/Down counter is periodically loaded with this value if periodic mode is selected (GPT_CTRL[3]=1). LOAD writes during Up/Down counter timeout events are delayed until the event has passed.	0x0	R/W

16-BIT CURRENT TIMER VALUE REGISTER

Address: 0x40006004, Reset: 0x0000, Name: GPT_CURR_VAL

Table 109. Bit Descriptions for GPT_CURR_VAL

Bits	Bit Name	Settings	Description	Reset	Access
[15:0]	GPT_CURR_VAL		Current Count. Reflects the current Up/Down counter value. Value delayed two PCLK cycles due to clock synchronizers.	0x0	R

CONTROL REGISTER

Address: 0x40006008, Reset: 0x000A, Name: GPT_CTRL

Table 110. Bit Descriptions for GPT_CTRL

Bits	Bit Name	Settings	Description	Reset	Access
[15:8]	RESERVED		Reserved.	0x0	R
7	GPT_RLD		Reload Control. RLD is only used for periodic mode; this bit allows the user to select whether the Up/Down counter should be reset only on a timeout event or also when GPTCLRI[0] is set	0x0	R/W
		1	Resets the Up/down Counter When GPTCLRI[0] is Set.		
		0	Up/Down Counter is Only Reset on a Timeout Event.		
6	GPT_CLK_SEL		Clock Select. Used to select a timer clock from the two available clock sources	0x0	R/W
		0	PCLK.		
		1	32KHz Clock Generated from AON.		
5	RESERVED		Reserved	0x0	R
4	GPT_EN		Timer Enable. Used to enable and disable the timer. Clearing this bit resets the timer, including the GPTVAL register.	0x0	R/W
		0	DIS. Timer is Disabled (Default).		
		1	EN. Timer is Enabled.		
3	GPT_MODE		Timer Mode. This bit is used to control whether the timer runs in periodic or free running mode. In periodic mode the up/down counter starts at the defined LOAD value (T0LD); in free running mode the up/down counter starts at 0x0000 or 0xFFFF depending on whether the timer is counting up or down.	0x1	R/W
		0	FREERUN.Timer Runs in Free Running Mode.		
		1	PERIODIC. Timer Runs in Periodic Mode (Default).		
2	GPT_UP_DOWN		Count up. Used to control whether the timer increments (counts up) or decrements (counts down) the Up/Down counter.	0x0	R/W
		0	DIS. Timer is Set to Count Down (Default).		
		1	EN. Timer is Set to Count up.		

Bits	Bit Name	Settings	Description	Reset	Access
[1:0]	GPT_PRE_SCALER		Prescaler. Controls the prescaler division factor applied to the timer's selected clock. If CLK source 0 or CLK Source 1 are selected then Prescaler value 0 means divide by 4, else it means divide by 1.	0x2	R/W
		00	Source_clock / [1 or 4].		
		01	Source_clock / 16.		
		10	Source_clock / 256.		
		11	Source_clock / 32,768.		

CLEAR INTERRUPT REGISTER

Address: 0x4000600C, Reset: 0x0000, Name: GPT_INT_CLR

Table 111. Bit Descriptions for GPT_INT_CLR

Bits	Bit Name	Settings	Description	Reset	Access
[15:1]	RESERVED		Reserved.	0x0	R
0	GPT_INT_TMOUT		Clear Timeout Interrupt. This bit is used to clear a timeout interrupt.	0x0	R/W1C

STATUS REGISTER

Address: 0x4000601C, Reset: 0x0000, Name: GPT_STATUS

Table 112. Bit Descriptions for GPT_STATUS

Bits	Bit Name	Settings	Description	Reset	Access
[15:8]	RESERVED		Reserved.	0x0	R
7	GPT_STATUS_INT_CLR_DONE		GPTCLRI Synchronization. This bit is set automatically when the user sets GPTCLRI[0]=1. It is cleared automatically when the clear interrupt request has crossed clock domains and taken effect in the timer clock domain	0x0	R
		0	CLR. the Interrupt is Cleared in the Timer Clock Domain.		
		1	SET. GPTCLRI[0] is Being Updated in the Timer Clock Domain.		
6	GPT_STATUS_CTRL_BUSY		Timer Busy. This bit informs the user that a write to GPT_CTRL is still crossing into the timer clock domain. This bit should be checked after writing GPT_CTRL and further writes should be suppressed until this bit is cleared.	0x0	R
		0	Timer Ready to Receive Commands to GPT_CTRL.		
		1	Timer Not Ready to Receive Commands to GPT_CTRL.		
[5:1]	RESERVED		Reserved.	0x0	R
0	GPT_STATUS_TMOUT		Timeout Event Occurred. This bit set automatically when the value of the counter reaches zero while counting down or reaches full scale when counting up. This bit is cleared when GPTCLRI[0] is set by the user.	0x0	R
		0	No Timeout Event Has Occurred.		
		1	a Timeout Event Has Occurred.		

REGISTER SUMMARY: WATCHDOG TIMER REGISTER MAP WDT

Table 113. WDT(WDT) Register Summary

Address	Name	Description	Reset	Access
0x40006800	LD	Watchdog Timer Load Value.	0x1000	R/W
0x40006804	VALS	Current Count Value.	0x1000	R
0x40006808	CON	Watchdog Timer Control Register.	0x00C9	R/W
0x4000680C	CLRI	Refresh Watchdog Register.	0x0000	W
0x40006818	STA	Timer Status.	0x0000	R
0x4000681C	MINLD	Minimum Load Value.	0x0800	R/W

REGISTER DETAILS: WATCHDOG TIMER

WATCHDOG TIMER LOAD VALUE REGISTER

Address: 0x40006800, Reset: 0x1000, Name: LD

Load value.

Table 114. Bit Descriptions for LD

Bits	Bit Name	Settings	Description	Reset	Access
[15:0]	WDT_LD_VAL		WDT Load Value. User programmable value. This is the value that the counter will start from and count down to 0.	0x1000	R/W

CURRENT COUNT VALUE REGISTER

Address: 0x40006804, Reset: 0x1000, Name: VALS

When read, returns the counter's value

Table 115. Bit Descriptions for VALS

Bits	Bit Name	Settings	Description	Reset	Access
[15:0]	WDT_CURR_VAL		Current WDT Count Value. Read only register.	0x1000	R

WATCHDOG TIMER CONTROL REGISTER

Address: 0x40006808, Reset: 0x00C9, Name: CON

Table 116. Bit Descriptions for CON

Bits	Bit Name	Settings	Description	Reset	Access
[15:11]	RESERVED		Reserved.	0x0	R
10	WDT_INT_EN		WDT Interrupt Enable.	0x0	R/W
9	WDT_MIN_LD_EN		Timer Window Control. When enabled, if user refreshes the Timer before counter reaches value in MIN_LOAD , reset/IRQ will occur.	0x0	R/W
8	WDT_CLKDIV2		WDT Clock Divided by 2.	0x0	R/W
7	SPARE		Reserved. Added to Match with the Signal "spare7" in RTL code. Reset Value 'h1.	0x1	R/W
6	WDT_MODE		Timer Mode Select.	0x1	R/W
		0	Free-running Mode. Note that in free running mode timer wraps around at 0x1000		
		1	Periodic Mode. In this mode, counter counts from WDTLD value down to 0		
5	WDT_EN		Timer Enable.	0x0	R/W
4	RESERVED		Reserved.	0x0	R
[3:2]	WDT_PRE_SCALER		Prescaler.	0x2	R/W
		00	Source Clock/1.		
		01	Source Clock/16.		
		10	Source Clock/256.		
		11	Source Clock/4096.		
1	WDT_TO_BEHAVE		WDT Interrupt Enable.	0x0	R/W
		0	Watchdog Timer Timeout Creates a Reset.		
		1	Timeout Creates an Interrupt Instead of Reset.		
0	RESERVED		Reserved.	0x1	R/W

REFRESH WATCHDOG REGISTER

Address: 0x4000680C, Reset: 0x0000, Name: CLRI

Clear Watchdog

Table 117. Bit Descriptions for CLRI

Bits	Bit Name	Settings	Description	Reset	Access
[15:0]	WDT_CLR		Refresh Register. User writes 0xCCCC to reset/reload/restart the counter or clear IRQ. A write of any other value causes a watchdog reset/IRQ. Write only, reads 0.	0x0	W

TIMER STATUS REGISTER

Address: 0x40006818, Reset: 0x0000, Name: STA

Timer Status

Table 118. Bit Descriptions for STA

Bits	Bit Name	Settings	Description	Reset	Access
[15:7]	RESERVED		Reserved.	0x0	R
6	WDT_MIN_LD_WR_BUSY		WDTMINLD Write Status.	0x0	R
5	WDT_RST_CTRL_WR_BUSY		Reset Type Status. This status bit indicates if WDTRCR.RSTCON has been written to. WDTRCR.RSTCON is a One time programmable register.	0x0	R
4	WDT_LOCK		Lock Status.	0x0	R
		0	Timer Operation Not Locked.		
		1	Timer Enabled and Locked. Set automatically in hardware when WDTCON[5] has been set by user code.		
3	WDT_CFG_WR_BUSY		WDTCON Write Status.	0x0	R
2	WDT_LD_VAL_WR_BUSY		WDTVAL Write Status.	0x0	R
1	WDT_CLR_WR_BUSY		WDTCLRI Write Status.	0x0	R
0	WDT_IRQ		WDT Interrupt. Set to 1 when watchdog timer interrupt occurs.	0x0	R
		0	Watchdog Timer Interrupt Not Pending.		
		1	Watchdog Timer Interrupt Pending. Cleared by writing to WDTCLRI register.		

MINIMUM LOAD VALUE REGISTER

Address: 0x4000681C, Reset: 0x0800, Name: MINLD

Watchdog Timer Minimum Time-out period. Lower Window limit.

Table 119. Bit Descriptions for MINLD

Bits	Bit Name	Settings	Description	Reset	Access
[15:0]	WDT_MIN_LD		WDT Min Load Value. If software writes to WDTCLRI before counter reaches MIN_LOAD value, WDT reset/IRQ will happen.	0x800	R/W

REGISTER SUMMARY: AUDIO DATA PATH

Table 120. Audio data path Register Summary

Address	Name	Description	Reset	Access
0x4000C000	VENDOR_ID	ADI Vendor ID.	0x41	R
0x4000C001	DEVICE_ID1	Device ID.	0x60	R
0x4000C002	DEVICE_ID2	Device ID.	0x18	R
0x4000C003	REVISION	Revision Code.	0x01	R
0x4000C004	ADC_DAC_HP_PWR	ADC, DAC, Headphone Power Controls.	0x00	R/W
0x4000C005	PLL_PGA_PWR	PLL, Mic Bias, and PGA Power Controls.	0x02	R/W
0x4000C006	DMIC_PWR	Digital Mic Power Controls.	0x00	R/W
0x4000C007	SAI_CLK_PWR	Serial Port, PDM Output, and DMIC CLK Power Controls.	0x00	R/W
0x4000C008	DSP_PWR	DSP Power Controls.	0x00	R/W
0x4000C009	ASRC_PWR	ASRC Power Controls.	0x00	R/W
0x4000C00A	FINT_PWR	Interpolator Power Controls.	0x00	R/W
0x4000C00B	FDEC_PWR	Decimator Power Controls.	0x00	R/W
0x4000C00C	KEEP_CM	State Retention Controls.	0x01	R/W
0x4000C00D	MEM_RETAIN	Retention Control for ADP and SOC Memories.	0x00	R/W
0x4000C00E	CHIP_PWR	Chip Power Control.	0x00	R/W
0x4000C010	CLK_CTRL1	Clock Control.	0xC8	R/W
0x4000C011	CLK_CTRL2	PLL Input Divider.	0x00	R/W
0x4000C012	CLK_CTRL3	PLL Feedback Integer Divider (LSBs).	0x04	R/W
0x4000C013	CLK_CTRL4	PLL Feedback Integer Divider (MSBs).	0x00	R/W
0x4000C014	CLK_CTRL5	PLL Fractional numerator value (LSBs).	0x00	R/W
0x4000C015	CLK_CTRL6	PLL Fractional numerator value (MSBs).	0x00	R/W
0x4000C016	CLK_CTRL7	PLL Fractional denominator (LSBs).	0x00	R/W
0x4000C017	CLK_CTRL8	PLL Fractional denominator (MSBs).	0x00	R/W
0x4000C018	CLK_CTRL9	PLL Update.	0x00	R/W
0x4000C019	CLK_CTRL10	TDSP Clock Rate, BUS(AHB/APB) Clock Rate and FFSRAM2 Rate.	0xC0	R/W
0x4000C01A	CLK_CTRL11	AON and UART_CTRL Clock Rate.	0x00	R/W
0x4000C01B	CLK_CTRL12	Frequency Multiplier Enable, Ratio.	0x00	R/W
0x4000C01C	CLK_CTRL13	PLL Frequency Index and ADC Frequency Index.	0x13	R/W
0x4000C01D	CLK_CTRL14	ADP_Clock_Enable, Low byte.	0xFF	R/W
0x4000C01E	CLK_CTRL15	ADP_Clock_Enable, High byte.	0xFF	R/W
0x4000C020	ADC_CTRL1	ADC Sample Rate Control.	0x22	R/W
0x4000C021	ADC_CTRL2	ADC IBias Controls.	0x00	R/W
0x4000C022	ADC_CTRL3	ADC IBias Controls.	0x00	R/W
0x4000C023	ADC_CTRL4	ADC HPF Control.	0x00	R/W
0x4000C024	ADC_CTRL5	ADC Mute and Compensation Control.	0x40	R/W
0x4000C025	ADC_CTRL6	Analog Input Pre-Charge Time.	0x86	R/W
0x4000C026	ADC_CTRL7	Analog Input Pre-Charge Time.	0x00	R/W
0x4000C027	ADC_MUTES	ADC Channel Mutes.	0x00	R/W
0x4000C028	ADC0_VOL	ADC Channel 0 Volume Control.	0x40	R/W
0x4000C029	ADC1_VOL	ADC Channel 1 Volume Control.	0x40	R/W
0x4000C02A	ADC2_VOL	ADC Channel 2 Volume Control.	0x40	R/W
0x4000C02B	ADC_DITHER_LEV	ADC Dithering level.	0x02	R/W
0x4000C030	PGA0_CTRL1	PGA Channel 0 Gain Control LSB's, Mute, Boost, Slew.	0x00	R/W
0x4000C031	PGA0_CTRL2	PGA Channel 0 Gain Control MSB's.	0x00	R/W
0x4000C032	PGA1_CTRL1	PGA Channel 1 Gain Control LSB's, Mute, Boost, Slew.	0x00	R/W
0x4000C033	PGA1_CTRL2	PGA Channel 1 Gain Control MSB's.	0x00	R/W

Address	Name	Description	Reset	Access
0x4000C034	PGA2_CTRL1	PGA Channel 2 Gain Control LSB's, Mute, Boost, Slew.	0x00	R/W
0x4000C035	PGA2_CTRL2	PGA Channel 2 Gain Control MSB's.	0x00	R/W
0x4000C036	PGA_CTRL1	PGA Slew Rate and Gain Link.	0x00	R/W
0x4000C037	PGA_CTRL2	PGA RIN and Power Mode.	0x00	R/W
0x4000C040	DMIC_CTRL1	DMIC clock rate control.	0x03	R/W
0x4000C041	DMIC_CTRL2	DMIC Channels 0 and 1 rate, order, mapping, and edge control.	0x01	R/W
0x4000C042	DMIC_CTRL3	DMIC Channels 2 and 3 rate, order, mapping, and edge control.	0x01	R/W
0x4000C043	DMIC_CTRL4	DMIC Volume Options.	0x04	R/W
0x4000C044	DMIC_MUTES	DMIC Channel Mute Controls.	0x00	R/W
0x4000C045	DMIC_VOL0	DMIC Channel 0 Volume Control.	0x40	R/W
0x4000C046	DMIC_VOL1	DMIC Channel 1 Volume Control.	0x40	R/W
0x4000C047	DMIC_VOL2	DMIC Channel 2 Volume Control.	0x40	R/W
0x4000C048	DMIC_VOL3	DMIC Channel 3 Volume Control.	0x40	R/W
0x4000C049	DMIC_CTRL5	DMIC Channels 4 and 5 rate, order, mapping, and edge control.	0x01	R/W
0x4000C04A	DMIC_CTRL6	DMIC Channels 6 and 7 rate, order, mapping, and edge control.	0x01	R/W
0x4000C04B	DMIC_CTRL7	DMIC Clock Map, DMIC Clock 1 Source Pin Select.	0x00	R/W
0x4000C04C	DMIC_VOL4	DMIC Channel 4 Volume Control.	0x40	R/W
0x4000C04D	DMIC_VOL5	DMIC Channel 5 Volume Control.	0x40	R/W
0x4000C04E	DMIC_VOL6	DMIC Channel 6 Volume Control.	0x40	R/W
0x4000C04F	DMIC_VOL7	DMIC Channel 7 Volume Control.	0x40	R/W
0x4000C050	DAC_CTRL1	DAC Sample Rate, Filtering, and Power Controls.	0x02	R/W
0x4000C051	DAC_CTRL2	DAC Volume Lunk, HPF, and Mute Controls.	0x44	R/W
0x4000C052	DAC_VOL0	DAC Channel 0 Volume.	0x40	R/W
0x4000C053	DAC_ROUTE0	DAC Channel 0 Routing.	0x00	R/W
0x4000C060	HP_CTRL	Headphone control.	0x00	R/W
0x4000C061	HP_LVMODE_CTRL1	HP Low voltage Mode Enable, CM Enable.	0x00	R/W
0x4000C062	HP_LVMODE_CTRL2	HP Low Voltage Auto Switch Mode, Delay, CM Delay.	0x30	R/W
0x4000C063	HP_LVMODE_CTRL3	HP Low Voltage Auto Switch(Go).	0x00	R/W
0x4000C066	HPLDO_CTRL	HPLDO CTRL.	0x00	R/W
0x4000C06C	PB_CTRL	HP Low voltage mode control.	0x00	R/W
0x4000C070	PMU_CTRL1	Memory Power Control.	0x00	R/W
0x4000C071	PMU_CTRL2	CM Control.	0x00	R/W
0x4000C072	PMU_CTRL3	IRQ Wakeup Control.	0x00	R/W
0x4000C073	PMU_CTRL4	IRQ Wakeup Control.	0x00	R/W
0x4000C074	PMU_CTRL5	DLDO Control.	0x81	R/W
0x4000C080	FDEC_CTRL1	Fast to Slow Decimator Sample Rates Channels 0 and 1.	0x25	R/W
0x4000C081	FDEC_CTRL2	Fast to Slow Decimator Sample Rates Channels 2 and 3.	0x25	R/W
0x4000C082	FDEC_CTRL3	Fast to Slow Decimator Sample Rates Channels 4 and 5.	0x25	R/W
0x4000C083	FDEC_CTRL4	Fast to Slow Decimator Sample Rates Channels 6 and 7.	0x25	R/W
0x4000C084	FDEC_ROUTE0	Fast to Slow Decimator Channel 0 Input Routing.	0x00	R/W
0x4000C085	FDEC_ROUTE1	Fast to Slow Decimator Channel 1 Input Routing.	0x00	R/W
0x4000C086	FDEC_ROUTE2	Fast to Slow Decimator Channel 2 Input Routing.	0x00	R/W
0x4000C087	FDEC_ROUTE3	Fast to Slow Decimator Channel 3 Input Routing.	0x00	R/W
0x4000C088	FDEC_ROUTE4	Fast to Slow Decimator Channel 4 Input Routing.	0x00	R/W
0x4000C089	FDEC_ROUTE5	Fast to Slow Decimator Channel 5 Input Routing.	0x00	R/W
0x4000C08A	FDEC_ROUTE6	Fast to Slow Decimator Channel 6 Input Routing.	0x00	R/W

Address	Name	Description	Reset	Access
0x4000C08B	FDEC_ROUTE7	Fast to Slow Decimator Channel 7 Input Routing.	0x00	R/W
0x4000C090	FINT_CTRL1	Slow to Fast Interpolator Sample Rates Channels 0/1.	0x52	R/W
0x4000C091	FINT_CTRL2	Slow to Fast Interpolator Sample Rates Channels 2/3.	0x52	R/W
0x4000C092	FINT_CTRL3	Slow to Fast Interpolator Sample Rates Channels 4/5.	0x52	R/W
0x4000C093	FINT_CTRL4	Slow to Fast Interpolator Sample Rates Channels 6/7.	0x52	R/W
0x4000C094	FINT_ROUTE0	Slow to Fast Interpolator Channel 0 Input Routing.	0x00	R/W
0x4000C095	FINT_ROUTE1	Slow to Fast Interpolator Channel 1 Input Routing.	0x00	R/W
0x4000C096	FINT_ROUTE2	Slow to Fast Interpolator Channel 2 Input Routing.	0x00	R/W
0x4000C097	FINT_ROUTE3	Slow to Fast Interpolator Channel 3 Input Routing.	0x00	R/W
0x4000C098	FINT_ROUTE4	Slow to Fast Interpolator Channel 4 Input Routing.	0x00	R/W
0x4000C099	FINT_ROUTE5	Slow to Fast Interpolator Channel 5 Input Routing.	0x00	R/W
0x4000C09A	FINT_ROUTE6	Slow to Fast Interpolator Channel 6 Input Routing.	0x00	R/W
0x4000C09B	FINT_ROUTE7	Slow to Fast Interpolator Channel 7 Input Routing.	0x00	R/W
0x4000C0A0	ASRCI_CTRL	Input ASRC control, source, and rate selection.	0x02	R/W
0x4000C0A1	ASRCI_ROUTE01	Input ASRC Channel 0 and 1 Input Routing.	0x00	R/W
0x4000C0A2	ASRCI_ROUTE23	Input ASRC Channel 2 adn 3 Input Routing.	0x00	R/W
0x4000C0A3	ASRCO_CTRL	Output ASRC Control.	0x02	R/W
0x4000C0A4	ASRCO_ROUTE0	Output ASRC Channel 0 Input Routing.	0x00	R/W
0x4000C0A5	ASRCO_ROUTE1	Output ASRC Channel 1 Input Routing.	0x00	R/W
0x4000C0A6	ASRCO_ROUTE2	Output ASRC Channel 2 Input Routing.	0x00	R/W
0x4000C0A7	ASRCO_ROUTE3	Output ASRC Channel 3 Input Routing.	0x00	R/W
0x4000C0B0	FDSP_RUN	FastDSP Run.	0x00	R/W
0x4000C0B1	FDSP_CTRL1	FastDSP current bank and bank ramping controls.	0x70	R/W
0x4000C0B2	FDSP_CTRL2	FastDSP bank ramping stop point.	0x3F	R/W
0x4000C0B3	FDSP_CTRL3	FastDSP bank copying.	0x00	W
0x4000C0B4	FDSP_CTRL4	FastDSP frame rate source.	0x00	R/W
0x4000C0B5	FDSP_CTRL5	FastDSP fixed rate division MSBs.	0x7F	R/W
0x4000C0B6	FDSP_CTRL6	FastDSP fixed rate division LSBs.	0x00	R/W
0x4000C0B7	FDSP_CTRL7	FastDSP Modulo N counter for lower rate conditional execution.	0x00	R/W
0x4000C0B8	FDSP_CTRL8	FastDSP generic conditional execution registers.	0x00	R/W
0x4000C0B9	FDSP_SL_ADDR	Fast DSP Safeload Address.	0x00	R/W
0x4000C0BA	FDSP_SL_P0_0	FastDSP Safeload Parameter 0 Value.	0x00	R/W
0x4000C0BB	FDSP_SL_P0_1	FastDSP Safeload Parameter 0 Value.	0x00	R/W
0x4000C0BC	FDSP_SL_P0_2	FastDSP Safeload Parameter 0 Value.	0x00	R/W
0x4000C0BD	FDSP_SL_P0_3	FastDSP Safeload Parameter 0 Value.	0x00	R/W
0x4000C0BE	FDSP_SL_P1_0	FastDSP Safeload Parameter 1 Value.	0x00	R/W
0x4000C0BF	FDSP_SL_P1_1	FastDSP Safeload Parameter 1 Value.	0x00	R/W
0x4000C0C0	FDSP_SL_P1_2	FastDSP Safeload Parameter 1 Value.	0x00	R/W
0x4000C0C1	FDSP_SL_P1_3	FastDSP Safeload Parameter 1 Value.	0x00	R/W
0x4000C0C2	FDSP_SL_P2_0	FastDSP Safeload Parameter 2 Value.	0x00	R/W
0x4000C0C3	FDSP_SL_P2_1	FastDSP Safeload Parameter 2 Value.	0x00	R/W
0x4000C0C4	FDSP_SL_P2_2	FastDSP Safeload Parameter 2 Value.	0x00	R/W
0x4000C0C5	FDSP_SL_P2_3	FastDSP Safeload Parameter 2 Value.	0x00	R/W
0x4000C0C6	FDSP_SL_P3_0	FastDSP Safeload Parameter 3 Value.	0x00	R/W
0x4000C0C7	FDSP_SL_P3_1	FastDSP Safeload Parameter 3 Value.	0x00	R/W
0x4000C0C8	FDSP_SL_P3_2	FastDSP Safeload Parameter 3 Value.	0x00	R/W
0x4000C0C9	FDSP_SL_P3_3	FastDSP Safeload Parameter 3 Value.	0x00	R/W
0x4000C0CA	FDSP_SL_P4_0	FastDSP Safeload Parameter 4 Value.	0x00	R/W
0x4000C0CB	FDSP_SL_P4_1	FastDSP Safeload Parameter 4 Value.	0x00	R/W

Address	Name	Description	Reset	Access
0x4000C0CC	FDSP_SL_P4_2	FastDSP Safeload Parameter 4 Value.	0x00	R/W
0x4000C0CD	FDSP_SL_P4_3	FastDSP Safeload Parameter 4 Value.	0x00	R/W
0x4000C0CE	FDSP_SL_UPDATE	FastDSP Safeload Update.	0x00	W
0x4000C0CF	FDSP_ONZ_MASK0	FastDSP Safeload Parameter 0 Value.	0x00	R/W
0x4000C0D0	FDSP_ONZ_MASK1	FastDSP Safeload Parameter 0 Value.	0x00	R/W
0x4000C0D1	FDSP_ONZ_MASK2	FastDSP Safeload Parameter 0 Value.	0x00	R/W
0x4000C0D2	EQ_CFG	EQ Configure.	0x00	R/W
0x4000C0D3	EQ_ROUTE	EQ Routing.	0x00	R/W
0x4000C0D4	TDSP_SOFT_RESET	Tensilica DSP Software Reset.	0x00	W
0x4000C0D5	TDSP_ALTVEC_EN	Tensilica DSP Alternative Reset Vector Enable.	0x00	R/W
0x4000C0D8	TDSP_ALTVEC_ADDR0	Tensilica DSP Alternative Reset Vector Address, Byte0.	0x00	R/W
0x4000C0D9	TDSP_ALTVEC_ADDR1	Tensilica DSP Alternative Reset Vector Address, Byte1.	0x00	R/W
0x4000C0DA	TDSP_ALTVEC_ADDR2	Tensilica DSP Alternative Reset Vector Address, Byte2.	0x00	R/W
0x4000C0DB	TDSP_ALTVEC_ADDR3	Tensilica DSP Alternative Reset Vector Address, Byte3.	0x00	R/W
0x4000C0DC	TDSP_RUN	Tensilica DSP Run.	0x01	R/W
0x4000C0E0	SPT0_CTRL1	Serial Port 0 Control 1.	0x00	R/W
0x4000C0E1	SPT0_CTRL2	Serial Port 0 Control 2.	0x00	R/W
0x4000C0E2	SPT0_CTRL3	Serial Port 0 Control 2.	0x00	R/W
0x4000C0E3	SPT0_ROUTE0	Serial Port 0 Output Routing Slot 0 (left).	0x10	R/W
0x4000C0E4	SPT0_ROUTE1	Serial Port 0 Output Routing Slot 1 (right).	0x11	R/W
0x4000C0E5	SPT0_ROUTE2	Serial Port 0 Output Routing Slot 2.	0x3F	R/W
0x4000C0E6	SPT0_ROUTE3	Serial Port 0 Output Routing Slot 3.	0x3F	R/W
0x4000C0E7	SPT0_ROUTE4	Serial Port 0 Output Routing Slot 4.	0x3F	R/W
0x4000C0E8	SPT0_ROUTE5	Serial Port 0 Output Routing Slot 5.	0x3F	R/W
0x4000C0E9	SPT0_ROUTE6	Serial Port 0 Output Routing Slot 6.	0x3F	R/W
0x4000C0EA	SPT0_ROUTE7	Serial Port 0 Output Routing Slot 7.	0x3F	R/W
0x4000C0EB	SPT0_ROUTE8	Serial Port 0 Output Routing Slot 8.	0x3F	R/W
0x4000C0EC	SPT0_ROUTE9	Serial Port 0 Output Routing Slot 9.	0x3F	R/W
0x4000C0ED	SPT0_ROUTE10	Serial Port 0 Output Routing Slot 10.	0x3F	R/W
0x4000C0EE	SPT0_ROUTE11	Serial Port 0 Output Routing Slot 11.	0x3F	R/W
0x4000C0EF	SPT0_ROUTE12	Serial Port 0 Output Routing Slot 12.	0x3F	R/W
0x4000C0F0	SPT0_ROUTE13	Serial Port 0 Output Routing Slot 13.	0x3F	R/W
0x4000C0F1	SPT0_ROUTE14	Serial Port 0 Output Routing Slot 14.	0x3F	R/W
0x4000C0F2	SPT0_ROUTE15	Serial Port 0 Output Routing Slot 15.	0x3F	R/W
0x4000C0F3	SPT1_CTRL1	Serial Port 1 Control 1.	0x00	R/W
0x4000C0F4	SPT1_CTRL2	Serial Port 1 Control 2.	0x00	R/W
0x4000C0F5	SPT1_CTRL3	Serial Port 1 Control 2.	0x00	R/W
0x4000C0F6	SPT1_ROUTE0	Serial Port 1 Output Routing Slot 0 (left).	0x10	R/W
0x4000C0F7	SPT1_ROUTE1	Serial Port 1 Output Routing Slot 1 (right).	0x11	R/W
0x4000C0F8	SPT1_ROUTE2	Serial Port 1 Output Routing Slot 2.	0x3F	R/W
0x4000C0F9	SPT1_ROUTE3	Serial Port 1 Output Routing Slot 3.	0x3F	R/W
0x4000C0FA	SPT1_ROUTE4	Serial Port 1 Output Routing Slot 4.	0x3F	R/W
0x4000C0FB	SPT1_ROUTE5	Serial Port 1 Output Routing Slot 5.	0x3F	R/W
0x4000C0FC	SPT1_ROUTE6	Serial Port 1 Output Routing Slot 6.	0x3F	R/W
0x4000C0FD	SPT1_ROUTE7	Serial Port 1 Output Routing Slot 7.	0x3F	R/W
0x4000C0FE	SPT1_ROUTE8	Serial Port 1 Output Routing Slot 8.	0x3F	R/W
0x4000C0FF	SPT1_ROUTE9	Serial Port 1 Output Routing Slot 9.	0x3F	R/W
0x4000C100	SPT1_ROUTE10	Serial Port 1 Output Routing Slot 10.	0x3F	R/W
0x4000C101	SPT1_ROUTE11	Serial Port 1 Output Routing Slot 11.	0x3F	R/W
0x4000C102	SPT1_ROUTE12	Serial Port 1 Output Routing Slot 12.	0x3F	R/W

Address	Name	Description	Reset	Access
0x4000C103	SPT1_ROUTE13	Serial Port 1 Output Routing Slot 13.	0x3F	R/W
0x4000C104	SPT1_ROUTE14	Serial Port 1 Output Routing Slot 14.	0x3F	R/W
0x4000C105	SPT1_ROUTE15	Serial Port 1 Output Routing Slot 15.	0x3F	R/W
0x4000C118	PDM_CTRL1	PDM Sample Rate and Filtering Control.	0x02	R/W
0x4000C119	PDM_CTRL2	PDM muting, high-pass, and volume options.	0xC4	R/W
0x4000C11A	PDM_VOL0	PDM Output Channel 0 Volume.	0x40	R/W
0x4000C11B	PDM_VOL1	PDM Output Channel 1 Volume.	0x40	R/W
0x4000C11C	PDM_ROUTE0	PDM Output Channel 0 Routing.	0x00	R/W
0x4000C11D	PDM_ROUTE1	PDM Output Channel 1 Routing.	0x01	R/W
0x4000C121	MP_CTRL1	MultiPurpose Pins 0/1 Mode Select.	0x00	R/W
0x4000C122	MP_CTRL2	MultiPurpose Pins 2/3 Mode Select.	0x00	R/W
0x4000C123	MP_CTRL3	MultiPurpose Pins 4/5 Mode Select.	0x00	R/W
0x4000C124	MP_CTRL4	MultiPurpose Pins 6/7 Mode Select.	0x00	R/W
0x4000C125	MP_CTRL5	MultiPurpose Pins 8/9 Mode Select.	0x00	R/W
0x4000C126	MP_CTRL6	MultiPurpose Pin 10/11 Mode Select.	0x00	R/W
0x4000C127	MP_CTRL7	MultiPurpose Pin 12/13 Mode Select.	0x00	R/W
0x4000C128	MP_CTRL8	MultiPurpose Pin 14/15 Mode Select.	0x00	R/W
0x4000C129	MP_CTRL9	MultiPurpose Pin 16/17 Mode Select.	0x00	R/W
0x4000C12A	MP_CTRL10	MultiPurpose Pin 18/19 Mode Select.	0x00	R/W
0x4000C12B	MP_CTRL11	MultiPurpose Pin 20/21 Mode Select.	0x00	R/W
0x4000C12C	MP_CTRL12	MultiPurpose Pin 22/23 Mode Select.	0x00	R/W
0x4000C12D	MP_CTRL13	MultiPurpose Pin 24/25 Mode Select.	0x00	R/W
0x4000C12E	MP_DB_CTRL	General Purpose Input Debounce Control and IRQ Input Debounce Control.	0x00	R/W
0x4000C12F	MP_MCLKO_RATE	MCLKO Rate Selection.	0x01	R/W
0x4000C130	MP_GPIO_CTRL1	General Purpose Outputs Control Pins 0-7.	0x00	R/W
0x4000C131	MP_GPIO_CTRL2	General Purpose Outputs Control Pins 8-15.	0x00	R/W
0x4000C132	MP_GPIO_CTRL3	General Purpose Outputs Control Pins 16-23.	0x00	R/W
0x4000C133	MP_GPIO_CTRL4	General Purpose Outputs Control Pins 24-25.	0x00	R/W
0x4000C134	DMIC_CLK_CTRL	DMIC_CLK Pin Controls.	0x05	R/W
0x4000C135	DMIC01_CTRL	DMIC01 Pin Controls.	0x05	R/W
0x4000C136	DMIC23_CTRL	DMIC23 Pin Controls.	0x05	R/W
0x4000C137	BCLK0_CTRL	BCLK_0 Pin Controls.	0x05	R/W
0x4000C138	FSYNC0_CTRL	FSYNC_0 Pin Controls.	0x05	R/W
0x4000C139	SDATAO0_CTRL	SDATAO_0 Pin Control.	0x05	R/W
0x4000C13A	SDATAI0_CTRL	SDATAI_0 Pin Controls.	0x05	R/W
0x4000C13B	BCLK1_CTRL	BCLK_1 Pin Controls.	0x05	R/W
0x4000C13C	FSYNC1_CTRL	FSYNC_1 Pin Controls.	0x05	R/W
0x4000C13D	SDATAO1_CTRL	SDATAO_1 Pin Controls.	0x05	R/W
0x4000C13E	SDATAI1_CTRL	SDATAI_1 Pin Controls.	0x05	R/W
0x4000C13F	QSPIM_CLK_CTRL	QSPIM_CLK Pin Controls.	0x05	R/W
0x4000C140	QSPIM_CS_CTRL	QSPIM_CS Pin Controls.	0x05	R/W
0x4000C141	QSPIM_SDIO0_CTRL	QSPIM_SDIO0 Pin Controls.	0x05	R/W
0x4000C142	QSPIM_SDIO1_CTRL	QSPIM_SDIO1 Pin Controls.	0x05	R/W
0x4000C143	QSPIM_SDIO2_CTRL	QSPIM_SDIO2 Pin Controls.	0x05	R/W
0x4000C144	QSPIM_SDIO3_CTRL	QSPIM_SDIO3 Pin Controls.	0x25	R/W
0x4000C145	UART_COMM_TX_CTRL	UART_COMM_TX Pin Controls.	0x25	R/W
0x4000C146	UART_COMM_RX_CTRL	UART_COMM_TX Pin Controls.	0x25	R/W
0x4000C147	SELFBOOT_CTRL	SELFBOOT Pin Controls.	0x05	R/W
0x4000C148	IRQ_CTRL	SELFBOOT Pin Controls.	0x05	R/W

Address	Name	Description	Reset	Access
0x4000C149	ROM_BOOT_MODE_CTRL	SELFBOOT Pin Controls.	0x05	R/W
0x4000C14A	TCK_CTRL	SELFBOOT Pin Controls.	0x15	R/W
0x4000C14B	TMS_CTRL	SELFBOOT Pin Controls.	0x35	R/W
0x4000C14C	TDO_CTRL	SELFBOOT Pin Controls.	0x05	R/W
0x4000C14D	TDI_CTRL	SELFBOOT Pin Controls.	0x35	R/W
0x4000C14E	I2C_SPI_CTRL	SDA/MISO Pin Controls.	0x00	R/W
0x4000C150	IRQ_CTRL1	IRQ signaling and clearing.	0x00	R/W
0x4000C151	IRQ1_MASK1	IRQ1 Masking.	0x7F	R/W
0x4000C152	IRQ1_MASK2	IRQ1 Masking.	0xF7	R/W
0x4000C153	IRQ1_MASK3	IRQ1 Masking.	0x3F	R/W
0x4000C154	IRQ1_MASK4	IRQ1 Masking.	0x3F	R/W
0x4000C155	IRQ1_MASK5	IRQ1 Masking.	0x0F	R/W
0x4000C156	IRQ2_MASK1	IRQ2 Masking.	0x7F	R/W
0x4000C157	IRQ2_MASK2	IRQ2 Masking.	0xF7	R/W
0x4000C158	IRQ2_MASK3	IRQ2 Masking.	0x3F	R/W
0x4000C159	IRQ2_MASK4	IRQ2 Masking.	0x3F	R/W
0x4000C15A	IRQ2_MASK5	IRQ2 Masking.	0x0F	R/W
0x4000C15B	IRQ3_MASK1	IRQ3 Masking.	0x7F	R/W
0x4000C15C	IRQ3_MASK2	IRQ3 Masking.	0xF7	R/W
0x4000C15D	IRQ3_MASK3	IRQ3 Masking.	0x3F	R/W
0x4000C15E	IRQ3_MASK4	IRQ3 Masking.	0x3F	R/W
0x4000C15F	IRQ3_MASK5	IRQ3 Masking.	0x0F	R/W
0x4000C160	IRQ4_MASK1	IRQ4 Masking.	0x7F	R/W
0x4000C161	IRQ4_MASK2	IRQ4 Masking.	0xF7	R/W
0x4000C162	IRQ4_MASK3	IRQ4 Masking.	0x3F	R/W
0x4000C163	IRQ4_MASK4	IRQ4 Masking.	0x3F	R/W
0x4000C164	IRQ4_MASK5	IRQ4 Masking.	0x0F	R/W
0x4000C165	SW_INT	Software Interrupts which can be set by external host or TDSP.	0x00	W
0x4000C168	MP_INPUT_IRQ_CTRL1	MP IRQ clearing.	0x00	R/W
0x4000C169	MP_INPUT_IRQ_CTRL2	MP IRQ clearing.	0x00	R/W
0x4000C16A	MP_INPUT_IRQ1_MASK1	MP IRQ1 Masking.	0xFF	R/W
0x4000C16B	MP_INPUT_IRQ1_MASK2	MP IRQ1 Masking.	0xFF	R/W
0x4000C16C	MP_INPUT_IRQ1_MASK3	MP IRQ1 Masking.	0xFF	R/W
0x4000C16D	MP_INPUT_IRQ1_MASK4	MP IRQ1 Masking.	0x03	R/W
0x4000C16E	MP_INPUT_IRQ2_MASK1	MP IRQ1 Masking.	0xFF	R/W
0x4000C16F	MP_INPUT_IRQ2_MASK2	MP IRQ1 Masking.	0xFF	R/W
0x4000C170	MP_INPUT_IRQ2_MASK3	MP IRQ1 Masking.	0xFF	R/W
0x4000C171	MP_INPUT_IRQ2_MASK4	MP IRQ1 Masking.	0x03	R/W
0x4000C172	MP_INPUT_IRQ3_MASK1	MP IRQ1 Masking.	0xFF	R/W
0x4000C173	MP_INPUT_IRQ3_MASK2	MP IRQ1 Masking.	0xFF	R/W
0x4000C174	MP_INPUT_IRQ3_MASK3	MP IRQ1 Masking.	0xFF	R/W
0x4000C175	MP_INPUT_IRQ3_MASK4	MP IRQ1 Masking.	0x03	R/W
0x4000C200	RESETS	Chip Resets.	0x00	W
0x4000C400	READ_LAMBDA	FastDSP Current Lambda.	0x3F	R
0x4000C401	STATUS1	Chip Status 1.	0x00	R
0x4000C402	STATUS2	Chip Status 2.	0x00	R
0x4000C403	SELFBOOT_STATUS	Tensilica DSP Self-boot Indicator.	0x00	R
0x4000C404	EQ_STATUS	EQ Status.	0x00	R
0x4000C405	FDSP_STATUS	FastDSP ONZ Status.	0x00	R

Address	Name	Description	Reset	Access
0x4000C406	PMU_STATUS1	Power Mode Status, DLDO Scale Busy, CM Delay Counter Done.	0x00	R
0x4000C407	PMU_STATUS2	Memory Retention Status.	0x00	R
0x4000C408	PMU_STATUS3	ADP Memory Shutdown Status.	0x00	R
0x4000C409	PMU_STATUS4	SOC Memory Shutdown Status.	0x00	R
0x4000C40A	TDSP_MODE_STATUS	TDSP Mode Status.	0x00	R
0x4000C40B	TDSP_ERROR_STATUS	TDSP Exception/Error Status.	0x00	R
0x4000C40C	TDSP_FAULT_INFO1	TDSP Fault Information.	0x00	R
0x4000C40D	TDSP_FAULT_INFO2	TDSP Fault Information.	0x00	R
0x4000C40E	TDSP_FAULT_INFO3	TDSP Fault Information.	0x00	R
0x4000C40F	TDSP_FAULT_INFO4	TDSP Fault Information.	0x00	R
0x4000C410	TDSP_GPO1	TDSP General Purpose Output Bit 0~7.	0x00	R
0x4000C411	TDSP_GPO2	TDSP General Purpose Output Bit 8~15.	0x00	R
0x4000C412	TDSP_GPO3	TDSP General Purpose Output Bit 16~23.	0x00	R
0x4000C413	TDSP_GPO4	TDSP General Purpose Output Bit 24~31.	0x00	R
0x4000C414	IRQ1_STATUS1	IRQ1 Status 1.	0x00	R
0x4000C415	IRQ1_STATUS2	IRQ1 Status 2.	0x00	R
0x4000C416	IRQ1_STATUS3	IRQ1 Status 3.	0x00	R
0x4000C417	IRQ1_STATUS4	IRQ1 Status 4.	0x00	R
0x4000C418	IRQ1_STATUS5	IRQ1 Status 5.	0x00	R
0x4000C419	IRQ2_STATUS1	IRQ2 Status 1.	0x00	R
0x4000C41A	IRQ2_STATUS2	IRQ2 Status 2.	0x00	R
0x4000C41B	IRQ2_STATUS3	IRQ2 Status 3.	0x00	R
0x4000C41C	IRQ2_STATUS4	IRQ2 Status 4.	0x00	R
0x4000C41D	IRQ2_STATUS5	IRQ2 Status 5.	0x00	R
0x4000C41E	IRQ3_STATUS1	IRQ3 Status 1.	0x00	R
0x4000C41F	IRQ3_STATUS2	IRQ3 Status 2.	0x00	R
0x4000C420	IRQ3_STATUS3	IRQ3 Status 3.	0x00	R
0x4000C421	IRQ3_STATUS4	IRQ3 Status 4.	0x00	R
0x4000C422	IRQ3_STATUS5	IRQ3 Status 5.	0x00	R
0x4000C423	IRQ4_STATUS1	IRQ4 Status 1.	0x00	R
0x4000C424	IRQ4_STATUS2	IRQ4 Status 2.	0x00	R
0x4000C425	IRQ4_STATUS3	IRQ4 Status 3.	0x00	R
0x4000C426	IRQ4_STATUS4	IRQ4 Status 4.	0x00	R
0x4000C427	IRQ4_STATUS5	IRQ4 Status 5.	0x00	R
0x4000C428	MP_INPUT_IRQ1_STATUS1	Multi Purpose IRQ1 Status 1.	0x00	R
0x4000C429	MP_INPUT_IRQ1_STATUS2	Multi Purpose IRQ1 Status 2.	0x00	R
0x4000C42A	MP_INPUT_IRQ1_STATUS3	Multi Purpose IRQ1 Status 3.	0x00	R
0x4000C42B	MP_INPUT_IRQ1_STATUS4	Multi Purpose IRQ1 Status 3.	0x00	R
0x4000C42C	MP_INPUT_IRQ2_STATUS1	Multi Purpose IRQ2 Status 1.	0x00	R
0x4000C42D	MP_INPUT_IRQ2_STATUS2	Multi Purpose IRQ2 Status 2.	0x00	R
0x4000C42E	MP_INPUT_IRQ2_STATUS3	Multi Purpose IRQ2 Status 3.	0x00	R
0x4000C42F	MP_INPUT_IRQ2_STATUS4	Multi Purpose IRQ1 Status 3.	0x00	R
0x4000C430	MP_INPUT_IRQ3_STATUS1	Multi Purpose IRQ3 Status 1.	0x00	R
0x4000C431	MP_INPUT_IRQ3_STATUS2	Multi Purpose IRQ3 Status 2.	0x00	R
0x4000C432	MP_INPUT_IRQ3_STATUS3	Multi Purpose IRQ3 Status 3.	0x00	R
0x4000C433	MP_INPUT_IRQ3_STATUS4	Multi Purpose IRQ1 Status 3.	0x00	R
0x4000C434	GPI1	General Purpose Input Read 0-7.	0x00	R
0x4000C435	GPI2	General Purpose Input Read 8-15.	0x00	R
0x4000C436	GPI3	General Purpose Input Read 16-20.	0x00	R

Address	Name	Description	Reset	Access
0x4000C437	GPI4	General Purpose Input Read 16-20.	0x00	R
0x4000C438	CTRL_PORT_MODE	Control Port Mode.	0x00	R
0x4000CC04	DAC_NOISE_CTRL2	DAC/PLL Test Modes.	0x00	R/W
0x4000CC12	DAC_NOISE_CTRL1	DAC Noise Control1.	0x00	R/W

REGISTER DETAILS: AUDIO DATA PATH

ADI VENDOR ID REGISTER

Address: 0x4000C000, Reset: 0x41, Name: VENDOR_ID

Table 121. Bit Descriptions for VENDOR_ID

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	VENDOR		ADI Vendor ID.	0x41	R

DEVICE ID REGISTER

Address: 0x4000C001, Reset: 0x60, Name: DEVICE_ID1

Table 122. Bit Descriptions for DEVICE_ID1

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	DEVICE1		Device ID 1.	0x60	R

DEVICE ID REGISTER

Address: 0x4000C002, Reset: 0x18, Name: DEVICE_ID2

Table 123. Bit Descriptions for DEVICE_ID2

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	DEVICE2		Device ID 2.	0x18	R

REVISION CODE REGISTER

Address: 0x4000C003, Reset: 0x01, Name: REVISION

Table 124. Bit Descriptions for REVISION

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	REV		Revision ID.	0x1	R

ADC, DAC, HEADPHONE POWER CONTROLS REGISTER

Address: 0x4000C004, Reset: 0x00, Name: ADC_DAC_HP_PWR

Table 125. Bit Descriptions for ADC_DAC_HP_PWR

Bits	Bit Name	Settings	Description	Reset	Access
[7:5]	RESERVED		Reserved.	0x0	R
4	PB0_EN		Playback Path (DAC/HP) Channel 0 Enable.	0x0	R/W
		0	DAC and Headphone/Line-Out Channel 0 Powered Off.		
		1	DAC and Headphone/Line-Out Channel 0 Powered On.		
3	RESERVED		Reserved.	0x0	R
2	ADC2_EN		ADC Channel 2 Enable.	0x0	R/W
		0	ADC Channel 2 Powered Off.		
		1	ADC Channel 2 Powered On.		

Bits	Bit Name	Settings	Description	Reset	Access
1	ADC1_EN		ADC Channel 1 Enable.	0x0	R/W
		0	ADC Channel 1 Powered Off.		
		1	ADC Channel 1 Powered On.		
0	ADC0_EN		ADC Channel 0 Enable.	0x0	R/W
		0	ADC Channel 0 Powered Off.		
		1	ADC Channel 0 Powered On.		

PLL AND PGA POWER CONTROLS REGISTER

Address: 0x4000C005, Reset: 0x02, Name: PLL_PGA_PWR

Table 126. Bit Descriptions for PLL_PGA_PWR

Bits	Bit Name	Settings	Description	Reset	Access
7	RESERVED		Reserved.	0x0	R
6	PGA2_EN		Note that the PGA inverts the signal going through it.	0x0	R/W
		0	PGA2 powered down.		
		1	PGA2 powered up with slewing.		
5	PGA1_EN		Note that the PGA inverts the signal going through it.	0x0	R/W
		0	PGA1 powered down.		
		1	PGA1 powered up with slewing.		
4	PGA0_EN		Note that the PGA inverts the signal going through it.	0x0	R/W
		0	PGA0 powered down.		
		1	PGA0 powered up with slewing.		
[3:2]	RESERVED		Reserved.	0x0	R
1	XTAL_EN		Crystal Oscillator Enable.	0x1	R/W
		0	Crystal oscillator powered off.		
		1	Crystal oscillator powered on.		
0	PLL_EN		PLL Enable.	0x0	R/W
		0	PLL Powered Off.		
		1	PLL Powered On.		

DIGITAL MIC POWER CONTROLS REGISTER

Address: 0x4000C006, Reset: 0x00, Name: DMIC_PWR

Table 127. Bit Descriptions for DMIC_PWR

Bits	Bit Name	Settings	Description	Reset	Access
7	DMIC7_EN		Digital Mic Channel 7 Enable.	0x0	R/W
		0	Digital Mic Channel 7 Powered Off.		
		1	Digital Mic Channel 7 Powered On.		
6	DMIC6_EN		Digital Mic Channel 6 Enable.	0x0	R/W
		0	Digital Mic Channel 6 Powered Off.		
		1	Digital Mic Channel 6 Powered On.		
5	DMIC5_EN		Digital Mic Channel 5 Enable.	0x0	R/W
		0	Digital Mic Channel 5 Powered Off.		
		1	Digital Mic Channel 5 Powered On.		
4	DMIC4_EN		Digital Mic Channel 4 Enable.	0x0	R/W
		0	Digital Mic Channel 4 Powered Off.		
		1	Digital Mic Channel 4 Powered On.		

Bits	Bit Name	Settings	Description	Reset	Access
3	DMIC3_EN		Digital Mic Channel 3 Enable.	0x0	R/W
		0	Digital Mic Channel 3 Powered Off.		
		1	Digital Mic Channel 3 Powered On.		
2	DMIC2_EN		Digital Mic Channel 2 Enable.	0x0	R/W
		0	Digital Mic Channel 2 Powered Off.		
		1	Digital Mic Channel 2 Powered On.		
1	DMIC1_EN		Digital Mic Channel 1 Enable.	0x0	R/W
		0	Digital Mic Channel 1 Powered Off.		
		1	Digital Mic Channel 1 Powered On.		
0	DMIC0_EN		Digital Mic Channel 0 Enable.	0x0	R/W
		0	Digital Mic Channel 0 Powered Off.		
		1	Digital Mic Channel 0 Powered On.		

SERIAL PORT, PDM OUTPUT, AND DMIC CLK POWER CONTROLS REGISTER

Address: 0x4000C007, Reset: 0x00, Name: SAI_CLK_PWR

Table 128. Bit Descriptions for SAI_CLK_PWR

Bits	Bit Name	Settings	Description	Reset	Access
7	PDM1_EN		PDM Output Channel 1 Enable.	0x0	R/W
		0	PDM Output Channel 1 Powered Off.		
		1	PDM Output Channel 1 Powered On.		
6	PDM0_EN		PDM Output Channel 0 Enable.	0x0	R/W
		0	PDM Output Channel 0 Powered Off.		
		1	PDM Output Channel 0 Powered On.		
5	DMIC_CLK1_EN		Digital Mic Clock 1 Enable.	0x0	R/W
		0	Digital Mic Clock 1 Powered Off.		
		1	Digital Mic Clock 1 Powered On.		
4	DMIC_CLK_EN		Digital Mic Clock Enable.	0x0	R/W
		0	Digital Mic Clock Powered Off.		
		1	Digital Mic Clock Powered On.		
3	SPT1_OUT_EN		Serial Audio Port 1 Output Side Enable.	0x0	R/W
		0	Serial Audio Port 1 Output Side Powered Off.		
		1	Serial Audio Port 1 Output Side Powered On.		
2	SPT1_IN_EN		Serial Audio Port 1 Input Side Enable.	0x0	R/W
		0	Serial Audio Port 1 Input Side Powered Off.		
		1	Serial Audio Port 1 Input Side Powered On.		
1	SPT0_OUT_EN		Serial Audio Port 0 Output Side Enable.	0x0	R/W
		0	Serial Audio Port 0 Output Side Powered Off.		
		1	Serial Audio Port 0 Output Side Powered On.		
0	SPT0_IN_EN		Serial Audio Port 0 Input Side Enable.	0x0	R/W
		0	Serial Audio Port 0 Input Side Powered Off.		
		1	Serial Audio Port 0 Input Side Powered On.		

DSP POWER CONTROLS REGISTER

Address: 0x4000C008, Reset: 0x00, Name: DSP_PWR

Table 129. Bit Descriptions for DSP_PWR

Bits	Bit Name	Settings	Description	Reset	Access
[7:1]	RESERVED		Reserved.	0x0	R
0	FDSP_EN		FastDSP Enable.	0x0	R/W
		0	FastDSP Powered Off.		
		1	FastDSP Powered On.		

ASRC POWER CONTROLS REGISTER

Address: 0x4000C009, Reset: 0x00, Name: ASRC_PWR

Table 130. Bit Descriptions for ASRC_PWR

Bits	Bit Name	Settings	Description	Reset	Access
7	ASRCO3_EN		Output Asynchronous Sample Rate Converter Channel 3 Enable.	0x0	R/W
		0	Output Asynchronous Sample Rate Converter Channel 3 Powered Off.		
		1	Output Asynchronous Sample Rate Converter Channel 3 Powered On.		
6	ASRCO2_EN		Output Asynchronous Sample Rate Converter Channel 2 Enable.	0x0	R/W
		0	Output Asynchronous Sample Rate Converter Channel 2 Powered Off.		
		1	Output Asynchronous Sample Rate Converter Channel 2 Powered On.		
5	ASRCO1_EN		Output Asynchronous Sample Rate Converter Channel 1 Enable.	0x0	R/W
		0	Output Asynchronous Sample Rate Converter Channel 1 Powered Off.		
		1	Output Asynchronous Sample Rate Converter Channel 1 Powered On.		
4	ASRCO0_EN		Output Asynchronous Sample Rate Converter Channel 0 Enable.	0x0	R/W
		0	Output Asynchronous Sample Rate Converter Channel 0 Powered Off.		
		1	Output Asynchronous Sample Rate Converter Channel 0 Powered On.		
3	ASRCI3_EN		Input Asynchronous Sample Rate Converter Channel 3 Enable.	0x0	R/W
		0	Input Asynchronous Sample Rate Converter Channel 3 Powered Off.		
		1	Input Asynchronous Sample Rate Converter Channel 3 Powered On.		
2	ASRCI2_EN		Input Asynchronous Sample Rate Converter Channel 2 Enable.	0x0	R/W
		0	Input Asynchronous Sample Rate Converter Channel 2 Powered Off.		
		1	Input Asynchronous Sample Rate Converter Channel 2 Powered On.		
1	ASRCI1_EN		Input Asynchronous Sample Rate Converter Channel 1 Enable.	0x0	R/W
		0	Input Asynchronous Sample Rate Converter Channel 1 Powered Off.		
		1	Input Asynchronous Sample Rate Converter Channel 1 Powered On.		
0	ASRCI0_EN		Input Asynchronous Sample Rate Converter Channel 0 Enable.	0x0	R/W
		0	Input Asynchronous Sample Rate Converter Channel 0 Powered Off.		
		1	Input Asynchronous Sample Rate Converter Channel 0 Powered On.		

INTERPOLATOR POWER CONTROLS REGISTER

Address: 0x4000C00A, Reset: 0x00, Name: FINT_PWR

Table 131. Bit Descriptions for FINT_PWR

Bits	Bit Name	Settings	Description	Reset	Access
7	FINT7_EN		Interpolation Channel 7 Enable.	0x0	R/W
		0	Interpolation Channel 7 Powered Off.		
		1	Interpolation Channel 7 Powered On.		
6	FINT6_EN		Interpolation Channel 6 Enable.	0x0	R/W
		0	Interpolation Channel 6 Powered Off.		
		1	Interpolation Channel 6 Powered On.		
5	FINT5_EN		Interpolation Channel 5 Enable.	0x0	R/W
		0	Interpolation Channel 5 Powered Off.		
		1	Interpolation Channel 5 Powered On.		
4	FINT4_EN		Interpolation Channel 4 Enable.	0x0	R/W
		0	Interpolation Channel 4 Powered Off.		
		1	Interpolation Channel 4 Powered On.		
3	FINT3_EN		Interpolation Channel 3 Enable.	0x0	R/W
		1	Interpolation Channel 3 Powered On.		
		0	Interpolation Channel 3 Powered Off.		
2	FINT2_EN		Interpolation Channel 2 Enable.	0x0	R/W
		0	Interpolation Channel 2 Powered Off.		
		1	Interpolation Channel 2 Powered On.		
1	FINT1_EN		Interpolation Channel 1 Enable.	0x0	R/W
		0	Interpolation Channel 1 Powered Off.		
		1	Interpolation Channel 1 Powered On.		
0	FINT0_EN		Interpolation Channel 0 Enable.	0x0	R/W
		0	Interpolation Channel 0 Powered Off.		
		1	Interpolation Channel 0 Powered On.		

DECIMATOR POWER CONTROLS REGISTER

Address: 0x4000C00B, Reset: 0x00, Name: FDEC_PWR

Table 132. Bit Descriptions for FDEC_PWR

Bits	Bit Name	Settings	Description	Reset	Access
7	FDEC7_EN		Decimation Channel 7 Enable.	0x0	R/W
		0	Decimator Channel 7 Powered Off.		
		1	Decimator Channel 7 Powered On.		
6	FDEC6_EN		Decimation Channel 6 Enable.	0x0	R/W
		0	Decimator Channel 6 Powered Off.		
		1	Decimator Channel 6 Powered On.		
5	FDEC5_EN		Decimation Channel 5 Enable.	0x0	R/W
		0	Decimator Channel 5 Powered Off.		
		1	Decimator Channel 5 Powered On.		
4	FDEC4_EN		Decimation Channel 4 Enable.	0x0	R/W
		0	Decimator Channel 4 Powered Off.		
		1	Decimator Channel 4 Powered On.		
3	FDEC3_EN		Decimation Channel 3 Enable.	0x0	R/W
		0	Decimator Channel 3 Powered Off.		
		1	Decimator Channel 3 Powered On.		

Bits	Bit Name	Settings	Description	Reset	Access
2	FDEC2_EN		Decimation Channel 2 Enable.	0x0	R/W
		0	Decimator Channel 2 Powered Off.		
		1	Decimator Channel 2 Powered On.		
1	FDEC1_EN		Decimation Channel 1 Enable.	0x0	R/W
		0	Decimator Channel 1 Powered Off.		
		1	Decimator Channel 1 Powered On.		
0	FDEC0_EN		Decimation Channel 0 Enable.	0x0	R/W
		0	Decimator Channel 0 Powered Off.		
		1	Decimator Channel 0 Powered On.		

STATE RETENTION CONTROLS REGISTER

Address: 0x4000C00C, Reset: 0x01, Name: KEEP_CM

Table 133. Bit Descriptions for KEEP_CM

Bits	Bit Name	Settings	Description	Reset	Access
[7:1]	RESERVED		Reserved.	0x0	R
0	CM_KEEP_ALIVE		Common Mode (CM) output keep alive during power down.	0x1	R/W
		0	Common Mode (CM) output will turn off when PWR_MODE=2'b00. This allows for lower shutdown power, but longer start-up timing.		
		1	Common Mode (CM) output will stay on when PWR_MODE=2'b00. This allows for faster start-up timing, but larger shutdown power.		

RETENTION CONTROL FOR ADP AND SOC MEMORIES REGISTER

Address: 0x4000C00D, Reset: 0x00, Name: MEM_RETAIN

Table 134. Bit Descriptions for MEM_RETAIN

Bits	Bit Name	Settings	Description	Reset	Access
[7:2]	SOC_MEM_RETAIN		State Retention control for SOC power domain memories. Bit[2]: Controls IRAM0 retention. Bit[3]: Controls DRAM0 retention. Bit[4]: Controls DRAM1 retention. Bit[5]: Controls SYSRAM retention. Bit[6]: Controls FFSRAM0 retention. Bit[7]: Controls FFSRAM1 retention. 1'b1 -- retention 1'b0 -- no retention	0x0	R/W
[1:0]	ADP_MEM_RETAIN		State Retention control for ADP power domain memories. Bit[0]: controls FDSP and EQ instruction/ parameter memory retention. Bit[1]: controls FFSRAM2 memory retention. 1'b1 -- retention 1'b0 -- no retention	0x0	R/W

CHIP POWER CONTROL REGISTER

Address: 0x4000C00E, Reset: 0x00, Name: CHIP_PWR

Table 135. Bit Descriptions for CHIP_PWR

Bits	Bit Name	Settings	Description	Reset	Access
[7:5]	RESERVED		Reserved.	0x0	R
4	CM_STARTUP_OVER		Disables high power CM startup boost mode.	0x0	R/W
		0	CM pin fast charge enabled.		
		1	CM pin fast charge disabled.		
3	RESERVED		Reserved.	0x0	R

Bits	Bit Name	Settings	Description	Reset	Access
2	MASTER_BLOCK_EN		Master block level enable. Gates block level enabling of all blocks except PLL, XTAL, FDSP, and TDSP.	0x0	R/W
		0	All blocks are disabled.		
		1	All blocks that have their respective block enable set are enabled.		
[1:0]	PWR_MODE		Power modes for PMU control.	0x0	R/W
		00	Hibernate2 Mode: ADP and SOC domain are powered off.		
		01	Hibernate1 Mode: SOC domain powered off, ADP domain powered on.		
		10	Hibernate0 Mode: SOC domain powered on, ADP domain powered off.		
		11	Active Mode: SOC and ADP domain are both powered on.		

CLOCK CONTROL REGISTER

Address: 0x4000C010, Reset: 0xC8, Name: CLK_CTRL1

Table 136. Bit Descriptions for CLK_CTRL1

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	SYNC_SOURCE		Source for phase synchronization signal to phase align multiple chips.	0x3	R/W
		00	LRCLK_0 signal used for phase synchronization.		
		01	LRCLK_1 signal used for phase synchronization.		
		10	Input ASRC used for phase synchronization signal. Used when LRCLK signal is async to core clock.		
		11	Phase synchronization signal internally generated.		
5	RESERVED		Reserved.	0x0	R
4	PLL_TYPE		Type of PLL (integer/fractional).	0x0	R/W
		0	Integer PLL.		
		1	Fractional PLL.		
3	XTAL_MODE		MCLK/Crystal Oscillator Mode.	0x1	R/W
		0	Logic level MCLK input used.		
		1	Crystal Oscillator used.		
[2:0]	PLL_SOURCE		PLL source clock selection.	0x0	R/W
		000	MCLKIN pin or XTAL is PLL source.		
		001	FSYNC0 pin is PLL source.		
		010	BCLK0 pin is PLL source.		
		011	FSYNC1 pin is PLL source.		
		100	BCLK1 pin is PLL source.		

PLL INPUT DIVIDER REGISTER

Address: 0x4000C011, Reset: 0x00, Name: CLK_CTRL2

Table 137. Bit Descriptions for CLK_CTRL2

Bits	Bit Name	Settings	Description	Reset	Access
[7:3]	RESERVED		Reserved.	0x0	R
[2:0]	PLL_INPUT_PRESCALER		PLL_INPUT_PRESCALER is the input divider rate. Note: 3'b000 or 3'b001 both means the input divider ratio is 1.	0x0	R/W

PLL FEEDBACK INTEGER DIVIDER (LSBS) REGISTER

Address: 0x4000C012, Reset: 0x04, Name: CLK_CTRL3

Table 138. Bit Descriptions for CLK_CTRL3

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	PLL_INTEGER_DIVIDER[7:0]		Feedback divider rate (integer mode).	0x4	R/W

PLL FEEDBACK INTEGER DIVIDER (MSBS) REGISTER

Address: 0x4000C013, Reset: 0x00, Name: CLK_CTRL4

Table 139. Bit Descriptions for CLK_CTRL4

Bits	Bit Name	Settings	Description	Reset	Access
[7:5]	RESERVED		Reserved.	0x0	R
[4:0]	PLL_INTEGER_DIVIDER[12:8]		Feedback divider rate (integer mode).	0x0	R/W

PLL FRACTIONAL NUMERATOR VALUE (LSBS) REGISTER

Address: 0x4000C014, Reset: 0x00, Name: CLK_CTRL5

Table 140. Bit Descriptions for CLK_CTRL5

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	PLL_NUMERATOR[7:0]		PLL numerator.	0x0	R/W

PLL FRACTIONAL NUMERATOR VALUE (MSBS) REGISTER

Address: 0x4000C015, Reset: 0x00, Name: CLK_CTRL6

Table 141. Bit Descriptions for CLK_CTRL6

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	PLL_NUMERATOR[15:8]		PLL numerator.	0x0	R/W

PLL FRACTIONAL DENOMINATOR (LSBS) REGISTER

Address: 0x4000C016, Reset: 0x00, Name: CLK_CTRL7

Table 142. Bit Descriptions for CLK_CTRL7

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	PLL_DENOMINATOR[7:0]		PLL denominator.	0x0	R/W

PLL FRACTIONAL DENOMINATOR (MSBS) REGISTER

Address: 0x4000C017, Reset: 0x00, Name: CLK_CTRL8

Table 143. Bit Descriptions for CLK_CTRL8

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	PLL_DENOMINATOR[15:8]		PLL denominator.	0x0	R/W

PLL UPDATE REGISTER

Address: 0x4000C018, Reset: 0x00, Name: CLK_CTRL9

Table 144. Bit Descriptions for CLK_CTRL9

Bits	Bit Name	Settings	Description	Reset	Access
[7:1]	RESERVED		Reserved.	0x0	R
0	PLL_UPDATE		Update PLL configuration.	0x0	R/W1T
		0	Writing of zero will do nothing.		
		1	Writing of one will update all PLL configuration settings.		

TDSP CLOCK RATE, BUS(AHB/APB) CLOCK RATE AND FFSRAM2 RATE REGISTER

Address: 0x4000C019, Reset: 0xC0, Name: CLK_CTRL10

Table 145. Bit Descriptions for CLK_CTRL10

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	FFSRAM_RATE		FIFO SRAM Access Rate Control.	0x3	R/W
		0	SRAM access rate is AHB_CLK rate;.		
		1	SRAM access rate is 1/2 AHB_CLK rate;.		
		10	SRAM access rate is 1/3 AHB_CLK rate;.		
		11	SRAM access rate is 1/4 AHB_CLK rate.		
[5:3]	BUS_CLK_RATE		AHB/APB Bus Clock Rate Selection. The ratio of mclk-frequency over AHB/APB-frequency. Please note that: There are requirements that: 1. The TDSP frequency and BUS frequency must keep 1:1, 2:1, 3:1, or 4:1 relationship. 2. If the TDSP and SoC subsystem need to talk to ADP subsystem, the BUS frequency must be at 24.576MHz, 49.152MHz, 73.728MHz, or 98.304MHz.	0x0	R/W
		0	Divided by 1.		
		1	Divided by 2.		
		10	Divided by 3.		
		11	Divided by 4.		
		100	Divided by 6.		
		101	Divided by 8.		
		110	Divided by 12.		
		111	Divided by 16.		
[2:0]	TDSP_CLK_RATE		Tensilica-DSP Clock Rate Selection. The ratio of mclk-frequency over Tensilica-DSP-frequency. Please note that: There are requirements that: 1. The TDSP frequency and BUS frequency must keep 1:1, 2:1, 3:1, or 4:1 relationship. 2. If the TDSP and SoC subsystem need to talk to ADP subsystem, the BUS frequency must be at 24.576MHz, 49.152MHz, 73.728MHz, or 98.304MHz.	0x0	R/W
		0	Divided by 1.		
		1	Divided by 2.		
		10	Divided by 3.		
		11	Divided by 4.		
		100	Divided by 6.		
		101	Divided by 8.		
		110	Divided by 12.		
		111	Divided by 16.		

AON AND UART_CTRL CLOCK RATE REGISTER

Address: 0x4000C01A, Reset: 0x00, Name: CLK_CTRL11

Table 146. Bit Descriptions for CLK_CTRL11

Bits	Bit Name	Settings	Description	Reset	Access
7	RESERVED		Reserved.	0x0	R
[6:4]	UART_CTRL_CLK_RATE		UART (Ctrl_Port) Clock Rate Selection. UART (Ctrl_Port) Clock Rate Selection The ratio of PLL-frequency or Frequency-Multiplier output frequency over UART-Clock There is requirement that the UART-Clock frequency cannot go over than 24.576MHz.	0x0	R/W
		0	Divided by 1.		
		1	Divided by 2.		
		10	Divided by 3.		
		11	Divided by 4.		
		100	Divided by 6.		
		101	Divided by 8.		
		110	Divided by 12.		
		111	Divided by 16.		
[3:0]	RESERVED		Reserved.	0x0	R

FREQUENCY MULTIPLIER ENABLE, RATIO REGISTER

Address: 0x4000C01B, Reset: 0x00, Name: CLK_CTRL12

Table 147. Bit Descriptions for CLK_CTRL12

Bits	Bit Name	Settings	Description	Reset	Access
[7:1]	RESERVED		Reserved.	0x0	R
0	FREQ_MULT_EN		Frequency Multiplier Enable.	0x0	R/W
		0	Disable Frequency Multiplier.		
		1	Enable Frequency Multiplier.		

PLL FREQUENCY INDEX AND ADC FREQUENCY INDEX REGISTER

Address: 0x4000C01C, Reset: 0x13, Name: CLK_CTRL13

Table 148. Bit Descriptions for CLK_CTRL13

Bits	Bit Name	Settings	Description	Reset	Access
7	PLL_FM_BYPASS		PLL or Frequency-Multiplier Bypass Control.	0x0	R/W
		0	PLL/FM Output is source of main chip clock.		
		1	PLL/FM is bypassed. Main chip clock sourced directly from PLL_SOURCE setting and must be 24.576MHz.		
[6:5]	RESERVED		Reserved.	0x0	R
4	ADC_FREQ		ADC and DAC Running Clock Frequency Selection.	0x1	R/W
		0	ADC and DAC runs at 2.048MHz.		
		1	ADC and DAC runs at 6.144MHz.		
[3:2]	RESERVED		Reserved.	0x0	R
[1:0]	MCLK_FREQ_INDEX		MCLK Clock Frequency Index.	0x3	R/W
		0	MCLK Frequency at 24.576MHz.		
		1	MCLK Frequency at 49.152MHz.		
		2	MCLK Frequency at 73.728MHz.		
		3	MCLK Frequency at 98.304MHz.		

ADP_CLOCK_ENABLE, LOW BYTE REGISTER

Address: 0x4000C01D, Reset: 0xFF, Name: CLK_CTRL14

Table 149. Bit Descriptions for CLK_CTRL14

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	ADP_CLOCK_EN[7:0]		ADP Clock Enable. Bit[0]:ADP_CLK_EN, ADP overall clock enable. 1'b0: Disable; 1'b1: Enable. Bit[1]:FDSP_CLK_EN, 1'b0: Disable; 1'b1: Enable. Bit[2]:PDM_CLK_EN, 1'b0: Disable; 1'b1: Enable. Bit[3]:EQ_CLK_EN, 1'b0: Disable; 1'b1: Enable. Bit[4]:PCTRL_CLK_EN,1'b0: Disable; 1'b1: Enable. Bit[5]:ADC_CLK_EN, 1'b0: Disable; 1'b1: Enable. Bit[6]:DMIC_CLK_EN, 1'b0: Disable; 1'b1: Enable. Bit[7]:SPT_CLK_EN, 1'b0: Disable; 1'b1: Enable. Bit[8]:DGEN_CLK_EN, 1'b0: Disable; 1'b1: Enable. Bit[9]:ASRCI_CLK_EN, 1'b0: Disable; 1'b1: Enable. Bit[10]:ASRCO_CLK_EN, 1'b0: Disable; 1'b1: Enable. Bit[11]:FDEC_CLK_EN, 1'b0: Disable; 1'b1: Enable. Bit[12]:FINT_CLK_EN,1'b0: Disable; 1'b1: Enable. Bit[13]:DAC_CLK_EN, 1'b0: Disable; 1'b1: Enable. Bit[14]:DBUF_CLK_EN, 1'b0: Disable; 1'b1: Enable. Bit[15]:DSYNC_CLK_EN, 1'b0: Disable; 1'b1: Enable.	0xFF	R/W

ADP_CLOCK_ENABLE, HIGH BYTE REGISTER

Address: 0x4000C01E, Reset: 0xFF, Name: CLK_CTRL15

Table 150. Bit Descriptions for CLK_CTRL15

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	ADP_CLOCK_EN[15:8]		ADP Clock Enable. Bit[0]:ADP_CLK_EN, ADP overall clock enable. 1'b0: Disable; 1'b1: Enable. Bit[1]:FDSP_CLK_EN, 1'b0: Disable; 1'b1: Enable. Bit[2]:PDM_CLK_EN, 1'b0: Disable; 1'b1: Enable. Bit[3]:EQ_CLK_EN, 1'b0: Disable; 1'b1: Enable. Bit[4]:PCTRL_CLK_EN,1'b0: Disable; 1'b1: Enable. Bit[5]:ADC_CLK_EN, 1'b0: Disable; 1'b1: Enable. Bit[6]:DMIC_CLK_EN, 1'b0: Disable; 1'b1: Enable. Bit[7]:SPT_CLK_EN, 1'b0: Disable; 1'b1: Enable. Bit[8]:DGEN_CLK_EN, 1'b0: Disable; 1'b1: Enable. Bit[9]:ASRCI_CLK_EN, 1'b0: Disable; 1'b1: Enable. Bit[10]:ASRCO_CLK_EN, 1'b0: Disable; 1'b1: Enable. Bit[11]:FDEC_CLK_EN, 1'b0: Disable; 1'b1: Enable. Bit[12]:FINT_CLK_EN,1'b0: Disable; 1'b1: Enable. Bit[13]:DAC_CLK_EN, 1'b0: Disable; 1'b1: Enable. Bit[14]:DBUF_CLK_EN, 1'b0: Disable; 1'b1: Enable. Bit[15]:DSYNC_CLK_EN, 1'b0: Disable; 1'b1: Enable.	0xFF	R/W

ADC SAMPLE RATE CONTROL REGISTER

Address: 0x4000C020, Reset: 0x22, Name: ADC_CTRL1

Table 151. Bit Descriptions for ADC_CTRL1

Bits	Bit Name	Settings	Description	Reset	Access
7	ADC2_DEC_ORDER		ADC Channels 2 Decimation filter order.	0x0	R/W
		0	Lower order decimation filter : Lower delay.		
		1	Higher order decimation filter : Higher delay.		
[6:4]	ADC2_FS		ADC Channels 2 Sample Rate Selection.	0x2	R/W
		000	12kHz Sample Rate.		
		001	24kHz Sample Rate.		
		010	48kHz Sample Rate.		
		011	96kHz Sample Rate.		

Bits	Bit Name	Settings	Description	Reset	Access
		100	192kHz Sample Rate.		
		101	384kHz Sample Rate.		
		110	768kHz Sample Rate.		
3	ADC01_DEC_ORDER		ADC Channels 0 and 1 Decimation filter order.	0x0	R/W
		0	Lower Order Decimation Filter : Lower Delay.		
		1	Lower Order Decimation Filter : Higher Delay.		
[2:0]	ADC01_FS		ADC Channels 0 and 1 Sample Rate Selection.	0x2	R/W
		000	12kHz Sample Rate.		
		001	24kHz Sample Rate.		
		010	48kHz Sample Rate.		
		011	96kHz Sample Rate.		
		100	192kHz Sample Rate.		
		101	384kHz Sample Rate.		
		110	768kHz Sample Rate.		

ADC I_{BIAS} CONTROLS REGISTER

Address: 0x4000C021, Reset: 0x00, Name: ADC_CTRL2

Table 152. Bit Descriptions for ADC_CTRL2

Bits	Bit Name	Settings	Description	Reset	Access
7	RESERVED		Reserved.	0x0	R
[6:4]	ADC1_IBIAS		ADC Channels 1 bias current setting. Higher bias currents result in higher performance.	0x0	R/W
		000	Normal operation (default).		
		010	Enhanced performance.		
		011	Extreme power saving.		
3	RESERVED		Reserved.	0x0	R
[2:0]	ADC0_IBIAS		ADC Channels 0 bias current setting. Higher bias currents result in higher performance.	0x0	R/W
		000	Normal operation (default).		
		010	Enhanced performance.		
		011	Extreme power saving.		

ADC I_{BIAS} CONTROLS REGISTER

Address: 0x4000C022, Reset: 0x00, Name: ADC_CTRL3

Table 153. Bit Descriptions for ADC_CTRL3

Bits	Bit Name	Settings	Description	Reset	Access
[7:3]	RESERVED		Reserved.	0x0	R
[2:0]	ADC2_IBIAS		ADC Channels 2 bias current setting. Higher bias currents result in higher performance.	0x0	R/W
		000	Normal operation (default).		
		010	Enhanced performance.		
		011	Extreme power saving.		

ADC HPF CONTROL REGISTER

Address: 0x4000C023, Reset: 0x00, Name: ADC_CTRL4

Table 154. Bit Descriptions for ADC_CTRL4

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
[5:4]	ADC2_HPF_EN		ADC Channel 2 enable high-pass filter.	0x0	R/W
		00	ADC High Pass Filter Off.		
		01	ADC High Pass Filter, Cut Off Frequency is 1Hz.		
		10	ADC High Pass Filter, Cut Off Frequency is 4Hz.		
		11	ADC High Pass Filter, Cut Off Frequency is 8Hz.		
[3:2]	ADC1_HPF_EN		ADC Channel 1 enable high-pass filter.	0x0	R/W
		00	ADC High Pass Filter Off.		
		01	ADC High Pass Filter, Cut Off Frequency is 1Hz.		
		10	ADC High Pass Filter, Cut Off Frequency is 4Hz.		
		11	ADC High Pass Filter, Cut Off Frequency is 8Hz.		
[1:0]	ADC0_HPF_EN		ADC Channel 0 enable high-pass filter.	0x0	R/W
		00	ADC High Pass Filter Off.		
		01	ADC High Pass Filter, Cut Off Frequency is 1Hz.		
		10	ADC High Pass Filter, Cut Off Frequency is 4Hz.		
		11	ADC High Pass Filter, Cut Off Frequency is 8Hz.		

ADC MUTE AND COMPENSATION CONTROL REGISTER

Address: 0x4000C024, Reset: 0x40, Name: ADC_CTRL5

Table 155. Bit Descriptions for ADC_CTRL5

Bits	Bit Name	Settings	Description	Reset	Access
7	RESERVED		Reserved.	0x0	R
6	ADC_VOL_ZC		ADC Volume Zero Cross Control.	0x1	R/W
		0	Volume change occurs at any time.		
		1	Volume change only occurs at zero crossing.		
5	ADC_VOL_LINK		ADC Volume Link.	0x0	R/W
		0	Each ADC channel uses its respective volume value.		
		1	All ADC channels use Channel 0 volume value.		
4	ADC_HARD_VOL		ADC Hard Volume.	0x0	R/W
		0	Soft Volume Ramping.		
		1	Hard/Immediate Volume Change.		
[3:2]	RESERVED		Reserved.	0x0	R
1	ADC2_FCOMP		ADC Channels 2 Frequency Response Compensation.	0x0	R/W
		0	High frequency response not compensated (lower delay).		
		1	High frequency response compensated (higher delay).		
0	ADC01_FCOMP		ADC Channels 0 and 1 Frequency Response Compensation.	0x0	R/W
		0	High frequency response not compensated (lower delay).		
		1	High frequency response compensated (higher delay).		

ANALOG INPUT PRECHARGE TIME REGISTER

Address: 0x4000C025, Reset: 0x86, Name: ADC_CTRL6

Table 156. Bit Descriptions for ADC_CTRL6

Bits	Bit Name	Settings	Description	Reset	Access
7	ADC_CHOP_EN		enable chopping in the feedback dac.	0x1	R/W
		0	ADC Chop Off.		
		1	ADC Chop On.		
[6:4]	RESERVED		Reserved.	0x0	R
[3:0]	ADC_AIN_CHRG_TIME		Analog Inputs pre-charge time selection. Controls the amount of time the pre-charge circuit is used to charge up the coupling caps. The time used depends on the value of the cap used and the required start-up time of the ADC.	0x6	R/W
		0x0	No Precharge.		
		0x1	5 ms Precharge.		
		0x2	10 ms Precharge.		
		0x3	20 ms Precharge.		
		0x4	30 ms Precharge.		
		0x5	40 ms Precharge.		
		0x6	50 ms Precharge.		
		0x7	60 ms Precharge.		
		0x8	80 ms Precharge.		
		0x9	100 ms Precharge.		
		0xA	125 ms Precharge.		
		0xB	150 ms Precharge.		
		0xC	200 ms Precharge.		
		0xD	250 ms Precharge.		
		0xE	300 ms Precharge.		
		0xF	400 ms Precharge.		

ANALOG INPUT PRECHARGE TIME REGISTER

Address: 0x4000C026, Reset: 0x00, Name: ADC_CTRL7

Table 157. Bit Descriptions for ADC_CTRL7

Bits	Bit Name	Settings	Description	Reset	Access
7	RESERVED		Reserved.	0x0	R
6	AIN2_INPUT_MODE		Analog Input Channel 2 (ADC2+PGA2) operating mode.	0x0	R/W
		0	differential input.		
		1	single ended input.		
5	AIN1_INPUT_MODE		Analog Input Channel 1 (ADC1+PGA1) operating mode.	0x0	R/W
		0	differential input.		
		1	single ended input.		
4	AIN0_INPUT_MODE		Analog Input Channel 0 (ADC0+PGA0) operating mode.	0x0	R/W
		0	differential input.		
		1	single ended input.		
[3:1]	RESERVED		Reserved.	0x0	R
0	ADC_VOC_WKUP		ADC Voice Wakeup Mode. More details in ADC Working Mode	0x0	R/W
		0	Disable Voice Wakeup Mode.		
		1	Enable Voice Wakeup Mode.		

ADC CHANNEL MUTES REGISTER

Address: 0x4000C027, Reset: 0x00, Name: ADC_MUTES

Table 158. Bit Descriptions for ADC_MUTES

Bits	Bit Name	Settings	Description	Reset	Access
[7:3]	RESERVED		Reserved.	0x0	R
2	ADC2_MUTE		ADC Channel 2 Mute Control.	0x0	R/W
		0	ADC Unmuted.		
		1	ADC Muted.		
1	ADC1_MUTE		ADC Channel 1 Mute Control.	0x0	R/W
		0	ADC Unmuted.		
		1	ADC Muted.		
0	ADC0_MUTE		ADC Channel 0 Mute Control.	0x0	R/W
		0	ADC Unmuted.		
		1	ADC Muted.		

ADC CHANNEL 0 VOLUME CONTROL REGISTER

Address: 0x4000C028, Reset: 0x40, Name: ADC0_VOL

Table 159. Bit Descriptions for ADC0_VOL

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	ADC0_VOL		ADC Channel 0 Volume Control.	0x40	R/W
		00000000	+24dB.		
		00000001	+23.625dB.		
		00000010	+23.25dB.		
		00000011	+22.875dB.		
		00000100-00111111	+22.5dB to +0.375dB.		
		01000000	0.		
		01000001-11111101	-0.375dB to -70.875dB.		
		11111110	-71.25dB.		
		11111111	Mute.		

ADC CHANNEL 1 VOLUME CONTROL REGISTER

Address: 0x4000C029, Reset: 0x40, Name: ADC1_VOL

Table 160. Bit Descriptions for ADC1_VOL

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	ADC1_VOL		ADC Channel 1 Volume Control.	0x40	R/W
		00000000	+24dB.		
		00000001	+23.625dB.		
		00000010	+23.25dB.		
		00000011	+22.875dB.		
		00000100-00111111	+22.5dB to +0.375dB.		
		01000000	0.		

Bits	Bit Name	Settings	Description	Reset	Access
		01000001-11111101	-0.375dB to -70.875dB.		
		11111110	-71.25dB.		
		11111111	Mute.		

ADC CHANNEL 2 VOLUME CONTROL REGISTER

Address: 0x4000C02A, Reset: 0x40, Name: ADC2_VOL

Table 161. Bit Descriptions for ADC2_VOL

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	ADC2_VOL		ADC Channel 2 Volume Control.	0x40	R/W
		00000000	+24dB.		
		00000001	+23.625dB.		
		00000010	+23.25dB.		
		00000011	+22.875dB.		
		00000100-00111111	+22.5dB to +0.375dB.		
		01000000	0.		
		01000001-11111101	-0.375dB to -70.875dB.		
		11111110	-71.25dB.		
		11111111	Mute.		

ADC DITHERING LEVEL REGISTER

Address: 0x4000C02B, Reset: 0x02, Name: ADC_DITHER_LEV

Table 162. Bit Descriptions for ADC_DITHER_LEV

Bits	Bit Name	Settings	Description	Reset	Access
[7:3]	RESERVED		Reserved.	0x0	R
[2:0]	ADC_DITHER_LEV		enable dither and set dither offset size.	0x2	R/W
		000	Dither Off.		
		001	1/4 LSB.		
		010	1/2 LSB.		

PGA CHANNEL 0 GAIN CONTROL LSBS, MUTE, BOOST, SLEW REGISTER

Address: 0x4000C030, Reset: 0x00, Name: PGA0_CTRL1

Table 163. Bit Descriptions for PGA0_CTRL1

Bits	Bit Name	Settings	Description	Reset	Access
7	PGA0_SLEW_DIS		PGA Channel 0 Gain Slew Disable.	0x0	R/W
		0	PGA slew enabled.		
		1	PGA slew disabled.		
[6:5]	RESERVED		Reserved.	0x0	R
[4:0]	PGA0_GAIN[4:0]		PGA Channel 0 Gain Control.	0x0	R/W
		00000	0 dB.		
		00001	+0.0234375 dB.		
		00010-11110	+0.0468750 dB to +0.7031250 dB.		
		11111	+0.7265625 dB.		

PGA CHANNEL 0 GAIN CONTROL MSBS REGISTER

Address: 0x4000C031, Reset: 0x00, Name: PGA0_CTRL2

Table 164. Bit Descriptions for PGA0_CTRL2

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
[5:0]	PGA0_GAIN[10:5]		PGA Channel 0 Gain Control.	0x0	R/W
		00000000000	0 dB.		
		00000100000	+0.75 dB.		
		00001000000-01111100000	+1.5dB to + 23.25 dB.		
		10000000000	+24 dB.		

PGA CHANNEL 1 GAIN CONTROL LSBS, MUTE, BOOST, SLEW REGISTER

Address: 0x4000C032, Reset: 0x00, Name: PGA1_CTRL1

Table 165. Bit Descriptions for PGA1_CTRL1

Bits	Bit Name	Settings	Description	Reset	Access
7	PGA1_SLEW_DIS		PGA Channel 1 Gain Slew Disable.	0x0	R/W
		0	PGA slew enabled.		
		1	PGA slew disabled.		
[6:5]	RESERVED		Reserved.	0x0	R
[4:0]	PGA1_GAIN[4:0]		PGA Channel 1 Gain Control.	0x0	R/W
		00000	0 dB.		
		00001	+0.0234375 dB.		
		00010-11110	+0.0468750 dB to +0.7031250 dB.		
		11111	+0.7265625 dB.		

PGA CHANNEL 1 GAIN CONTROL MSBS REGISTER

Address: 0x4000C033, Reset: 0x00, Name: PGA1_CTRL2

Table 166. Bit Descriptions for PGA1_CTRL2

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
[5:0]	PGA1_GAIN[10:5]		PGA Channel 1 Gain Control.	0x0	R/W
		00000000000	0 dB.		
		00000100000	+0.75 dB.		
		00001000000-01111100000	+1.5dB to + 23.25 dB.		
		10000000000	+24 dB.		

PGA CHANNEL 2 GAIN CONTROL LSBS, MUTE, BOOST, SLEW REGISTER

Address: 0x4000C034, Reset: 0x00, Name: PGA2_CTRL1

Table 167. Bit Descriptions for PGA2_CTRL1

Bits	Bit Name	Settings	Description	Reset	Access
7	PGA2_SLEW_DIS		PGA Channel 2 Gain Slew Disable.	0x0	R/W
		0	PGA slew enabled.		
		1	PGA slew disabled.		
[6:5]	RESERVED		Reserved.	0x0	R
[4:0]	PGA2_GAIN[4:0]		PGA Channel 2 Gain Control.	0x0	R/W
		00000	0 dB.		
		00001	+0.0234375 dB.		
		00010-11110	+0.0468750 dB to +0.7031250 dB.		
		11111	+0.7265625 dB.		

PGA CHANNEL 2 GAIN CONTROL MSBS REGISTER

Address: 0x4000C035, Reset: 0x00, Name: PGA2_CTRL2

Table 168. Bit Descriptions for PGA2_CTRL2

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
[5:0]	PGA2_GAIN[10:5]		PGA Channel 2 Gain Control.	0x0	R/W
		00000000000	0 dB.		
		00000100000	+0.75 dB.		
		00001000000-01111100000	+1.5dB to + 23.25 dB.		
		10000000000	+24 dB.		

PGA SLEW RATE AND GAIN LINK REGISTER

Address: 0x4000C036, Reset: 0x00, Name: PGA_CTRL1

Table 169. Bit Descriptions for PGA_CTRL1

Bits	Bit Name	Settings	Description	Reset	Access
[7:5]	RESERVED		Reserved.	0x0	R
4	PGA_GAIN_LINK		PGA Gain Link.	0x0	R/W
		0	Each PGA channel uses its respective gain value.		
		1	All PGA channels use Channel 0 gain value.		
[3:2]	RESERVED		Reserved.	0x0	R
[1:0]	PGA_SLEW_RATE		Controls how fast the PGA is slewed when changing gain.	0x0	R/W
		00	2.2 dB/ms.		
		01	1.1 dB/ms.		
		10	0.5 dB/ms.		

PGA RIN AND POWER MODE REGISTER

Address: 0x4000C037, Reset: 0x00, Name: PGA_CTRL2

Table 170. Bit Descriptions for PGA_CTRL2

Bits	Bit Name	Settings	Description	Reset	Access
7	RESERVED		Reserved.	0x0	R
6	PGA2_POWER_MODE		Select PGA2 Power Mode.	0x0	R/W
		0	PGA Channel 0 is in normal operation mode.		
		1	PGA Channel 0 is in enhanced performance mode.		
5	PGA1_POWER_MODE		Select PGA1 Power Mode.	0x0	R/W
		0	PGA Channel 0 is in normal operation mode.		
		1	PGA Channel 0 is in enhanced performance mode.		
4	PGA0_POWER_MODE		Select PGA0 Power Mode.	0x0	R/W
		0	PGA Channel 0 is in normal operation mode.		
		1	PGA Channel 0 is in enhanced performance mode.		
3	RESERVED		Reserved.	0x0	R
2	PGA2_RIN		Select PGA2 Input Resistance Mode.	0x0	R/W
		0	PGA Channel 2 has high input resistance.		
		1	PGA Channel 2 has low input resistance.		
1	PGA1_RIN		Select PGA1 Input Resistance Mode.	0x0	R/W
		0	PGA Channel 1 has high input resistance.		
		1	PGA Channel 1 has low input resistance.		
0	PGA0_RIN		Select PGA0 Input Resistance Mode.	0x0	R/W
		0	PGA Channel 0 has high input resistance.		
		1	PGA Channel 0 has low input resistance.		

DMIC CLOCK RATE CONTROL REGISTER

Address: 0x4000C040, Reset: 0x03, Name: DMIC_CTRL1

Table 171. Bit Descriptions for DMIC_CTRL1

Bits	Bit Name	Settings	Description	Reset	Access
7	RESERVED		Reserved.	0x0	R
[6:4]	DMIC_CLK1_RATE		Digital Mic Clock 1 Rate.	0x3	R/W
		0	384kHz Clock Rate.		
		1	768kHz Clock Rate.		
		10	1536kHz Clock Rate.		
		11	3072kHz Clock Rate.		
		100	6144kHz Clock Rate.		
3	RESERVED		Reserved.	0x0	R
[2:0]	DMIC_CLK_RATE		Digital Mic Clock 0 Rate.	0x3	R/W
		0	384kHz Clock Rate.		
		1	768kHz Clock Rate.		
		10	1536kHz Clock Rate.		
		11	3072kHz Clock Rate.		
		100	6144kHz Clock Rate.		

DMIC CHANNEL 0 AND CHANNEL 1 RATE, ORDER, MAPPING, AND EDGE CONTROL REGISTER

Address: 0x4000C041, Reset: 0x01, Name: DMIC_CTRL2

Table 172. Bit Descriptions for DMIC_CTRL2

Bits	Bit Name	Settings	Description	Reset	Access
7	DMIC01_DAT_WIRE		Selects DMIC0/1 Share Same Data Wire or Not.	0x0	R/W
		0	0 DMIC0/1 share MP1 pin, DMIC0 rising edge, DMIC1 falling edge.		
		1	1 DMIC0 uses MP1 rising edge, DMIC1 uses MP24 falling edge.		
6	DMIC01_FCOMP		DMIC Channels 0 and 1 Frequency Response Compensation.	0x0	R/W
		0	High frequency response not compensated (lower delay).		
		1	High frequency response compensated (higher delay).		
[5:4]	DMIC01_HPF_EN		DMIC Channels 0 and 1 High-Pass Filter Enable.	0x0	R/W
		0	High Pass Filter Off.		
		1	High Pass Filter, Cut Off Freq. 1Hz.		
		10	High Pass Filter, Cut Off Freq. 4Hz.		
		11	High Pass Filter, Cut Off Freq. 8Hz.		
3	DMIC01_DEC_ORDER		DMIC Channels 0 and 1 Decimation Filter Order.	0x0	R/W
		0	4th order decimation filter.		
		1	5th order decimation filter.		
[2:0]	DMIC01_FS		Digital Microphone Channels 0 and 1 Output Sample Rate.	0x1	R/W
		000	12kHz Sample Rate.		
		001	24kHz Sample Rate.		
		010	48kHz Sample Rate.		
		011	96kHz Sample Rate.		
		100	192kHz Sample Rate.		
		101	384kHz Sample Rate.		
		110	768kHz Sample Rate.		

DMIC CHANNEL 2 AND CHANNEL 3 RATE, ORDER, MAPPING, AND EDGE CONTROL REGISTER

Address: 0x4000C042, Reset: 0x01, Name: DMIC_CTRL3

Table 173. Bit Descriptions for DMIC_CTRL3

Bits	Bit Name	Settings	Description	Reset	Access
7	DMIC23_DAT_WIRE		Selects DMIC2/3 Share Same Data Wire or Not.	0x0	R/W
		0	0 DMIC2/3 share MP2 pin, DMIC2 rising edge, DMIC3 falling edge.		
		1	1 DMIC2 uses MP2 rising edge, DMIC3 uses MP25 falling edge.		
6	DMIC23_FCOMP		DMIC Channels 2 and 3 Frequency Response Compensation.	0x0	R/W
		0	High frequency response not compensated (lower delay).		
		1	High frequency response compensated (higher delay).		
[5:4]	DMIC23_HPF_EN		DMIC Channels 2 and 3 High-Pass Filter Enable.	0x0	R/W
		0	High Pass Filter Off.		
		1	High Pass Filter, Cut Off Freq. 1Hz.		
		10	High Pass Filter, Cut Off Freq. 4Hz.		
		11	High Pass Filter, Cut Off Freq. 8Hz.		
3	DMIC23_DEC_ORDER		DMIC Channels 2 and 3 Decimation Filter Order.	0x0	R/W
		0	4th order decimation filter.		
		1	5th order decimation filter.		
[2:0]	DMIC23_FS		Digital Microphone Channels 2 and 3 Output Sample Rate.	0x1	R/W
		000	12kHz Sample Rate.		
		001	24kHz Sample Rate.		
		010	48kHz Sample Rate.		

Bits	Bit Name	Settings	Description	Reset	Access
		011	96kHz Sample Rate.		
		100	192kHz Sample Rate.		
		101	384kHz Sample Rate.		
		110	768kHz Sample Rate.		

DMIC VOLUME OPTIONS REGISTER

Address: 0x4000C043, Reset: 0x04, Name: DMIC_CTRL4

Table 174. Bit Descriptions for DMIC_CTRL4

Bits	Bit Name	Settings	Description	Reset	Access
[7:3]	RESERVED		Reserved.	0x0	R
2	DMIC_VOL_ZC		Digital Mic Volume Zero Cross Control.	0x1	R/W
		0	Volume change occurs at any time.		
		1	Volume change only occurs at zero crossing.		
1	DMIC_VOL_LINK		Digital Mic Volume Link.	0x0	R/W
		0	Each DMIC channel uses its respective volume value.		
		1	All DMIC channels use Channel 0 volume value.		
0	DMIC_HARD_VOL		Digital Mic Hard Volume.	0x0	R/W
		0	Soft Volume Ramping.		
		1	Hard/Immediate Volume Change.		

DMIC CHANNEL MUTE CONTROLS REGISTER

Address: 0x4000C044, Reset: 0x00, Name: DMIC_MUTES

Table 175. Bit Descriptions for DMIC_MUTES

Bits	Bit Name	Settings	Description	Reset	Access
7	DMIC7_MUTE		DMIC Channel 7 Mute Control.	0x0	R/W
		0	DMIC Unmuted.		
		1	DMIC Muted.		
6	DMIC6_MUTE		DMIC Channel 6 Mute Control.	0x0	R/W
		0	DMIC Unmuted.		
		1	DMIC Muted.		
5	DMIC5_MUTE		DMIC Channel 5 Mute Control.	0x0	R/W
		0	DMIC Unmuted.		
		1	DMIC Muted.		
4	DMIC4_MUTE		DMIC Channel 4 Mute Control.	0x0	R/W
		0	DMIC Unmuted.		
		1	DMIC Muted.		
3	DMIC3_MUTE		DMIC Channel 3 Mute Control.	0x0	R/W
		0	DMIC Unmuted.		
		1	DMIC Muted.		
2	DMIC2_MUTE		DMIC Channel 2 Mute Control.	0x0	R/W
		0	DMIC Unmuted.		
		1	DMIC Muted.		

Bits	Bit Name	Settings	Description	Reset	Access
1	DMIC1_MUTE		DMIC Channel 1 Mute Control.	0x0	R/W
		0	DMIC Unmuted.		
		1	DMIC Muted.		
0	DMIC0_MUTE		DMIC Channel 0 Mute Control.	0x0	R/W
		0	DMIC Unmuted.		
		1	DMIC Muted.		

DMIC CHANNEL 0 VOLUME CONTROL REGISTER

Address: 0x4000C045, Reset: 0x40, Name: DMIC_VOL0

Table 176. Bit Descriptions for DMIC_VOL0

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	DMIC0_VOL		Digital Mic Channel 0 Volume Control.	0x40	R/W
		00000000	+24dB.		
		00000001	+23.625dB.		
		00000010	+23.35dB.		
		00000011	+22.875dB.		
		00000100-00111111	+22.5dB to +0.375dB.		
		01000000	0.		
		01000001-11111101	-0.375dB to -70.875dB.		
		11111110	-71.25dB.		
		11111111	Mute.		

DMIC CHANNEL 1 VOLUME CONTROL REGISTER

Address: 0x4000C046, Reset: 0x40, Name: DMIC_VOL1

Table 177. Bit Descriptions for DMIC_VOL1

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	DMIC1_VOL		Digital Mic Channel 1 Volume Control.	0x40	R/W
		00000000	+24dB.		
		00000001	+23.625dB.		
		00000010	+23.35dB.		
		00000011	+22.875dB.		
		00000100-00111111	+22.5dB to +0.375dB.		
		01000000	0.		
		01000001-11111101	-0.375dB to -70.875dB.		
		11111110	-71.25dB.		
		11111111	Mute.		

DMIC CHANNEL 2 VOLUME CONTROL REGISTER

Address: 0x4000C047, Reset: 0x40, Name: DMIC_VOL2

Table 178. Bit Descriptions for DMIC_VOL2

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	DMIC2_VOL		Digital Mic Channel 2 Volume Control.	0x40	R/W
		00000000	+24dB.		
		00000001	+23.625dB.		
		00000010	+23.35dB.		
		00000011	+22.875dB.		
		00000100-00111111	+22.5dB to +0.375dB.		
		01000000	0.		
		01000001-11111101	-0.375dB to -70.875dB.		
		11111110	-71.25dB.		
		11111111	Mute.		

DMIC CHANNEL 3 VOLUME CONTROL REGISTER

Address: 0x4000C048, Reset: 0x40, Name: DMIC_VOL3

Table 179. Bit Descriptions for DMIC_VOL3

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	DMIC3_VOL		Digital Mic Channel 3 Volume Control.	0x40	R/W
		00000000	+24dB.		
		00000001	+23.625dB.		
		00000010	+23.35dB.		
		00000011	+22.875dB.		
		00000100-00111111	+22.5dB to +0.375dB.		
		01000000	0.		
		01000001-11111101	-0.375dB to -70.875dB.		
		11111110	-71.25dB.		
		11111111	Mute.		

DMIC CHANNEL 4 AND CHANNEL 5 RATE, ORDER, MAPPING, AND EDGE CONTROL REGISTER

Address: 0x4000C049, Reset: 0x01, Name: DMIC_CTRL5

Table 180. Bit Descriptions for DMIC_CTRL5

Bits	Bit Name	Settings	Description	Reset	Access
7	DMIC45_EDGE		Selects clock edge for channels 4 and 5.	0x0	R/W
		0	0 rising edge, 1 falling edge.		
		1	1 rising edge, 0 falling edge.		
6	DMIC45_FCOMP		DMIC Channels 4 and 5 Frequency Response Compensation.	0x0	R/W
		0	High frequency response not compensated (lower delay).		
		1	High frequency response compensated (higher delay).		
5	DMIC45_DEC_ORDER		DMIC Channels 4 and 5 Decimation Filter Order.	0x0	R/W
		0	4th order decimation filter.		
		1	5th order decimation filter.		

Bits	Bit Name	Settings	Description	Reset	Access
[4:3]	DMIC45_HPF_EN		DMIC Channels 4 and 5 High-Pass Filter Enable.	0x0	R/W
		0	High Pass Filter Off.		
		1	High Pass Filter, Cut Off Freq. 1Hz.		
		10	High Pass Filter, Cut Off Freq. 4Hz.		
		11	High Pass Filter, Cut Off Freq. 8Hz.		
[2:0]	DMIC45_FS		Digital Microphone Channels 4 and 5 Output Sample Rate.	0x1	R/W
		000	12kHz Sample Rate.		
		001	24kHz Sample Rate.		
		010	48kHz Sample Rate.		
		011	96kHz Sample Rate.		
		100	192kHz Sample Rate.		
		101	384kHz Sample Rate.		
		110	768kHz Sample Rate.		

DMIC CHANNEL 6 AND CHANNEL 7 RATE, ORDER, MAPPING, AND EDGE CONTROL REGISTER

Address: 0x4000C04A, Reset: 0x01, Name: DMIC_CTRL6

Table 181. Bit Descriptions for DMIC_CTRL6

Bits	Bit Name	Settings	Description	Reset	Access
7	DMIC67_EDGE		Selects clock edge for channels 6 and 7.	0x0	R/W
		0	0 rising edge, 1 falling edge.		
		1	1 rising edge, 0 falling edge.		
6	DMIC67_FCOMP		DMIC Channels 6 and 7 Frequency Response Compensation.	0x0	R/W
		0	High frequency response not compensated (lower delay).		
		1	High frequency response compensated (higher delay).		
5	DMIC67_DEC_ORDER		DMIC Channels 6 and 7 Decimation Filter Order.	0x0	R/W
		0	4th order decimation filter.		
		1	5th order decimation filter.		
[4:3]	DMIC67_HPF_EN		DMIC Channels 6 and 7 High-Pass Filter Enable.	0x0	R/W
		0	High Pass Filter Off.		
		1	High Pass Filter, Cut Off Freq. 1Hz.		
		10	High Pass Filter, Cut Off Freq. 4Hz.		
		11	High Pass Filter, Cut Off Freq. 8Hz.		
[2:0]	DMIC67_FS		Digital Microphone Channels 6 and 7 Output Sample Rate.	0x1	R/W
		000	12kHz Sample Rate.		
		001	24kHz Sample Rate.		
		010	48kHz Sample Rate.		
		011	96kHz Sample Rate.		
		100	192kHz Sample Rate.		
		101	384kHz Sample Rate.		
		110	768kHz Sample Rate.		

DMIC CLOCK MAP, DMIC CLOCK 1 SOURCE PIN SELECT REGISTER

Address: 0x4000C04B, Reset: 0x00, Name: DMIC_CTRL7

Table 182. Bit Descriptions for DMIC_CTRL7

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
[5:4]	DMIC_CLK1_PIN_SRC		DMIC Clock 1 Pin Source Select.	0x0	R/W
		00	DMIC Clock 1 Comes From MP7 Pin.		
		01	DMIC Clock 1 Comes From MP11 Pin.		
		10	DMIC Clock 1 Comes From MP17 Pin.		
		11	DMIC Clock 1 Comes From MP22 Pin.		
3	DMIC67_MAP		DMIC Channels 6 and 7 Clock Mapping.	0x0	R/W
		0	Digital Mic channels use DMIC_CLK0.		
		1	Digital Mic channels use DMIC_CLK1.		
2	DMIC45_MAP		DMIC Channels 4 and 5 Clock Mapping.	0x0	R/W
		0	Digital Mic channels use DMIC_CLK0.		
		1	Digital Mic channels use DMIC_CLK1.		
1	DMIC23_MAP		DMIC Channels 2 and 3 Clock Mapping.	0x0	R/W
		0	Digital Mic channels use DMIC_CLK0.		
		1	Digital Mic channels use DMIC_CLK1.		
0	DMIC01_MAP		DMIC Channels 0 and 1 Clock Mapping.	0x0	R/W
		0	Digital Mic channels use DMIC_CLK0.		
		1	Digital Mic channels use DMIC_CLK1.		

DMIC CHANNEL 4 VOLUME CONTROL REGISTER

Address: 0x4000C04C, Reset: 0x40, Name: DMIC_VOL4

Table 183. Bit Descriptions for DMIC_VOL4

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	DMIC4_VOL		Digital Mic Channel 4 Volume Control.	0x40	R/W
		00000000	+24dB.		
		00000001	+23.625dB.		
		00000010	+23.35dB.		
		00000011	+22.875dB.		
		00000100-00111111	+22.5dB to +0.375dB.		
		01000000	0.		
		01000001-11111101	-0.375dB to -70.875dB.		
		11111110	-71.25dB.		
		11111111	Mute.		

DMIC CHANNEL 5 VOLUME CONTROL REGISTER

Address: 0x4000C04D, Reset: 0x40, Name: DMIC_VOL5

Table 184. Bit Descriptions for DMIC_VOL5

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	DMIC5_VOL		Digital Mic Channel 5 Volume Control.	0x40	R/W
		00000000	+24dB.		
		00000001	+23.625dB.		
		00000010	+23.35dB.		
		00000011	+22.875dB.		
		00000100-00111111	+22.5dB to +0.375dB.		
		01000000	0.		
		01000001-11111101	-0.375dB to -70.875dB.		
		11111110	-71.25dB.		
		11111111	Mute.		

DMIC CHANNEL 6 VOLUME CONTROL REGISTER

Address: 0x4000C04E, Reset: 0x40, Name: DMIC_VOL6

Table 185. Bit Descriptions for DMIC_VOL6

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	DMIC6_VOL		Digital Mic Channel 6 Volume Control.	0x40	R/W
		00000000	+24dB.		
		00000001	+23.625dB.		
		00000010	+23.35dB.		
		00000011	+22.875dB.		
		00000100-00111111	+22.5dB to +0.375dB.		
		01000000	0.		
		01000001-11111101	-0.375dB to -70.875dB.		
		11111110	-71.25dB.		
		11111111	Mute.		

DMIC CHANNEL 7 VOLUME CONTROL REGISTER

Address: 0x4000C04F, Reset: 0x40, Name: DMIC_VOL7

Table 186. Bit Descriptions for DMIC_VOL7

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	DMIC7_VOL		Digital Mic Channel 7 Volume Control.	0x40	R/W
		00000000	+24dB.		
		00000001	+23.625dB.		
		00000010	+23.35dB.		
		00000011	+22.875dB.		
		00000100-00111111	+22.5dB to +0.375dB.		
		01000000	0.		

Bits	Bit Name	Settings	Description	Reset	Access
		01000001-11111101	-0.375dB to -70.875dB.		
		11111110	-71.25dB.		
		11111111	Mute.		

DAC SAMPLE RATE, FILTERING, AND POWER CONTROLS REGISTER

Address: 0x4000C050, Reset: 0x02, Name: DAC_CTRL1

Table 187. Bit Descriptions for DAC_CTRL1

Bits	Bit Name	Settings	Description	Reset	Access
7	DAC_MORE_FILT		DAC Additional Interpolation Filtering Selection.	0x0	R/W
		0	Less interpolation filtering : Lower delay.		
		1	More interpolation filtering : Higher delay.		
6	DAC_LPM		DAC Low Power Mode Enable.	0x0	R/W
		0	DAC Low Power Mode Off (6.144MHz).		
		1	DAC Low Power Mode On (3.072MHz).		
[5:4]	RESERVED		Reserved.	0x0	R/W
3	DAC_FCOMP		DAC Frequency Response Compensation.	0x0	R/W
		0	High frequency response not compensated (lower delay).		
		1	High frequency response compensated for samples rates of 192kHz or lower when DAC_MORE_FILT=1. (higher delay).		
[2:0]	DAC_FS		DAC Path Sample Rate Selection.	0x2	R/W
		000	12kHz Sample Rate.		
		001	24kHz Sample Rate.		
		010	48kHz Sample Rate.		
		011	96kHz Sample Rate.		
		100	192kHz Sample Rate.		
		101	384kHz Sample Rate.		
		110	768kHz Sample Rate.		

DAC VOLUME LUNK, HPF, AND MUTE CONTROLS REGISTER

Address: 0x4000C051, Reset: 0x44, Name: DAC_CTRL2

Table 188. Bit Descriptions for DAC_CTRL2

Bits	Bit Name	Settings	Description	Reset	Access
7	DAC0_FORCE_MUTE		DAC Channel Force Mute Control.	0x0	R/W
		0	DAC Unmuted.		
		1	DAC Muted.		
6	DAC0_MUTE		DAC Channel 0 Mute Control.	0x1	R/W
		0	DAC Unmuted.		
		1	DAC Muted.		
5	RESERVED		Reserved.	0x0	R
4	DAC0_HPF_EN		DAC Channel 0 enable high-pass filter.	0x0	R/W
		0	ADC High Pass Filter Off.		
		1	ADC High Pass Filter On.		
3	DAC_LPM_II		DAC Low Power Mode 2 Enable.	0x0	R/W
		0	DAC Low Power Mode 2 Off.		
		1	DAC Low Power Mode 2 On. Reduced output activity.		

Bits	Bit Name	Settings	Description	Reset	Access
2	DAC_VOL_ZC		DAC Volume Zero Cross Control.	0x1	R/W
		0	Volume change occurs at any time.		
		1	Volume change only occurs at zero crossing.		
1	DAC_HARD_VOL		DAC Hard Volume.	0x0	R/W
		0	Soft Volume Ramping.		
		1	Hard/Immediate Volume Change.		
0	RESERVED		Reserved.	0x0	R

DAC CHANNEL 0 VOLUME REGISTER

Address: 0x4000C052, Reset: 0x40, Name: DAC_VOL0

Table 189. Bit Descriptions for DAC_VOL0

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	DAC0_VOL		DAC Channel 0 Volume Control.	0x40	R/W
		00000000	+24dB.		
		00000001	+23.625dB.		
		00000010	+23.35dB.		
		00000011	+22.875dB.		
		00000100-00111111	+22.5dB to +0.375dB.		
		01000000	0.		
		01000001-11111101	-0.375dB to -70.875dB.		
		11111110	-71.25dB.		
		11111111	Mute.		

DAC CHANNEL 0 ROUTING REGISTER

Address: 0x4000C053, Reset: 0x00, Name: DAC_ROUTE0

Table 190. Bit Descriptions for DAC_ROUTE0

Bits	Bit Name	Settings	Description	Reset	Access
7	RESERVED		Reserved.	0x0	R
[6:0]	DAC0_ROUTE		DAC Channel 0 Input Routing.	0x0	R/W
		00	Serial Port 0 Channel 0.		
		01	Serial Port 0 Channel 1.		
		02	Serial Port 0 Channel 2.		
		03	Serial Port 0 Channel 3.		
		04	Serial Port 0 Channel 4.		
		05	Serial Port 0 Channel 5.		
		06	Serial Port 0 Channel 6.		
		07	Serial Port 0 Channel 7.		
		08	Serial Port 0 Channel 8.		
		09	Serial Port 0 Channel 9.		
		10	Serial Port 0 Channel 10.		
		11	Serial Port 0 Channel 11.		
		12	Serial Port 0 Channel 12.		
		13	Serial Port 0 Channel 13.		
		14	Serial Port 0 Channel 14.		
		15	Serial Port 0 Channel 15.		
		16	Serial Port 1 Channel 0.		
		17	Serial Port 1 Channel 1.		

Bits	Bit Name	Settings	Description	Reset	Access
		18	Serial Port 1 Channel 2.		
		19	Serial Port 1 Channel 3.		
		20	Serial Port 1 Channel 4.		
		21	Serial Port 1 Channel 5.		
		22	Serial Port 1 Channel 6.		
		23	Serial Port 1 Channel 7.		
		24	Serial Port 1 Channel 8.		
		25	Serial Port 1 Channel 9.		
		26	Serial Port 1 Channel 10.		
		27	Serial Port 1 Channel 11.		
		28	Serial Port 1 Channel 12.		
		29	Serial Port 1 Channel 13.		
		30	Serial Port 1 Channel 14.		
		31	Serial Port 1 Channel 15.		
		32	FastDSP Channel 0.		
		33	FastDSP Channel 1.		
		34	FastDSP Channel 2.		
		35	FastDSP Channel 3.		
		36	FastDSP Channel 4.		
		37	FastDSP Channel 5.		
		38	FastDSP Channel 6.		
		39	FastDSP Channel 7.		
		40	FastDSP Channel 8.		
		41	FastDSP Channel 9.		
		42	FastDSP Channel 10.		
		43	FastDSP Channel 11.		
		44	FastDSP Channel 12.		
		45	FastDSP Channel 13.		
		46	FastDSP Channel 14.		
		47	FastDSP Channel 15.		
		48	Tensilica DSP Channel 0.		
		49	Tensilica DSP Channel 1.		
		50	Tensilica DSP Channel 2.		
		51	Tensilica DSP Channel 3.		
		52	Tensilica DSP Channel 4.		
		53	Tensilica DSP Channel 5.		
		54	Tensilica DSP Channel 6.		
		55	Tensilica DSP Channel 7.		
		56	Tensilica DSP Channel 8.		
		57	Tensilica DSP Channel 9.		
		58	Tensilica DSP Channel 10.		
		59	Tensilica DSP Channel 11.		
		60	Tensilica DSP Channel 12.		
		61	Tensilica DSP Channel 13.		
		62	Tensilica DSP Channel 14.		
		63	Tensilica DSP Channel 15.		
		64	Input ASRC Channel 0.		
		65	Input ASRC Channel 1.		
		66	Input ASRC Channel 2.		
		67	Input ASRC Channel 3.		
		68	ADC Channel 0.		

Bits	Bit Name	Settings	Description	Reset	Access
		69	ADC Channel 1.		
		70	ADC Channel 2.		
		71	DMIC Channel 0.		
		72	DMIC Channel 1.		
		73	DMIC Channel 2.		
		74	DMIC Channel 3.		
		75	EQ0.		
		76	FINT Channel 0.		
		77	FINT Channel 1.		
		78	FINT Channel 2.		
		79	FINT Channel 3.		
		80	FINT Channel 4.		
		81	FINT Channel 5.		
		82	FINT Channel 6.		
		83	FINT Channel 7.		
		87	DMIC Channel 4.		
		88	DMIC Channel 5.		
		89	DMIC Channel 6.		
		90	DMIC Channel 7.		

HEADPHONE CONTROL REGISTER

Address: 0x4000C060, Reset: 0x00, Name: HP_CTRL

Table 191. Bit Descriptions for HP_CTRL

Bits	Bit Name	Settings	Description	Reset	Access
[7:1]	RESERVED		Reserved.	0x0	R
0	HP0_MODE		Headphone Channel 0 Output Mode.	0x0	R/W
		0	HPOUTP0/HPOUTN0 in headphone mode.		
		1	HPOUTP0/HPOUTN0 in line-out mode.		

HP LOW VOLTAGE MODE ENABLE, CM ENABLE REGISTER

Address: 0x4000C061, Reset: 0x00, Name: HP_LVMODE_CTRL1

Table 192. Bit Descriptions for HP_LVMODE_CTRL1

Bits	Bit Name	Settings	Description	Reset	Access
[7:2]	RESERVED		Reserved.	0x0	R
1	HP_LVMODE_CM_EN		Setting the Common Mode Voltage of the Headphone Driver.	0x0	R/W
		0	Common Mode Voltage is set to 0.85V.		
		1	Common Mode Voltage is set to HPVDDL/2.		
0	HP_LVMODE_EN		Enable the Low Voltage Mode of Headphone Driver. The Bit shall be used together with HP_LVMODE_SWITCH bit with a certain configuration sequence. Please check the OUTPUT SIGNAL PATHS chapter in datasheet for the details.	0x0	R/W
		0	Auxiliary circuit for LVM disabled.		
		1	Auxiliary circuit for LVM enabled.		

HP LOW VOLTAGE AUTO SWITCH MODE, DELAY, CM DELAY REGISTER

Address: 0x4000C062, Reset: 0x30, Name: HP_LVMODE_CTRL2

Table 193. Bit Descriptions for HP_LVMODE_CTRL2

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
[5:4]	HP_LVMODE_CM_DLY		CM voltage ready wait time.	0x3	R/W
		0	1.25ms.		
		1	2.5ms.		
		10	5ms.		
		11	10ms.		
[3:1]	HP_LVMODE_AUTO_DLY		Switch to low voltage HP mode, auto counting mode delay timer. Define the time between output stage switched and unmute in auto transition mode	0x0	R/W
		0	10us.		
		1	20us.		
		10	40us.		
		11	80us.		
		100	160us.		
		101	320us.		
		110	640us.		
		111	1.28ms.		
0	HP_LVMODE_AUTOSW_MODE		When set to 1'b1, it means when switching between NVM and LVM, the playback path will be muted first and then HP supply switches.	0x0	R/W
		0	Switch between NVM and LVM is controlled by user manually. User will manually control the mute and unmute procedure.		
		1	Switch between NVM and LVM is handled by internal FSM. When HP_LVMODE_HOTSW_MODE=0, playback path will be muted and then the HP power supply switches. When HP_LVMODE_HOTSW_MODE=1,the HP power supply switches between NVM and LVM when the output signal is crossing zero.		

HP LOW VOLTAGE AUTO SWITCH (GO) REGISTER

Address: 0x4000C063, Reset: 0x00, Name: HP_LVMODE_CTRL3

Table 194. Bit Descriptions for HP_LVMODE_CTRL3

Bits	Bit Name	Settings	Description	Reset	Access
[7:1]	RESERVED		Reserved.	0x0	R
0	HP_LVMODE_SWITCH		Control the HPAMP working under LVM or NVM. The Bit shall be used together with HP_LVMODE_EN bit with a certain configuration sequence. Please check the OUTPUT SIGNAL PATHS chapter in datasheet for the details.	0x0	R/W
		0	Normal Voltage Mode.		
		1	Low Voltage Mode.		

HPLDO CTRL REGISTER

Address: 0x4000C066, Reset: 0x00, Name: HPLDO_CTRL

Table 195. Bit Descriptions for HPLDO_CTRL

Bits	Bit Name	Settings	Description	Reset	Access
[7:4]	RESERVED		Reserved.	0x0	R
[3:2]	HPLDO_VOUT_SET		HPLDO VOUT Value.	0x0	R/W
		0x0	1.1V.		
		0x1	1.15V.		
		0x2	1.2V.		
		0x3	1.05V.		
1	HPLDO_BYPASS		HPLDO Bypass.	0x0	R/W
		0x0	HPLDO NO BYPASS.		
		0x1	HPLDO BYPASS.		
0	HPLDO_EN		HPLDO Enable.	0x0	R/W
		0x0	HPLDO DISABLE.		
		0x1	HPLDO ENABLE.		

ANALOG BACKEND POWER CONTROL

Address: 0x4000C06C, Reset: 0x00, Name: PB_CTRL

Table 196. Bit Descriptions for PB_CTRL

Bits	Bit Name	Settings	Description	Reset	Access
[7:2]	RESERVED		Reserved.	0x0	R
[1:0]	PB_POWER_CTRL		Analog Backend Power Ctrl. Please set the DAC_LOW_NOISE_EN1 bit and DAC_LOW_NOISE_EN2 Bit for Better Performance when configured as ENHANCE_MODE.	0x0	R/W
		00	Normal Mode.		
		10	Enhance Mode.		

MEMORY POWER CONTROL REGISTER

Address: 0x4000C070, Reset: 0x00, Name: PMU_CTRL1

Table 197. Bit Descriptions for PMU_CTRL1

Bits	Bit Name	Settings	Description	Reset	Access
[7:2]	RESERVED		Reserved.	0x0	R
[1:0]	MEM_VOLT_MODE		Memory voltage mode.	0x0	R/W
		0	LP Mode: If digital power supply is 0.85V or 0.9V, memory voltage mode should be set to LP Mode.		
		1	MP Mode: If digital power supply is 0.95V or 1.0V, memory voltage mode should be set to MP Mode.		
		10	HP Mode: If digital power supply is 1.05V or 1.1V, memory voltage mode should be set to HP Mode. 2'b11 also means HP mode.		

CM CONTROL REGISTER

Address: 0x4000C071, Reset: 0x00, Name: PMU_CTRL2

Table 198. Bit Descriptions for PMU_CTRL2

Bits	Bit Name	Settings	Description	Reset	Access
[7:5]	RESERVED		Reserved.	0x0	R
4	TDSP_WAIT_MODE_BYP		HiFi3z can enter sleep (wait) mode by WAITI instruction, this gate off the majority of the clocks in the processor. If TDSP_WAIT_MODE_BYP=0, When bit[1] of register PWR_MODE is 1'b0, SOC domain will be powered off when WAITI instruction is executed and Pwait Mode is asserted. If TDSP_WAIT_MODE_BYP=1, SOC domain will be powered off directly when bit[1] of register PWR_MODE is 1'b0.	0x0	R/W
		0	When configuring PWR_MODE[1]=1'b0, SOC domain will be powered off after WAITI instruction is executed.		
		1	When configuring PWR_MODE[1]=1'b0, SOC domain will be powered off directly.		
[3:1]	RESERVED		Reserved.	0x0	R
0	STARTUP_DLYCNT_BYP		Start up Delay Counter Bypass.	0x0	R/W
		0	Do Not Bypass StartUp Delay Counter.		
		1	Bypass StartUp Delay Counter.		

IRQ WAKEUP CONTROL REGISTER

Address: 0x4000C072, Reset: 0x00, Name: PMU_CTRL3

Table 199. Bit Descriptions for PMU_CTRL3

Bits	Bit Name	Settings	Description	Reset	Access
7	RESERVED		Reserved.	0x0	R
[6:0]	IRQ_WKUP_EN		IRQ Wakeup Enable for PMU. Bit[0]: Enable IRQ1 as PMU Wakeup source. Bit[1]: Enable IRQ2 as PMU Wakeup source. Bit[2]: Enable IRQ3 as PMU Wakeup source. Bit[3]: Enable IRQ4 as PMU Wakeup source. Bit[4]: Enable Multipurposed Input IRQ1 as PMU Wakeup source. Bit[5]: Enable Multipurposed Input IRQ2 as PMU Wakeup source. Bit[6]: Enable Multipurposed Input IRQ3 as PMU Wakeup source.	0x0	R/W

IRQ WAKEUP CONTROL REGISTER

Address: 0x4000C073, Reset: 0x00, Name: PMU_CTRL4

Table 200. Bit Descriptions for PMU_CTRL4

Bits	Bit Name	Settings	Description	Reset	Access
7	DLY_UNIT_SEL		Wake up delay time unit.	0x0	R/W
		0	Delay Unit is 5us.		
		1	Delay Unit is 667us.		
[6:0]	WAKEUP_DLY		Wait time after gated domain is brought up before clocks enabled, unit is decided by DLY_UNIT_SEL.	0x0	R/W

DLDO CONTROL REGISTER

Address: 0x4000C074, Reset: 0x81, Name: PMU_CTRL5

Table 201. Bit Descriptions for PMU_CTRL5

Bits	Bit Name	Settings	Description	Reset	Access
[7:3]	RESERVED		Reserved.	0x10	R/W
[2:0]	DLDO_CTRL		DLDO Output Voltage.	0x1	R/W
		000	Digital LDO output is 0.85V.		
		001	Digital LDO output is 0.9V.		
		010	Digital LDO output is 0.95V.		
		011	Digital LDO output is 1.0V.		
		100	Digital LDO output is 1.05V.		
		101	Digital LDO output is 1.1V.		

FAST TO SLOW DECIMATOR SAMPLE RATES CHANNEL 0 AND CHANNEL 1 REGISTER

Address: 0x4000C080, Reset: 0x25, Name: FDEC_CTRL1

Table 202. Bit Descriptions for FDEC_CTRL1

Bits	Bit Name	Settings	Description	Reset	Access
[7:4]	FDEC01_OUT_FS		Decimator Channels 0/1 Output Sampling Rate.	0x2	R/W
		0000	12kHz Sample Rate.		
		0001	24kHz Sample Rate.		
		0010	48kHz Sample Rate.		
		0011	96kHz Sample Rate.		
		0100	192kHz Sample Rate.		
		0101	384kHz Sample Rate.		
		1000	8kHz Sample Rate.		
		1001	16kHz Sample Rate.		
3	RESERVED		Reserved.	0x0	R
[2:0]	FDEC01_IN_FS		Decimator Channels 0/1 Input Sampling Rate.	0x5	R/W
		001	24kHz Sample Rate.		
		010	48kHz Sample Rate.		
		011	96kHz Sample Rate.		
		100	192kHz Sample Rate.		
		101	384kHz Sample Rate.		
		110	768kHz Sample Rate.		

FAST TO SLOW DECIMATOR SAMPLE RATES CHANNEL 2 AND CHANNEL 3 REGISTER

Address: 0x4000C081, Reset: 0x25, Name: FDEC_CTRL2

Table 203. Bit Descriptions for FDEC_CTRL2

Bits	Bit Name	Settings	Description	Reset	Access
[7:4]	FDEC23_OUT_FS		Decimator Channels 2/3 Output Sampling Rate.	0x2	R/W
		0000	12kHz Sample Rate.		
		0001	24kHz Sample Rate.		
		0010	48kHz Sample Rate.		
		0011	96kHz Sample Rate.		
		0100	192kHz Sample Rate.		

Bits	Bit Name	Settings	Description	Reset	Access
		0101	384kHz Sample Rate.		
		1000	8kHz Sample Rate.		
		1001	16kHz Sample Rate.		
3	RESERVED		Reserved.	0x0	R
[2:0]	FDEC23_IN_FS		Decimator Channels 2/3 Input Sampling Rate.	0x5	R/W
		001	24kHz Sample Rate.		
		010	48kHz Sample Rate.		
		011	96kHz Sample Rate.		
		100	192kHz Sample Rate.		
		101	384kHz Sample Rate.		
		110	768kHz Sample Rate.		

FAST TO SLOW DECIMATOR SAMPLE RATES CHANNEL 4 AND CHANNEL 5 REGISTER

Address: 0x4000C082, Reset: 0x25, Name: FDEC_CTRL3

Table 204. Bit Descriptions for FDEC_CTRL3

Bits	Bit Name	Settings	Description	Reset	Access
[7:4]	FDEC45_OUT_FS		Decimator Channels 4/5 Output Sampling Rate.	0x2	R/W
		0000	12kHz Sample Rate.		
		0001	24kHz Sample Rate.		
		0010	48kHz Sample Rate.		
		0011	96kHz Sample Rate.		
		0100	192kHz Sample Rate.		
		0101	384kHz Sample Rate.		
		1000	8kHz Sample Rate.		
		1001	16kHz Sample Rate.		
3	RESERVED		Reserved.	0x0	R
[2:0]	FDEC45_IN_FS		Decimator Channels 4/5 Input Sampling Rate.	0x5	R/W
		001	24kHz Sample Rate.		
		010	48kHz Sample Rate.		
		011	96kHz Sample Rate.		
		100	192kHz Sample Rate.		
		101	384kHz Sample Rate.		
		110	768kHz Sample Rate.		

FAST TO SLOW DECIMATOR SAMPLE RATES CHANNEL 6 AND CHANNEL 7 REGISTER

Address: 0x4000C083, Reset: 0x25, Name: FDEC_CTRL4

Table 205. Bit Descriptions for FDEC_CTRL4

Bits	Bit Name	Settings	Description	Reset	Access
[7:4]	FDEC67_OUT_FS		Decimator Channels 6/7 Output Sampling Rate.	0x2	R/W
		0000	12kHz Sample Rate.		
		0001	24kHz Sample Rate.		
		0010	48kHz Sample Rate.		
		0011	96kHz Sample Rate.		
		0100	192kHz Sample Rate.		
		0101	384kHz Sample Rate.		
		1000	8kHz Sample Rate.		
		1001	16kHz Sample Rate.		

Bits	Bit Name	Settings	Description	Reset	Access
3	RESERVED		Reserved.	0x0	R
[2:0]	FDEC67_IN_FS		Decimator Channels 6/7 Input Sampling Rate.	0x5	R/W
		001	24kHz Sample Rate.		
		010	48kHz Sample Rate.		
		011	96kHz Sample Rate.		
		100	192kHz Sample Rate.		
		101	384kHz Sample Rate.		
		110	768kHz Sample Rate.		

FAST TO SLOW DECIMATOR CHANNEL 0 INPUT ROUTING REGISTER

Address: 0x4000C084, Reset: 0x00, Name: FDEC_ROUTE0

Table 206. Bit Descriptions for FDEC_ROUTE0

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
[5:0]	FDEC0_ROUTE		Fast to Slow Decimator Channel 0 Input Routing.	0x0	R/W
		00	FastDSP Channel 0.		
		01	FastDSP Channel 1.		
		02	FastDSP Channel 2.		
		03	FastDSP Channel 3.		
		04	FastDSP Channel 4.		
		05	FastDSP Channel 5.		
		06	FastDSP Channel 6.		
		07	FastDSP Channel 7.		
		08	FastDSP Channel 8.		
		09	FastDSP Channel 9.		
		10	FastDSP Channel 10.		
		11	FastDSP Channel 11.		
		12	FastDSP Channel 12.		
		13	FastDSP Channel 13.		
		14	FastDSP Channel 14.		
		15	FastDSP Channel 15.		
		16	Tensilica DSP Channel 0.		
		17	Tensilica DSP Channel 1.		
		18	Tensilica DSP Channel 2.		
		19	Tensilica DSP Channel 3.		
		20	Tensilica DSP Channel 4.		
		21	Tensilica DSP Channel 5.		
		22	Tensilica DSP Channel 6.		
		23	Tensilica DSP Channel 7.		
		24	Tensilica DSP Channel 8.		
		25	Tensilica DSP Channel 9.		
		26	Tensilica DSP Channel 10.		
		27	Tensilica DSP Channel 11.		
		28	Tensilica DSP Channel 12.		
		29	Tensilica DSP Channel 13.		
		30	Tensilica DSP Channel 14.		
		31	Tensilica DSP Channel 15.		
		32	Input ASRC Channel 0.		
		33	Input ASRC Channel 1.		

Bits	Bit Name	Settings	Description	Reset	Access
		34	Input ASRC Channel 2.		
		35	Input ASRC Channel 3.		
		36	ADC Channel 0.		
		37	ADC Channel 1.		
		38	ADC Channel 2.		
		39	DMIC Channel 0.		
		40	DMIC Channel 1.		
		41	DMIC Channel 2.		
		42	DMIC Channel 3.		
		43	EQ0.		
		47	DMIC Channel 4.		
		48	DMIC Channel 5.		
		49	DMIC Channel 6.		
		50	DMIC Channel 7.		

FAST TO SLOW DECIMATOR CHANNEL 1 INPUT ROUTING REGISTER

Address: 0x4000C085, Reset: 0x00, Name: FDEC_ROUTE1

Table 207. Bit Descriptions for FDEC_ROUTE1

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
[5:0]	FDEC0_ROUTE		Fast to Slow Decimator Channel 0 Input Routing.	0x0	R/W
		00	FastDSP Channel 0.		
		01	FastDSP Channel 1.		
		02	FastDSP Channel 2.		
		03	FastDSP Channel 3.		
		04	FastDSP Channel 4.		
		05	FastDSP Channel 5.		
		06	FastDSP Channel 6.		
		07	FastDSP Channel 7.		
		08	FastDSP Channel 8.		
		09	FastDSP Channel 9.		
		10	FastDSP Channel 10.		
		11	FastDSP Channel 11.		
		12	FastDSP Channel 12.		
		13	FastDSP Channel 13.		
		14	FastDSP Channel 14.		
		15	FastDSP Channel 15.		
		16	Tensilica DSP Channel 0.		
		17	Tensilica DSP Channel 1.		
		18	Tensilica DSP Channel 2.		
		19	Tensilica DSP Channel 3.		
		20	Tensilica DSP Channel 4.		
		21	Tensilica DSP Channel 5.		
		22	Tensilica DSP Channel 6.		
		23	Tensilica DSP Channel 7.		
		24	Tensilica DSP Channel 8.		
		25	Tensilica DSP Channel 9.		
		26	Tensilica DSP Channel 10.		
		27	Tensilica DSP Channel 11.		

Bits	Bit Name	Settings	Description	Reset	Access
		28	Tensilica DSP Channel 12.		
		29	Tensilica DSP Channel 13.		
		30	Tensilica DSP Channel 14.		
		31	Tensilica DSP Channel 15.		
		32	Input ASRC Channel 0.		
		33	Input ASRC Channel 1.		
		34	Input ASRC Channel 2.		
		35	Input ASRC Channel 3.		
		36	ADC Channel 0.		
		37	ADC Channel 1.		
		38	ADC Channel 2.		
		39	DMIC Channel 0.		
		40	DMIC Channel 1.		
		41	DMIC Channel 2.		
		42	DMIC Channel 3.		
		43	EQ0.		
		47	DMIC Channel 4.		
		48	DMIC Channel 5.		
		49	DMIC Channel 6.		
		50	DMIC Channel 7.		

FAST TO SLOW DECIMATOR CHANNEL 2 INPUT ROUTING REGISTER

Address: 0x4000C086, Reset: 0x00, Name: FDEC_ROUTE2

Table 208. Bit Descriptions for FDEC_ROUTE2

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
[5:0]	FDEC0_ROUTE		Fast to Slow Decimator Channel 0 Input Routing.	0x0	R/W
		00	FastDSP Channel 0.		
		01	FastDSP Channel 1.		
		02	FastDSP Channel 2.		
		03	FastDSP Channel 3.		
		04	FastDSP Channel 4.		
		05	FastDSP Channel 5.		
		06	FastDSP Channel 6.		
		07	FastDSP Channel 7.		
		08	FastDSP Channel 8.		
		09	FastDSP Channel 9.		
		10	FastDSP Channel 10.		
		11	FastDSP Channel 11.		
		12	FastDSP Channel 12.		
		13	FastDSP Channel 13.		
		14	FastDSP Channel 14.		
		15	FastDSP Channel 15.		
		16	Tensilica DSP Channel 0.		
		17	Tensilica DSP Channel 1.		
		18	Tensilica DSP Channel 2.		
		19	Tensilica DSP Channel 3.		
		20	Tensilica DSP Channel 4.		
		21	Tensilica DSP Channel 5.		

Bits	Bit Name	Settings	Description	Reset	Access
		22	Tensilica DSP Channel 6.		
		23	Tensilica DSP Channel 7.		
		24	Tensilica DSP Channel 8.		
		25	Tensilica DSP Channel 9.		
		26	Tensilica DSP Channel 10.		
		27	Tensilica DSP Channel 11.		
		28	Tensilica DSP Channel 12.		
		29	Tensilica DSP Channel 13.		
		30	Tensilica DSP Channel 14.		
		31	Tensilica DSP Channel 15.		
		32	Input ASRC Channel 0.		
		33	Input ASRC Channel 1.		
		34	Input ASRC Channel 2.		
		35	Input ASRC Channel 3.		
		36	ADC Channel 0.		
		37	ADC Channel 1.		
		38	ADC Channel 2.		
		39	DMIC Channel 0.		
		40	DMIC Channel 1.		
		41	DMIC Channel 2.		
		42	DMIC Channel 3.		
		43	EQ0.		
		47	DMIC Channel 4.		
		48	DMIC Channel 5.		
		49	DMIC Channel 6.		
		50	DMIC Channel 7.		

FAST TO SLOW DECIMATOR CHANNEL 3 INPUT ROUTING REGISTER

Address: 0x4000C087, Reset: 0x00, Name: FDEC_ROUTE3

Table 209. Bit Descriptions for FDEC_ROUTE3

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
[5:0]	FDEC0_ROUTE		Fast to Slow Decimator Channel 0 Input Routing.	0x0	R/W
		00	FastDSP Channel 0.		
		01	FastDSP Channel 1.		
		02	FastDSP Channel 2.		
		03	FastDSP Channel 3.		
		04	FastDSP Channel 4.		
		05	FastDSP Channel 5.		
		06	FastDSP Channel 6.		
		07	FastDSP Channel 7.		
		08	FastDSP Channel 8.		
		09	FastDSP Channel 9.		
		10	FastDSP Channel 10.		
		11	FastDSP Channel 11.		
		12	FastDSP Channel 12.		
		13	FastDSP Channel 13.		
		14	FastDSP Channel 14.		
		15	FastDSP Channel 15.		

Bits	Bit Name	Settings	Description	Reset	Access
		16	Tensilica DSP Channel 0.		
		17	Tensilica DSP Channel 1.		
		18	Tensilica DSP Channel 2.		
		19	Tensilica DSP Channel 3.		
		20	Tensilica DSP Channel 4.		
		21	Tensilica DSP Channel 5.		
		22	Tensilica DSP Channel 6.		
		23	Tensilica DSP Channel 7.		
		24	Tensilica DSP Channel 8.		
		25	Tensilica DSP Channel 9.		
		26	Tensilica DSP Channel 10.		
		27	Tensilica DSP Channel 11.		
		28	Tensilica DSP Channel 12.		
		29	Tensilica DSP Channel 13.		
		30	Tensilica DSP Channel 14.		
		31	Tensilica DSP Channel 15.		
		32	Input ASRC Channel 0.		
		33	Input ASRC Channel 1.		
		34	Input ASRC Channel 2.		
		35	Input ASRC Channel 3.		
		36	ADC Channel 0.		
		37	ADC Channel 1.		
		38	ADC Channel 2.		
		39	DMIC Channel 0.		
		40	DMIC Channel 1.		
		41	DMIC Channel 2.		
		42	DMIC Channel 3.		
		43	EQ0.		
		47	DMIC Channel 4.		
		48	DMIC Channel 5.		
		49	DMIC Channel 6.		
		50	DMIC Channel 7.		

FAST TO SLOW DECIMATOR CHANNEL 4 INPUT ROUTING REGISTER

Address: 0x4000C088, Reset: 0x00, Name: FDEC_ROUTE4

Table 210. Bit Descriptions for FDEC_ROUTE4

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
[5:0]	FDEC0_ROUTE		Fast to Slow Decimator Channel 0 Input Routing.	0x0	R/W
		00	FastDSP Channel 0.		
		01	FastDSP Channel 1.		
		02	FastDSP Channel 2.		
		03	FastDSP Channel 3.		
		04	FastDSP Channel 4.		
		05	FastDSP Channel 5.		
		06	FastDSP Channel 6.		
		07	FastDSP Channel 7.		
		08	FastDSP Channel 8.		
		09	FastDSP Channel 9.		

Bits	Bit Name	Settings	Description	Reset	Access
		10	FastDSP Channel 10.		
		11	FastDSP Channel 11.		
		12	FastDSP Channel 12.		
		13	FastDSP Channel 13.		
		14	FastDSP Channel 14.		
		15	FastDSP Channel 15.		
		16	Tensilica DSP Channel 0.		
		17	Tensilica DSP Channel 1.		
		18	Tensilica DSP Channel 2.		
		19	Tensilica DSP Channel 3.		
		20	Tensilica DSP Channel 4.		
		21	Tensilica DSP Channel 5.		
		22	Tensilica DSP Channel 6.		
		23	Tensilica DSP Channel 7.		
		24	Tensilica DSP Channel 8.		
		25	Tensilica DSP Channel 9.		
		26	Tensilica DSP Channel 10.		
		27	Tensilica DSP Channel 11.		
		28	Tensilica DSP Channel 12.		
		29	Tensilica DSP Channel 13.		
		30	Tensilica DSP Channel 14.		
		31	Tensilica DSP Channel 15.		
		32	Input ASRC Channel 0.		
		33	Input ASRC Channel 1.		
		34	Input ASRC Channel 2.		
		35	Input ASRC Channel 3.		
		36	ADC Channel 0.		
		37	ADC Channel 1.		
		38	ADC Channel 2.		
		39	DMIC Channel 0.		
		40	DMIC Channel 1.		
		41	DMIC Channel 2.		
		42	DMIC Channel 3.		
		43	EQ0.		
		47	DMIC Channel 4.		
		48	DMIC Channel 5.		
		49	DMIC Channel 6.		
		50	DMIC Channel 7.		

FAST TO SLOW DECIMATOR CHANNEL 5 INPUT ROUTING REGISTER

Address: 0x4000C089, Reset: 0x00, Name: FDEC_ROUTE5

Table 211. Bit Descriptions for FDEC_ROUTE5

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
[5:0]	FDEC0_ROUTE		Fast to Slow Decimator Channel 0 Input Routing.	0x0	R/W
		00	FastDSP Channel 0.		
		01	FastDSP Channel 1.		
		02	FastDSP Channel 2.		
		03	FastDSP Channel 3.		

Bits	Bit Name	Settings	Description	Reset	Access
		04	FastDSP Channel 4.		
		05	FastDSP Channel 5.		
		06	FastDSP Channel 6.		
		07	FastDSP Channel 7.		
		08	FastDSP Channel 8.		
		09	FastDSP Channel 9.		
		10	FastDSP Channel 10.		
		11	FastDSP Channel 11.		
		12	FastDSP Channel 12.		
		13	FastDSP Channel 13.		
		14	FastDSP Channel 14.		
		15	FastDSP Channel 15.		
		16	Tensilica DSP Channel 0.		
		17	Tensilica DSP Channel 1.		
		18	Tensilica DSP Channel 2.		
		19	Tensilica DSP Channel 3.		
		20	Tensilica DSP Channel 4.		
		21	Tensilica DSP Channel 5.		
		22	Tensilica DSP Channel 6.		
		23	Tensilica DSP Channel 7.		
		24	Tensilica DSP Channel 8.		
		25	Tensilica DSP Channel 9.		
		26	Tensilica DSP Channel 10.		
		27	Tensilica DSP Channel 11.		
		28	Tensilica DSP Channel 12.		
		29	Tensilica DSP Channel 13.		
		30	Tensilica DSP Channel 14.		
		31	Tensilica DSP Channel 15.		
		32	Input ASRC Channel 0.		
		33	Input ASRC Channel 1.		
		34	Input ASRC Channel 2.		
		35	Input ASRC Channel 3.		
		36	ADC Channel 0.		
		37	ADC Channel 1.		
		38	ADC Channel 2.		
		39	DMIC Channel 0.		
		40	DMIC Channel 1.		
		41	DMIC Channel 2.		
		42	DMIC Channel 3.		
		43	EQ0.		
		47	DMIC Channel 4.		
		48	DMIC Channel 5.		
		49	DMIC Channel 6.		
		50	DMIC Channel 7.		

FAST TO SLOW DECIMATOR CHANNEL 6 INPUT ROUTING REGISTER

Address: 0x4000C08A, Reset: 0x00, Name: FDEC_ROUTE6

Table 212. Bit Descriptions for FDEC_ROUTE6

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
[5:0]	FDEC0_ROUTE		Fast to Slow Decimator Channel 0 Input Routing.	0x0	R/W
		00	FastDSP Channel 0.		
		01	FastDSP Channel 1.		
		02	FastDSP Channel 2.		
		03	FastDSP Channel 3.		
		04	FastDSP Channel 4.		
		05	FastDSP Channel 5.		
		06	FastDSP Channel 6.		
		07	FastDSP Channel 7.		
		08	FastDSP Channel 8.		
		09	FastDSP Channel 9.		
		10	FastDSP Channel 10.		
		11	FastDSP Channel 11.		
		12	FastDSP Channel 12.		
		13	FastDSP Channel 13.		
		14	FastDSP Channel 14.		
		15	FastDSP Channel 15.		
		16	Tensilica DSP Channel 0.		
		17	Tensilica DSP Channel 1.		
		18	Tensilica DSP Channel 2.		
		19	Tensilica DSP Channel 3.		
		20	Tensilica DSP Channel 4.		
		21	Tensilica DSP Channel 5.		
		22	Tensilica DSP Channel 6.		
		23	Tensilica DSP Channel 7.		
		24	Tensilica DSP Channel 8.		
		25	Tensilica DSP Channel 9.		
		26	Tensilica DSP Channel 10.		
		27	Tensilica DSP Channel 11.		
		28	Tensilica DSP Channel 12.		
		29	Tensilica DSP Channel 13.		
		30	Tensilica DSP Channel 14.		
		31	Tensilica DSP Channel 15.		
		32	Input ASRC Channel 0.		
		33	Input ASRC Channel 1.		
		34	Input ASRC Channel 2.		
		35	Input ASRC Channel 3.		
		36	ADC Channel 0.		
		37	ADC Channel 1.		
		38	ADC Channel 2.		
		39	DMIC Channel 0.		
		40	DMIC Channel 1.		

Bits	Bit Name	Settings	Description	Reset	Access
		41	DMIC Channel 2.		
		42	DMIC Channel 3.		
		43	EQ0.		
		47	DMIC Channel 4.		
		48	DMIC Channel 5.		
		49	DMIC Channel 6.		
		50	DMIC Channel 7.		

FAST TO SLOW DECIMATOR CHANNEL 7 INPUT ROUTING REGISTER

Address: 0x4000C08B, Reset: 0x00, Name: FDEC_ROUTE7

Table 213. Bit Descriptions for FDEC_ROUTE7

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
[5:0]	FDEC0_ROUTE		Fast to Slow Decimator Channel 0 Input Routing.	0x0	R/W
		00	FastDSP Channel 0.		
		01	FastDSP Channel 1.		
		02	FastDSP Channel 2.		
		03	FastDSP Channel 3.		
		04	FastDSP Channel 4.		
		05	FastDSP Channel 5.		
		06	FastDSP Channel 6.		
		07	FastDSP Channel 7.		
		08	FastDSP Channel 8.		
		09	FastDSP Channel 9.		
		10	FastDSP Channel 10.		
		11	FastDSP Channel 11.		
		12	FastDSP Channel 12.		
		13	FastDSP Channel 13.		
		14	FastDSP Channel 14.		
		15	FastDSP Channel 15.		
		16	Tensilica DSP Channel 0.		
		17	Tensilica DSP Channel 1.		
		18	Tensilica DSP Channel 2.		
		19	Tensilica DSP Channel 3.		
		20	Tensilica DSP Channel 4.		
		21	Tensilica DSP Channel 5.		
		22	Tensilica DSP Channel 6.		
		23	Tensilica DSP Channel 7.		
		24	Tensilica DSP Channel 8.		
		25	Tensilica DSP Channel 9.		
		26	Tensilica DSP Channel 10.		
		27	Tensilica DSP Channel 11.		
		28	Tensilica DSP Channel 12.		
		29	Tensilica DSP Channel 13.		
		30	Tensilica DSP Channel 14.		
		31	Tensilica DSP Channel 15.		
		32	Input ASRC Channel 0.		
		33	Input ASRC Channel 1.		
		34	Input ASRC Channel 2.		

Bits	Bit Name	Settings	Description	Reset	Access
		35	Input ASRC Channel 3.		
		36	ADC Channel 0.		
		37	ADC Channel 1.		
		38	ADC Channel 2.		
		39	DMIC Channel 0.		
		40	DMIC Channel 1.		
		41	DMIC Channel 2.		
		42	DMIC Channel 3.		
		43	EQ0.		
		47	DMIC Channel 4.		
		48	DMIC Channel 5.		
		49	DMIC Channel 6.		
		50	DMIC Channel 7.		

SLOW TO FAST INTERPOLATOR SAMPLE RATES CHANNEL 0/CHANNEL 1 REGISTER

Address: 0x4000C090, Reset: 0x52, Name: FINT_CTRL1

Table 214. Bit Descriptions for FINT_CTRL1

Bits	Bit Name	Settings	Description	Reset	Access
7	RESERVED		Reserved.	0x0	R
[6:4]	FINT01_OUT_FS		Interpolator Channel 0/Channel 1 Output Sampling Rate.	0x5	R/W
		001	24kHz Sample Rate.		
		010	48kHz Sample Rate.		
		011	96kHz Sample Rate.		
		100	192kHz Sample Rate.		
		101	384kHz Sample Rate.		
		110	768kHz Sample Rate.		
[3:0]	FINT01_IN_FS		Interpolator Channel 0/Channel 1 Input Sampling Rate.	0x2	R/W
		0000	12kHz Sample Rate.		
		0001	24kHz Sample Rate.		
		0010	48kHz Sample Rate.		
		0011	96kHz Sample Rate.		
		0100	192kHz Sample Rate.		
		0101	384kHz Sample Rate.		
		1000	8kHz Sample Rate.		
		1001	16kHz Sample Rate.		

SLOW TO FAST INTERPOLATOR SAMPLE RATES CHANNEL 2/CHANNEL 3 REGISTER

Address: 0x4000C091, Reset: 0x52, Name: FINT_CTRL2

Table 215. Bit Descriptions for FINT_CTRL2

Bits	Bit Name	Settings	Description	Reset	Access
7	RESERVED		Reserved.	0x0	R
[6:4]	FINT23_OUT_FS		Interpolator Channel 2/Channel 3 Output Sampling Rate.	0x5	R/W
		001	24kHz Sample Rate.		
		010	48kHz Sample Rate.		
		011	96kHz Sample Rate.		

Bits	Bit Name	Settings	Description	Reset	Access
		100	192kHz Sample Rate.		
		101	384kHz Sample Rate.		
		110	768kHz Sample Rate.		
[3:0]	FINT23_IN_FS		Interpolator Channel 2/Channel 3 Input Sampling Rate.	0x2	R/W
		0000	12kHz Sample Rate.		
		0001	24kHz Sample Rate.		
		0010	48kHz Sample Rate.		
		0011	96kHz Sample Rate.		
		0100	192kHz Sample Rate.		
		0101	384kHz Sample Rate.		
		1000	8kHz Sample Rate.		
		1001	16kHz Sample Rate.		

SLOW TO FAST INTERPOLATOR SAMPLE RATES CHANNEL 4/CHANNEL 5 REGISTER

Address: 0x4000C092, Reset: 0x52, Name: FINT_CTRL3

Table 216. Bit Descriptions for FINT_CTRL3

Bits	Bit Name	Settings	Description	Reset	Access
7	RESERVED		Reserved.	0x0	R
[6:4]	FINT45_OUT_FS		Interpolator Channel 4/Channel 5 Output Sampling Rate.	0x5	R/W
		001	24kHz Sample Rate.		
		010	48kHz Sample Rate.		
		011	96kHz Sample Rate.		
		100	192kHz Sample Rate.		
		101	384kHz Sample Rate.		
		110	768kHz Sample Rate.		
[3:0]	FINT45_IN_FS		Interpolator Channel 4/Channel 5 Input Sampling Rate.	0x2	R/W
		0000	12kHz Sample Rate.		
		0001	24kHz Sample Rate.		
		0010	48kHz Sample Rate.		
		0011	96kHz Sample Rate.		
		0100	192kHz Sample Rate.		
		0101	384kHz Sample Rate.		
		1000	8kHz Sample Rate.		
		1001	16kHz Sample Rate.		

SLOW TO FAST INTERPOLATOR SAMPLE RATES CHANNEL 6/CHANNEL 7 REGISTER

Address: 0x4000C093, Reset: 0x52, Name: FINT_CTRL4

Table 217. Bit Descriptions for FINT_CTRL4

Bits	Bit Name	Settings	Description	Reset	Access
7	RESERVED		Reserved.	0x0	R
[6:4]	FINT67_OUT_FS		Interpolator Channel 6/Channel 7 Output Sampling Rate.	0x5	R/W
		001	24kHz Sample Rate.		
		010	48kHz Sample Rate.		
		011	96kHz Sample Rate.		
		100	192kHz Sample Rate.		
		101	384kHz Sample Rate.		
		110	768kHz Sample Rate.		

Bits	Bit Name	Settings	Description	Reset	Access
[3:0]	FINT67_IN_FS		Interpolator Channel 6/Channel 7 Input Sampling Rate.	0x2	R/W
		0000	12kHz Sample Rate.		
		0001	24kHz Sample Rate.		
		0010	48kHz Sample Rate.		
		0011	96kHz Sample Rate.		
		0100	192kHz Sample Rate.		
		0101	384kHz Sample Rate.		
		1000	8kHz Sample Rate.		
		1001	16kHz Sample Rate.		

SLOW TO FAST INTERPOLATOR CHANNEL 0 INPUT ROUTING REGISTER

Address: 0x4000C094, Reset: 0x00, Name: FINT_ROUTE0

Table 218. Bit Descriptions for FINT_ROUTE0

Bits	Bit Name	Settings	Description	Reset	Access
7	RESERVED		Reserved.	0x0	R
[6:0]	FINT0_ROUTE		Slow to Fast Interpolator Channel 0 Input Routing.	0x0	R/W
		0	Serial Port 0 Channel 0.		
		1	Serial Port 0 Channel 1.		
		2	Serial Port 0 Channel 2.		
		3	Serial Port 0 Channel 3.		
		4	Serial Port 0 Channel 4.		
		5	Serial Port 0 Channel 5.		
		6	Serial Port 0 Channel 6.		
		7	Serial Port 0 Channel 7.		
		8	Serial Port 0 Channel 8.		
		9	Serial Port 0 Channel 9.		
		10	Serial Port 0 Channel 10.		
		11	Serial Port 0 Channel 11.		
		12	Serial Port 0 Channel 12.		
		13	Serial Port 0 Channel 13.		
		14	Serial Port 0 Channel 14.		
		15	Serial Port 0 Channel 15.		
		16	Serial Port 1 Channel 0.		
		17	Serial Port 1 Channel 1.		
		18	Serial Port 1 Channel 2.		
		19	Serial Port 1 Channel 3.		
		20	Serial Port 1 Channel 4.		
		21	Serial Port 1 Channel 5.		
		22	Serial Port 1 Channel 6.		
		23	Serial Port 1 Channel 7.		
		24	Serial Port 1 Channel 8.		
		25	Serial Port 1 Channel 9.		
		26	Serial Port 1 Channel 10.		
		27	Serial Port 1 Channel 11.		
		28	Serial Port 1 Channel 12.		
		29	Serial Port 1 Channel 13.		
		30	Serial Port 1 Channel 14.		
		31	Serial Port 1 Channel 15.		
		32	FastDSP Channel 0.		

Bits	Bit Name	Settings	Description	Reset	Access
		33	FastDSP Channel 1.		
		34	FastDSP Channel 2.		
		35	FastDSP Channel 3.		
		36	FastDSP Channel 4.		
		37	FastDSP Channel 5.		
		38	FastDSP Channel 6.		
		39	FastDSP Channel 7.		
		40	FastDSP Channel 8.		
		41	FastDSP Channel 9.		
		42	FastDSP Channel 10.		
		43	FastDSP Channel 11.		
		44	FastDSP Channel 12.		
		45	FastDSP Channel 13.		
		46	FastDSP Channel 14.		
		47	FastDSP Channel 15.		
		48	Tensilica DSP Channel 0.		
		49	Tensilica DSP Channel 1.		
		50	Tensilica DSP Channel 2.		
		51	Tensilica DSP Channel 3.		
		52	Tensilica DSP Channel 4.		
		53	Tensilica DSP Channel 5.		
		54	Tensilica DSP Channel 6.		
		55	Tensilica DSP Channel 7.		
		56	Tensilica DSP Channel 8.		
		57	Tensilica DSP Channel 9.		
		58	Tensilica DSP Channel 10.		
		59	Tensilica DSP Channel 11.		
		60	Tensilica DSP Channel 12.		
		61	Tensilica DSP Channel 13.		
		62	Tensilica DSP Channel 14.		
		63	Tensilica DSP Channel 15.		
		64	Input ASRC Channel 0.		
		65	Input ASRC Channel 1.		
		66	Input ASRC Channel 2.		
		67	Input ASRC Channel 3.		
		68	ADC Channel 0.		
		69	ADC Channel 1.		
		70	ADC Channel 2.		
		71	DMIC Channel 0.		
		72	DMIC Channel 1.		
		73	DMIC Channel 2.		
		74	DMIC Channel 3.		
		75	EQ0.		
		79	DMIC Channel 4.		
		80	DMIC Channel 5.		
		81	DMIC Channel 6.		
		82	DMIC Channel 7.		

SLOW TO FAST INTERPOLATOR CHANNEL 1 INPUT ROUTING REGISTER

Address: 0x4000C095, Reset: 0x00, Name: FINT_ROUTE1

Table 219. Bit Descriptions for FINT_ROUTE1

Bits	Bit Name	Settings	Description	Reset	Access
7	RESERVED		Reserved.	0x0	R
[6:0]	FINT0_ROUTE		Slow to Fast Interpolator Channel 0 Input Routing.	0x0	R/W
		0	Serial Port 0 Channel 0.		
		1	Serial Port 0 Channel 1.		
		2	Serial Port 0 Channel 2.		
		3	Serial Port 0 Channel 3.		
		4	Serial Port 0 Channel 4.		
		5	Serial Port 0 Channel 5.		
		6	Serial Port 0 Channel 6.		
		7	Serial Port 0 Channel 7.		
		8	Serial Port 0 Channel 8.		
		9	Serial Port 0 Channel 9.		
		10	Serial Port 0 Channel 10.		
		11	Serial Port 0 Channel 11.		
		12	Serial Port 0 Channel 12.		
		13	Serial Port 0 Channel 13.		
		14	Serial Port 0 Channel 14.		
		15	Serial Port 0 Channel 15.		
		16	Serial Port 1 Channel 0.		
		17	Serial Port 1 Channel 1.		
		18	Serial Port 1 Channel 2.		
		19	Serial Port 1 Channel 3.		
		20	Serial Port 1 Channel 4.		
		21	Serial Port 1 Channel 5.		
		22	Serial Port 1 Channel 6.		
		23	Serial Port 1 Channel 7.		
		24	Serial Port 1 Channel 8.		
		25	Serial Port 1 Channel 9.		
		26	Serial Port 1 Channel 10.		
		27	Serial Port 1 Channel 11.		
		28	Serial Port 1 Channel 12.		
		29	Serial Port 1 Channel 13.		
		30	Serial Port 1 Channel 14.		
		31	Serial Port 1 Channel 15.		
		32	FastDSP Channel 0.		
		33	FastDSP Channel 1.		
		34	FastDSP Channel 2.		
		35	FastDSP Channel 3.		
		36	FastDSP Channel 4.		
		37	FastDSP Channel 5.		
		38	FastDSP Channel 6.		
		39	FastDSP Channel 7.		
		40	FastDSP Channel 8.		
		41	FastDSP Channel 9.		
		42	FastDSP Channel 10.		
		43	FastDSP Channel 11.		

Bits	Bit Name	Settings	Description	Reset	Access
		44	FastDSP Channel 12.		
		45	FastDSP Channel 13.		
		46	FastDSP Channel 14.		
		47	FastDSP Channel 15.		
		48	Tensilica DSP Channel 0.		
		49	Tensilica DSP Channel 1.		
		50	Tensilica DSP Channel 2.		
		51	Tensilica DSP Channel 3.		
		52	Tensilica DSP Channel 4.		
		53	Tensilica DSP Channel 5.		
		54	Tensilica DSP Channel 6.		
		55	Tensilica DSP Channel 7.		
		56	Tensilica DSP Channel 8.		
		57	Tensilica DSP Channel 9.		
		58	Tensilica DSP Channel 10.		
		59	Tensilica DSP Channel 11.		
		60	Tensilica DSP Channel 12.		
		61	Tensilica DSP Channel 13.		
		62	Tensilica DSP Channel 14.		
		63	Tensilica DSP Channel 15.		
		64	Input ASRC Channel 0.		
		65	Input ASRC Channel 1.		
		66	Input ASRC Channel 2.		
		67	Input ASRC Channel 3.		
		68	ADC Channel 0.		
		69	ADC Channel 1.		
		70	ADC Channel 2.		
		71	DMIC Channel 0.		
		72	DMIC Channel 1.		
		73	DMIC Channel 2.		
		74	DMIC Channel 3.		
		75	EQ0.		
		79	DMIC Channel 4.		
		80	DMIC Channel 5.		
		81	DMIC Channel 6.		
		82	DMIC Channel 7.		

SLOW TO FAST INTERPOLATOR CHANNEL 2 INPUT ROUTING REGISTER

Address: 0x4000C096, Reset: 0x00, Name: FINT_ROUTE2

Table 220. Bit Descriptions for FINT_ROUTE2

Bits	Bit Name	Settings	Description	Reset	Access
7	RESERVED		Reserved.	0x0	R
[6:0]	FINT0_ROUTE		Slow to Fast Interpolator Channel 0 Input Routing.	0x0	R/W
		0	Serial Port 0 Channel 0.		
		1	Serial Port 0 Channel 1.		
		2	Serial Port 0 Channel 2.		
		3	Serial Port 0 Channel 3.		
		4	Serial Port 0 Channel 4.		
		5	Serial Port 0 Channel 5.		

Bits	Bit Name	Settings	Description	Reset	Access
		6	Serial Port 0 Channel 6.		
		7	Serial Port 0 Channel 7.		
		8	Serial Port 0 Channel 8.		
		9	Serial Port 0 Channel 9.		
		10	Serial Port 0 Channel 10.		
		11	Serial Port 0 Channel 11.		
		12	Serial Port 0 Channel 12.		
		13	Serial Port 0 Channel 13.		
		14	Serial Port 0 Channel 14.		
		15	Serial Port 0 Channel 15.		
		16	Serial Port 1 Channel 0.		
		17	Serial Port 1 Channel 1.		
		18	Serial Port 1 Channel 2.		
		19	Serial Port 1 Channel 3.		
		20	Serial Port 1 Channel 4.		
		21	Serial Port 1 Channel 5.		
		22	Serial Port 1 Channel 6.		
		23	Serial Port 1 Channel 7.		
		24	Serial Port 1 Channel 8.		
		25	Serial Port 1 Channel 9.		
		26	Serial Port 1 Channel 10.		
		27	Serial Port 1 Channel 11.		
		28	Serial Port 1 Channel 12.		
		29	Serial Port 1 Channel 13.		
		30	Serial Port 1 Channel 14.		
		31	Serial Port 1 Channel 15.		
		32	FastDSP Channel 0.		
		33	FastDSP Channel 1.		
		34	FastDSP Channel 2.		
		35	FastDSP Channel 3.		
		36	FastDSP Channel 4.		
		37	FastDSP Channel 5.		
		38	FastDSP Channel 6.		
		39	FastDSP Channel 7.		
		40	FastDSP Channel 8.		
		41	FastDSP Channel 9.		
		42	FastDSP Channel 10.		
		43	FastDSP Channel 11.		
		44	FastDSP Channel 12.		
		45	FastDSP Channel 13.		
		46	FastDSP Channel 14.		
		47	FastDSP Channel 15.		
		48	Tensilica DSP Channel 0.		
		49	Tensilica DSP Channel 1.		
		50	Tensilica DSP Channel 2.		
		51	Tensilica DSP Channel 3.		
		52	Tensilica DSP Channel 4.		
		53	Tensilica DSP Channel 5.		
		54	Tensilica DSP Channel 6.		
		55	Tensilica DSP Channel 7.		
		56	Tensilica DSP Channel 8.		

Bits	Bit Name	Settings	Description	Reset	Access
		57	Tensilica DSP Channel 9.		
		58	Tensilica DSP Channel 10.		
		59	Tensilica DSP Channel 11.		
		60	Tensilica DSP Channel 12.		
		61	Tensilica DSP Channel 13.		
		62	Tensilica DSP Channel 14.		
		63	Tensilica DSP Channel 15.		
		64	Input ASRC Channel 0.		
		65	Input ASRC Channel 1.		
		66	Input ASRC Channel 2.		
		67	Input ASRC Channel 3.		
		68	ADC Channel 0.		
		69	ADC Channel 1.		
		70	ADC Channel 2.		
		71	DMIC Channel 0.		
		72	DMIC Channel 1.		
		73	DMIC Channel 2.		
		74	DMIC Channel 3.		
		75	EQ0.		
		79	DMIC Channel 4.		
		80	DMIC Channel 5.		
		81	DMIC Channel 6.		
		82	DMIC Channel 7.		

SLOW TO FAST INTERPOLATOR CHANNEL 3 INPUT ROUTING REGISTER

Address: 0x4000C097, Reset: 0x00, Name: FINT_ROUTE3

Bits	Bit Name	Settings	Description	Reset	Access
7	RESERVED		Reserved.	0x0	R
[6:0]	FINT0_ROUTE		Slow to Fast Interpolator Channel 0 Input Routing.	0x0	R/W
		0	Serial Port 0 Channel 0.		
		1	Serial Port 0 Channel 1.		
		2	Serial Port 0 Channel 2.		
		3	Serial Port 0 Channel 3.		
		4	Serial Port 0 Channel 4.		
		5	Serial Port 0 Channel 5.		
		6	Serial Port 0 Channel 6.		
		7	Serial Port 0 Channel 7.		
		8	Serial Port 0 Channel 8.		
		9	Serial Port 0 Channel 9.		
		10	Serial Port 0 Channel 10.		
		11	Serial Port 0 Channel 11.		
		12	Serial Port 0 Channel 12.		
		13	Serial Port 0 Channel 13.		
		14	Serial Port 0 Channel 14.		
		15	Serial Port 0 Channel 15.		
		16	Serial Port 1 Channel 0.		
		17	Serial Port 1 Channel 1.		
		18	Serial Port 1 Channel 2.		
		19	Serial Port 1 Channel 3.		

		20	Serial Port 1 Channel 4.		
		21	Serial Port 1 Channel 5.		
		22	Serial Port 1 Channel 6.		
		23	Serial Port 1 Channel 7.		
		24	Serial Port 1 Channel 8.		
		25	Serial Port 1 Channel 9.		
		26	Serial Port 1 Channel 10.		
		27	Serial Port 1 Channel 11.		
		28	Serial Port 1 Channel 12.		
		29	Serial Port 1 Channel 13.		
		30	Serial Port 1 Channel 14.		
		31	Serial Port 1 Channel 15.		
		32	FastDSP Channel 0.		
		33	FastDSP Channel 1.		
		34	FastDSP Channel 2.		
		35	FastDSP Channel 3.		
		36	FastDSP Channel 4.		
		37	FastDSP Channel 5.		
		38	FastDSP Channel 6.		
		39	FastDSP Channel 7.		
		40	FastDSP Channel 8.		
		41	FastDSP Channel 9.		
		42	FastDSP Channel 10.		
		43	FastDSP Channel 11.		
		44	FastDSP Channel 12.		
		45	FastDSP Channel 13.		
		46	FastDSP Channel 14.		
		47	FastDSP Channel 15.		
		48	Tensilica DSP Channel 0.		
		49	Tensilica DSP Channel 1.		
		50	Tensilica DSP Channel 2.		
		51	Tensilica DSP Channel 3.		
		52	Tensilica DSP Channel 4.		
		53	Tensilica DSP Channel 5.		
		54	Tensilica DSP Channel 6.		
		55	Tensilica DSP Channel 7.		
		56	Tensilica DSP Channel 8.		
		57	Tensilica DSP Channel 9.		
		58	Tensilica DSP Channel 10.		
		59	Tensilica DSP Channel 11.		
		60	Tensilica DSP Channel 12.		
		61	Tensilica DSP Channel 13.		
		62	Tensilica DSP Channel 14.		
		63	Tensilica DSP Channel 15.		
		64	Input ASRC Channel 0.		
		65	Input ASRC Channel 1.		
		66	Input ASRC Channel 2.		
		67	Input ASRC Channel 3.		
		68	ADC Channel 0.		
		69	ADC Channel 1.		
		70	ADC Channel 2.		

		71	DMIC Channel 0.		
		72	DMIC Channel 1.		
		73	DMIC Channel 2.		
		74	DMIC Channel 3.		
		75	EQ0.		
		79	DMIC Channel 4.		
		80	DMIC Channel 5.		
		81	DMIC Channel 6.		
		82	DMIC Channel 7.		

SLOW TO FAST INTERPOLATOR CHANNEL 4 INPUT ROUTING REGISTER

Address: 0x4000C098, Reset: 0x00, Name: FINT_ROUTE4

Table 221. Bit Descriptions for FINT_ROUTE4

Bits	Bit Name	Settings	Description	Reset	Access
7	RESERVED		Reserved.	0x0	R
[6:0]	FINT0_ROUTE		Slow to Fast Interpolator Channel 0 Input Routing.	0x0	R/W
		0	Serial Port 0 Channel 0.		
		1	Serial Port 0 Channel 1.		
		2	Serial Port 0 Channel 2.		
		3	Serial Port 0 Channel 3.		
		4	Serial Port 0 Channel 4.		
		5	Serial Port 0 Channel 5.		
		6	Serial Port 0 Channel 6.		
		7	Serial Port 0 Channel 7.		
		8	Serial Port 0 Channel 8.		
		9	Serial Port 0 Channel 9.		
		10	Serial Port 0 Channel 10.		
		11	Serial Port 0 Channel 11.		
		12	Serial Port 0 Channel 12.		
		13	Serial Port 0 Channel 13.		
		14	Serial Port 0 Channel 14.		
		15	Serial Port 0 Channel 15.		
		16	Serial Port 1 Channel 0.		
		17	Serial Port 1 Channel 1.		
		18	Serial Port 1 Channel 2.		
		19	Serial Port 1 Channel 3.		
		20	Serial Port 1 Channel 4.		
		21	Serial Port 1 Channel 5.		
		22	Serial Port 1 Channel 6.		
		23	Serial Port 1 Channel 7.		
		24	Serial Port 1 Channel 8.		
		25	Serial Port 1 Channel 9.		
		26	Serial Port 1 Channel 10.		
		27	Serial Port 1 Channel 11.		
		28	Serial Port 1 Channel 12.		
		29	Serial Port 1 Channel 13.		
		30	Serial Port 1 Channel 14.		
		31	Serial Port 1 Channel 15.		
		32	FastDSP Channel 0.		
		33	FastDSP Channel 1.		

Bits	Bit Name	Settings	Description	Reset	Access
		34	FastDSP Channel 2.		
		35	FastDSP Channel 3.		
		36	FastDSP Channel 4.		
		37	FastDSP Channel 5.		
		38	FastDSP Channel 6.		
		39	FastDSP Channel 7.		
		40	FastDSP Channel 8.		
		41	FastDSP Channel 9.		
		42	FastDSP Channel 10.		
		43	FastDSP Channel 11.		
		44	FastDSP Channel 12.		
		45	FastDSP Channel 13.		
		46	FastDSP Channel 14.		
		47	FastDSP Channel 15.		
		48	Tensilica DSP Channel 0.		
		49	Tensilica DSP Channel 1.		
		50	Tensilica DSP Channel 2.		
		51	Tensilica DSP Channel 3.		
		52	Tensilica DSP Channel 4.		
		53	Tensilica DSP Channel 5.		
		54	Tensilica DSP Channel 6.		
		55	Tensilica DSP Channel 7.		
		56	Tensilica DSP Channel 8.		
		57	Tensilica DSP Channel 9.		
		58	Tensilica DSP Channel 10.		
		59	Tensilica DSP Channel 11.		
		60	Tensilica DSP Channel 12.		
		61	Tensilica DSP Channel 13.		
		62	Tensilica DSP Channel 14.		
		63	Tensilica DSP Channel 15.		
		64	Input ASRC Channel 0.		
		65	Input ASRC Channel 1.		
		66	Input ASRC Channel 2.		
		67	Input ASRC Channel 3.		
		68	ADC Channel 0.		
		69	ADC Channel 1.		
		70	ADC Channel 2.		
		71	DMIC Channel 0.		
		72	DMIC Channel 1.		
		73	DMIC Channel 2.		
		74	DMIC Channel 3.		
		75	EQ0.		
		79	DMIC Channel 4.		
		80	DMIC Channel 5.		
		81	DMIC Channel 6.		
		82	DMIC Channel 7.		

SLOW TO FAST INTERPOLATOR CHANNEL 5 INPUT ROUTING REGISTER

Address: 0x4000C099, Reset: 0x00, Name: FINT_ROUTE5

Table 222. Bit Descriptions for FINT_ROUTE5

Bits	Bit Name	Settings	Description	Reset	Access
7	RESERVED		Reserved.	0x0	R
[6:0]	FINT0_ROUTE		Slow to Fast Interpolator Channel 0 Input Routing.	0x0	R/W
		0	Serial Port 0 Channel 0.		
		1	Serial Port 0 Channel 1.		
		2	Serial Port 0 Channel 2.		
		3	Serial Port 0 Channel 3.		
		4	Serial Port 0 Channel 4.		
		5	Serial Port 0 Channel 5.		
		6	Serial Port 0 Channel 6.		
		7	Serial Port 0 Channel 7.		
		8	Serial Port 0 Channel 8.		
		9	Serial Port 0 Channel 9.		
		10	Serial Port 0 Channel 10.		
		11	Serial Port 0 Channel 11.		
		12	Serial Port 0 Channel 12.		
		13	Serial Port 0 Channel 13.		
		14	Serial Port 0 Channel 14.		
		15	Serial Port 0 Channel 15.		
		16	Serial Port 1 Channel 0.		
		17	Serial Port 1 Channel 1.		
		18	Serial Port 1 Channel 2.		
		19	Serial Port 1 Channel 3.		
		20	Serial Port 1 Channel 4.		
		21	Serial Port 1 Channel 5.		
		22	Serial Port 1 Channel 6.		
		23	Serial Port 1 Channel 7.		
		24	Serial Port 1 Channel 8.		
		25	Serial Port 1 Channel 9.		
		26	Serial Port 1 Channel 10.		
		27	Serial Port 1 Channel 11.		
		28	Serial Port 1 Channel 12.		
		29	Serial Port 1 Channel 13.		
		30	Serial Port 1 Channel 14.		
		31	Serial Port 1 Channel 15.		
		32	FastDSP Channel 0.		
		33	FastDSP Channel 1.		
		34	FastDSP Channel 2.		
		35	FastDSP Channel 3.		
		36	FastDSP Channel 4.		
		37	FastDSP Channel 5.		
		38	FastDSP Channel 6.		
		39	FastDSP Channel 7.		
		40	FastDSP Channel 8.		
		41	FastDSP Channel 9.		
		42	FastDSP Channel 10.		
		43	FastDSP Channel 11.		

Bits	Bit Name	Settings	Description	Reset	Access
		44	FastDSP Channel 12.		
		45	FastDSP Channel 13.		
		46	FastDSP Channel 14.		
		47	FastDSP Channel 15.		
		48	Tensilica DSP Channel 0.		
		49	Tensilica DSP Channel 1.		
		50	Tensilica DSP Channel 2.		
		51	Tensilica DSP Channel 3.		
		52	Tensilica DSP Channel 4.		
		53	Tensilica DSP Channel 5.		
		54	Tensilica DSP Channel 6.		
		55	Tensilica DSP Channel 7.		
		56	Tensilica DSP Channel 8.		
		57	Tensilica DSP Channel 9.		
		58	Tensilica DSP Channel 10.		
		59	Tensilica DSP Channel 11.		
		60	Tensilica DSP Channel 12.		
		61	Tensilica DSP Channel 13.		
		62	Tensilica DSP Channel 14.		
		63	Tensilica DSP Channel 15.		
		64	Input ASRC Channel 0.		
		65	Input ASRC Channel 1.		
		66	Input ASRC Channel 2.		
		67	Input ASRC Channel 3.		
		68	ADC Channel 0.		
		69	ADC Channel 1.		
		70	ADC Channel 2.		
		71	DMIC Channel 0.		
		72	DMIC Channel 1.		
		73	DMIC Channel 2.		
		74	DMIC Channel 3.		
		75	EQ0.		
		79	DMIC Channel 4.		
		80	DMIC Channel 5.		
		81	DMIC Channel 6.		
		82	DMIC Channel 7.		

SLOW TO FAST INTERPOLATOR CHANNEL 6 INPUT ROUTING REGISTER

Address: 0x4000C09A, Reset: 0x00, Name: FINT_ROUTE6

Table 223. Bit Descriptions for FINT_ROUTE6

Bits	Bit Name	Settings	Description	Reset	Access
7	RESERVED		Reserved.	0x0	R
[6:0]	FINT0_ROUTE		Slow to Fast Interpolator Channel 0 Input Routing.	0x0	R/W
		0	Serial Port 0 Channel 0.		
		1	Serial Port 0 Channel 1.		
		2	Serial Port 0 Channel 2.		
		3	Serial Port 0 Channel 3.		
		4	Serial Port 0 Channel 4.		
		5	Serial Port 0 Channel 5.		

Bits	Bit Name	Settings	Description	Reset	Access
		6	Serial Port 0 Channel 6.		
		7	Serial Port 0 Channel 7.		
		8	Serial Port 0 Channel 8.		
		9	Serial Port 0 Channel 9.		
		10	Serial Port 0 Channel 10.		
		11	Serial Port 0 Channel 11.		
		12	Serial Port 0 Channel 12.		
		13	Serial Port 0 Channel 13.		
		14	Serial Port 0 Channel 14.		
		15	Serial Port 0 Channel 15.		
		16	Serial Port 1 Channel 0.		
		17	Serial Port 1 Channel 1.		
		18	Serial Port 1 Channel 2.		
		19	Serial Port 1 Channel 3.		
		20	Serial Port 1 Channel 4.		
		21	Serial Port 1 Channel 5.		
		22	Serial Port 1 Channel 6.		
		23	Serial Port 1 Channel 7.		
		24	Serial Port 1 Channel 8.		
		25	Serial Port 1 Channel 9.		
		26	Serial Port 1 Channel 10.		
		27	Serial Port 1 Channel 11.		
		28	Serial Port 1 Channel 12.		
		29	Serial Port 1 Channel 13.		
		30	Serial Port 1 Channel 14.		
		31	Serial Port 1 Channel 15.		
		32	FastDSP Channel 0.		
		33	FastDSP Channel 1.		
		34	FastDSP Channel 2.		
		35	FastDSP Channel 3.		
		36	FastDSP Channel 4.		
		37	FastDSP Channel 5.		
		38	FastDSP Channel 6.		
		39	FastDSP Channel 7.		
		40	FastDSP Channel 8.		
		41	FastDSP Channel 9.		
		42	FastDSP Channel 10.		
		43	FastDSP Channel 11.		
		44	FastDSP Channel 12.		
		45	FastDSP Channel 13.		
		46	FastDSP Channel 14.		
		47	FastDSP Channel 15.		
		48	Tensilica DSP Channel 0.		
		49	Tensilica DSP Channel 1.		
		50	Tensilica DSP Channel 2.		
		51	Tensilica DSP Channel 3.		
		52	Tensilica DSP Channel 4.		
		53	Tensilica DSP Channel 5.		
		54	Tensilica DSP Channel 6.		
		55	Tensilica DSP Channel 7.		
		56	Tensilica DSP Channel 8.		

Bits	Bit Name	Settings	Description	Reset	Access
		57	Tensilica DSP Channel 9.		
		58	Tensilica DSP Channel 10.		
		59	Tensilica DSP Channel 11.		
		60	Tensilica DSP Channel 12.		
		61	Tensilica DSP Channel 13.		
		62	Tensilica DSP Channel 14.		
		63	Tensilica DSP Channel 15.		
		64	Input ASRC Channel 0.		
		65	Input ASRC Channel 1.		
		66	Input ASRC Channel 2.		
		67	Input ASRC Channel 3.		
		68	ADC Channel 0.		
		69	ADC Channel 1.		
		70	ADC Channel 2.		
		71	DMIC Channel 0.		
		72	DMIC Channel 1.		
		73	DMIC Channel 2.		
		74	DMIC Channel 3.		
		75	EQ0.		
		79	DMIC Channel 4.		
		80	DMIC Channel 5.		
		81	DMIC Channel 6.		
		82	DMIC Channel 7.		

SLOW TO FAST INTERPOLATOR CHANNEL 7 INPUT ROUTING REGISTER

Address: 0x4000C09B, Reset: 0x00, Name: FINT_ROUTE7

Table 224. Bit Descriptions for FINT_ROUTE7

Bits	Bit Name	Settings	Description	Reset	Access
7	RESERVED		Reserved.	0x0	R
[6:0]	FINT0_ROUTE		Slow to Fast Interpolator Channel 0 Input Routing.	0x0	R/W
		0	Serial Port 0 Channel 0.		
		1	Serial Port 0 Channel 1.		
		2	Serial Port 0 Channel 2.		
		3	Serial Port 0 Channel 3.		
		4	Serial Port 0 Channel 4.		
		5	Serial Port 0 Channel 5.		
		6	Serial Port 0 Channel 6.		
		7	Serial Port 0 Channel 7.		
		8	Serial Port 0 Channel 8.		
		9	Serial Port 0 Channel 9.		
		10	Serial Port 0 Channel 10.		
		11	Serial Port 0 Channel 11.		
		12	Serial Port 0 Channel 12.		
		13	Serial Port 0 Channel 13.		
		14	Serial Port 0 Channel 14.		
		15	Serial Port 0 Channel 15.		
		16	Serial Port 1 Channel 0.		
		17	Serial Port 1 Channel 1.		
		18	Serial Port 1 Channel 2.		

Bits	Bit Name	Settings	Description	Reset	Access
		19	Serial Port 1 Channel 3.		
		20	Serial Port 1 Channel 4.		
		21	Serial Port 1 Channel 5.		
		22	Serial Port 1 Channel 6.		
		23	Serial Port 1 Channel 7.		
		24	Serial Port 1 Channel 8.		
		25	Serial Port 1 Channel 9.		
		26	Serial Port 1 Channel 10.		
		27	Serial Port 1 Channel 11.		
		28	Serial Port 1 Channel 12.		
		29	Serial Port 1 Channel 13.		
		30	Serial Port 1 Channel 14.		
		31	Serial Port 1 Channel 15.		
		32	FastDSP Channel 0.		
		33	FastDSP Channel 1.		
		34	FastDSP Channel 2.		
		35	FastDSP Channel 3.		
		36	FastDSP Channel 4.		
		37	FastDSP Channel 5.		
		38	FastDSP Channel 6.		
		39	FastDSP Channel 7.		
		40	FastDSP Channel 8.		
		41	FastDSP Channel 9.		
		42	FastDSP Channel 10.		
		43	FastDSP Channel 11.		
		44	FastDSP Channel 12.		
		45	FastDSP Channel 13.		
		46	FastDSP Channel 14.		
		47	FastDSP Channel 15.		
		48	Tensilica DSP Channel 0.		
		49	Tensilica DSP Channel 1.		
		50	Tensilica DSP Channel 2.		
		51	Tensilica DSP Channel 3.		
		52	Tensilica DSP Channel 4.		
		53	Tensilica DSP Channel 5.		
		54	Tensilica DSP Channel 6.		
		55	Tensilica DSP Channel 7.		
		56	Tensilica DSP Channel 8.		
		57	Tensilica DSP Channel 9.		
		58	Tensilica DSP Channel 10.		
		59	Tensilica DSP Channel 11.		
		60	Tensilica DSP Channel 12.		
		61	Tensilica DSP Channel 13.		
		62	Tensilica DSP Channel 14.		
		63	Tensilica DSP Channel 15.		
		64	Input ASRC Channel 0.		
		65	Input ASRC Channel 1.		
		66	Input ASRC Channel 2.		
		67	Input ASRC Channel 3.		
		68	ADC Channel 0.		
		69	ADC Channel 1.		

Bits	Bit Name	Settings	Description	Reset	Access
		70	ADC Channel 2.		
		71	DMIC Channel 0.		
		72	DMIC Channel 1.		
		73	DMIC Channel 2.		
		74	DMIC Channel 3.		
		75	EQ0.		
		79	DMIC Channel 4.		
		80	DMIC Channel 5.		
		81	DMIC Channel 6.		
		82	DMIC Channel 7.		

INPUT ASRC CONTROL, SOURCE, AND RATE SELECTION REGISTER

Address: 0x4000C0A0, Reset: 0x02, Name: ASRCI_CTRL

Table 225. Bit Descriptions for ASRCI_CTRL

Bits	Bit Name	Settings	Description	Reset	Access
7	ASRCI_MORE_FILT		Input ASRC Additional Filtering Enable. This can enable additional filtering within the ASRC which can provide higher performance under some conditions.	0x0	R/W
		0	No additional voice band filter.		
		1	Voice Band Filter On.		
6	ASRCI_VFILT		Input ASRC Voice Filter Enable.	0x0	R/W
		0	Voice-Filter Off.		
		1	Voice-Filter On.		
5	ASRCI_LPM		Input ASRC Low Power Mode Selection.	0x0	R/W
		0	High-Performance Mode.		
		1	Low-Power Mode.		
4	ASRCI_SOURCE		Input ASRC Source.	0x0	R/W
		0	Serial Audio Port 0 source for Input ASRC.		
		1	Serial Audio Port 1 source for Input ASRC.		
3	ASRCI_LPM_II		Input ASRC Low Power Mode Selection II. Even lower power.	0x0	R/W
		0	High-Performance Mode.		
		1	Low-Power Mode.		
[2:0]	ASRCI_OUT_FS		Input ASRC Sample Rate Selection.	0x2	R/W
		000	12kHz Sample Rate.		
		001	24kHz Sample Rate.		
		010	48kHz Sample Rate.		
		011	96kHz Sample Rate.		
		100	192kHz Sample Rate.		

INPUT ASRC CHANNEL 0 AND CHANNEL 1 INPUT ROUTING REGISTER

Address: 0x4000C0A1, Reset: 0x00, Name: ASRCI_ROUTE01

Table 226. Bit Descriptions for ASRCI_ROUTE01

Bits	Bit Name	Settings	Description	Reset	Access
[7:4]	ASRCI1_ROUTE		Input ASRC Channel 1 routing.	0x0	R/W
		0	Serial Port Channel 0.		
		1	Serial Port Channel 1.		
		2	Serial Port Channel 2.		

Bits	Bit Name	Settings	Description	Reset	Access
		3	Serial Port Channel 3.		
		4	Serial Port Channel 4.		
		5	Serial Port Channel 5.		
		6	Serial Port Channel 6.		
		7	Serial Port Channel 7.		
		8	Serial Port Channel 8.		
		9	Serial Port Channel 9.		
		10	Serial Port Channel 10.		
		11	Serial Port Channel 11.		
		12	Serial Port Channel 12.		
		13	Serial Port Channel 13.		
		14	Serial Port Channel 14.		
		15	Serial Port Channel 15.		
[3:0]	ASRCI0_ROUTE		Input ASRC Channel 0 routing.	0x0	R/W
		0	Serial Port Channel 0.		
		1	Serial Port Channel 1.		
		2	Serial Port Channel 2.		
		3	Serial Port Channel 3.		
		4	Serial Port Channel 4.		
		5	Serial Port Channel 5.		
		6	Serial Port Channel 6.		
		7	Serial Port Channel 7.		
		8	Serial Port Channel 8.		
		9	Serial Port Channel 9.		
		10	Serial Port Channel 10.		
		11	Serial Port Channel 11.		
		12	Serial Port Channel 12.		
		13	Serial Port Channel 13.		
		14	Serial Port Channel 14.		
		15	Serial Port Channel 15.		

INPUT ASRC CHANNEL 2 AND CHANNEL 3 INPUT ROUTING REGISTER

Address: 0x4000C0A2, Reset: 0x00, Name: ASRCI_ROUTE23

Table 227. Bit Descriptions for ASRCI_ROUTE23

Bits	Bit Name	Settings	Description	Reset	Access
[7:4]	ASRCI3_ROUTE		Input ASRC Channel 3 routing.	0x0	R/W
		0	Serial Port Channel 0.		
		1	Serial Port Channel 1.		
		2	Serial Port Channel 2.		
		3	Serial Port Channel 3.		
		4	Serial Port Channel 4.		
		5	Serial Port Channel 5.		
		6	Serial Port Channel 6.		
		7	Serial Port Channel 7.		
		8	Serial Port Channel 8.		
		9	Serial Port Channel 9.		
		10	Serial Port Channel 10.		
		11	Serial Port Channel 11.		
		12	Serial Port Channel 12.		

Bits	Bit Name	Settings	Description	Reset	Access
		13	Serial Port Channel 13.		
		14	Serial Port Channel 14.		
		15	Serial Port Channel 15.		
[3:0]	ASRCI2_ROUTE		Input ASRC Channel 2 routing.	0x0	R/W
		0	Serial Port Channel 0.		
		1	Serial Port Channel 1.		
		2	Serial Port Channel 2.		
		3	Serial Port Channel 3.		
		4	Serial Port Channel 4.		
		5	Serial Port Channel 5.		
		6	Serial Port Channel 6.		
		7	Serial Port Channel 7.		
		8	Serial Port Channel 8.		
		9	Serial Port Channel 9.		
		10	Serial Port Channel 10.		
		11	Serial Port Channel 11.		
		12	Serial Port Channel 12.		
		13	Serial Port Channel 13.		
		14	Serial Port Channel 14.		
		15	Serial Port Channel 15.		

OUTPUT ASRC CONTROL REGISTER

Address: 0x4000C0A3, Reset: 0x02, Name: ASRCO_CTRL

Table 228. Bit Descriptions for ASRCO_CTRL

Bits	Bit Name	Settings	Description	Reset	Access
7	ASRCO_MORE_FILT		Output ASRC Additional Filtering Enable. This can enable additional filtering within the ASRC which can provide higher performance under some conditions.	0x0	R/W
		0	No additional voice band filter.		
		1	Voice Band Filter On.		
6	ASRCO_VFILT		Output ASRC Voice Filter Enable.	0x0	R/W
		0	Voice-Filter Off.		
		1	Voice-Filter On.		
5	ASRCO_LPM		Output ASRC Low Power Mode Selection.	0x0	R/W
		0	High-Performance Mode.		
		1	Low-Power Mode.		
4	ASRCO_SAI_SEL		Output ASRC Serial Port External Rate Source Selection.	0x0	R/W
		0	Use Serial Port 0 as rate source.		
		1	Use Serial Port 1 as rate source.		
3	ASRCO_LPM_II		Output ASRC Low Power Mode Selection. Even lower power.	0x0	R/W
		0	High-Performance Mode.		
		1	Low-Power Mode.		
[2:0]	ASRCO_IN_FS		Output ASRC Input Sample Rate Selection.	0x2	R/W
		000	12kHz Sample Rate.		
		001	24kHz Sample Rate.		
		010	48kHz Sample Rate.		
		011	96kHz Sample Rate.		
		100	192kHz Sample Rate.		

OUTPUT ASRC CHANNEL 0 INPUT ROUTING REGISTER

Address: 0x4000C0A4, Reset: 0x00, Name: ASRCO_ROUTE0

Table 229. Bit Descriptions for ASRCO_ROUTE0

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
[5:0]	ASRCO0_ROUTE		Output ASRC Channel 0 Input Routing.	0x0	R/W
		0	FastDSP Channel 0.		
		1	FastDSP Channel 1.		
		2	FastDSP Channel 2.		
		3	FastDSP Channel 3.		
		4	FastDSP Channel 4.		
		5	FastDSP Channel 5.		
		6	FastDSP Channel 6.		
		7	FastDSP Channel 7.		
		8	FastDSP Channel 8.		
		9	FastDSP Channel 9.		
		10	FastDSP Channel 10.		
		11	FastDSP Channel 11.		
		12	FastDSP Channel 12.		
		13	FastDSP Channel 13.		
		14	FastDSP Channel 14.		
		15	FastDSP Channel 15.		
		16	Tensilica DSP Channel 0.		
		17	Tensilica DSP Channel 1.		
		18	Tensilica DSP Channel 2.		
		19	Tensilica DSP Channel 3.		
		20	Tensilica DSP Channel 4.		
		21	Tensilica DSP Channel 5.		
		22	Tensilica DSP Channel 6.		
		23	Tensilica DSP Channel 7.		
		24	Tensilica DSP Channel 8.		
		25	Tensilica DSP Channel 9.		
		26	Tensilica DSP Channel 10.		
		27	Tensilica DSP Channel 11.		
		28	Tensilica DSP Channel 12.		
		29	Tensilica DSP Channel 13.		
		30	Tensilica DSP Channel 14.		
		31	Tensilica DSP Channel 15.		
		32	ADC Channel 0.		
		33	ADC Channel 1.		
		34	ADC Channel 2.		
		35	DMIC Channel 0.		
		36	DMIC Channel 1.		
		37	DMIC Channel 2.		
		38	DMIC Channel 3.		
		39	Fast to Slow Decimator Channel 0.		
		40	Fast to Slow Decimator Channel 1.		
		41	Fast to Slow Decimator Channel 2.		
		42	Fast to Slow Decimator Channel 3.		
		43	Fast to Slow Decimator Channel 4.		

Bits	Bit Name	Settings	Description	Reset	Access
		44	Fast to Slow Decimator Channel 5.		
		45	Fast to Slow Decimator Channel 6.		
		46	Fast to Slow Decimator Channel 7.		
		47	EQ0.		
		51	DMIC Channel 4.		
		52	DMIC Channel 5.		
		53	DMIC Channel 6.		
		54	DMIC Channel 7.		

OUTPUT ASRC CHANNEL 1 INPUT ROUTING REGISTER

Address: 0x4000C0A5, Reset: 0x00, Name: ASRCO_ROUTE1

Table 230. Bit Descriptions for ASRCO_ROUTE1

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
[5:0]	ASRCO0_ROUTE		Output ASRC Channel 0 Input Routing.	0x0	R/W
		0	FastDSP Channel 0.		
		1	FastDSP Channel 1.		
		2	FastDSP Channel 2.		
		3	FastDSP Channel 3.		
		4	FastDSP Channel 4.		
		5	FastDSP Channel 5.		
		6	FastDSP Channel 6.		
		7	FastDSP Channel 7.		
		8	FastDSP Channel 8.		
		9	FastDSP Channel 9.		
		10	FastDSP Channel 10.		
		11	FastDSP Channel 11.		
		12	FastDSP Channel 12.		
		13	FastDSP Channel 13.		
		14	FastDSP Channel 14.		
		15	FastDSP Channel 15.		
		16	Tensilica DSP Channel 0.		
		17	Tensilica DSP Channel 1.		
		18	Tensilica DSP Channel 2.		
		19	Tensilica DSP Channel 3.		
		20	Tensilica DSP Channel 4.		
		21	Tensilica DSP Channel 5.		
		22	Tensilica DSP Channel 6.		
		23	Tensilica DSP Channel 7.		
		24	Tensilica DSP Channel 8.		
		25	Tensilica DSP Channel 9.		
		26	Tensilica DSP Channel 10.		
		27	Tensilica DSP Channel 11.		
		28	Tensilica DSP Channel 12.		
		29	Tensilica DSP Channel 13.		
		30	Tensilica DSP Channel 14.		
		31	Tensilica DSP Channel 15.		
		32	ADC Channel 0.		
		33	ADC Channel 1.		

Bits	Bit Name	Settings	Description	Reset	Access
		34	ADC Channel 2.		
		35	DMIC Channel 0.		
		36	DMIC Channel 1.		
		37	DMIC Channel 2.		
		38	DMIC Channel 3.		
		39	Fast to Slow Decimator Channel 0.		
		40	Fast to Slow Decimator Channel 1.		
		41	Fast to Slow Decimator Channel 2.		
		42	Fast to Slow Decimator Channel 3.		
		43	Fast to Slow Decimator Channel 4.		
		44	Fast to Slow Decimator Channel 5.		
		45	Fast to Slow Decimator Channel 6.		
		46	Fast to Slow Decimator Channel 7.		
		47	EQ0.		
		51	DMIC Channel 4.		
		52	DMIC Channel 5.		
		53	DMIC Channel 6.		
		54	DMIC Channel 7.		

OUTPUT ASRC CHANNEL 2 INPUT ROUTING REGISTER

Address: 0x4000C0A6, Reset: 0x00, Name: ASRCO_ROUTE2

Table 231. Bit Descriptions for ASRCO_ROUTE2

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
[5:0]	ASRCO0_ROUTE		Output ASRC Channel 0 Input Routing.	0x0	R/W
		0	FastDSP Channel 0.		
		1	FastDSP Channel 1.		
		2	FastDSP Channel 2.		
		3	FastDSP Channel 3.		
		4	FastDSP Channel 4.		
		5	FastDSP Channel 5.		
		6	FastDSP Channel 6.		
		7	FastDSP Channel 7.		
		8	FastDSP Channel 8.		
		9	FastDSP Channel 9.		
		10	FastDSP Channel 10.		
		11	FastDSP Channel 11.		
		12	FastDSP Channel 12.		
		13	FastDSP Channel 13.		
		14	FastDSP Channel 14.		
		15	FastDSP Channel 15.		
		16	Tensilica DSP Channel 0.		
		17	Tensilica DSP Channel 1.		
		18	Tensilica DSP Channel 2.		
		19	Tensilica DSP Channel 3.		
		20	Tensilica DSP Channel 4.		
		21	Tensilica DSP Channel 5.		
		22	Tensilica DSP Channel 6.		
		23	Tensilica DSP Channel 7.		

Bits	Bit Name	Settings	Description	Reset	Access
		24	Tensilica DSP Channel 8.		
		25	Tensilica DSP Channel 9.		
		26	Tensilica DSP Channel 10.		
		27	Tensilica DSP Channel 11.		
		28	Tensilica DSP Channel 12.		
		29	Tensilica DSP Channel 13.		
		30	Tensilica DSP Channel 14.		
		31	Tensilica DSP Channel 15.		
		32	ADC Channel 0.		
		33	ADC Channel 1.		
		34	ADC Channel 2.		
		35	DMIC Channel 0.		
		36	DMIC Channel 1.		
		37	DMIC Channel 2.		
		38	DMIC Channel 3.		
		39	Fast to Slow Decimator Channel 0.		
		40	Fast to Slow Decimator Channel 1.		
		41	Fast to Slow Decimator Channel 2.		
		42	Fast to Slow Decimator Channel 3.		
		43	Fast to Slow Decimator Channel 4.		
		44	Fast to Slow Decimator Channel 5.		
		45	Fast to Slow Decimator Channel 6.		
		46	Fast to Slow Decimator Channel 7.		
		47	EQ0.		
		51	DMIC Channel 4.		
		52	DMIC Channel 5.		
		53	DMIC Channel 6.		
		54	DMIC Channel 7.		

OUTPUT ASRC CHANNEL 3 INPUT ROUTING REGISTER

Address: 0x4000C0A7, Reset: 0x00, Name: ASRCO_ROUTE3

Table 232. Bit Descriptions for ASRCO_ROUTE3

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
[5:0]	ASRCO0_ROUTE		Output ASRC Channel 0 Input Routing.	0x0	R/W
		0	FastDSP Channel 0.		
		1	FastDSP Channel 1.		
		2	FastDSP Channel 2.		
		3	FastDSP Channel 3.		
		4	FastDSP Channel 4.		
		5	FastDSP Channel 5.		
		6	FastDSP Channel 6.		
		7	FastDSP Channel 7.		
		8	FastDSP Channel 8.		
		9	FastDSP Channel 9.		
		10	FastDSP Channel 10.		
		11	FastDSP Channel 11.		
		12	FastDSP Channel 12.		
		13	FastDSP Channel 13.		

Bits	Bit Name	Settings	Description	Reset	Access
		14	FastDSP Channel 14.		
		15	FastDSP Channel 15.		
		16	Tensilica DSP Channel 0.		
		17	Tensilica DSP Channel 1.		
		18	Tensilica DSP Channel 2.		
		19	Tensilica DSP Channel 3.		
		20	Tensilica DSP Channel 4.		
		21	Tensilica DSP Channel 5.		
		22	Tensilica DSP Channel 6.		
		23	Tensilica DSP Channel 7.		
		24	Tensilica DSP Channel 8.		
		25	Tensilica DSP Channel 9.		
		26	Tensilica DSP Channel 10.		
		27	Tensilica DSP Channel 11.		
		28	Tensilica DSP Channel 12.		
		29	Tensilica DSP Channel 13.		
		30	Tensilica DSP Channel 14.		
		31	Tensilica DSP Channel 15.		
		32	ADC Channel 0.		
		33	ADC Channel 1.		
		34	ADC Channel 2.		
		35	DMIC Channel 0.		
		36	DMIC Channel 1.		
		37	DMIC Channel 2.		
		38	DMIC Channel 3.		
		39	Fast to Slow Decimator Channel 0.		
		40	Fast to Slow Decimator Channel 1.		
		41	Fast to Slow Decimator Channel 2.		
		42	Fast to Slow Decimator Channel 3.		
		43	Fast to Slow Decimator Channel 4.		
		44	Fast to Slow Decimator Channel 5.		
		45	Fast to Slow Decimator Channel 6.		
		46	Fast to Slow Decimator Channel 7.		
		47	EQ0.		
		51	DMIC Channel 4.		
		52	DMIC Channel 5.		
		53	DMIC Channel 6.		
		54	DMIC Channel 7.		

FASTDSP RUN REGISTER

Address: 0x4000C0B0, Reset: 0x00, Name: FDSP_RUN

Table 233. Bit Descriptions for FDSP_RUN

Bits	Bit Name	Settings	Description	Reset	Access
[7:1]	RESERVED		Reserved.	0x0	R
0	FDSP_RUN		Allow FDSP to run with go signal.	0x0	R/W
		0	FastDSP has no go signal. Not running, but Memories can be loaded if FDSP_EN=1.		
		1	FastDSP has go signal and is running.		

FASTDSP CURRENT BANK AND BANK RAMPING CONTROLS REGISTER

Address: 0x4000C0B1, Reset: 0x70, Name: FDSP_CTRL1

Table 234. Bit Descriptions for FDSP_CTRL1

Bits	Bit Name	Settings	Description	Reset	Access
[7:4]	FDSP_RAMP_RATE		FastDSP Parameter Bank Ramp Rate of Change. Determines time to complete full ramp from one bank to another.	0x7	R/W
		0	0.02s Ramp.		
		1	0.04s Ramp.		
		2	0.06s Ramp.		
		3	0.08s Ramp.		
		4	0.1s Ramp.		
		5	0.15s Ramp.		
		6	0.2s Ramp.		
		7	0.25s Ramp.		
		8	0.3s Ramp.		
		9	0.5s Ramp.		
		10	0.75s Ramp.		
		11	1s Ramp.		
		12	1.25s Ramp.		
		13	1.5s Ramp.		
		14	1.75s Ramp.		
		15	2s Ramp.		
3	FDSP_ZERO_STATE		Zeroes the state of the FastDSP data memory during bank switching. When switching active parameter banks between two settings, zeroing the state of the bank prevents the new filter settings from being active on old data that is recirculating in filters. Zeroing the state may prevent filter instability or unwanted noises upon bank switching.	0x0	R/W
		0	Do not zero state during bank switch.		
		1	Zero state during back switch.		
2	FDSP_RAMP_MODE		FastDSP Parameter Bank Ramp Mode.	0x0	R/W
		0	Parameters will linearly ramp when current bank is changed.		
		1	Parameters will instantly change when current bank is changed.		
[1:0]	FDSP_BANK_SEL		FastDSP Current Parameter Bank Selection.	0x0	R/W
		0	FDSP uses Parameter Bank A.		
		1	FDSP uses Parameter Bank B.		
		10	FDSP uses Parameter Bank C.		

FASTDSP BANK RAMPING STOP POINT REGISTER

Address: 0x4000C0B2, Reset: 0x3F, Name: FDSP_CTRL2

Table 235. Bit Descriptions for FDSP_CTRL2

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
[5:0]	FDSP_LAMBDA		FastDSP Bank Switch Ramp Stop Point. Lambda is a 6-Bit value representing the point along the linear interpolation curve between two banks at which the bank ramp switch will stop. 0=((63/64)*A + (1/64)*B), 1 =((62/64)*A + (2/64)*B), …, 62 = ((1/64)*A + (63/64)*B), 63=B (default) Lambda can be updated "on the fly" via control interface. In order to complete a bank switch a value of 63 (default setting) must be set. Actual current ramp point (current_lambda : 0-63) can be read via a status register. When this reaches 63 the bank switch is complete and current parameters used will match current bank. Actual step size of linear interpolation will be ~12-bits (4096 steps) Parameters in banks that are being ramped between should not change during a bank switch	0x3F	R/W
		0	Bank switch parameter ramp will stop at 1/64 of full ramp.		
		1	Bank switch parameter ramp will stop at 2/64 of full ramp.		
		2-61	…		
		62	Bank switch parameter ramp will stop at 63/64 of full ramp.		
		63	Bank switch parameter ramp will complete ramp to current bank.		

FASTDSP BANK COPYING REGISTER

Address: 0x4000C0B3, Reset: 0x00, Name: FDSP_CTRL3

Table 236. Bit Descriptions for FDSP_CTRL3

Bits	Bit Name	Settings	Description	Reset	Access
7	FDSP_COPY_DONE_CLEAR		FastDSP Copy Parameter Done Clear.	0x0	W
		0	Normal Operation.		
		1	Writing of 1 will clear FDSP_BANK_COPY_DONE bit.		
6	RESERVED		Reserved.	0x0	R
5	FDSP_COPY_CB		FastDSP Copy Parameter Bank C to Bank B.	0x0	W
		0	Normal Operation.		
		1	Writing of 1 will copy bank.		
4	FDSP_COPY_CA		FastDSP Copy Parameter Bank C to Bank A.	0x0	W
		0	Normal Operation.		
		1	Writing of 1 will copy bank.		
3	FDSP_COPY_BC		FastDSP Copy Parameter Bank B to Bank C.	0x0	W
		0	Normal Operation.		
		1	Writing of 1 will copy bank.		
2	FDSP_COPY_BA		FastDSP Copy Parameter Bank B to Bank A.	0x0	W
		0	Normal Operation.		
		1	Writing of 1 will copy bank.		
1	FDSP_COPY_AC		FastDSP Copy Parameter Bank A to Bank C.	0x0	W
		0	Normal Operation.		
		1	Writing of 1 will copy bank.		
0	FDSP_COPY_AB		FastDSP Copy Parameter Bank A to Bank B.	0x0	W
		0	Normal Operation.		
		1	Writing of 1 will copy bank.		

FASTDSP FRAME RATE SOURCE REGISTER

Address: 0x4000C0B4, Reset: 0x00, Name: FDSP_CTRL4

Table 237. Bit Descriptions for FDSP_CTRL4

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
5	FDSP_ONZ_SOURCE		FastDSP Output Non-zero source.	0x0	R/W
		0	Channel 14.		
		1	Channel 15.		
4	FDSP_EXP_ATK_SPEED		FastDSP Expander attack/ramp-down speed.	0x0	R/W
		0	Normal ramp down speed.		
		1	Force expander to ramp down at 1/1024 speed for longer attack times.		
[3:0]	FDSP_RATE_SOURCE		FastDSP frame rate source.	0x0	R/W
		0	ADC01.		
		1	ADC2.		
		2	DMIC01.		
		3	DMIC23.		
		6	SAI0.		
		7	SAI1.		
		10	FINT01.		
		11	FINT23.		
		12	FINT45.		
		13	FINT67.		
		14	ASRCI.		
		15	FIXED.		

FASTDSP FIXED RATE DIVISION MSBS REGISTER

Address: 0x4000C0B5, Reset: 0x7F, Name: FDSP_CTRL5

Table 238. Bit Descriptions for FDSP_CTRL5

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	FDSP_RATE_DIV[7:0]		FastDSP go signal division. Number of 24.576MHz clock cycles between go signal is FDSP_RATE_DIV minus 1 when FDSP_RATE_SOURCE set to fixed.	0x7F	R/W

FASTDSP FIXED RATE DIVISION LSBS REGISTER

Address: 0x4000C0B6, Reset: 0x00, Name: FDSP_CTRL6

Table 239. Bit Descriptions for FDSP_CTRL6

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	FDSP_RATE_DIV[15:8]		FastDSP go signal division. Number of 24.576MHz clock cycles between go signal is FDSP_RATE_DIV minus 1 when FDSP_RATE_SOURCE set to fixed.	0x0	R/W

FASTDSP MODULO N COUNTER FOR LOWER RATE CONDITIONAL EXECUTION REGISTER

Address: 0x4000C0B7, Reset: 0x00, Name: FDSP_CTRL7

Table 240. Bit Descriptions for FDSP_CTRL7

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
[5:0]	FDSP_MOD_N		FastDSP Modulo N counter reset for conditional execution.	0x0	R/W

FASTDSP GENERIC CONDITIONAL EXECUTION REGISTERS

Address: 0x4000C0B8, Reset: 0x00, Name: FDSP_CTRL8

Table 241. Bit Descriptions for FDSP_CTRL8

Bits	Bit Name	Settings	Description	Reset	Access
7	FDSP_REG_COND7		FastDSP Generic Register for Conditional Execution. The value of this register can be used for conditional instruction execution in the FastDSP	0x0	R/W
		0	Conditional Register is 0.		
		1	Conditional Register is 1.		
6	FDSP_REG_COND6		FastDSP Generic Register for Conditional Execution. The value of this register can be used for conditional instruction execution in the FastDSP	0x0	R/W
		0	Conditional Register is 0.		
		1	Conditional Register is 1.		
5	FDSP_REG_COND5		FastDSP Generic Register for Conditional Execution. The value of this register can be used for conditional instruction execution in the FastDSP	0x0	R/W
		0	Conditional Register is 0.		
		1	Conditional Register is 1.		
4	FDSP_REG_COND4		FastDSP Generic Register for Conditional Execution. The value of this register can be used for conditional instruction execution in the FastDSP	0x0	R/W
		0	Conditional Register is 0.		
		1	Conditional Register is 1.		
3	FDSP_REG_COND3		FastDSP Generic Register for Conditional Execution. The value of this register can be used for conditional instruction execution in the FastDSP	0x0	R/W
		0	Conditional Register is 0.		
		1	Conditional Register is 1.		
2	FDSP_REG_COND2		FastDSP Generic Register for Conditional Execution. The value of this register can be used for conditional instruction execution in the FastDSP	0x0	R/W
		0	Conditional Register is 0.		
		1	Conditional Register is 1.		
1	FDSP_REG_COND1		FastDSP Generic Register for Conditional Execution. The value of this register can be used for conditional instruction execution in the FastDSP	0x0	R/W
		0	Conditional Register is 0.		
		1	Conditional Register is 1.		
0	FDSP_REG_COND0		FastDSP Generic Register for Conditional Execution. The value of this register can be used for conditional instruction execution in the FastDSP	0x0	R/W
		0	Conditional Register is 0.		
		1	Conditional Register is 1.		

FAST DSP SAFELOAD ADDRESS REGISTER

Address: 0x4000C0B9, Reset: 0x00, Name: FDSP_SL_ADDR

Table 242. Bit Descriptions for FDSP_SL_ADDR

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
[5:0]	FDSP_SL_ADDR		FastDSP Safeload Instruction Number.	0x0	R/W

FASTDSP SAFELOAD PARAMETER 0 VALUE REGISTER

Address: 0x4000C0BA, Reset: 0x00, Name: FDSP_SL_P0_0

Table 243. Bit Descriptions for FDSP_SL_P0_0

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	FDSP_SL_P0[7:0]		FastDSP Safeload Parameter 0 (B0 coeff) value to be written.	0x0	R/W

FASTDSP SAFELOAD PARAMETER 0 VALUE REGISTER

Address: 0x4000C0BB, Reset: 0x00, Name: FDSP_SL_P0_1

Table 244. Bit Descriptions for FDSP_SL_P0_1

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	FDSP_SL_P0[15:8]		FastDSP Safeload Parameter 0 (B0 coeff) value to be written.	0x0	R/W

FASTDSP SAFELOAD PARAMETER 0 VALUE REGISTER

Address: 0x4000C0BC, Reset: 0x00, Name: FDSP_SL_P0_2

Table 245. Bit Descriptions for FDSP_SL_P0_2

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	FDSP_SL_P0[23:16]		FastDSP Safeload Parameter 0 (B0 coeff) value to be written.	0x0	R/W

FASTDSP SAFELOAD PARAMETER 0 VALUE REGISTER

Address: 0x4000C0BD, Reset: 0x00, Name: FDSP_SL_P0_3

Table 246. Bit Descriptions for FDSP_SL_P0_3

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	FDSP_SL_P0[31:24]		FastDSP Safeload Parameter 0 (B0 coeff) value to be written.	0x0	R/W

FASTDSP SAFELOAD PARAMETER 1 VALUE REGISTER

Address: 0x4000C0BE, Reset: 0x00, Name: FDSP_SL_P1_0

Table 247. Bit Descriptions for FDSP_SL_P1_0

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	FDSP_SL_P1[7:0]		FastDSP Safeload Parameter 1 (B1 coeff) value to be written.	0x0	R/W

FASTDSP SAFELOAD PARAMETER 1 VALUE REGISTER

Address: 0x4000C0BF, Reset: 0x00, Name: FDSP_SL_P1_1

Table 248. Bit Descriptions for FDSP_SL_P1_1

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	FDSP_SL_P1[15:8]		FastDSP Safeload Parameter 1 (B1 coeff) value to be written.	0x0	R/W

FASTDSP SAFELOAD PARAMETER 1 VALUE REGISTER

Address: 0x4000C0C0, Reset: 0x00, Name: FDSP_SL_P1_2

Table 249. Bit Descriptions for FDSP_SL_P1_2

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	FDSP_SL_P1[23:16]		FastDSP Safeload Parameter 1 (B1 coeff) value to be written.	0x0	R/W

FASTDSP SAFELOAD PARAMETER 1 VALUE REGISTER

Address: 0x4000C0C1, Reset: 0x00, Name: FDSP_SL_P1_3

Table 250. Bit Descriptions for FDSP_SL_P1_3

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	FDSP_SL_P1[31:24]		FastDSP Safeload Parameter 1 (B1 coeff) value to be written.	0x0	R/W

FASTDSP SAFELOAD PARAMETER 2 VALUE REGISTER

Address: 0x4000C0C2, Reset: 0x00, Name: FDSP_SL_P2_0

Table 251. Bit Descriptions for FDSP_SL_P2_0

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	FDSP_SL_P2[7:0]		FastDSP Safeload Parameter 2 (B2 coeff) value to be written.	0x0	R/W

FASTDSP SAFELOAD PARAMETER 2 VALUE REGISTER

Address: 0x4000C0C3, Reset: 0x00, Name: FDSP_SL_P2_1

Table 252. Bit Descriptions for FDSP_SL_P2_1

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	FDSP_SL_P2[15:8]		FastDSP Safeload Parameter 2 (B2 coeff) value to be written.	0x0	R/W

FASTDSP SAFELOAD PARAMETER 2 VALUE REGISTER

Address: 0x4000C0C4, Reset: 0x00, Name: FDSP_SL_P2_2

Table 253. Bit Descriptions for FDSP_SL_P2_2

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	FDSP_SL_P2[23:16]		FastDSP Safeload Parameter 2 (B2 coeff) value to be written.	0x0	R/W

FASTDSP SAFELOAD PARAMETER 2 VALUE REGISTER

Address: 0x4000C0C5, Reset: 0x00, Name: FDSP_SL_P2_3

Table 254. Bit Descriptions for FDSP_SL_P2_3

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	FDSP_SL_P2[31:24]		FastDSP Safeload Parameter 2 (B2 coeff) value to be written.	0x0	R/W

FASTDSP SAFELOAD PARAMETER 3 VALUE REGISTER

Address: 0x4000C0C6, Reset: 0x00, Name: FDSP_SL_P3_0

Table 255. Bit Descriptions for FDSP_SL_P3_0

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	FDSP_SL_P3[7:0]		FastDSP Safeload Parameter 3 (A1 coeff) value to be written.	0x0	R/W

FASTDSP SAFELOAD PARAMETER 3 VALUE REGISTER

Address: 0x4000C0C7, Reset: 0x00, Name: FDSP_SL_P3_1

Table 256. Bit Descriptions for FDSP_SL_P3_1

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	FDSP_SL_P3[15:8]		FastDSP Safeload Parameter 3 (A1 coeff) value to be written.	0x0	R/W

FASTDSP SAFELOAD PARAMETER 3 VALUE REGISTER

Address: 0x4000C0C8, Reset: 0x00, Name: FDSP_SL_P3_2

Table 257. Bit Descriptions for FDSP_SL_P3_2

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	FDSP_SL_P3[23:16]		FastDSP Safeload Parameter 3 (A1 coeff) value to be written.	0x0	R/W

FASTDSP SAFELOAD PARAMETER 3 VALUE REGISTER

Address: 0x4000C0C9, Reset: 0x00, Name: FDSP_SL_P3_3

Table 258. Bit Descriptions for FDSP_SL_P3_3

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	FDSP_SL_P3[31:24]		FastDSP Safeload Parameter 3 (A1 coeff) value to be written.	0x0	R/W

FASTDSP SAFELOAD PARAMETER 4 VALUE REGISTER

Address: 0x4000C0CA, Reset: 0x00, Name: FDSP_SL_P4_0

Table 259. Bit Descriptions for FDSP_SL_P4_0

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	FDSP_SL_P4[7:0]		FastDSP Safeload Parameter 4 (A2 coeff) value to be written.	0x0	R/W

FASTDSP SAFELOAD PARAMETER 4 VALUE REGISTER

Address: 0x4000C0CB, Reset: 0x00, Name: FDSP_SL_P4_1

Table 260. Bit Descriptions for FDSP_SL_P4_1

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	FDSP_SL_P4[15:8]		FastDSP Safeload Parameter 4 (A2 coeff) value to be written.	0x0	R/W

FASTDSP SAFELOAD PARAMETER 4 VALUE REGISTER

Address: 0x4000C0CC, Reset: 0x00, Name: FDSP_SL_P4_2

Table 261. Bit Descriptions for FDSP_SL_P4_2

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	FDSP_SL_P4[23:16]		FastDSP Safeload Parameter 4 (A2 coeff) value to be written.	0x0	R/W

FASTDSP SAFELOAD PARAMETER 4 VALUE REGISTER

Address: 0x4000C0CD, Reset: 0x00, Name: FDSP_SL_P4_3

Table 262. Bit Descriptions for FDSP_SL_P4_3

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	FDSP_SL_P4[31:24]		FastDSP Safeload Parameter 4 (A2 coeff) value to be written.	0x0	R/W

FASTDSP SAFELOAD UPDATE REGISTER

Address: 0x4000C0CE, Reset: 0x00, Name: FDSP_SL_UPDATE

Table 263. Bit Descriptions for FDSP_SL_UPDATE

Bits	Bit Name	Settings	Description	Reset	Access
[7:1]	RESERVED		Reserved.	0x0	R
0	FDSP_SL_UPDATE		FastDSP Safeload Update. Writing a 1 to this register will write the parameter values in the FDSP_SL_Px registers to the addresses in the current bank associated with the instruction number in the FDSP_SL_ADDR register at the beginning of the next frame.	0x0	W
		0	No Action.		
		1	Writing of 1 will cause update of safeload parameters at the beginning of next frame.		

FASTDSP SAFELOAD PARAMETER 0 VALUE REGISTER

Address: 0x4000C0CF, Reset: 0x00, Name: FDSP_ONZ_MASK0

Table 264. Bit Descriptions for FDSP_ONZ_MASK0

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	FDSP_ONZ_MASK[7:0]		FastDSP Safeload Parameter 0 (B0 coeff) value to be written.	0x0	R/W

FASTDSP SAFELOAD PARAMETER 0 VALUE REGISTER

Address: 0x4000C0D0, Reset: 0x00, Name: FDSP_ONZ_MASK1

Table 265. Bit Descriptions for FDSP_ONZ_MASK1

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	FDSP_ONZ_MASK[15:8]		FastDSP Safeload Parameter 0 (B0 coeff) value to be written.	0x0	R/W

FASTDSP SAFELOAD PARAMETER 0 VALUE REGISTER

Address: 0x4000C0D1, Reset: 0x00, Name: FDSP_ONZ_MASK2

Table 266. Bit Descriptions for FDSP_ONZ_MASK2

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	FDSP_ONZ_MASK[23:16]		FastDSP Safeload Parameter 0 (B0 coeff) value to be written.	0x0	R/W

EQ CONFIGURE REGISTER

Address: 0x4000C0D2, Reset: 0x00, Name: EQ_CFG

Table 267. Bit Descriptions for EQ_CFG

Bits	Bit Name	Settings	Description	Reset	Access
[7:5]	RESERVED		Reserved.	0x0	R
4	EQ_CLEAR		Write 1 to clear data memory inside EQ.	0x0	W
		0	No Action.		
		1	Writing of 1 will cause clear of eq memories.		
[3:2]	RESERVED		Reserved.	0x0	R
1	EQ_BANK_SEL		Indicate which ram to be selected as parameter from two ram.	0x0	R/W
		0	EQ Selects Bank0 Parameter.		
		1	EQ Selects Bank1 Parameter.		
0	EQ_RUN		Control EQ is running or not.	0x0	R/W
		0	EQ engine is not running.		
		1	EQ engine is running.		

EQ ROUTING REGISTER

Address: 0x4000C0D3, Reset: 0x00, Name: EQ_ROUTE

Table 268. Bit Descriptions for EQ_ROUTE

Bits	Bit Name	Settings	Description	Reset	Access
7	RESERVED		Reserved.	0x0	R
[6:0]	EQ_ROUTE		EQ Input Routing.	0x0	R/W
		00	Serial Port 0 Channel 0.		
		01	Serial Port 0 Channel 1.		
		02	Serial Port 0 Channel 2.		
		03	Serial Port 0 Channel 3.		
		04	Serial Port 0 Channel 4.		
		05	Serial Port 0 Channel 5.		
		06	Serial Port 0 Channel 6.		
		07	Serial Port 0 Channel 7.		
		08	Serial Port 0 Channel 8.		
		09	Serial Port 0 Channel 9.		
		10	Serial Port 0 Channel 10.		
		11	Serial Port 0 Channel 11.		
		12	Serial Port 0 Channel 12.		
		13	Serial Port 0 Channel 13.		
		14	Serial Port 0 Channel 14.		
		15	Serial Port 0 Channel 15.		
		16	Serial Port 1 Channel 0.		
		17	Serial Port 1 Channel 1.		

Bits	Bit Name	Settings	Description	Reset	Access
		18	Serial Port 1 Channel 2.		
		19	Serial Port 1 Channel 3.		
		20	Serial Port 1 Channel 4.		
		21	Serial Port 1 Channel 5.		
		22	Serial Port 1 Channel 6.		
		23	Serial Port 1 Channel 7.		
		24	Serial Port 1 Channel 8.		
		25	Serial Port 1 Channel 9.		
		26	Serial Port 1 Channel 10.		
		27	Serial Port 1 Channel 11.		
		28	Serial Port 1 Channel 12.		
		29	Serial Port 1 Channel 13.		
		30	Serial Port 1 Channel 14.		
		31	Serial Port 1 Channel 15.		
		32	FastDSP Channel 0.		
		33	FastDSP Channel 1.		
		34	FastDSP Channel 2.		
		35	FastDSP Channel 3.		
		36	FastDSP Channel 4.		
		37	FastDSP Channel 5.		
		38	FastDSP Channel 6.		
		39	FastDSP Channel 7.		
		40	FastDSP Channel 8.		
		41	FastDSP Channel 9.		
		42	FastDSP Channel 10.		
		43	FastDSP Channel 11.		
		44	FastDSP Channel 12.		
		45	FastDSP Channel 13.		
		46	FastDSP Channel 14.		
		47	FastDSP Channel 15.		
		48	Tensilica DSP Channel 0.		
		49	Tensilica DSP Channel 1.		
		50	Tensilica DSP Channel 2.		
		51	Tensilica DSP Channel 3.		
		52	Tensilica DSP Channel 4.		
		53	Tensilica DSP Channel 5.		
		54	Tensilica DSP Channel 6.		
		55	Tensilica DSP Channel 7.		
		56	Tensilica DSP Channel 8.		
		57	Tensilica DSP Channel 9.		
		58	Tensilica DSP Channel 10.		
		59	Tensilica DSP Channel 11.		
		60	Tensilica DSP Channel 12.		
		61	Tensilica DSP Channel 13.		
		62	Tensilica DSP Channel 14.		
		63	Tensilica DSP Channel 15.		
		64	Input ASRC Channel 0.		
		65	Input ASRC Channel 1.		
		66	Input ASRC Channel 2.		
		67	Input ASRC Channel 3.		
		68	ADC Channel 0.		

Bits	Bit Name	Settings	Description	Reset	Access
		69	ADC Channel 1.		
		70	ADC Channel 2.		
		71	DMIC Channel 0.		
		72	DMIC Channel 1.		
		73	DMIC Channel 2.		
		74	DMIC Channel 3.		
		75	DMIC Channel 4.		
		76	DMIC Channel 5.		
		77	DMIC Channel 6.		
		78	DMIC Channel 7.		

TENSILICA DSP SOFTWARE RESET REGISTER

Address: 0x4000C0D4, Reset: 0x00, Name: TDSP_SOFT_RESET

Table 269. Bit Descriptions for TDSP_SOFT_RESET

Bits	Bit Name	Settings	Description	Reset	Access
[7:1]	RESERVED		Reserved.	0x0	R
0	TDSP_SOFT_RESET		Software reset for Tensilica DSP.	0x0	W
		0	N/A.		
		1	Write once to soft reset.		

TENSILICA DSP ALTERNATIVE RESET VECTOR ENABLE REGISTER

Address: 0x4000C0D5, Reset: 0x00, Name: TDSP_ALTVEC_EN

Table 270. Bit Descriptions for TDSP_ALTVEC_EN

Bits	Bit Name	Settings	Description	Reset	Access
[7:1]	RESERVED		Reserved.	0x0	R
0	TDSP_ALTVEC_EN		Control where TDSP boots from.	0x0	R/W
		0	Tensilica core will always boot from ADI internal ROM start address.		
		1	Tensilica core will boot from user specific address after the core comes out of reset. The user specific address is specified in MMR 'TDSP_ALTVEC_ADDR'.		

TENSILICA DSP ALTERNATIVE RESET VECTOR ADDRESS, BYTE 0 REGISTER

Address: 0x4000C0D8, Reset: 0x00, Name: TDSP_ALTVEC_ADDR0

Table 271. Bit Descriptions for TDSP_ALTVEC_ADDR0

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	TDSP_ALTVEC_ADDR[7:0]		This register specifies the boot address when TDSP_ALTVEC_EN=1'b1.	0x0	R/W

TENSILICA DSP ALTERNATIVE RESET VECTOR ADDRESS, BYTE 1 REGISTER

Address: 0x4000C0D9, Reset: 0x00, Name: TDSP_ALTVEC_ADDR1

Table 272. Bit Descriptions for TDSP_ALTVEC_ADDR1

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	TDSP_ALTVEC_ADDR[15:8]		This register specifies the boot address when TDSP_ALTVEC_EN=1'b1.	0x0	R/W

TENSILICA DSP ALTERNATIVE RESET VECTOR ADDRESS, BYTE 2 REGISTER

Address: 0x4000C0DA, Reset: 0x00, Name: TDSP_ALTVEC_ADDR2

Table 273. Bit Descriptions for TDSP_ALTVEC_ADDR2

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	TDSP_ALTVEC_ADDR[23:16]		This register specifies the boot address when TDSP_ALTVEC_EN=1'b1.	0x0	R/W

TENSILICA DSP ALTERNATIVE RESET VECTOR ADDRESS, BYTE 3 REGISTER

Address: 0x4000C0DB, Reset: 0x00, Name: TDSP_ALTVEC_ADDR3

Table 274. Bit Descriptions for TDSP_ALTVEC_ADDR3

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	TDSP_ALTVEC_ADDR[31:24]		This register specifies the boot address when TDSP_ALTVEC_EN=1'b1.	0x0	R/W

TENSILICA DSP RUN REGISTER

Address: 0x4000C0DC, Reset: 0x01, Name: TDSP_RUN

Table 275. Bit Descriptions for TDSP_RUN

Bits	Bit Name	Settings	Description	Reset	Access
[7:1]	RESERVED		Reserved.	0x0	R
0	TDSP_RUN		Control TDSP is running or stalled.	0x1	R/W
		0	TDSP core is being stalled.		
		1	TDSP core is running.		

SERIAL PORT 0 CONTROL 1 REGISTER

Address: 0x4000C0E0, Reset: 0x00, Name: SPT0_CTRL1

Table 276. Bit Descriptions for SPT0_CTRL1

Bits	Bit Name	Settings	Description	Reset	Access
7	RESERVED		Reserved.	0x0	R/W
6	SPT0_TRI_STATE		Serial port output - tri-state enable.	0x0	R/W
		1	tri-state enabled.		
		0	tri-state disabled.		
[5:4]	SPT0_SLOT_WIDTH		Serial port - selects slot width.	0x0	R/W
		00	32 BCLK's per slot.		
		01	16 BCLK's per slot.		
		10	24 BCLK's per slot.		
[3:1]	SPT0_DATA_FORMAT		Serial port - selects data format.	0x0	R/W
		001	Left Justified,delay by 0.		
		000	Typical I2S mode, delay by 1.		
		010	Delay by 8.		
		011	Delay by 12.		
		100	Delay by 16.		
0	SPT0_SAI_MODE		Serial port - selects SAI mode.	0x0	R/W
		0	STEREO (I2S, LJ, RJ).		
		1	TDM.		

SERIAL PORT 0 CONTROL 2 REGISTER

Address: 0x4000C0E1, Reset: 0x00, Name: SPT0_CTRL2

Table 277. Bit Descriptions for SPT0_CTRL2

Bits	Bit Name	Settings	Description	Reset	Access
[7:4]	RESERVED		Reserved.	0x0	R
3	SPT0_BCLK_POL		Serial port - selects bclk polarity.	0x0	R/W
		0	Capture on rising edge.		
		1	Capture on falling egde.		
[2:0]	SPT0_BCLK_SRC		Serial port - selects bclk source and rate.	0x0	R/W
		000	BCLK is from external source.		
		001	Generate BCLK at 3.072MHz.		
		010	Generate BCLK at 6.144MHz.		
		011	Generate BCLK at 12.288MHz.		
		100	Generate BCLK at 24.576MHz.		

SERIAL PORT 0 CONTROL 2 REGISTER

Address: 0x4000C0E2, Reset: 0x00, Name: SPT0_CTRL3

Table 278. Bit Descriptions for SPT0_CTRL3

Bits	Bit Name	Settings	Description	Reset	Access
[7:5]	RESERVED		Reserved.	0x0	R
4	SPT0_LRCLK_POL		Serial port - selects lrclk polarity.	0x0	R/W
		0	Normal Polarity.		
		1	Inverted Polarity.		
[3:0]	SPT0_LRCLK_SRC		Serial port - selects lrclk source and rate.	0x0	R/W
		0000	lrclk is from external.		
		0001	generate lrclk with 48KHz.		
		0010	generate lrclk with 96KHz.		
		0011	generate lrclk with 192KHz.		
		0100	generate lrclk with 12kHz.		
		0101	generate lrclk with 24KHz.		
		0110	generate lrclk with 384KHz.		
		0111	generate lrclk with 768KHz.		
		1000	generate lrclk with 8KHz.		
		1001	generate lrclk with 16KHz.		

SERIAL PORT 0 OUTPUT ROUTING SLOT 0 (LEFT) REGISTER

Address: 0x4000C0E3, Reset: 0x10, Name: SPT0_ROUTE0

Table 279. Bit Descriptions for SPT0_ROUTE0

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
[5:0]	SPT0_OUT_ROUTE0		Serial port output route slot 0 (left).	0x10	R/W
		0	FastDSP Channel 0.		
		1	FastDSP Channel 1.		
		2	FastDSP Channel 2.		
		3	FastDSP Channel 3.		
		4	FastDSP Channel 4.		

Bits	Bit Name	Settings	Description	Reset	Access
		5	FastDSP Channel 5.		
		6	FastDSP Channel 6.		
		7	FastDSP Channel 7.		
		8	FastDSP Channel 8.		
		9	FastDSP Channel 9.		
		10	FastDSP Channel 10.		
		11	FastDSP Channel 11.		
		12	FastDSP Channel 12.		
		13	FastDSP Channel 13.		
		14	FastDSP Channel 14.		
		15	FastDSP Channel 15.		
		16	Tensilica DSP Channel 0.		
		17	Tensilica DSP Channel 1.		
		18	Tensilica DSP Channel 2.		
		19	Tensilica DSP Channel 3.		
		20	Tensilica DSP Channel 4.		
		21	Tensilica DSP Channel 5.		
		22	Tensilica DSP Channel 6.		
		23	Tensilica DSP Channel 7.		
		24	Tensilica DSP Channel 8.		
		25	Tensilica DSP Channel 9.		
		26	Tensilica DSP Channel 10.		
		27	Tensilica DSP Channel 11.		
		28	Tensilica DSP Channel 12.		
		29	Tensilica DSP Channel 13.		
		30	Tensilica DSP Channel 14.		
		31	Tensilica DSP Channel 15.		
		32	Output ASRC Channel 0.		
		33	Output ASRC Channel 1.		
		34	Output ASRC Channel 2.		
		35	Output ASRC Channel 3.		
		36	ADC Channel 0.		
		37	ADC Channel 1.		
		38	ADC Channel 2.		
		39	DMIC Channel 0.		
		40	DMIC Channel 1.		
		41	DMIC Channel 2.		
		42	DMIC Channel 3.		
		43	Fast to Slow Decimator Channel 0.		
		44	Fast to Slow Decimator Channel 1.		
		45	Fast to Slow Decimator Channel 2.		
		46	Fast to Slow Decimator Channel 3.		
		47	Fast to Slow Decimator Channel 4.		
		48	Fast to Slow Decimator Channel 5.		
		49	Fast to Slow Decimator Channel 6.		

Bits	Bit Name	Settings	Description	Reset	Access
		50	Fast to Slow Decimator Channel 7.		
		51	EQ0.		
		55	DMIC Channel 4.		
		56	DMIC Channel 5.		
		57	DMIC Channel 6.		
		58	DMIC Channel 7.		
		63	No Output. Slot not used.		

SERIAL PORT 0 OUTPUT ROUTING SLOT 1 (RIGHT) REGISTER

Address: 0x4000C0E4, Reset: 0x11, Name: SPT0_ROUTE1

Table 280. Bit Descriptions for SPT0_ROUTE1

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
[5:0]	SPT0_OUT_ROUTE1		Serial port output route slot 1 (right).	0x11	R/W
		0	FastDSP Channel 0.		
		1	FastDSP Channel 1.		
		2	FastDSP Channel 2.		
		3	FastDSP Channel 3.		
		4	FastDSP Channel 4.		
		5	FastDSP Channel 5.		
		6	FastDSP Channel 6.		
		7	FastDSP Channel 7.		
		8	FastDSP Channel 8.		
		9	FastDSP Channel 9.		
		10	FastDSP Channel 10.		
		11	FastDSP Channel 11.		
		12	FastDSP Channel 12.		
		13	FastDSP Channel 13.		
		14	FastDSP Channel 14.		
		15	FastDSP Channel 15.		
		16	Tensilica DSP Channel 0.		
		17	Tensilica DSP Channel 1.		
		18	Tensilica DSP Channel 2.		
		19	Tensilica DSP Channel 3.		
		20	Tensilica DSP Channel 4.		
		21	Tensilica DSP Channel 5.		
		22	Tensilica DSP Channel 6.		
		23	Tensilica DSP Channel 7.		
		24	Tensilica DSP Channel 8.		
		25	Tensilica DSP Channel 9.		
		26	Tensilica DSP Channel 10.		
		27	Tensilica DSP Channel 11.		
		28	Tensilica DSP Channel 12.		
		29	Tensilica DSP Channel 13.		
		30	Tensilica DSP Channel 14.		
		31	Tensilica DSP Channel 15.		
		32	Output ASRC Channel 0.		
		33	Output ASRC Channel 1.		
		34	Output ASRC Channel 2.		

Bits	Bit Name	Settings	Description	Reset	Access
		35	Output ASRC Channel 3.		
		36	ADC Channel 0.		
		37	ADC Channel 1.		
		38	ADC Channel 2.		
		39	DMIC Channel 0.		
		40	DMIC Channel 1.		
		41	DMIC Channel 2.		
		42	DMIC Channel 3.		
		43	Fast to Slow Decimator Channel 0.		
		44	Fast to Slow Decimator Channel 1.		
		45	Fast to Slow Decimator Channel 2.		
		46	Fast to Slow Decimator Channel 3.		
		47	Fast to Slow Decimator Channel 4.		
		48	Fast to Slow Decimator Channel 5.		
		49	Fast to Slow Decimator Channel 6.		
		50	Fast to Slow Decimator Channel 7.		
		51	EQ0.		
		55	DMIC Channel 4.		
		56	DMIC Channel 5.		
		57	DMIC Channel 6.		
		58	DMIC Channel 7.		
		63	No Output. Slot not used.		

SERIAL PORT 0 OUTPUT ROUTING SLOT 2 REGISTER

Address: 0x4000C0E5, Reset: 0x3F, Name: SPT0_ROUTE2

Table 281. Bit Descriptions for SPT0_ROUTE2

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
[5:0]	SPT0_OUT_ROUTE2		Serial port output route slot 2.	0x3F	R/W
		0	FastDSP Channel 0.		
		1	FastDSP Channel 1.		
		2	FastDSP Channel 2.		
		3	FastDSP Channel 3.		
		4	FastDSP Channel 4.		
		5	FastDSP Channel 5.		
		6	FastDSP Channel 6.		
		7	FastDSP Channel 7.		
		8	FastDSP Channel 8.		
		9	FastDSP Channel 9.		
		10	FastDSP Channel 10.		
		11	FastDSP Channel 11.		
		12	FastDSP Channel 12.		
		13	FastDSP Channel 13.		
		14	FastDSP Channel 14.		
		15	FastDSP Channel 15.		
		16	Tensilica DSP Channel 0.		
		17	Tensilica DSP Channel 1.		
		18	Tensilica DSP Channel 2.		
		19	Tensilica DSP Channel 3.		

Bits	Bit Name	Settings	Description	Reset	Access
		20	Tensilica DSP Channel 4.		
		21	Tensilica DSP Channel 5.		
		22	Tensilica DSP Channel 6.		
		23	Tensilica DSP Channel 7.		
		24	Tensilica DSP Channel 8.		
		25	Tensilica DSP Channel 9.		
		26	Tensilica DSP Channel 10.		
		27	Tensilica DSP Channel 11.		
		28	Tensilica DSP Channel 12.		
		29	Tensilica DSP Channel 13.		
		30	Tensilica DSP Channel 14.		
		31	Tensilica DSP Channel 15.		
		32	Output ASRC Channel 0.		
		33	Output ASRC Channel 1.		
		34	Output ASRC Channel 2.		
		35	Output ASRC Channel 3.		
		36	ADC Channel 0.		
		37	ADC Channel 1.		
		38	ADC Channel 2.		
		39	DMIC Channel 0.		
		40	DMIC Channel 1.		
		41	DMIC Channel 2.		
		42	DMIC Channel 3.		
		43	Fast to Slow Decimator Channel 0.		
		44	Fast to Slow Decimator Channel 1.		
		45	Fast to Slow Decimator Channel 2.		
		46	Fast to Slow Decimator Channel 3.		
		47	Fast to Slow Decimator Channel 4.		
		48	Fast to Slow Decimator Channel 5.		
		49	Fast to Slow Decimator Channel 6.		
		50	Fast to Slow Decimator Channel 7.		
		51	EQ0.		
		55	DMIC Channel 4.		
		56	DMIC Channel 5.		
		57	DMIC Channel 6.		
		58	DMIC Channel 7.		
		63	No Output. Slot not used.		

SERIAL PORT 0 OUTPUT ROUTING SLOT 3 REGISTER

Address: 0x4000C0E6, Reset: 0x3F, Name: SPT0_ROUTE3

Table 282. Bit Descriptions for SPT0_ROUTE3

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
[5:0]	SPT0_OUT_ROUTE3		Serial port output route slot 3.	0x3F	R/W
		0	FastDSP Channel 0.		
		1	FastDSP Channel 1.		
		2	FastDSP Channel 2.		
		3	FastDSP Channel 3.		
		4	FastDSP Channel 4.		

Bits	Bit Name	Settings	Description	Reset	Access
		5	FastDSP Channel 5.		
		6	FastDSP Channel 6.		
		7	FastDSP Channel 7.		
		8	FastDSP Channel 8.		
		9	FastDSP Channel 9.		
		10	FastDSP Channel 10.		
		11	FastDSP Channel 11.		
		12	FastDSP Channel 12.		
		13	FastDSP Channel 13.		
		14	FastDSP Channel 14.		
		15	FastDSP Channel 15.		
		16	Tensilica DSP Channel 0.		
		17	Tensilica DSP Channel 1.		
		18	Tensilica DSP Channel 2.		
		19	Tensilica DSP Channel 3.		
		20	Tensilica DSP Channel 4.		
		21	Tensilica DSP Channel 5.		
		22	Tensilica DSP Channel 6.		
		23	Tensilica DSP Channel 7.		
		24	Tensilica DSP Channel 8.		
		25	Tensilica DSP Channel 9.		
		26	Tensilica DSP Channel 10.		
		27	Tensilica DSP Channel 11.		
		28	Tensilica DSP Channel 12.		
		29	Tensilica DSP Channel 13.		
		30	Tensilica DSP Channel 14.		
		31	Tensilica DSP Channel 15.		
		32	Output ASRC Channel 0.		
		33	Output ASRC Channel 1.		
		34	Output ASRC Channel 2.		
		35	Output ASRC Channel 3.		
		36	ADC Channel 0.		
		37	ADC Channel 1.		
		38	ADC Channel 2.		
		39	DMIC Channel 0.		
		40	DMIC Channel 1.		
		41	DMIC Channel 2.		
		42	DMIC Channel 3.		
		43	Fast to Slow Decimator Channel 0.		
		44	Fast to Slow Decimator Channel 1.		
		45	Fast to Slow Decimator Channel 2.		
		46	Fast to Slow Decimator Channel 3.		
		47	Fast to Slow Decimator Channel 4.		
		48	Fast to Slow Decimator Channel 5.		
		49	Fast to Slow Decimator Channel 6.		

Bits	Bit Name	Settings	Description	Reset	Access
		50	Fast to Slow Decimator Channel 7.		
		51	EQ0.		
		55	DMIC Channel 4.		
		56	DMIC Channel 5.		
		57	DMIC Channel 6.		
		58	DMIC Channel 7.		
		63	No Output. Slot not used.		

SERIAL PORT 0 OUTPUT ROUTING SLOT 4 REGISTER

Address: 0x4000C0E7, Reset: 0x3F, Name: SPT0_ROUTE4

Table 283. Bit Descriptions for SPT0_ROUTE4

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
[5:0]	SPT0_OUT_ROUTE4		Serial port output route slot 4.	0x3F	R/W
		0	FastDSP Channel 0.		
		1	FastDSP Channel 1.		
		2	FastDSP Channel 2.		
		3	FastDSP Channel 3.		
		4	FastDSP Channel 4.		
		5	FastDSP Channel 5.		
		6	FastDSP Channel 6.		
		7	FastDSP Channel 7.		
		8	FastDSP Channel 8.		
		9	FastDSP Channel 9.		
		10	FastDSP Channel 10.		
		11	FastDSP Channel 11.		
		12	FastDSP Channel 12.		
		13	FastDSP Channel 13.		
		14	FastDSP Channel 14.		
		15	FastDSP Channel 15.		
		16	Tensilica DSP Channel 0.		
		17	Tensilica DSP Channel 1.		
		18	Tensilica DSP Channel 2.		
		19	Tensilica DSP Channel 3.		
		20	Tensilica DSP Channel 4.		
		21	Tensilica DSP Channel 5.		
		22	Tensilica DSP Channel 6.		
		23	Tensilica DSP Channel 7.		
		24	Tensilica DSP Channel 8.		
		25	Tensilica DSP Channel 9.		
		26	Tensilica DSP Channel 10.		
		27	Tensilica DSP Channel 11.		
		28	Tensilica DSP Channel 12.		
		29	Tensilica DSP Channel 13.		
		30	Tensilica DSP Channel 14.		
		31	Tensilica DSP Channel 15.		
		32	Output ASRC Channel 0.		
		33	Output ASRC Channel 1.		
		34	Output ASRC Channel 2.		

Bits	Bit Name	Settings	Description	Reset	Access
		35	Output ASRC Channel 3.		
		36	ADC Channel 0.		
		37	ADC Channel 1.		
		38	ADC Channel 2.		
		39	DMIC Channel 0.		
		40	DMIC Channel 1.		
		41	DMIC Channel 2.		
		42	DMIC Channel 3.		
		43	Fast to Slow Decimator Channel 0.		
		44	Fast to Slow Decimator Channel 1.		
		45	Fast to Slow Decimator Channel 2.		
		46	Fast to Slow Decimator Channel 3.		
		47	Fast to Slow Decimator Channel 4.		
		48	Fast to Slow Decimator Channel 5.		
		49	Fast to Slow Decimator Channel 6.		
		50	Fast to Slow Decimator Channel 7.		
		51	EQ0.		
		55	DMIC Channel 4.		
		56	DMIC Channel 5.		
		57	DMIC Channel 6.		
		58	DMIC Channel 7.		
		63	No Output. Slot not used.		

SERIAL PORT 0 OUTPUT ROUTING SLOT 5 REGISTER

Address: 0x4000C0E8, Reset: 0x3F, Name: SPT0_ROUTE5

Table 284. Bit Descriptions for SPT0_ROUTE5

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
[5:0]	SPT0_OUT_ROUTE5		Serial port output route slot 5.	0x3F	R/W
		0	FastDSP Channel 0.		
		1	FastDSP Channel 1.		
		2	FastDSP Channel 2.		
		3	FastDSP Channel 3.		
		4	FastDSP Channel 4.		
		5	FastDSP Channel 5.		
		6	FastDSP Channel 6.		
		7	FastDSP Channel 7.		
		8	FastDSP Channel 8.		
		9	FastDSP Channel 9.		
		10	FastDSP Channel 10.		
		11	FastDSP Channel 11.		
		12	FastDSP Channel 12.		
		13	FastDSP Channel 13.		
		14	FastDSP Channel 14.		
		15	FastDSP Channel 15.		
		16	Tensilica DSP Channel 0.		
		17	Tensilica DSP Channel 1.		
		18	Tensilica DSP Channel 2.		
		19	Tensilica DSP Channel 3.		

Bits	Bit Name	Settings	Description	Reset	Access
		20	Tensilica DSP Channel 4.		
		21	Tensilica DSP Channel 5.		
		22	Tensilica DSP Channel 6.		
		23	Tensilica DSP Channel 7.		
		24	Tensilica DSP Channel 8.		
		25	Tensilica DSP Channel 9.		
		26	Tensilica DSP Channel 10.		
		27	Tensilica DSP Channel 11.		
		28	Tensilica DSP Channel 12.		
		29	Tensilica DSP Channel 13.		
		30	Tensilica DSP Channel 14.		
		31	Tensilica DSP Channel 15.		
		32	Output ASRC Channel 0.		
		33	Output ASRC Channel 1.		
		34	Output ASRC Channel 2.		
		35	Output ASRC Channel 3.		
		36	ADC Channel 0.		
		37	ADC Channel 1.		
		38	ADC Channel 2.		
		39	DMIC Channel 0.		
		40	DMIC Channel 1.		
		41	DMIC Channel 2.		
		42	DMIC Channel 3.		
		43	Fast to Slow Decimator Channel 0.		
		44	Fast to Slow Decimator Channel 1.		
		45	Fast to Slow Decimator Channel 2.		
		46	Fast to Slow Decimator Channel 3.		
		47	Fast to Slow Decimator Channel 4.		
		48	Fast to Slow Decimator Channel 5.		
		49	Fast to Slow Decimator Channel 6.		
		50	Fast to Slow Decimator Channel 7.		
		51	EQ0.		
		55	DMIC Channel 4.		
		56	DMIC Channel 5.		
		57	DMIC Channel 6.		
		58	DMIC Channel 7.		
		63	No Output. Slot not used.		

SERIAL PORT 0 OUTPUT ROUTING SLOT 6 REGISTER

Address: 0x4000C0E9, Reset: 0x3F, Name: SPT0_ROUTE6

Table 285. Bit Descriptions for SPT0_ROUTE6

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
[5:0]	SPT0_OUT_ROUTE6		Serial port output route slot 6.	0x3F	R/W
		0	FastDSP Channel 0.		
		1	FastDSP Channel 1.		
		2	FastDSP Channel 2.		
		3	FastDSP Channel 3.		
		4	FastDSP Channel 4.		

Bits	Bit Name	Settings	Description	Reset	Access
		5	FastDSP Channel 5.		
		6	FastDSP Channel 6.		
		7	FastDSP Channel 7.		
		8	FastDSP Channel 8.		
		9	FastDSP Channel 9.		
		10	FastDSP Channel 10.		
		11	FastDSP Channel 11.		
		12	FastDSP Channel 12.		
		13	FastDSP Channel 13.		
		14	FastDSP Channel 14.		
		15	FastDSP Channel 15.		
		16	Tensilica DSP Channel 0.		
		17	Tensilica DSP Channel 1.		
		18	Tensilica DSP Channel 2.		
		19	Tensilica DSP Channel 3.		
		20	Tensilica DSP Channel 4.		
		21	Tensilica DSP Channel 5.		
		22	Tensilica DSP Channel 6.		
		23	Tensilica DSP Channel 7.		
		24	Tensilica DSP Channel 8.		
		25	Tensilica DSP Channel 9.		
		26	Tensilica DSP Channel 10.		
		27	Tensilica DSP Channel 11.		
		28	Tensilica DSP Channel 12.		
		29	Tensilica DSP Channel 13.		
		30	Tensilica DSP Channel 14.		
		31	Tensilica DSP Channel 15.		
		32	Output ASRC Channel 0.		
		33	Output ASRC Channel 1.		
		34	Output ASRC Channel 2.		
		35	Output ASRC Channel 3.		
		36	ADC Channel 0.		
		37	ADC Channel 1.		
		38	ADC Channel 2.		
		39	DMIC Channel 0.		
		40	DMIC Channel 1.		
		41	DMIC Channel 2.		
		42	DMIC Channel 3.		
		43	Fast to Slow Decimator Channel 0.		
		44	Fast to Slow Decimator Channel 1.		
		45	Fast to Slow Decimator Channel 2.		
		46	Fast to Slow Decimator Channel 3.		
		47	Fast to Slow Decimator Channel 4.		
		48	Fast to Slow Decimator Channel 5.		
		49	Fast to Slow Decimator Channel 6.		

Bits	Bit Name	Settings	Description	Reset	Access
		50	Fast to Slow Decimator Channel 7.		
		51	EQ0.		
		55	DMIC Channel 4.		
		56	DMIC Channel 5.		
		57	DMIC Channel 6.		
		58	DMIC Channel 7.		
		63	No Output. Slot not used.		

SERIAL PORT 0 OUTPUT ROUTING SLOT 7 REGISTER

Address: 0x4000C0EA, Reset: 0x3F, Name: SPT0_ROUTE7

Table 286. Bit Descriptions for SPT0_ROUTE7

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
[5:0]	SPT0_OUT_ROUTE7		Serial port output route slot 7.	0x3F	R/W
		0	FastDSP Channel 0.		
		1	FastDSP Channel 1.		
		2	FastDSP Channel 2.		
		3	FastDSP Channel 3.		
		4	FastDSP Channel 4.		
		5	FastDSP Channel 5.		
		6	FastDSP Channel 6.		
		7	FastDSP Channel 7.		
		8	FastDSP Channel 8.		
		9	FastDSP Channel 9.		
		10	FastDSP Channel 10.		
		11	FastDSP Channel 11.		
		12	FastDSP Channel 12.		
		13	FastDSP Channel 13.		
		14	FastDSP Channel 14.		
		15	FastDSP Channel 15.		
		16	Tensilica DSP Channel 0.		
		17	Tensilica DSP Channel 1.		
		18	Tensilica DSP Channel 2.		
		19	Tensilica DSP Channel 3.		
		20	Tensilica DSP Channel 4.		
		21	Tensilica DSP Channel 5.		
		22	Tensilica DSP Channel 6.		
		23	Tensilica DSP Channel 7.		
		24	Tensilica DSP Channel 8.		
		25	Tensilica DSP Channel 9.		
		26	Tensilica DSP Channel 10.		
		27	Tensilica DSP Channel 11.		
		28	Tensilica DSP Channel 12.		
		29	Tensilica DSP Channel 13.		
		30	Tensilica DSP Channel 14.		
		31	Tensilica DSP Channel 15.		
		32	Output ASRC Channel 0.		
		33	Output ASRC Channel 1.		
		34	Output ASRC Channel 2.		

Bits	Bit Name	Settings	Description	Reset	Access
		35	Output ASRC Channel 3.		
		36	ADC Channel 0.		
		37	ADC Channel 1.		
		38	ADC Channel 2.		
		39	DMIC Channel 0.		
		40	DMIC Channel 1.		
		41	DMIC Channel 2.		
		42	DMIC Channel 3.		
		43	Fast to Slow Decimator Channel 0.		
		44	Fast to Slow Decimator Channel 1.		
		45	Fast to Slow Decimator Channel 2.		
		46	Fast to Slow Decimator Channel 3.		
		47	Fast to Slow Decimator Channel 4.		
		48	Fast to Slow Decimator Channel 5.		
		49	Fast to Slow Decimator Channel 6.		
		50	Fast to Slow Decimator Channel 7.		
		51	EQ0.		
		55	DMIC Channel 4.		
		56	DMIC Channel 5.		
		57	DMIC Channel 6.		
		58	DMIC Channel 7.		
		63	No Output. Slot not used.		

SERIAL PORT 0 OUTPUT ROUTING SLOT 8 REGISTER

Address: 0x4000C0EB, Reset: 0x3F, Name: SPT0_ROUTE8

Table 287. Bit Descriptions for SPT0_ROUTE8

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
[5:0]	SPT0_OUT_ROUTE8		Serial port output route slot 8.	0x3F	R/W
		0	FastDSP Channel 0.		
		1	FastDSP Channel 1.		
		2	FastDSP Channel 2.		
		3	FastDSP Channel 3.		
		4	FastDSP Channel 4.		
		5	FastDSP Channel 5.		
		6	FastDSP Channel 6.		
		7	FastDSP Channel 7.		
		8	FastDSP Channel 8.		
		9	FastDSP Channel 9.		
		10	FastDSP Channel 10.		
		11	FastDSP Channel 11.		
		12	FastDSP Channel 12.		
		13	FastDSP Channel 13.		
		14	FastDSP Channel 14.		
		15	FastDSP Channel 15.		
		16	Tensilica DSP Channel 0.		
		17	Tensilica DSP Channel 1.		
		18	Tensilica DSP Channel 2.		
		19	Tensilica DSP Channel 3.		

Bits	Bit Name	Settings	Description	Reset	Access
		20	Tensilica DSP Channel 4.		
		21	Tensilica DSP Channel 5.		
		22	Tensilica DSP Channel 6.		
		23	Tensilica DSP Channel 7.		
		24	Tensilica DSP Channel 8.		
		25	Tensilica DSP Channel 9.		
		26	Tensilica DSP Channel 10.		
		27	Tensilica DSP Channel 11.		
		28	Tensilica DSP Channel 12.		
		29	Tensilica DSP Channel 13.		
		30	Tensilica DSP Channel 14.		
		31	Tensilica DSP Channel 15.		
		32	Output ASRC Channel 0.		
		33	Output ASRC Channel 1.		
		34	Output ASRC Channel 2.		
		35	Output ASRC Channel 3.		
		36	ADC Channel 0.		
		37	ADC Channel 1.		
		38	ADC Channel 2.		
		39	DMIC Channel 0.		
		40	DMIC Channel 1.		
		41	DMIC Channel 2.		
		42	DMIC Channel 3.		
		43	Fast to Slow Decimator Channel 0.		
		44	Fast to Slow Decimator Channel 1.		
		45	Fast to Slow Decimator Channel 2.		
		46	Fast to Slow Decimator Channel 3.		
		47	Fast to Slow Decimator Channel 4.		
		48	Fast to Slow Decimator Channel 5.		
		49	Fast to Slow Decimator Channel 6.		
		50	Fast to Slow Decimator Channel 7.		
		51	EQ0.		
		55	DMIC Channel 4.		
		56	DMIC Channel 5.		
		57	DMIC Channel 6.		
		58	DMIC Channel 7.		
		63	No Output. Slot not used.		

SERIAL PORT 0 OUTPUT ROUTING SLOT 9 REGISTER

Address: 0x4000C0EC, Reset: 0x3F, Name: SPT0_ROUTE9

Table 288. Bit Descriptions for SPT0_ROUTE9

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
[5:0]	SPT0_OUT_ROUTE9		Serial port output route slot 9.	0x3F	R/W
		0	FastDSP Channel 0.		
		1	FastDSP Channel 1.		
		2	FastDSP Channel 2.		
		3	FastDSP Channel 3.		
		4	FastDSP Channel 4.		

Bits	Bit Name	Settings	Description	Reset	Access
		5	FastDSP Channel 5.		
		6	FastDSP Channel 6.		
		7	FastDSP Channel 7.		
		8	FastDSP Channel 8.		
		9	FastDSP Channel 9.		
		10	FastDSP Channel 10.		
		11	FastDSP Channel 11.		
		12	FastDSP Channel 12.		
		13	FastDSP Channel 13.		
		14	FastDSP Channel 14.		
		15	FastDSP Channel 15.		
		16	Tensilica DSP Channel 0.		
		17	Tensilica DSP Channel 1.		
		18	Tensilica DSP Channel 2.		
		19	Tensilica DSP Channel 3.		
		20	Tensilica DSP Channel 4.		
		21	Tensilica DSP Channel 5.		
		22	Tensilica DSP Channel 6.		
		23	Tensilica DSP Channel 7.		
		24	Tensilica DSP Channel 8.		
		25	Tensilica DSP Channel 9.		
		26	Tensilica DSP Channel 10.		
		27	Tensilica DSP Channel 11.		
		28	Tensilica DSP Channel 12.		
		29	Tensilica DSP Channel 13.		
		30	Tensilica DSP Channel 14.		
		31	Tensilica DSP Channel 15.		
		32	Output ASRC Channel 0.		
		33	Output ASRC Channel 1.		
		34	Output ASRC Channel 2.		
		35	Output ASRC Channel 3.		
		36	ADC Channel 0.		
		37	ADC Channel 1.		
		38	ADC Channel 2.		
		39	DMIC Channel 0.		
		40	DMIC Channel 1.		
		41	DMIC Channel 2.		
		42	DMIC Channel 3.		
		43	Fast to Slow Decimator Channel 0.		
		44	Fast to Slow Decimator Channel 1.		
		45	Fast to Slow Decimator Channel 2.		
		46	Fast to Slow Decimator Channel 3.		
		47	Fast to Slow Decimator Channel 4.		
		48	Fast to Slow Decimator Channel 5.		
		49	Fast to Slow Decimator Channel 6.		

Bits	Bit Name	Settings	Description	Reset	Access
		50	Fast to Slow Decimator Channel 7.		
		51	EQ0.		
		55	DMIC Channel 4.		
		56	DMIC Channel 5.		
		57	DMIC Channel 6.		
		58	DMIC Channel 7.		
		63	No Output. Slot not used.		

SERIAL PORT 0 OUTPUT ROUTING SLOT 10 REGISTER

Address: 0x4000C0ED, Reset: 0x3F, Name: SPT0_ROUTE10

Table 289. Bit Descriptions for SPT0_ROUTE10

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
[5:0]	SPT0_OUT_ROUTE10		Serial port output route slot 10.	0x3F	R/W
		0	FastDSP Channel 0.		
		1	FastDSP Channel 1.		
		2	FastDSP Channel 2.		
		3	FastDSP Channel 3.		
		4	FastDSP Channel 4.		
		5	FastDSP Channel 5.		
		6	FastDSP Channel 6.		
		7	FastDSP Channel 7.		
		8	FastDSP Channel 8.		
		9	FastDSP Channel 9.		
		10	FastDSP Channel 10.		
		11	FastDSP Channel 11.		
		12	FastDSP Channel 12.		
		13	FastDSP Channel 13.		
		14	FastDSP Channel 14.		
		15	FastDSP Channel 15.		
		16	Tensilica DSP Channel 0.		
		17	Tensilica DSP Channel 1.		
		18	Tensilica DSP Channel 2.		
		19	Tensilica DSP Channel 3.		
		20	Tensilica DSP Channel 4.		
		21	Tensilica DSP Channel 5.		
		22	Tensilica DSP Channel 6.		
		23	Tensilica DSP Channel 7.		
		24	Tensilica DSP Channel 8.		
		25	Tensilica DSP Channel 9.		
		26	Tensilica DSP Channel 10.		
		27	Tensilica DSP Channel 11.		
		28	Tensilica DSP Channel 12.		
		29	Tensilica DSP Channel 13.		
		30	Tensilica DSP Channel 14.		
		31	Tensilica DSP Channel 15.		
		32	Output ASRC Channel 0.		
		33	Output ASRC Channel 1.		
		34	Output ASRC Channel 2.		

Bits	Bit Name	Settings	Description	Reset	Access
		35	Output ASRC Channel 3.		
		36	ADC Channel 0.		
		37	ADC Channel 1.		
		38	ADC Channel 2.		
		39	DMIC Channel 0.		
		40	DMIC Channel 1.		
		41	DMIC Channel 2.		
		42	DMIC Channel 3.		
		43	Fast to Slow Decimator Channel 0.		
		44	Fast to Slow Decimator Channel 1.		
		45	Fast to Slow Decimator Channel 2.		
		46	Fast to Slow Decimator Channel 3.		
		47	Fast to Slow Decimator Channel 4.		
		48	Fast to Slow Decimator Channel 5.		
		49	Fast to Slow Decimator Channel 6.		
		50	Fast to Slow Decimator Channel 7.		
		51	EQ0.		
		55	DMIC Channel 4.		
		56	DMIC Channel 5.		
		57	DMIC Channel 6.		
		58	DMIC Channel 7.		
		63	No Output. Slot not used.		

SERIAL PORT 0 OUTPUT ROUTING SLOT 11 REGISTER

Address: 0x4000C0EE, Reset: 0x3F, Name: SPT0_ROUTE11

Table 290. Bit Descriptions for SPT0_ROUTE11

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
[5:0]	SPT0_OUT_ROUTE11		Serial port output route slot 11.	0x3F	R/W
		0	FastDSP Channel 0.		
		1	FastDSP Channel 1.		
		2	FastDSP Channel 2.		
		3	FastDSP Channel 3.		
		4	FastDSP Channel 4.		
		5	FastDSP Channel 5.		
		6	FastDSP Channel 6.		
		7	FastDSP Channel 7.		
		8	FastDSP Channel 8.		
		9	FastDSP Channel 9.		
		10	FastDSP Channel 10.		
		11	FastDSP Channel 11.		
		12	FastDSP Channel 12.		
		13	FastDSP Channel 13.		
		14	FastDSP Channel 14.		
		15	FastDSP Channel 15.		
		16	Tensilica DSP Channel 0.		
		17	Tensilica DSP Channel 1.		
		18	Tensilica DSP Channel 2.		
		19	Tensilica DSP Channel 3.		

Bits	Bit Name	Settings	Description	Reset	Access
		20	Tensilica DSP Channel 4.		
		21	Tensilica DSP Channel 5.		
		22	Tensilica DSP Channel 6.		
		23	Tensilica DSP Channel 7.		
		24	Tensilica DSP Channel 8.		
		25	Tensilica DSP Channel 9.		
		26	Tensilica DSP Channel 10.		
		27	Tensilica DSP Channel 11.		
		28	Tensilica DSP Channel 12.		
		29	Tensilica DSP Channel 13.		
		30	Tensilica DSP Channel 14.		
		31	Tensilica DSP Channel 15.		
		32	Output ASRC Channel 0.		
		33	Output ASRC Channel 1.		
		34	Output ASRC Channel 2.		
		35	Output ASRC Channel 3.		
		36	ADC Channel 0.		
		37	ADC Channel 1.		
		38	ADC Channel 2.		
		39	DMIC Channel 0.		
		40	DMIC Channel 1.		
		41	DMIC Channel 2.		
		42	DMIC Channel 3.		
		43	Fast to Slow Decimator Channel 0.		
		44	Fast to Slow Decimator Channel 1.		
		45	Fast to Slow Decimator Channel 2.		
		46	Fast to Slow Decimator Channel 3.		
		47	Fast to Slow Decimator Channel 4.		
		48	Fast to Slow Decimator Channel 5.		
		49	Fast to Slow Decimator Channel 6.		
		50	Fast to Slow Decimator Channel 7.		
		51	EQ0.		
		55	DMIC Channel 4.		
		56	DMIC Channel 5.		
		57	DMIC Channel 6.		
		58	DMIC Channel 7.		
		63	No Output. Slot not used.		

SERIAL PORT 0 OUTPUT ROUTING SLOT 12 REGISTER

Address: 0x4000C0EF, Reset: 0x3F, Name: SPT0_ROUTE12

Table 291. Bit Descriptions for SPT0_ROUTE12

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
[5:0]	SPT0_OUT_ROUTE12		Serial port output route slot 12.	0x3F	R/W
		0	FastDSP Channel 0.		
		1	FastDSP Channel 1.		
		2	FastDSP Channel 2.		
		3	FastDSP Channel 3.		
		4	FastDSP Channel 4.		

Bits	Bit Name	Settings	Description	Reset	Access
		5	FastDSP Channel 5.		
		6	FastDSP Channel 6.		
		7	FastDSP Channel 7.		
		8	FastDSP Channel 8.		
		9	FastDSP Channel 9.		
		10	FastDSP Channel 10.		
		11	FastDSP Channel 11.		
		12	FastDSP Channel 12.		
		13	FastDSP Channel 13.		
		14	FastDSP Channel 14.		
		15	FastDSP Channel 15.		
		16	Tensilica DSP Channel 0.		
		17	Tensilica DSP Channel 1.		
		18	Tensilica DSP Channel 2.		
		19	Tensilica DSP Channel 3.		
		20	Tensilica DSP Channel 4.		
		21	Tensilica DSP Channel 5.		
		22	Tensilica DSP Channel 6.		
		23	Tensilica DSP Channel 7.		
		24	Tensilica DSP Channel 8.		
		25	Tensilica DSP Channel 9.		
		26	Tensilica DSP Channel 10.		
		27	Tensilica DSP Channel 11.		
		28	Tensilica DSP Channel 12.		
		29	Tensilica DSP Channel 13.		
		30	Tensilica DSP Channel 14.		
		31	Tensilica DSP Channel 15.		
		32	Output ASRC Channel 0.		
		33	Output ASRC Channel 1.		
		34	Output ASRC Channel 2.		
		35	Output ASRC Channel 3.		
		36	ADC Channel 0.		
		37	ADC Channel 1.		
		38	ADC Channel 2.		
		39	DMIC Channel 0.		
		40	DMIC Channel 1.		
		41	DMIC Channel 2.		
		42	DMIC Channel 3.		
		43	Fast to Slow Decimator Channel 0.		
		44	Fast to Slow Decimator Channel 1.		
		45	Fast to Slow Decimator Channel 2.		
		46	Fast to Slow Decimator Channel 3.		
		47	Fast to Slow Decimator Channel 4.		
		48	Fast to Slow Decimator Channel 5.		
		49	Fast to Slow Decimator Channel 6.		

Bits	Bit Name	Settings	Description	Reset	Access
		50	Fast to Slow Decimator Channel 7.		
		51	EQ0.		
		55	DMIC Channel 4.		
		56	DMIC Channel 5.		
		57	DMIC Channel 6.		
		58	DMIC Channel 7.		
		63	No Output. Slot not used.		

SERIAL PORT 0 OUTPUT ROUTING SLOT 13 REGISTER

Address: 0x4000C0F0, Reset: 0x3F, Name: SPT0_ROUTE13

Table 292. Bit Descriptions for SPT0_ROUTE13

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
[5:0]	SPT0_OUT_ROUTE13		Serial port output route slot 13.	0x3F	R/W
		0	FastDSP Channel 0.		
		1	FastDSP Channel 1.		
		2	FastDSP Channel 2.		
		3	FastDSP Channel 3.		
		4	FastDSP Channel 4.		
		5	FastDSP Channel 5.		
		6	FastDSP Channel 6.		
		7	FastDSP Channel 7.		
		8	FastDSP Channel 8.		
		9	FastDSP Channel 9.		
		10	FastDSP Channel 10.		
		11	FastDSP Channel 11.		
		12	FastDSP Channel 12.		
		13	FastDSP Channel 13.		
		14	FastDSP Channel 14.		
		15	FastDSP Channel 15.		
		16	Tensilica DSP Channel 0.		
		17	Tensilica DSP Channel 1.		
		18	Tensilica DSP Channel 2.		
		19	Tensilica DSP Channel 3.		
		20	Tensilica DSP Channel 4.		
		21	Tensilica DSP Channel 5.		
		22	Tensilica DSP Channel 6.		
		23	Tensilica DSP Channel 7.		
		24	Tensilica DSP Channel 8.		
		25	Tensilica DSP Channel 9.		
		26	Tensilica DSP Channel 10.		
		27	Tensilica DSP Channel 11.		
		28	Tensilica DSP Channel 12.		
		29	Tensilica DSP Channel 13.		
		30	Tensilica DSP Channel 14.		
		31	Tensilica DSP Channel 15.		
		32	Output ASRC Channel 0.		
		33	Output ASRC Channel 1.		
		34	Output ASRC Channel 2.		

Bits	Bit Name	Settings	Description	Reset	Access
		35	Output ASRC Channel 3.		
		36	ADC Channel 0.		
		37	ADC Channel 1.		
		38	ADC Channel 2.		
		39	DMIC Channel 0.		
		40	DMIC Channel 1.		
		41	DMIC Channel 2.		
		42	DMIC Channel 3.		
		43	Fast to Slow Decimator Channel 0.		
		44	Fast to Slow Decimator Channel 1.		
		45	Fast to Slow Decimator Channel 2.		
		46	Fast to Slow Decimator Channel 3.		
		47	Fast to Slow Decimator Channel 4.		
		48	Fast to Slow Decimator Channel 5.		
		49	Fast to Slow Decimator Channel 6.		
		50	Fast to Slow Decimator Channel 7.		
		51	EQ0.		
		55	DMIC Channel 4.		
		56	DMIC Channel 5.		
		57	DMIC Channel 6.		
		58	DMIC Channel 7.		
		63	No Output. Slot not used.		

SERIAL PORT 0 OUTPUT ROUTING SLOT 14 REGISTER

Address: 0x4000C0F1, Reset: 0x3F, Name: SPT0_ROUTE14

Table 293. Bit Descriptions for SPT0_ROUTE14

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
[5:0]	SPT0_OUT_ROUTE14		Serial port output route slot 14.	0x3F	R/W
		0	FastDSP Channel 0.		
		1	FastDSP Channel 1.		
		2	FastDSP Channel 2.		
		3	FastDSP Channel 3.		
		4	FastDSP Channel 4.		
		5	FastDSP Channel 5.		
		6	FastDSP Channel 6.		
		7	FastDSP Channel 7.		
		8	FastDSP Channel 8.		
		9	FastDSP Channel 9.		
		10	FastDSP Channel 10.		
		11	FastDSP Channel 11.		
		12	FastDSP Channel 12.		
		13	FastDSP Channel 13.		
		14	FastDSP Channel 14.		
		15	FastDSP Channel 15.		
		16	Tensilica DSP Channel 0.		
		17	Tensilica DSP Channel 1.		
		18	Tensilica DSP Channel 2.		
		19	Tensilica DSP Channel 3.		

Bits	Bit Name	Settings	Description	Reset	Access
		20	Tensilica DSP Channel 4.		
		21	Tensilica DSP Channel 5.		
		22	Tensilica DSP Channel 6.		
		23	Tensilica DSP Channel 7.		
		24	Tensilica DSP Channel 8.		
		25	Tensilica DSP Channel 9.		
		26	Tensilica DSP Channel 10.		
		27	Tensilica DSP Channel 11.		
		28	Tensilica DSP Channel 12.		
		29	Tensilica DSP Channel 13.		
		30	Tensilica DSP Channel 14.		
		31	Tensilica DSP Channel 15.		
		32	Output ASRC Channel 0.		
		33	Output ASRC Channel 1.		
		34	Output ASRC Channel 2.		
		35	Output ASRC Channel 3.		
		36	ADC Channel 0.		
		37	ADC Channel 1.		
		38	ADC Channel 2.		
		39	DMIC Channel 0.		
		40	DMIC Channel 1.		
		41	DMIC Channel 2.		
		42	DMIC Channel 3.		
		43	Fast to Slow Decimator Channel 0.		
		44	Fast to Slow Decimator Channel 1.		
		45	Fast to Slow Decimator Channel 2.		
		46	Fast to Slow Decimator Channel 3.		
		47	Fast to Slow Decimator Channel 4.		
		48	Fast to Slow Decimator Channel 5.		
		49	Fast to Slow Decimator Channel 6.		
		50	Fast to Slow Decimator Channel 7.		
		51	EQ0.		
		55	DMIC Channel 4.		
		56	DMIC Channel 5.		
		57	DMIC Channel 6.		
		58	DMIC Channel 7.		
		63	No Output. Slot not used.		

SERIAL PORT 0 OUTPUT ROUTING SLOT 15 REGISTER

Address: 0x4000C0F2, Reset: 0x3F, Name: SPT0_ROUTE15

Table 294. Bit Descriptions for SPT0_ROUTE15

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
[5:0]	SPT0_OUT_ROUTE15		Serial port output route slot 15.	0x3F	R/W
		0	FastDSP Channel 0.		
		1	FastDSP Channel 1.		
		2	FastDSP Channel 2.		
		3	FastDSP Channel 3.		
		4	FastDSP Channel 4.		

Bits	Bit Name	Settings	Description	Reset	Access
		5	FastDSP Channel 5.		
		6	FastDSP Channel 6.		
		7	FastDSP Channel 7.		
		8	FastDSP Channel 8.		
		9	FastDSP Channel 9.		
		10	FastDSP Channel 10.		
		11	FastDSP Channel 11.		
		12	FastDSP Channel 12.		
		13	FastDSP Channel 13.		
		14	FastDSP Channel 14.		
		15	FastDSP Channel 15.		
		16	Tensilica DSP Channel 0.		
		17	Tensilica DSP Channel 1.		
		18	Tensilica DSP Channel 2.		
		19	Tensilica DSP Channel 3.		
		20	Tensilica DSP Channel 4.		
		21	Tensilica DSP Channel 5.		
		22	Tensilica DSP Channel 6.		
		23	Tensilica DSP Channel 7.		
		24	Tensilica DSP Channel 8.		
		25	Tensilica DSP Channel 9.		
		26	Tensilica DSP Channel 10.		
		27	Tensilica DSP Channel 11.		
		28	Tensilica DSP Channel 12.		
		29	Tensilica DSP Channel 13.		
		30	Tensilica DSP Channel 14.		
		31	Tensilica DSP Channel 15.		
		32	Output ASRC Channel 0.		
		33	Output ASRC Channel 1.		
		34	Output ASRC Channel 2.		
		35	Output ASRC Channel 3.		
		36	ADC Channel 0.		
		37	ADC Channel 1.		
		38	ADC Channel 2.		
		39	DMIC Channel 0.		
		40	DMIC Channel 1.		
		41	DMIC Channel 2.		
		42	DMIC Channel 3.		
		43	Fast to Slow Decimator Channel 0.		
		44	Fast to Slow Decimator Channel 1.		
		45	Fast to Slow Decimator Channel 2.		
		46	Fast to Slow Decimator Channel 3.		
		47	Fast to Slow Decimator Channel 4.		
		48	Fast to Slow Decimator Channel 5.		
		49	Fast to Slow Decimator Channel 6.		

Bits	Bit Name	Settings	Description	Reset	Access
		50	Fast to Slow Decimator Channel 7.		
		51	EQ0.		
		55	DMIC Channel 4.		
		56	DMIC Channel 5.		
		57	DMIC Channel 6.		
		58	DMIC Channel 7.		
		63	No Output. Slot not used.		

SERIAL PORT 1 CONTROL 1 REGISTER

Address: 0x4000C0F3, Reset: 0x00, Name: SPT1_CTRL1

Table 295. Bit Descriptions for SPT1_CTRL1

Bits	Bit Name	Settings	Description	Reset	Access
7	RESERVED		Reserved.	0x0	R/W
6	SPT1_TRI_STATE		Serial port output - tri-state enable.	0x0	R/W
		1	tri-state enabled.		
		0	tri-state disabled.		
[5:4]	SPT1_SLOT_WIDTH		Serial port - selects slot width.	0x0	R/W
		00	32 BCLK's per slot.		
		01	16 BCLK's per slot.		
		10	24 BCLK's per slot.		
[3:1]	SPT1_DATA_FORMAT		Serial port - selects data format.	0x0	R/W
		001	Left Justified,delay by 0.		
		000	Typical I2S mode, delay by 1.		
		010	Delay by 8.		
		011	Delay by 12.		
		100	Delay by 16.		
0	SPT1_SAI_MODE		Serial port - selects SAI mode.	0x0	R/W
		0	STEREO (I2S, LJ, RJ).		
		1	TDM.		

SERIAL PORT 1 CONTROL 2 REGISTER

Address: 0x4000C0F4, Reset: 0x00, Name: SPT1_CTRL2

Table 296. Bit Descriptions for SPT1_CTRL2

Bits	Bit Name	Settings	Description	Reset	Access
[7:4]	RESERVED		Reserved.	0x0	R
3	SPT1_BCLK_POL		Serial port - selects bclk polarity.	0x0	R/W
		0	Capture on rising edge.		
		1	Capture on falling egde.		
[2:0]	SPT1_BCLK_SRC		Serial port - selects bclk source and rate.	0x0	R/W
		000	BCLK is from external source.		
		001	Generate BCLK at 3.072MHz.		
		010	Generate BCLK at 6.144MHz.		
		011	Generate BCLK at 12.288MHz.		
		100	Generate BCLK at 24.576MHz.		

SERIAL PORT 1 CONTROL 2 REGISTER

Address: 0x4000C0F5, Reset: 0x00, Name: SPT1_CTRL3

Table 297. Bit Descriptions for SPT1_CTRL3

Bits	Bit Name	Settings	Description	Reset	Access
[7:5]	RESERVED		Reserved.	0x0	R
4	SPT1_LRCLK_POL		Serial port - selects lrclk polarity.	0x0	R/W
		0	Normal Polarity.		
		1	Inverted Polarity.		
[3:0]	SPT1_LRCLK_SRC		Serial port - selects lrclk source and rate.	0x0	R/W
		0000	lrclk is from external.		
		0001	generate lrclk with 48KHz.		
		0010	generate lrclk with 96KHz.		
		0011	generate lrclk with 192KHz.		
		0100	generate lrclk with 12kHz.		
		0101	generate lrclk with 24KHz.		
		0110	generate lrclk with 384KHz.		
		0111	generate lrclk with 768KHz.		
		1000	generate lrclk with 8KHz.		
		1001	generate lrclk with 16KHz.		

SERIAL PORT 1 OUTPUT ROUTING SLOT 0 (LEFT) REGISTER

Address: 0x4000C0F6, Reset: 0x10, Name: SPT1_ROUTE0

Table 298. Bit Descriptions for SPT1_ROUTE0

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
[5:0]	SPT1_OUT_ROUTE0		Serial port output route slot 0 (left).	0x10	R/W
		0	FastDSP Channel 0.		
		1	FastDSP Channel 1.		
		2	FastDSP Channel 2.		
		3	FastDSP Channel 3.		
		4	FastDSP Channel 4.		
		5	FastDSP Channel 5.		
		6	FastDSP Channel 6.		
		7	FastDSP Channel 7.		
		8	FastDSP Channel 8.		
		9	FastDSP Channel 9.		
		10	FastDSP Channel 10.		
		11	FastDSP Channel 11.		
		12	FastDSP Channel 12.		
		13	FastDSP Channel 13.		
		14	FastDSP Channel 14.		
		15	FastDSP Channel 15.		
		16	Tensilica DSP Channel 0.		
		17	Tensilica DSP Channel 1.		
		18	Tensilica DSP Channel 2.		
		19	Tensilica DSP Channel 3.		
		20	Tensilica DSP Channel 4.		
		21	Tensilica DSP Channel 5.		
		22	Tensilica DSP Channel 6.		

Bits	Bit Name	Settings	Description	Reset	Access
		23	Tensilica DSP Channel 7.		
		24	Tensilica DSP Channel 8.		
		25	Tensilica DSP Channel 9.		
		26	Tensilica DSP Channel 10.		
		27	Tensilica DSP Channel 11.		
		28	Tensilica DSP Channel 12.		
		29	Tensilica DSP Channel 13.		
		30	Tensilica DSP Channel 14.		
		31	Tensilica DSP Channel 15.		
		32	Output ASRC Channel 0.		
		33	Output ASRC Channel 1.		
		34	Output ASRC Channel 2.		
		35	Output ASRC Channel 3.		
		36	ADC Channel 0.		
		37	ADC Channel 1.		
		38	ADC Channel 2.		
		39	DMIC Channel 0.		
		40	DMIC Channel 1.		
		41	DMIC Channel 2.		
		42	DMIC Channel 3.		
		43	Fast to Slow Decimator Channel 0.		
		44	Fast to Slow Decimator Channel 1.		
		45	Fast to Slow Decimator Channel 2.		
		46	Fast to Slow Decimator Channel 3.		
		47	Fast to Slow Decimator Channel 4.		
		48	Fast to Slow Decimator Channel 5.		
		49	Fast to Slow Decimator Channel 6.		
		50	Fast to Slow Decimator Channel 7.		
		51	EQ0.		
		55	DMIC Channel 4.		
		56	DMIC Channel 5.		
		57	DMIC Channel 6.		
		58	DMIC Channel 7.		
		63	No Output. Slot not used.		

SERIAL PORT 1 OUTPUT ROUTING SLOT 1 (RIGHT) REGISTER

Address: 0x4000C0F7, Reset: 0x11, Name: SPT1_ROUTE1

Table 299. Bit Descriptions for SPT1_ROUTE1

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
[5:0]	SPT1_OUT_ROUTE1		Serial port output route slot 1 (right).	0x11	R/W
		0	FastDSP Channel 0.		
		1	FastDSP Channel 1.		
		2	FastDSP Channel 2.		
		3	FastDSP Channel 3.		
		4	FastDSP Channel 4.		
		5	FastDSP Channel 5.		
		6	FastDSP Channel 6.		
		7	FastDSP Channel 7.		

Bits	Bit Name	Settings	Description	Reset	Access
		8	FastDSP Channel 8.		
		9	FastDSP Channel 9.		
		10	FastDSP Channel 10.		
		11	FastDSP Channel 11.		
		12	FastDSP Channel 12.		
		13	FastDSP Channel 13.		
		14	FastDSP Channel 14.		
		15	FastDSP Channel 15.		
		16	Tensilica DSP Channel 0.		
		17	Tensilica DSP Channel 1.		
		18	Tensilica DSP Channel 2.		
		19	Tensilica DSP Channel 3.		
		20	Tensilica DSP Channel 4.		
		21	Tensilica DSP Channel 5.		
		22	Tensilica DSP Channel 6.		
		23	Tensilica DSP Channel 7.		
		24	Tensilica DSP Channel 8.		
		25	Tensilica DSP Channel 9.		
		26	Tensilica DSP Channel 10.		
		27	Tensilica DSP Channel 11.		
		28	Tensilica DSP Channel 12.		
		29	Tensilica DSP Channel 13.		
		30	Tensilica DSP Channel 14.		
		31	Tensilica DSP Channel 15.		
		32	Output ASRC Channel 0.		
		33	Output ASRC Channel 1.		
		34	Output ASRC Channel 2.		
		35	Output ASRC Channel 3.		
		36	ADC Channel 0.		
		37	ADC Channel 1.		
		38	ADC Channel 2.		
		39	DMIC Channel 0.		
		40	DMIC Channel 1.		
		41	DMIC Channel 2.		
		42	DMIC Channel 3.		
		43	Fast to Slow Decimator Channel 0.		
		44	Fast to Slow Decimator Channel 1.		
		45	Fast to Slow Decimator Channel 2.		
		46	Fast to Slow Decimator Channel 3.		
		47	Fast to Slow Decimator Channel 4.		
		48	Fast to Slow Decimator Channel 5.		
		49	Fast to Slow Decimator Channel 6.		
		50	Fast to Slow Decimator Channel 7.		
		51	EQ0.		
		55	DMIC Channel 4.		
		56	DMIC Channel 5.		
		57	DMIC Channel 6.		
		58	DMIC Channel 7.		
		63	No Output. Slot not used.		

SERIAL PORT 1 OUTPUT ROUTING SLOT 2 REGISTER

Address: 0x4000C0F8, Reset: 0x3F, Name: SPT1_ROUTE2

Table 300. Bit Descriptions for SPT1_ROUTE2

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
[5:0]	SPT1_OUT_ROUTE2		Serial port output route slot 2.	0x3F	R/W
		0	FastDSP Channel 0.		
		1	FastDSP Channel 1.		
		2	FastDSP Channel 2.		
		3	FastDSP Channel 3.		
		4	FastDSP Channel 4.		
		5	FastDSP Channel 5.		
		6	FastDSP Channel 6.		
		7	FastDSP Channel 7.		
		8	FastDSP Channel 8.		
		9	FastDSP Channel 9.		
		10	FastDSP Channel 10.		
		11	FastDSP Channel 11.		
		12	FastDSP Channel 12.		
		13	FastDSP Channel 13.		
		14	FastDSP Channel 14.		
		15	FastDSP Channel 15.		
		16	Tensilica DSP Channel 0.		
		17	Tensilica DSP Channel 1.		
		18	Tensilica DSP Channel 2.		
		19	Tensilica DSP Channel 3.		
		20	Tensilica DSP Channel 4.		
		21	Tensilica DSP Channel 5.		
		22	Tensilica DSP Channel 6.		
		23	Tensilica DSP Channel 7.		
		24	Tensilica DSP Channel 8.		
		25	Tensilica DSP Channel 9.		
		26	Tensilica DSP Channel 10.		
		27	Tensilica DSP Channel 11.		
		28	Tensilica DSP Channel 12.		
		29	Tensilica DSP Channel 13.		
		30	Tensilica DSP Channel 14.		
		31	Tensilica DSP Channel 15.		
		32	Output ASRC Channel 0.		
		33	Output ASRC Channel 1.		
		34	Output ASRC Channel 2.		
		35	Output ASRC Channel 3.		
		36	ADC Channel 0.		
		37	ADC Channel 1.		
		38	ADC Channel 2.		
		39	DMIC Channel 0.		
		40	DMIC Channel 1.		
		41	DMIC Channel 2.		
		42	DMIC Channel 3.		
		43	Fast to Slow Decimator Channel 0.		

Bits	Bit Name	Settings	Description	Reset	Access
		44	Fast to Slow Decimator Channel 1.		
		45	Fast to Slow Decimator Channel 2.		
		46	Fast to Slow Decimator Channel 3.		
		47	Fast to Slow Decimator Channel 4.		
		48	Fast to Slow Decimator Channel 5.		
		49	Fast to Slow Decimator Channel 6.		
		50	Fast to Slow Decimator Channel 7.		
		51	EQ0.		
		55	DMIC Channel 4.		
		56	DMIC Channel 5.		
		57	DMIC Channel 6.		
		58	DMIC Channel 7.		
		63	No Output. Slot not used.		

SERIAL PORT 1 OUTPUT ROUTING SLOT 3 REGISTER

Address: 0x4000C0F9, Reset: 0x3F, Name: SPT1_ROUTE3

Table 301. Bit Descriptions for SPT1_ROUTE3

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
[5:0]	SPT1_OUT_ROUTE3		Serial port output route slot 3.	0x3F	R/W
		0	FastDSP Channel 0.		
		1	FastDSP Channel 1.		
		2	FastDSP Channel 2.		
		3	FastDSP Channel 3.		
		4	FastDSP Channel 4.		
		5	FastDSP Channel 5.		
		6	FastDSP Channel 6.		
		7	FastDSP Channel 7.		
		8	FastDSP Channel 8.		
		9	FastDSP Channel 9.		
		10	FastDSP Channel 10.		
		11	FastDSP Channel 11.		
		12	FastDSP Channel 12.		
		13	FastDSP Channel 13.		
		14	FastDSP Channel 14.		
		15	FastDSP Channel 15.		
		16	Tensilica DSP Channel 0.		
		17	Tensilica DSP Channel 1.		
		18	Tensilica DSP Channel 2.		
		19	Tensilica DSP Channel 3.		
		20	Tensilica DSP Channel 4.		
		21	Tensilica DSP Channel 5.		
		22	Tensilica DSP Channel 6.		
		23	Tensilica DSP Channel 7.		
		24	Tensilica DSP Channel 8.		
		25	Tensilica DSP Channel 9.		
		26	Tensilica DSP Channel 10.		
		27	Tensilica DSP Channel 11.		
		28	Tensilica DSP Channel 12.		

Bits	Bit Name	Settings	Description	Reset	Access
		29	Tensilica DSP Channel 13.		
		30	Tensilica DSP Channel 14.		
		31	Tensilica DSP Channel 15.		
		32	Output ASRC Channel 0.		
		33	Output ASRC Channel 1.		
		34	Output ASRC Channel 2.		
		35	Output ASRC Channel 3.		
		36	ADC Channel 0.		
		37	ADC Channel 1.		
		38	ADC Channel 2.		
		39	DMIC Channel 0.		
		40	DMIC Channel 1.		
		41	DMIC Channel 2.		
		42	DMIC Channel 3.		
		43	Fast to Slow Decimator Channel 0.		
		44	Fast to Slow Decimator Channel 1.		
		45	Fast to Slow Decimator Channel 2.		
		46	Fast to Slow Decimator Channel 3.		
		47	Fast to Slow Decimator Channel 4.		
		48	Fast to Slow Decimator Channel 5.		
		49	Fast to Slow Decimator Channel 6.		
		50	Fast to Slow Decimator Channel 7.		
		51	EQ0.		
		55	DMIC Channel 4.		
		56	DMIC Channel 5.		
		57	DMIC Channel 6.		
		58	DMIC Channel 7.		
		63	No Output. Slot not used.		

SERIAL PORT 1 OUTPUT ROUTING SLOT 4 REGISTER

Address: 0x4000C0FA, Reset: 0x3F, Name: SPT1_ROUTE4

Table 302. Bit Descriptions for SPT1_ROUTE4

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
[5:0]	SPT1_OUT_ROUTE4		Serial port output route slot 4.	0x3F	R/W
		0	FastDSP Channel 0.		
		1	FastDSP Channel 1.		
		2	FastDSP Channel 2.		
		3	FastDSP Channel 3.		
		4	FastDSP Channel 4.		
		5	FastDSP Channel 5.		
		6	FastDSP Channel 6.		
		7	FastDSP Channel 7.		
		8	FastDSP Channel 8.		
		9	FastDSP Channel 9.		
		10	FastDSP Channel 10.		
		11	FastDSP Channel 11.		
		12	FastDSP Channel 12.		
		13	FastDSP Channel 13.		

Bits	Bit Name	Settings	Description	Reset	Access
		14	FastDSP Channel 14.		
		15	FastDSP Channel 15.		
		16	Tensilica DSP Channel 0.		
		17	Tensilica DSP Channel 1.		
		18	Tensilica DSP Channel 2.		
		19	Tensilica DSP Channel 3.		
		20	Tensilica DSP Channel 4.		
		21	Tensilica DSP Channel 5.		
		22	Tensilica DSP Channel 6.		
		23	Tensilica DSP Channel 7.		
		24	Tensilica DSP Channel 8.		
		25	Tensilica DSP Channel 9.		
		26	Tensilica DSP Channel 10.		
		27	Tensilica DSP Channel 11.		
		28	Tensilica DSP Channel 12.		
		29	Tensilica DSP Channel 13.		
		30	Tensilica DSP Channel 14.		
		31	Tensilica DSP Channel 15.		
		32	Output ASRC Channel 0.		
		33	Output ASRC Channel 1.		
		34	Output ASRC Channel 2.		
		35	Output ASRC Channel 3.		
		36	ADC Channel 0.		
		37	ADC Channel 1.		
		38	ADC Channel 2.		
		39	DMIC Channel 0.		
		40	DMIC Channel 1.		
		41	DMIC Channel 2.		
		42	DMIC Channel 3.		
		43	Fast to Slow Decimator Channel 0.		
		44	Fast to Slow Decimator Channel 1.		
		45	Fast to Slow Decimator Channel 2.		
		46	Fast to Slow Decimator Channel 3.		
		47	Fast to Slow Decimator Channel 4.		
		48	Fast to Slow Decimator Channel 5.		
		49	Fast to Slow Decimator Channel 6.		
		50	Fast to Slow Decimator Channel 7.		
		51	EQ0.		
		55	DMIC Channel 4.		
		56	DMIC Channel 5.		
		57	DMIC Channel 6.		
		58	DMIC Channel 7.		
		63	No Output. Slot not used.		

SERIAL PORT 1 OUTPUT ROUTING SLOT 5 REGISTER

Address: 0x4000C0FB, Reset: 0x3F, Name: SPT1_ROUTE5

Table 303. Bit Descriptions for SPT1_ROUTE5

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
[5:0]	SPT1_OUT_ROUTE5		Serial port output route slot 5.	0x3F	R/W
		0	FastDSP Channel 0.		
		1	FastDSP Channel 1.		
		2	FastDSP Channel 2.		
		3	FastDSP Channel 3.		
		4	FastDSP Channel 4.		
		5	FastDSP Channel 5.		
		6	FastDSP Channel 6.		
		7	FastDSP Channel 7.		
		8	FastDSP Channel 8.		
		9	FastDSP Channel 9.		
		10	FastDSP Channel 10.		
		11	FastDSP Channel 11.		
		12	FastDSP Channel 12.		
		13	FastDSP Channel 13.		
		14	FastDSP Channel 14.		
		15	FastDSP Channel 15.		
		16	Tensilica DSP Channel 0.		
		17	Tensilica DSP Channel 1.		
		18	Tensilica DSP Channel 2.		
		19	Tensilica DSP Channel 3.		
		20	Tensilica DSP Channel 4.		
		21	Tensilica DSP Channel 5.		
		22	Tensilica DSP Channel 6.		
		23	Tensilica DSP Channel 7.		
		24	Tensilica DSP Channel 8.		
		25	Tensilica DSP Channel 9.		
		26	Tensilica DSP Channel 10.		
		27	Tensilica DSP Channel 11.		
		28	Tensilica DSP Channel 12.		
		29	Tensilica DSP Channel 13.		
		30	Tensilica DSP Channel 14.		
		31	Tensilica DSP Channel 15.		
		32	Output ASRC Channel 0.		
		33	Output ASRC Channel 1.		
		34	Output ASRC Channel 2.		
		35	Output ASRC Channel 3.		
		36	ADC Channel 0.		
		37	ADC Channel 1.		
		38	ADC Channel 2.		
		39	DMIC Channel 0.		
		40	DMIC Channel 1.		
		41	DMIC Channel 2.		
		42	DMIC Channel 3.		
		43	Fast to Slow Decimator Channel 0.		

Bits	Bit Name	Settings	Description	Reset	Access
		44	Fast to Slow Decimator Channel 1.		
		45	Fast to Slow Decimator Channel 2.		
		46	Fast to Slow Decimator Channel 3.		
		47	Fast to Slow Decimator Channel 4.		
		48	Fast to Slow Decimator Channel 5.		
		49	Fast to Slow Decimator Channel 6.		
		50	Fast to Slow Decimator Channel 7.		
		51	EQ0.		
		55	DMIC Channel 4.		
		56	DMIC Channel 5.		
		57	DMIC Channel 6.		
		58	DMIC Channel 7.		
		63	No Output. Slot not used.		

SERIAL PORT 1 OUTPUT ROUTING SLOT 6 REGISTER

Address: 0x4000C0FC, Reset: 0x3F, Name: SPT1_ROUTE6

Table 304. Bit Descriptions for SPT1_ROUTE6

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
[5:0]	SPT1_OUT_ROUTE6		Serial port output route slot 6.	0x3F	R/W
		0	FastDSP Channel 0.		
		1	FastDSP Channel 1.		
		2	FastDSP Channel 2.		
		3	FastDSP Channel 3.		
		4	FastDSP Channel 4.		
		5	FastDSP Channel 5.		
		6	FastDSP Channel 6.		
		7	FastDSP Channel 7.		
		8	FastDSP Channel 8.		
		9	FastDSP Channel 9.		
		10	FastDSP Channel 10.		
		11	FastDSP Channel 11.		
		12	FastDSP Channel 12.		
		13	FastDSP Channel 13.		
		14	FastDSP Channel 14.		
		15	FastDSP Channel 15.		
		16	Tensilica DSP Channel 0.		
		17	Tensilica DSP Channel 1.		
		18	Tensilica DSP Channel 2.		
		19	Tensilica DSP Channel 3.		
		20	Tensilica DSP Channel 4.		
		21	Tensilica DSP Channel 5.		
		22	Tensilica DSP Channel 6.		
		23	Tensilica DSP Channel 7.		
		24	Tensilica DSP Channel 8.		
		25	Tensilica DSP Channel 9.		
		26	Tensilica DSP Channel 10.		
		27	Tensilica DSP Channel 11.		
		28	Tensilica DSP Channel 12.		

Bits	Bit Name	Settings	Description	Reset	Access
		29	Tensilica DSP Channel 13.		
		30	Tensilica DSP Channel 14.		
		31	Tensilica DSP Channel 15.		
		32	Output ASRC Channel 0.		
		33	Output ASRC Channel 1.		
		34	Output ASRC Channel 2.		
		35	Output ASRC Channel 3.		
		36	ADC Channel 0.		
		37	ADC Channel 1.		
		38	ADC Channel 2.		
		39	DMIC Channel 0.		
		40	DMIC Channel 1.		
		41	DMIC Channel 2.		
		42	DMIC Channel 3.		
		43	Fast to Slow Decimator Channel 0.		
		44	Fast to Slow Decimator Channel 1.		
		45	Fast to Slow Decimator Channel 2.		
		46	Fast to Slow Decimator Channel 3.		
		47	Fast to Slow Decimator Channel 4.		
		48	Fast to Slow Decimator Channel 5.		
		49	Fast to Slow Decimator Channel 6.		
		50	Fast to Slow Decimator Channel 7.		
		51	EQ0.		
		55	DMIC Channel 4.		
		56	DMIC Channel 5.		
		57	DMIC Channel 6.		
		58	DMIC Channel 7.		
		63	No Output. Slot not used.		

SERIAL PORT 1 OUTPUT ROUTING SLOT 7 REGISTER

Address: 0x4000C0FD, Reset: 0x3F, Name: SPT1_ROUTE7

Table 305. Bit Descriptions for SPT1_ROUTE7

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
[5:0]	SPT1_OUT_ROUTE7		Serial port output route slot 7.	0x3F	R/W
		0	FastDSP Channel 0.		
		1	FastDSP Channel 1.		
		2	FastDSP Channel 2.		
		3	FastDSP Channel 3.		
		4	FastDSP Channel 4.		
		5	FastDSP Channel 5.		
		6	FastDSP Channel 6.		
		7	FastDSP Channel 7.		
		8	FastDSP Channel 8.		
		9	FastDSP Channel 9.		
		10	FastDSP Channel 10.		
		11	FastDSP Channel 11.		
		12	FastDSP Channel 12.		
		13	FastDSP Channel 13.		

Bits	Bit Name	Settings	Description	Reset	Access
		14	FastDSP Channel 14.		
		15	FastDSP Channel 15.		
		16	Tensilica DSP Channel 0.		
		17	Tensilica DSP Channel 1.		
		18	Tensilica DSP Channel 2.		
		19	Tensilica DSP Channel 3.		
		20	Tensilica DSP Channel 4.		
		21	Tensilica DSP Channel 5.		
		22	Tensilica DSP Channel 6.		
		23	Tensilica DSP Channel 7.		
		24	Tensilica DSP Channel 8.		
		25	Tensilica DSP Channel 9.		
		26	Tensilica DSP Channel 10.		
		27	Tensilica DSP Channel 11.		
		28	Tensilica DSP Channel 12.		
		29	Tensilica DSP Channel 13.		
		30	Tensilica DSP Channel 14.		
		31	Tensilica DSP Channel 15.		
		32	Output ASRC Channel 0.		
		33	Output ASRC Channel 1.		
		34	Output ASRC Channel 2.		
		35	Output ASRC Channel 3.		
		36	ADC Channel 0.		
		37	ADC Channel 1.		
		38	ADC Channel 2.		
		39	DMIC Channel 0.		
		40	DMIC Channel 1.		
		41	DMIC Channel 2.		
		42	DMIC Channel 3.		
		43	Fast to Slow Decimator Channel 0.		
		44	Fast to Slow Decimator Channel 1.		
		45	Fast to Slow Decimator Channel 2.		
		46	Fast to Slow Decimator Channel 3.		
		47	Fast to Slow Decimator Channel 4.		
		48	Fast to Slow Decimator Channel 5.		
		49	Fast to Slow Decimator Channel 6.		
		50	Fast to Slow Decimator Channel 7.		
		51	EQ0.		
		55	DMIC Channel 4.		
		56	DMIC Channel 5.		
		57	DMIC Channel 6.		
		58	DMIC Channel 7.		
		63	No Output. Slot not used.		

SERIAL PORT 1 OUTPUT ROUTING SLOT 8 REGISTER

Address: 0x4000C0FE, Reset: 0x3F, Name: SPT1_ROUTE8

Table 306. Bit Descriptions for SPT1_ROUTE8

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
[5:0]	SPT1_OUT_ROUTE8		Serial port output route slot 8.	0x3F	R/W
		0	FastDSP Channel 0.		
		1	FastDSP Channel 1.		
		2	FastDSP Channel 2.		
		3	FastDSP Channel 3.		
		4	FastDSP Channel 4.		
		5	FastDSP Channel 5.		
		6	FastDSP Channel 6.		
		7	FastDSP Channel 7.		
		8	FastDSP Channel 8.		
		9	FastDSP Channel 9.		
		10	FastDSP Channel 10.		
		11	FastDSP Channel 11.		
		12	FastDSP Channel 12.		
		13	FastDSP Channel 13.		
		14	FastDSP Channel 14.		
		15	FastDSP Channel 15.		
		16	Tensilica DSP Channel 0.		
		17	Tensilica DSP Channel 1.		
		18	Tensilica DSP Channel 2.		
		19	Tensilica DSP Channel 3.		
		20	Tensilica DSP Channel 4.		
		21	Tensilica DSP Channel 5.		
		22	Tensilica DSP Channel 6.		
		23	Tensilica DSP Channel 7.		
		24	Tensilica DSP Channel 8.		
		25	Tensilica DSP Channel 9.		
		26	Tensilica DSP Channel 10.		
		27	Tensilica DSP Channel 11.		
		28	Tensilica DSP Channel 12.		
		29	Tensilica DSP Channel 13.		
		30	Tensilica DSP Channel 14.		
		31	Tensilica DSP Channel 15.		
		32	Output ASRC Channel 0.		
		33	Output ASRC Channel 1.		
		34	Output ASRC Channel 2.		
		35	Output ASRC Channel 3.		
		36	ADC Channel 0.		
		37	ADC Channel 1.		
		38	ADC Channel 2.		
		39	DMIC Channel 0.		
		40	DMIC Channel 1.		
		41	DMIC Channel 2.		
		42	DMIC Channel 3.		
		43	Fast to Slow Decimator Channel 0.		

Bits	Bit Name	Settings	Description	Reset	Access
		44	Fast to Slow Decimator Channel 1.		
		45	Fast to Slow Decimator Channel 2.		
		46	Fast to Slow Decimator Channel 3.		
		47	Fast to Slow Decimator Channel 4.		
		48	Fast to Slow Decimator Channel 5.		
		49	Fast to Slow Decimator Channel 6.		
		50	Fast to Slow Decimator Channel 7.		
		51	EQ0.		
		55	DMIC Channel 4.		
		56	DMIC Channel 5.		
		57	DMIC Channel 6.		
		58	DMIC Channel 7.		
		63	No Output. Slot not used.		

SERIAL PORT 1 OUTPUT ROUTING SLOT 9 REGISTER

Address: 0x4000C0FF, Reset: 0x3F, Name: SPT1_ROUTE9

Table 307. Bit Descriptions for SPT1_ROUTE9

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
[5:0]	SPT1_OUT_ROUTE9		Serial port output route slot 9.	0x3F	R/W
		0	FastDSP Channel 0.		
		1	FastDSP Channel 1.		
		2	FastDSP Channel 2.		
		3	FastDSP Channel 3.		
		4	FastDSP Channel 4.		
		5	FastDSP Channel 5.		
		6	FastDSP Channel 6.		
		7	FastDSP Channel 7.		
		8	FastDSP Channel 8.		
		9	FastDSP Channel 9.		
		10	FastDSP Channel 10.		
		11	FastDSP Channel 11.		
		12	FastDSP Channel 12.		
		13	FastDSP Channel 13.		
		14	FastDSP Channel 14.		
		15	FastDSP Channel 15.		
		16	Tensilica DSP Channel 0.		
		17	Tensilica DSP Channel 1.		
		18	Tensilica DSP Channel 2.		
		19	Tensilica DSP Channel 3.		
		20	Tensilica DSP Channel 4.		
		21	Tensilica DSP Channel 5.		
		22	Tensilica DSP Channel 6.		
		23	Tensilica DSP Channel 7.		
		24	Tensilica DSP Channel 8.		
		25	Tensilica DSP Channel 9.		
		26	Tensilica DSP Channel 10.		
		27	Tensilica DSP Channel 11.		
		28	Tensilica DSP Channel 12.		

Bits	Bit Name	Settings	Description	Reset	Access
		29	Tensilica DSP Channel 13.		
		30	Tensilica DSP Channel 14.		
		31	Tensilica DSP Channel 15.		
		32	Output ASRC Channel 0.		
		33	Output ASRC Channel 1.		
		34	Output ASRC Channel 2.		
		35	Output ASRC Channel 3.		
		36	ADC Channel 0.		
		37	ADC Channel 1.		
		38	ADC Channel 2.		
		39	DMIC Channel 0.		
		40	DMIC Channel 1.		
		41	DMIC Channel 2.		
		42	DMIC Channel 3.		
		43	Fast to Slow Decimator Channel 0.		
		44	Fast to Slow Decimator Channel 1.		
		45	Fast to Slow Decimator Channel 2.		
		46	Fast to Slow Decimator Channel 3.		
		47	Fast to Slow Decimator Channel 4.		
		48	Fast to Slow Decimator Channel 5.		
		49	Fast to Slow Decimator Channel 6.		
		50	Fast to Slow Decimator Channel 7.		
		51	EQ0.		
		55	DMIC Channel 4.		
		56	DMIC Channel 5.		
		57	DMIC Channel 6.		
		58	DMIC Channel 7.		
		63	No Output. Slot not used.		

SERIAL PORT 1 OUTPUT ROUTING SLOT 10 REGISTER

Address: 0x4000C100, Reset: 0x3F, Name: SPT1_ROUTE10

Table 308. Bit Descriptions for SPT1_ROUTE10

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
[5:0]	SPT1_OUT_ROUTE10		Serial port output route slot 10.	0x3F	R/W
		0	FastDSP Channel 0.		
		1	FastDSP Channel 1.		
		2	FastDSP Channel 2.		
		3	FastDSP Channel 3.		
		4	FastDSP Channel 4.		
		5	FastDSP Channel 5.		
		6	FastDSP Channel 6.		
		7	FastDSP Channel 7.		
		8	FastDSP Channel 8.		
		9	FastDSP Channel 9.		
		10	FastDSP Channel 10.		
		11	FastDSP Channel 11.		
		12	FastDSP Channel 12.		
		13	FastDSP Channel 13.		

Bits	Bit Name	Settings	Description	Reset	Access
		14	FastDSP Channel 14.		
		15	FastDSP Channel 15.		
		16	Tensilica DSP Channel 0.		
		17	Tensilica DSP Channel 1.		
		18	Tensilica DSP Channel 2.		
		19	Tensilica DSP Channel 3.		
		20	Tensilica DSP Channel 4.		
		21	Tensilica DSP Channel 5.		
		22	Tensilica DSP Channel 6.		
		23	Tensilica DSP Channel 7.		
		24	Tensilica DSP Channel 8.		
		25	Tensilica DSP Channel 9.		
		26	Tensilica DSP Channel 10.		
		27	Tensilica DSP Channel 11.		
		28	Tensilica DSP Channel 12.		
		29	Tensilica DSP Channel 13.		
		30	Tensilica DSP Channel 14.		
		31	Tensilica DSP Channel 15.		
		32	Output ASRC Channel 0.		
		33	Output ASRC Channel 1.		
		34	Output ASRC Channel 2.		
		35	Output ASRC Channel 3.		
		36	ADC Channel 0.		
		37	ADC Channel 1.		
		38	ADC Channel 2.		
		39	DMIC Channel 0.		
		40	DMIC Channel 1.		
		41	DMIC Channel 2.		
		42	DMIC Channel 3.		
		43	Fast to Slow Decimator Channel 0.		
		44	Fast to Slow Decimator Channel 1.		
		45	Fast to Slow Decimator Channel 2.		
		46	Fast to Slow Decimator Channel 3.		
		47	Fast to Slow Decimator Channel 4.		
		48	Fast to Slow Decimator Channel 5.		
		49	Fast to Slow Decimator Channel 6.		
		50	Fast to Slow Decimator Channel 7.		
		51	EQ0.		
		55	DMIC Channel 4.		
		56	DMIC Channel 5.		
		57	DMIC Channel 6.		
		58	DMIC Channel 7.		
		63	No Output. Slot not used.		

SERIAL PORT 1 OUTPUT ROUTING SLOT 11 REGISTER

Address: 0x4000C101, Reset: 0x3F, Name: SPT1_ROUTE11

Table 309. Bit Descriptions for SPT1_ROUTE11

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
[5:0]	SPT1_OUT_ROUTE11		Serial port output route slot 11.	0x3F	R/W
		0	FastDSP Channel 0.		
		1	FastDSP Channel 1.		
		2	FastDSP Channel 2.		
		3	FastDSP Channel 3.		
		4	FastDSP Channel 4.		
		5	FastDSP Channel 5.		
		6	FastDSP Channel 6.		
		7	FastDSP Channel 7.		
		8	FastDSP Channel 8.		
		9	FastDSP Channel 9.		
		10	FastDSP Channel 10.		
		11	FastDSP Channel 11.		
		12	FastDSP Channel 12.		
		13	FastDSP Channel 13.		
		14	FastDSP Channel 14.		
		15	FastDSP Channel 15.		
		16	Tensilica DSP Channel 0.		
		17	Tensilica DSP Channel 1.		
		18	Tensilica DSP Channel 2.		
		19	Tensilica DSP Channel 3.		
		20	Tensilica DSP Channel 4.		
		21	Tensilica DSP Channel 5.		
		22	Tensilica DSP Channel 6.		
		23	Tensilica DSP Channel 7.		
		24	Tensilica DSP Channel 8.		
		25	Tensilica DSP Channel 9.		
		26	Tensilica DSP Channel 10.		
		27	Tensilica DSP Channel 11.		
		28	Tensilica DSP Channel 12.		
		29	Tensilica DSP Channel 13.		
		30	Tensilica DSP Channel 14.		
		31	Tensilica DSP Channel 15.		
		32	Output ASRC Channel 0.		
		33	Output ASRC Channel 1.		
		34	Output ASRC Channel 2.		
		35	Output ASRC Channel 3.		
		36	ADC Channel 0.		
		37	ADC Channel 1.		
		38	ADC Channel 2.		
		39	DMIC Channel 0.		
		40	DMIC Channel 1.		
		41	DMIC Channel 2.		
		42	DMIC Channel 3.		
		43	Fast to Slow Decimator Channel 0.		

Bits	Bit Name	Settings	Description	Reset	Access
		44	Fast to Slow Decimator Channel 1.		
		45	Fast to Slow Decimator Channel 2.		
		46	Fast to Slow Decimator Channel 3.		
		47	Fast to Slow Decimator Channel 4.		
		48	Fast to Slow Decimator Channel 5.		
		49	Fast to Slow Decimator Channel 6.		
		50	Fast to Slow Decimator Channel 7.		
		51	EQ0.		
		55	DMIC Channel 4.		
		56	DMIC Channel 5.		
		57	DMIC Channel 6.		
		58	DMIC Channel 7.		
		63	No Output. Slot not used.		

SERIAL PORT 1 OUTPUT ROUTING SLOT 12 REGISTER

Address: 0x4000C102, Reset: 0x3F, Name: SPT1_ROUTE12

Table 310. Bit Descriptions for SPT1_ROUTE12

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
[5:0]	SPT1_OUT_ROUTE12		Serial port output route slot 12.	0x3F	R/W
		0	FastDSP Channel 0.		
		1	FastDSP Channel 1.		
		2	FastDSP Channel 2.		
		3	FastDSP Channel 3.		
		4	FastDSP Channel 4.		
		5	FastDSP Channel 5.		
		6	FastDSP Channel 6.		
		7	FastDSP Channel 7.		
		8	FastDSP Channel 8.		
		9	FastDSP Channel 9.		
		10	FastDSP Channel 10.		
		11	FastDSP Channel 11.		
		12	FastDSP Channel 12.		
		13	FastDSP Channel 13.		
		14	FastDSP Channel 14.		
		15	FastDSP Channel 15.		
		16	Tensilica DSP Channel 0.		
		17	Tensilica DSP Channel 1.		
		18	Tensilica DSP Channel 2.		
		19	Tensilica DSP Channel 3.		
		20	Tensilica DSP Channel 4.		
		21	Tensilica DSP Channel 5.		
		22	Tensilica DSP Channel 6.		
		23	Tensilica DSP Channel 7.		
		24	Tensilica DSP Channel 8.		
		25	Tensilica DSP Channel 9.		
		26	Tensilica DSP Channel 10.		
		27	Tensilica DSP Channel 11.		
		28	Tensilica DSP Channel 12.		

Bits	Bit Name	Settings	Description	Reset	Access
		29	Tensilica DSP Channel 13.		
		30	Tensilica DSP Channel 14.		
		31	Tensilica DSP Channel 15.		
		32	Output ASRC Channel 0.		
		33	Output ASRC Channel 1.		
		34	Output ASRC Channel 2.		
		35	Output ASRC Channel 3.		
		36	ADC Channel 0.		
		37	ADC Channel 1.		
		38	ADC Channel 2.		
		39	DMIC Channel 0.		
		40	DMIC Channel 1.		
		41	DMIC Channel 2.		
		42	DMIC Channel 3.		
		43	Fast to Slow Decimator Channel 0.		
		44	Fast to Slow Decimator Channel 1.		
		45	Fast to Slow Decimator Channel 2.		
		46	Fast to Slow Decimator Channel 3.		
		47	Fast to Slow Decimator Channel 4.		
		48	Fast to Slow Decimator Channel 5.		
		49	Fast to Slow Decimator Channel 6.		
		50	Fast to Slow Decimator Channel 7.		
		51	EQ0.		
		55	DMIC Channel 4.		
		56	DMIC Channel 5.		
		57	DMIC Channel 6.		
		58	DMIC Channel 7.		
		63	No Output. Slot not used.		

SERIAL PORT 1 OUTPUT ROUTING SLOT 13 REGISTER

Address: 0x4000C103, Reset: 0x3F, Name: SPT1_ROUTE13

Table 311. Bit Descriptions for SPT1_ROUTE13

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
[5:0]	SPT1_OUT_ROUTE13		Serial port output route slot 13.	0x3F	R/W
		0	FastDSP Channel 0.		
		1	FastDSP Channel 1.		
		2	FastDSP Channel 2.		
		3	FastDSP Channel 3.		
		4	FastDSP Channel 4.		
		5	FastDSP Channel 5.		
		6	FastDSP Channel 6.		
		7	FastDSP Channel 7.		
		8	FastDSP Channel 8.		
		9	FastDSP Channel 9.		
		10	FastDSP Channel 10.		
		11	FastDSP Channel 11.		
		12	FastDSP Channel 12.		
		13	FastDSP Channel 13.		

Bits	Bit Name	Settings	Description	Reset	Access
		14	FastDSP Channel 14.		
		15	FastDSP Channel 15.		
		16	Tensilica DSP Channel 0.		
		17	Tensilica DSP Channel 1.		
		18	Tensilica DSP Channel 2.		
		19	Tensilica DSP Channel 3.		
		20	Tensilica DSP Channel 4.		
		21	Tensilica DSP Channel 5.		
		22	Tensilica DSP Channel 6.		
		23	Tensilica DSP Channel 7.		
		24	Tensilica DSP Channel 8.		
		25	Tensilica DSP Channel 9.		
		26	Tensilica DSP Channel 10.		
		27	Tensilica DSP Channel 11.		
		28	Tensilica DSP Channel 12.		
		29	Tensilica DSP Channel 13.		
		30	Tensilica DSP Channel 14.		
		31	Tensilica DSP Channel 15.		
		32	Output ASRC Channel 0.		
		33	Output ASRC Channel 1.		
		34	Output ASRC Channel 2.		
		35	Output ASRC Channel 3.		
		36	ADC Channel 0.		
		37	ADC Channel 1.		
		38	ADC Channel 2.		
		39	DMIC Channel 0.		
		40	DMIC Channel 1.		
		41	DMIC Channel 2.		
		42	DMIC Channel 3.		
		43	Fast to Slow Decimator Channel 0.		
		44	Fast to Slow Decimator Channel 1.		
		45	Fast to Slow Decimator Channel 2.		
		46	Fast to Slow Decimator Channel 3.		
		47	Fast to Slow Decimator Channel 4.		
		48	Fast to Slow Decimator Channel 5.		
		49	Fast to Slow Decimator Channel 6.		
		50	Fast to Slow Decimator Channel 7.		
		51	EQ0.		
		55	DMIC Channel 4.		
		56	DMIC Channel 5.		
		57	DMIC Channel 6.		
		58	DMIC Channel 7.		
		63	No Output. Slot not used.		

SERIAL PORT 1 OUTPUT ROUTING SLOT 14 REGISTER

Address: 0x4000C104, Reset: 0x3F, Name: SPT1_ROUTE14

Table 312. Bit Descriptions for SPT1_ROUTE14

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
[5:0]	SPT1_OUT_ROUTE14		Serial port output route slot 14.	0x3F	R/W
		0	FastDSP Channel 0.		
		1	FastDSP Channel 1.		
		2	FastDSP Channel 2.		
		3	FastDSP Channel 3.		
		4	FastDSP Channel 4.		
		5	FastDSP Channel 5.		
		6	FastDSP Channel 6.		
		7	FastDSP Channel 7.		
		8	FastDSP Channel 8.		
		9	FastDSP Channel 9.		
		10	FastDSP Channel 10.		
		11	FastDSP Channel 11.		
		12	FastDSP Channel 12.		
		13	FastDSP Channel 13.		
		14	FastDSP Channel 14.		
		15	FastDSP Channel 15.		
		16	Tensilica DSP Channel 0.		
		17	Tensilica DSP Channel 1.		
		18	Tensilica DSP Channel 2.		
		19	Tensilica DSP Channel 3.		
		20	Tensilica DSP Channel 4.		
		21	Tensilica DSP Channel 5.		
		22	Tensilica DSP Channel 6.		
		23	Tensilica DSP Channel 7.		
		24	Tensilica DSP Channel 8.		
		25	Tensilica DSP Channel 9.		
		26	Tensilica DSP Channel 10.		
		27	Tensilica DSP Channel 11.		
		28	Tensilica DSP Channel 12.		
		29	Tensilica DSP Channel 13.		
		30	Tensilica DSP Channel 14.		
		31	Tensilica DSP Channel 15.		
		32	Output ASRC Channel 0.		
		33	Output ASRC Channel 1.		
		34	Output ASRC Channel 2.		
		35	Output ASRC Channel 3.		
		36	ADC Channel 0.		
		37	ADC Channel 1.		
		38	ADC Channel 2.		
		39	DMIC Channel 0.		
		40	DMIC Channel 1.		
		41	DMIC Channel 2.		
		42	DMIC Channel 3.		
		43	Fast to Slow Decimator Channel 0.		

Bits	Bit Name	Settings	Description	Reset	Access
		44	Fast to Slow Decimator Channel 1.		
		45	Fast to Slow Decimator Channel 2.		
		46	Fast to Slow Decimator Channel 3.		
		47	Fast to Slow Decimator Channel 4.		
		48	Fast to Slow Decimator Channel 5.		
		49	Fast to Slow Decimator Channel 6.		
		50	Fast to Slow Decimator Channel 7.		
		51	EQ0.		
		55	DMIC Channel 4.		
		56	DMIC Channel 5.		
		57	DMIC Channel 6.		
		58	DMIC Channel 7.		
		63	No Output. Slot not used.		

SERIAL PORT 1 OUTPUT ROUTING SLOT 15 REGISTER

Address: 0x4000C105, Reset: 0x3F, Name: SPT1_ROUTE15

Table 313. Bit Descriptions for SPT1_ROUTE15

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
[5:0]	SPT1_OUT_ROUTE15		Serial port output route slot 15.	0x3F	R/W
		0	FastDSP Channel 0.		
		1	FastDSP Channel 1.		
		2	FastDSP Channel 2.		
		3	FastDSP Channel 3.		
		4	FastDSP Channel 4.		
		5	FastDSP Channel 5.		
		6	FastDSP Channel 6.		
		7	FastDSP Channel 7.		
		8	FastDSP Channel 8.		
		9	FastDSP Channel 9.		
		10	FastDSP Channel 10.		
		11	FastDSP Channel 11.		
		12	FastDSP Channel 12.		
		13	FastDSP Channel 13.		
		14	FastDSP Channel 14.		
		15	FastDSP Channel 15.		
		16	Tensilica DSP Channel 0.		
		17	Tensilica DSP Channel 1.		
		18	Tensilica DSP Channel 2.		
		19	Tensilica DSP Channel 3.		
		20	Tensilica DSP Channel 4.		
		21	Tensilica DSP Channel 5.		
		22	Tensilica DSP Channel 6.		
		23	Tensilica DSP Channel 7.		
		24	Tensilica DSP Channel 8.		
		25	Tensilica DSP Channel 9.		
		26	Tensilica DSP Channel 10.		
		27	Tensilica DSP Channel 11.		
		28	Tensilica DSP Channel 12.		

Bits	Bit Name	Settings	Description	Reset	Access
		29	Tensilica DSP Channel 13.		
		30	Tensilica DSP Channel 14.		
		31	Tensilica DSP Channel 15.		
		32	Output ASRC Channel 0.		
		33	Output ASRC Channel 1.		
		34	Output ASRC Channel 2.		
		35	Output ASRC Channel 3.		
		36	ADC Channel 0.		
		37	ADC Channel 1.		
		38	ADC Channel 2.		
		39	DMIC Channel 0.		
		40	DMIC Channel 1.		
		41	DMIC Channel 2.		
		42	DMIC Channel 3.		
		43	Fast to Slow Decimator Channel 0.		
		44	Fast to Slow Decimator Channel 1.		
		45	Fast to Slow Decimator Channel 2.		
		46	Fast to Slow Decimator Channel 3.		
		47	Fast to Slow Decimator Channel 4.		
		48	Fast to Slow Decimator Channel 5.		
		49	Fast to Slow Decimator Channel 6.		
		50	Fast to Slow Decimator Channel 7.		
		51	EQ0.		
		55	DMIC Channel 4.		
		56	DMIC Channel 5.		
		57	DMIC Channel 6.		
		58	DMIC Channel 7.		
		63	No Output. Slot not used.		

PDM SAMPLE RATE AND FILTERING CONTROL REGISTER

Address: 0x4000C118, Reset: 0x02, Name: PDM_CTRL1

Table 314. Bit Descriptions for PDM_CTRL1

Bits	Bit Name	Settings	Description	Reset	Access
7	PDM_MORE_FILT		PDM Output Additional Interpolation Filtering Selection.	0x0	R/W
		0	Less interpolation filtering : Lower delay.		
		1	More interpolation filtering : Higher delay.		
[6:5]	RESERVED		Reserved.	0x0	R
4	PDM_RATE		PDM Output Rate.	0x0	R/W
		0	6.144MHz PDM Output Rate.		
		1	3.072MHz PDM Output Rate.		
3	PDM_FCOMP		PDM Output Frequency Response Compensation.	0x0	R/W
		0	High frequency response not compensated (lower delay).		
		1	High frequency response compensated for samples rates of 192kHz or lower when DAC_MORE_FILT=1. (higher delay).		
[2:0]	PDM_FS		PDM Output Path Sample Rate Selection.	0x2	R/W
		000	12kHz Sample Rate.		
		001	24kHz Sample Rate.		
		010	48kHz Sample Rate.		
		011	96kHz Sample Rate.		

Bits	Bit Name	Settings	Description	Reset	Access
		100	192kHz Sample Rate.		
		101	384kHz Sample Rate.		
		110	768kHz Sample Rate.		

PDM MUTING, HIGH-PASS, AND VOLUME OPTIONS REGISTER

Address: 0x4000C119, Reset: 0xC4, Name: PDM_CTRL2

Table 315. Bit Descriptions for PDM_CTRL2

Bits	Bit Name	Settings	Description	Reset	Access
7	PDM1_MUTE		PDM Output Channel 1 Mute Control.	0x1	R/W
		0	PDM Output Unmuted.		
		1	PDM Output Muted.		
6	PDM0_MUTE		PDM Output Channel 0 Mute Control.	0x1	R/W
		0	PDM Output Unmuted.		
		1	PDM Output Muted.		
5	PDM1_HPF_EN		PDM Output Channel 1 enable high-pass filter.	0x0	R/W
		0	PDM Output High Pass Filter Off.		
		1	PDM Output High Pass Filter On.		
4	PDM0_HPF_EN		PDM Output 0 enable high-pass filter.	0x0	R/W
		0	PDM Output High Pass Filter Off.		
		1	PDM Output High Pass Filter On.		
3	RESERVED		Reserved.	0x0	R
2	PDM_VOL_ZC		PDM Output Volume Zero Cross Control.	0x1	R/W
		0	Volume change occurs at any time.		
		1	Volume change only occurs at zero crossing.		
1	PDM_HARD_VOL		PDM Output Hard Volume.	0x0	R/W
		0	Soft Volume Ramping.		
		1	Hard/Immediate Volume Change.		
0	PDM_VOL_LINK		PDM Output Volume Link.	0x0	R/W
		0	Each ADC channel uses its respective volume value.		
		1	All ADC channels use Channel 0 volume value.		

PDM OUTPUT CHANNEL 0 VOLUME REGISTER

Address: 0x4000C11A, Reset: 0x40, Name: PDM_VOL0

Table 316. Bit Descriptions for PDM_VOL0

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	PDM0_VOL		PDM Output Channel 0 Volume Control.	0x40	R/W
		00000000	+24dB.		
		00000001	+23.625dB.		
		00000010	+23.35dB.		
		00000011	+22.875dB.		
		00000100-00111111	+22.5dB to +0.375dB.		
		01000000	0.		
		01000001-11111101	-0.375dB to -70.875dB.		
		11111110	-71.25dB.		
		11111111	Mute.		

PDM OUTPUT CHANNEL 1 VOLUME REGISTER

Address: 0x4000C11B, Reset: 0x40, Name: PDM_VOL1

Table 317. Bit Descriptions for PDM_VOL1

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	PDM1_VOL		PDM Output Channel 1 Volume Control.	0x40	R/W
		00000000	+24dB.		
		00000001	+23.625dB.		
		00000010	+23.35dB.		
		00000011	+22.875dB.		
		00000100-00111111	+22.5dB to +0.375dB.		
		01000000	0.		
		01000001-11111101	-0.375dB to -70.875dB.		
		11111110	-71.25dB.		
		11111111	Mute.		

PDM OUTPUT CHANNEL 0 ROUTING REGISTER

Address: 0x4000C11C, Reset: 0x00, Name: PDM_ROUTE0

Table 318. Bit Descriptions for PDM_ROUTE0

Bits	Bit Name	Settings	Description	Reset	Access
7	RESERVED		Reserved.	0x0	R
[6:0]	PDM0_ROUTE		PDM Output Channel 0 Input Routing.	0x0	R/W
		0	Serial Port 0 Channel 0.		
		1	Serial Port 0 Channel 1.		
		2	Serial Port 0 Channel 2.		
		3	Serial Port 0 Channel 3.		
		4	Serial Port 0 Channel 4.		
		5	Serial Port 0 Channel 5.		
		6	Serial Port 0 Channel 6.		
		7	Serial Port 0 Channel 7.		
		8	Serial Port 0 Channel 8.		
		9	Serial Port 0 Channel 9.		
		10	Serial Port 0 Channel 10.		
		11	Serial Port 0 Channel 11.		
		12	Serial Port 0 Channel 12.		
		13	Serial Port 0 Channel 13.		
		14	Serial Port 0 Channel 14.		
		15	Serial Port 0 Channel 15.		
		16	Serial Port 1 Channel 0.		
		17	Serial Port 1 Channel 1.		
		18	Serial Port 1 Channel 2.		
		19	Serial Port 1 Channel 3.		
		20	Serial Port 1 Channel 4.		
		21	Serial Port 1 Channel 5.		
		22	Serial Port 1 Channel 6.		
		23	Serial Port 1 Channel 7.		
		24	Serial Port 1 Channel 8.		

Bits	Bit Name	Settings	Description	Reset	Access
		25	Serial Port 1 Channel 9.		
		26	Serial Port 1 Channel 10.		
		27	Serial Port 1 Channel 11.		
		28	Serial Port 1 Channel 12.		
		29	Serial Port 1 Channel 13.		
		30	Serial Port 1 Channel 14.		
		31	Serial Port 1 Channel 15.		
		32	FastDSP Channel 0.		
		33	FastDSP Channel 1.		
		34	FastDSP Channel 2.		
		35	FastDSP Channel 3.		
		36	FastDSP Channel 4.		
		37	FastDSP Channel 5.		
		38	FastDSP Channel 6.		
		39	FastDSP Channel 7.		
		40	FastDSP Channel 8.		
		41	FastDSP Channel 9.		
		42	FastDSP Channel 10.		
		43	FastDSP Channel 11.		
		44	FastDSP Channel 12.		
		45	FastDSP Channel 13.		
		46	FastDSP Channel 14.		
		47	FastDSP Channel 15.		
		48	Tensilica DSP Channel 0.		
		49	Tensilica DSP Channel 1.		
		50	Tensilica DSP Channel 2.		
		51	Tensilica DSP Channel 3.		
		52	Tensilica DSP Channel 4.		
		53	Tensilica DSP Channel 5.		
		54	Tensilica DSP Channel 6.		
		55	Tensilica DSP Channel 7.		
		56	Tensilica DSP Channel 8.		
		57	Tensilica DSP Channel 9.		
		58	Tensilica DSP Channel 10.		
		59	Tensilica DSP Channel 11.		
		60	Tensilica DSP Channel 12.		
		61	Tensilica DSP Channel 13.		
		62	Tensilica DSP Channel 14.		
		63	Tensilica DSP Channel 15.		
		64	Input ASRC Channel 0.		
		65	Input ASRC Channel 1.		
		66	Input ASRC Channel 2.		
		67	Input ASRC Channel 3.		
		68	ADC Channel 0.		
		69	ADC Channel 1.		
		70	ADC Channel 2.		
		71	DMIC Channel 0.		
		72	DMIC Channel 1.		
		73	DMIC Channel 2.		
		74	DMIC Channel 3.		
		75	EQ0.		

Bits	Bit Name	Settings	Description	Reset	Access
		76	FINT Channel 0.		
		77	FINT Channel 1.		
		78	FINT Channel 2.		
		79	FINT Channel 3.		
		80	FINT Channel 4.		
		81	FINT Channel 5.		
		82	FINT Channel 6.		
		83	FINT Channel 7.		
		87	DMIC Channel 4.		
		88	DMIC Channel 5.		
		89	DMIC Channel 6.		
		90	DMIC Channel 7.		

PDM OUTPUT CHANNEL 1 ROUTING REGISTER

Address: 0x4000C11D, Reset: 0x01, Name: PDM_ROUTE1

Table 319. Bit Descriptions for PDM_ROUTE1

Bits	Bit Name	Settings	Description	Reset	Access
7	RESERVED		Reserved.	0x0	R
[6:0]	PDM1_ROUTE		PDM Output Channel 1 Input Routing.	0x1	R/W
		0	Serial Port 0 Channel 0.		
		1	Serial Port 0 Channel 1.		
		2	Serial Port 0 Channel 2.		
		3	Serial Port 0 Channel 3.		
		4	Serial Port 0 Channel 4.		
		5	Serial Port 0 Channel 5.		
		6	Serial Port 0 Channel 6.		
		7	Serial Port 0 Channel 7.		
		8	Serial Port 0 Channel 8.		
		9	Serial Port 0 Channel 9.		
		10	Serial Port 0 Channel 10.		
		11	Serial Port 0 Channel 11.		
		12	Serial Port 0 Channel 12.		
		13	Serial Port 0 Channel 13.		
		14	Serial Port 0 Channel 14.		
		15	Serial Port 0 Channel 15.		
		16	Serial Port 1 Channel 0.		
		17	Serial Port 1 Channel 1.		
		18	Serial Port 1 Channel 2.		
		19	Serial Port 1 Channel 3.		
		20	Serial Port 1 Channel 4.		
		21	Serial Port 1 Channel 5.		
		22	Serial Port 1 Channel 6.		
		23	Serial Port 1 Channel 7.		
		24	Serial Port 1 Channel 8.		
		25	Serial Port 1 Channel 9.		
		26	Serial Port 1 Channel 10.		
		27	Serial Port 1 Channel 11.		
		28	Serial Port 1 Channel 12.		
		29	Serial Port 1 Channel 13.		

Bits	Bit Name	Settings	Description	Reset	Access
		30	Serial Port 1 Channel 14.		
		31	Serial Port 1 Channel 15.		
		32	FastDSP Channel 0.		
		33	FastDSP Channel 1.		
		34	FastDSP Channel 2.		
		35	FastDSP Channel 3.		
		36	FastDSP Channel 4.		
		37	FastDSP Channel 5.		
		38	FastDSP Channel 6.		
		39	FastDSP Channel 7.		
		40	FastDSP Channel 8.		
		41	FastDSP Channel 9.		
		42	FastDSP Channel 10.		
		43	FastDSP Channel 11.		
		44	FastDSP Channel 12.		
		45	FastDSP Channel 13.		
		46	FastDSP Channel 14.		
		47	FastDSP Channel 15.		
		48	Tensilica DSP Channel 0.		
		49	Tensilica DSP Channel 1.		
		50	Tensilica DSP Channel 2.		
		51	Tensilica DSP Channel 3.		
		52	Tensilica DSP Channel 4.		
		53	Tensilica DSP Channel 5.		
		54	Tensilica DSP Channel 6.		
		55	Tensilica DSP Channel 7.		
		56	Tensilica DSP Channel 8.		
		57	Tensilica DSP Channel 9.		
		58	Tensilica DSP Channel 10.		
		59	Tensilica DSP Channel 11.		
		60	Tensilica DSP Channel 12.		
		61	Tensilica DSP Channel 13.		
		62	Tensilica DSP Channel 14.		
		63	Tensilica DSP Channel 15.		
		64	Input ASRC Channel 0.		
		65	Input ASRC Channel 1.		
		66	Input ASRC Channel 2.		
		67	Input ASRC Channel 3.		
		68	ADC Channel 0.		
		69	ADC Channel 1.		
		70	ADC Channel 2.		
		71	DMIC Channel 0.		
		72	DMIC Channel 1.		
		73	DMIC Channel 2.		
		74	DMIC Channel 3.		
		75	EQ0.		
		76	FINT Channel 0.		
		77	FINT Channel 1.		
		78	FINT Channel 2.		
		79	FINT Channel 3.		
		80	FINT Channel 4.		

Bits	Bit Name	Settings	Description	Reset	Access
		81	FINT Channel 5.		
		82	FINT Channel 6.		
		83	FINT Channel 7.		
		87	DMIC Channel 4.		
		88	DMIC Channel 5.		
		89	DMIC Channel 6.		
		90	DMIC Channel 7.		

MULTIPURPOSE PIN 0/PIN 1 MODE SELECT REGISTER

Address: 0x4000C121, Reset: 0x00, Name: MP_CTRL1

Table 320. Bit Descriptions for MP_CTRL1

Bits	Bit Name	Settings	Description	Reset	Access
[7:4]	MP1_MODE		Multipurpose Pin 1 Mode Selection (DMIC0_1).	0x0	R/W
		0x0	Normal Operation.		
		0x1	General Purpose Input.		
		0x2	General Purpose Output from GPIOx_OUT register.		
		0x3	General Purpose Output from Tensilica DSP.		
		0x4	MCLK Output.		
		0x5	IRQ1 Output.		
		0x6	IRQ2 Output.		
		0x7	IRQ3 Output.		
		0x8	IRQ4 Output.		
		0x9	Interrupt Input.		
		0xA	PDM Clock Output.		
		0xB	PDM Data Output.		
		0xC	Digital Mic Channels 4-5 Input.		
		0xD	Digital Mic Channels 6-7 Input.		
[3:0]	MP0_MODE		Multipurpose Pin 0 Mode Selection (DMIC_CLK).	0x0	R/W
		0x0	Normal Operation.		
		0x1	General Purpose Input.		
		0x2	General Purpose Output from GPIOx_OUT register.		
		0x3	General Purpose Output from Tensilica DSP.		
		0x4	MCLK Output.		
		0x5	IRQ1 Output.		
		0x6	IRQ2 Output.		
		0x7	IRQ3 Output.		
		0x8	IRQ4 Output.		
		0x9	Interrupt Input.		
		0xA	PDM Clock Output.		
		0xB	PDM Data Output.		
		0xC	Digital Mic Channels 4-5 Input.		
		0xD	Digital Mic Channels 6-7 Input.		

MULTIPURPOSE PIN 2/PIN 3 MODE SELECT REGISTER

Address: 0x4000C122, Reset: 0x00, Name: MP_CTRL2

Table 321. Bit Descriptions for MP_CTRL2

Bits		Bit Name	Settings	Description	Reset	Access
[7:4]		MP3_MODE		Multipurpose Pin 3 Mode Selection (BCLK_0).	0x0	R/W
			0x0	Normal Operation.		
			0x1	General Purpose Input.		
			0x2	General Purpose Output from GPIOx_OUT register.		
			0x3	General Purpose Output from Tensilica DSP.		
			0x4	MCLK Output.		
			0x5	IRQ1 Output.		
			0x6	IRQ2 Output.		
			0x7	IRQ3 Output.		
			0x8	IRQ4 Output.		
			0x9	Interrupt Input.		
			0xA	PDM Clock Output.		
			0xB	PDM Data Output.		
			0xC	Digital Mic Channels 4-5 Input.		
			0xD	Digital Mic Channels 6-7 Input.		
[3:0]		MP2_MODE		Multipurpose Pin 2 Mode Selection (DMIC2_3).	0x0	R/W
			0x0	Normal Operation.		
			0x1	General Purpose Input.		
			0x2	General Purpose Output from GPIOx_OUT register.		
			0x3	General Purpose Output from Tensilica DSP.		
			0x4	MCLK Output.		
			0x5	IRQ1 Output.		
			0x6	IRQ2 Output.		
			0x7	IRQ3 Output.		
			0x8	IRQ4 Output.		
			0x9	Interrupt Input.		
			0xA	PDM Clock Output.		
			0xB	PDM Data Output.		
			0xC	Digital Mic Channels 4-5 Input.		
			0xD	Digital Mic Channels 6-7 Input.		

MULTIPURPOSE PIN 4/PIN 5 MODE SELECT REGISTER

Address: 0x4000C123, Reset: 0x00, Name: MP_CTRL3

Table 322. Bit Descriptions for MP_CTRL3

Bits	Bit Name	Settings	Description	Reset	Access
[7:4]	MP5_MODE		Multipurpose Pin 5 Mode Selection (SDATAO_0).	0x0	R/W
		0x0	Normal Operation.		
		0x1	General Purpose Input.		
		0x2	General Purpose Output from GPIOx_OUT register.		
		0x3	General Purpose Output from Tensilica DSP.		
		0x4	MCLK Output.		
		0x5	IRQ1 Output.		
		0x6	IRQ2 Output.		
		0x7	IRQ3 Output.		
		0x8	IRQ4 Output.		

Bits	Bit Name	Settings	Description	Reset	Access
		0x9	Interrupt Input.		
		0xA	PDM Clock Output.		
		0xB	PDM Data Output.		
		0xC	Digital Mic Channels 4-5 Input.		
		0xD	Digital Mic Channels 6-7 Input.		
[3:0]	MP4_MODE		Multipurpose Pin 4 Mode Selection (FSYNC_0).	0x0	R/W
		0x0	Normal Operation.		
		0x1	General Purpose Input.		
		0x2	General Purpose Output from GPIOx_OUT register.		
		0x3	General Purpose Output from Tensilica DSP.		
		0x4	MCLK Output.		
		0x5	IRQ1 Output.		
		0x6	IRQ2 Output.		
		0x7	IRQ3 Output.		
		0x8	IRQ4 Output.		
		0x9	Interrupt Input.		
		0xA	PDM Clock Output.		
		0xB	PDM Data Output.		
		0xC	Digital Mic Channels 4-5 Input.		
		0xD	Digital Mic Channels 6-7 Input.		

MULTIPURPOSE PIN 6/PIN 7 MODE SELECT REGISTER

Address: 0x4000C124, Reset: 0x00, Name: MP_CTRL4

Table 323. Bit Descriptions for MP_CTRL4

Bits	Bit Name	Settings	Description	Reset	Access
[7:4]	MP7_MODE		Multipurpose Pin 7 Mode Selection (BCLK_1).	0x0	R/W
		0x0	Normal Operation.		
		0x1	General Purpose Input.		
		0x2	General Purpose Output from GPIOx_OUT register.		
		0x3	General Purpose Output from Tensilica DSP.		
		0x4	MCLK Output.		
		0x5	IRQ1 Output.		
		0x6	IRQ2 Output.		
		0x7	IRQ3 Output.		
		0x8	IRQ4 Output.		
		0x9	Interrupt Input.		
		0xA	PDM Clock Output.		
		0xB	PDM Data Output.		
		0xC	Digital Mic Channels 4-5 Input.		
		0xD	Digital Mic Channels 6-7 Input.		
		0xE	Digital Mic Clock Output 1.		
[3:0]	MP6_MODE		Multipurpose Pin 6 Mode Selection (SDATAI_0).	0x0	R/W
		0x0	Normal Operation.		
		0x1	General Purpose Input.		
		0x2	General Purpose Output from GPIOx_OUT register.		
		0x3	General Purpose Output from Tensilica DSP.		
		0x4	MCLK Output.		
		0x5	IRQ1 Output.		
		0x6	IRQ2 Output.		

Bits	Bit Name	Settings	Description	Reset	Access
		0x7	IRQ3 Output.		
		0x8	IRQ4 Output.		
		0x9	Interrupt Input.		
		0xA	PDM Clock Output.		
		0xB	PDM Data Output.		
		0xC	Digital Mic Channels 4-5 Input.		
		0xD	Digital Mic Channels 6-7 Input.		

MULTIPURPOSE PIN 8/PIN 9 MODE SELECT REGISTER

Address: 0x4000C125, Reset: 0x00, Name: MP_CTRL5

Table 324. Bit Descriptions for MP_CTRL5

Bits	Bit Name	Settings	Description	Reset	Access
[7:4]	MP9_MODE		Multipurpose Pin 9 Mode Selection (SDATAO_1).	0x0	R/W
		0x0	Normal Operation.		
		0x1	General Purpose Input.		
		0x2	General Purpose Output from GPIOx_OUT register.		
		0x3	General Purpose Output from Tensilica DSP.		
		0x4	Inverted DMIC_CLK0 Output.		
		0x5	IRQ1 Output.		
		0x6	IRQ2 Output.		
		0x7	IRQ3 Output.		
		0x8	IRQ4 Output.		
		0x9	Interrupt Input.		
		0xA	PDM Clock Output.		
		0xB	PDM Data Output.		
		0xC	Digital Mic Channels 4-5 Input.		
		0xD	Digital Mic Channels 6-7 Input.		
[3:0]	MP8_MODE		Multipurpose Pin 8 Mode Selection (FSYNC_1).	0x0	R/W
		0x0	Normal Operation.		
		0x1	General Purpose Input.		
		0x2	General Purpose Output from GPIOx_OUT register.		
		0x3	General Purpose Output from Tensilica DSP.		
		0x4	Inverted DMIC_CLK0 Output.		
		0x5	IRQ1 Output.		
		0x6	IRQ2 Output.		
		0x7	IRQ3 Output.		
		0x8	IRQ4 Output.		
		0x9	Interrupt Input.		
		0xA	PDM Clock Output.		
		0xB	PDM Data Output.		
		0xC	Digital Mic Channels 4-5 Input.		
		0xD	Digital Mic Channels 6-7 Input.		

MULTIPURPOSE PIN 10/PIN 11 MODE SELECT REGISTER

Address: 0x4000C126, Reset: 0x00, Name: MP_CTRL6

Table 325. Bit Descriptions for MP_CTRL6

Bits	Bit Name	Settings	Description	Reset	Access
[7:4]	MP11_MODE		Multipurpose Pin 11 Mode Selection (QSPIM_CLK).	0x0	R/W
		0x0	Normal Operation.		
		0x1	General Purpose Input.		
		0x2	General Purpose Output from GPIOx_OUT register.		
		0x3	General Purpose Output from Tensilica DSP.		
		0x4	MCLK Output.		
		0x5	IRQ1 Output.		
		0x6	IRQ2 Output.		
		0x7	IRQ3 Output.		
		0x8	IRQ4 Output.		
		0x9	Interrupt Input.		
		0xA	PDM Clock Output.		
		0xB	PDM Data Output.		
		0xC	Digital Mic Channels 4-5 Input.		
		0xD	Digital Mic Channels 6-7 Input.		
		0xE	Digital Mic Clock Output 1.		
[3:0]	MP10_MODE		Multipurpose Pin 10 Mode Selection (SDATAI_1).	0x0	R/W
		0x0	Normal Operation.		
		0x1	General Purpose Input.		
		0x2	General Purpose Output from GPIOx_OUT register.		
		0x3	General Purpose Output from Tensilica DSP.		
		0x4	Inverted DMIC_CLK0 Output.		
		0x5	IRQ1 Output.		
		0x6	IRQ2 Output.		
		0x7	IRQ3 Output.		
		0x8	IRQ4 Output.		
		0x9	Interrupt Input.		
		0xA	PDM Clock Output.		
		0xB	PDM Data Output.		
		0xC	Digital Mic Channels 4-5 Input.		
		0xD	Digital Mic Channels 6-7 Input.		

MULTIPURPOSE PIN 12/PIN 13 MODE SELECT REGISTER

Address: 0x4000C127, Reset: 0x00, Name: MP_CTRL7

Table 326. Bit Descriptions for MP_CTRL7

Bits	Bit Name	Settings	Description	Reset	Access
[7:4]	MP13_MODE		Multipurpose Pin 13 Mode Selection (QSPIM_SDIO0).	0x0	R/W
		0x0	Normal Operation.		
		0x1	General Purpose Input.		
		0x2	General Purpose Output from GPIOx_OUT register.		
		0x3	General Purpose Output from Tensilica DSP.		
		0x4	MCLK Output.		
		0x5	IRQ1 Output.		
		0x6	IRQ2 Output.		
		0x7	IRQ3 Output.		

Bits	Bit Name	Settings	Description	Reset	Access
		0x8	IRQ4 Output.		
		0x9	Interrupt Input.		
		0xA	PDM Clock Output.		
		0xB	PDM Data Output.		
		0xC	Digital Mic Channels 4-5 Input.		
		0xD	Digital Mic Channels 6-7 Input.		
[3:0]	MP12_MODE		Multipurpose Pin 12 Mode Selection (QSPIM_CS).	0x0	R/W
		0x0	Normal Operation.		
		0x1	General Purpose Input.		
		0x2	General Purpose Output from GPIOx_OUT register.		
		0x3	General Purpose Output from Tensilica DSP.		
		0x4	MCLK Output.		
		0x5	IRQ1 Output.		
		0x6	IRQ2 Output.		
		0x7	IRQ3 Output.		
		0x8	IRQ4 Output.		
		0x9	Interrupt Input.		
		0xA	PDM Clock Output.		
		0xB	PDM Data Output.		
		0xC	Digital Mic Channels 4-5 Input.		
		0xD	Digital Mic Channels 6-7 Input.		

MULTIPURPOSE PIN 14/PIN 15 MODE SELECT REGISTER

Address: 0x4000C128, Reset: 0x00, Name: MP_CTRL8

Table 327. Bit Descriptions for MP_CTRL8

Bits	Bit Name	Settings	Description	Reset	Access
[7:4]	MP15_MODE		Multipurpose Pin 15 Mode Selection (QSPIM_SDIO2).	0x0	R/W
		0x0	Normal Operation.		
		0x1	General Purpose Input.		
		0x2	General Purpose Output from GPIOx_OUT register.		
		0x3	General Purpose Output from Tensilica DSP.		
		0x4	MCLK Output.		
		0x5	IRQ1 Output.		
		0x6	IRQ2 Output.		
		0x7	IRQ3 Output.		
		0x8	IRQ4 Output.		
		0x9	Interrupt Input.		
		0xA	PDM Clock Output.		
		0xB	PDM Data Output.		
		0xC	Digital Mic Channels 4-5 Input.		
		0xD	Digital Mic Channels 6-7 Input.		
[3:0]	MP14_MODE		Multipurpose Pin 14 Mode Selection (QSPIM_SDIO1).	0x0	R/W
		0x0	Normal Operation.		
		0x1	General Purpose Input.		
		0x2	General Purpose Output from GPIOx_OUT register.		
		0x3	General Purpose Output from Tensilica DSP.		
		0x4	MCLK Output.		
		0x5	IRQ1 Output.		
		0x6	IRQ2 Output.		

Bits	Bit Name	Settings	Description	Reset	Access
		0x7	IRQ3 Output.		
		0x8	IRQ4 Output.		
		0x9	Interrupt Input.		
		0xA	PDM Clock Output.		
		0xB	PDM Data Output.		
		0xC	Digital Mic Channels 4-5 Input.		
		0xD	Digital Mic Channels 6-7 Input.		

MULTIPURPOSE PIN 16/PIN 17 MODE SELECT REGISTER

Address: 0x4000C129, Reset: 0x00, Name: MP_CTRL9

Table 328. Bit Descriptions for MP_CTRL9

Bits	Bit Name	Settings	Description	Reset	Access
[7:4]	MP17_MODE		Multipurpose Pin 17 Mode Selection (UART_COMM_TX).	0x0	R/W
		0x0	Normal Operation.		
		0x1	General Purpose Input.		
		0x2	General Purpose Output from GPIOx_OUT register.		
		0x3	General Purpose Output from Tensilica DSP.		
		0x4	MCLK Output.		
		0x5	IRQ1 Output.		
		0x6	IRQ2 Output.		
		0x7	IRQ3 Output.		
		0x8	IRQ4 Output.		
		0x9	Interrupt Input.		
		0xA	PDM Clock Output.		
		0xB	PDM Data Output.		
		0xC	Digital Mic Channels 4-5 Input.		
		0xD	Digital Mic Channels 6-7 Input.		
		0xE	Digital Mic Clock Output 1.		
[3:0]	MP16_MODE		Multipurpose Pin 16 Mode Selection (QSPIM_SDIO3).	0x0	R/W
		0x0	Normal Operation.		
		0x1	General Purpose Input.		
		0x2	General Purpose Output from GPIOx_OUT register.		
		0x3	General Purpose Output from Tensilica DSP.		
		0x4	MCLK Output.		
		0x5	IRQ1 Output.		
		0x6	IRQ2 Output.		
		0x7	IRQ3 Output.		
		0x8	IRQ4 Output.		
		0x9	Interrupt Input.		
		0xA	PDM Clock Output.		
		0xB	PDM Data Output.		
		0xC	Digital Mic Channels 4-5 Input.		
		0xD	Digital Mic Channels 6-7 Input.		

MULTIPURPOSE PIN 18/PIN 19 MODE SELECT REGISTER

Address: 0x4000C12A, Reset: 0x00, Name: MP_CTRL10

Table 329. Bit Descriptions for MP_CTRL10

Bits	Bit Name	Settings	Description	Reset	Access
[7:4]	MP19_MODE		Multipurpose Pin 19 Mode Selection (SELFBOOT).	0x0	R/W
		0x0	Normal Operation.		
		0x1	General Purpose Input.		
		0x2	General Purpose Output from GPIOx_OUT register.		
		0x3	General Purpose Output from Tensilica DSP.		
		0x4	MCLK Output.		
		0x5	IRQ1 Output.		
		0x6	IRQ2 Output.		
		0x7	IRQ3 Output.		
		0x8	IRQ4 Output.		
		0x9	Interrupt Input.		
		0xA	PDM Clock Output.		
		0xB	PDM Data Output.		
		0xC	Digital Mic Channels 4-5 Input.		
		0xD	Digital Mic Channels 6-7 Input.		
[3:0]	MP18_MODE		Multipurpose Pin 18 Mode Selection (UART_COMM_RX).	0x0	R/W
		0x0	Normal Operation.		
		0x1	General Purpose Input.		
		0x2	General Purpose Output from GPIOx_OUT register.		
		0x3	General Purpose Output from Tensilica DSP.		
		0x4	MCLK Output.		
		0x5	IRQ1 Output.		
		0x6	IRQ2 Output.		
		0x7	IRQ3 Output.		
		0x8	IRQ4 Output.		
		0x9	Interrupt Input.		
		0xA	PDM Clock Output.		
		0xB	PDM Data Output.		
		0xC	Digital Mic Channels 4-5 Input.		
		0xD	Digital Mic Channels 6-7 Input.		

MULTIPURPOSE PIN 20/PIN 21 MODE SELECT REGISTER

Address: 0x4000C12B, Reset: 0x00, Name: MP_CTRL11

Table 330. Bit Descriptions for MP_CTRL11

Bits	Bit Name	Settings	Description	Reset	Access
[7:4]	MP21_MODE		Multipurpose Pin 21 Mode Selection (ROM_BOOT_MODE).	0x0	R/W
		0x0	Normal Operation.		
		0x1	General Purpose Input.		
		0x2	General Purpose Output from GPIOx_OUT register.		
		0x3	General Purpose Output from Tensilica DSP.		
		0x4	MCLK Output.		
		0x5	IRQ1 Output.		
		0x6	IRQ2 Output.		
		0x7	IRQ3 Output.		
		0x8	IRQ4 Output.		

Bits	Bit Name	Settings	Description	Reset	Access
		0x9	Interrupt Input.		
		0xA	PDM Clock Output.		
		0xB	PDM Data Output.		
		0xC	Digital Mic Channels 4-5 Input.		
		0xD	Digital Mic Channels 6-7 Input.		
[3:0]	MP20_MODE		Multipurpose Pin 20 Mode Selection (IRQ).	0x0	R/W
		0x0	Normal Operation.		
		0x1	General Purpose Input.		
		0x2	General Purpose Output from GPIOx_OUT register.		
		0x3	General Purpose Output from Tensilica DSP.		
		0x4	MCLK Output.		
		0x5	IRQ1 Output.		
		0x6	IRQ2 Output.		
		0x7	IRQ3 Output.		
		0x8	IRQ4 Output.		
		0x9	Interrupt Input.		
		0xA	PDM Clock Output.		
		0xB	PDM Data Output.		
		0xC	Digital Mic Channels 4-5 Input.		
		0xD	Digital Mic Channels 6-7 Input.		

MULTIPURPOSE PIN 22/PIN 23 MODE SELECT REGISTER

Address: 0x4000C12C, Reset: 0x00, Name: MP_CTRL12

Table 331. Bit Descriptions for MP_CTRL12

Bits	Bit Name	Settings	Description	Reset	Access
[7:4]	MP23_MODE		Multipurpose Pin 23 Mode Selection (TMS).	0x0	R/W
		0x0	Normal Operation.		
		0x1	General Purpose Input.		
		0x2	General Purpose Output from GPIOx_OUT register.		
		0x3	General Purpose Output from Tensilica DSP.		
		0x4	MCLK Output.		
		0x5	IRQ1 Output.		
		0x6	IRQ2 Output.		
		0x7	IRQ3 Output.		
		0x8	IRQ4 Output.		
		0x9	Interrupt Input.		
		0xA	PDM Clock Output.		
		0xB	PDM Data Output.		
		0xC	Digital Mic Channels 4-5 Input.		
		0xD	Digital Mic Channels 6-7 Input.		
[3:0]	MP22_MODE		Multipurpose Pin 22 Mode Selection (TCK).	0x0	R/W
		0x0	Normal Operation.		
		0x1	General Purpose Input.		
		0x2	General Purpose Output from GPIOx_OUT register.		
		0x3	General Purpose Output from Tensilica DSP.		
		0x4	Inverted DMIC_CLK0 Output.		
		0x5	IRQ1 Output.		
		0x6	IRQ2 Output.		
		0x7	IRQ3 Output.		

Bits	Bit Name	Settings	Description	Reset	Access
		0x8	IRQ4 Output.		
		0x9	Interrupt Input.		
		0xA	PDM Clock Output.		
		0xB	PDM Data Output.		
		0xC	Digital Mic Channels 4-5 Input.		
		0xD	Digital Mic Channels 6-7 Input.		
		0xE	Digital Mic Clock Output 1.		

MULTIPURPOSE PIN 24/PIN 25 MODE SELECT REGISTER

Address: 0x4000C12D, Reset: 0x00, Name: MP_CTRL13

Table 332. Bit Descriptions for MP_CTRL13

Bits	Bit Name	Settings	Description	Reset	Access
[7:4]	MP25_MODE		Multipurpose Pin 25 Mode Selection (TDI).	0x0	R/W
		0x0	Normal Operation.		
		0x1	General Purpose Input.		
		0x2	General Purpose Output from GPIOx_OUT register.		
		0x3	General Purpose Output from Tensilica DSP.		
		0x4	MCLK Output.		
		0x5	IRQ1 Output.		
		0x6	IRQ2 Output.		
		0x7	IRQ3 Output.		
		0x8	IRQ4 Output.		
		0x9	Interrupt Input.		
		0xA	PDM Clock Output.		
		0xB	PDM Data Output.		
		0xC	Digital Mic Channels 4-5 Input and DMIC 3 Input in Dual Port mode.		
		0xD	Digital Mic Channels 6-7 Input.		
[3:0]	MP24_MODE		Multipurpose Pin 24 Mode Selection (TDO).	0x0	R/W
		0x0	Normal Operation.		
		0x1	General Purpose Input.		
		0x2	General Purpose Output from GPIOx_OUT register.		
		0x3	General Purpose Output from Tensilica DSP.		
		0x4	MCLK Output.		
		0x5	IRQ1 Output.		
		0x6	IRQ2 Output.		
		0x7	IRQ3 Output.		
		0x8	IRQ4 Output.		
		0x9	Interrupt Input.		
		0xA	PDM Clock Output.		
		0xB	PDM Data Output.		
		0xC	Digital Mic Channels 4-5 Input and DMIC 1 Input in Dual Port mode.		
		0xD	Digital Mic Channels 6-7 Input.		

GENERAL-PURPOSE INPUT DEBOUNCE CONTROL AND IRQ INPUT DEBOUNCE CONTROL REGISTER

Address: 0x4000C12E, Reset: 0x00, Name: MP_DB_CTRL

Table 333. Bit Descriptions for MP_DB_CTRL

Bits	Bit Name	Settings	Description	Reset	Access
7	RESERVED		Reserved.	0x0	R
[6:4]	IRQ_IN_DB		IRQ Input Debounce.	0x0	R/W
		0	Interrupt Input without debounce.		
		1	Interrupt Input with debounce (1 us).		
		10	Interrupt Input with debounce (2 us).		
		11	Interrupt Input with debounce (3 us).		
		100	Interrupt Input with debounce (4 us).		
		101	Interrupt Input with debounce (5 us).		
		110	Interrupt Input with debounce (6 us).		
3	RESERVED		Reserved.	0x0	R
[2:0]	GPI_DB		General Purpose Input Debounce.	0x0	R/W
		0	GPIO Input without debounce.		
		1	GPIO Input with debounce (0.3 ms).		
		10	GPIO Input with debounce (0.6 ms).		
		11	GPIO Input with debounce (0.9 ms).		
		100	GPIO Input with debounce (5 ms).		
		101	GPIO Input with debounce (10 ms).		
		110	GPIO Input with debounce (20 ms).		

MCLKO RATE SELECTION REGISTER

Address: 0x4000C12F, Reset: 0x01, Name: MP_MCLKO_RATE

Table 334. Bit Descriptions for MP_MCLKO_RATE

Bits	Bit Name	Settings	Description	Reset	Access
[7:3]	RESERVED		Reserved.	0x0	R
[2:0]	MCLKO_RATE		MCLK output rate selection.	0x1	R/W
		0	MCLKO at 24.576MHz.		
		1	MCLKO at 12.288MHz.		
		10	MCLKO at 6.144MHz.		
		11	MCLKO at 3.072MHz.		
		100	MCLKO at 1.536MHz.		
		101	MCLKO at 768kHz.		
		110	MCLKO at 384kHz.		
		111	MCLKO at 192kHz.		

GENERAL-PURPOSE OUTPUTS CONTROL PIN 0 TO PIN 7 REGISTER

Address: 0x4000C130, Reset: 0x00, Name: MP_GPIO_CTRL1

Table 335. Bit Descriptions for MP_GPIO_CTRL1

Bits	Bit Name	Settings	Description	Reset	Access
7	GPIO7_OUT		GPIO7 Pin output setting.	0x0	R/W
		0	Pin set low when used as GP output.		
		1	Pin set high when used as GP output.		
6	GPIO6_OUT		GPIO6 Pin output setting.	0x0	R/W
		0	Pin set low when used as GP output.		
		1	Pin set high when used as GP output.		
5	GPIO5_OUT		GPIO5 Pin output setting.	0x0	R/W
		0	Pin set low when used as GP output.		
		1	Pin set high when used as GP output.		
4	GPIO4_OUT		GPIO4 Pin output setting.	0x0	R/W
		0	Pin set low when used as GP output.		
		1	Pin set high when used as GP output.		
3	GPIO3_OUT		GPIO3 Pin output setting.	0x0	R/W
		0	Pin set low when used as GP output.		
		1	Pin set high when used as GP output.		
2	GPIO2_OUT		GPIO2 Pin output setting.	0x0	R/W
		0	Pin set low when used as GP output.		
		1	Pin set high when used as GP output.		
1	GPIO1_OUT		GPIO1 Pin output setting.	0x0	R/W
		0	Pin set low when used as GP output.		
		1	Pin set high when used as GP output.		
0	GPIO0_OUT		GPIO0 Pin output setting.	0x0	R/W
		0	Pin set low when used as GP output.		
		1	Pin set high when used as GP output.		

GENERAL-PURPOSE OUTPUTS CONTROL PIN 8 TO PIN 15 REGISTER

Address: 0x4000C131, Reset: 0x00, Name: MP_GPIO_CTRL2

Table 336. Bit Descriptions for MP_GPIO_CTRL2

Bits	Bit Name	Settings	Description	Reset	Access
7	GPIO15_OUT		GPIO15 Pin output setting.	0x0	R/W
		0	Pin set low when used as GP output.		
		1	Pin set high when used as GP output.		
6	GPIO14_OUT		GPIO14 Pin output setting.	0x0	R/W
		0	Pin set low when used as GP output.		
		1	Pin set high when used as GP output.		
5	GPIO13_OUT		GPIO13 Pin output setting.	0x0	R/W
		0	Pin set low when used as GP output.		
		1	Pin set high when used as GP output.		
4	GPIO12_OUT		GPIO12 Pin output setting.	0x0	R/W
		0	Pin set low when used as GP output.		
		1	Pin set high when used as GP output.		
3	GPIO11_OUT		GPIO11 Pin output setting.	0x0	R/W
		0	Pin set low when used as GP output.		
		1	Pin set high when used as GP output.		

Bits	Bit Name	Settings	Description	Reset	Access
2	GPIO10_OUT		GPIO10 Pin output setting.	0x0	R/W
		0	Pin set low when used as GP output.		
		1	Pin set high when used as GP output.		
1	GPIO9_OUT		GPIO9 Pin output setting.	0x0	R/W
		0	Pin set low when used as GP output.		
		1	Pin set high when used as GP output.		
0	GPIO8_OUT		GPIO8 Pin output setting.	0x0	R/W
		0	Pin set low when used as GP output.		
		1	Pin set high when used as GP output.		

GENERAL-PURPOSE OUTPUTS CONTROL PIN 16 TO PIN 23 REGISTER

Address: 0x4000C132, Reset: 0x00, Name: MP_GPIO_CTRL3

Table 337. Bit Descriptions for MP_GPIO_CTRL3

Bits	Bit Name	Settings	Description	Reset	Access
7	GPIO23_OUT		GPIO23 Pin output setting.	0x0	R/W
		0	Pin set low when used as GP output.		
		1	Pin set high when used as GP output.		
6	GPIO22_OUT		GPIO22 Pin output setting.	0x0	R/W
		0	Pin set low when used as GP output.		
		1	Pin set high when used as GP output.		
5	GPIO21_OUT		GPIO21 Pin output setting.	0x0	R/W
		0	Pin set low when used as GP output.		
		1	Pin set high when used as GP output.		
4	GPIO20_OUT		GPIO20 Pin output setting.	0x0	R/W
		0	Pin set low when used as GP output.		
		1	Pin set high when used as GP output.		
3	GPIO19_OUT		GPIO19 Pin output setting.	0x0	R/W
		0	Pin set low when used as GP output.		
		1	Pin set high when used as GP output.		
2	GPIO18_OUT		GPIO18 Pin output setting.	0x0	R/W
		0	Pin set low when used as GP output.		
		1	Pin set high when used as GP output.		
1	GPIO17_OUT		GPIO17 Pin output setting.	0x0	R/W
		0	Pin set low when used as GP output.		
		1	Pin set high when used as GP output.		
0	GPIO16_OUT		GPIO16 Pin output setting.	0x0	R/W
		0	Pin set low when used as GP output.		
		1	Pin set high when used as GP output.		

GENERAL-PURPOSE OUTPUTS CONTROL PIN 24 TO PIN 25 REGISTER

Address: 0x4000C133, Reset: 0x00, Name: MP_GPIO_CTRL4

Table 338. Bit Descriptions for MP_GPIO_CTRL4

Bits	Bit Name	Settings	Description	Reset	Access
[7:2]	RESERVED		Reserved.	0x0	R
1	GPIO25_OUT		GPIO25 Pin output setting.	0x0	R/W
		0	Pin set low when used as GP output.		
		1	Pin set high when used as GP output.		

Bits	Bit Name	Settings	Description	Reset	Access
0	GPIO24_OUT		GPIO24 Pin output setting.	0x0	R/W
		0	Pin set low when used as GP output.		
		1	Pin set high when used as GP output.		

DMIC_CLK PIN CONTROLS REGISTER

Address: 0x4000C134, Reset: 0x05, Name: DMIC_CLK_CTRL

Table 339. Bit Descriptions for DMIC_CLK_CTRL

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
5	DMIC_CLK_PULL_SEL		DMIC_CLK Pin Weak Pull-Up/Down Selection.	0x0	R/W
		0	Weak pull-down when enabled.		
		1	Weak pull-up when enabled.		
4	DMIC_CLK_PULL_EN		DMIC_CLK Pin Weak Pull-Up/Down Enable.	0x0	R/W
		0	No pull-up or pull-down.		
		1	Weak pull-up or pull-down set by SEL register.		
3	RESERVED		Reserved.	0x0	R
2	DMIC_CLK_SLEW		DMIC_CLK Pin Slew Rate. Determines the slew rate of the pin when used as an output.	0x1	R/W
		0	Fast Slew Rate.		
		1	Slow Slew Rate.		
[1:0]	DMIC_CLK_DRIVE		DMIC_CLK Pin Drive Strength. Determines the drive strength of the pin when used as an output.	0x1	R/W
		0	2mA Output Drive.		
		1	4mA Output Drive.		
		10	8mA Output Drive.		
		11	12mA Output Drive.		

DMIC01 PIN CONTROLS REGISTER

Address: 0x4000C135, Reset: 0x05, Name: DMIC01_CTRL

Table 340. Bit Descriptions for DMIC01_CTRL

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
5	DMIC01_PULL_SEL		DMIC01 Pin Weak Pull-Up/Down Selection.	0x0	R/W
		0	Weak pull-down when enabled.		
		1	Weak pull-up when enabled.		
4	DMIC01_PULL_EN		DMIC01 Pin Weak Pull-Up/Down Enable.	0x0	R/W
		0	No pull-up or pull-down.		
		1	Weak pull-up or pull-down set by SEL register.		
3	RESERVED		Reserved.	0x0	R
2	DMIC01_SLEW		DMIC01 Pin Slew Rate. Determines the slew rate of the pin when used as an output.	0x1	R/W
		0	Fast Slew Rate.		
		1	Slow Slew Rate.		

Bits	Bit Name	Settings	Description	Reset	Access
[1:0]	DMIC01_DRIVE		DMIC01 Pin Drive Strength. Determines the drive strength of the pin when used as an output.	0x1	R/W
		0	2mA Output Drive.		
		1	4mA Output Drive.		
		10	8mA Output Drive.		
		11	12mA Output Drive.		

DMIC23 PIN CONTROLS REGISTER

Address: 0x4000C136, Reset: 0x05, Name: DMIC23_CTRL

Table 341. Bit Descriptions for DMIC23_CTRL

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
5	DMIC23_PULL_SEL		DMIC23 Pin Weak Pull-Up/Down Selection.	0x0	R/W
		0	Weak pull-down when enabled.		
		1	Weak pull-up when enabled.		
4	DMIC23_PULL_EN		DMIC23 Pin Weak Pull-Up/Down Enable.	0x0	R/W
		0	No pull-up or pull-down.		
		1	Weak pull-up or pull-down set by SEL register.		
3	RESERVED		Reserved.	0x0	R
2	DMIC23_SLEW		DMIC23 Pin Slew Rate. Determines the slew rate of the pin when used as an output.	0x1	R/W
		0	Fast Slew Rate.		
		1	Slow Slew Rate.		
[1:0]	DMIC23_DRIVE		DMIC23 Pin Drive Strength. Determines the drive strength of the pin when used as an output.	0x1	R/W
		0	2mA Output Drive.		
		1	4mA Output Drive.		
		10	8mA Output Drive.		
		11	12mA Output Drive.		

BCLK_0 PIN CONTROLS REGISTER

Address: 0x4000C137, Reset: 0x05, Name: BCLK0_CTRL

Table 342. Bit Descriptions for BCLK0_CTRL

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
5	BCLK0_PULL_SEL		BCLK_0 Pin Weak Pull-Up/Down Selection.	0x0	R/W
		0	Weak pull-down when enabled.		
		1	Weak pull-up when enabled.		
4	BCLK0_PULL_EN		BCLK_0 Pin Weak Pull-Up/Down Enable.	0x0	R/W
		0	No pull-up or pull-down.		
		1	Weak pull-up or pull-down set by SEL register.		
3	RESERVED		Reserved.	0x0	R
2	BCLK0_SLEW		BCLK_0 Pin Slew Rate. Determines the slew rate of the pin when used as an output.	0x1	R/W
		0	Fast Slew Rate.		
		1	Slow Slew Rate.		

Bits	Bit Name	Settings	Description	Reset	Access
[1:0]	BCLK0_DRIVE		BCLK_0 Pin Drive Strength. Determines the drive strength of the pin when used as an output.	0x1	R/W
		0	2mA Output Drive.		
		1	4mA Output Drive.		
		10	8mA Output Drive.		
		11	12mA Output Drive.		

FSYNC_0 PIN CONTROLS REGISTER

Address: 0x4000C138, Reset: 0x05, Name: FSYNC0_CTRL

Table 343. Bit Descriptions for FSYNC0_CTRL

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
5	FSYNC0_PULL_SEL		FSYNC_0 Pin Weak Pull-Up/Down Selection.	0x0	R/W
		0	Weak pull-down when enabled.		
		1	Weak pull-up when enabled.		
4	FSYNC0_PULL_EN		FSYNC_0 Pin Weak Pull-Up/Down Enable.	0x0	R/W
		0	No pull-up or pull-down.		
		1	Weak pull-up or pull-down set by SEL register.		
3	RESERVED		Reserved.	0x0	R
2	FSYNC0_SLEW		FSYNC_0 Pin Slew Rate. Determines the slew rate of the pin when used as an output.	0x1	R/W
		0	Fast Slew Rate.		
		1	Slow Slew Rate.		
[1:0]	FSYNC0_DRIVE		FSYNC_0 Pin Drive Strength. Determines the drive strength of the pin when used as an output.	0x1	R/W
		0	2mA Output Drive.		
		1	4mA Output Drive.		
		10	8mA Output Drive.		
		11	12mA Output Drive.		

SDATAO_0 PIN CONTROL REGISTER

Address: 0x4000C139, Reset: 0x05, Name: SDATAO0_CTRL

Table 344. Bit Descriptions for SDATAO0_CTRL

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
5	SDATAO0_PULL_SEL		SDATAO_0 Pin Weak Pull-Up/Down Selection.	0x0	R/W
		0	Weak pull-down when enabled.		
		1	Weak pull-up when enabled.		
4	SDATAO0_PULL_EN		SDATAO_0 Pin Weak Pull-Up/Down Enable.	0x0	R/W
		0	No pull-up or pull-down.		
		1	Weak pull-up or pull-down set by SEL register.		
3	RESERVED		Reserved.	0x0	R
2	SDATAO0_SLEW		SDATAO_0 Pin Slew Rate. Determines the slew rate of the pin when used as an output.	0x1	R/W
		0	Fast Slew Rate.		
		1	Slow Slew Rate.		

Bits	Bit Name	Settings	Description	Reset	Access
[1:0]	SDATAO0_DRIVE		SDATAO0 Pin Drive Strength. Determines the drive strength of the pin when used as an output.	0x1	R/W
		0	2mA Output Drive.		
		1	4mA Output Drive.		
		10	8mA Output Drive.		
		11	12mA Output Drive.		

SDATAI_0 PIN CONTROLS REGISTER

Address: 0x4000C13A, Reset: 0x05, Name: SDATAI0_CTRL

Table 345. Bit Descriptions for SDATAI0_CTRL

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
5	SDATAI0_PULL_SEL		SDATAI_0 Pin Weak Pull-Up/Down Selection.	0x0	R/W
		0	Weak pull-down when enabled.		
		1	Weak pull-up when enabled.		
4	SDATAI0_PULL_EN		SDATAI_0 Pin Weak Pull-Up/Down Enable.	0x0	R/W
		0	No pull-up or pull-down.		
		1	Weak pull-up or pull-down set by SEL register.		
3	RESERVED		Reserved.	0x0	R
2	SDATAI0_SLEW		SDATAI_0 Pin Slew Rate. Determines the slew rate of the pin when used as an output.	0x1	R/W
		0	Fast Slew Rate.		
		1	Slow Slew Rate.		
[1:0]	SDATAI0_DRIVE		SDATAI_0 Pin Drive Strength. Determines the drive strength of the pin when used as an output.	0x1	R/W
		0	2mA Output Drive.		
		1	4mA Output Drive.		
		10	8mA Output Drive.		
		11	12mA Output Drive.		

BCLK_1 PIN CONTROLS REGISTER

Address: 0x4000C13B, Reset: 0x05, Name: BCLK1_CTRL

Table 346. Bit Descriptions for BCLK1_CTRL

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
5	BCLK1_PULL_SEL		BCLK_1 Pin Weak Pull-Up/Down Selection.	0x0	R/W
		0	Weak pull-down when enabled.		
		1	Weak pull-up when enabled.		
4	BCLK1_PULL_EN		BCLK_1 Pin Weak Pull-Up/Down Enable.	0x0	R/W
		0	No pull-up or pull-down.		
		1	Weak pull-up or pull-down set by SEL register.		
3	RESERVED		Reserved.	0x0	R
2	BCLK1_SLEW		BCLK_1 Pin Slew Rate. Determines the slew rate of the pin when used as an output.	0x1	R/W
		0	Fast Slew Rate.		
		1	Slow Slew Rate.		

Bits	Bit Name	Settings	Description	Reset	Access
[1:0]	BCLK1_DRIVE		BCLK_1 Pin Drive Strength. Determines the drive strength of the pin when used as an output.	0x1	R/W
		0	2mA Output Drive.		
		1	4mA Output Drive.		
		10	8mA Output Drive.		
		11	12mA Output Drive.		

FSYNC_1 PIN CONTROLS REGISTER

Address: 0x4000C13C, Reset: 0x05, Name: FSYNC1_CTRL

Table 347. Bit Descriptions for FSYNC1_CTRL

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
5	FSYNC1_PULL_SEL		FSYNC_1 Pin Weak Pull-Up/Down Selection.	0x0	R/W
		0	Weak pull-down when enabled.		
		1	Weak pull-up when enabled.		
4	FSYNC1_PULL_EN		FSYNC_1 Pin Weak Pull-Up/Down Enable.	0x0	R/W
		0	No pull-up or pull-down.		
		1	Weak pull-up or pull-down set by SEL register.		
3	RESERVED		Reserved.	0x0	R
2	FSYNC1_SLEW		FSYNC_1 Pin Slew Rate. Determines the slew rate of the pin when used as an output.	0x1	R/W
		0	Fast Slew Rate.		
		1	Slow Slew Rate.		
[1:0]	FSYNC1_DRIVE		FSYNC_1 Pin Drive Strength. Determines the drive strength of the pin when used as an output.	0x1	R/W
		0	2mA Output Drive.		
		1	4mA Output Drive.		
		10	8mA Output Drive.		
		11	12mA Output Drive.		

SDATAO_1 PIN CONTROLS REGISTER

Address: 0x4000C13D, Reset: 0x05, Name: SDATAO1_CTRL

Table 348. Bit Descriptions for SDATAO1_CTRL

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
5	SDATAO1_PULL_SEL		SDATAO_1 Pin Weak Pull-Up/Down Selection.	0x0	R/W
		0	Weak pull-down when enabled.		
		1	Weak pull-up when enabled.		
4	SDATAO1_PULL_EN		SDATAO_1 Pin Weak Pull-Up/Down Enable.	0x0	R/W
		0	No pull-up or pull-down.		
		1	Weak pull-up or pull-down set by SEL register.		
3	RESERVED		Reserved.	0x0	R
2	SDATAO1_SLEW		SDATAO_1 Pin Slew Rate. Determines the slew rate of the pin when used as an output.	0x1	R/W
		0	Fast Slew Rate.		
		1	Slow Slew Rate.		

Bits	Bit Name	Settings	Description	Reset	Access
[1:0]	SDATAO1_DRIVE		SDATAO_0 Pin Drive Strength. Determines the drive strength of the pin when used as an output.	0x1	R/W
		0	2mA Output Drive.		
		1	4mA Output Drive.		
		10	8mA Output Drive.		
		11	12mA Output Drive.		

SDATAI_1 PIN CONTROLS REGISTER

Address: 0x4000C13E, Reset: 0x05, Name: SDATAI1_CTRL

Table 349. Bit Descriptions for SDATAI1_CTRL

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
5	SDATAI1_PULL_SEL		SDATAI_1 Pin Weak Pull-Up/Down Selection.	0x0	R/W
		0	Weak pull-down when enabled.		
		1	Weak pull-up when enabled.		
4	SDATAI1_PULL_EN		SDATAI_1 Pin Weak Pull-Up/Down Enable.	0x0	R/W
		0	No pull-up or pull-down.		
		1	Weak pull-up or pull-down set by SEL register.		
3	RESERVED		Reserved.	0x0	R
2	SDATAI1_SLEW		SDATAI_1 Pin Slew Rate. Determines the slew rate of the pin when used as an output.	0x1	R/W
		0	Fast Slew Rate.		
		1	Slow Slew Rate.		
[1:0]	SDATAI1_DRIVE		SDATAI_1 Pin Drive Strength. Determines the drive strength of the pin when used as an output.	0x1	R/W
		0	2mA Output Drive.		
		1	4mA Output Drive.		
		10	8mA Output Drive.		
		11	12mA Output Drive.		

QSPIM_CLK PIN CONTROLS REGISTER

Address: 0x4000C13F, Reset: 0x05, Name: QSPIM_CLK_CTRL

Table 350. Bit Descriptions for QSPIM_CLK_CTRL

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
5	QSPIM_CLK_PULL_SEL		QSPIM_CLK Pin Weak Pull-Up/Down Selection.	0x0	R/W
		0	Weak pull-down when enabled.		
		1	Weak pull-up when enabled.		
4	QSPIM_CLK_PULL_EN		QSPIM_CLK Pin Weak Pull-Up/Down Enable.	0x0	R/W
		0	No pull-up or pull-down.		
		1	Weak pull-up or pull-down set by SEL register.		
3	RESERVED		Reserved.	0x0	R
2	QSPIM_CLK_SLEW		QSPIM_CLK Pin Slew Rate. Determines the slew rate of the pin when used as an output.	0x1	R/W
		0	Fast Slew Rate.		
		1	Slow Slew Rate.		

Bits	Bit Name	Settings	Description	Reset	Access
[1:0]	QSPIM_CLK_DRIVE		QSPIM_CLK Pin Drive Strength. Determines the drive strength of the pin when used as an output.	0x1	R/W
		0	2mA Output Drive.		
		1	4mA Output Drive.		
		10	8mA Output Drive.		
		11	12mA Output Drive.		

QSPIM_CS PIN CONTROLS REGISTER

Address: 0x4000C140, Reset: 0x05, Name: QSPIM_CS_CTRL

Table 351. Bit Descriptions for QSPIM_CS_CTRL

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
5	QSPIM_CS_PULL_SEL		QSPIM_CS Pin Weak Pull-Up/Down Selection.	0x0	R/W
		0	Weak pull-down when enabled.		
		1	Weak pull-up when enabled.		
4	QSPIM_CS_PULL_EN		QSPIM_CS Pin Weak Pull-Up/Down Enable.	0x0	R/W
		0	No pull-up or pull-down.		
		1	Weak pull-up or pull-down set by SEL register.		
3	RESERVED		Reserved.	0x0	R
2	QSPIM_CS_SLEW		QSPIM_CS Pin Slew Rate. Determines the slew rate of the pin when used as an output.	0x1	R/W
		0	Fast Slew Rate.		
		1	Slow Slew Rate.		
[1:0]	QSPIM_CS_DRIVE		QSPIM_CS Pin Drive Strength. Determines the drive strength of the pin when used as an output.	0x1	R/W
		0	2mA Output Drive.		
		1	4mA Output Drive.		
		10	8mA Output Drive.		
		11	12mA Output Drive.		

QSPIM_SDIO0 PIN CONTROLS REGISTER

Address: 0x4000C141, Reset: 0x05, Name: QSPIM_SDIO0_CTRL

Table 352. Bit Descriptions for QSPIM_SDIO0_CTRL

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
5	QSPIM_SDIO0_PULL_SEL		QSPIM_SDIO0 Pin Weak Pull-Up/Down Selection.	0x0	R/W
		0	Weak pull-down when enabled.		
		1	Weak pull-up when enabled.		
4	QSPIM_SDIO0_PULL_EN		QSPIM_SDIO0 Pin Weak Pull-Up/Down Enable.	0x0	R/W
		0	No pull-up or pull-down.		
		1	Weak pull-up or pull-down set by SEL register.		
3	RESERVED		Reserved.	0x0	R
2	QSPIM_SDIO0_SLEW		QSPIM_SDIO0 Pin Slew Rate. Determines the slew rate of the pin when used as an output.	0x1	R/W
		0	Fast Slew Rate.		
		1	Slow Slew Rate.		

Bits	Bit Name	Settings	Description	Reset	Access
[1:0]	QSPIM_SDIO0_DRIVE		QSPIM_SDIO0 Pin Drive Strength. Determines the drive strength of the pin when used as an output.	0x1	R/W
		0	2mA Output Drive.		
		1	4mA Output Drive.		
		10	8mA Output Drive.		
		11	12mA Output Drive.		

QSPIM_SDIO1 PIN CONTROLS REGISTER

Address: 0x4000C142, Reset: 0x05, Name: QSPIM_SDIO1_CTRL

Table 353. Bit Descriptions for QSPIM_SDIO1_CTRL

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
5	QSPIM_SDIO1_PULL_SEL		QSPIM_SDIO1 Pin Weak Pull-Up/Down Selection.	0x0	R/W
		0	Weak pull-down when enabled.		
		1	Weak pull-up when enabled.		
4	QSPIM_SDIO1_PULL_EN		QSPIM_SDIO1 Pin Weak Pull-Up/Down Enable.	0x0	R/W
		0	No pull-up or pull-down.		
		1	Weak pull-up or pull-down set by SEL register.		
3	RESERVED		Reserved.	0x0	R
2	QSPIM_SDIO1_SLEW		QSPIM_SDIO1 Pin Slew Rate. Determines the slew rate of the pin when used as an output.	0x1	R/W
		0	Fast Slew Rate.		
		1	Slow Slew Rate.		
[1:0]	QSPIM_SDIO1_DRIVE		QSPIM_SDIO1 Pin Drive Strength. Determines the drive strength of the pin when used as an output.	0x1	R/W
		0	2mA Output Drive.		
		1	4mA Output Drive.		
		10	8mA Output Drive.		
		11	12mA Output Drive.		

QSPIM_SDIO2 PIN CONTROLS REGISTER

Address: 0x4000C143, Reset: 0x05, Name: QSPIM_SDIO2_CTRL

Table 354. Bit Descriptions for QSPIM_SDIO2_CTRL

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
5	QSPIM_SDIO2_PULL_SEL		QSPIM_SDIO2 Pin Weak Pull-Up/Down Selection.	0x0	R/W
		0	Weak pull-down when enabled.		
		1	Weak pull-up when enabled.		
4	QSPIM_SDIO2_PULL_EN		QSPIM_SDIO2 Pin Weak Pull-Up/Down Enable.	0x0	R/W
		0	No pull-up or pull-down.		
		1	Weak pull-up or pull-down set by SEL register.		
3	RESERVED		Reserved.	0x0	R
2	QSPIM_SDIO2_SLEW		QSPIM_SDIO2 Pin Slew Rate. Determines the slew rate of the pin when used as an output.	0x1	R/W
		0	Fast Slew Rate.		
		1	Slow Slew Rate.		

Bits	Bit Name	Settings	Description	Reset	Access
[1:0]	QSPIM_SDIO2_DRIVE		QSPIM_SDIO2 Pin Drive Strength. Determines the drive strength of the pin when used as an output.	0x1	R/W
		0	2mA Output Drive.		
		1	4mA Output Drive.		
		10	8mA Output Drive.		
		11	12mA Output Drive.		

QSPIM_SDIO3 PIN CONTROLS REGISTER

Address: 0x4000C144, Reset: 0x25, Name: QSPIM_SDIO3_CTRL

Table 355. Bit Descriptions for QSPIM_SDIO3_CTRL

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
5	QSPIM_SDIO3_PULL_SEL		QSPIM_SDIO3 Pin Slew Rate. Determines the slew rate of the pin when used as an output.	0x1	R/W
		0	Fast Slew Rate.		
		1	Slow Slew Rate.		
4	QSPIM_SDIO3_PULL_EN		QSPIM_SDIO3 Pin Weak Pull-Up/Down Enable.	0x0	R/W
		0	No pull-up or pull-down.		
		1	Weak pull-up or pull-down set by SEL register.		
3	RESERVED		Reserved.	0x0	R
2	QSPIM_SDIO3_SLEW		QSPIM_SDIO3 Pin Slew Rate. Determines the slew rate of the pin when used as an output.	0x1	R/W
		0	Fast Slew Rate.		
		1	Slow Slew Rate.		
[1:0]	QSPIM_SDIO3_DRIVE		QSPIM_SDIO3 Pin Drive Strength. Determines the drive strength of the pin when used as an output.	0x1	R/W
		0	2mA Output Drive.		
		1	4mA Output Drive.		
		10	8mA Output Drive.		
		11	12mA Output Drive.		

UART_COMM_TX PIN CONTROLS REGISTER

Address: 0x4000C145, Reset: 0x25, Name: UART_COMM_TX_CTRL

Table 356. Bit Descriptions for UART_COMM_TX_CTRL

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
5	UART_COMM_TX_PULL_SEL		UART_COMM_TX Pin Slew Rate. Determines the slew rate of the pin when used as an output.	0x1	R/W
		0	Fast Slew Rate.		
		1	Slow Slew Rate.		
4	UART_COMM_TX_PULL_EN		UART_COMM_TX Pin Weak Pull-Up/Down Enable.	0x0	R/W
		0	No pull-up or pull-down.		
		1	Weak pull-up or pull-down set by SEL register.		
3	RESERVED		Reserved.	0x0	R
2	UART_COMM_TX_SLEW		UART_COMM_TX Pin Slew Rate. Determines the slew rate of the pin when used as an output.	0x1	R/W
		0	Fast Slew Rate.		
		1	Slow Slew Rate.		

Bits	Bit Name	Settings	Description	Reset	Access
[1:0]	UART_COMM_TX_DRIVE		UART_COMM_TX Pin Drive Strength. Determines the drive strength of the pin when used as an output.	0x1	R/W
		0	2mA Output Drive.		
		1	4mA Output Drive.		
		10	8mA Output Drive.		
		11	12mA Output Drive.		

UART_COMM_TX PIN CONTROLS REGISTER

Address: 0x4000C146, Reset: 0x25, Name: UART_COMM_RX_CTRL

Table 357. Bit Descriptions for UART_COMM_RX_CTRL

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
5	UART_COMM_RX_PULL_SEL		UART_COMM_RX Pin Slew Rate. Determines the slew rate of the pin when used as an output.	0x1	R/W
		0	Fast Slew Rate.		
		1	Slow Slew Rate.		
4	UART_COMM_RX_PULL_EN		UART_COMM_RX Pin Weak Pull-Up/Down Enable.	0x0	R/W
		0	No pull-up or pull-down.		
		1	Weak pull-up or pull-down set by SEL register.		
3	RESERVED		Reserved.	0x0	R
2	UART_COMM_RX_SLEW		UART_COMM_RX Pin Slew Rate. Determines the slew rate of the pin when used as an output.	0x1	R/W
		0	Fast Slew Rate.		
		1	Slow Slew Rate.		
[1:0]	UART_COMM_RX_DRIVE		UART_COMM_RX Pin Drive Strength. Determines the drive strength of the pin when used as an output.	0x1	R/W
		0	2mA Output Drive.		
		1	4mA Output Drive.		
		10	8mA Output Drive.		
		11	12mA Output Drive.		

SELFBOOT PIN CONTROLS REGISTER

Address: 0x4000C147, Reset: 0x05, Name: SELFBOOT_CTRL

Table 358. Bit Descriptions for SELFBOOT_CTRL

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
5	SELFBOOT_PULL_SEL		SELFBOOT Pin Weak Pull-Up/Down Selection.	0x0	R/W
		0	Weak pull-down when enabled.		
		1	Weak pull-up when enabled.		
4	SELFBOOT_PULL_EN		SELFBOOT Pin Weak Pull-Up/Down Enable.	0x0	R/W
		0	No pull-up or pull-down.		
		1	Weak pull-up or pull-down set by SEL register.		
3	RESERVED		Reserved.	0x0	R
2	SELFBOOT_SLEW		SELFBOOT Pin Slew Rate. Determines the slew rate of the pin when used as an output.	0x1	R/W
		0	Fast Slew Rate.		
		1	Slow Slew Rate.		

Bits	Bit Name	Settings	Description	Reset	Access
[1:0]	SELFBOOT_DRIVE		SELFBOOT Pin Drive Strength. Determines the drive strength of the pin when used as an output.	0x1	R/W
		0	2mA Output Drive.		
		1	4mA Output Drive.		
		10	8mA Output Drive.		
		11	12mA Output Drive.		

SELFBOOT PIN CONTROLS REGISTER

Address: 0x4000C148, Reset: 0x05, Name: IRQ_CTRL

Table 359. Bit Descriptions for IRQ_CTRL

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
5	IRQ_PULL_SEL		IRQ Pin Weak Pull-Up/Down Selection.	0x0	R/W
		0	Weak pull-down when enabled.		
		1	Weak pull-up when enabled.		
4	IRQ_PULL_EN		IRQ Pin Weak Pull-Up/Down Enable.	0x0	R/W
		0	No pull-up or pull-down.		
		1	Weak pull-up or pull-down set by SEL register.		
3	RESERVED		Reserved.	0x0	R
2	IRQ_SLEW		IRQ Pin Slew Rate. Determines the slew rate of the pin when used as an output.	0x1	R/W
		0	Fast Slew Rate.		
		1	Slow Slew Rate.		
[1:0]	IRQ_DRIVE		IRQ Pin Drive Strength. Determines the drive strength of the pin when used as an output.	0x1	R/W
		0	2mA Output Drive.		
		1	4mA Output Drive.		
		10	8mA Output Drive.		
		11	12mA Output Drive.		

SELFBOOT PIN CONTROLS REGISTER

Address: 0x4000C149, Reset: 0x05, Name: ROM_BOOT_MODE_CTRL

Table 360. Bit Descriptions for ROM_BOOT_MODE_CTRL

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
5	ROM_BOOT_MODE_PULL_SEL		ROM_BOOT_MODE Pin Weak Pull-Up/Down Selection.	0x0	R/W
		0	Weak pull-down when enabled.		
		1	Weak pull-up when enabled.		
4	ROM_BOOT_MODE_PULL_EN		ROM_BOOT_MODE Pin Weak Pull-Up/Down Enable.	0x0	R/W
		0	No pull-up or pull-down.		
		1	Weak pull-up or pull-down set by SEL register.		
3	RESERVED		Reserved.	0x0	R
2	ROM_BOOT_MODE_SLEW		ROM_BOOT_MODE Pin Slew Rate. Determines the slew rate of the pin when used as an output.	0x1	R/W
		0	Fast Slew Rate.		
		1	Slow Slew Rate.		

Bits	Bit Name	Settings	Description	Reset	Access
[1:0]	ROM_BOOT_MODE_DRIVE		ROM_BOOT_MODE Pin Drive Strength. Determines the drive strength of the pin when used as an output.	0x1	R/W
		0	2mA Output Drive.		
		1	4mA Output Drive.		
		10	8mA Output Drive.		
		11	12mA Output Drive.		

SELFBOOT PIN CONTROLS REGISTER

Address: 0x4000C14A, Reset: 0x15, Name: TCK_CTRL

Table 361. Bit Descriptions for TCK_CTRL

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
5	TCK_PULL_SEL		TCK Pin Weak Pull-Up/Down Selection.	0x0	R/W
		0	Weak pull-down when enabled.		
		1	Weak pull-up when enabled.		
4	TCK_PULL_EN		TCK Pin Weak Pull-Up/Down Enable.	0x1	R/W
		0	No pull-up or pull-down.		
		1	Weak pull-up or pull-down set by SEL register.		
3	RESERVED		Reserved.	0x0	R
2	TCK_SLEW		TCK Pin Slew Rate. Determines the slew rate of the pin when used as an output.	0x1	R/W
		0	Fast Slew Rate.		
		1	Slow Slew Rate.		
[1:0]	TCK_DRIVE		TCK Pin Drive Strength. Determines the drive strength of the pin when used as an output.	0x1	R/W
		0	2mA Output Drive.		
		1	4mA Output Drive.		
		10	8mA Output Drive.		
		11	12mA Output Drive.		

SELFBOOT PIN CONTROLS REGISTER

Address: 0x4000C14B, Reset: 0x35, Name: TMS_CTRL

Table 362. Bit Descriptions for TMS_CTRL

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
5	TMS_PULL_SEL		TMS Pin Weak Pull-Up/Down Selection.	0x1	R/W
		0	Weak pull-down when enabled.		
		1	Weak pull-up when enabled.		
4	TMS_PULL_EN		TMS Pin Weak Pull-Up/Down Enable.	0x1	R/W
		0	No pull-up or pull-down.		
		1	Weak pull-up or pull-down set by SEL register.		
3	RESERVED		Reserved.	0x0	R
2	TMS_SLEW		TMS Pin Slew Rate. Determines the slew rate of the pin when used as an output.	0x1	R/W
		0	Fast Slew Rate.		
		1	Slow Slew Rate.		

Bits	Bit Name	Settings	Description	Reset	Access
[1:0]	TMS_DRIVE		TMS Pin Drive Strength. Determines the drive strength of the pin when used as an output.	0x1	R/W
		0	2mA Output Drive.		
		1	4mA Output Drive.		
		10	8mA Output Drive.		
		11	12mA Output Drive.		

SELFBOOT PIN CONTROLS REGISTER

Address: 0x4000C14C, Reset: 0x05, Name: TDO_CTRL

Table 363. Bit Descriptions for TDO_CTRL

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
5	TDO_PULL_SEL		TDO Pin Weak Pull-Up/Down Selection.	0x0	R/W
		0	Weak pull-down when enabled.		
		1	Weak pull-up when enabled.		
4	TDO_PULL_EN		TDO Pin Weak Pull-Up/Down Enable.	0x0	R/W
		0	No pull-up or pull-down.		
		1	Weak pull-up or pull-down set by SEL register.		
3	RESERVED		Reserved.	0x0	R
2	TDO_SLEW		TDO Pin Slew Rate. Determines the slew rate of the pin when used as an output.	0x1	R/W
		0	Fast Slew Rate.		
		1	Slow Slew Rate.		
[1:0]	TDO_DRIVE		TDO Pin Drive Strength. Determines the drive strength of the pin when used as an output.	0x1	R/W
		0	2mA Output Drive.		
		1	4mA Output Drive.		
		10	8mA Output Drive.		
		11	12mA Output Drive.		

SELFBOOT PIN CONTROLS REGISTER

Address: 0x4000C14D, Reset: 0x35, Name: TDI_CTRL

Table 364. Bit Descriptions for TDI_CTRL

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
5	TDI_PULL_SEL		TDI Pin Weak Pull-Up/Down Selection.	0x1	R/W
		0	Weak pull-down when enabled.		
		1	Weak pull-up when enabled.		
4	TDI_PULL_EN		TDI Pin Weak Pull-Up/Down Enable.	0x1	R/W
		0	No pull-up or pull-down.		
		1	Weak pull-up or pull-down set by SEL register.		
3	RESERVED		Reserved.	0x0	R
2	TDI_SLEW		TDI Pin Slew Rate. Determines the slew rate of the pin when used as an output.	0x1	R/W
		0	Fast Slew Rate.		
		1	Slow Slew Rate.		

Bits	Bit Name	Settings	Description	Reset	Access
[1:0]	TDI_DRIVE		TDI Pin Drive Strength. Determines the drive strength of the pin when used as an output.	0x1	R/W
		0	2mA Output Drive.		
		1	4mA Output Drive.		
		10	8mA Output Drive.		
		11	12mA Output Drive.		

SDA/MISO PIN CONTROLS REGISTER

Address: 0x4000C14E, Reset: 0x00, Name: I2C_SPI_CTRL

Table 365. Bit Descriptions for I2C_SPI_CTRL

Bits	Bit Name	Settings	Description	Reset	Access
[7:2]	RESERVED		Reserved.	0x0	R
1	SCL_SCLK_DRIVE		SCL/SCLK Output Pin Drive Strength.	0x0	R/W
		0	4mA drive strength.		
		1	20mA drive strength. May be required for fast mode plus I2C operation.		
0	SDA_MISO_DRIVE		SDA/MISO Output Pin Drive Strength.	0x0	R/W
		0	4mA drive strength.		
		1	20mA drive strength. May be required for fast mode plus I2C operation.		

IRQ SIGNALING AND CLEARING REGISTER

Address: 0x4000C150, Reset: 0x00, Name: IRQ_CTRL1

Table 366. Bit Descriptions for IRQ_CTRL1

Bits	Bit Name	Settings	Description	Reset	Access
7	IRQ4_PINOUT_POL		IRQ4 pin output polarity control.	0x0	R/W
		0	Active Low Interrupt Signaling.		
		1	Active High Interrupt Signaling.		
6	IRQ3_PINOUT_POL		IRQ3 pin output polarity control.	0x0	R/W
		0	Active Low Interrupt Signaling.		
		1	Active High Interrupt Signaling.		
5	IRQ2_PINOUT_POL		IRQ2 pin output polarity control.	0x0	R/W
		0	Active Low Interrupt Signaling.		
		1	Active High Interrupt Signaling.		
4	IRQ1_PINOUT_POL		IRQ1 pin output polarity control.	0x0	R/W
		0	Active Low Interrupt Signaling on Pin.		
		1	Active High Interrupt Signaling on Pin.		
3	IRQ4_CLEAR		Write one to clear IRQ4.	0x0	R/W1T
		0	Not Clear.		
		1	Write once to clear IRQ4.		
2	IRQ3_CLEAR		Write one to clear IRQ 3.	0x0	R/W1T
		0	Not Clear.		
		1	Write once to clear IRQ4.		
1	IRQ2_CLEAR		Write one to clear IRQ 2.	0x0	R/W1T
		0	Not Clear.		
		1	Write once to clear IRQ2.		
0	IRQ1_CLEAR		Write once to clear IRQ 1.	0x0	R/W1T
		0	Not Clear.		
		1	Write once to clear IRQ1.		

IRQ1 MASKING REGISTER

Address: 0x4000C151, Reset: 0x7F, Name: IRQ1_MASK1

Table 367. Bit Descriptions for IRQ1_MASK1

Bits	Bit Name	Settings	Description	Reset	Access
7	RESERVED		Reserved.	0x0	R
6	IRQ1_AVDD_UVW_MASK		Mask AVDD undervoltage warning to IRQ1.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		
5	IRQ1_PLL_UNLOCKED_MASK		Mask PLL Locked to Unlocked Transition to IRQ1.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		
4	IRQ1_PLL_LOCKED_MASK		Mask PLL Unlocked to Locked Transition to IRQ1.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		
3	IRQ1_ADC2_CLIP_MASK		Mask ADC Channel 2 Clipping to IRQ1.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		
2	IRQ1_ADC1_CLIP_MASK		Mask ADC Channel 1 Clipping to IRQ1.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		
1	IRQ1_ADC0_CLIP_MASK		Mask ADC Channel 0 Clipping to IRQ1.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		
0	IRQ1_DAC0_CLIP_MASK		Mask DAC Channel 0 Clipping to IRQ1.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		

IRQ1 MASKING REGISTER

Address: 0x4000C152, Reset: 0xF7, Name: IRQ1_MASK2

Table 368. Bit Descriptions for IRQ1_MASK2

Bits	Bit Name	Settings	Description	Reset	Access
7	IRQ1_ASRCO_UNLOCKED_MASK		Mask Output ASRC Locked to Unlocked Transition to IRQ.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		
6	IRQ1_ASRCO_LOCKED_MASK		Mask Output ASRC Unlocked to Locked Transition to IRQ.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		
5	IRQ1_ASRCI_UNLOCKED_MASK		Mask Input ASRC Locked to Unlocked Transition to IRQ.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		
4	IRQ1_ASRCI_LOCKED_MASK		Mask Input ASRC Unlocked to Locked Transition to IRQ.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		
3	RESERVED		Reserved.	0x0	R
2	IRQ1_EQ_CLEAR_DONE_MASK		EQ Memory Clear Done Mask.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		

Bits	Bit Name	Settings	Description	Reset	Access
1	IRQ1_FDSP_MASK_ONZ_MASK		FastDSP Masked Output Non Zero Indicator Mask.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		
0	IRQ1_PRAMP_MASK		Parameter ramp complete transition interrupt Mask.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		

IRQ1 MASKING REGISTER

Address: 0x4000C153, Reset: 0x3F, Name: IRQ1_MASK3

Table 369. Bit Descriptions for IRQ1_MASK3

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
5	IRQ1_DLDO_DVS_DONE_MASK		DLDO DVS Done Mask to IRQ1.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		
4	IRQ1_POWER_UP_COMPLETE_MASK		Mask Power Up Not Finished to Completed Transition to IRQ1.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		
3	IRQ1_SPT1_UNLOCKED_MASK		Mask SPT1 Locked to Unlocked Transition to IRQ1.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		
2	IRQ1_SPT0_UNLOCKED_MASK		Mask SPT0 Locked to Unlocked Transition to IRQ1.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		
1	IRQ1_SYNC_UNLOCKED_MASK		Mask Sync_Lock Locked to Unlocked Transition to IRQ1.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		
0	IRQ1_SYNC_LOCKED_MASK		Mask Sync Lock Unlocked to Locked Transition to IRQ1.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		

IRQ1 MASKING REGISTER

Address: 0x4000C154, Reset: 0x3F, Name: IRQ1_MASK4

Table 370. Bit Descriptions for IRQ1_MASK4

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
5	IRQ1_FFSRAM2_IRQ_MASK		FFSRAM2 Interrupt output to IRQ1 Mask.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		
4	IRQ1_DATA_SYNC_IRQ_MASK		DATA SYNC Interrupt to IRQ1 Mask.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		
3	IRQ1_SW_INT3_MASK		Software Interrupt 3 to IRQ1 MASK.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		

Bits	Bit Name	Settings	Description	Reset	Access
2	IRQ1_SW_INT2_MASK		Software Interrupt 2 to IRQ1 MASK.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		
1	IRQ1_SW_INT1_MASK		Software Interrupt 1 to IRQ1 MASK.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		
0	IRQ1_SW_INT0_MASK		Software Interrupt 0 to IRQ1 MASK.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		

IRQ1 MASKING REGISTER

Address: 0x4000C155, Reset: 0x0F, Name: IRQ1_MASK5

Table 371. Bit Descriptions for IRQ1_MASK5

Bits	Bit Name	Settings	Description	Reset	Access
[7:4]	RESERVED		Reserved.	0x0	R
3	IRQ1_WDT_IRQ_MASK		Watchdog Timer Interrupt to IRQ1 Mask.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		
2	IRQ1_AHB_BUSY_MASK		AHB Busy to IRQ1 Mask.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		
1	IRQ1_AHB_ERR_MASK		AHB Error to IRQ1 Mask.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		
0	IRQ1_TDSP_ERR_MASK		TDSP Error to IRQ1 Mask.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		

IRQ2 MASKING REGISTER

Address: 0x4000C156, Reset: 0x7F, Name: IRQ2_MASK1

Table 372. Bit Descriptions for IRQ2_MASK1

Bits	Bit Name	Settings	Description	Reset	Access
7	RESERVED		Reserved.	0x0	R
6	IRQ2_AVDD_UVW_MASK		Mask AVDD undervoltage warning to IRQ2.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		
5	IRQ2_PLL_UNLOCKED_MASK		Mask PLL Locked to Unlocked Transition to IRQ2.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		
4	IRQ2_PLL_LOCKED_MASK		Mask PLL Unlocked to Locked Transition to IRQ2.	0x1	R/W
		0	Fault will cause IRQ.		
		1	Fault is masked and will not cause IRQ.		
3	IRQ2_ADC2_CLIP_MASK		Mask ADC Channel 2 Clipping to IRQ2.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		

Bits	Bit Name	Settings	Description	Reset	Access
2	IRQ2_ADC1_CLIP_MASK		Mask ADC Channel 1 Clipping to IRQ2.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		
1	IRQ2_ADC0_CLIP_MASK		Mask ADC Channel 0 Clipping to IRQ2.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		
0	IRQ2_DAC0_CLIP_MASK		Mask DAC Channel 0 Clipping to IRQ2.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		

IRQ2 MASKING REGISTER

Address: 0x4000C157, Reset: 0xF7, Name: IRQ2_MASK2

Table 373. Bit Descriptions for IRQ2_MASK2

Bits	Bit Name	Settings	Description	Reset	Access
7	IRQ2_ASRCO_UNLOCKED_MASK		Mask Output ASRC Locked to Unlocked Transition to IRQ2.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		
6	IRQ2_ASRCO_LOCKED_MASK		Mask Output ASRC Unlocked to Locked Transition to IRQ2.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		
5	IRQ2_ASRCI_UNLOCKED_MASK		Mask Input ASRC Locked to Unlocked Transition to IRQ2.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		
4	IRQ2_ASRCI_LOCKED_MASK		Mask Input ASRC Unlocked to Locked Transition to IRQ2.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		
3	RESERVED		Reserved.	0x0	R
2	IRQ2_EQ_CLEAR_DONE_MASK		EQ Memory Clear Done Mask.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		
1	IRQ2_FDSP_MASK_ONZ_MASK		FastDSP Masked Output Non Zero Indicator Mask.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		
0	IRQ2_PRAMP_MASK		Parameter ramp complete transition interrupt Mask to IRQ2.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		

IRQ2 MASKING REGISTER

Address: 0x4000C158, Reset: 0x3F, Name: IRQ2_MASK3

Table 374. Bit Descriptions for IRQ2_MASK3

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
5	IRQ2_DLDO_DVS_DONE_MASK		DLDO DVS Done Mask to IRQ2.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		

Bits	Bit Name	Settings	Description	Reset	Access
4	IRQ2_POWER_UP_COMPLETE_MASK		Mask Power Up Not Finished to Completed Transition to IRQ2.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		
3	IRQ2_SPT1_UNLOCKED_MASK		Mask SPT1 Locked to Unlocked Transition to IRQ2.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		
2	IRQ2_SPT0_UNLOCKED_MASK		Mask SPT0 Locked to Unlocked Transition to IRQ2.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		
1	IRQ2_SYNC_UNLOCKED_MASK		Mask PLL Locked to Unlocked Transition to IRQ2.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		
0	IRQ2_SYNC_LOCKED_MASK		Mask PLL Unlocked to Locked Transition to IRQ2.	0x1	R/W
		0	Fault will cause IRQ.		
		1	Fault is masked and will not cause IRQ.		

IRQ2 MASKING REGISTER

Address: 0x4000C159, Reset: 0x3F, Name: IRQ2_MASK4

Table 375. Bit Descriptions for IRQ2_MASK4

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
5	IRQ2_FFSRAM2_IRQ_MASK		FFSRAM2 IRQ to IRQ2 Mask.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		
4	IRQ2_DATA_SYNC_IRQ_MASK		DATA SYNC IRQ to IRQ2 Mask.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		
3	IRQ2_SW_INT3_MASK		Software Interrupt 3 to IRQ2 MASK.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		
2	IRQ2_SW_INT2_MASK		Software Interrupt 2 to IRQ2 MASK.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		
1	IRQ2_SW_INT1_MASK		Software Interrupt 1 to IRQ2 MASK.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		
0	IRQ2_SW_INT0_MASK		Software Interrupt 0 to IRQ2 MASK.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		

IRQ2 MASKING REGISTER

Address: 0x4000C15A, Reset: 0x0F, Name: IRQ2_MASK5

Table 376. Bit Descriptions for IRQ2_MASK5

Bits	Bit Name	Settings	Description	Reset	Access
[7:4]	RESERVED		Reserved.	0x0	R
3	IRQ2_WDT_IRQ_MASK		Mask Watchdog Timer Interrupt Transition to IRQ2.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		
2	IRQ2_AHB_BUSY_MASK		Mask AHB BUSY FROM LOW to HIGH Transition to IRQ2.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		
1	IRQ2_AHB_ERR_MASK		Mask AHB Error FROM LOW to HIGH Transition to IRQ2.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		
0	IRQ2_TDSP_ERR_MASK		Mask TDSP Error FROM LOW to HIGH Transition to IRQ2.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		

IRQ3 MASKING REGISTER

Address: 0x4000C15B, Reset: 0x7F, Name: IRQ3_MASK1

Table 377. Bit Descriptions for IRQ3_MASK1

Bits	Bit Name	Settings	Description	Reset	Access
7	RESERVED		Reserved.	0x0	R
6	IRQ3_AVDD_UVW_MASK		Mask AVDD undervoltage warning to IRQ3.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		
5	IRQ3_PLL_UNLOCKED_MASK		Mask PLL Locked to Unlocked Transition to IRQ3.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		
4	IRQ3_PLL_LOCKED_MASK		Mask PLL Unlocked to Locked Transition to IRQ3.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		
3	IRQ3_ADC2_CLIP_MASK		Mask ADC Channel 2 Clipping to IRQ3.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		
2	IRQ3_ADC1_CLIP_MASK		Mask ADC Channel 1 Clipping to IRQ3.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		
1	IRQ3_ADC0_CLIP_MASK		Mask ADC Channel 0 Clipping to IRQ3.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		
0	IRQ3_DAC0_CLIP_MASK		Mask DAC Channel 0 Clipping to IRQ3.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		

IRQ3 MASKING REGISTER

Address: 0x4000C15C, Reset: 0xF7, Name: IRQ3_MASK2

Table 378. Bit Descriptions for IRQ3_MASK2

Bits	Bit Name	Settings	Description	Reset	Access
7	IRQ3_ASRCO_UNLOCKED_MASK		Mask Output ASRC Locked to Unlocked Transition to IRQ3.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		
6	IRQ3_ASRCO_LOCKED_MASK		Mask Output ASRC Unlocked to Locked Transition to IRQ3.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		
5	IRQ3_ASRCI_UNLOCKED_MASK		Mask Input ASRC Locked to Unlocked Transition to IRQ3.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		
4	IRQ3_ASRCI_LOCKED_MASK		Mask Input ASRC Unlocked to Locked Transition to IRQ3.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		
3	RESERVED		Reserved.	0x0	R
2	IRQ3_EQ_CLEAR_DONE_MASK		EQ Memory Clear Done Mask.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		
1	IRQ3_FDSP_MASK_ONZ_MASK		FastDSP Masked Output Non Zero Indicator Mask.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		
0	IRQ3_PRAMP_MASK		Parameter ramp complete transition interrupt Mask to IRQ3.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		

IRQ3 MASKING REGISTER

Address: 0x4000C15D, Reset: 0x3F, Name: IRQ3_MASK3

Table 379. Bit Descriptions for IRQ3_MASK3

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
5	IRQ3_DLDO_DVS_DONE_MASK		DLDO DVS Done Mask to IRQ3.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		
4	IRQ3_POWER_UP_COMPLETE_MASK		Mask Power Up Not Finished to Completed Transition to IRQ3.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		
3	IRQ3_SPT1_UNLOCKED_MASK		Mask SPT1 Locked to Unlocked Transition to IRQ3.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		
2	IRQ3_SPT0_UNLOCKED_MASK		Mask SPT0 Locked to Unlocked Transition to IRQ3.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		

Bits	Bit Name	Settings	Description	Reset	Access
1	IRQ3_SYNC_UNLOCKED_MASK		Mask Sync_Lock Locked to Unlocked Transition to IRQ3.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		
0	IRQ3_SYNC_LOCKED_MASK		Mask Sync Lock Unlocked to Locked Transition to IRQ3.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		

IRQ3 MASKING REGISTER

Address: 0x4000C15E, Reset: 0x3F, Name: IRQ3_MASK4

Table 380. Bit Descriptions for IRQ3_MASK4

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
5	IRQ3_FFSRAM2_IRQ_MASK		FFSRAM2 IRQ to IRQ3 Mask.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		
4	IRQ3_DATA_SYNC_IRQ_MASK		DATA SYNC IRQ to IRQ3 Mask.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		
3	IRQ3_SW_INT3_MASK		Software Interrupt 3 to IRQ3 MASK.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		
2	IRQ3_SW_INT2_MASK		Software Interrupt 2 to IRQ3 MASK.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		
1	IRQ3_SW_INT1_MASK		Software Interrupt 1 to IRQ3 MASK.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		
0	IRQ3_SW_INT0_MASK		Software Interrupt 0 to IRQ3 MASK.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		

IRQ3 MASKING REGISTER

Address: 0x4000C15F, Reset: 0x0F, Name: IRQ3_MASK5

Table 381. Bit Descriptions for IRQ3_MASK5

Bits	Bit Name	Settings	Description	Reset	Access
[7:4]	RESERVED		Reserved.	0x0	R
3	IRQ3_WDT_IRQ_MASK		Mask Watchdog Timer Interrupt Transition to IRQ3.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		
2	IRQ3_AHB_BUSY_MASK		Mask AHB BUSY FROM LOW to HIGH Transition to IRQ3.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		

Bits	Bit Name	Settings	Description	Reset	Access
1	IRQ3_AHB_ERR_MASK		Mask AHB Error FROM LOW to HIGH Transition to IRQ3.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		
0	IRQ3_TDSP_ERR_MASK		Mask TDSP Error FROM LOW to HIGH Transition to IRQ3.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		

IRQ4 MASKING REGISTER

Address: 0x4000C160, Reset: 0x7F, Name: IRQ4_MASK1

Table 382. Bit Descriptions for IRQ4_MASK1

Bits	Bit Name	Settings	Description	Reset	Access
7	RESERVED		Reserved.	0x0	R
6	IRQ4_AVDD_UVW_MASK		Mask AVDD undervoltage warning to IRQ4.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		
5	IRQ4_PLL_UNLOCKED_MASK		Mask PLL Locked to Unlocked Transition to IRQ4.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		
4	IRQ4_PLL_LOCKED_MASK		Mask PLL Unlocked to Locked Transition to IRQ4.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		
3	IRQ4_ADC2_CLIP_MASK		Mask ADC Channel 2 Clipping to IRQ4.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		
2	IRQ4_ADC1_CLIP_MASK		Mask ADC Channel 1 Clipping to IRQ4.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		
1	IRQ4_ADC0_CLIP_MASK		Mask ADC Channel 0 Clipping to IRQ4.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		
0	IRQ4_DAC0_CLIP_MASK		Mask DAC Channel 0 Clipping to IRQ4.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		

IRQ4 MASKING REGISTER

Address: 0x4000C161, Reset: 0xF7, Name: IRQ4_MASK2

Table 383. Bit Descriptions for IRQ4_MASK2

Bits	Bit Name	Settings	Description	Reset	Access
7	IRQ4_ASRCO_UNLOCKED_MASK		Mask Output ASRC Locked to Unlocked Transition to IRQ4.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		
6	IRQ4_ASRCO_LOCKED_MASK		Mask Output ASRC Unlocked to Locked Transition to IRQ4.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		
5	IRQ4_ASRCI_UNLOCKED_MASK		Mask Input ASRC Locked to Unlocked Transition to IRQ4.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		

Bits	Bit Name	Settings	Description	Reset	Access
4	IRQ4_ASRCI_LOCKED_MASK		Mask Input ASRC Unlocked to Locked Transition to IRQ4.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		
3	RESERVED		Reserved.	0x0	R
2	IRQ4_EQ_CLEAR_DONE_MASK		EQ Memory Clear Done Mask.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		
1	IRQ4_FDSP_MASK_ONZ_MASK		FastDSP Masked Output Non Zero Indicator Mask.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		
0	IRQ4_PRAMP_MASK		Parameter ramp complete transition interrupt Mask to IRQ4.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		

IRQ4 MASKING REGISTER

Address: 0x4000C162, Reset: 0x3F, Name: IRQ4_MASK3

Table 384. Bit Descriptions for IRQ4_MASK3

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
5	IRQ4_DLDO_DVS_DONE_MASK		DLDO DVS Done Mask to IRQ4.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		
4	IRQ4_POWER_UP_COMPLETE_MASK		Mask Power Up Not Finished to Completed Transition to IRQ4.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		
3	IRQ4_SPT1_UNLOCKED_MASK		Mask SPT1 Locked to Unlocked Transition to IRQ4.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		
2	IRQ4_SPT0_UNLOCKED_MASK		Mask SPT0 Locked to Unlocked Transition to IRQ4.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		
1	IRQ4_SYNC_UNLOCKED_MASK		Mask Sync_Lock Locked to Unlocked Transition to IRQ4.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		
0	IRQ4_SYNC_LOCKED_MASK		Mask Sync Lock Unlocked to Locked Transition to IRQ4.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		

IRQ4 MASKING REGISTER

Address: 0x4000C163, Reset: 0x3F, Name: IRQ4_MASK4

Table 385. Bit Descriptions for IRQ4_MASK4

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
5	IRQ4_FFSRAM2_IRQ_MASK		FFSRAM2 IRQ to IRQ4 Mask.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		
4	IRQ4_DATA_SYNC_IRQ_MASK		DATA SYNC IRQ to IRQ4 Mask.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		
3	IRQ4_SW_INT3_MASK		Software Interrupt 3 to IRQ4 MASK.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		
2	IRQ4_SW_INT2_MASK		Software Interrupt 2 to IRQ4 MASK.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		
1	IRQ4_SW_INT1_MASK		Software Interrupt 1 to IRQ4 MASK.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		
0	IRQ4_SW_INT0_MASK		Software Interrupt 0 to IRQ4 MASK.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		

IRQ4 MASKING REGISTER

Address: 0x4000C164, Reset: 0x0F, Name: IRQ4_MASK5

Table 386. Bit Descriptions for IRQ4_MASK5

Bits	Bit Name	Settings	Description	Reset	Access
[7:4]	RESERVED		Reserved.	0x0	R
3	IRQ4_WDT_IRQ_MASK		Mask Watchdog Timer Interrupt Transition to IRQ4.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		
2	IRQ4_AHB_BUSY_MASK		Mask AHB BUSY FROM LOW to HIGH Transition to IRQ4.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		
1	IRQ4_AHB_ERR_MASK		Mask AHB Error FROM LOW to HIGH Transition to IRQ4.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		
0	IRQ4_TDSP_ERR_MASK		Mask TDSP Error FROM LOW to HIGH Transition to IRQ4.	0x1	R/W
		0	Event will cause IRQ.		
		1	Event is masked and will not cause IRQ.		

SOFTWARE INTERRUPTS WHICH CAN BE SET BY THE EXTERNAL HOST OR TDSP REGISTER

Address: 0x4000C165, Reset: 0x00, Name: SW_INT

Table 387. Bit Descriptions for SW_INT

Bits	Bit Name	Settings	Description	Reset	Access
[7:4]	RESERVED		Reserved.	0x0	R
3	SW_INT3		Tensilica DSP Trigger Interrupt 3.	0x0	W
		0	Writing of 0 has no effect.		
		1	Writing of 1 will trigger software interrupt.		
2	SW_INT2		Tensilica DSP Trigger Interrupt 2.	0x0	W
		0	Writing of 0 has no effect.		
		1	Writing of 1 will trigger software interrupt.		
1	SW_INT1		Tensilica DSP Trigger Interrupt 1.	0x0	W
		0	Writing of 0 has no effect.		
		1	Writing of 1 will trigger software interrupt.		
0	SW_INT0		Tensilica DSP Trigger Interrupt 0.	0x0	W
		0	Writing of 0 has no effect.		
		1	Writing of 1 will trigger software interrupt.		

MP IRQ CLEARING REGISTER

Address: 0x4000C168, Reset: 0x00, Name: MP_INPUT_IRQ_CTRL1

Table 388. Bit Descriptions for MP_INPUT_IRQ_CTRL1

Bits	Bit Name	Settings	Description	Reset	Access
[7:3]	RESERVED		Reserved.	0x0	R
2	MP_INPUT_IRQ3_CLEAR		Write once to clear MP IRQ 3.	0x0	R/W1T
		0	Not Clear.		
		1	Write once to clear IRQ3.		
1	MP_INPUT_IRQ2_CLEAR		Write once to clear MP IRQ 2.	0x0	R/W1T
		0	Not Clear.		
		1	Write once to clear IRQ2.		
0	MP_INPUT_IRQ1_CLEAR		Write once to clear MP IRQ 1.	0x0	R/W1T
		0	Not Clear.		
		1	Write 1 to clear IRQ1.		

MP IRQ CLEARING REGISTER

Address: 0x4000C169, Reset: 0x00, Name: MP_INPUT_IRQ_CTRL2

Table 389. Bit Descriptions for MP_INPUT_IRQ_CTRL2

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
[5:4]	MP_INPUT_IRQ3_MODE		MP IRQ3 Trigger Mode.	0x0	R/W
		00	Low Active.		
		01	High Active.		
		10	Rising Edge.		
		11	Falling Edge.		

Bits	Bit Name	Settings	Description	Reset	Access
[3:2]	MP_INPUT_IRQ2_MODE		MP IRQ2 Trigger Mode.	0x0	R/W
		00	Low Active.		
		01	High Active.		
		10	Rising Edge.		
		11	Falling Edge.		
[1:0]	MP_INPUT_IRQ1_MODE		MP IRQ1 Trigger Mode.	0x0	R/W
		00	Low Active.		
		01	High Active.		
		10	Rising Edge.		
		11	Falling Edge.		

MP IRQ1 MASKING REGISTER

Address: 0x4000C16A, Reset: 0xFF, Name: MP_INPUT_IRQ1_MASK1

Table 390. Bit Descriptions for MP_INPUT_IRQ1_MASK1

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	MP_INPUT_IRQ1_MASK[7:0]		Mask Multipurpose IRQ input (IRQ_IN_0 to IRQ_IN_25) Transition to MP_INPUT_IRQ1_TDSP.	0xFF	R/W

MP IRQ1 MASKING REGISTER

Address: 0x4000C16B, Reset: 0xFF, Name: MP_INPUT_IRQ1_MASK2

Table 391. Bit Descriptions for MP_INPUT_IRQ1_MASK2

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	MP_INPUT_IRQ1_MASK[15:8]		Mask Multipurpose IRQ input (IRQ_IN_0 to IRQ_IN_25) Transition to MP_INPUT_IRQ1_TDSP.	0xFF	R/W

MP IRQ1 MASKING REGISTER

Address: 0x4000C16C, Reset: 0xFF, Name: MP_INPUT_IRQ1_MASK3

Table 392. Bit Descriptions for MP_INPUT_IRQ1_MASK3

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	MP_INPUT_IRQ1_MASK[23:16]		Mask Multipurpose IRQ input (IRQ_IN_0 to IRQ_IN_25) Transition to MP_INPUT_IRQ1_TDSP.	0xFF	R/W

MP IRQ1 MASKING REGISTER

Address: 0x4000C16D, Reset: 0x03, Name: MP_INPUT_IRQ1_MASK4

Table 393. Bit Descriptions for MP_INPUT_IRQ1_MASK4

Bits	Bit Name	Settings	Description	Reset	Access
[7:2]	RESERVED		Reserved.	0x0	R
[1:0]	MP_INPUT_IRQ1_MASK[25:24]		Mask Multipurpose IRQ input (IRQ_IN_0 to IRQ_IN_25) Transition to MP_INPUT_IRQ1_TDSP.	0x3	R/W

MP IRQ1 MASKING REGISTER

Address: 0x4000C16E, Reset: 0xFF, Name: MP_INPUT_IRQ2_MASK1

Table 394. Bit Descriptions for MP_INPUT_IRQ2_MASK1

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	MP_INPUT_IRQ2_MASK[7:0]		Mask Multipurpose IRQ input (IRQ_IN_0 to IRQ_IN_25) Transition to MP_INPUT_IRQ1_TDSP.	0xFF	R/W

MP IRQ1 MASKING REGISTER

Address: 0x4000C16F, Reset: 0xFF, Name: MP_INPUT_IRQ2_MASK2

Table 395. Bit Descriptions for MP_INPUT_IRQ2_MASK2

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	MP_INPUT_IRQ2_MASK[15:8]		Mask Multipurpose IRQ input (IRQ_IN_0 to IRQ_IN_25) Transition to MP_INPUT_IRQ1_TDSP.	0xFF	R/W

MP IRQ1 MASKING REGISTER

Address: 0x4000C170, Reset: 0xFF, Name: MP_INPUT_IRQ2_MASK3

Table 396. Bit Descriptions for MP_INPUT_IRQ2_MASK3

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	MP_INPUT_IRQ2_MASK[23:16]		Mask Multipurpose IRQ input (IRQ_IN_0 to IRQ_IN_25) Transition to MP_INPUT_IRQ1_TDSP.	0xFF	R/W

MP IRQ1 MASKING REGISTER

Address: 0x4000C171, Reset: 0x03, Name: MP_INPUT_IRQ2_MASK4

Table 397. Bit Descriptions for MP_INPUT_IRQ2_MASK4

Bits	Bit Name	Settings	Description	Reset	Access
[7:2]	RESERVED		Reserved.	0x0	R
[1:0]	MP_INPUT_IRQ2_MASK[25:24]		Mask Multipurpose IRQ input (IRQ_IN_0 to IRQ_IN_25) Transition to MP_INPUT_IRQ1_TDSP.	0x3	R/W

MP IRQ1 MASKING REGISTER

Address: 0x4000C172, Reset: 0xFF, Name: MP_INPUT_IRQ3_MASK1

Table 398. Bit Descriptions for MP_INPUT_IRQ3_MASK1

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	MP_INPUT_IRQ3_MASK[7:0]		Mask Multipurpose IRQ input (IRQ_IN_0 to IRQ_IN_25) Transition to MP_INPUT_IRQ3_TDSP.	0xFF	R/W

MP IRQ1 MASKING REGISTER

Address: 0x4000C173, Reset: 0xFF, Name: MP_INPUT_IRQ3_MASK2

Table 399. Bit Descriptions for MP_INPUT_IRQ3_MASK2

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	MP_INPUT_IRQ3_MASK[15:8]		Mask Multipurpose IRQ input (IRQ_IN_0 to IRQ_IN_25) Transition to MP_INPUT_IRQ3_TDSP.	0xFF	R/W

MP IRQ1 MASKING REGISTER

Address: 0x4000C174, Reset: 0xFF, Name: MP_INPUT_IRQ3_MASK3

Table 400. Bit Descriptions for MP_INPUT_IRQ3_MASK3

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	MP_INPUT_IRQ3_MASK[23:16]		Mask Multipurpose IRQ input (IRQ_IN_0 to IRQ_IN_25) Transition to MP_INPUT_IRQ3_TDSP.	0xFF	R/W

MP IRQ1 MASKING REGISTER

Address: 0x4000C175, Reset: 0x03, Name: MP_INPUT_IRQ3_MASK4

Table 401. Bit Descriptions for MP_INPUT_IRQ3_MASK4

Bits	Bit Name	Settings	Description	Reset	Access
[7:2]	RESERVED		Reserved.	0x0	R
[1:0]	MP_INPUT_IRQ3_MASK[25:24]		Mask Multipurpose IRQ input (IRQ_IN_0 to IRQ_IN_25) Transition to MP_INPUT_IRQ3_TDSP.	0x3	R/W

FASTDSP WORKING MODE REGISTER

Address: 0x4000C1D0, Reset: 0x00, Name: FDSP_MODE

Table 402. Bit Descriptions for FDSP_MODE

Bits	Bit Name	Settings	Description	Reset	Access
[7:1]	RESERVED		Reserved.	0x0	R
0	FDSP_MODE		FastDSP Working Mode.	0x0	R/W
		0	Normal Mode, FDSP Maximum Support 64 Instructions.		
		1	Turbo Mode, FDSP Maximum Support 96 Instructions.		

CHIP RESETS REGISTER

Address: 0x4000C200, Reset: 0x00, Name: RESETS

Table 403. Bit Descriptions for RESETS

Bits	Bit Name	Settings	Description	Reset	Access
[7:5]	RESERVED		Reserved.	0x0	R
4	SOFT_RESET		Software reset not including register settigns.	0x0	W
		0	N/A.		
		1	Write once to soft reset.		
[3:1]	RESERVED		Reserved.	0x0	R
0	SOFT_FULL_RESET		Software reset of entire IC.	0x0	W
		0	N/A.		
		1	Write once to soft full reset.		

FASTDSP CURRENT LAMBDA REGISTER

Address: 0x4000C400, Reset: 0x3F, Name: READ_LAMBDA

Table 404. Bit Descriptions for READ_LAMBDA

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
[5:0]	FDSP_CURRENT_LAMBDA		FastDSP Bank Switch Ramp Current Lambda Status. Lambda is a 6-Bit value representing the point along the linear interpolation curve between two banks at which the bank ramp switch will stop. 0=((63/64)*A + (1/64)*B), 1 =((62/64)*A + (2/64)*B), …, 62 = ((1/64)*A + (63/64)*B), 63=B (default) Lambda can be updated "on the fly" via control interface. In order to complete a bank switch a value of 63 (default setting) must be set. Actual current ramp point (current_lambda : 0-63) can be read via a status register. When this reaches 63 the bank switch is complete and current parameters used will match current bank. Actual step size of linear interpolation will be ~12-bits (4096 steps) Parameters in banks that are being ramped between should not change during a bank switch	0x3F	R
		0	Bank switch parameter ramp is at 1/64 of full ramp.		
		1	Bank switch parameter ramp is at 2/64 of full ramp.		
		…	…		
		62	Bank switch parameter ramp is at 63/64 of full ramp.		
		63	Bank switch parameter ramp is complete.		

CHIP STATUS 1 REGISTER

Address: 0x4000C401, Reset: 0x00, Name: STATUS1

Table 405. Bit Descriptions for STATUS1

Bits	Bit Name	Settings	Description	Reset	Access
[7:5]	RESERVED		Reserved.	0x0	R
4	AVDD_UVW		AVDD Under-Voltage Warning.	0x0	R
		0	Normal Operation.		
		1	Undervoltage on AVDD detected.		
3	ADC2_CLIP		ADC Channel 2 Clip Detector.	0x0	R
		0	Normal Operation.		
		1	Amplifier Clipping Detected.		
2	ADC1_CLIP		ADC Channel 1 Clip Detector.	0x0	R
		0	Normal Operation.		
		1	Amplifier Clipping Detected.		
1	ADC0_CLIP		ADC Channel 0 Clip Detector.	0x0	R
		0	Normal Operation.		
		1	Amplifier Clipping Detected.		
0	DAC0_CLIP		DAC Channel 0 Clip Detector.	0x0	R
		0	Normal Operation.		
		1	Clipping Detected.		

CHIP STATUS 2 REGISTER

Address: 0x4000C402, Reset: 0x00, Name: STATUS2

Table 406. Bit Descriptions for STATUS2

Bits	Bit Name	Settings	Description	Reset	Access
7	POWER_UP_COMPLETE		Status of power domain power up caused by POWER_EN=1.	0x0	R
6	SYNC_LOCK		Read the multi-chip synchronization lock status.	0x0	R
5	SPT1_LOCK		Read the Serial Port 1 lock status.	0x0	R
4	SPT0_LOCK		Read the Serial Port 0 lock status.	0x0	R
3	ASRCO_LOCK		Output ASRCI Lock Status.	0x0	R
		0	ASRC currently unlocked.		
		1	ASRC currently locked.		
2	ASRCI_LOCK		Input ASRCI Lock Status.	0x0	R
		0	ASRC currently unlocked.		
		1	ASRC currently locked.		
1	FM_CLK_READY		Frequency Multiplier Output Clock Ready.	0x0	R
		0	PLL is not locked.		
		1	PLL is locked.		
0	PLL_LOCK		Read the PLL lock status.	0x0	R
		0	PLL is not locked.		
		1	PLL is locked.		

TENSILICA DSP SELF-BOOT INDICATOR REGISTER

Address: 0x4000C403, Reset: 0x00, Name: SELFBOOT_STATUS

Table 407. Bit Descriptions for SELFBOOT_STATUS

Bits	Bit Name	Settings	Description	Reset	Access
[7:2]	RESERVED		Reserved.	0x0	R
1	SELFBOOT_UNPACK_MODE		Self-boot Unpack Mode Indicator.	0x0	R
		0	Flash Code Load Program Code From Flash to RAM.		
		1	ROM Code Load Program Code from Flash to RAM.		
0	SELFBOOT		Self-boot Indicator.	0x0	R
		0			
		1			

EQ STATUS REGISTER

Address: 0x4000C404, Reset: 0x00, Name: EQ_STATUS

Table 408. Bit Descriptions for EQ_STATUS

Bits	Bit Name	Settings	Description	Reset	Access
[7:1]	RESERVED		Reserved.	0x0	R
0	EQ_CLEAR_DONE		EQ clear done.	0x0	R
		1	EQ clear done.		
		0	EQ no clear done.		

FASTDSP ONZ STATUS REGISTER

Address: 0x4000C405, Reset: 0x00, Name: FDSP_STATUS

Table 409. Bit Descriptions for FDSP_STATUS

Bits	Bit Name	Settings	Description	Reset	Access
[7:5]	RESERVED		Reserved.	0x0	R
4	FDSP_BANK_COPY_DONE		FastDSP Bank Switch Ramp Current Lambda Status. FDSP bank copy indication.	0x0	R
		0	Event not happened.		
		1	Event happened.		
[3:1]	RESERVED		Reserved.	0x0	R
0	FDSP_MASK_ONZ		Masked FastDSP Output Not Zero Indication.	0x0	R
		0	Event not happened.		
		1	Event happened.		

POWER MODE STATUS, DLDO SCALE BUSY, CM DELAY COUNTER DONE REGISTER

Address: 0x4000C406, Reset: 0x00, Name: PMU_STATUS1

Table 410. Bit Descriptions for PMU_STATUS1

Bits	Bit Name	Settings	Description	Reset	Access
7	STARTUP_DLYCNT_DONE		Indicate CM Delay Counter Done.	0x0	R
6	RESERVED		Reserved.	0x0	R
5	DLDO_DVS_DONE		Indicate DLDO dynamic voltage scaling is done.	0x0	R
4	DLDO_SCALE_BUSY		Indicate DLDO voltage scaling is on-going.	0x0	R
[3:2]	MEM_VOLT_STA		Status of mem_volt_mode. mem_volt_mode may take some time to take effect, this bit indicates real status of mem_volt_mode	0x0	R
		0	LP Mode.		
		1	MP Mode.		
		10	HP Mode. 2'b11 also mean HP mode.		
[1:0]	POWER_MODE_STA		Power Mode Status.	0x0	R
		0			
		1			
		10			
		11			

MEMORY RETENTION STATUS REGISTER

Address: 0x4000C407, Reset: 0x00, Name: PMU_STATUS2

Table 411. Bit Descriptions for PMU_STATUS2

Bits	Bit Name	Settings	Description	Reset	Access
[7:3]	RESERVED		Reserved.	0x0	R
2	SOC_MEM_DEEPSLEEP		SOC memories hibernate status.	0x0	R
1	RESERVED		Reserved.	0x0	R
0	ADP_MEM_DEEPSLEEP		ADP memories hibernate status.	0x0	R

ADP MEMORY SHUTDOWN STATUS REGISTER

Address: 0x4000C408, Reset: 0x00, Name: PMU_STATUS3

Table 412. Bit Descriptions for PMU_STATUS3

Bits	Bit Name	Settings	Description	Reset	Access
[7:3]	RESERVED		Reserved.	0x0	R
[2:0]	ADP_MEM_SHUTDOWN		ADP memories shutdown status. [0]: FDSP instr/ parameter memory; EQ instr/ param memory [1]: FFSRAM2	0x0	R

SOC MEMORY SHUTDOWN STATUS REGISTER

Address: 0x4000C409, Reset: 0x00, Name: PMU_STATUS4

Table 413. Bit Descriptions for PMU_STATUS4

Bits	Bit Name	Settings	Description	Reset	Access
7	RESERVED		Reserved.	0x0	R
[6:0]	SOC_MEM_SHUTDOWN		SOC memories shutdown status. [0]: IRAM0 [1]: DRAM0 [2]: DRAM1 [3]: SYSRAM [4]: FFSRAM0 [5]: FFSRAM1	0x0	R

TDSP MODE STATUS REGISTER

Address: 0x4000C40A, Reset: 0x00, Name: TDSP_MODE_STATUS

Table 414. Bit Descriptions for TDSP_MODE_STATUS

Bits	Bit Name	Settings	Description	Reset	Access
[7:4]	RESERVED		Reserved.	0x0	R
3	TDSP_DBG_MODE		TDSP Debug Mode Status.	0x0	R
2	TDSP_XOCD_MODE		TDSP XOCD (Xtensa On-Chip-Debug) Mode Status.	0x0	R
1	TDSP_IRAM0_LDST		TDSP IRAM0 Load/Store Status.	0x0	R
0	TDSP_PWAIT_MODE		TDSP PWaitMode Status.	0x0	R

TDSP EXCEPTION/ERROR STATUS REGISTER

Address: 0x4000C40B, Reset: 0x00, Name: TDSP_ERROR_STATUS

Table 415. Bit Descriptions for TDSP_ERROR_STATUS

Bits	Bit Name	Settings	Description	Reset	Access
[7:2]	RESERVED		Reserved.	0x0	R
1	TDSP_PFATAL_ERROR		TDSP PFatalError Status.	0x0	R
0	TDSP_DOUB_EXCP		TDSP Double-Exception Status.	0x0	R

TDSP FAULT INFORMATION REGISTER

Address: 0x4000C40C, Reset: 0x00, Name: TDSP_FAULT_INFO1

Table 416. Bit Descriptions for TDSP_FAULT_INFO1

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	TDSP_PFAULT_INFO[7:0]		TDSP PFatalError Information.	0x0	R

TDSP FAULT INFORMATION REGISTER

Address: 0x4000C40D, Reset: 0x00, Name: TDSP_FAULT_INFO2

Table 417. Bit Descriptions for TDSP_FAULT_INFO2

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	TDSP_PFAULT_INFO[15:8]		TDSP PFatalError Information.	0x0	R

TDSP FAULT INFORMATION REGISTER

Address: 0x4000C40E, Reset: 0x00, Name: TDSP_FAULT_INFO3

Table 418. Bit Descriptions for TDSP_FAULT_INFO3

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	TDSP_PFAULT_INFO[23:16]		TDSP PFatalError Information.	0x0	R

TDSP FAULT INFORMATION REGISTER

Address: 0x4000C40F, Reset: 0x00, Name: TDSP_FAULT_INFO4

Table 419. Bit Descriptions for TDSP_FAULT_INFO4

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	TDSP_PFAULT_INFO[31:24]		TDSP PFatalError Information.	0x0	R

TDSP GENERAL PURPOSE OUTPUT BIT 0 TO BIT 7 REGISTER

Address: 0x4000C410, Reset: 0x00, Name: TDSP_GPO1

Table 420. Bit Descriptions for TDSP_GPO1

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	TDSP_GPO[7:0]			0x0	R

TDSP GENERAL PURPOSE OUTPUT BIT 8 TO BIT 15 REGISTER

Address: 0x4000C411, Reset: 0x00, Name: TDSP_GPO2

Table 421. Bit Descriptions for TDSP_GPO2

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	TDSP_GPO[15:8]			0x0	R

TDSP GENERAL PURPOSE OUTPUT BIT 16 TO BIT 23 REGISTER

Address: 0x4000C412, Reset: 0x00, Name: TDSP_GPO3

Table 422. Bit Descriptions for TDSP_GPO3

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	TDSP_GPO[23:16]			0x0	R

TDSP GENERAL PURPOSE OUTPUT BIT 24 TO BIT 31 REGISTER

Address: 0x4000C413, Reset: 0x00, Name: TDSP_GPO4

Table 423. Bit Descriptions for TDSP_GPO4

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	TDSP_GPO[31:24]			0x0	R

IRQ1 STATUS 1 REGISTER

Address: 0x4000C414, Reset: 0x00, Name: IRQ1_STATUS1

Table 424. Bit Descriptions for IRQ1_STATUS1

Bits	Bit Name	Settings	Description	Reset	Access
7	RESERVED		Reserved.	0x0	R
6	IRQ1_AVDD_UVW		AVDD Undervoltage Warning Detected to IRQ1.	0x0	R
		0	Interrupt was not triggered.		
		1	AVDD undervoltage warning was detected.		
5	IRQ1_PLL_UNLOCKED		PLL Locked to Unlocked Transition Detected to IRQ1.	0x0	R
		0	Interrupt was not triggered by PLL unlock event.		
		1	PLL locked to unlocked transition was detected.		
4	IRQ1_PLL_LOCKED		PLL Unlocked to Locked Transition Detected to IRQ1.	0x0	R
		0	Interrupt was not triggered by PLL lock event.		
		1	PLL unlocked to locked transition was detected.		
3	IRQ1_ADC2_CLIP		ADC Channel 2 Clipping Detected to IRQ1.	0x0	R
		0	Interrupt was not triggered.		
		1	Clipping was detected.		
2	IRQ1_ADC1_CLIP		ADC Channel 1 Clipping Detected to IRQ1.	0x0	R
		0	Interrupt was not triggered.		
		1	Clipping was detected.		
1	IRQ1_ADC0_CLIP		ADC Channel 0 Clipping Detected to IRQ1.	0x0	R
		0	Interrupt was not triggered.		
		1	Clipping was detected.		
0	IRQ1_DAC0_CLIP		DAC Channel 0 Clipping Detected to IRQ1.	0x0	R
		0	Interrupt was not triggered.		
		1	Clipping was detected.		

IRQ1 STATUS 2 REGISTER

Address: 0x4000C415, Reset: 0x00, Name: IRQ1_STATUS2

Table 425. Bit Descriptions for IRQ1_STATUS2

Bits	Bit Name	Settings	Description	Reset	Access
7	IRQ1_ASRCO_UNLOCKED		Output ASRC Locked to Unlocked Transition Detected to IRQ1.	0x0	R
		0	Interrupt was not triggered.		
		1	Locked to unlocked transition was detected.		
6	IRQ1_ASRCO_LOCKED		Output ASRC Unlocked to Locked Transition Detected to IRQ1.	0x0	R
		0	Interrupt was not triggered.		
		1	Unlocked to locked transition was detected.		
5	IRQ1_ASRCI_UNLOCKED		Input ASRC Locked to Unlocked Transition Detected to IRQ1.	0x0	R
		0	Interrupt was not triggered.		
		1	Locked to unlocked transition was detected.		

Bits	Bit Name	Settings	Description	Reset	Access
4	IRQ1_ASRCI_LOCKED		Input ASRC Unlocked to Locked Transition Detected to IRQ1.	0x0	R
		0	Interrupt was not triggered.		
		1	Unlocked to locked transition was detected.		
3	RESERVED		Reserved.	0x0	R
2	IRQ1_EQ_CLEAR_DONE		EQ Memory Clear Done interrupt to IRQ1.	0x0	R
		0	Interrupt was not triggered.		
		1	Interrupt was triggered.		
1	IRQ1_FDSP_MASK_ONZ		FastDSP Masked Output Non Zero interrupt to IRQ1.	0x0	R
		0	Interrupt was not triggered.		
		1	Interrupt was triggered.		
0	IRQ1_PRAMP		Parameter ramp complete interrupt to IRQ1.	0x0	R
		0	Interrupt was not triggered.		
		1	Interrupt was triggered.		

IRQ1 STATUS 3 REGISTER

Address: 0x4000C416, Reset: 0x00, Name: IRQ1_STATUS3

Table 426. Bit Descriptions for IRQ1_STATUS3

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
5	IRQ1_DLDO_DVS_DONE		Dynamic Voltage Scaling Done to IRQ1.	0x0	R
		0	Interrupt was not triggered.		
		1	DLDO Scaling Done was detected.		
4	IRQ1_POWER_UP_COMPLETE		Power Up Not Finished to Completed Transition Detected to IRQ1.	0x0	R
		0	Interrupt was not triggered by Power up complete event.		
		1	Power up complete transition was detected.		
3	IRQ1_SPT1_UNLOCKED		SPT1 Locked to Unlocked Transition to IRQ1.	0x0	R
		0	Interrupt was not triggered.		
		1	Interrupt was triggered.		
2	IRQ1_SPT0_UNLOCKED		SPT0 Locked to Unlocked Transition to IRQ1.	0x0	R
		0	Interrupt was not triggered.		
		1	Interrupt was triggered.		
1	IRQ1_SYNC_UNLOCKED		Sync Locked to Unlocked Transition Detected to IRQ1.	0x0	R
		0	Interrupt was not triggered.		
		1	Interrupt was triggered..		
0	IRQ1_SYNC_LOCKED		Sync Unlocked to Locked Transition Detected to IRQ1.	0x0	R
		0	Interrupt was not triggered.		
		1	Interrupt was triggered.		

IRQ1 STATUS 4 REGISTER

Address: 0x4000C417, Reset: 0x00, Name: IRQ1_STATUS4

Table 427. Bit Descriptions for IRQ1_STATUS4

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
5	IRQ1_FFSRAM2_IRQ		FFSRAM2 Interrupt output to IRQ1.	0x0	R
		0	Interrupt was not triggered.		
		1	Interrupt was triggered.		

Bits	Bit Name	Settings	Description	Reset	Access
4	IRQ1_DATA_SYNC_IRQ		Data Sync Interrupt to IRQ1.	0x0	R
		0	Interrupt was not triggered.		
		1	Interrupt was triggered.		
3	IRQ1_SW_INT3		TDSP Interrupt 3 to IRQ1.	0x0	R
		0	Interrupt was not triggered.		
		1	Interrupt was triggered.		
2	IRQ1_SW_INT2		TDSP Interrupt 2 to IRQ1.	0x0	R
		0	Interrupt was not triggered.		
		1	Interrupt was triggered.		
1	IRQ1_SW_INT1		TDSP Interrupt 1 to IRQ1.	0x0	R
		0	Interrupt was not triggered.		
		1	Interrupt was triggered.		
0	IRQ1_SW_INT0		TDSP Interrupt 0 to IRQ1.	0x0	R
		0	Interrupt was not triggered.		
		1	Interrupt was triggered.		

IRQ1 STATUS 5 REGISTER

Address: 0x4000C418, Reset: 0x00, Name: IRQ1_STATUS5

Table 428. Bit Descriptions for IRQ1_STATUS5

Bits	Bit Name	Settings	Description	Reset	Access
[7:4]	RESERVED		Reserved.	0x0	R
3	IRQ1_WDT_IRQ		Watchdog Timer Interrupt to IRQ1.	0x0	R
		0	Interrupt was not triggered.		
		1	Interrupt was triggered.		
2	IRQ1_AHB_BUSY		AHB BUSY FROM LOW to HIGH Transition Detected to IRQ1.	0x0	R
		0	Interrupt was not triggered.		
		1	Interrupt was triggered.		
1	IRQ1_AHB_ERR		AHB Error FROM LOW to HIGH Transition Detected to IRQ1.	0x0	R
		0	Interrupt was not triggered.		
		1	Interrupt was triggered.		
0	IRQ1_TDSP_ERR		TDSP Error FROM LOW to HIGH Transition Detected to IRQ1.	0x0	R
		0	Interrupt was not triggered.		
		1	Interrupt was triggered.		

IRQ2 STATUS 1 REGISTER

Address: 0x4000C419, Reset: 0x00, Name: IRQ2_STATUS1

Table 429. Bit Descriptions for IRQ2_STATUS1

Bits	Bit Name	Settings	Description	Reset	Access
7	RESERVED		Reserved.	0x0	R
6	IRQ2_AVDD_UVW		AVDD Undervoltage Warning Detected to IRQ2.	0x0	R
		0	Interrupt was not triggered.		
		1	AVDD undervoltage warning was detected.		
5	IRQ2_PLL_UNLOCKED		PLL Locked to Unlocked Transition Detected to IRQ2.	0x0	R
		0	Interrupt was not triggered by PLL unlock event.		
		1	PLL locked to unlocked transition was detected.		

Bits	Bit Name	Settings	Description	Reset	Access
4	IRQ2_PLL_LOCKED		PLL Unlocked to Locked Transition Detected to IRQ2.	0x0	R
		0	Interrupt was not triggered by PLL lock event.		
		1	PLL unlocked to locked transition was detected.		
3	IRQ2_ADC2_CLIP		ADC Channel 2 Clipping Detected to IRQ2.	0x0	R
		0	Interrupt was not triggered.		
		1	Clipping was detected.		
2	IRQ2_ADC1_CLIP		ADC Channel 1 Clipping Detected to IRQ2.	0x0	R
		0	Interrupt was not triggered.		
		1	Clipping was detected.		
1	IRQ2_ADC0_CLIP		ADC Channel 0 Clipping Detected to IRQ2.	0x0	R
		0	Interrupt was not triggered.		
		1	Clipping was detected.		
0	IRQ2_DAC0_CLIP		DAC Channel 0 Clipping Detected to IRQ2.	0x0	R
		0	Interrupt was not triggered.		
		1	Clipping was detected.		

IRQ2 STATUS 2 REGISTER

Address: 0x4000C41A, Reset: 0x00, Name: IRQ2_STATUS2

Table 430. Bit Descriptions for IRQ2_STATUS2

Bits	Bit Name	Settings	Description	Reset	Access
7	IRQ2_ASRCO_UNLOCKED		Output ASRC Locked to Unlocked Transition Detected to IRQ2.	0x0	R
		0	Interrupt was not triggered.		
		1	Locked to unlocked transition was detected.		
6	IRQ2_ASRCO_LOCKED		Output ASRC Unlocked to Locked Transition Detected to IRQ2.	0x0	R
		0	Interrupt was not triggered.		
		1	Unlocked to locked transition was detected.		
5	IRQ2_ASRCI_UNLOCKED		Input ASRC Locked to Unlocked Transition Detected to IRQ2.	0x0	R
		0	Interrupt was not triggered.		
		1	Locked to unlocked transition was detected.		
4	IRQ2_ASRCI_LOCKED		Input ASRC Unlocked to Locked Transition Detected to IRQ2.	0x0	R
		0	Interrupt was not triggered.		
		1	Unlocked to locked transition was detected.		
3	RESERVED		Reserved.	0x0	R
2	IRQ2_EQ_CLEAR_DONE		EQ Memory Clear Done interrupt to IRQ2.	0x0	R
		0	Interrupt was not triggered.		
		1	Interrupt was triggered.		
1	IRQ2_FDSP_MASK_ONZ		FastDSP Masked Output Non Zero interrupt to IRQ2.	0x0	R
		0	Interrupt was not triggered.		
		1	Interrupt was triggered.		
0	IRQ2_PRAMP		Parameter ramp complete interrupt to IRQ2.	0x0	R
		0	Interrupt was not triggered.		
		1	Interrupt was triggered.		

IRQ2 STATUS 3 REGISTER

Address: 0x4000C41B, Reset: 0x00, Name: IRQ2_STATUS3

Table 431. Bit Descriptions for IRQ2_STATUS3

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
5	IRQ2_DLDO_DVS_DONE		Dynamic Voltage Scaling Done to IRQ2.	0x0	R
		0	Interrupt was not triggered.		
		1	DLDO Scaling Done was detected.		
4	IRQ2_POWER_UP_COMPLETE		Power Up Not Finished to Completed Transition Detected to IRQ2.	0x0	R
		0	Interrupt was not triggered by Power up complete event.		
		1	Power up complete transition was detected.		
3	IRQ2_SPT1_UNLOCKED		SPT1 Locked to Unlocked Transition to IRQ2.	0x0	R
		0	Interrupt was not triggered.		
		1	Interrupt was triggered.		
2	IRQ2_SPT0_UNLOCKED		SPT0 Locked to Unlocked Transition to IRQ2.	0x0	R
		0	Interrupt was not triggered.		
		1	Interrupt was triggered.		
1	IRQ2_SYNC_UNLOCKED		Sync Locked to Unlocked Transition Detected to IRQ2.	0x0	R
		0	Interrupt was not triggered.		
		1	Interrupt was triggered..		
0	IRQ2_SYNC_LOCKED		Sync Unlocked to Locked Transition Detected to IRQ2.	0x0	R
		0	Interrupt was not triggered.		
		1	Interrupt was triggered.		

IRQ2 STATUS 4 REGISTER

Address: 0x4000C41C, Reset: 0x00, Name: IRQ2_STATUS4

Table 432. Bit Descriptions for IRQ2_STATUS4

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
5	IRQ2_FFSRAM2_IRQ		FFSRAM2 Interrupt output to IRQ2.	0x0	R
		0	Interrupt was not triggered.		
		1	Interrupt was triggered.		
4	IRQ2_DATA_SYNC_IRQ		Data Sync Interrupt to IRQ2.	0x0	R
		0	Interrupt was not triggered.		
		1	Interrupt was triggered.		
3	IRQ2_SW_INT3		TDSP Interrupt 3 to IRQ2.	0x0	R
		0	Interrupt was not triggered.		
		1	Interrupt was triggered.		
2	IRQ2_SW_INT2		TDSP Interrupt 2 to IRQ2.	0x0	R
		0	Interrupt was not triggered.		
		1	Interrupt was triggered.		
1	IRQ2_SW_INT1		TDSP Interrupt 1 to IRQ2.	0x0	R
		0	Interrupt was not triggered.		
		1	Interrupt was triggered.		
0	IRQ2_SW_INT0		TDSP Interrupt 0 to IRQ2.	0x0	R
		0	Interrupt was not triggered.		
		1	Interrupt was triggered.		

IRQ2 STATUS 5 REGISTER

Address: 0x4000C41D, Reset: 0x00, Name: IRQ2_STATUS5

Table 433. Bit Descriptions for IRQ2_STATUS5

Bits	Bit Name	Settings	Description	Reset	Access
[7:4]	RESERVED		Reserved.	0x0	R
3	IRQ2_WDT_IRQ		Watchdog Timer Interrupt to IRQ2.	0x0	R
		0	Interrupt was not triggered.		
		1	Interrupt was triggered.		
2	IRQ2_AHB_BUSY		AHB BUSY FROM LOW to HIGH Transition Detected to IRQ2.	0x0	R
		0	Interrupt was not triggered.		
		1	Interrupt was triggered.		
1	IRQ2_AHB_ERR		AHB Error FROM LOW to HIGH Transition Detected to IRQ2.	0x0	R
		0	Interrupt was not triggered.		
		1	Interrupt was triggered.		
0	IRQ2_TDSP_ERR		TDSP Error FROM LOW to HIGH Transition Detected to IRQ2.	0x0	R
		0	Interrupt was not triggered.		
		1	Interrupt was triggered.		

IRQ3 STATUS 1 REGISTER

Address: 0x4000C41E, Reset: 0x00, Name: IRQ3_STATUS1

Table 434. Bit Descriptions for IRQ3_STATUS1

Bits	Bit Name	Settings	Description	Reset	Access
7	RESERVED		Reserved.	0x0	R
6	IRQ3_AVDD_UVW		AVDD Undervoltage Warning Detected to IRQ3.	0x0	R
		0	Interrupt was not triggered.		
		1	AVDD undervoltage warning was detected.		
5	IRQ3_PLL_UNLOCKED		PLL Locked to Unlocked Transition Detected to IRQ3.	0x0	R
		0	Interrupt was not triggered by PLL unlock event.		
		1	PLL locked to unlocked transition was detected.		
4	IRQ3_PLL_LOCKED		PLL Unlocked to Locked Transition Detected to IRQ3.	0x0	R
		0	Interrupt was not triggered by PLL lock event.		
		1	PLL unlocked to locked transition was detected.		
3	IRQ3_ADC2_CLIP		ADC Channel 2 Clipping Detected to IRQ3.	0x0	R
		0	Interrupt was not triggered.		
		1	Clipping was detected.		
2	IRQ3_ADC1_CLIP		ADC Channel 1 Clipping Detected to IRQ3.	0x0	R
		0	Interrupt was not triggered.		
		1	Clipping was detected.		
1	IRQ3_ADC0_CLIP		ADC Channel 0 Clipping Detected to IRQ3.	0x0	R
		0	Interrupt was not triggered.		
		1	Clipping was detected.		
0	IRQ3_DAC0_CLIP		DAC Channel 0 Clipping Detected to IRQ3.	0x0	R
		0	Interrupt was not triggered.		
		1	Clipping was detected.		

IRQ3 STATUS 2 REGISTER

Address: 0x4000C41F, Reset: 0x00, Name: IRQ3_STATUS2

Table 435. Bit Descriptions for IRQ3_STATUS2

Bits	Bit Name	Settings	Description	Reset	Access
7	IRQ3_ASRCO_UNLOCKED		Output ASRC Locked to Unlocked Transition Detected to IRQ3.	0x0	R
		0	Interrupt was not triggered.		
		1	Locked to unlocked transition was detected.		
6	IRQ3_ASRCO_LOCKED		Output ASRC Unlocked to Locked Transition Detected to IRQ3.	0x0	R
		0	Interrupt was not triggered.		
		1	Unlocked to locked transition was detected.		
5	IRQ3_ASRCI_UNLOCKED		Input ASRC Locked to Unlocked Transition Detected to IRQ3.	0x0	R
		0	Interrupt was not triggered.		
		1	Locked to unlocked transition was detected.		
4	IRQ3_ASRCI_LOCKED		Input ASRC Unlocked to Locked Transition Detected to IRQ3.	0x0	R
		0	Interrupt was not triggered.		
		1	Unlocked to locked transition was detected.		
3	RESERVED		Reserved.	0x0	R
2	IRQ3_EQ_CLEAR_DONE		EQ Memory Clear Done interrupt to IRQ3.	0x0	R
		0	Interrupt was not triggered.		
		1	Interrupt was triggered.		
1	IRQ3_FDSP_MASK_ONZ		FastDSP Masked Output Non Zero interrupt to IRQ3.	0x0	R
		0	Interrupt was not triggered.		
		1	Interrupt was triggered.		
0	IRQ3_PRAMP		Parameter ramp complete interrupt to IRQ3.	0x0	R
		0	Interrupt was not triggered.		
		1	Interrupt was triggered.		

IRQ3 STATUS 3 REGISTER

Address: 0x4000C420, Reset: 0x00, Name: IRQ3_STATUS3

Table 436. Bit Descriptions for IRQ3_STATUS3

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
5	IRQ3_DLDO_DVS_DONE		Dynamic Voltage Scaling Done to IRQ3.	0x0	R
		0	Interrupt was not triggered.		
		1	DLDO Scaling Done was detected.		
4	IRQ3_POWER_UP_COMPLETE		Power Up Not Finished to Completed Transition Detected to IRQ3.	0x0	R
		0	Interrupt was not triggered by Power up complete event.		
		1	Power up complete transition was detected.		
3	IRQ3_SPT1_UNLOCKED		SPT1 Locked to Unlocked Transition to IRQ3.	0x0	R
		0	Interrupt was not triggered.		
		1	Interrupt was triggered.		
2	IRQ3_SPT0_UNLOCKED		SPT0 Locked to Unlocked Transition to IRQ3.	0x0	R
		0	Interrupt was not triggered.		
		1	Interrupt was triggered.		

Bits	Bit Name	Settings	Description	Reset	Access
1	IRQ3_SYNC_UNLOCKED		Sync Locked to Unlocked Transition Detected to IRQ3.	0x0	R
		0	Interrupt was not triggered.		
		1	Interrupt was triggered..		
0	IRQ3_SYNC_LOCKED		Sync Unlocked to Locked Transition Detected to IRQ3.	0x0	R
		0	Interrupt was not triggered.		
		1	Interrupt was triggered.		

IRQ3 STATUS 4 REGISTER

Address: 0x4000C421, Reset: 0x00, Name: IRQ3_STATUS4

Table 437. Bit Descriptions for IRQ3_STATUS4

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
5	IRQ3_FFSRAM2_IRQ		FFSRAM2 Interrupt output to IRQ3.	0x0	R
		0	Interrupt was not triggered.		
		1	Interrupt was triggered.		
4	IRQ3_DATA_SYNC_IRQ		Data Sync Interrupt to IRQ3.	0x0	R
		0	Interrupt was not triggered.		
		1	Interrupt was triggered.		
3	IRQ3_SW_INT3		TDSP Interrupt 3 to IRQ3.	0x0	R
		0	Interrupt was not triggered.		
		1	Interrupt was triggered.		
2	IRQ3_SW_INT2		TDSP Interrupt 2 to IRQ3.	0x0	R
		0	Interrupt was not triggered.		
		1	Interrupt was triggered.		
1	IRQ3_SW_INT1		TDSP Interrupt 1 to IRQ3.	0x0	R
		0	Interrupt was not triggered.		
		1	Interrupt was triggered.		
0	IRQ3_SW_INT0		TDSP Interrupt 0 to IRQ3.	0x0	R
		0	Interrupt was not triggered.		
		1	Interrupt was triggered.		

IRQ3 STATUS 5 REGISTER

Address: 0x4000C422, Reset: 0x00, Name: IRQ3_STATUS5

Table 438. Bit Descriptions for IRQ3_STATUS5

Bits	Bit Name	Settings	Description	Reset	Access
[7:4]	RESERVED		Reserved.	0x0	R
3	IRQ3_WDT_IRQ		Watchdog Timer Interrupt to IRQ3.	0x0	R
		0	Interrupt was not triggered.		
		1	Interrupt was triggered.		
2	IRQ3_AHB_BUSY		AHB BUSY FROM LOW to HIGH Transition Detected to IRQ3.	0x0	R
		0	Interrupt was not triggered.		
		1	Interrupt was triggered.		

Bits	Bit Name	Settings	Description	Reset	Access
1	IRQ3_AHB_ERR		AHB Error FROM LOW to HIGH Transition Detected to IRQ3.	0x0	R
		0	Interrupt was not triggered.		
		1	Interrupt was triggered.		
0	IRQ3_TDSP_ERR		TDSP Error FROM LOW to HIGH Transition Detected to IRQ3.	0x0	R
		0	Interrupt was not triggered.		
		1	Interrupt was triggered.		

IRQ4 STATUS 1 REGISTER

Address: 0x4000C423, Reset: 0x00, Name: IRQ4_STATUS1

Table 439. Bit Descriptions for IRQ4_STATUS1

Bits	Bit Name	Settings	Description	Reset	Access
7	RESERVED		Reserved.	0x0	R
6	IRQ4_AVDD_UVW		AVDD Undervoltage Warning Detected to IRQ4.	0x0	R
		0	Interrupt was not triggered.		
		1	AVDD undervoltage warning was detected.		
5	IRQ4_PLL_UNLOCKED		PLL Locked to Unlocked Transition Detected to IRQ4.	0x0	R
		0	Interrupt was not triggered by PLL unlock event.		
		1	PLL locked to unlocked transition was detected.		
4	IRQ4_PLL_LOCKED		PLL Unlocked to Locked Transition Detected to IRQ4.	0x0	R
		0	Interrupt was not triggered by PLL lock event.		
		1	PLL unlocked to locked transition was detected.		
3	IRQ4_ADC2_CLIP		ADC Channel 2 Clipping Detected to IRQ4.	0x0	R
		0	Interrupt was not triggered.		
		1	Clipping was detected.		
2	IRQ4_ADC1_CLIP		ADC Channel 1 Clipping Detected to IRQ4.	0x0	R
		0	Interrupt was not triggered.		
		1	Clipping was detected.		
1	IRQ4_ADC0_CLIP		ADC Channel 0 Clipping Detected to IRQ4.	0x0	R
		0	Interrupt was not triggered.		
		1	Clipping was detected.		
0	IRQ4_DAC0_CLIP		DAC Channel 0 Clipping Detected to IRQ4.	0x0	R
		0	Interrupt was not triggered.		
		1	Clipping was detected.		

IRQ4 STATUS 2 REGISTER

Address: 0x4000C424, Reset: 0x00, Name: IRQ4_STATUS2

Table 440. Bit Descriptions for IRQ4_STATUS2

Bits	Bit Name	Settings	Description	Reset	Access
7	IRQ4_ASRCO_UNLOCKED		Output ASRC Locked to Unlocked Transition Detected to IRQ4.	0x0	R
		0	Interrupt was not triggered.		
		1	Locked to unlocked transition was detected.		
6	IRQ4_ASRCO_LOCKED		Output ASRC Unlocked to Locked Transition Detected to IRQ4.	0x0	R
		0	Interrupt was not triggered.		
		1	Unlocked to locked transition was detected.		
5	IRQ4_ASRCI_UNLOCKED		Input ASRC Locked to Unlocked Transition Detected to IRQ4.	0x0	R
		0	Interrupt was not triggered.		
		1	Locked to unlocked transition was detected.		

Bits	Bit Name	Settings	Description	Reset	Access
4	IRQ4_ASRCI_LOCKED		Input ASRC Unlocked to Locked Transition Detected to IRQ4.	0x0	R
		0	Interrupt was not triggered.		
		1	Unlocked to locked transition was detected.		
3	RESERVED		Reserved.	0x0	R
2	IRQ4_EQ_CLEAR_DONE		EQ Memory Clear Done interrupt to IRQ4.	0x0	R
		0	Interrupt was not triggered.		
		1	Interrupt was triggered.		
1	IRQ4_FDSP_MASK_ONZ		FastDSP Masked Output Non Zero interrupt to IRQ4.	0x0	R
		0	Interrupt was not triggered.		
		1	Interrupt was triggered.		
0	IRQ4_PRAMP		Parameter ramp complete interrupt to IRQ4.	0x0	R
		0	Interrupt was not triggered.		
		1	Interrupt was triggered.		

IRQ4 STATUS 3 REGISTER

Address: 0x4000C425, Reset: 0x00, Name: IRQ4_STATUS3

Table 441. Bit Descriptions for IRQ4_STATUS3

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
5	IRQ4_DLDO_DVS_DONE		Dynamic Voltage Scaling Done to IRQ4.	0x0	R
		0	Interrupt was not triggered.		
		1	DLDO Scaling Done was detected.		
4	IRQ4_POWER_UP_COMPLETE		Power Up Not Finished to Completed Transition Detected to IRQ4.	0x0	R
		0	Interrupt was not triggered by Power up complete event.		
		1	Power up complete transition was detected.		
3	IRQ4_SPT1_UNLOCKED		SPT1 Locked to Unlocked Transition to IRQ4.	0x0	R
		0	Interrupt was not triggered.		
		1	Interrupt was triggered.		
2	IRQ4_SPT0_UNLOCKED		SPT0 Locked to Unlocked Transition to IRQ4.	0x0	R
		0	Interrupt was not triggered.		
		1	Interrupt was triggered.		
1	IRQ4_SYNC_UNLOCKED		Sync Locked to Unlocked Transition Detected to IRQ4.	0x0	R
		0	Interrupt was not triggered.		
		1	Interrupt was triggered..		
0	IRQ4_SYNC_LOCKED		Sync Unlocked to Locked Transition Detected to IRQ4.	0x0	R
		0	Interrupt was not triggered.		
		1	Interrupt was triggered.		

IRQ4 STATUS 4 REGISTER

Address: 0x4000C426, Reset: 0x00, Name: IRQ4_STATUS4

Table 442. Bit Descriptions for IRQ4_STATUS4

Bits	Bit Name	Settings	Description	Reset	Access
[7:6]	RESERVED		Reserved.	0x0	R
5	IRQ4_FFSRAM2_IRQ		FFSRAM2 Interrupt output to IRQ4.	0x0	R
		0	Interrupt was not triggered.		
		1	Interrupt was triggered.		

Bits	Bit Name	Settings	Description	Reset	Access
4	IRQ4_DATA_SYNC_IRQ		Data Sync Interrupt to IRQ4.	0x0	R
		0	Interrupt was not triggered.		
		1	Interrupt was triggered.		
3	IRQ4_SW_INT3		TDSP Interrupt 3 to IRQ4.	0x0	R
		0	Interrupt was not triggered.		
		1	Interrupt was triggered.		
2	IRQ4_SW_INT2		TDSP Interrupt 2 to IRQ4.	0x0	R
		0	Interrupt was not triggered.		
		1	Interrupt was triggered.		
1	IRQ4_SW_INT1		TDSP Interrupt 1 to IRQ4.	0x0	R
		0	Interrupt was not triggered.		
		1	Interrupt was triggered.		
0	IRQ4_SW_INT0		TDSP Interrupt 0 to IRQ4.	0x0	R
		0	Interrupt was not triggered.		
		1	Interrupt was triggered.		

IRQ4 STATUS 5 REGISTER

Address: 0x4000C427, Reset: 0x00, Name: IRQ4_STATUS5

Table 443. Bit Descriptions for IRQ4_STATUS5

Bits	Bit Name	Settings	Description	Reset	Access
[7:4]	RESERVED		Reserved.	0x0	R
3	IRQ4_WDT_IRQ		Watchdog Timer Interrupt to IRQ4.	0x0	R
		0	Interrupt was not triggered.		
		1	Interrupt was triggered.		
2	IRQ4_AHB_BUSY		AHB BUSY FROM LOW to HIGH Transition Detected to IRQ4.	0x0	R
		0	Interrupt was not triggered.		
		1	Interrupt was triggered.		
1	IRQ4_AHB_ERR		AHB Error FROM LOW to HIGH Transition Detected to IRQ4.	0x0	R
		0	Interrupt was not triggered.		
		1	Interrupt was triggered.		
0	IRQ4_TDSP_ERR		TDSP Error FROM LOW to HIGH Transition Detected to IRQ4.	0x0	R
		0	Interrupt was not triggered.		
		1	Interrupt was triggered.		

MULTIPURPOSE IRQ1 STATUS 1 REGISTER

Address: 0x4000C428, Reset: 0x00, Name: MP_INPUT_IRQ1_STATUS1

Table 444. Bit Descriptions for MP_INPUT_IRQ1_STATUS1

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	MP_INPUT_IRQ1_STATUS[7:0]		MP IRQ 1 input status.	0x0	R

MULTIPURPOSE IRQ1 STATUS 2 REGISTER

Address: 0x4000C429, Reset: 0x00, Name: MP_INPUT_IRQ1_STATUS2

Table 445. Bit Descriptions for MP_INPUT_IRQ1_STATUS2

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	MP_INPUT_IRQ1_STATUS[15:8]		MP IRQ 1 input status.	0x0	R

MULTIPURPOSE IRQ1 STATUS 3 REGISTER

Address: 0x4000C42A, Reset: 0x00, Name: MP_INPUT_IRQ1_STATUS3

Table 446. Bit Descriptions for MP_INPUT_IRQ1_STATUS3

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	MP_INPUT_IRQ1_STATUS[23:16]		MP IRQ 1 input status.	0x0	R

MULTIPURPOSE IRQ1 STATUS 3 REGISTER

Address: 0x4000C42B, Reset: 0x00, Name: MP_INPUT_IRQ1_STATUS4

Table 447. Bit Descriptions for MP_INPUT_IRQ1_STATUS4

Bits	Bit Name	Settings	Description	Reset	Access
[7:2]	RESERVED		Reserved.	0x0	R
[1:0]	MP_INPUT_IRQ1_STATUS[25:24]		MP IRQ 1 input status.	0x0	R

MULTIPURPOSE IRQ2 STATUS 1 REGISTER

Address: 0x4000C42C, Reset: 0x00, Name: MP_INPUT_IRQ2_STATUS1

Table 448. Bit Descriptions for MP_INPUT_IRQ2_STATUS1

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	MP_INPUT_IRQ2_STATUS[7:0]		MP IRQ 2 input status.	0x0	R

MULTIPURPOSE IRQ2 STATUS 2 REGISTER

Address: 0x4000C42D, Reset: 0x00, Name: MP_INPUT_IRQ2_STATUS2

Table 449. Bit Descriptions for MP_INPUT_IRQ2_STATUS2

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	MP_INPUT_IRQ2_STATUS[15:8]		MP IRQ 2 input status.	0x0	R

MULTIPURPOSE IRQ2 STATUS 3 REGISTER

Address: 0x4000C42E, Reset: 0x00, Name: MP_INPUT_IRQ2_STATUS3

Table 450. Bit Descriptions for MP_INPUT_IRQ2_STATUS3

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	MP_INPUT_IRQ2_STATUS[23:16]		MP IRQ 2 input status.	0x0	R

MULTIPURPOSE IRQ1 STATUS 3 REGISTER

Address: 0x4000C42F, Reset: 0x00, Name: MP_INPUT_IRQ2_STATUS4

Table 451. Bit Descriptions for MP_INPUT_IRQ2_STATUS4

Bits	Bit Name	Settings	Description	Reset	Access
[7:2]	RESERVED		Reserved.	0x0	R
[1:0]	MP_INPUT_IRQ2_STATUS[25:24]		MP IRQ 2 input status.	0x0	R

MULTIPURPOSE IRQ3 STATUS 1 REGISTER

Address: 0x4000C430, Reset: 0x00, Name: MP_INPUT_IRQ3_STATUS1

Table 452. Bit Descriptions for MP_INPUT_IRQ3_STATUS1

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	MP_INPUT_IRQ3_IRQ_STATUS[7:0]		MP IRQ 3 input status.	0x0	R

MULTIPURPOSE IRQ3 STATUS 2 REGISTER

Address: 0x4000C431, Reset: 0x00, Name: MP_INPUT_IRQ3_STATUS2

Table 453. Bit Descriptions for MP_INPUT_IRQ3_STATUS2

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	MP_INPUT_IRQ3_IRQ_STATUS[15:8]		MP IRQ 3 input status.	0x0	R

MULTIPURPOSE IRQ3 STATUS 3 REGISTER

Address: 0x4000C432, Reset: 0x00, Name: MP_INPUT_IRQ3_STATUS3

Table 454. Bit Descriptions for MP_INPUT_IRQ3_STATUS3

Bits	Bit Name	Settings	Description	Reset	Access
[7:0]	MP_INPUT_IRQ3_IRQ_STATUS[23:16]		MP IRQ 3 input status.	0x0	R

MULTIPURPOSE IRQ1 STATUS 3 REGISTER

Address: 0x4000C433, Reset: 0x00, Name: MP_INPUT_IRQ3_STATUS4

Table 455. Bit Descriptions for MP_INPUT_IRQ3_STATUS4

Bits	Bit Name	Settings	Description	Reset	Access
[7:2]	RESERVED		Reserved.	0x0	R
[1:0]	MP_INPUT_IRQ3_IRQ_STATUS[25:24]		MP IRQ 3 input status.	0x0	R

GENERAL-PURPOSE INPUT READ 0 TO 7 REGISTER

Address: 0x4000C434, Reset: 0x00, Name: GPI1

Table 456. Bit Descriptions for GPI1

Bits	Bit Name	Settings	Description	Reset	Access
7	GPIO7_IN		GPIO7 pin input reading.	0x0	R
		0	GPIO7 pin is low.		
		1	GPIO7 pin is high.		
6	GPIO6_IN		GPIO6 pin input reading.	0x0	R
		0	GPIO6 pin is low.		
		1	GPIO6 pin is high.		
5	GPIO5_IN		GPIO5 pin input reading.	0x0	R
		0	GPIO5 pin is low.		
		1	GPIO5 pin is high.		
4	GPIO4_IN		GPIO4 pin input reading.	0x0	R
		0	GPIO4 pin is low.		
		1	GPIO4 pin is high.		

Bits	Bit Name	Settings	Description	Reset	Access
3	GPIO3_IN		GPIO3 pin input reading.	0x0	R
		0	GPIO3 pin is low.		
		1	GPIO3 pin is high.		
2	GPIO2_IN		GPIO2 pin input reading.	0x0	R
		0	GPIO2 pin is low.		
		1	GPIO2 pin is high.		
1	GPIO1_IN		GPIO1 pin input reading.	0x0	R
		0	GPIO1 pin is low.		
		1	GPIO1 pin is high.		
0	GPIO0_IN		GPIO0 pin input reading.	0x0	R
		0	GPIO0 pin is low.		
		1	GPIO0 pin is high.		

GENERAL-PURPOSE INPUT READ 8 TO 15 REGISTER

Address: 0x4000C435, Reset: 0x00, Name: GPI2

Table 457. Bit Descriptions for GPI2

Bits	Bit Name	Settings	Description	Reset	Access
7	GPIO15_IN		GPIO15 pin input reading.	0x0	R
		0	GPIO15 pin is low.		
		1	GPIO15 pin is high.		
6	GPIO14_IN		GPIO14 pin input reading.	0x0	R
		0	GPIO14 pin is low.		
		1	GPIO14 pin is high.		
5	GPIO13_IN		GPIO13 pin input reading.	0x0	R
		0	GPIO13 pin is low.		
		1	GPIO13 pin is high.		
4	GPIO12_IN		GPIO12 pin input reading.	0x0	R
		0	GPIO12 pin is low.		
		1	GPIO12 pin is high.		
3	GPIO11_IN		GPIO11 pin input reading.	0x0	R
		0	GPIO11 pin is low.		
		1	GPIO11 pin is high.		
2	GPIO10_IN		GPIO10 pin input reading.	0x0	R
		0	GPIO10 pin is low.		
		1	GPIO10 pin is high.		
1	GPIO9_IN		GPIO9 pin input reading.	0x0	R
		0	GPIO9 pin is low.		
		1	GPIO9 pin is high.		
0	GPIO8_IN		GPIO8 pin input reading.	0x0	R
		0	GPIO8 pin is low.		
		1	GPIO8 pin is high.		

GENERAL-PURPOSE INPUT READ 16 TO 20 REGISTER

Address: 0x4000C436, Reset: 0x00, Name: GPI3

Table 458. Bit Descriptions for GPI3

Bits	Bit Name	Settings	Description	Reset	Access
7	GPIO23_IN		GPIO23 pin input reading.	0x0	R
		0	GPIO23 pin is low.		
		1	GPIO23 pin is high.		
6	GPIO22_IN		GPIO22 pin input reading.	0x0	R
		0	GPIO22 pin is low.		
		1	GPIO22 pin is high.		
5	GPIO21_IN		GPIO21 pin input reading.	0x0	R
		0	GPIO21 pin is low.		
		1	GPIO21 pin is high.		
4	GPIO20_IN		GPIO20 pin input reading.	0x0	R
		0	GPIO20 pin is low.		
		1	GPIO20 pin is high.		
3	GPIO19_IN		GPIO19 pin input reading.	0x0	R
		0	GPIO19 pin is low.		
		1	GPIO19 pin is high.		
2	GPIO18_IN		GPIO18 pin input reading.	0x0	R
		0	GPIO18 pin is low.		
		1	GPIO18 pin is high.		
1	GPIO17_IN		GPIO17 pin input reading.	0x0	R
		0	GPIO17 pin is low.		
		1	GPIO17 pin is high.		
0	GPIO16_IN		GPIO16 pin input reading.	0x0	R
		0	GPIO16 pin is low.		
		1	GPIO16 pin is high.		

GENERAL-PURPOSE INPUT READ 16 TO 20 REGISTER

Address: 0x4000C437, Reset: 0x00, Name: GPI4

Table 459. Bit Descriptions for GPI4

Bits	Bit Name	Settings	Description	Reset	Access
[7:2]	RESERVED		Reserved.	0x0	R
1	GPIO25_IN		GPIO25 pin input reading.	0x0	R
		0	GPIO25 pin is low.		
		1	GPIO25 pin is high.		
0	GPIO24_IN		GPIO24 pin input reading.	0x0	R
		0	GPIO24 pin is low.		
		1	GPIO24 pin is high.		

CONTROL PORT MODE REGISTER

Address: 0x4000C438, Reset: 0x00, Name: CTRL_PORT_MODE

Table 460. Bit Descriptions for CTRL_PORT_MODE

Bits	Bit Name	Settings	Description	Reset	Access
[7:2]	RESERVED		Reserved.	0x0	R
[1:0]	CTRL_PORT_MODE		Control Port Mode.	0x0	R
		0	Control Interface Operates at I2C Mode.		
		1	Control Interface Operates at SPI Mode.		
		10	Control Interface Operates at UART Mode.		

DAC NOISE CONTROL 2 REGISTER

Address: 0x4000CC04, Reset: 0x00, Name: DAC_NOISE_CTRL2

Table 461. Bit Descriptions for DAC_NOISE_CTRL2

Bits	Bit Name	Settings	Description	Reset	Access
[7:2]	RESERVED		Reserved.	0x0	R/W
1	DAC_LOW_NOISE_EN2		Should work with DAC_NOISE_CTRL1 to achieve the lowest DAC noise.	0x0	R/W
		0	Normal DAC Noise.		
		1	Lower DAC Noise.		
0	RESERVED		Reserved.	0x0	R/W

DAC NOISE CONTROL 1 REGISTER

Address: 0x4000CC12, Reset: 0x00, Name: DAC_NOISE_CTRL1

Table 462. Bit Descriptions for DAC_NOISE_CTRL1

Bits	Bit Name	Settings	Description	Reset	Access
[7:5]	RESERVED		Reserved.	0x0	R
4	DAC_LOW_NOISE_EN1		Should work with DAC_LOW_NOISE_EN2 to achieve the lowest DAC noise.	0x0	R/W
		0	Normal DAC Noise.		
		1	Lower DAC Noise.		
[3:0]	RESERVED		Reserved.	0x0	R/W

REGISTER SUMMARY: FIFOSRAM0_CONFIGURATION

Table 463. FFSRAM0_CFG (FIFO_SRAM) Register Summary

Address	Name	Description	Reset	Access
0x40010000	FF0_PORT		0x00000000	R/W
0x40010004	FF0_INIT		0x00000000	R/W
0x40010008	FF0_CSR		0x00000000	R/W
0x4001000C	FF0_PTR		0x00000000	R
0x40010010	FF0_INT		0x00000001	R/W
0x40010014	FF0_TH		0x00000000	R/W
0x40010020	FF1_PORT		0x00000000	R/W
0x40010024	FF1_INIT		0x00000000	R/W
0x40010028	FF1_CSR		0x00000000	R/W
0x4001002C	FF1_PTR		0x00000000	R
0x40010030	FF1_INT		0x00000001	R/W
0x40010034	FF1_TH		0x00000000	R/W
0x40010040	FF2_PORT		0x00000000	R/W
0x40010044	FF2_INIT		0x00000000	R/W
0x40010048	FF2_CSR		0x00000000	R/W
0x4001004C	FF2_PTR		0x00000000	R
0x40010050	FF2_INT		0x00000001	R/W
0x40010054	FF2_TH		0x00000000	R/W
0x40010060	FF3_PORT		0x00000000	R/W
0x40010064	FF3_INIT		0x00000000	R/W
0x40010068	FF3_CSR		0x00000000	R/W
0x4001006C	FF3_PTR		0x00000000	R
0x40010070	FF3_INT		0x00000001	R/W
0x40010074	FF3_TH		0x00000000	R/W

REGISTER DETAILS: FIFOSRAM0_CONFIGURATION

Address: 0x40010000, Reset: 0x00000000, Name: FF0_PORT

Table 464. Bit Descriptions for FF0_PORT

Bits	Bit Name	Settings	Description	Reset	Access
[31:0]	FIFO0_PORT			0x0	R/W

Address: 0x40010004, Reset: 0x00000000, Name: FF0_INIT

Table 465. Bit Descriptions for FF0_INIT

Bits	Bit Name	Settings	Description	Reset	Access
[31:17]	RESERVED		Reserved.	0x0	R
16	FIFO0_WIDTH			0x0	R/W
[15:8]	FIFO0_LEN			0x0	R/W
[7:0]	FIFO0_BASE_ADDR			0x0	R/W

Address: 0x40010008, Reset: 0x00000000, Name: FF0_CSR

Table 466. Bit Descriptions for FF0_CSR

Bits	Bit Name	Settings	Description	Reset	Access
[31:19]	RESERVED		Reserved.	0x0	R
18	FIFO0_DLINE_EN			0x0	R/W
17	FIFO0_EN			0x0	R/W
16	FIFO0_FLUSH			0x0	W
[15:0]	FIFO0_USED			0x0	R

Address: 0x4001000C, Reset: 0x00000000, Name: FF0_PTR

Table 467. Bit Descriptions for FF0_PTR

Bits	Bit Name	Settings	Description	Reset	Access
[31:16]	FIFO0_RD_PTR			0x0	R
[15:0]	FIFO0_WR_PTR			0x0	R

Address: 0x40010010, Reset: 0x00000001, Name: FF0_INT

Table 468. Bit Descriptions for FF0_INT

Bits	Bit Name	Settings	Description	Reset	Access
[31:22]	RESERVED		Reserved.	0x0	R
21	FIFO0_OV_CLR			0x0	W
20	FIFO0_UD_CLR			0x0	W
19	FIFO0_ALMF_CLR			0x0	W
18	FIFO0_ALME_CLR			0x0	W
17	FIFO0_FULL_CLR			0x0	W
16	FIFO0_EMPTY_CLR			0x0	W
[15:14]	RESERVED		Reserved.	0x0	R
13	FIFO0_OV_EN			0x0	R/W
12	FIFO0_UD_EN			0x0	R/W
11	FIFO0_ALMF_EN			0x0	R/W

Bits	Bit Name	Settings	Description	Reset	Access
10	FIFO0_ALME_EN			0x0	R/W
9	FIFO0_FULL_EN			0x0	R/W
8	FIFO0_EMPTY_EN			0x0	R/W
[7:6]	RESERVED		Reserved.	0x0	R
5	FIFO0_OV_STATUS			0x0	R
4	FIFO0_UD_STATUS			0x0	R
3	FIFO0_ALMF_STATUS			0x0	R
2	FIFO0_ALME_STATUS			0x0	R
1	FIFO0_FULL_STATUS			0x0	R
0	FIFO0_EMPTY_STATUS			0x1	R

Address: 0x40010014, Reset: 0x00000000, Name: FF0_TH

Table 469. Bit Descriptions for FF0_TH

Bits	Bit Name	Settings	Description	Reset	Access
[31:16]	FIFO0_DLINE_LEN			0x0	R/W
[15:8]	FIFO0_ALME_LEN			0x0	R/W
[7:0]	FIFO0_ALMF_LEN			0x0	R/W

Address: 0x40010020, Reset: 0x00000000, Name: FF1_PORT

Table 470. Bit Descriptions for FF1_PORT

Bits	Bit Name	Settings	Description	Reset	Access
[31:0]	FIFO1_PORT			0x0	R/W

Address: 0x40010024, Reset: 0x00000000, Name: FF1_INIT

Table 471. Bit Descriptions for FF1_INIT

Bits	Bit Name	Settings	Description	Reset	Access
[31:17]	RESERVED		Reserved.	0x0	R
16	FIFO1_WIDTH			0x0	R/W
[15:8]	FIFO1_LEN			0x0	R/W
[7:0]	FIFO1_BASE_ADDR			0x0	R/W

Address: 0x40010028, Reset: 0x00000000, Name: FF1_CSR

Table 472. Bit Descriptions for FF1_CSR

Bits	Bit Name	Settings	Description	Reset	Access
[31:19]	RESERVED		Reserved.	0x0	R
18	FIFO1_DLINE_EN			0x0	R/W
17	FIFO1_EN			0x0	R/W
16	FIFO1_FLUSH			0x0	W
[15:0]	FIFO1_USED			0x0	R

Address: 0x4001002C, Reset: 0x00000000, Name: FF1_PTR

Table 473. Bit Descriptions for FF1_PTR

Bits	Bit Name	Settings	Description	Reset	Access
[31:16]	FIFO1_RD_PTR			0x0	R
[15:0]	FIFO1_WR_PTR			0x0	R

Address: 0x40010030, Reset: 0x00000001, Name: FF1_INT

Table 474. Bit Descriptions for FF1_INT

Bits	Bit Name	Settings	Description	Reset	Access
[31:22]	RESERVED		Reserved.	0x0	R
21	FIFO1_OV_CLR			0x0	W
20	FIFO1_UD_CLR			0x0	W
19	FIFO1_ALMF_CLR			0x0	W
18	FIFO1_ALME_CLR			0x0	W
17	FIFO1_FULL_CLR			0x0	W
16	FIFO1_EMPTY_CLR			0x0	W
[15:14]	RESERVED		Reserved.	0x0	R
13	FIFO1_OV_EN			0x0	R/W
12	FIFO1_UD_EN			0x0	R/W
11	FIFO1_ALMF_EN			0x0	R/W
10	FIFO1_ALME_EN			0x0	R/W
9	FIFO1_FULL_EN			0x0	R/W
8	FIFO1_EMPTY_EN			0x0	R/W
[7:6]	RESERVED		Reserved.	0x0	R
5	FIFO1_OV_STATUS			0x0	R
4	FIFO1_UD_STATUS			0x0	R
3	FIFO1_ALMF_STATUS			0x0	R
2	FIFO1_ALME_STATUS			0x0	R
1	FIFO1_FULL_STATUS			0x0	R
0	FIFO1_EMPTY_STATUS			0x1	R

Address: 0x40010034, Reset: 0x00000000, Name: FF1_TH

Table 475. Bit Descriptions for FF1_TH

Bits	Bit Name	Settings	Description	Reset	Access
[31:16]	FIFO1_DLINE_LEN			0x0	R/W
[15:8]	FIFO1_ALME_LEN			0x0	R/W
[7:0]	FIFO1_ALMF_LEN			0x0	R/W

Address: 0x40010040, Reset: 0x00000000, Name: FF2_PORT

Table 476. Bit Descriptions for FF2_PORT

Bits	Bit Name	Settings	Description	Reset	Access
[31:0]	FIFO2_PORT			0x0	R/W

Address: 0x40010044, Reset: 0x00000000, Name: FF2_INIT

Table 477. Bit Descriptions for FF2_INIT

Bits	Bit Name	Settings	Description	Reset	Access
[31:17]	RESERVED		Reserved.	0x0	R
16	FIFO2_WIDTH			0x0	R/W
[15:8]	FIFO2_LEN			0x0	R/W
[7:0]	FIFO2_BASE_ADDR			0x0	R/W

Address: 0x40010048, Reset: 0x00000000, Name: FF2_CSR

Table 478. Bit Descriptions for FF2_CSR

Bits	Bit Name	Settings	Description	Reset	Access
[31:19]	RESERVED		Reserved.	0x0	R
18	FIFO2_DLINE_EN			0x0	R/W
17	FIFO2_EN			0x0	R/W
16	FIFO2_FLUSH			0x0	W
[15:0]	FIFO2_USED			0x0	R

Address: 0x4001004C, Reset: 0x00000000, Name: FF2_PTR

Table 479. Bit Descriptions for FF2_PTR

Bits	Bit Name	Settings	Description	Reset	Access
[31:16]	FIFO2_RD_PTR			0x0	R
[15:0]	FIFO2_WR_PTR			0x0	R

Address: 0x40010050, Reset: 0x00000001, Name: FF2_INT

Table 480. Bit Descriptions for FF2_INT

Bits	Bit Name	Settings	Description	Reset	Access
[31:22]	RESERVED		Reserved.	0x0	R
21	FIFO2_OV_CLR			0x0	W
20	FIFO2_UD_CLR			0x0	W
19	FIFO2_ALMF_CLR			0x0	W
18	FIFO2_ALME_CLR			0x0	W
17	FIFO2_FULL_CLR			0x0	W
16	FIFO2_EMPTY_CLR			0x0	W
[15:14]	RESERVED		Reserved.	0x0	R
13	FIFO2_OV_EN			0x0	R/W
12	FIFO2_UD_EN			0x0	R/W
11	FIFO2_ALMF_EN			0x0	R/W
10	FIFO2_ALME_EN			0x0	R/W
9	FIFO2_FULL_EN			0x0	R/W
8	FIFO2_EMPTY_EN			0x0	R/W
[7:6]	RESERVED		Reserved.	0x0	R
5	FIFO2_OV_STATUS			0x0	R
4	FIFO2_UD_STATUS			0x0	R
3	FIFO2_ALMF_STATUS			0x0	R

Bits	Bit Name	Settings	Description	Reset	Access
2	FIFO2_ALME_STATUS			0x0	R
1	FIFO2_FULL_STATUS			0x0	R
0	FIFO2_EMPTY_STATUS			0x1	R

Address: 0x40010054, Reset: 0x00000000, Name: FF2_TH

Table 481. Bit Descriptions for FF2_TH

Bits	Bit Name	Settings	Description	Reset	Access
[31:16]	FIFO2_DLINE_LEN			0x0	R/W
[15:8]	FIFO2_ALME_LEN			0x0	R/W
[7:0]	FIFO2_ALMF_LEN			0x0	R/W

Address: 0x40010060, Reset: 0x00000000, Name: FF3_PORT

Table 482. Bit Descriptions for FF3_PORT

Bits	Bit Name	Settings	Description	Reset	Access
[31:0]	FIFO3_PORT			0x0	R/W

Address: 0x40010064, Reset: 0x00000000, Name: FF3_INIT

Table 483. Bit Descriptions for FF3_INIT

Bits	Bit Name	Settings	Description	Reset	Access
[31:17]	RESERVED		Reserved.	0x0	R
16	FIFO3_WIDTH			0x0	R/W
[15:8]	FIFO3_LEN			0x0	R/W
[7:0]	FIFO3_BASE_ADDR			0x0	R/W

Address: 0x40010068, Reset: 0x00000000, Name: FF3_CSR

able 484. Bit Descriptions for FF3_CSR

Bits	Bit Name	Settings	Description	Reset	Access
[31:19]	RESERVED		Reserved.	0x0	R
18	FIFO3_DLINE_EN			0x0	R/W
17	FIFO3_EN			0x0	R/W
16	FIFO3_FLUSH			0x0	W
[15:0]	FIFO3_USED			0x0	R

Address: 0x4001006C, Reset: 0x00000000, Name: FF3_PTR

Table 485. Bit Descriptions for FF3_PTR

Bits	Bit Name	Settings	Description	Reset	Access
[31:16]	FIFO3_RD_PTR			0x0	R
[15:0]	FIFO3_WR_PTR			0x0	R

Address: 0x40010070, Reset: 0x00000001, Name: FF3_INT

Table 486. Bit Descriptions for FF3_INT

Bits	Bit Name	Settings	Description	Reset	Access
[31:22]	RESERVED		Reserved.	0x0	R
21	FIFO3_OV_CLR			0x0	W
20	FIFO3_UD_CLR			0x0	W
19	FIFO3_ALMF_CLR			0x0	W
18	FIFO3_ALME_CLR			0x0	W
17	FIFO3_FULL_CLR			0x0	W
16	FIFO3_EMPTY_CLR			0x0	W
[15:14]	RESERVED		Reserved.	0x0	R
13	FIFO3_OV_EN			0x0	R/W
12	FIFO3_UD_EN			0x0	R/W
11	FIFO3_ALMF_EN			0x0	R/W
10	FIFO3_ALME_EN			0x0	R/W
9	FIFO3_FULL_EN			0x0	R/W
8	FIFO3_EMPTY_EN			0x0	R/W
[7:6]	RESERVED		Reserved.	0x0	R
5	FIFO3_OV_STATUS			0x0	R
4	FIFO3_UD_STATUS			0x0	R
3	FIFO3_ALMF_STATUS			0x0	R
2	FIFO3_ALME_STATUS			0x0	R
1	FIFO3_FULL_STATUS			0x0	R
0	FIFO3_EMPTY_STATUS			0x1	R

Address: 0x40010074, Reset: 0x00000000, Name: FF3_TH

Table 487. Bit Descriptions for FF3_TH

Bits	Bit Name	Settings	Description	Reset	Access
[31:16]	FIFO3_DLINE_LEN			0x0	R/W
[15:8]	FIFO3_ALME_LEN			0x0	R/W
[7:0]	FIFO3_ALMF_LEN			0x0	R/W

REGISTER SUMMARY: FIFOSRAM1_CONFIGURATION

Table 488. FFSRAM1_CFG(FIFO_SRAM) Register Summary

Address	Name	Description	Reset	Access
0x40020000	FF0_PORT		0x00000000	R/W
0x40020004	FF0_INIT		0x00000000	R/W
0x40020008	FF0_CSR		0x00000000	R/W
0x4002000C	FF0_PTR		0x00000000	R
0x40020010	FF0_INT		0x00000001	R/W
0x40020014	FF0_TH		0x00000000	R/W
0x40020020	FF1_PORT		0x00000000	R/W
0x40020024	FF1_INIT		0x00000000	R/W
0x40020028	FF1_CSR		0x00000000	R/W
0x4002002C	FF1_PTR		0x00000000	R
0x40020030	FF1_INT		0x00000001	R/W
0x40020034	FF1_TH		0x00000000	R/W
0x40020040	FF2_PORT		0x00000000	R/W
0x40020044	FF2_INIT		0x00000000	R/W
0x40020048	FF2_CSR		0x00000000	R/W
0x4002004C	FF2_PTR		0x00000000	R
0x40020050	FF2_INT		0x00000001	R/W
0x40020054	FF2_TH		0x00000000	R/W
0x40020060	FF3_PORT		0x00000000	R/W
0x40020064	FF3_INIT		0x00000000	R/W
0x40020068	FF3_CSR		0x00000000	R/W
0x4002006C	FF3_PTR		0x00000000	R
0x40020070	FF3_INT		0x00000001	R/W
0x40020074	FF3_TH		0x00000000	R/W

REGISTER DETAILS: FIFOSRAM1_CONFIGURATION

Address: 0x40020000, Reset: 0x00000000, Name: FF0_PORT

Table 489. Bit Descriptions for FF0_PORT

Bits	Bit Name	Settings	Description	Reset	Access
[31:0]	FIFO0_PORT			0x0	R/W

Address: 0x40020004, Reset: 0x00000000, Name: FF0_INIT

Table 490. Bit Descriptions for FF0_INIT

Bits	Bit Name	Settings	Description	Reset	Access
[31:17]	RESERVED		Reserved.	0x0	R
16	FIFO0_WIDTH			0x0	R/W
[15:8]	FIFO0_LEN			0x0	R/W
[7:0]	FIFO0_BASE_ADDR			0x0	R/W

Address: 0x40020008, Reset: 0x00000000, Name: FF0_CSR

Table 491. Bit Descriptions for FF0_CSR

Bits	Bit Name	Settings	Description	Reset	Access
[31:19]	RESERVED		Reserved.	0x0	R
18	FIFO0_DLINE_EN			0x0	R/W
17	FIFO0_EN			0x0	R/W
16	FIFO0_FLUSH			0x0	W
[15:0]	FIFO0_USED			0x0	R

Address: 0x4002000C, Reset: 0x00000000, Name: FF0_PTR

Table 492. Bit Descriptions for FF0_PTR

Bits	Bit Name	Settings	Description	Reset	Access
[31:16]	FIFO0_RD_PTR			0x0	R
[15:0]	FIFO0_WR_PTR			0x0	R

Address: 0x40020010, Reset: 0x00000001, Name: FF0_INT

Table 493. Bit Descriptions for FF0_INT

Bits	Bit Name	Settings	Description	Reset	Access
[31:22]	RESERVED		Reserved.	0x0	R
21	FIFO0_OV_CLR			0x0	W
20	FIFO0_UD_CLR			0x0	W
19	FIFO0_ALMF_CLR			0x0	W
18	FIFO0_ALME_CLR			0x0	W
17	FIFO0_FULL_CLR			0x0	W
16	FIFO0_EMPTY_CLR			0x0	W
[15:14]	RESERVED		Reserved.	0x0	R
13	FIFO0_OV_EN			0x0	R/W
12	FIFO0_UD_EN			0x0	R/W
11	FIFO0_ALMF_EN			0x0	R/W

Bits	Bit Name	Settings	Description	Reset	Access
10	FIFO0_ALME_EN			0x0	R/W
9	FIFO0_FULL_EN			0x0	R/W
8	FIFO0_EMPTY_EN			0x0	R/W
[7:6]	RESERVED		Reserved.	0x0	R
5	FIFO0_OV_STATUS			0x0	R
4	FIFO0_UD_STATUS			0x0	R
3	FIFO0_ALMF_STATUS			0x0	R
2	FIFO0_ALME_STATUS			0x0	R
1	FIFO0_FULL_STATUS			0x0	R
0	FIFO0_EMPTY_STATUS			0x1	R

Address: 0x40020014, Reset: 0x00000000, Name: FF0_TH

Table 494. Bit Descriptions for FF0_TH

Bits	Bit Name	Settings	Description	Reset	Access
[31:16]	FIFO0_DLINE_LEN			0x0	R/W
[15:8]	FIFO0_ALME_LEN			0x0	R/W
[7:0]	FIFO0_ALMF_LEN			0x0	R/W

Address: 0x40020020, Reset: 0x00000000, Name: FF1_PORT

Table 495. Bit Descriptions for FF1_PORT

Bits	Bit Name	Settings	Description	Reset	Access
[31:0]	FIFO1_PORT			0x0	R/W

Address: 0x40020024, Reset: 0x00000000, Name: FF1_INIT

Table 496. Bit Descriptions for FF1_INIT

Bits	Bit Name	Settings	Description	Reset	Access
[31:17]	RESERVED		Reserved.	0x0	R
16	FIFO1_WIDTH			0x0	R/W
[15:8]	FIFO1_LEN			0x0	R/W
[7:0]	FIFO1_BASE_ADDR			0x0	R/W

Address: 0x40020028, Reset: 0x00000000, Name: FF1_CSR

Table 497. Bit Descriptions for FF1_CSR

Bits	Bit Name	Settings	Description	Reset	Access
[31:19]	RESERVED		Reserved.	0x0	R
18	FIFO1_DLINE_EN			0x0	R/W
17	FIFO1_EN			0x0	R/W
16	FIFO1_FLUSH			0x0	W
[15:0]	FIFO1_USED			0x0	R

Address: 0x4002002C, Reset: 0x00000000, Name: FF1_PTR

Table 498. Bit Descriptions for FF1_PTR

Bits	Bit Name	Settings	Description	Reset	Access
[31:16]	FIFO1_RD_PTR			0x0	R
[15:0]	FIFO1_WR_PTR			0x0	R

Address: 0x40020030, Reset: 0x00000001, Name: FF1_INT

Table 499. Bit Descriptions for FF1_INT

Bits	Bit Name	Settings	Description	Reset	Access
[31:22]	RESERVED		Reserved.	0x0	R
21	FIFO1_OV_CLR			0x0	W
20	FIFO1_UD_CLR			0x0	W
19	FIFO1_ALMF_CLR			0x0	W
18	FIFO1_ALME_CLR			0x0	W
17	FIFO1_FULL_CLR			0x0	W
16	FIFO1_EMPTY_CLR			0x0	W
[15:14]	RESERVED		Reserved.	0x0	R
13	FIFO1_OV_EN			0x0	R/W
12	FIFO1_UD_EN			0x0	R/W
11	FIFO1_ALMF_EN			0x0	R/W
10	FIFO1_ALME_EN			0x0	R/W
9	FIFO1_FULL_EN			0x0	R/W
8	FIFO1_EMPTY_EN			0x0	R/W
[7:6]	RESERVED		Reserved.	0x0	R
5	FIFO1_OV_STATUS			0x0	R
4	FIFO1_UD_STATUS			0x0	R
3	FIFO1_ALMF_STATUS			0x0	R
2	FIFO1_ALME_STATUS			0x0	R
1	FIFO1_FULL_STATUS			0x0	R
0	FIFO1_EMPTY_STATUS			0x1	R

Address: 0x40020034, Reset: 0x00000000, Name: FF1_TH

Table 500. Bit Descriptions for FF1_TH

Bits	Bit Name	Settings	Description	Reset	Access
[31:16]	FIFO1_DLINE_LEN			0x0	R/W
[15:8]	FIFO1_ALME_LEN			0x0	R/W
[7:0]	FIFO1_ALMF_LEN			0x0	R/W

Address: 0x40020040, Reset: 0x00000000, Name: FF2_PORT

Table 501. Bit Descriptions for FF2_PORT

Bits	Bit Name	Settings	Description	Reset	Access
[31:0]	FIFO2_PORT			0x0	R/W

Address: 0x40020044, Reset: 0x00000000, Name: FF2_INIT

Table 502. Bit Descriptions for FF2_INIT

Bits	Bit Name	Settings	Description	Reset	Access
[31:17]	RESERVED		Reserved.	0x0	R
16	FIFO2_WIDTH			0x0	R/W
[15:8]	FIFO2_LEN			0x0	R/W
[7:0]	FIFO2_BASE_ADDR			0x0	R/W

Address: 0x40020048, Reset: 0x00000000, Name: FF2_CSR

Table 503. Bit Descriptions for FF2_CSR

Bits	Bit Name	Settings	Description	Reset	Access
[31:19]	RESERVED		Reserved.	0x0	R
18	FIFO2_DLINE_EN			0x0	R/W
17	FIFO2_EN			0x0	R/W
16	FIFO2_FLUSH			0x0	W
[15:0]	FIFO2_USED			0x0	R

Address: 0x4002004C, Reset: 0x00000000, Name: FF2_PTR

Table 504. Bit Descriptions for FF2_PTR

Bits	Bit Name	Settings	Description	Reset	Access
[31:16]	FIFO2_RD_PTR			0x0	R
[15:0]	FIFO2_WR_PTR			0x0	R

Address: 0x40020050, Reset: 0x00000001, Name: FF2_INT

Table 505. Bit Descriptions for FF2_INT

Bits	Bit Name	Settings	Description	Reset	Access
[31:22]	RESERVED		Reserved.	0x0	R
21	FIFO2_OV_CLR			0x0	W
20	FIFO2_UD_CLR			0x0	W
19	FIFO2_ALMF_CLR			0x0	W
18	FIFO2_ALME_CLR			0x0	W
17	FIFO2_FULL_CLR			0x0	W
16	FIFO2_EMPTY_CLR			0x0	W
[15:14]	RESERVED		Reserved.	0x0	R
13	FIFO2_OV_EN			0x0	R/W
12	FIFO2_UD_EN			0x0	R/W
11	FIFO2_ALMF_EN			0x0	R/W
10	FIFO2_ALME_EN			0x0	R/W
9	FIFO2_FULL_EN			0x0	R/W
8	FIFO2_EMPTY_EN			0x0	R/W
[7:6]	RESERVED		Reserved.	0x0	R
5	FIFO2_OV_STATUS			0x0	R
4	FIFO2_UD_STATUS			0x0	R
3	FIFO2_ALMF_STATUS			0x0	R

Bits	Bit Name	Settings	Description	Reset	Access
2	FIFO2_ALME_STATUS			0x0	R
1	FIFO2_FULL_STATUS			0x0	R
0	FIFO2_EMPTY_STATUS			0x1	R

Address: 0x40020054, Reset: 0x00000000, Name: FF2_TH

Table 506. Bit Descriptions for FF2_TH

Bits	Bit Name	Settings	Description	Reset	Access
[31:16]	FIFO2_DLINE_LEN			0x0	R/W
[15:8]	FIFO2_ALME_LEN			0x0	R/W
[7:0]	FIFO2_ALMF_LEN			0x0	R/W

Address: 0x40020060, Reset: 0x00000000, Name: FF3_PORT

Table 507. Bit Descriptions for FF3_PORT

Bits	Bit Name	Settings	Description	Reset	Access
[31:0]	FIFO3_PORT			0x0	R/W

Address: 0x40020064, Reset: 0x00000000, Name: FF3_INIT

Table 508. Bit Descriptions for FF3_INIT

Bits	Bit Name	Settings	Description	Reset	Access
[31:17]	RESERVED		Reserved.	0x0	R
16	FIFO3_WIDTH			0x0	R/W
[15:8]	FIFO3_LEN			0x0	R/W
[7:0]	FIFO3_BASE_ADDR			0x0	R/W

Address: 0x40020068, Reset: 0x00000000, Name: FF3_CSR

Table 509. Bit Descriptions for FF3_CSR

Bits	Bit Name	Settings	Description	Reset	Access
[31:19]	RESERVED		Reserved.	0x0	R
18	FIFO3_DLINE_EN			0x0	R/W
17	FIFO3_EN			0x0	R/W
16	FIFO3_FLUSH			0x0	W
[15:0]	FIFO3_USED			0x0	R

Address: 0x4002006C, Reset: 0x00000000, Name: FF3_PTR

Table 510. Bit Descriptions for FF3_PTR

Bits	Bit Name	Settings	Description	Reset	Access
[31:16]	FIFO3_RD_PTR			0x0	R
[15:0]	FIFO3_WR_PTR			0x0	R

Address: 0x40020070, Reset: 0x00000001, Name: FF3_INT

Table 511. Bit Descriptions for FF3_INT

Bits	Bit Name	Settings	Description	Reset	Access
[31:22]	RESERVED		Reserved.	0x0	R
21	FIFO3_OV_CLR			0x0	W
20	FIFO3_UD_CLR			0x0	W
19	FIFO3_ALMF_CLR			0x0	W
18	FIFO3_ALME_CLR			0x0	W
17	FIFO3_FULL_CLR			0x0	W
16	FIFO3_EMPTY_CLR			0x0	W
[15:14]	RESERVED		Reserved.	0x0	R
13	FIFO3_OV_EN			0x0	R/W
12	FIFO3_UD_EN			0x0	R/W
11	FIFO3_ALMF_EN			0x0	R/W
10	FIFO3_ALME_EN			0x0	R/W
9	FIFO3_FULL_EN			0x0	R/W
8	FIFO3_EMPTY_EN			0x0	R/W
[7:6]	RESERVED		Reserved.	0x0	R
5	FIFO3_OV_STATUS			0x0	R
4	FIFO3_UD_STATUS			0x0	R
3	FIFO3_ALMF_STATUS			0x0	R
2	FIFO3_ALME_STATUS			0x0	R
1	FIFO3_FULL_STATUS			0x0	R
0	FIFO3_EMPTY_STATUS			0x1	R

Address: 0x40020074, Reset: 0x00000000, Name: FF3_TH

Table 512. Bit Descriptions for FF3_TH

Bits	Bit Name	Settings	Description	Reset	Access
[31:16]	FIFO3_DLINE_LEN			0x0	R/W
[15:8]	FIFO3_ALME_LEN			0x0	R/W
[7:0]	FIFO3_ALMF_LEN			0x0	R/W

REGISTER SUMMARY: FIFOSRAM2_CONFIGURATION

Table 513. FFSRAM2_CFG (FIFO_SRAM) Register Summary

Address	Name	Description	Reset	Access
0x40030000	FF0_PORT		0x00000000	R/W
0x40030004	FF0_INIT		0x00000000	R/W
0x40030008	FF0_CSR		0x00000000	R/W
0x4003000C	FF0_PTR		0x00000000	R
0x40030010	FF0_INT		0x00000001	R/W
0x40030014	FF0_TH		0x00000000	R/W
0x40030020	FF1_PORT		0x00000000	R/W
0x40030024	FF1_INIT		0x00000000	R/W
0x40030028	FF1_CSR		0x00000000	R/W
0x4003002C	FF1_PTR		0x00000000	R
0x40030030	FF1_INT		0x00000001	R/W
0x40030034	FF1_TH		0x00000000	R/W
0x40030040	FF2_PORT		0x00000000	R/W
0x40030044	FF2_INIT		0x00000000	R/W
0x40030048	FF2_CSR		0x00000000	R/W
0x4003004C	FF2_PTR		0x00000000	R
0x40030050	FF2_INT		0x00000001	R/W
0x40030054	FF2_TH		0x00000000	R/W
0x40030060	FF3_PORT		0x00000000	R/W
0x40030064	FF3_INIT		0x00000000	R/W
0x40030068	FF3_CSR		0x00000000	R/W
0x4003006C	FF3_PTR		0x00000000	R
0x40030070	FF3_INT		0x00000001	R/W
0x40030074	FF3_TH		0x00000000	R/W

REGISTER DETAILS: FIFOSRAM2_CONFIGURATION

Address: 0x40030000, Reset: 0x00000000, Name: FF0_PORT

Table 514. Bit Descriptions for FF0_PORT

Bits	Bit Name	Settings	Description	Reset	Access
[31:0]	FIFO0_PORT			0x0	R/W

Address: 0x40030004, Reset: 0x00000000, Name: FF0_INIT

Table 515. Bit Descriptions for FF0_INIT

Bits	Bit Name	Settings	Description	Reset	Access
[31:17]	RESERVED		Reserved.	0x0	R
16	FIFO0_WIDTH			0x0	R/W
[15:8]	FIFO0_LEN			0x0	R/W
[7:0]	FIFO0_BASE_ADDR			0x0	R/W

Address: 0x40030008, Reset: 0x00000000, Name: FF0_CSR

Table 516. Bit Descriptions for FF0_CSR

Bits	Bit Name	Settings	Description	Reset	Access
[31:19]	RESERVED		Reserved.	0x0	R
18	FIFO0_DLINE_EN			0x0	R/W
17	FIFO0_EN			0x0	R/W
16	FIFO0_FLUSH			0x0	W
[15:0]	FIFO0_USED			0x0	R

Address: 0x4003000C, Reset: 0x00000000, Name: FF0_PTR

Table 517. Bit Descriptions for FF0_PTR

Bits	Bit Name	Settings	Description	Reset	Access
[31:16]	FIFO0_RD_PTR			0x0	R
[15:0]	FIFO0_WR_PTR			0x0	R

Address: 0x40030010, Reset: 0x00000001, Name: FF0_INT

Table 518. Bit Descriptions for FF0_INT

Bits	Bit Name	Settings	Description	Reset	Access
[31:22]	RESERVED		Reserved.	0x0	R
21	FIFO0_OV_CLR			0x0	W
20	FIFO0_UD_CLR			0x0	W
19	FIFO0_ALMF_CLR			0x0	W
18	FIFO0_ALME_CLR			0x0	W
17	FIFO0_FULL_CLR			0x0	W
16	FIFO0_EMPTY_CLR			0x0	W
[15:14]	RESERVED		Reserved.	0x0	R
13	FIFO0_OV_EN			0x0	R/W
12	FIFO0_UD_EN			0x0	R/W
11	FIFO0_ALMF_EN			0x0	R/W

Bits	Bit Name	Settings	Description	Reset	Access
10	FIFO0_ALME_EN			0x0	R/W
9	FIFO0_FULL_EN			0x0	R/W
8	FIFO0_EMPTY_EN			0x0	R/W
[7:6]	RESERVED		Reserved.	0x0	R
5	FIFO0_OV_STATUS			0x0	R
4	FIFO0_UD_STATUS			0x0	R
3	FIFO0_ALMF_STATUS			0x0	R
2	FIFO0_ALME_STATUS			0x0	R
1	FIFO0_FULL_STATUS			0x0	R
0	FIFO0_EMPTY_STATUS			0x1	R

Address: 0x40030014, Reset: 0x00000000, Name: FF0_TH

Table 519. Bit Descriptions for FF0_TH

Bits	Bit Name	Settings	Description	Reset	Access
[31:16]	FIFO0_DLINE_LEN			0x0	R/W
[15:8]	FIFO0_ALME_LEN			0x0	R/W
[7:0]	FIFO0_ALMF_LEN			0x0	R/W

Address: 0x40030020, Reset: 0x00000000, Name: FF1_PORT

Table 520. Bit Descriptions for FF1_PORT

Bits	Bit Name	Settings	Description	Reset	Access
[31:0]	FIFO1_PORT			0x0	R/W

Address: 0x40030024, Reset: 0x00000000, Name: FF1_INIT

Table 521. Bit Descriptions for FF1_INIT

Bits	Bit Name	Settings	Description	Reset	Access
[31:17]	RESERVED		Reserved.	0x0	R
16	FIFO1_WIDTH			0x0	R/W
[15:8]	FIFO1_LEN			0x0	R/W
[7:0]	FIFO1_BASE_ADDR			0x0	R/W

Address: 0x40030028, Reset: 0x00000000, Name: FF1_CSR

Table 522. Bit Descriptions for FF1_CSR

Bits	Bit Name	Settings	Description	Reset	Access
[31:19]	RESERVED		Reserved.	0x0	R
18	FIFO1_DLINE_EN			0x0	R/W
17	FIFO1_EN			0x0	R/W
16	FIFO1_FLUSH			0x0	W
[15:0]	FIFO1_USED			0x0	R

Address: 0x4003002C, Reset: 0x00000000, Name: FF1_PTR

Table 523. Bit Descriptions for FF1_PTR

Bits	Bit Name	Settings	Description	Reset	Access
[31:16]	FIFO1_RD_PTR			0x0	R
[15:0]	FIFO1_WR_PTR			0x0	R

Address: 0x40030030, Reset: 0x00000001, Name: FF1_INT

Table 524. Bit Descriptions for FF1_INT

Bits	Bit Name	Settings	Description	Reset	Access
[31:22]	RESERVED		Reserved.	0x0	R
21	FIFO1_OV_CLR			0x0	W
20	FIFO1_UD_CLR			0x0	W
19	FIFO1_ALMF_CLR			0x0	W
18	FIFO1_ALME_CLR			0x0	W
17	FIFO1_FULL_CLR			0x0	W
16	FIFO1_EMPTY_CLR			0x0	W
[15:14]	RESERVED		Reserved.	0x0	R
13	FIFO1_OV_EN			0x0	R/W
12	FIFO1_UD_EN			0x0	R/W
11	FIFO1_ALMF_EN			0x0	R/W
10	FIFO1_ALME_EN			0x0	R/W
9	FIFO1_FULL_EN			0x0	R/W
8	FIFO1_EMPTY_EN			0x0	R/W
[7:6]	RESERVED		Reserved.	0x0	R
5	FIFO1_OV_STATUS			0x0	R
4	FIFO1_UD_STATUS			0x0	R
3	FIFO1_ALMF_STATUS			0x0	R
2	FIFO1_ALME_STATUS			0x0	R
1	FIFO1_FULL_STATUS			0x0	R
0	FIFO1_EMPTY_STATUS			0x1	R

Address: 0x40030034, Reset: 0x00000000, Name: FF1_TH

Table 525. Bit Descriptions for FF1_TH

Bits	Bit Name	Settings	Description	Reset	Access
[31:16]	FIFO1_DLINE_LEN			0x0	R/W
[15:8]	FIFO1_ALME_LEN			0x0	R/W
[7:0]	FIFO1_ALMF_LEN			0x0	R/W

Address: 0x40030040, Reset: 0x00000000, Name: FF2_PORT

Table 526. Bit Descriptions for FF2_PORT

Bits	Bit Name	Settings	Description	Reset	Access
[31:0]	FIFO2_PORT			0x0	R/W

Address: 0x40030044, Reset: 0x00000000, Name: FF2_INIT

Table 527. Bit Descriptions for FF2_INIT

Bits	Bit Name	Settings	Description	Reset	Access
[31:17]	RESERVED		Reserved.	0x0	R
16	FIFO2_WIDTH			0x0	R/W
[15:8]	FIFO2_LEN			0x0	R/W
[7:0]	FIFO2_BASE_ADDR			0x0	R/W

Address: 0x40030048, Reset: 0x00000000, Name: FF2_CSR

Table 528. Bit Descriptions for FF2_CSR

Bits	Bit Name	Settings	Description	Reset	Access
[31:19]	RESERVED		Reserved.	0x0	R
18	FIFO2_DLINE_EN			0x0	R/W
17	FIFO2_EN			0x0	R/W
16	FIFO2_FLUSH			0x0	W
[15:0]	FIFO2_USED			0x0	R

Address: 0x4003004C, Reset: 0x00000000, Name: FF2_PTR

Table 529. Bit Descriptions for FF2_PTR

Bits	Bit Name	Settings	Description	Reset	Access
[31:16]	FIFO2_RD_PTR			0x0	R
[15:0]	FIFO2_WR_PTR			0x0	R

Address: 0x40030050, Reset: 0x00000001, Name: FF2_INT

Table 530. Bit Descriptions for FF2_INT

Bits	Bit Name	Settings	Description	Reset	Access
[31:22]	RESERVED		Reserved.	0x0	R
21	FIFO2_OV_CLR			0x0	W
20	FIFO2_UD_CLR			0x0	W
19	FIFO2_ALMF_CLR			0x0	W
18	FIFO2_ALME_CLR			0x0	W
17	FIFO2_FULL_CLR			0x0	W
16	FIFO2_EMPTY_CLR			0x0	W
[15:14]	RESERVED		Reserved.	0x0	R
13	FIFO2_OV_EN			0x0	R/W
12	FIFO2_UD_EN			0x0	R/W
11	FIFO2_ALMF_EN			0x0	R/W
10	FIFO2_ALME_EN			0x0	R/W
9	FIFO2_FULL_EN			0x0	R/W
8	FIFO2_EMPTY_EN			0x0	R/W
[7:6]	RESERVED		Reserved.	0x0	R
5	FIFO2_OV_STATUS			0x0	R
4	FIFO2_UD_STATUS			0x0	R
3	FIFO2_ALMF_STATUS			0x0	R

Bits	Bit Name	Settings	Description	Reset	Access
2	FIFO2_ALME_STATUS			0x0	R
1	FIFO2_FULL_STATUS			0x0	R
0	FIFO2_EMPTY_STATUS			0x1	R

Address: 0x40030054, Reset: 0x00000000, Name: FF2_TH

Table 531. Bit Descriptions for FF2_TH

Bits	Bit Name	Settings	Description	Reset	Access
[31:16]	FIFO2_DLINE_LEN			0x0	R/W
[15:8]	FIFO2_ALME_LEN			0x0	R/W
[7:0]	FIFO2_ALMF_LEN			0x0	R/W

Address: 0x40030060, Reset: 0x00000000, Name: FF3_PORT

Table 532. Bit Descriptions for FF3_PORT

Bits	Bit Name	Settings	Description	Reset	Access
[31:0]	FIFO3_PORT			0x0	R/W

Address: 0x40030064, Reset: 0x00000000, Name: FF3_INIT

Table 533. Bit Descriptions for FF3_INIT

Bits	Bit Name	Settings	Description	Reset	Access
[31:17]	RESERVED		Reserved.	0x0	R
16	FIFO3_WIDTH			0x0	R/W
[15:8]	FIFO3_LEN			0x0	R/W
[7:0]	FIFO3_BASE_ADDR			0x0	R/W

Address: 0x40030068, Reset: 0x00000000, Name: FF3_CSR

Table 534. Bit Descriptions for FF3_CSR

Bits	Bit Name	Settings	Description	Reset	Access
[31:19]	RESERVED		Reserved.	0x0	R
18	FIFO3_DLINE_EN			0x0	R/W
17	FIFO3_EN			0x0	R/W
16	FIFO3_FLUSH			0x0	W
[15:0]	FIFO3_USED			0x0	R

Address: 0x4003006C, Reset: 0x00000000, Name: FF3_PTR

Table 535. Bit Descriptions for FF3_PTR

Bits	Bit Name	Settings	Description	Reset	Access
[31:16]	FIFO3_RD_PTR			0x0	R
[15:0]	FIFO3_WR_PTR			0x0	R

Address: 0x40030070, Reset: 0x00000001, Name: FF3_INT

Table 536. Bit Descriptions for FF3_INT

Bits	Bit Name	Settings	Description	Reset	Access
[31:22]	RESERVED		Reserved.	0x0	R
21	FIFO3_OV_CLR			0x0	W
20	FIFO3_UD_CLR			0x0	W
19	FIFO3_ALMF_CLR			0x0	W
18	FIFO3_ALME_CLR			0x0	W
17	FIFO3_FULL_CLR			0x0	W
16	FIFO3_EMPTY_CLR			0x0	W
[15:14]	RESERVED		Reserved.	0x0	R
13	FIFO3_OV_EN			0x0	R/W
12	FIFO3_UD_EN			0x0	R/W
11	FIFO3_ALMF_EN			0x0	R/W
10	FIFO3_ALME_EN			0x0	R/W
9	FIFO3_FULL_EN			0x0	R/W
8	FIFO3_EMPTY_EN			0x0	R/W
[7:6]	RESERVED		Reserved.	0x0	R
5	FIFO3_OV_STATUS			0x0	R
4	FIFO3_UD_STATUS			0x0	R
3	FIFO3_ALMF_STATUS			0x0	R
2	FIFO3_ALME_STATUS			0x0	R
1	FIFO3_FULL_STATUS			0x0	R
0	FIFO3_EMPTY_STATUS			0x1	R

Address: 0x40030074, Reset: 0x00000000, Name: FF3_TH

Table 537. Bit Descriptions for FF3_TH

Bits	Bit Name	Settings	Description	Reset	Access
[31:16]	FIFO3_DLINE_LEN			0x0	R/W
[15:8]	FIFO3_ALME_LEN			0x0	R/W
[7:0]	FIFO3_ALMF_LEN			0x0	R/W

REGISTER SUMMARY: DATA TRANSPORTATION BETWEEN DATAPATH AND TDSP

Table 538. DTSYNC (DATA_SYNC) Register Summary

Address	Name	Description	Reset	Access
0x40040300	DS_CTRL		0x000001FF	R/W
0x40040304	DS_DMA_CTRL		0x00000000	R/W
0x40040308	DS_OUT_CTRL		0x0000FFFF	R/W
0x4004030C	DS_INT_MASK		0x0003FFFF	R/W
0x40040310	DS_INT_STATUS		0x00000000	R/W
0x40040314	DS_INT_CLR_CFG		0x0003FFFF	R/W
0x40040340	DS_RDY2DMA_SEL0		0x0000000F	R/W
0x40040344	DS_RDY2DMA_SEL1		0x0000000F	R/W
0x40040348	DS_RDY2DMA_SEL2		0x0000000F	R/W
0x4004034C	DS_RDY2DMA_SEL3		0x0000000F	R/W
0x40040350	DS_RDY2DMA_SEL4		0x0000000F	R/W
0x40040354	DS_RDY2DMA_SEL5		0x0000000F	R/W
0x40040358	DS_RDY2DMA_SEL6		0x0000000F	R/W
0x4004035C	DS_RDY2DMA_SEL7		0x0000000F	R/W
0x40040360	DS_RDY2DMA_SEL8		0x0000000F	R/W
0x40040364	DS_RDY2DMA_SEL9		0x0000000F	R/W
0x40040368	DS_RDY2DMA_SEL10		0x0000000F	R/W
0x4004036C	DS_RDY2DMA_SEL11		0x0000000F	R/W
0x40040370	DS_RDY2DMA_SEL12		0x0000000F	R/W
0x40040374	DS_RDY2DMA_SEL13		0x0000000F	R/W
0x40040378	DS_RDY2DMA_SEL14		0x0000000F	R/W
0x4004037C	DS_RDY2DMA_SEL15		0x0000000F	R/W
0x40040380	DS_RDY2OUT_SEL0		0x0000001F	R/W
0x40040384	DS_RDY2OUT_SEL1		0x0000001F	R/W
0x40040388	DS_RDY2OUT_SEL2		0x0000001F	R/W
0x4004038C	DS_RDY2OUT_SEL3		0x0000001F	R/W
0x40040390	DS_RDY2OUT_SEL4		0x0000001F	R/W
0x40040394	DS_RDY2OUT_SEL5		0x0000001F	R/W
0x40040398	DS_RDY2OUT_SEL6		0x0000001F	R/W
0x4004039C	DS_RDY2OUT_SEL7		0x0000001F	R/W
0x400403A0	DS_RDY2OUT_SEL8		0x0000001F	R/W
0x400403A4	DS_RDY2OUT_SEL9		0x0000001F	R/W
0x400403A8	DS_RDY2OUT_SEL10		0x0000001F	R/W
0x400403AC	DS_RDY2OUT_SEL11		0x0000001F	R/W
0x400403B0	DS_RDY2OUT_SEL12		0x0000001F	R/W
0x400403B4	DS_RDY2OUT_SEL13		0x0000001F	R/W
0x400403B8	DS_RDY2OUT_SEL14		0x0000001F	R/W
0x400403BC	DS_RDY2OUT_SEL15		0x0000001F	R/W
0x400403C0	DS_FIFO_CFG0		0x00000000	R/W
0x400403C4	DS_FIFO_CFG1		0x00000000	R/W
0x400403C8	DS_FIFO_CFG2		0x00000000	R/W
0x400403CC	DS_FIFO_CFG3		0x00000000	R/W
0x400403D0	DS_FIFO_CFG4		0x00000000	R/W
0x400403D4	DS_FIFO_CFG5		0x00000000	R/W
0x400403D8	DS_FIFO_CFG6		0x00000000	R/W
0x400403DC	DS_FIFO_CFG7		0x00000000	R/W

Address	Name	Description	Reset	Access
0x400403E0	DS_FIFO_CFG8		0x00000000	R/W
0x400403E4	DS_FIFO_CFG9		0x00000000	R/W
0x400403E8	DS_FIFO_CFG10		0x00000000	R/W
0x400403EC	DS_FIFO_CFG11		0x00000000	R/W

REGISTER DETAILS: DATA TRANSPORTATION BETWEEN DATAPATH AND TDSP

Address: 0x40040300, Reset: 0x000001FF, Name: DS_CTRL

Table 539. Bit Descriptions for DS_CTRL

Bits	Bit Name	Settings	Description	Reset	Access
[31:25]	RESERVED		Reserved.	0x0	R
24	LT_EN		Enable TIE lookup table interface. 0 is disable; 1 is enable.	0x0	R/W
[23:16]	LT_RDY_WIDTH		Data Ready Signal Width (Unit is Core Clock Period) for TIE Lookup table Reading. When it is zero, data ready signal will be auto clear through reading dedicated address of lookup table.	0x0	R/W
[15:0]	RDY_RATE_DIV		Rate Division for Generating Ready Signal from 24M clock in Data Sync. Default is for 48K.	0x1FF	R/W

Address: 0x40040304, Reset: 0x00000000, Name: DS_DMA_CTRL

Table 540. Bit Descriptions for DS_DMA_CTRL

Bits	Bit Name	Settings	Description	Reset	Access
[31:16]	RESERVED		Reserved.	0x0	R
[15:0]	DMA_REQ_EN		Enable DMA Channels for Data Sync AHB Bus Interface. [0:15] Corresponding to DAM Channel 0 to Channel 15.	0x0	R/W

Address: 0x40040308, Reset: 0x0000FFFF, Name: DS_OUT_CTRL

Table 541. Bit Descriptions for DS_OUT_CTRL

Bits	Bit Name	Settings	Description	Reset	Access
[31:16]	RESERVED		Reserved.	0x0	R
[15:0]	RESYNC2RDY_EN		Output Data Resync to Ready when Enable. [0:15] Corresponding to Output Data Channel 0 to Channel 15.	0xFFFF	R/W

Address: 0x4004030C, Reset: 0x0003FFFF, Name: DS_INT_MASK

Table 542. Bit Descriptions for DS_INT_MASK

Bits	Bit Name	Settings	Description	Reset	Access
[31:18]	RESERVED		Reserved.	0x0	R
17	INT_MASK_FIXED		Mask interrupt from data ready of fixed which is generated by Data Sync.	0x1	R/W
16	INT_MASK_ME		Mask interrupt from data ready of MAC Engine.	0x1	R/W
15	INT_MASK_FDSP		Mask interrupt from data ready of FDSP.	0x1	R/W
14	INT_MASK_FINT67		Mask interrupt from data ready of FINT6 and FINT7.	0x1	R/W
13	INT_MASK_FINT45		Mask interrupt from data ready of FINT4 and FINT5.	0x1	R/W
12	INT_MASK_FINT23		Mask interrupt from data ready of FINT2 and FINT3.	0x1	R/W
11	INT_MASK_FINT01		Mask interrupt from data ready of FINT0 and FINT1.	0x1	R/W
10	INT_MASK_FDEC67		Mask interrupt from data ready of FDEC6 and FDEC7.	0x1	R/W
9	INT_MASK_FDEC45		Mask interrupt from data ready of FDEC4 and FDEC5.	0x1	R/W
8	INT_MASK_FDEC23		Mask interrupt from data ready of FDEC2 and FDEC3.	0x1	R/W
7	INT_MASK_FDEC01		Mask interrupt from data ready of FDEC0 and FDEC1.	0x1	R/W
6	INT_MASK_SPT1		Mask interrupt from data ready of SPT1.	0x1	R/W
5	INT_MASK_SPT0		Mask interrupt from data ready of SPT0.	0x1	R/W
4	INT_MASK_ASRCI		Mask interrupt from data ready of ASRCI.	0x1	R/W
3	INT_MASK_DMIC23		Mask interrupt from data ready of DMIC2 and DMIC3.	0x1	R/W

Bits	Bit Name	Settings	Description	Reset	Access
2	INT_MASK_DMIC01		Mask interrupt from data ready of DMIC0 and DMIC1.	0x1	R/W
1	INT_MASK_ADC23		Mask interrupt from data ready of ADC2.	0x1	R/W
0	INT_MASK_ADC01		Mask interrupt from data ready of ADC0 and ADC1.	0x1	R/W

Address: 0x40040310, Reset: 0x00000000, Name: DS_INT_STATUS

Table 543. Bit Descriptions for DS_INT_STATUS

Bits	Bit Name	Settings	Description	Reset	Access
[31:18]	RESERVED		Reserved.	0x0	R
17	INT_STATUS_FIXED		This bit is set to 1 when data ready(fixed).Write 1 to this bit to clear the interrupt. This can be also reset automatically when it is read if the associated aclear bit is set.	0x0	R/W1C
16	INT_STATUS_ME		This bit is set to 1 when data ready of MAC Engine.Write 1 to this bit to clear the interrupt. This can be also reset automatically when it is read if the associated aclear bit is set.	0x0	R/W1C
15	INT_STATUS_FDSP		This bit is set to 1 when data ready of FDSP.Write 1 to this bit to clear the interrupt. This can be also reset automatically when it is read if the associated aclear bit is set.	0x0	R/W1C
14	INT_STATUS_FINT67		This bit is set to 1 when data ready of FINT6 and FINT7.Write 1 to this bit to clear the interrupt. This can be also reset automatically when it is read if the associated aclear bit is set.	0x0	R/W1C
13	INT_STATUS_FINT45		This bit is set to 1 when data ready of FINT4 and FINT5.Write 1 to this bit to clear the interrupt. This can be also reset automatically when it is read if the associated aclear bit is set.	0x0	R/W1C
12	INT_STATUS_FINT23		This bit is set to 1 when data ready of FINT2 and FINT3.Write 1 to this bit to clear the interrupt. This can be also reset automatically when it is read if the associated aclear bit is set.	0x0	R/W1C
11	INT_STATUS_FINT01		This bit is set to 1 when data ready of FINT0 and FINT1.Write 1 to this bit to clear the interrupt. This can be also reset automatically when it is read if the associated aclear bit is set.	0x0	R/W1C
10	INT_STATUS_FDEC67		This bit is set to 1 when data ready of FDEC6 and FDEC7.Write 1 to this bit to clear the interrupt. This can be also reset automatically when it is read if the associated aclear bit is set.	0x0	R/W1C
9	INT_STATUS_FDEC45		This bit is set to 1 when data ready of FDEC4 and FDEC5. Write 1 to this bit to clear the interrupt. This can be also reset automatically when it is read if the associated aclear bit is set.	0x0	R/W1C
8	INT_STATUS_FDEC23		This bit is set to 1 when data ready of FDEC2 and FDEC3.Write 1 to this bit to clear the interrupt. This can be also reset automatically when it is read if the associated aclear bit is set.	0x0	R/W1C
7	INT_STATUS_FDEC01		This bit is set to 1 when data ready of FDEC0 and FDEC1.Write 1 to this bit to clear the interrupt. This can be also reset automatically when it is read if the associated aclear bit is set.	0x0	R/W1C
6	INT_STATUS_SPT1		This bit is set to 1 when data ready of SPT1.Write 1 to this bit to clear the interrupt. This can be also reset automatically when it is read if the associated aclear bit is set.	0x0	R/W1C
5	INT_STATUS_SPT0		This bit is set to 1 when data ready of SPT0.Write 1 to this bit to clear the interrupt. This can be also reset automatically when it is read if the associated aclear bit is set.	0x0	R/W1C
4	INT_STATUS_ASRCI		This bit is set to 1 when data ready of ASRCI.Write 1 to this bit to clear the interrupt. This can be also reset automatically when it is read if the associated aclear bit is set.	0x0	R/W1C
3	INT_STATUS_DMIC23		This bit is set to 1 when data ready of DMIC2 and DMIC3.Write 1 to this bit to clear the interrupt. This can be also reset automatically when it is read if the associated aclear bit is set.	0x0	R/W1C

Bits	Bit Name	Settings	Description	Reset	Access
2	INT_STATUS_DMIC01		This bit is set to 1 when data ready of DMIC0 and DMIC1.Write 1 to this bit to clear the interrupt. This can be also reset automatically when it is read if the associated aclear bit is set.	0x0	R/W1C
1	INT_STATUS_ADC23		This bit is set to 1 when data ready of ADC2.Write 1 to this bit to clear the interrupt. This can be also reset automatically when it is read if the associated aclear bit is set.	0x0	R/W1C
0	INT_STATUS_ADC01		This bit is set to 1 when data ready of ADC0 and ADC1.Write 1 to this bit to clear the interrupt. This can be also reset automatically when it is read if the associated aclear bit is set.	0x0	R/W1C

Address: 0x40040314, Reset: 0x0003FFFF, Name: DS_INT_CLR_CFG

Table 544. Bit Descriptions for DS_INT_CLR_CFG

Bits	Bit Name	Settings	Description	Reset	Access
[31:18]	RESERVED		Reserved.	0x0	R
17	INT_ACLR_FIXED		Set this bit to 1 automatic clearing of the interrupt when any audio data is read.	0x1	R/W
16	INT_ACLR_ME		Set this bit to 1 automatic clearing of the interrupt when related audio data is read.	0x1	R/W
15	INT_ACLR_FDSP		Set this bit to 1 automatic clearing of the interrupt when related audio data is read.	0x1	R/W
14	INT_ACLR_FINT67		Set this bit to 1 automatic clearing of the interrupt when related audio data is read.	0x1	R/W
13	INT_ACLR_FINT45		Set this bit to 1 automatic clearing of the interrupt when related audio data is read.	0x1	R/W
12	INT_ACLR_FINT23		Set this bit to 1 automatic clearing of the interrupt when related audio data is read.	0x1	R/W
11	INT_ACLR_FINT01		Set this bit to 1 automatic clearing of the interrupt when related audio data is read.	0x1	R/W
10	INT_ACLR_FDEC67		Set this bit to 1 automatic clearing of the interrupt when related audio data is read.	0x1	R/W
9	INT_ACLR_FDEC45		Set this bit to 1 automatic clearing of the interrupt when related audio data is read.	0x1	R/W
8	INT_ACLR_FDEC23		Set this bit to 1 automatic clearing of the interrupt when related audio data is read.	0x1	R/W
7	INT_ACLR_FDEC01		Set this bit to 1 automatic clearing of the interrupt when related audio data is read.	0x1	R/W
6	INT_ACLR_SPT1		Set this bit to 1 automatic clearing of the interrupt when related audio data is read.	0x1	R/W
5	INT_ACLR_SPT0		Set this bit to 1 automatic clearing of the interrupt when related audio data is read.	0x1	R/W
4	INT_ACLR_ASRCI		Set this bit to 1 automatic clearing of the interrupt when related audio data is read.	0x1	R/W
3	INT_ACLR_DMIC23		Set this bit to 1 automatic clearing of the interrupt when related audio data is read.	0x1	R/W
2	INT_ACLR_DMIC01		Set this bit to 1 automatic clearing of the interrupt when related audio data is read.	0x1	R/W
1	INT_ACLR_ADC23		Set this bit to 1 automatic clearing of the interrupt when related audio data is read.	0x1	R/W
0	INT_ACLR_ADC01		Set this bit to 1 automatic clearing of the interrupt when related audio data is read.	0x1	R/W

Address: 0x40040340, Reset: 0x0000000F, Name: DS_RDY2DMA_SEL0

Table 545. Bit Descriptions for DS_RDY2DMA_SEL0

Bits	Bit Name	Settings	Description	Reset	Access
[31:7]	RESERVED		Reserved.	0x0	R
[6:0]	RDY2DMA_SEL0		Select Ready Signal for DMA Channel 0.	0xF	R/W

Address: 0x40040344, Reset: 0x0000000F, Name: DS_RDY2DMA_SEL1

Table 546. Bit Descriptions for DS_RDY2DMA_SEL1

Bits	Bit Name	Settings	Description	Reset	Access
[31:7]	RESERVED		Reserved.	0x0	R
[6:0]	RDY2DMA_SEL1		Select Ready Signal for DMA Channel 1.	0xF	R/W

Address: 0x40040348, Reset: 0x0000000F, Name: DS_RDY2DMA_SEL2

Table 547. Bit Descriptions for DS_RDY2DMA_SEL2

Bits	Bit Name	Settings	Description	Reset	Access
[31:7]	RESERVED		Reserved.	0x0	R
[6:0]	RDY2DMA_SEL2		Select Ready Signal for DMA Channel 2.	0xF	R/W

Address: 0x4004034C, Reset: 0x0000000F, Name: DS_RDY2DMA_SEL3

Table 548. Bit Descriptions for DS_RDY2DMA_SEL3

Bits	Bit Name	Settings	Description	Reset	Access
[31:7]	RESERVED		Reserved.	0x0	R
[6:0]	RDY2DMA_SEL3		Select Ready Signal for DMA Channel 3.	0xF	R/W

Address: 0x40040350, Reset: 0x0000000F, Name: DS_RDY2DMA_SEL4

Table 549. Bit Descriptions for DS_RDY2DMA_SEL4

Bits	Bit Name	Settings	Description	Reset	Access
[31:7]	RESERVED		Reserved.	0x0	R
[6:0]	RDY2DMA_SEL4		Select Ready Signal for DMA Channel 4.	0xF	R/W

Address: 0x40040354, Reset: 0x0000000F, Name: DS_RDY2DMA_SEL5

Table 550. Bit Descriptions for DS_RDY2DMA_SEL5

Bits	Bit Name	Settings	Description	Reset	Access
[31:7]	RESERVED		Reserved.	0x0	R
[6:0]	RDY2DMA_SEL5		Select Ready Signal for DMA Channel 5.	0xF	R/W

Address: 0x40040358, Reset: 0x0000000F, Name: DS_RDY2DMA_SEL6

Table 551. Bit Descriptions for DS_RDY2DMA_SEL6

Bits	Bit Name	Settings	Description	Reset	Access
[31:7]	RESERVED		Reserved.	0x0	R
[6:0]	RDY2DMA_SEL6		Select Ready Signal for DMA Channel 6.	0xF	R/W

Address: 0x4004035C, Reset: 0x0000000F, Name: DS_RDY2DMA_SEL7

Table 552. Bit Descriptions for DS_RDY2DMA_SEL7

Bits	Bit Name	Settings	Description	Reset	Access
[31:7]	RESERVED		Reserved.	0x0	R
[6:0]	RDY2DMA_SEL7		Select Ready Signal for DMA Channel 7.	0xF	R/W

Address: 0x40040360, Reset: 0x0000000F, Name: DS_RDY2DMA_SEL8

Table 553. Bit Descriptions for DS_RDY2DMA_SEL8

Bits	Bit Name	Settings	Description	Reset	Access
[31:7]	RESERVED		Reserved.	0x0	R
[6:0]	RDY2DMA_SEL8		Select Ready Signal for DMA Channel 8.	0xF	R/W

Address: 0x40040364, Reset: 0x0000000F, Name: DS_RDY2DMA_SEL9

Table 554. Bit Descriptions for DS_RDY2DMA_SEL9

Bits	Bit Name	Settings	Description	Reset	Access
[31:7]	RESERVED		Reserved.	0x0	R
[6:0]	RDY2DMA_SEL9		Select Ready Signal for DMA Channel 9.	0xF	R/W

Address: 0x40040368, Reset: 0x0000000F, Name: DS_RDY2DMA_SEL10

Table 555. Bit Descriptions for DS_RDY2DMA_SEL10

Bits	Bit Name	Settings	Description	Reset	Access
[31:7]	RESERVED		Reserved.	0x0	R
[6:0]	RDY2DMA_SEL10		Select Ready Signal for DMA Channel 10.	0xF	R/W

Address: 0x4004036C, Reset: 0x0000000F, Name: DS_RDY2DMA_SEL11

Table 556. Bit Descriptions for DS_RDY2DMA_SEL11

Bits	Bit Name	Settings	Description	Reset	Access
[31:7]	RESERVED		Reserved.	0x0	R
[6:0]	RDY2DMA_SEL11		Select Ready Signal for DMA Channel 11.	0xF	R/W

Address: 0x40040370, Reset: 0x0000000F, Name: DS_RDY2DMA_SEL12

Table 557. Bit Descriptions for DS_RDY2DMA_SEL12

Bits	Bit Name	Settings	Description	Reset	Access
[31:7]	RESERVED		Reserved.	0x0	R
[6:0]	RDY2DMA_SEL12		Select Ready Signal for DMA Channel 12.	0xF	R/W

Address: 0x40040374, Reset: 0x0000000F, Name: DS_RDY2DMA_SEL13

Table 558. Bit Descriptions for DS_RDY2DMA_SEL13

Bits	Bit Name	Settings	Description	Reset	Access
[31:7]	RESERVED		Reserved.	0x0	R
[6:0]	RDY2DMA_SEL13		Select Ready Signal for DMA Channel 13.	0xF	R/W

Address: 0x40040378, Reset: 0x0000000F, Name: DS_RDY2DMA_SEL14

Table 559. Bit Descriptions for DS_RDY2DMA_SEL14

Bits	Bit Name	Settings	Description	Reset	Access
[31:7]	RESERVED		Reserved.	0x0	R
[6:0]	RDY2DMA_SEL14		Select Ready Signal for DMA Channel 14.	0xF	R/W

Address: 0x4004037C, Reset: 0x0000000F, Name: DS_RDY2DMA_SEL15

Table 560. Bit Descriptions for DS_RDY2DMA_SEL15

Bits	Bit Name	Settings	Description	Reset	Access
[31:7]	RESERVED		Reserved.	0x0	R
[6:0]	RDY2DMA_SEL15		Select Ready Signal for DMA Channel 15.	0xF	R/W

Address: 0x40040380, Reset: 0x0000001F, Name: DS_RDY2OUT_SEL0

Table 561. Bit Descriptions for DS_RDY2OUT_SEL0

Bits	Bit Name	Settings	Description	Reset	Access
[31:5]	RESERVED		Reserved.	0x0	R
[4:0]	RDY2OUT_SEL0		Select Ready Signal for Output Channel 0.	0x1F	R/W

Address: 0x40040384, Reset: 0x0000001F, Name: DS_RDY2OUT_SEL1

Table 562. Bit Descriptions for DS_RDY2OUT_SEL1

Bits	Bit Name	Settings	Description	Reset	Access
[31:5]	RESERVED		Reserved.	0x0	R
[4:0]	RDY2OUT_SEL1		Select Ready Signal for Output Channel 1.	0x1F	R/W

Address: 0x40040388, Reset: 0x0000001F, Name: DS_RDY2OUT_SEL2

Table 563. Bit Descriptions for DS_RDY2OUT_SEL2

Bits	Bit Name	Settings	Description	Reset	Access
[31:5]	RESERVED		Reserved.	0x0	R
[4:0]	RDY2OUT_SEL2		Select Ready Signal for Output Channel 2.	0x1F	R/W

Address: 0x4004038C, Reset: 0x0000001F, Name: DS_RDY2OUT_SEL3

Table 564. Bit Descriptions for DS_RDY2OUT_SEL3

Bits	Bit Name	Settings	Description	Reset	Access
[31:5]	RESERVED		Reserved.	0x0	R
[4:0]	RDY2OUT_SEL3		Select Ready Signal for Output Channel 3.	0x1F	R/W

Address: 0x40040390, Reset: 0x0000001F, Name: DS_RDY2OUT_SEL4

Table 565. Bit Descriptions for DS_RDY2OUT_SEL4

Bits	Bit Name	Settings	Description	Reset	Access
[31:5]	RESERVED		Reserved.	0x0	R
[4:0]	RDY2OUT_SEL4		Select Ready Signal for Output Channel 4.	0x1F	R/W

Address: 0x40040394, Reset: 0x0000001F, Name: DS_RDY2OUT_SEL5

Table 566. Bit Descriptions for DS_RDY2OUT_SEL5

Bits	Bit Name	Settings	Description	Reset	Access
[31:5]	RESERVED		Reserved.	0x0	R
[4:0]	RDY2OUT_SEL5		Select Ready Signal for Output Channel 5.	0x1F	R/W

Address: 0x40040398, Reset: 0x0000001F, Name: DS_RDY2OUT_SEL6

Table 567. Bit Descriptions for DS_RDY2OUT_SEL6

Bits	Bit Name	Settings	Description	Reset	Access
[31:5]	RESERVED		Reserved.	0x0	R
[4:0]	RDY2OUT_SEL6		Select Ready Signal for Output Channel 6.	0x1F	R/W

Address: 0x4004039C, Reset: 0x0000001F, Name: DS_RDY2OUT_SEL7

Table 568. Bit Descriptions for DS_RDY2OUT_SEL7

Bits	Bit Name	Settings	Description	Reset	Access
[31:5]	RESERVED		Reserved.	0x0	R
[4:0]	RDY2OUT_SEL7		Select Ready Signal for Output Channel 7.	0x1F	R/W

Address: 0x400403A0, Reset: 0x0000001F, Name: DS_RDY2OUT_SEL8

Table 569. Bit Descriptions for DS_RDY2OUT_SEL8

Bits	Bit Name	Settings	Description	Reset	Access
[31:5]	RESERVED		Reserved.	0x0	R
[4:0]	RDY2OUT_SEL8		Select Ready Signal for Output Channel 8.	0x1F	R/W

Address: 0x400403A4, Reset: 0x0000001F, Name: DS_RDY2OUT_SEL9

Table 570. Bit Descriptions for DS_RDY2OUT_SEL9

Bits	Bit Name	Settings	Description	Reset	Access
[31:5]	RESERVED		Reserved.	0x0	R
[4:0]	RDY2OUT_SEL9		Select Ready Signal for Output Channel 9.	0x1F	R/W

Address: 0x400403A8, Reset: 0x0000001F, Name: DS_RDY2OUT_SEL10

Table 571. Bit Descriptions for DS_RDY2OUT_SEL10

Bits	Bit Name	Settings	Description	Reset	Access
[31:5]	RESERVED		Reserved.	0x0	R
[4:0]	RDY2OUT_SEL10		Select Ready Signal for Output Channel 10.	0x1F	R/W

Address: 0x400403AC, Reset: 0x0000001F, Name: DS_RDY2OUT_SEL11

Table 572. Bit Descriptions for DS_RDY2OUT_SEL11

Bits	Bit Name	Settings	Description	Reset	Access
[31:5]	RESERVED		Reserved.	0x0	R
[4:0]	RDY2OUT_SEL11		Select Ready Signal for Output Channel 11.	0x1F	R/W

Address: 0x400403B0, Reset: 0x0000001F, Name: DS_RDY2OUT_SEL12

Table 573. Bit Descriptions for DS_RDY2OUT_SEL12

Bits	Bit Name	Settings	Description	Reset	Access
[31:5]	RESERVED		Reserved.	0x0	R
[4:0]	RDY2OUT_SEL12		Select Ready Signal for Output Channel 12.	0x1F	R/W

Address: 0x400403B4, Reset: 0x0000001F, Name: DS_RDY2OUT_SEL13

Table 574. Bit Descriptions for DS_RDY2OUT_SEL13

Bits	Bit Name	Settings	Description	Reset	Access
[31:5]	RESERVED		Reserved.	0x0	R
[4:0]	RDY2OUT_SEL13		Select Ready Signal for Output Channel 13.	0x1F	R/W

Address: 0x400403B8, Reset: 0x0000001F, Name: DS_RDY2OUT_SEL14

Table 575. Bit Descriptions for DS_RDY2OUT_SEL14

Bits	Bit Name	Settings	Description	Reset	Access
[31:5]	RESERVED		Reserved.	0x0	R
[4:0]	RDY2OUT_SEL14		Select Ready Signal for Output Channel 14.	0x1F	R/W

Address: 0x400403BC, Reset: 0x0000001F, Name: DS_RDY2OUT_SEL15

Table 576. Bit Descriptions for DS_RDY2OUT_SEL15

Bits	Bit Name	Settings	Description	Reset	Access
[31:5]	RESERVED		Reserved.	0x0	R
[4:0]	RDY2OUT_SEL15		Select Ready Signal for Output Channel 15.	0x1F	R/W

Address: 0x400403C0, Reset: 0x00000000, Name: DS_FIFO_CFG0

Table 577. Bit Descriptions for DS_FIFO_CFG0

Bits	Bit Name	Settings	Description	Reset	Access
[31:18]	RESERVED		Reserved.	0x0	R
17	FIFO_CFG_REN0		Enable to Read FIFO Data to DST.	0x0	R/W
16	FIFO_CFG_WEN0		Enable to Write SRC Data to FIFO.	0x0	R/W
[15:12]	RESERVED		Reserved.	0x0	R
[11:8]	FIFO_CFG_DST0		Select One of 16 Destination Audio Sources.	0x0	R/W
7	RESERVED		Reserved.	0x0	R
[6:0]	FIFO_CFG_SRC0		Select One of 75 Audio Sources.	0x0	R/W

Address: 0x400403C4, Reset: 0x00000000, Name: DS_FIFO_CFG1

Table 578. Bit Descriptions for DS_FIFO_CFG1

Bits	Bit Name	Settings	Description	Reset	Access
[31:18]	RESERVED		Reserved.	0x0	R
17	FIFO_CFG_REN1		Enable to Read FIFO Data to DST.	0x0	R/W
16	FIFO_CFG_WEN1		Enable to Write SRC Data to FIFO.	0x0	R/W
[15:12]	RESERVED		Reserved.	0x0	R

Bits	Bit Name	Settings	Description	Reset	Access
[11:8]	FIFO_CFG_DST1		Select One of 16 Destination Audio Sources.	0x0	R/W
7	RESERVED		Reserved.	0x0	R
[6:0]	FIFO_CFG_SRC1		Select One of 75 Audio Sources.	0x0	R/W

Address: 0x400403C8, Reset: 0x00000000, Name: DS_FIFO_CFG2

Table 579. Bit Descriptions for DS_FIFO_CFG2

Bits	Bit Name	Settings	Description	Reset	Access
[31:18]	RESERVED		Reserved.	0x0	R
17	FIFO_CFG_REN2		Enable to Read FIFO Data to DST.	0x0	R/W
16	FIFO_CFG_WEN2		Enable to Write SRC Data to FIFO.	0x0	R/W
[15:12]	RESERVED		Reserved.	0x0	R
[11:8]	FIFO_CFG_DST2		Select One of 16 Destination Audio Sources.	0x0	R/W
7	RESERVED		Reserved.	0x0	R
[6:0]	FIFO_CFG_SRC2		Select One of 75 Audio Sources.	0x0	R/W

Address: 0x400403CC, Reset: 0x00000000, Name: DS_FIFO_CFG3

Table 580. Bit Descriptions for DS_FIFO_CFG3

Bits	Bit Name	Settings	Description	Reset	Access
[31:18]	RESERVED		Reserved.	0x0	R
17	FIFO_CFG_REN3		Enable to Read FIFO Data to DST.	0x0	R/W
16	FIFO_CFG_WEN3		Enable to Write SRC Data to FIFO.	0x0	R/W
[15:12]	RESERVED		Reserved.	0x0	R
[11:8]	FIFO_CFG_DST3		Select One of 16 Destination Audio Sources.	0x0	R/W
7	RESERVED		Reserved.	0x0	R
[6:0]	FIFO_CFG_SRC3		Select One of 75 Audio Sources.	0x0	R/W

Address: 0x400403D0, Reset: 0x00000000, Name: DS_FIFO_CFG4

Table 581. Bit Descriptions for DS_FIFO_CFG4

Bits	Bit Name	Settings	Description	Reset	Access
[31:18]	RESERVED		Reserved.	0x0	R
17	FIFO_CFG_REN4		Enable to Read FIFO Data to DST.	0x0	R/W
16	FIFO_CFG_WEN4		Enable to Write SRC Data to FIFO.	0x0	R/W
[15:12]	RESERVED		Reserved.	0x0	R
[11:8]	FIFO_CFG_DST4		Select One of 16 Destination Audio Sources.	0x0	R/W
7	RESERVED		Reserved.	0x0	R
[6:0]	FIFO_CFG_SRC4		Select One of 75 Audio Sources.	0x0	R/W

Address: 0x400403D4, Reset: 0x00000000, Name: DS_FIFO_CFG5

Table 582. Bit Descriptions for DS_FIFO_CFG5

Bits	Bit Name	Settings	Description	Reset	Access
[31:18]	RESERVED		Reserved.	0x0	R
17	FIFO_CFG_REN5		Enable to Read FIFO Data to DST.	0x0	R/W
16	FIFO_CFG_WEN5		Enable to Write SRC Data to FIFO.	0x0	R/W
[15:12]	RESERVED		Reserved.	0x0	R

Bits	Bit Name	Settings	Description	Reset	Access
[11:8]	FIFO_CFG_DST5		Select One of 16 Destination Audio Sources.	0x0	R/W
7	RESERVED		Reserved.	0x0	R
[6:0]	FIFO_CFG_SRC5		Select One of 75 Audio Sources.	0x0	R/W

Address: 0x400403D8, Reset: 0x00000000, Name: DS_FIFO_CFG6

Table 583. Bit Descriptions for DS_FIFO_CFG6

Bits	Bit Name	Settings	Description	Reset	Access
[31:18]	RESERVED		Reserved.	0x0	R
17	FIFO_CFG_REN6		Enable to Read FIFO Data to DST.	0x0	R/W
16	FIFO_CFG_WEN6		Enable to Write SRC Data to FIFO.	0x0	R/W
[15:12]	RESERVED		Reserved.	0x0	R
[11:8]	FIFO_CFG_DST6		Select One of 16 Destination Audio Sources.	0x0	R/W
7	RESERVED		Reserved.	0x0	R
[6:0]	FIFO_CFG_SRC6		Select One of 75 Audio Sources.	0x0	R/W

Address: 0x400403DC, Reset: 0x00000000, Name: DS_FIFO_CFG7

Table 584. Bit Descriptions for DS_FIFO_CFG7

Bits	Bit Name	Settings	Description	Reset	Access
[31:18]	RESERVED		Reserved.	0x0	R
17	FIFO_CFG_REN7		Enable to Read FIFO Data to DST.	0x0	R/W
16	FIFO_CFG_WEN7		Enable to Write SRC Data to FIFO.	0x0	R/W
[15:12]	RESERVED		Reserved.	0x0	R
[11:8]	FIFO_CFG_DST7		Select One of 16 Destination Audio Sources.	0x0	R/W
7	RESERVED		Reserved.	0x0	R
[6:0]	FIFO_CFG_SRC7		Select One of 75 Audio Sources.	0x0	R/W

Address: 0x400403E0, Reset: 0x00000000, Name: DS_FIFO_CFG8

Table 585. Bit Descriptions for DS_FIFO_CFG8

Bits	Bit Name	Settings	Description	Reset	Access
[31:18]	RESERVED		Reserved.	0x0	R
17	FIFO_CFG_REN8		Enable to Read FIFO Data to DST.	0x0	R/W
16	FIFO_CFG_WEN8		Enable to Write SRC Data to FIFO.	0x0	R/W
[15:12]	RESERVED		Reserved.	0x0	R
[11:8]	FIFO_CFG_DST8		Select One of 16 Destination Audio Sources.	0x0	R/W
7	RESERVED		Reserved.	0x0	R
[6:0]	FIFO_CFG_SRC8		Select One of 75 Audio Sources.	0x0	R/W

Address: 0x400403E4, Reset: 0x00000000, Name: DS_FIFO_CFG9

Table 586. Bit Descriptions for DS_FIFO_CFG9

Bits	Bit Name	Settings	Description	Reset	Access
[31:18]	RESERVED		Reserved.	0x0	R
17	FIFO_CFG_REN9		Enable to Read FIFO Data to DST.	0x0	R/W
16	FIFO_CFG_WEN9		Enable to Write SRC Data to FIFO.	0x0	R/W
[15:12]	RESERVED		Reserved.	0x0	R

Bits	Bit Name	Settings	Description	Reset	Access
[11:8]	FIFO_CFG_DST9		Select One of 16 Destination Audio Sources.	0x0	R/W
7	RESERVED		Reserved.	0x0	R
[6:0]	FIFO_CFG_SRC9		Select One of 75 Audio Sources.	0x0	R/W

Address: 0x400403E8, Reset: 0x00000000, Name: DS_FIFO_CFG10

Table 587. Bit Descriptions for DS_FIFO_CFG10

Bits	Bit Name	Settings	Description	Reset	Access
[31:18]	RESERVED		Reserved.	0x0	R
17	FIFO_CFG_REN10		Enable to Read FIFO Data to DST.	0x0	R/W
16	FIFO_CFG_WEN10		Enable to Write SRC Data to FIFO.	0x0	R/W
[15:12]	RESERVED		Reserved.	0x0	R
[11:8]	FIFO_CFG_DST10		Select One of 16 Destination Audio Sources.	0x0	R/W
7	RESERVED		Reserved.	0x0	R
[6:0]	FIFO_CFG_SRC10		Select One of 75 Audio Sources.	0x0	R/W

Address: 0x400403EC, Reset: 0x00000000, Name: DS_FIFO_CFG11

Table 588. Bit Descriptions for DS_FIFO_CFG11

Bits	Bit Name	Settings	Description	Reset	Access
[31:18]	RESERVED		Reserved.	0x0	R
17	FIFO_CFG_REN11		Enable to Read FIFO Data to DST.	0x0	R/W
16	FIFO_CFG_WEN11		Enable to Write SRC Data to FIFO.	0x0	R/W
[15:12]	RESERVED		Reserved.	0x0	R
[11:8]	FIFO_CFG_DST11		Select One of 16 Destination Audio Sources.	0x0	R/W
7	RESERVED		Reserved.	0x0	R
[6:0]	FIFO_CFG_SRC11		Select One of 75 Audio Sources.	0x0	R/W

SYSTEM BLOCK DIAGRAM

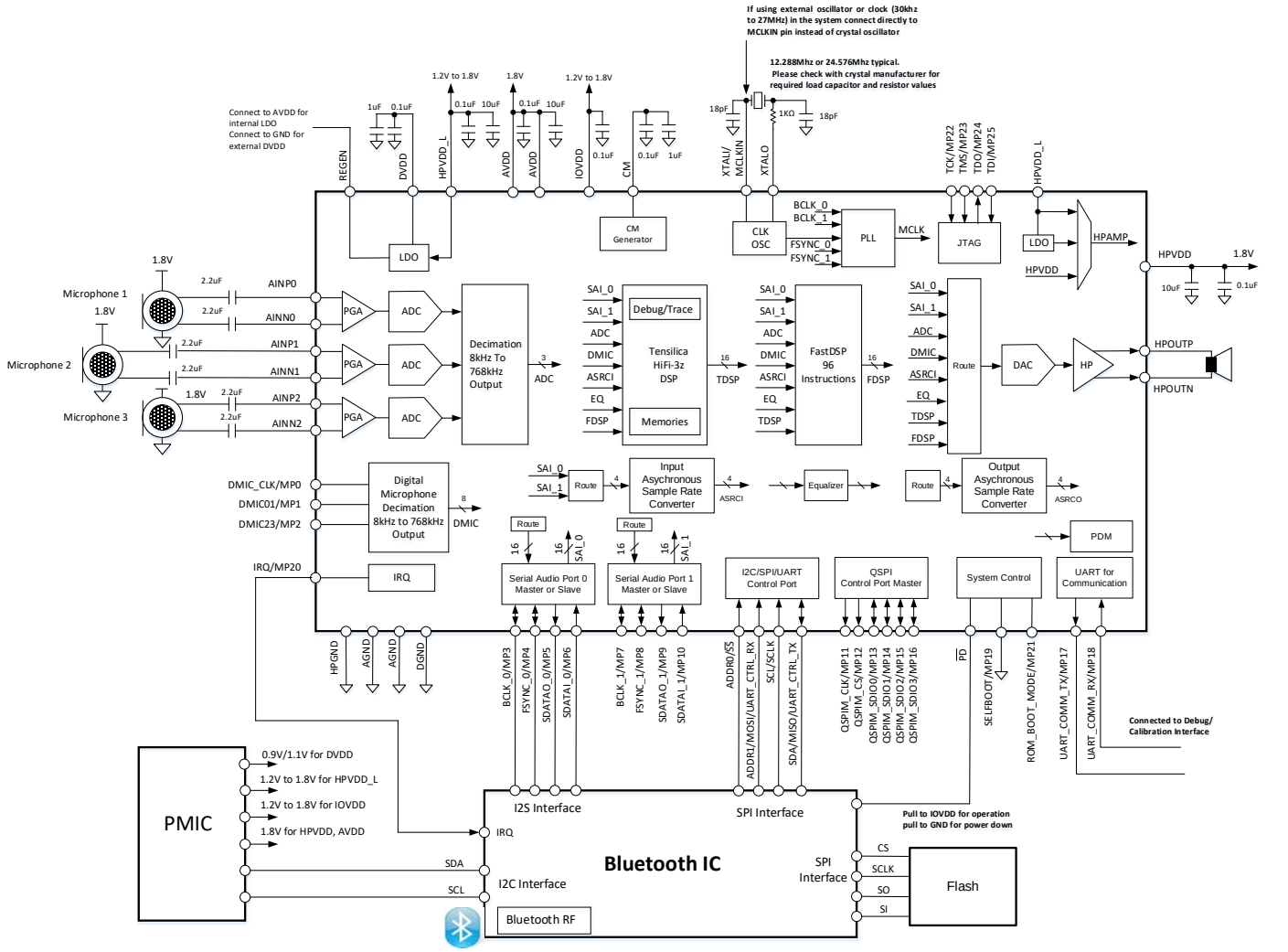


Figure 27. ADAU186x System Block Diagram with Analog Microphones, Non-Self-Boot Mode

ESD Caution

ESD (electrostatic discharge) sensitive device. Charged devices and circuit boards can discharge without detection. Although this product features patented or proprietary protection circuitry, damage may occur on devices subjected to high energy ESD. Therefore, proper ESD precautions should be taken to avoid performance degradation or loss of functionality.

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UG2257-7/24(0)

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